

An Introduction to Chromatographic Separations

There are very few, if any, methods for chemical analysis that are specific for a single chemical species. At best, analytical methods are selective for a few species or a class of species. As a result, the separation of the analyte from potential interferences is quite often a vital step in analytical procedures. Until the middle of the twentieth century, analytical separations were largely carried out by such classical methods as precipitation, distillation, and extraction. Today, however, analytical separations are most commonly carried out by **chromatography** and **electrophoresis**, particularly with samples that are multicomponent and complex.

Chromatography comprises a group of powerful separation methods that find applications in all branches of science. Chromatography was invented and named by the Russian botanist Mikhail Tswett shortly after the turn of the last century. He used the technique to separate various plant pigments such as chlorophylls and xanthophylls by passing solutions of these compounds through a glass column packed with finely divided calcium carbonate. The separated species appeared as colored bands on the column, which accounts for the name he chose for the method (Greek *chroma* meaning “color” and *graphein* meaning “writing”).

The applications of chromatography have grown explosively in the last half century, due not only to the development of several new types of chromatographic techniques but also to the growing need by scientists for better methods for characterizing complex mixtures. The tremendous impact of these methods on science is attested by the 1952 Nobel Prize in Chemistry that was awarded to A. J. P. Martin and R. L. M. Synge for their discoveries in the field. Many of the Nobel Prizes awarded since that time have been based on work in which chromatography played a vital role.

26A GENERAL DESCRIPTION OF CHROMATOGRAPHY

Chromatography encompasses a diverse and important group of methods that allow the separation, identification, and determination of closely related components of complex mixtures; many of these separations are impossible by other means.¹ In all chromatographic separations the sample is dissolved in a *mobile*



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¹Some general references on chromatography include J. M. Miller, *Chromatography: Concepts and Contrasts*, 2nd ed., New York: Wiley, 2009; *Chromatography: Fundamentals and Applications of Chromatography and Related Differential Migration Methods*, 6th ed., E. F. Heftman, ed., Amsterdam, Boston: Elsevier, 2004; C. F. Poole, *The Essence of Chromatography*, Amsterdam, Boston: Elsevier, 2003; J. Cazes and R. P. W. Scott, *Chromatography Theory*, New York: Dekker, 2002; A. Braithwaite and F. J. Smith, *Chromatographic Methods*, 5th ed., London: Blackie, 1996; R. P. W. Scott, *Techniques and Practice of Chromatography*, New York: Marcel Dekker, 1995; J. C. Giddings, *Unified Separation Science*, New York: Wiley, 1991.

TABLE 26-1 Classification of Column Chromatographic Methods

General Classification	Specific Method	Stationary Phase	Type of Equilibrium
1. Gas chromatography (GC)	a. Gas-liquid chromatography (GLC)	Liquid adsorbed or bonded to a solid surface	Partition between gas and liquid
	b. Gas-solid	Solid	Adsorption
2. Liquid chromatography (LC)	a. Liquid-liquid (partition)	Liquid adsorbed or bonded to a solid surface	Interaction with or partitioning between immiscible liquids*
	b. Liquid-solid or adsorption	Solid	Adsorption
	c. Ion exchange	Ion-exchange resin	Ion exchange
	d. Size exclusion	Liquid in interstices of a polymeric solid	Partition/sieving
	e. Affinity	Group-specific liquid bonded to a solid surface	Partition between surface liquid and mobile liquid
3. Supercritical fluid chromatography (SFC; mobile phase: supercritical fluid)		Organic species bonded to a solid surface	Partition between supercritical fluid and bonded surface

*In the early days of LC, partitioning (solubility) of the solute between immiscible phases was the major mode of retention. In modern LC, intermolecular interactions, including dispersion, dipole-dipole, hydrogen bonding, charge-transfer and ionic, are the major contributors to retention and selectivity.

phase, which may be a gas, a liquid, or a supercritical fluid. This mobile phase is then forced through an immiscible *stationary phase*, which is fixed in place in a column or on a solid surface. The two phases are chosen so that the components of the sample distribute themselves between the mobile and stationary phases to varying degrees. Those components strongly *retained* by the stationary phase move only slowly with the flow of mobile phase. In contrast, components that are weakly held by the stationary phase travel rapidly. Because of these differences in migration rates, sample components separate into discrete *bands*, or *zones*, that can be analyzed qualitatively and quantitatively.

26A-1 Classification of Chromatographic Methods

Chromatographic methods can be categorized in two ways. The first classification is based on the physical means by which the stationary and mobile phases are brought into contact. In *column chromatography*, the stationary phase is held in a narrow tube through which the mobile phase is forced under pressure. In *planar chromatography*, the stationary phase is supported on a flat plate or in the interstices of paper. In this arrangement, the mobile phase moves through the stationary phase by capillary action or under the influence of gravity. The discussion in this and the next three chapters focuses on column chromatography. Section 28I is devoted to planar methods. It is important to point out, however, that the equilibria on which the two types of chromatography are based are identical and that the theory developed for column chromatography is readily adapted to planar chromatography.

A more fundamental classification of chromatographic methods is one based on the types of mobile and stationary phases and the kinds of equilibria involved in the transfer of solutes between phases. Table 26-1 lists three general categories of chromatography: *gas chromatography* (GC), *liquid chromatography* (LC), and *supercritical fluid chromatography* (SFC). As the names imply, the mobile phases in the three techniques are gases, liquids, and supercritical fluids, respectively. As shown in column 2 of the table, several specific chromatographic methods fall into each of the first two general categories.

Note that only LC can be performed either in columns or on planar surfaces; GC and SFC, on the other hand, are restricted to column procedures so that the column walls enclose the mobile phase.

26A-2 Elution in Column Chromatography

Figure 26-1 shows schematically how two substances A and B are separated on a packed column by *elution*. The column consists of a narrow-bore tubing packed with a finely divided inert solid that holds the stationary phase on its surface. The mobile phase occupies the open spaces between the particles of the packing. Initially, a solution of the sample containing a mixture of A and B in the mobile phase is introduced at the head of the column as a narrow plug, as shown in Figure 26-1 at time t_0 . The two components distribute themselves between the mobile and the stationary phases as fresh mobile phase is added continuously and A and B are washed through the column. This process is called *elution*. In modern chromatography, continuous

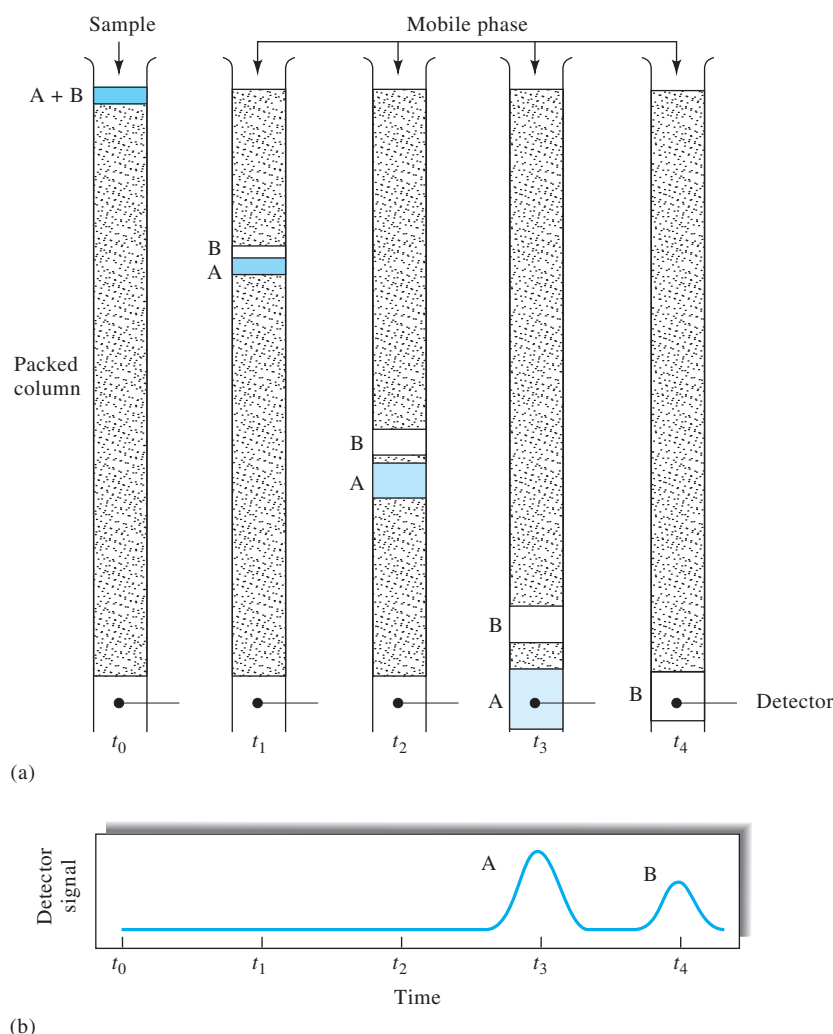


FIGURE 26-1 (a) Diagram showing the separation of a mixture of components A and B by column elution chromatography. (b) The output of the signal detector at the various stages of elution shown in (a).

addition is accomplished by means of a pump in LC and SFC or by the application of pressure in GC.

With the first introduction of fresh mobile phase, the *eluent*—the portion of the sample contained in the mobile phase—moves down the column, where it further distributes between the mobile phase and the stationary phase (time t_1). Interaction of the solute with the fresh mobile phase and the stationary phase takes place as the mobile phase washes the sample through the column.

As fresh mobile phase flows through the column, it carries solute molecules down the column in a continuous series of interactions with the two phases. Because solute movement can occur only in the mobile phase, the average *rate* at which a

solute zone migrates down the column *depends on the fraction of time it spends in the mobile phase*. This fraction is small for solutes strongly retained by the stationary phase (component B in Figure 26-1) and large when the solute resides mostly in the mobile phase (component A). Ideally, the resulting differences in rates cause the components in a mixture to separate into *bands*, or *zones*, along the length of the column (see time t_2 in Figure 26-1). Isolation of the separated species is then accomplished by passing a sufficient quantity of mobile phase through the column to cause the individual zones to pass out the end (to be *eluted* from the column), where they can be detected or collected (times t_3 and t_4 in Figure 26-1).

Analyte Dilution

Figure 26-1 illustrates an important general characteristic of the chromatographic process—namely, dilution of analytes always accompanies chromatographic separation. Thus, the



Simulation: Learn more about **elution chromatography** at www.tinyurl.com/skoogpia7

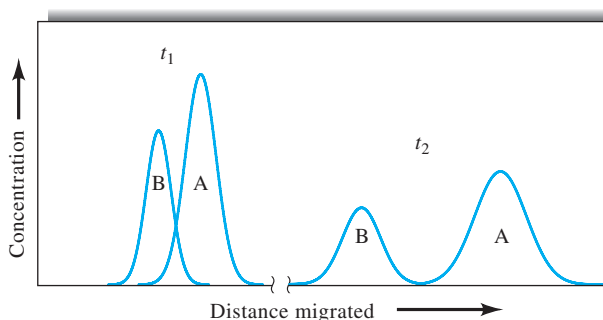


FIGURE 26-2 Concentration profiles of solute bands A and B at two different times in their migration down the column in Figure 26-1. The times t_1 and t_2 are indicated in Figure 26-1.

size of the original zone containing the analytes in the figure is noticeably smaller than either of the two zones that reach the detector, meaning that significant dilution of the analytes and band spreading have occurred while they were being separated. As a result, the detectors employed for separated analytes must often be more sensitive than would be required if the separation process were unnecessary.

Chromatograms

If a detector that responds to solute concentration is placed at the end of the column and its signal is plotted as function of time (or of volume of the added mobile phase), a series of peaks is obtained, as shown in the lower part of Figure 26-1. Such a plot, called a *chromatogram*, is useful for both qualitative and quantitative analysis. The positions of peaks on the time axis can be used to identify the components of the sample; the areas under the peaks provide a quantitative measure of the amount of each component.

Improving Column Performance

Figure 26-2 shows concentration profiles for the bands containing solutes A and B in Figure 26-1 at time t_1 and a later time t_2 .² Because species B is more strongly retained by the stationary phase than is A, B lags during the migration. Note that movement down the column increases the distance between the two bands. At the same time, however, broadening of both zones takes place, which lowers the efficiency of the column as a separating device. While band broadening is inevitable, conditions can ordinarily be found where it occurs more slowly than band separation. Thus, as shown in Figures 26-1 and 26-2, a clean separation of species is often possible provided the column is sufficiently long.

²Note that the relative positions of bands for A and B in the concentration profile shown in Figure 26-2 appear to be reversed from their positions in the lower part of Figure 26-1. The difference is that the abscissa represents distance along the column in Figure 26-2 but time in Figure 26-1. Thus, in Figure 26-1 the front of a peak lies to the left and the tail to the right; in Figure 26-2, the reverse is true.

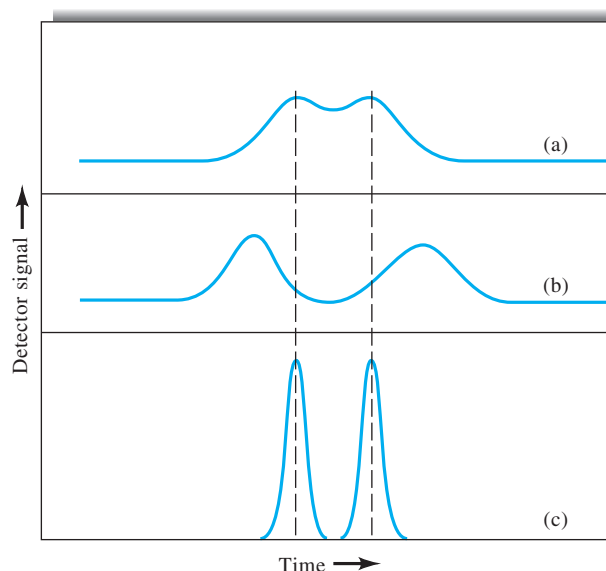


FIGURE 26-3 Two-component chromatogram illustrating two methods for improving separation: (a) original chromatogram with overlapping peaks, (b) improvement brought about by an increase in band separation, and (c) improvement brought about by a decrease in the widths.

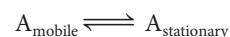
Several chemical and physical variables influence the rates of band separation and band broadening. Separations can often be improved by controlling variables that either (1) increase the rate of band separation or (2) decrease the rate of band spreading. These alternatives are illustrated in Figure 26-3. Variables that influence the relative migration rate of solute zones through a stationary phase are described in the next section. In Section 26C, we consider factors that influence the rate of zone broadening.

26B MIGRATION RATES OF SOLUTES

The effectiveness of a chromatographic column in separating two solutes depends in part on the relative rates at which the two species are eluted. These rates are determined by the magnitude of the equilibrium constants for the reactions by which the solutes distribute themselves between the mobile and stationary phases.

26B-1 Distribution Constants

Often, the distribution equilibria of chromatography are represented by relatively straightforward equations that describe the transfer of an analyte between the mobile and stationary phases. Thus, for the solute species A, we may write



The equilibrium constant K_c for the distribution of species A between the two phases is called the *distribution constant*, which is defined as

$$K_c = \frac{(a_A)_S}{(a_A)_M} \quad (26-1)$$

where $(a_A)_S$ is the activity of solute A in the stationary phase and $(a_A)_M$ is the activity in the mobile phase. When concentrations are low or when nonionic species are present, activity coefficients (see Appendix 2) are nearly unity. Under these conditions, we often substitute for activities c_S , the molar analytical concentration of the solute in the stationary phase, and c_M , its molar analytical concentration in the mobile phase. Hence, we often write Equation 26-1 as

$$K_c = \frac{c_S}{c_M} = \frac{n_S/V_S}{n_M/V_M} \quad (26-2)$$

where n_S and n_M are the number of moles of analyte in the two phases and V_S and V_M are the volumes of the two phases. Ideally, the distribution constant, sometimes called a *partition ratio* or *partition coefficient*,³ is constant over a wide range of solute concentrations; that is, c_S is directly proportional to c_M . Chromatography in which Equation 26-2 applies is termed *linear chromatography* and results in such characteristics as symmetric Gaussian-type peaks and retention times independent of the amount of analyte injected. We limit our theoretical discussions here to linear chromatography.

Note that K_c is the fundamental quantity affecting distribution of components between phases and thus separations. By appropriate choice of the mobile phase, the stationary phase, or both, the distribution constant can be manipulated within limits. By adjusting the volume of a phase, we can alter the molar ratio in the two phases. The model presented here is useful even though the interactions with the two phases may not involve true partitioning.

26B-2 Retention Time

Although the distribution constant is fundamental to chromatographic separations, it is not readily measured. Instead, we can measure a quantity called the *retention time* that is a function of K_c . To see how this is done, let us refer to Figure 26-4, a simple chromatogram made up of two peaks. The small peak on the left is for a species that is *not* retained by the column. Often, the sample or the mobile phase contains an unretained species. When it does not, such a species may be added to aid in peak identification. The time t_M for the unretained species to reach

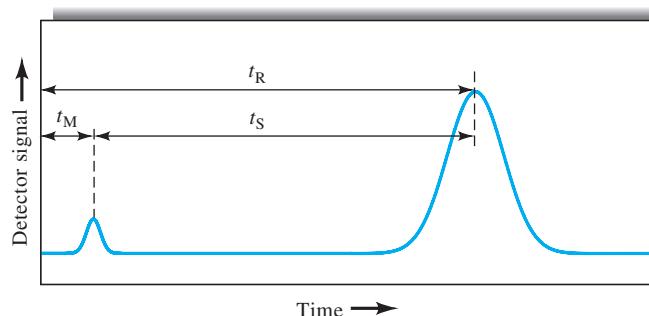


FIGURE 26-4 A typical chromatogram for a two-component mixture. The small peak on the left represents a solute that is not retained on the column and so reaches the detector almost immediately after elution is begun. Thus, its retention time t_M is approximately equal to the time required for a molecule of the mobile phase to pass through the column.

the detector is sometimes called the *dead*, or *void*, *time*. The dead time provides a measure of the average rate of migration of the mobile phase and is an important parameter in identifying analyte peaks. All components spend time t_M in the mobile phase. The larger peak on the right in Figure 26-4 is that of an analyte species. The time required for this zone to reach the detector after sample injection is called the *retention time* and is given the symbol t_R . The analyte has been retained because it spends a time t_S in the stationary phase. The retention time is then

$$t_R = t_S + t_M \quad (26-3)$$

The average linear rate of solute migration through the column $\bar{v}$ (usually cm/s) is

$$\bar{v} = \frac{L}{t_R} \quad (26-4)$$

where L is the length of the column packing. Similarly, the average linear velocity u of the mobile-phase molecules is

$$u = \frac{L}{t_M} \quad (26-5)$$

26B-3 The Relationship between Volumetric Flow Rate and Linear Flow Velocity

Experimentally, in chromatography the mobile-phase flow is usually characterized by the volumetric flow rate F (cm^3/min) at the column outlet. For an open tubular column, F is related to the linear velocity at the column outlet u_o

$$F = u_o A = u_o \times \pi r^2 \quad (26-6)$$

where A is the cross-sectional area of the tube (πr^2). For a packed column, the entire column volume is not available to the liquid and so Equation 26-6 must be modified to

$$F = \pi r^2 u_o \epsilon \quad (26-7)$$

³For recommendations of the IUPAC Commission on Analytical Nomenclature for chromatography, see L. S. Ettre, *Pure Appl. Chem.*, **1993**, 65, 819, DOI: 10.1351/pac199365040819. The committee recommends the use of the term distribution constant K_c rather than the older term “partition coefficient” or “partition ratio,” K . Be aware that both terms are found in the chromatographic literature.

where ε is the fraction of the total column volume available to the liquid (column porosity).

26B-4 The Relationship between Retention Time and Distribution Constant

To relate the rate of migration of a solute to its distribution constant, we express the rate as a fraction of the velocity of the mobile phase:

$$\bar{v} = u \times (\text{fraction of time solute spends in mobile phase})$$

This fraction, however, equals the average number of moles of solute in the mobile phase at any instant divided by the total number of moles of solute in the column:

$$\bar{v} = u \times \frac{\text{moles of solute in mobile phase}}{\text{total moles of solute}}$$

The total number of moles of solute in the mobile phase is equal to the molar concentration, c_M , of the solute in that phase multiplied by its volume, V_M . Similarly, the number of moles of solute in the stationary phase is given by the product of the concentration, c_S , of the solute in the stationary phase and its volume, V_S . Therefore, the total number of moles of solute is $c_S V_S + c_M V_M$, and

$$\bar{v} = u \times \frac{c_M V_M}{c_M V_M + c_S V_S} = u \times \frac{1}{1 + c_S V_S / c_M V_M}$$

Substitution of Equation 26-2 into this equation gives an expression for the rate of solute migration as a function of its distribution constant as well as a function of the volumes of the stationary and mobile phases:

$$\bar{v} = u \times \frac{1}{1 + K_A V_S / V_M} \quad (26-8)$$

The two volumes can be estimated from the method by which the column is prepared.

26B-5 The Rate of Solute Migration: The Retention Factor

The *retention factor* k is an important experimental quantity widely used to compare the migration rates of solutes in columns.⁴ The reason that k is so useful is that it does not depend on column geometry or on volumetric flow rate. This means that for a given combination of solute, mobile phase, and stationary phase any column of any geometry operated at any mobile-phase flow rate will give the same retention factor. For solute A, the retention factor k_A is defined as

$$k_A = \frac{K_A V_S}{V_M} \quad (26-9)$$

⁴In the older literature, this constant was called the capacity factor and symbolized by k' . In 1993, however, the IUPAC Commission on Analytical Nomenclature recommended that this constant be termed the *retention factor* and symbolized by k .

where K_A is the distribution constant for solute A. Substitution of Equation 26-9 into 26-8 yields

$$\bar{v} = u \times \frac{1}{1 + k_A} \quad (26-10)$$

To show how k_A can be derived from a chromatogram, we substitute Equations 26-4 and 26-5 into Equation 26-10:

$$\frac{L}{t_R} = \frac{L}{t_M} \times \frac{1}{1 + k_A} \quad (26-11)$$

This equation rearranges to

$$k_A = \frac{t_R - t_M}{t_M} \quad (26-12)$$

Note that the time spent in the stationary phase, $t_R - t_M$, is sometimes called the *adjusted retention time* and given the symbol t'_R .

As shown in Figure 26-4, t_R and t_M are readily obtained from a chromatogram. The retention factor is then found from these quantities and Equation 26-12. A retention factor much less than unity means that the solute emerges from the column at a time near that of the void time. When the retention factor is larger than perhaps 20 to 30, elution times become inordinately long. Ideally, separations are performed under conditions in which the retention factors for the solutes in a mixture lie in the range between 1 and 10. Optimization of separations (see Section 26D) is then done by optimizing k values for the components of interest.

26B-6 Relative Migration Rates: The Selectivity Factor

The *selectivity factor* α of a column for the two solutes A and B is defined as

$$\alpha = \frac{K_B}{K_A} \quad (26-13)$$

where K_B is the distribution constant for the more strongly retained species B and K_A is the constant for the less strongly held or more rapidly eluted species A. According to this definition, α is always greater than unity.

Substitution of Equation 26-9 and the analogous equation for solute B into Equation 26-13 provides a relationship between the selectivity factor for two solutes and their retention factors:

$$\alpha = \frac{k_B}{k_A} \quad (26-14)$$

where k_B and k_A are the retention factors for B and A, respectively. Substitution of Equation 26-12 for the two solutes into Equation 26-14 gives an expression that permits the determination of α from an experimental chromatogram:

$$\alpha = \frac{(t_R)_B - t_M}{(t_R)_A - t_M} \quad (26-15)$$

In Section 26D-2, we show how we use the selectivity and retention factors to compute the resolving power of a column.

26C BAND BROADENING AND COLUMN EFFICIENCY

The efficiency of a chromatographic column is affected by the amount of band broadening that occurs as a compound passes through the column. Before defining column efficiency in more quantitative terms, let us examine the reasons that bands become broader as they move down a column.

26C-1 The Rate Theory of Chromatography

The *rate theory* of chromatography describes the shapes and breadths of elution bands in quantitative terms based on a random-walk mechanism for the migration of molecules through a column. A detailed discussion of the rate theory is beyond the scope of this text. We can, however, give a qualitative picture of why bands broaden and what variables improve column efficiency.⁵

If you examine the chromatograms shown in this and the next chapter, you will see that the elution peaks look very much like the Gaussian or normal error curves (see Appendix 1). As shown in Appendix 1, Section 1B-1, normal error curves are rationalized by assuming that the uncertainty associated with any single measurement is the summation of a much larger number of small, individually undetectable and random uncertainties, each of which has an equal probability of being positive or negative. In a similar way, the typical Gaussian shape of a chromatographic band can be attributed to the additive combination of the random motions of the various molecules as they move down the column. We assume in the following discussion that a narrow zone has been introduced so that the injection width is not the limiting factor determining the overall width of the band that elutes. It is important to realize that the widths of eluting bands can never be *narrower* than the width of the injection zone.

It is instructive to consider a single solute molecule as it undergoes many thousands of transfers between the stationary and the mobile phases during elution. Residence time in either phase is highly irregular. Transfer from one phase to the other requires energy, and the molecule must acquire this energy from its surroundings. Thus, the residence time in a given phase may be transitory after some transfers and relatively long after others. Recall that movement down the column can occur *only while the molecule is in the mobile phase*. The result is that certain particles travel rapidly by virtue of their accidental inclusion in the mobile phase for most of the time while others lag because they happen to be incorporated in the stationary phase for a greater-than-average length of time. The result of these random individual processes is a symmetric spread of velocities around

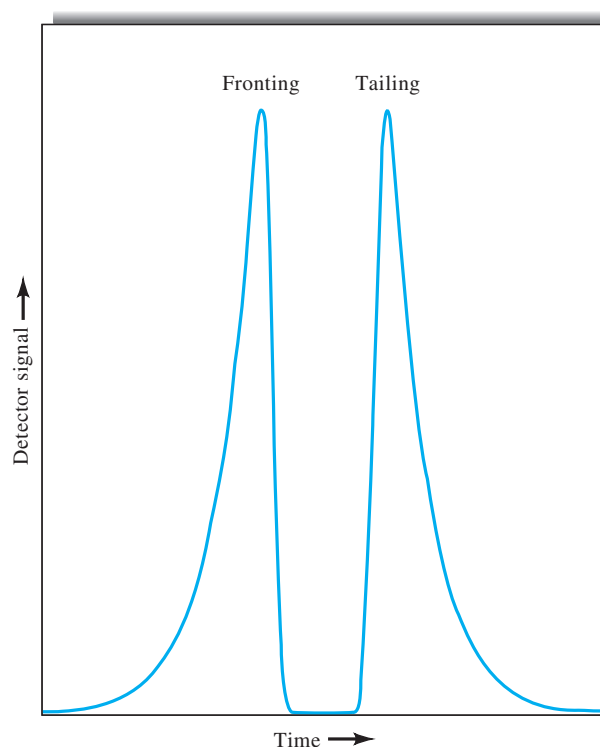


FIGURE 26-5 Illustration of fronting and tailing in chromatographic peaks.

the mean value, which represents the behavior of the average analyte molecule.

The breadth of a zone increases as it moves down the column because more time is allowed for spreading to occur. Thus, the zone breadth is directly related to residence time in the column and inversely related to the velocity of the mobile-phase flow.

As shown in Figure 26-5, some chromatographic peaks are nonideal and exhibit *tailing* or *fronting*. In the former case the tail of the peak, appearing to the right on the chromatogram, is drawn out and the front is steepened. With fronting, the reverse is the case. A common cause of tailing and fronting is a distribution constant that varies with concentration. Fronting also arises when the amount of sample introduced onto a column is too large. Distortions of this kind are undesirable because they lead to poorer separations and less-reproducible elution times. In the discussion that follows, tailing and fronting are assumed to be minimal.

26C-2 A Quantitative Description of Column Efficiency

Two related terms are widely used as quantitative measures of chromatographic column efficiency: (1) *plate height* H and (2) *plate count*, or *number of theoretical plates*, N . The two are related by the equation

$$N = \frac{L}{H} \quad (26-16)$$

⁵For more information, see J. C. Giddings, *Unified Separation Science*, New York: Wiley, 1991, pp. 94–96.

where L is the length (usually in centimeters) of the column packing. The efficiency of chromatographic columns increases as the plate count N becomes greater and as the plate height H becomes smaller. Enormous differences in efficiencies are encountered in columns as a result of differences in column type and in mobile and stationary phases. Efficiencies in terms of plate numbers can vary from a few hundred to several hundred thousand; plate heights ranging from a few tenths to one thousandth of a centimeter or smaller are not uncommon.

The genesis of the terms “plate height” and “number of theoretical plates” is a pioneering theoretical study of Martin and Synge in which they treated a chromatographic column as if it were similar to a distillation column made up of numerous discrete but contiguous narrow layers called *theoretical plates*.⁶ At each plate, equilibration of the solute between the mobile and stationary phases was assumed to take place. Movement of the solute down the column was then treated as a stepwise transfer of equilibrated mobile phase from one plate to the next.

The *plate theory* successfully accounts for the Gaussian shape of chromatographic peaks and their rate of movement down a column. The theory was ultimately abandoned in favor of the *rate theory*, however, because it fails to account for peak broadening in a mechanistic way. Nevertheless, the original terms for efficiency have been carried over to the rate theory. This nomenclature is perhaps unfortunate because it tends to perpetuate the myth that a column contains plates where equilibrium conditions exist. In fact, the equilibrium state can never be realized with the mobile phase in constant motion.

The Definition of Plate Height

As shown in Section a1B-1 of Appendix 1, the breadth of a Gaussian curve is described by its standard deviation σ or its variance σ^2 . Because chromatographic bands are often Gaussian or nearly Gaussian in shape, it is convenient to define the efficiency of a column in terms of variance per unit length of column. That is, the plate height H is given by

$$H = \frac{\sigma^2}{L} \quad (26-17)$$

This definition of column efficiency is illustrated in Figure 26-6a, which shows a column having a packing L cm in length. Above this schematic (Figure 26-6b) is a plot showing the distribution of molecules along the length of the column at the moment the analyte peak reaches the end of the packing (that is, at the retention time). The curve is Gaussian, and the locations of $L + 1\sigma$ and $L - 1\sigma$ are indicated as broken vertical lines. Note that L carries units of centimeters and σ^2 units of centimeters squared; thus H represents a linear distance in centimeters as well (Equation 26-17). In fact, the plate height can be thought of as the length of column that contains a fraction of the analyte

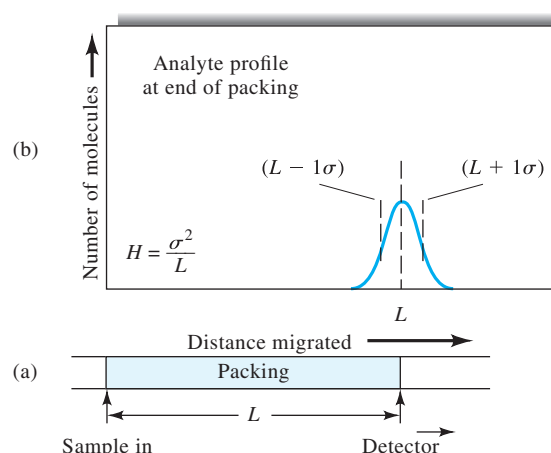


FIGURE 26-6 Definition of plate height $H = \sigma^2/L$. In (a) the column length is shown as the distance from the sample entrance point to the detector. In (b) the Gaussian distribution of sample molecules is shown.

that lies between L and $L - \sigma$. Because the area under a normal error curve bounded by plus or minus one standard deviation ($\pm 1\sigma$) is about 68% of the total area, the plate height, as defined, contains approximately 34% of the analyte.

The Experimental Evaluation of H and N

Figure 26-7 is a typical chromatogram with time as the abscissa. The variance of the solute peak, which can be obtained by a straightforward graphical procedure, has units of seconds squared and is usually designated as τ^2 to distinguish it from σ^2 , which has units of centimeters squared. The two standard deviations τ and σ are related by

$$\tau = \frac{\sigma}{L/t_R} \quad (26-18)$$

where L/t_R is the average linear velocity $\bar{v}$ of the solute in centimeters per second (Equation 26-4).

Figure 26-7 illustrates a method for approximating τ and σ from an experimental chromatogram. Tangents at the inflection points on the two sides of the chromatographic peak are extended to form a triangle with the baseline of the chromatogram. The area of this triangle can be shown to be approximately 96% of the total area under the peak if the peak is Gaussian. In Section a1B-1 of Appendix 1 it is shown that about 96% of the area under a Gaussian peak is included within plus or minus two standard deviations ($\pm 2\sigma$) of its maximum. Thus, the intercepts shown in Figure 26-7 occur at approximately plus or minus two standard deviations of the solute peak ($\pm 2\tau$) from the maximum, and $W = 4\tau$, where W is the magnitude of the base of the triangle. Substituting this relationship into Equation 26-18 and rearranging yields

$$\sigma = \frac{LW}{4t_R} \quad (26-19)$$

⁶A. J. P. Martin and R. L. M. Synge, *Biochem. J.*, **1941**, 35, 1358, DOI: 10.1042/bj0351358.

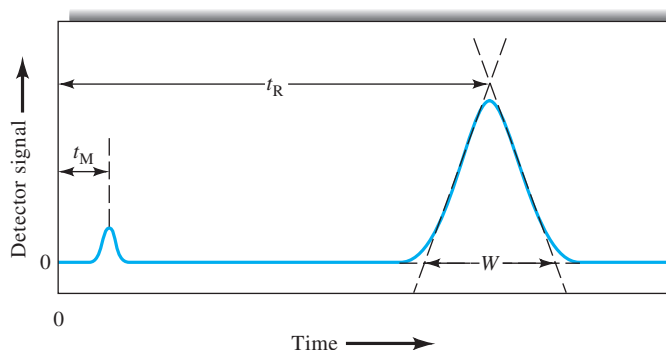


FIGURE 26-7 Determination of the number of plates $N = 16 \left(\frac{t_R}{W} \right)^2$.

Substitution of this equation for σ into Equation 26-17 gives

$$H = \frac{LW^2}{16t_R^2} \quad (26-20)$$

To obtain N , we substitute into Equation 26-16 and rearrange to get

$$N = 16 \left(\frac{t_R}{W} \right)^2 \quad (26-21)$$

Thus, N can be calculated from two time measurements, t_R and W ; to obtain H , the length of the column packing L must also be known. Note that these calculations are only approximate and assume Gaussian peak shapes.

Another method for approximating N , which some workers believe to be more reliable, is to determine $W_{1/2}$, the width of the peak at half its maximum height. The plate count is then given by

$$N = 5.54 \left(\frac{t_R}{W_{1/2}} \right)^2 \quad (26-22)$$

Because the experimental determinations of H and N described here are based on Gaussian chromatographic peaks, the calculations are considered estimates. More accurate methods for dealing with skewed Gaussian peaks have been described in the literature; they are based on finding the variance of the peaks using statistical second-moment calculations.⁷

The plate count N and the plate height H are widely used in the literature and by instrument manufacturers. They can be quite useful quantities in comparing separating power and efficiencies among columns. However, for these numbers to be meaningful in comparing columns, it is essential that they be determined with the *same compound*.

26C-3 Kinetic Variables Affecting Column Efficiency

Band broadening reflects a loss of column efficiency. The slower the rate of mass-transfer processes occurring while a solute migrates through a column, the broader the band at the column exit. Some of the variables that affect mass-transfer rates are controllable and can be exploited to improve separations. Table 26-2 lists the most important of these variables. Their effects on column efficiency, as measured by plate height H , are described in the paragraphs that follow.

The Effect of Mobile-Phase Flow Rate

The extent of band broadening depends on the length of time the mobile phase is in contact with the stationary phase, which in turn depends on the flow rate of the mobile phase. For this reason, efficiency studies have generally been carried out by determining H (by means of Equations 26-21 or 26-22 and Equation 26-16) as a function of mobile-phase velocity. The plots for LC

TABLE 26-2 Variables That Influence Column Efficiency

Variable	Symbol	Usual Units
Linear velocity of mobile phase	u	cm s^{-1}
Diffusion coefficient in mobile phase*	$D_M^\dagger$	$\text{cm}^2 \text{s}^{-1}$
Diffusion coefficient in stationary phase*	D_S	$\text{cm}^2 \text{s}^{-1}$
Retention factor (Equation 26-12)	k	unitless
Diameter of packing particles	d_p	cm
Thickness of liquid coating on stationary phase	d_f	cm

*Increases as temperature increases and viscosity decreases.

[†]In liquids, D_M values for small and medium-size molecules are on the order of $10^{-5} \text{ cm}^2 \text{s}^{-1}$; for gases, D_M values are $\sim 10^5$ times larger.

⁷J. P. Foley and J. G. Dorsey, *Anal. Chem.*, **1983**, 55, 730, DOI: 10.1021/ac00255a033; J. Chromatogr. Sci., **1984**, 22, 40, DOI: 10.1093/chromsci/22.1.40. For a spreadsheet approach to nonideal peak shapes, see S. R. Crouch and F. J. Holler, *Applications of Microsoft® Excel in Analytical Chemistry*, 3rd ed., Belmont, CA: Cengage Learning, 2017, pp. 391–393.

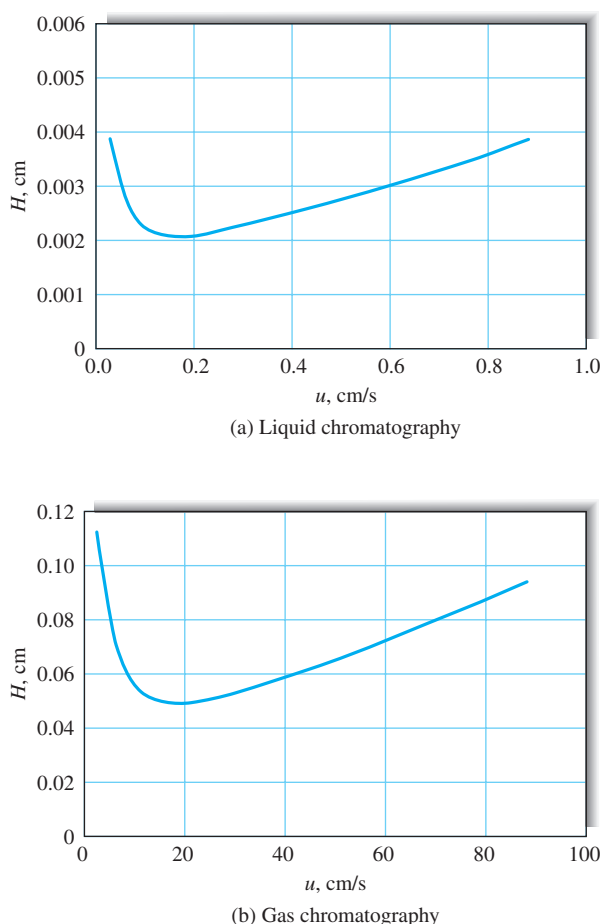


FIGURE 26-8 Effect of mobile-phase flow rate on plate height for (a) LC and (b) GC. Note very different flow rate and plate height scales.

and for GC shown in Figure 26-8 are typical of the data obtained from such studies. Although both show a minimum in H (or a maximum in efficiency) at low linear flow rates, the minimum for LC usually occurs at flow rates well below those for GC. Often these flow rates are so low that the minimum H is not observed for LC under normal operating conditions. The rate theory of chromatographic zone broadening, which is described in this section, accurately predicts the shape of a plot of H versus u , which is often called a *van Deemter plot* after the originator of the theory.

As suggested by Figure 26-8, flow rates for LC are significantly smaller than those used in GC. This means that gas chromatographic separations are typically completed in shorter times than are liquid chromatographic separations. Furthermore, as shown in the figure, plate heights for liquid chromatographic

columns are an order of magnitude or more smaller than those encountered with gas chromatographic columns. Offsetting this advantage, however, is the fact that LC columns longer than about 25 cm are impractical because of high pressure drops. On the other hand, GC columns may be 50 m long or more. The total number of plates is then greater, and thus overall column efficiency is usually superior with gas chromatographic columns. Thus, one comparison between GC and LC is that GC is capable of faster separations and higher-efficiency separations, although not necessarily both simultaneously.

Theory of Band Broadening

Over the last 40 years, an enormous amount of theoretical and experimental effort has been devoted to developing quantitative relationships describing the effects of the experimental variables listed in Table 26-2 on plate heights for various types of columns. Perhaps a dozen or more expressions for calculating plate height have been proposed and applied with various degrees of success. None of these is entirely adequate to explain the complex physical interactions and effects that lead to zone broadening and thus lower column efficiencies. Some of the equations, though imperfect, have been quite useful, however, in pointing the way toward improved column performance.

The efficiency of chromatographic columns can be approximated by the expression

$$H = A + \frac{B}{u} + C_s u + C_M u \quad (26-23)$$

where H is the plate height in centimeters and u is the linear velocity of the mobile phase in centimeters per second. The quantity A is a coefficient that describes *multiple path effects* (eddy diffusion) as discussed later, B is the *longitudinal diffusion coefficient*, and C_s and C_M are *mass-transfer coefficients* for the stationary and mobile phases, respectively. Equation 26-23 is equivalent to the well-known *van Deemter equation*, developed by Dutch chemical engineers in the 1950s and often used to describe chromatographic efficiency.

More recent studies have led to elaborations of the basic van Deemter expression, but it has been shown experimentally that Equation 26-23 is quite satisfactory in explaining column efficiency.⁸ Note that the van Deemter expression contains terms linearly and inversely proportional to, as well as independent of, the mobile-phase velocity.

Let us now examine in some detail the variables that affect the four terms in Equation 26-23. These terms are identified in Table 26-3 and described in the paragraphs that follow.⁹

⁸E. Katz, K. L. Ogan, and R. P. W. Scott, *J. Chromatogr.*, **1983**, 270, 51, DOI: 10.1016/S0021-9673(01)96351-4.

⁹For a spreadsheet approach to calculations using the van Deemter equation, see S. R. Crouch and F. J. Holler, *Applications of Microsoft® Excel in Analytical Chemistry*, 3rd ed., Belmont, CA: Cengage Learning, 2017, pp. 394–399.



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TABLE 26-3 Processes That Contribute to Band Broadening

Process	Term in Equation 26-23	Relationship to Column* and Analyte Properties
Multiple flow paths	A	$A = 2\lambda d_p$
Longitudinal diffusion	B/u	$\frac{B}{u} = \frac{2\gamma D_M}{u}$
Mass transfer to and from stationary phase	$C_S u$	$C_S u = \frac{f(k)d_f^2}{D_S} u$
Mass transfer in mobile phase	$C_M u$	$C_M u = \frac{f'(k)d_p^2}{D_M} u$

* u , D_S , D_M , d_p , d_f , and k are as defined in Table 26-2.

$f(k)$ and $f'(k)$ are functions of k .

λ and γ are constants that depend on quality of the packing.

B is the coefficient of longitudinal diffusion.

C_S and C_M are coefficients of mass transfer in stationary and mobile phase, respectively.

The Multipath Term A . Zone broadening in the mobile phase is due in part to the multitude of pathways by which a molecule (or ion) can find its way through a packed column. As shown in Figure 26-9, the length of these pathways may differ significantly; thus, the residence time in the column for molecules of the same species is also variable. Solute molecules then reach the end of the column over a time interval, which leads to a broadened band. This multiple path effect, which is sometimes called *eddy diffusion*, would be independent of solvent velocity if it were not partially offset by ordinary diffusion, which results in molecules being transferred from a stream following one pathway to a stream following another. If the velocity of flow is very low, a large number of these transfers will occur, and each molecule in its

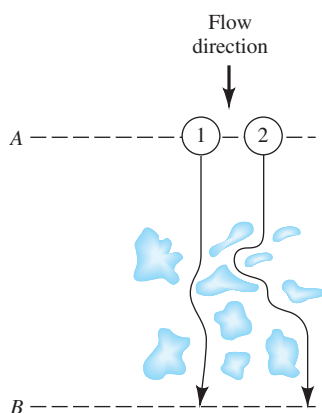


FIGURE 26-9 Typical pathways of two molecules during elution. Note that the distance traveled by molecule 2 is greater than that traveled by molecule 1. Thus, molecule 2 will arrive at B later than molecule 1.

movement down the column will sample numerous flow paths, spending a brief time in each. As a result, the rate at which each molecule moves down the column tends to approach that of the average. Thus, at low mobile-phase velocities, the molecules are not significantly dispersed by the multiple-path effect. At moderate or high velocities, however, sufficient time is not available for diffusion averaging to occur, and band broadening due to the different path lengths is observed. At sufficiently high velocities, the effect of eddy diffusion becomes independent of flow rate.

Superimposed on the eddy-diffusion effect is one that arises from stagnant pools of the mobile phase retained in the stationary phase. Thus, when a solid serves as the stationary phase, its pores are filled with *static* volumes of mobile phase. Solute molecules must then diffuse through these stagnant pools before transfer can occur between the *moving* mobile phase and the stationary phase. This situation applies not only to solid stationary phases but also to liquid stationary phases immobilized on porous solids because the immobilized liquid does not usually fully fill the pores.

The presence of stagnant pools of mobile phase slows the exchange process and produces a contribution to the plate height directly proportional to the mobile-phase velocity and inversely proportional to the diffusion coefficient for the solute in the mobile phase. An increase in internal volume then accompanies increases in particle size.

The Longitudinal Diffusion Term B/u . Diffusion is a process in which species migrate from a more concentrated part of a medium to a more dilute region. The rate of migration is proportional to the concentration difference between the regions and to the *diffusion coefficient* D_M of the species. The latter, which is a measure of the mobility of a substance in a given medium, is a constant for a given species equal to the velocity of migration under a unit concentration gradient.

In chromatography, longitudinal diffusion results in the migration of a solute from the concentrated center of a band to the more dilute regions on either side (that is, toward and opposed to the direction of flow). Longitudinal diffusion is a common source of band broadening in GC because gaseous molecules diffuse at relatively high rates. The phenomenon is of little significance in LC where diffusion rates are much smaller. The magnitude of the B term in Equation 26-23 is largely determined by the diffusion coefficient D_M of the analyte in the mobile phase and is directly proportional to this constant.

As shown by Equation 26-23, the contribution of longitudinal diffusion to plate height is inversely proportional to the linear velocity of the eluent. This relationship is not surprising because the analyte is in the column for a shorter period when the flow rate is high. Thus, diffusion from the center of the band to the two edges has less time to occur.

The initial decreases in H shown in both curves in Figure 26-8 are a direct result of the longitudinal diffusion. Note that the effect is much less pronounced in LC because of the much lower diffusion rates in a liquid mobile phase. The striking difference in plate heights shown by the two curves in Figure 26-8 can also be explained by considering the relative rates of

longitudinal diffusion in the two mobile phases. That is, diffusion coefficients in gaseous media are orders of magnitude larger than in liquids. Thus band broadening occurs to a much greater extent in GC than in LC.

The Stationary-Phase Mass-Transfer Term $C_s u$. When the stationary phase is an immobilized liquid, the mass-transfer coefficient is directly proportional to the square of the thickness of the film on the support particles d_f^2 and inversely proportional to the diffusion coefficient D_s of the solute in the film. These effects can be understood by realizing that both reduce the average frequency at which analyte molecules reach the liquid-liquid interface where transfer to the mobile phase can occur. That is, with thick films, molecules must on the average travel farther to reach the surface, and with smaller diffusion coefficients, they travel slower. The result is a slower rate of mass transfer and an increase in plate height.

When the stationary phase is a solid surface, the mass-transfer coefficient C_s is directly proportional to the time required for a species to be adsorbed or desorbed, which in turn is inversely proportional to the first-order rate constant for the processes.

The Mobile-Phase Mass-Transfer Term $C_M u$. The mass-transfer processes that occur in the mobile phase are sufficiently complex that we do not yet have a complete quantitative description. On the other hand, we have a good qualitative understanding of the variables affecting zone broadening from this cause, and this understanding has led to vast improvements in all types of chromatographic columns.

The mobile-phase mass-transfer coefficient C_M is known to be inversely proportional to the diffusion coefficient of the analyte in the mobile phase D_M . For packed columns, C_M is proportional to the square of the particle diameter of the packing material, d_p^2 . For open tubular columns, C_M is proportional to the square of the column diameter, d_c^2 .

The contribution of mobile-phase mass transfer to plate height is the product of the mass-transfer coefficient C_M (which

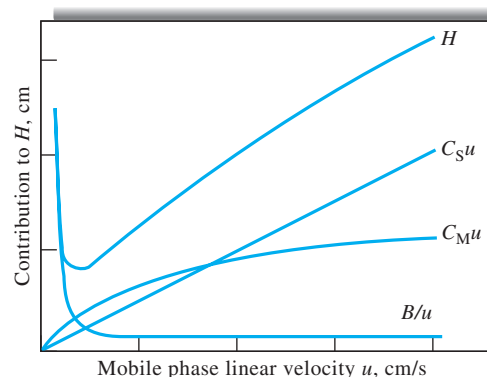


FIGURE 26-10 Contribution of various mass-transfer terms to plate height. $C_s u$ arises from the rate of mass transfer to and from the stationary phase, $C_M u$ comes from a limitation in the rate of mass transfer in the mobile phase, and B/u is associated with longitudinal diffusion.

is a function of solvent velocity) and the velocity of the solvent itself. Thus, the net contribution of $C_M u$ to plate height is not linear in u (see the curve labeled $C_M u$ in Figure 26-10) but bears a complex dependency on solvent velocity.

Effect of Mobile-Phase Velocity on Terms in Equation 26-23. Figure 26-10 shows the variation of the three terms in Equation 26-23 as a function of mobile-phase velocity. The top curve is the summation of these various effects. Note that there is an optimal flow rate at which the plate height is a minimum and the separation efficiency is a maximum.

Summary of Methods for Reducing Band Broadening. For packed columns, the most important variable that affects column efficiency is the diameter of the particles making up the packing. For open tubular columns, the diameter of the column itself is an important variable. The effect of particle diameter is demonstrated by the data shown in Figure 26-11 for GC.

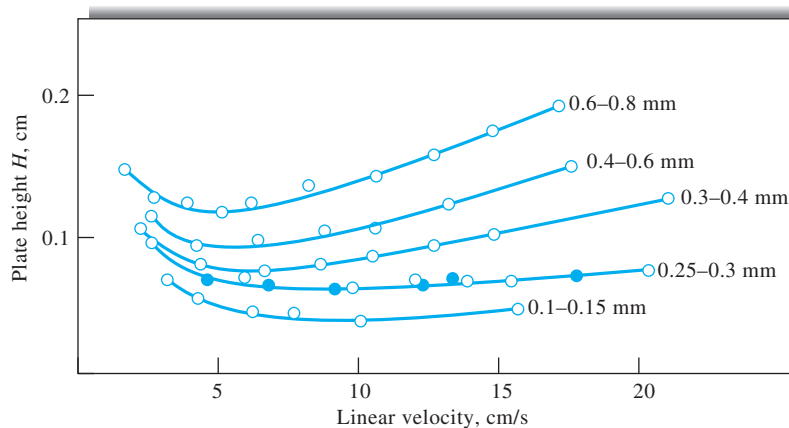


FIGURE 26-11 Effect of particle size on plate height for a packed GC column. The numbers to the right of each curve are particle diameters. (From J. Boheman and J. H. Purnell, in *Gas Chromatography 1958*, D. H. Desty, ed., New York: Academic Press, 1958. With permission.)

A similar plot for LC is shown in Figure 28-2. To take advantage of the effect of column diameter, narrower and narrower columns have been used in recent years.

With gaseous mobile phases, the rate of longitudinal diffusion can be reduced appreciably by lowering the temperature and thus the diffusion coefficient. When operating at low mobile-phase velocities where the B/u term is important (see Figure 26-10), lower temperatures can give smaller plate heights. This effect is usually not noticeable in LC because diffusion is slow enough so that the longitudinal diffusion term has little effect on overall plate height. Also, most chromatography is done with mobile-phase velocities higher than the optimal velocity so that mass-transfer terms (C terms) control column efficiency.

With liquid stationary phases, the thickness of the layer of adsorbed liquid should be minimized because C_s in Equation 26-23 is proportional to the square of this variable.

26D OPTIMIZATION OF COLUMN PERFORMANCE

A chromatographic separation is optimized by varying experimental conditions until the components of a mixture are separated cleanly in a minimum amount of time. Optimization experiments are aimed at either (1) reducing zone broadening or (2) altering relative migration rates of the components. As we have shown in Section 26C, zone broadening is increased by

those kinetic variables that increase the plate height of a column. Migration rates, on the other hand, are varied by changing those variables that affect retention and selectivity factors (Section 26B).

26D-1 Column Resolution

The *resolution* R_s of a column tells us how far apart two bands are relative to their widths. The resolution provides a quantitative measure of the ability of the column to separate two analytes. The significance of this term is illustrated in Figure 26-12, which consists of chromatograms for species A and B on three columns with different resolving powers. The resolution of each column is defined as

$$R_s = \frac{\Delta Z}{\frac{W_A}{2} + \frac{W_B}{2}} = \frac{2\Delta Z}{W_A + W_B} = \frac{2[(t_R)_B - (t_R)_A]}{W_A + W_B} \quad (26-24)$$

where all of the terms on the right side are as defined in the figure.

Note from Figure 26-12 that a resolution of 1.5 gives an essentially complete separation of A and B, but a resolution of 0.75 does not. At a resolution of 1.0, zone A contains about 4% B



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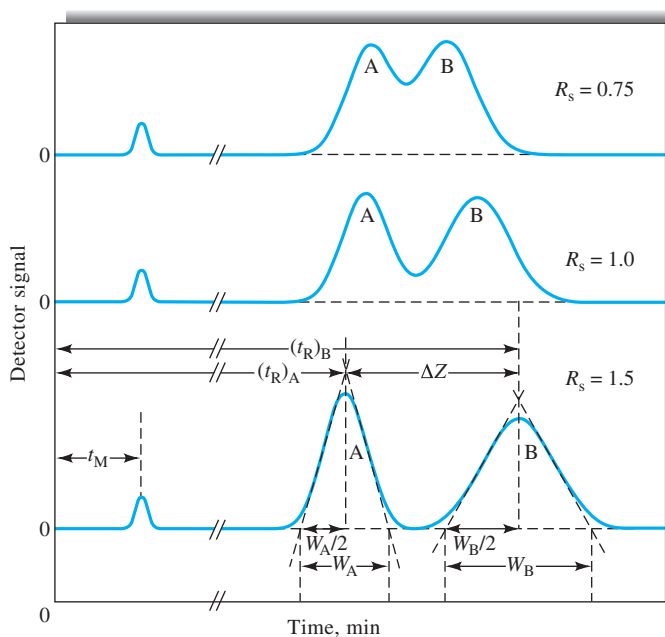


FIGURE 26-12 Separation at three resolution values: $R_s = 2\Delta Z/(W_A + W_B)$. The peak heights have been adjusted to be approximately the same in each case.

and zone B contains about 4% A. At a resolution of 1.5, the overlap is about 0.3%. The resolution for a given stationary phase can be improved by lengthening the column, which increases the number of plates. Adding more theoretical plates, however, comes at a cost of increasing the time required for separating the components.

26D-2 The Effect of Retention and Selectivity Factors on Resolution

It is useful to develop a mathematical relationship between the resolution of a column and the retention factors k_A and k_B for two solutes, the selectivity factor α , and the number of plates N making up the column. We will assume that we are dealing with two solutes A and B having retention times close enough to one another that we can assume

$$W_A = W_B \approx W$$

Equation 26-24 then takes the form

$$R_s = \frac{(t_R)_B - (t_R)_A}{W}$$

We solve Equation 26-21 for W in terms of $(t_R)_B$ and N and then substitute the resulting expression into the foregoing equation to obtain

$$R_s = \frac{(t_R)_B - (t_R)_A}{(t_R)_B} \times \frac{\sqrt{N}}{4}$$

Substituting Equation 26-12 into this equation and rearranging leads to an expression for R_s in terms of the retention factors for A and B. That is,

$$R_s = \frac{k_B - k_A}{1 + k_B} \times \frac{\sqrt{N}}{4}$$

Let us eliminate k_A from this expression by substituting Equation 26-14 and rearranging. Thus,

$$R_s = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_B}{1 + k_B} \right) \quad (26-25)$$

where k_B is the retention factor of the slower-moving species and α is the selectivity factor. Equation 26-25 can be rearranged to give the number of plates needed to realize a given resolution.

$$N = 16R_s^2 \left(\frac{\alpha}{\alpha - 1} \right)^2 \left(\frac{1 + k_B}{k_B} \right)^2 \quad (26-26)$$

Simplified forms of Equations 26-25 and 26-26 are sometimes encountered where these equations are applied to a pair of solutes whose distribution constants are similar enough to make their separation difficult. Thus, when $K_A \approx K_B$, it follows from Equation 26-9 that $k_A \approx k_B = k$ and from Equation 26-13,

$\alpha \rightarrow 1$. With these approximations, Equations 26-25 and 26-26 reduce to

$$R_s = \frac{\sqrt{N}}{4} (\alpha - 1) \left(\frac{k}{1 + k} \right) \quad (26-27)$$

$$N = 16R_s^2 \left(\frac{1}{\alpha - 1} \right)^2 \left(\frac{1 + k}{k} \right)^2 \quad (26-28)$$

where k is the average of k_A and k_B .

26D-3 The Effect of Resolution on Retention Time

Before considering in detail the significance of the four equations just derived, it is worthwhile to develop an equation for a related performance characteristic for a column, the time required to complete the separation of solutes A and B. The goal in chromatography is the highest possible resolution in the shortest possible time. Unfortunately, these goals tend to be incompatible, and a compromise between the two is usually necessary.

The completion time of a separation is determined by the velocity $\bar{v}_B$ of the slower-moving solute, as given in Equation 26-4. That is,

$$\bar{v}_B = \frac{L}{(t_R)_B}$$

Combining this expression with 26-10 and 26-16 yields, after rearranging

$$(t_R)_B = \frac{NH(1 + k_B)}{u}$$

where $(t_R)_B$ is the time required to bring the peak for B to the end of the column when the velocity of the mobile phase is u . When this equation is combined with Equation 26-26 and rearranged, we find that

$$(t_R)_B = \frac{16R_s^2 H}{u} \left(\frac{\alpha}{\alpha - 1} \right)^2 \frac{(1 + k_B)^3}{(k_B)^2} \quad (26-29)$$

Because H is a function of u and α is a function of k , such an equation is difficult to solve and use except when doing comparisons where several of the factors may cancel in a ratio.

Several of these relationships are applied in Example 26-1.

EXAMPLE 26-1

Substances A and B have retention times of 16.40 and 17.63 min, respectively, on a 30.0-cm column. An unretained species passes through the column in 1.30 min. The peak widths (at base) for A and B are 1.11 and 1.21 min, respectively. Calculate (a) the column resolution, (b) the average number of plates in the column, (c) the plate height, (d) the length of column required to achieve a resolution of 1.5, (e) the time required to elute substance B on the column that



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gives an R_s value of 1.5, and (f) the plate height required for a resolution of 1.5 on the original 30-cm column and in the original time.

Solution

- (a) Using Equation 26-24, we find

$$R_s = \frac{2(17.63 - 16.40)}{1.11 + 1.21} = 1.06$$

- (b) Equation 26-21 permits computation of N :

$$N = 16 \left(\frac{16.40}{1.11} \right)^2 = 3493 \quad \text{and} \quad N = 16 \left(\frac{17.63}{1.21} \right)^2 = 3397$$

$$N_{av} = \frac{3493 + 3397}{2} = 3445$$

- (c) $H = \frac{L}{N} = \frac{30.0}{3445} = 8.7 \times 10^{-3} \text{ cm}$

- (d) k and α do not change greatly with increasing N and L . Thus, substituting N_1 and N_2 into Equation 26-25 and dividing one of the resulting equations by the other yields

$$\frac{(R_s)_1}{(R_s)_2} = \frac{\sqrt{N_1}}{\sqrt{N_2}}$$

where the subscripts 1 and 2 refer to the original and longer columns, respectively. Substituting the appropriate values for N_1 , $(R_s)_1$, and $(R_s)_2$ gives

$$\frac{1.06}{1.5} = \frac{\sqrt{3445}}{\sqrt{N_2}} \\ N_2 = 3445 \left(\frac{1.5}{1.06} \right)^2 = 6.9 \times 10^3$$

But

$$L = NH = 6.9 \times 10^3 \times 8.7 \times 10^{-3} = 60 \text{ cm}$$

- (e) Substituting $(R_s)_1$ and $(R_s)_2$ into Equation 26-29 and dividing yields

$$\frac{(t_R)_1}{(t_R)_2} = \frac{(R_s)_1^2}{(R_s)_2^2} = \frac{17.63}{(t_R)_2} = \frac{(1.06)^2}{(1.5)^2} \\ (t_R)_2 = 35 \text{ min}$$

Thus, to obtain the improved resolution, the column length and, therefore, the separation time must be doubled.

- (f) Substituting H_1 and H_2 into Equation 26-29 and dividing one of the resulting equations by the second gives

$$\frac{(t_R)_B}{(t_R)_B} = \frac{(R_s)_1^2}{(R_s)_2^2} \times \frac{H_1}{H_2}$$

where the subscripts 1 and 2 refer to the original and the new plate heights, respectively. Rearranging gives

$$H_2 = H_1 \frac{(R_s)_1^2}{(R_s)_2^2} = 8.7 \times 10^{-3} \text{ cm} \times \frac{(1.06)^2}{(1.5)^2} \\ = 4.3 \times 10^{-3} \text{ cm}$$

Thus, to achieve a resolution of 1.5 in 17.63 min on a 30-cm column, the plate height would need to be halved.

26D-4 Variables That Affect Column Performance

Equations 26-25 and 26-29 are significant because they serve as guides to the choice of conditions that are likely to achieve the sometimes elusive goal of a clean separation in a minimum amount of time. An examination of these equations reveals that each is made up of three parts. The first, which is related to the kinetic effects that lead to band broadening, consists of $\sqrt{N}$ or H/u . The second and third terms are related to the thermodynamics of the constituents being separated—that is, to the relative magnitude of their distribution constants and the volumes of the mobile and stationary phases. The second term in Equations 26-25 and 26-29, which is the quotient containing α , is a selectivity term. For a given mobile-phase and stationary-phase combination, the α term depends solely on the properties of the two solutes. The third term, which is the quotient containing k_B , depends on the properties of both the solute and the column.

In seeking optimal conditions for achieving a desired separation, it must be kept in mind that the fundamental parameters α , k , and N (or H) can be adjusted more or less independently. Thus, α and k can be varied most easily by varying temperature or the composition of the mobile phase. Less conveniently, a different type of column packing can be used. As we have seen, it is possible to change N by changing the length of the column and to change H by altering the flow rate of the mobile phase, the particle size of the packing, the viscosity of the mobile phase (and thus D_M or D_S), and the thickness of the film of adsorbed liquid constituting the stationary phase (see Table 26-3).

Variation in N

An obvious way to improve resolution is to increase the number of plates in the column (Equation 26-25). As shown by Example 26-1, this approach usually requires more time to complete the separation unless the increase in N is accomplished by reducing H rather than by lengthening the column.

Variation in H

In Example 26-1f, we showed that resolution can be significantly improved at no cost in time if the plate height can be reduced. Table 26-3 reveals the variables available for accomplishing this end. Note that decreases in particle size of the packing lead to substantial improvements in H . For liquid mobile phases, where

B/u is usually negligible, reduced plate heights can also be achieved by reducing the solvent viscosity, thus increasing the diffusion coefficient in the mobile phase.

Variation in the Retention Factor

Often, a separation can be improved significantly by manipulating the retention factor k_B . Increases in k_B generally enhance resolution (but at the expense of elution time). To determine the optimal range of values for k_B , it is convenient to write Equation 26-25 in the form

$$R_s = \frac{Qk_B}{1 + k_B}$$

where $Q = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right)$, and Equation 26-29 as

$$(t_R)_B = Q' \frac{(1 + k_B)^3}{(k_B)^2}$$

where

$$Q' = \frac{16R_s^2 H}{u} \left(\frac{\alpha}{\alpha - 1} \right)^2$$

Figure 26-13 is a plot of R_s/Q and $(t_R)_B/Q'$ as a function of k_B , assuming that Q and Q' remain approximately constant. Note that values of k_B greater than about 10 should be avoided because they provide little increase in resolution but noticeably increase the time required for separations. The minimum in the

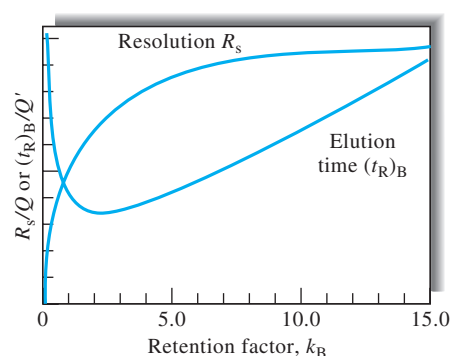


FIGURE 26-13 Effect of retention factor k_B on resolution R_s and elution time $(t_R)_B$. It is assumed that Q and Q' remain constant with variations in k_B .

elution time curve occurs at a value of k_B of about 2. The curves of Figure 26-13 suggest that the optimal value of k_B , taking into account both resolution and time, lies in the range of 1 to 5.

Usually, the easiest way to improve resolution is by optimizing k . For gaseous mobile phases, k can often be improved by temperature changes. For liquid mobile phases, changes in the solvent composition often permit manipulation of k to yield better separations. An example of the dramatic effect that relatively simple solvent changes can bring about is demonstrated in Figure 26-14. The four chromatograms show how modest variations in the methanol-to-water ratio convert unsatisfactory

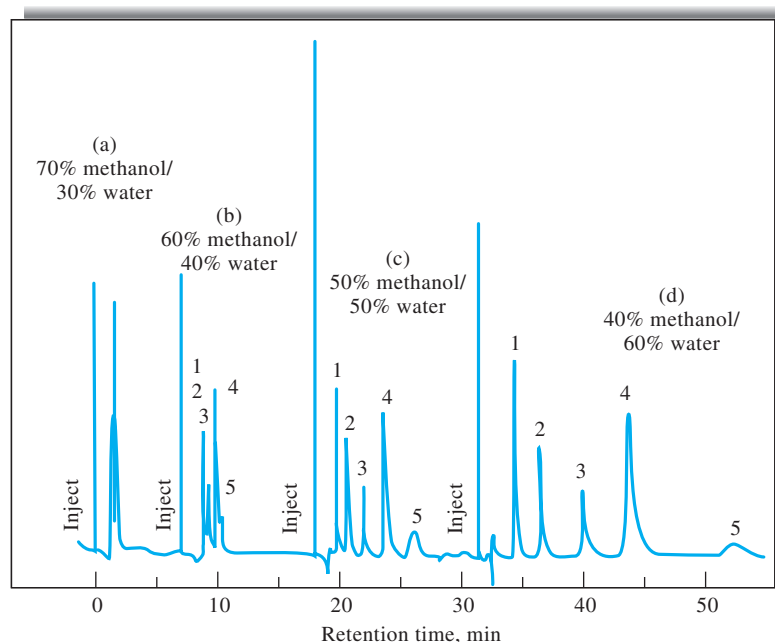


FIGURE 26-14 Effect of solvent variation on chromatograms. Analytes: (1) 9,10-anthraquinone; (2) 2-methyl-9, 10-anthraquinone; (3) 2-ethyl-9,10-anthraquinone; (4) 1,4-dimethyl-9, 10-anthraquinone; and (5) 2-*t*-butyl-9, 10-anthraquinone.

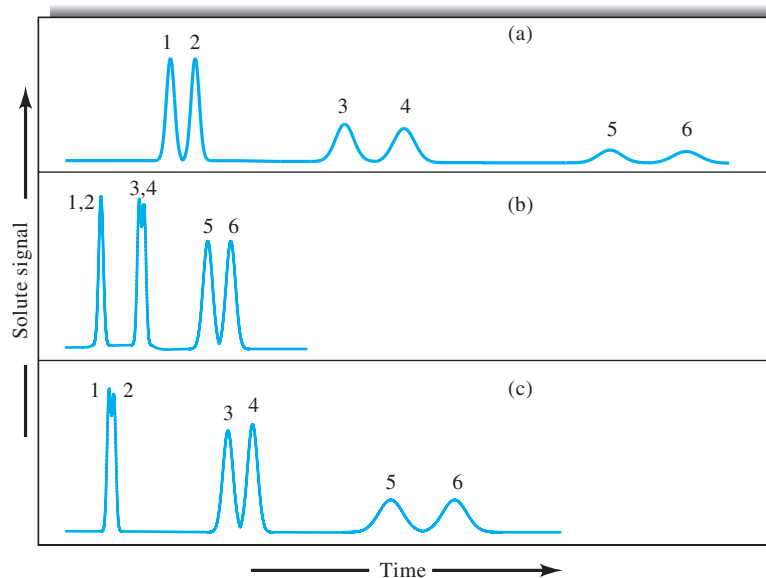


FIGURE 26-15 The general elution problem in chromatography.

chromatograms (a and b) to ones with well-separated peaks for each component (c and d). For most purposes, the chromatogram shown in (c) is best because it shows adequate resolution in minimum time. The retention factor is also influenced by the stationary-phase film thickness.

Variation in the Selectivity Factor

Optimizing k and increasing N are not sufficient to give a satisfactory separation of two solutes in a reasonable time when α approaches unity. When this occurs, a means must be found to increase α while maintaining k in the range of 1 to 10. Several options are available; in decreasing order of their desirability as determined by promise and convenience, the options are (1) changing the composition of the mobile phase, (2) changing the column temperature, (3) changing the composition of the stationary phase, and (4) using special chemical effects.

An example of the use of option 1 has been reported for the separation of anisole ($\text{C}_6\text{H}_5\text{OCH}_3$) and benzene.¹⁰ With a mobile phase that was a 50% mixture of water and methanol, k was 4.5 for anisole and 4.7 for benzene, and α was only 1.04. Substitution of an aqueous mobile phase containing 37% tetrahydrofuran gave k values of 3.9 and 4.7 and an α value of 1.20. Peak overlap was significant with the first solvent system and negligible with the second.

A less convenient but often highly effective method of improving α while maintaining values for k in their optimal range is to alter the chemical composition of the stationary phase. To take advantage of this option, most laboratories that carry out chromatographic separations frequently maintain several columns that can be interchanged conveniently.

Increases in temperature usually cause decreases in k and may affect α values in liquid-liquid and liquid-solid chromatography. In contrast, with ion-exchange chromatography, temperature effects can be large enough to make exploration of this option worthwhile before resorting to a change in column packing material.

A final method for enhancing resolution is to incorporate into the stationary phase a species that complexes or otherwise interacts with one or more components of the sample. A well-known example of the use of this option occurs when an adsorbent impregnated with a silver salt is used to improve the separation of olefins. The improvement results from the formation of complexes between the silver ions and unsaturated organic compounds.

26D-5 The General Elution Problem

Figure 26-15 shows hypothetical chromatograms for a six-component mixture made up of three pairs of components with widely different distribution constants and thus widely different retention factors. In chromatogram (a), conditions have been adjusted so that the retention factors for components 1 and 2 (k_1 and k_2) are in the optimal range of 1 to 5. The factors for the other components are far larger than the optimal, however. Thus, the bands corresponding to components 5 and 6 appear after a significant period of time has passed; furthermore, the bands are so broad that they may be difficult to identify unambiguously.

As shown in chromatogram (b), changing conditions to optimize the separation of components 5 and 6 bunches the peaks for the first four components to the point that their resolution is unsatisfactory. The total elution time is ideal, however.

¹⁰L. R. Snyder and J. J. Kirkland, *Introduction to Modern Liquid Chromatography*, 2nd ed., p. 75, New York: Wiley, 1979.

A third set of conditions, in which k values for components 3 and 4 are optimal, results in chromatogram (c). Again, separation of the other two pairs is not entirely satisfactory.

The phenomenon illustrated in Figure 26-15 is encountered often enough to be given a name, the *general elution problem*. A common solution to this problem is to change conditions that determine the values of k as the separation proceeds. These changes can be performed in a stepwise manner or continuously. Thus, for the mixture shown in Figure 26-15, conditions at the beginning of the separation could be those producing chromatogram (a). Immediately after the elution of components 1 and 2, conditions could be changed to those that are optimal for separating components 3 and 4, as in chromatogram (c). With the appearance of peaks for these components, the elution could be completed under the conditions used for producing chromatogram (b). Often, such a procedure leads to satisfactory separation of all the components of a mixture in minimal time.

For LC, variations in k can be achieved by varying the composition of the mobile phase during elution. Such a procedure is called *gradient elution* or *solvent programming*. Elution with constant mobile-phase composition is called *isocratic elution*. For GC, the temperature can be changed in a known fashion to bring about changes in k . This *temperature programming* mode can help achieve optimal conditions for many separations. These modes are discussed in more detail in Chapters 27 and 28.

26E SUMMARY OF CHROMATOGRAPHIC RELATIONSHIPS

The number of quantities, terms, and relationships used in chromatography is large and often confusing. Tables 26-4 and 26-5 summarize the most important definitions and equations that are used in the next three chapters.

26F APPLICATIONS OF CHROMATOGRAPHY

Chromatography has grown to be the premiere method for separating closely related chemical species. In addition, it can be used for qualitative identification and quantitative determination of separated species. This section considers some of the general characteristics of chromatography as a tool for completion of an analysis.

26F-1 Qualitative Analysis

A chromatogram provides only a single piece of qualitative information about each species in a sample, its retention time or its position on the stationary phase after a certain elution period. Additional data can, of course, be derived from chromatograms involving different mobile and stationary phases and various elution temperatures. Still, the amount of information revealed by chromatography is small compared with the amount provided by a single IR, nuclear magnetic resonance, or mass spectrum. Furthermore, spectral wavelength or frequency data can be determined with much higher precision than can their chromatographic counterpart t_R .

This should not be interpreted to mean that chromatography lacks important qualitative applications. Indeed, it is a widely used tool for recognizing the presence or absence of components of mixtures containing a limited number of possible species whose identities are known. For example, thirty or more amino acids in a protein hydrolysate can be identified with a relatively high degree of certainty by using chromatography. Even in such cases, confirmation of identity requires spectral or chemical investigation of the isolated components. Note, however, that positive spectroscopic identification would ordinarily be impossible on as complex a sample as a protein hydrolysate.

TABLE 26-4 Important Chromatographic Quantities and Relationships

Name	Symbol of Experimental Quantity	Determined From
Migration time, unretained species	t_M	Chromatogram (Figure 26-7)
Retention time, species A and B	$(t_R)_A, (t_R)_B$	Chromatogram (Figures 26-7 and 26-12)
Adjusted retention time for A	$(t'_R)_A$	$(t'_R)_A = (t_R)_A - t_M$
Peak widths for A and B	W_A, W_B	Chromatogram (Figures 26-7 and 26-12)
Length of column packing	L	Direct measurement
Volumetric flow rate	F	Direct measurement
Linear flow velocity	u	F and column dimensions (Equations 26-6 and 26-7)
Stationary-phase volume	V_S	Packing preparation data
Concentration of analyte in mobile and stationary phases	c_M, c_S	Analysis and preparation data

TABLE 26-5 Important Derived Quantities and Relationships

Name	Calculation of Derived Quantities	Relationship to Other Quantities
Linear mobile-phase velocity	$u = \frac{L}{t_M}$	
Volume of mobile phase	$V_M = t_M F$	
Retention factor	$k = \frac{t_R - t_M}{t_M}$	$k = \frac{KV_S}{V_M}$
Distribution constant	$K = \frac{kV_M}{V_S}$	$K = \frac{c_S}{c_M}$
Selectivity factor	$\alpha = \frac{(t_R)_B - t_M}{(t_R)_A - t_M}$	$\alpha = \frac{k_B}{k_A} = \frac{K_B}{K_A}$
Resolution	$R_s = \frac{2[(t_R)_B - (t_R)_A]}{W_A + W_B}$	$R_s = \frac{\sqrt{N}}{4} \left(\frac{\alpha - 1}{\alpha} \right) \left(\frac{k_B}{1 + k_B} \right)$
Number of plates	$N = 16 \left(\frac{t_R}{W} \right)^2$	$N = 16 R_s^2 \left(\frac{\alpha}{\alpha - 1} \right)^2 \left(\frac{1 + k_B}{k_B} \right)^2$
Plate height	$H = \frac{L}{N}$	
Retention time	$(t_R)_B = \frac{16 R_s^2 H}{u} \left(\frac{\alpha}{\alpha - 1} \right)^2 \frac{(1 + k_B)^3}{(k_B)^2}$	

without a preliminary chromatographic separation. Thus, chromatography is often an essential step in qualitative spectroscopic analyses.

It is important to note that, although chromatograms may not lead to positive identification of species present in a sample, they often provide sure evidence of the *absence* of certain compounds. Thus, if the sample does not produce a peak at the same retention time as a standard run under identical conditions, it can be assumed that the compound in question is either absent or is present at a concentration level below the detection limit of the procedure.

26F-2 Quantitative Analysis

Chromatography owes its rapid growth during the past four decades in part to its speed, simplicity, relatively low cost, and wide applicability as a tool for separations. It is doubtful, however, that chromatography would have become as widespread had it not been able to provide useful quantitative information about the separated species. It is important, therefore, to

discuss some of the quantitative aspects that apply to all types of chromatography.

Quantitative column chromatography is based on a comparison of either the height or the area of the analyte peak with that of one or more standards. For planar chromatography, the area covered by the separated species serves as the analytical variable. If conditions are properly controlled, these variables vary linearly with concentration.

Analyses Based on Peak Height

The height of a chromatographic peak is obtained by connecting the baselines on either side of the peak by a straight line and measuring the perpendicular distance from this line to the peak. This measurement can often be made with reasonably high precision. It is important to note, however, that peak heights are inversely related to peak widths. Thus, accurate results are obtained with peak heights only if variations in column conditions do not alter the peak widths during the period required to obtain chromatograms for sample and standards. The variables that must be controlled closely are column temperature, eluent flow rate, and rate of sample injection. In addition, care must be taken to avoid overloading the column. The effect of sample injection rate is particularly critical for the early peaks of a chromatogram. Relative errors of 5% to 10% due to this cause are not unusual with syringe injection.



Tutorial: Learn more about **quantitative chromatographic analysis** at www.tinyurl.com/skoogpia7

Analyses Based on Peak Areas

Peak areas are independent of broadening effects due to the variables mentioned in the previous paragraph. From this standpoint, therefore, areas are a more satisfactory analytical parameter than peak heights. On the other hand, peak heights are more easily measured and, for narrow peaks, more accurately determined. However, peak heights are affected by changes in retention times or column efficiency, but peak areas are not. Hence, peak areas are usually the preferred method of quantitation.

Most modern chromatographic instruments are equipped with computers or digital integrators that precisely estimate peak areas. If such equipment is not available, a manual estimate must be made. A simple method, which works well for symmetric peaks of reasonable widths, is to multiply the height of the peak by its width at one half the peak height. Older methods used a planimeter or a cut-and-weigh technique. In general, manual integration techniques provide areas that are reproducible to 2% to 5%; digital integrators are at least an order of magnitude more precise.¹¹

Calibration and Standards

In the most straightforward method for quantitative chromatographic analyses, a series of external-standard solutions is prepared that approximates the composition of the unknown. Chromatograms for the standards are then obtained and peak heights or areas are plotted as a function of concentration. A plot of the data should yield a straight line passing through the origin; determinations are based on this calibration curve. Frequent restandardization is necessary for highest accuracy.

The most important source of error in analyses by the method just described is usually the uncertainty in the volume

of sample; occasionally, the rate of injection is also a factor. Samples are usually small ($\sim 1 \mu\text{L}$), and the uncertainties associated with injection of a reproducible volume of this size with a microsyringe may amount to several percent relative. The situation is even worse in GC, where the sample must be injected into a heated sample port; here, evaporation from the needle tip may lead to large variations in the volume injected. Errors in sample volume can be reduced to perhaps 1% to 2% relative by means of autosamplers or a rotary sample valve such as that described in Chapter 27.

The Internal-Standard Method

The highest precision for quantitative chromatography is obtained using internal standards because the uncertainties introduced by sample injection are avoided. In this procedure, a carefully measured quantity of an internal-standard substance is introduced into each standard and sample, and the ratio of analyte to internal standard peak areas (or heights) serves as the analytical variable. For this method to be successful, the internal-standard peak must be well separated from the peaks of all other components of the sample ($R_s > 1.25$); the internal-standard peak should, on the other hand, appear close to the analyte peak. With a suitable internal standard, precisions of better than 1% relative can usually be achieved.

The Area-Normalization Method

Another approach that avoids the uncertainties associated with sample injection is the area-normalization method. Complete elution of all components of the sample is required. In the normalization method, the areas of all eluted peaks are computed; after correcting these areas for differences in the detector response to different compound types, the concentration of the analyte is found from the ratio of its area to the total area of all peaks. The following example illustrates the procedure.

EXAMPLE 26-2

The area normalization method was applied to the determination of normal-, secondary-, iso-, and tertiary-butyl alcohol. To determine the relative response factor for the alcohols, a standard solution of the alcohols was prepared and its gas chromatogram observed. The results were as follows:

Alcohol	Mass Taken, g	% Alcohol by Mass	Peak Area A, cm^2	% Area by Mass	Relative Response Factor F
<i>n</i> -Butyl	0.1731	24.61	3.023	8.141	1.000
<i>i</i> -Butyl	0.1964	27.92	3.074	9.083	1.116
<i>s</i> -Butyl	0.1514	21.52	3.112	6.915	0.849
<i>t</i> -Butyl	<u>0.1826</u>	<u>25.96</u>	<u>3.004</u>	8.642	1.062
$\Sigma \text{ mass} = 0.7035$		$\Sigma \% = 100.00$	$\Sigma A = 12.213$		

¹¹See N. A. Dyson, *Chromatographic Integration Methods*, 2nd ed., London: Royal Society of Chemistry, 1998.

The relative response factors were obtained by dividing the data in column 5 by 8.141, the first entry in column 5.

A sample containing only the four alcohols yielded the area data in the second column below. Calculate the percent of each alcohol present by mass.

Solution

The results are shown in column 4 below.

Alcohol	Peak Area, cm ²	Area $\times$ F	% Alcohol by Mass
<i>n</i> -Butyl	1.731	1.731	18.18
<i>i</i> -Butyl	3.753	4.188	43.99
<i>s</i> -Butyl	2.845	2.415	25.36
<i>t</i> -Butyl	1.117	1.186	12.46
		$\Sigma = 9.521$	99.99

Unfortunately, it is often not practical to arrange conditions so that all of the components of a mixture are eluted from a column in a reasonable period. As a result, the area-normalization method has limited applications.

QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

26-1 Define

- | | | |
|---------------------------|--------------------------|----------------------------|
| (a) elution | (e) retention time | (i) selectivity factor |
| (b) mobile phase | (f) retention factor | (j) plate height |
| (c) stationary phase | (g) volumetric flow rate | (k) column resolution |
| (d) distribution constant | (h) linear flow velocity | (l) longitudinal diffusion |

26-2 Describe the general elution problem.

26-3 List the variables that lead to zone broadening in chromatography.

26-4 What are the major differences between gas-liquid and liquid-liquid chromatography?

26-5 What are the differences between liquid-liquid and liquid-solid chromatography?

26-6 What variables are likely to affect the selectivity factor α for a pair of analytes?

26-7 Describe how the retention factor for a solute can be manipulated.

26-8 Describe a method for determining the number of plates in a column.

26-9 Name two general methods for improving the resolution of two substances on a chromatographic column.

26-10 Why does the minimum in a plot of plate height versus flow rate occur at lower flow rates with LC than with GC?

26-11 What is gradient elution?

26-12 List the variables in chromatography that lead to zone separation.

26-13 What would be the effect on a chromatographic peak of introducing the sample at too slow a rate?

*** 26-14** The following data are for a liquid chromatographic column

Length of packing	24.7 cm
Flow rate	0.313 mL/min
V_M	1.37 mL
V_S	0.164 mL

A chromatogram of a mixture of species A, B, C, and D provided the following data:

	Retention Time, min	Width of Peak Base (W), min
Nonretained	3.1	—
A	6.2	0.49
B	13.3	1.07
C	15.7	1.32
D	21.6	1.72

Calculate

- (a) the number of plates from each peak.
- (b) the mean and the standard deviation for N .
- (c) the plate height for the column.

*** 26-15** From the data in Problem 26-14, calculate for A, B, C, and D

- (a) the retention factor.
- (b) the distribution constant.

*** 26-16** From the data in Problem 26-14, calculate for species B and C

- (a) the resolution.
- (b) the selectivity factor α .
- (c) the length of column necessary to separate the two species with a resolution of 2.5.
- (d) the time required to separate the two species on the column in part (c).

*** 26-17** From the data in Problem 26-14, calculate for species C and D

- (a) the resolution.
- (b) the length of column necessary to separate the two species with a resolution of 2.5.



*** 26-18** The following data were obtained by gas-liquid chromatography on a 40-cm packed column:

Compound	t_R , min	W , min
Air	1.9	—
Methylcyclohexane	10.0	0.76
Methylcyclohexene	10.9	0.82
Toluene	13.4	1.06

Calculate

- (a) an average number of plates from the data.
- (b) the standard deviation for the average in (a).
- (c) an average plate height for the column.

>> QUESTIONS AND PROBLEMS (continued)



* 26-19 Referring to Problem 26-18, calculate the resolution for

- (a) methylcyclohexene and methylcyclohexane.
- (b) methylcyclohexene and toluene.
- (c) methylcyclohexane and toluene.



* 26-20 If a resolution of 1.5 is desired in separating methylcyclohexane and methylcyclohexene in Problem 26-18,

- (a) how many plates are required?
- (b) how long must the column be if the same packing is used?
- (c) what is the retention time for methylcyclohexene on the column of part (b)?



* 26-21 If V_S and V_M for the column in Problem 26-18 are 19.6 and 62.6 mL, respectively, and a nonretained air peak appears after 1.9 min, calculate

- (a) the retention factor for each compound.
- (b) the distribution constant for each compound.
- (c) the selectivity factor for methylcyclohexane and methylcyclohexene.



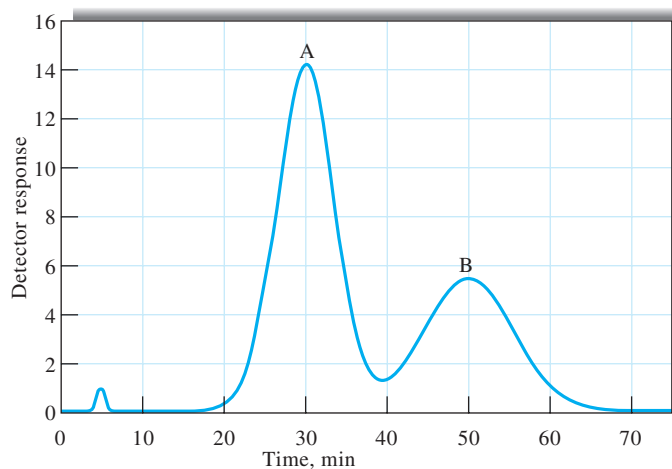
* 26-22 The relative areas for the five gas chromatographic peaks obtained in the separation of five steroids are given next. Also shown are the relative responses of the detector to the five compounds. Calculate the percentage of each component in the mixture.

Compound	Peak Area, Relative	Detector Response, Relative
Dehydroepiandrosterone	27.6	0.70
Estradiol	32.4	0.72
Estrone	47.1	0.75
Testosterone	40.6	0.73
Estriol	27.3	0.78

Challenge Problem

26-23 A chromatogram of a two-component mixture on a 25-cm packed LC column is shown in the following figure. The flow rate was 0.40 mL/min.

- (a) Find the times that components A and B spend in the stationary phase.
- (b) Find the retention times for A and B.
- (c) Determine the retention factors for the two components.
- (d) Find the full widths of each peak and the full width at half-maximum values.
- (e) Find the resolution of the two peaks.



- (f) Find the average number of plates for the column.
- (g) Find the average plate height.
- (h) What column length would be needed to achieve a resolution of 1.75?
- (i) What time would be required to achieve the resolution in part (h)?
- (j) Assume that the column length is fixed at 25 cm and the packing material is fixed. What measures could you take to increase the resolution to achieve baseline separation?
- (k) Are there any measures you could use to achieve a better separation in a shorter time with the same column as in part (j)?

High-Performance Liquid Chromatography

High-performance liquid chromatography (HPLC) is the most versatile and widely used type of elution chromatography. The technique is used by scientists for separating and determining species in a variety of organic, inorganic, and biological materials. In liquid chromatography, the mobile phase is a liquid solvent containing the sample as a mixture of solutes. The types of HPLC are often classified by separation mechanism or by the type of stationary phase. The varieties include (1) **partition, or liquid-liquid, chromatography**; (2) **adsorption, or liquid-solid chromatography**; (3) **ion-exchange, or ion, chromatography**; (4) **size-exclusion chromatography**; (5) **affinity chromatography**, and (6) **chiral chromatography**. Most of this chapter deals with column applications of these important types of chromatography. The final section, however, presents a brief description of planar liquid chromatography because this technique provides a simple and inexpensive way of determining likely optimal conditions for column separations.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

Early liquid chromatography (LC) was carried out in glass columns with diameters of 10–50 mm. The columns were packed with 50- to 500-cm lengths of solid particles coated with an adsorbed liquid that formed the stationary phase. To ensure reasonable flow rates through this type of stationary phase, the particle size of the solid was kept larger than 150–200 μm ; even then, flow rates were at best a few tenths of a milliliter per minute. Thus, separation times were long—often several hours. Attempts to speed up this classic procedure by applying vacuum or pressure were not effective because increases in flow rates tended to increase plate heights beyond the minimum in the typical plate height versus flow rate curve (see Figure 26-8a). Decreased efficiencies were the result.

Early in the development of LC, scientists realized that major increases in column efficiency could be achieved by decreasing the particle size of packings. It was not until the late 1960s, however, that the technology for producing and using packings with particle diameters as small as 3–10 μm was developed. This technology required sophisticated instruments operating at high pressures, which contrasted markedly with the simple glass columns of classic gravity-flow LC. The name *high-performance liquid chromatography* (HPLC) was originally used to distinguish these newer procedures from the original gravity-flow methods. Today, virtually all LC is done using pressurized flow, and we use the abbreviations LC and HPLC interchangeably.¹

Over the past two decades, the ever-increasing need for rapid and efficient separations has stimulated the development of ultra-high-performance (pressure) liquid chromatography (UHPLC), which relies on packings of sub-2- μm particles and pressures up to 20,000 psi. Since 2004, manufacturers of HPLC equipment have added UHPLC capabilities to many of their instruments.² The photo on page 695 is an example of such a unit.

¹For detailed discussions of HPLC, see L. R. Snyder, J. J. Kirkland, and J. W. Dolan, *Introduction to Modern Liquid Chromatography*, 3rd ed., Hoboken, NJ: Wiley, 2010; V. Meyer, *Practical High-Performance Liquid Chromatography*, 5th ed., Chichester, UK: Wiley 2010.

²S. Fekete et al., *Trends Anal. Chem.*, **2014**, 63, 2, DOI: 10.1016/j.trac.2014.08.007.

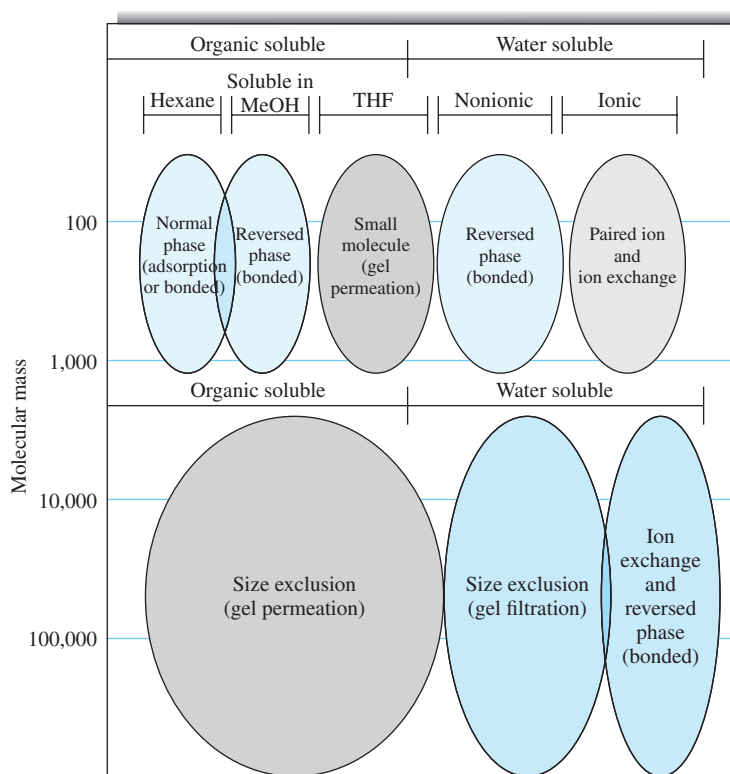


FIGURE 28-1 Applications of liquid chromatography. Methods can be chosen based on solubility and molecular mass. In most cases for nonionic small molecules ($M < 2000$), reversed-phase methods are suitable. Techniques toward the bottom of the diagram are best suited for species of high molecular mass ($M > 2000$). MeOH = methanol; THF = tetrahydrofuran (Adapted from *High Performance Liquid Chromatography*, 2nd ed., S. Lindsay and H. Barnes, eds. Copyright 1987, 1992, Thames Polytechnic, London, UK. New York: Wiley, 1992.)

28A SCOPE OF HPLC

LC is the most widely used of all of the analytical separation techniques. The reasons for the popularity of the method are its sensitivity, its ready adaptability to accurate quantitative determinations, its ease of automation, its suitability for separating nonvolatile species or thermally fragile ones, and above all, its widespread applicability to substances that are important to industry, to many fields of science, and to the public. Examples of such materials include amino acids, proteins, nucleic acids, hydrocarbons, carbohydrates, drugs, terpenoids, pesticides, antibiotics, steroids, metal-organic species, and a variety of inorganic substances.

Figure 28-1 reveals that the various liquid chromatographic procedures are complementary in their application. Thus, for solutes having molecular masses greater than 10,000, size-exclusion chromatography is often used, although it is now becoming possible to handle such compounds by reversed-phase chromatography as well. For lower-molecular-mass ionic species, ion-exchange chromatography is widely used. Small

polar but nonionic species are best handled by reversed-phase methods. In addition, this procedure is frequently useful for separating members of a homologous series. Adsorption chromatography was once used for separating nonpolar species, structural isomers, and compound classes such as aliphatic hydrocarbons from aliphatic alcohols. Because of problems with retention reproducibility and irreversible adsorption, adsorption chromatography with solid stationary phases has been largely replaced by normal-phase (bonded-phase) chromatography. Among the specialized forms of LC, affinity chromatography is widely used for isolation and preparation of biomolecules, and chiral chromatography is used for separating enantiomers.

28B COLUMN EFFICIENCY IN LC

The discussion on band broadening in Section 26C-3 is generally applicable to LC. Here, we illustrate the important effect of stationary-phase particle size and describe two additional sources of zone spreading that are sometimes of considerable importance in LC.

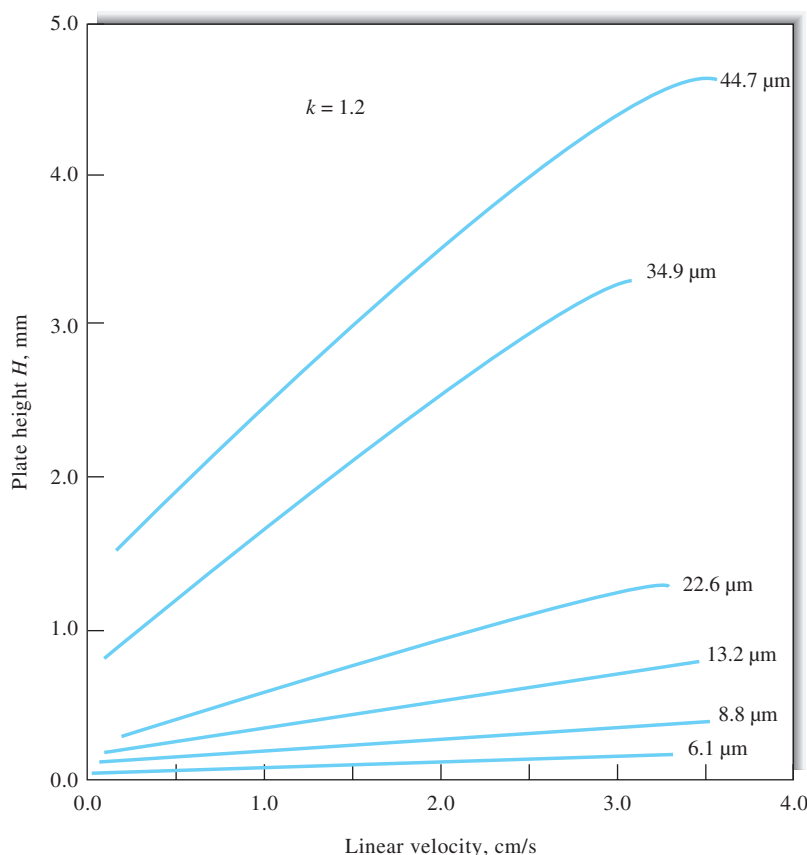


FIGURE 28-2 Effect of particle size of packing and flow rate on plate height H in LC. Column dimensions: 30 cm $\times$ 2.4 mm. Solute: N,N' -diethyl- p -aminoazobenzene. Mobile phase: mixture of hexane, methylene chloride, and isopropyl alcohol. (Adapted from R. E. Majors, *J. Chromatogr. Sci.*, **1973**, 11, 88, DOI: 10.1093/chromsci/11.2.88).

28B-1 Effects of Particle Size of Packings

The mobile-phase mass-transfer coefficient (see Table 26-3) reveals that C_M in Equation 26-23 is directly related to the square of the diameter d_p of the particles making up a packing. Because of this, the efficiency of an LC column should improve dramatically as the particle size decreases. Figure 28-2 is an experimental demonstration of this effect, where it is seen that a reduction of particle size from 45 to 6 μm results in a tenfold or more decrease in plate height. Note that none of the plots in this figure exhibits the minimum that is predicted by Equation 26-23. Such minima are, in fact, observable in LC (see Figure 26-8a) but usually at flow rates too low for most practical applications.

28B-2 Extracolumn Band Broadening in LC

In LC, significant band broadening sometimes occurs outside the column packing itself. This *extracolumn band broadening* occurs as the solute is carried through open tubes such as those found in the injection system, the detector region, and the piping connecting the various components of the system. Here, broadening arises from differences in flow rates between layers of liquid adjacent to the wall and the center of the tube. As a result, the center part of

a solute band moves more rapidly than the peripheral part. In gas chromatography (GC), extracolumn spreading is largely offset by diffusion. Diffusion in liquids, however, is significantly slower, and band broadening of this type often becomes noticeable.

It has been shown that the contribution of extracolumn effects H_{ex} to the total plate height is given by³

$$H_{\text{ex}} = \frac{\pi r^2 u}{24 D_M} \quad (28-1)$$

where u is the linear-flow velocity (cm/s), r is the radius of the tube (cm), and D_M is the diffusion coefficient of the solute in the mobile phase (cm^2/s).

Extracolumn broadening can become quite serious when small-bore columns are used. Here, the radius of the extracolumn components should be reduced to 0.010 inch or less, and the length of extracolumn tubing made as small as feasible to minimize this source of broadening.

³R. P. W. Scott and P. Kucera, *J. Chromatogr. Sci.*, **1971**, 9, 641, DOI: 10.1093/chromsci/9.11.641.

28C INSTRUMENTATION

Pumping pressures of several hundred atmospheres are required to achieve reasonable flow rates with packings of 3–10 μm , which are common in modern LC. Because of these high pressures, the equipment for HPLC tends to be more elaborate and expensive than equipment for other types of chromatography. Figure 28-3 is a diagram showing the important components of a typical LC instrument.

28C-1 Mobile-Phase Reservoirs and Solvent Treatment Systems

A modern LC apparatus is equipped with one or more glass reservoirs, each of which contains 500 mL or more of a solvent. Provisions are often included to remove dissolved gases and dust from the liquids. Dissolved gases can lead to irreproducible flow rates and band spreading; in addition, both bubbles and dust interfere with the performance of most detectors. Degassers may consist of a vacuum pumping system, a distillation system, a device for heating and stirring, or as shown in Figure 28-3, a system for *sparging*, in which the dissolved gases are swept out of solution by fine bubbles of an inert gas that is not soluble in the mobile phase. Often the systems also contain a means of filtering dust and particulate matter from the solvents to prevent these particles from damaging the pumping or injection systems or clogging the column. It is not necessary that the degassers and filters be integral parts of the HPLC system as shown in Figure 28-3. For example, a convenient way of treating solvents before introduction into the reservoir is to filter them through a millipore filter under vacuum. This treatment removes gases as well as suspended matter.

An elution with a single solvent or solvent mixture of constant composition is termed an *isocratic elution*. In *gradient elution*, two (and sometimes more) solvent systems that differ significantly in polarity are used and varied in composition during the separation. The ratio of the two solvents is varied in a preprogrammed way, sometimes continuously and sometimes in a series of steps. Modern HPLC instruments are often equipped with proportioning valves that introduce liquids from two or more reservoirs at ratios that can be varied continuously (Figure 28-3). The volume ratio of the solvents can be altered linearly or exponentially with time.

Figure 28-4 illustrates the advantage of a gradient eluent in the separation of a mixture of chlorobenzenes. Isocratic elution with a 50:50 (v/v) methanol-water solution yielded the curve in Figure 28-4b. The curve in Figure 28-4a is for gradient elution, which was initiated with a 40:60 mixture of the two solvents; the methanol concentration was then increased at the rate of 8%/min. Note that gradient elution shortened the time of separation significantly without sacrificing the resolution of the early peaks. Note also that gradient elution produces effects similar to those produced by temperature programming in GC (see Figure 27-7).

28C-2 Pumping Systems

The requirements for liquid chromatographic pumps include (1) the generation of pressures of up to 6000 psi (lb/in.²)



Simulation: Learn more about **liquid chromatography** at www.tinyurl.com/skoogpia7

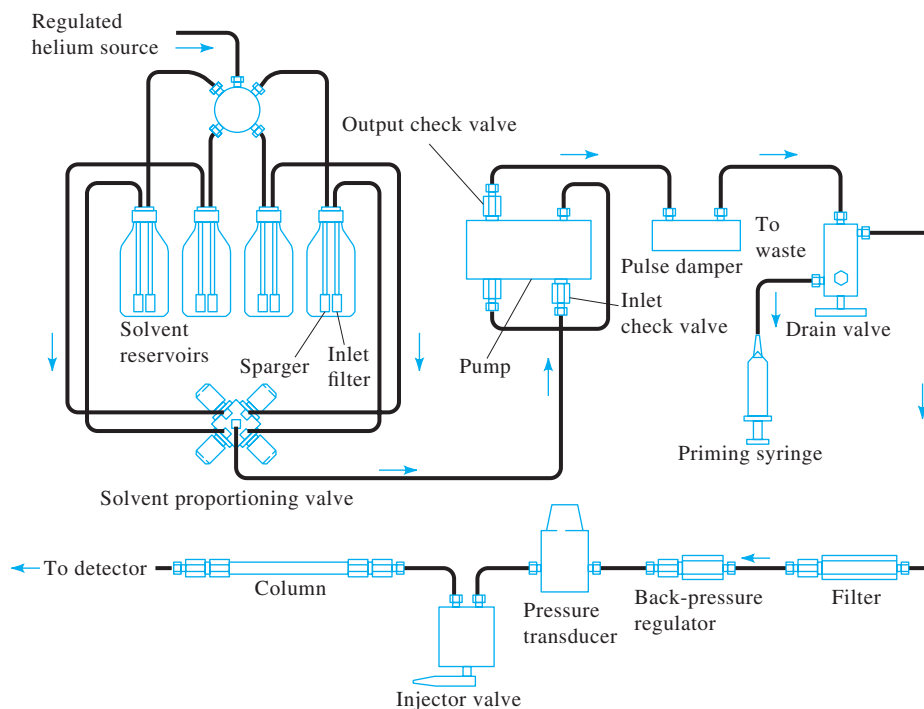


FIGURE 28-3 Block diagram showing components of a typical apparatus for HPLC. (Courtesy of PerkinElmer Corp., Norwalk, CT).

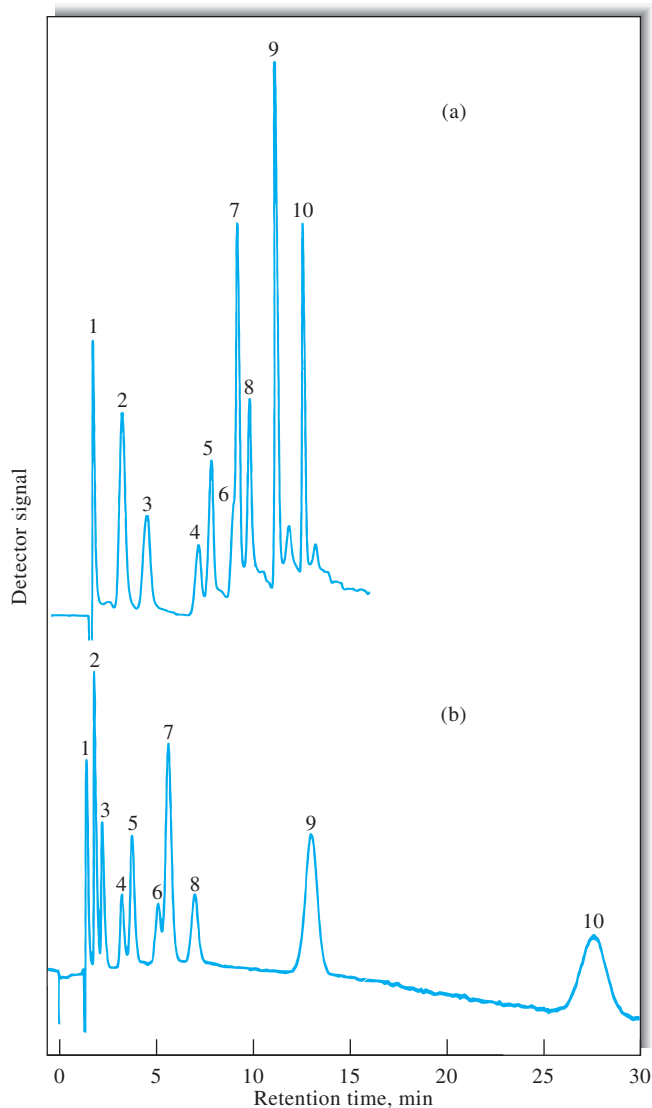


FIGURE 28-4 Improvement in separation effectiveness by (a) gradient elution and (b) isocratic elution. Column: 1 m $\times$ 2.1 mm inside-diameter, precision-bore stainless steel; packing: 1% Permaphase® ODS (C₁₈). Sample: 5 μ L of chlorinated benzenes in isopropanol. Detector: UV photometer (254 nm). Conditions: temperature, 60°C; pressure, 1200 psi. Peak identities: (1) benzene, (2) monochlorobenzene, (3) orthodichlorobenzene, (4) 1,2,3-trichlorobenzene, (5) 1,3,5-trichlorobenzene, (6) 1,2,4-trichlorobenzene, (7) 1,2,3,4-tetrachlorobenzene, (8) 1,2,4,5-tetrachlorobenzene, (9) pentachlorobenzene, (10) hexachlorobenzene. (Adapted from J. J. Kirkland, *Modern Practice of Liquid Chromatography*, p. 88, New York: Interscience, 1971. Reprinted by permission of John Wiley & Sons, Inc.)

or 414 bar,⁴ (2) pulse-free output, (3) flow rates ranging from 0.1 to 10 mL/min, (4) flow reproducibilities of 0.5% relative or better, and (5) resistance to corrosion by a variety of solvents. The high pressures generated by liquid chromatographic

pumps are not an explosion hazard because liquids are not very compressible. Thus, rupture of a component results only in solvent leakage. However, such leakage may constitute a fire or environmental hazard with some solvents.

Two major types of pumps are used in LC: the screw-driven syringe type and the reciprocating pump. Reciprocating pumps are used in almost all modern commercial chromatographs.

Reciprocating Pumps

Reciprocating pumps usually consist of a small chamber in which the solvent is pumped by the back and forth motion of a motor-driven piston (see Figure 28-5). Two ball check valves, which open and close alternately, control the flow of solvent into and out of a cylinder. The solvent is in direct contact with the piston. As an alternative, pressure may be transmitted to the solvent via a flexible diaphragm, which in turn is hydraulically pumped by a reciprocating piston. Reciprocating pumps have the disadvantage of producing a pulsed flow, which must be damped because the pulses appear as baseline noise on the chromatogram. Modern LC instruments use dual (or triple) pump heads or elliptical cams to minimize such pulsations. The advantages of reciprocating pumps include their small internal volume (35–400 μ L), their high output pressures (up to 10,000 psi, 20,000 psi in UHPLC systems), their adaptability to gradient elution, their large solvent capacities, and their relatively constant flow rates, which are largely independent of column back pressure and solvent viscosity.

Displacement Pumps

Displacement pumps usually consist of large, syringe-like chambers equipped with a plunger activated by a screw-driven mechanism powered by a stepping motor. Displacement pumps also produce a flow that tends to be independent of viscosity and back pressure. In addition, the output is pulse free. Disadvantages include limited solvent capacity ($\sim$ 250 mL), relatively low pressure limits ($\sim$ 10,000 psi), and considerable inconvenience when solvents must be changed.

Flow Control and Programming Systems

As part of their pumping systems, many commercial instruments are equipped with computer-controlled devices for measuring the flow rate by determining the pressure drop across a restrictor located at the pump outlet. Any difference in signal from a preset

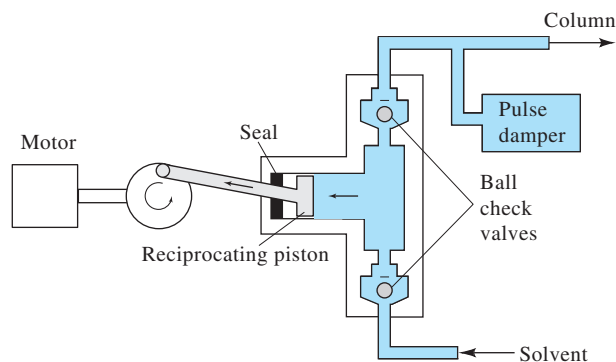


FIGURE 28-5 A reciprocating pump for HPLC.

⁴UHPLC systems require pressures up to 20,000 psi = 1361 atm = 137.9 MPa = 1,379 bar.

value is then used to increase or decrease the speed of the pump motor. Most instruments also have a means for varying the composition of the solvent either continuously or in a stepwise fashion. For example, the instrument shown in Figure 28-3 contains a proportioning valve that permits mixing of up to four solvents in a preprogrammed and continuously variable way.

28C-3 Sample-Injection Systems

Often, the limiting factor in the precision of liquid chromatographic measurements is the reproducibility with which samples can be introduced onto the column packing. The problem is exacerbated by band broadening, which accompanies a lengthy sample injection plug. Thus, sample volumes must be very small—a few tenths of a microliter to perhaps 500 μL . Furthermore, it is convenient to be able to introduce the sample without depressurizing the system.

The most widely used method of sample introduction in LC is based on sampling loops, such as that shown in Figures 28-6 and 27-5. These devices are often an integral part of liquid-chromatographic equipment and have interchangeable loops providing a choice of sample sizes from 1 μL to 100 μL or more. Sampling loops of this type permit the introduction of samples at pressures up to 7000 psi with relative standard deviations of a few tenths of a percent.

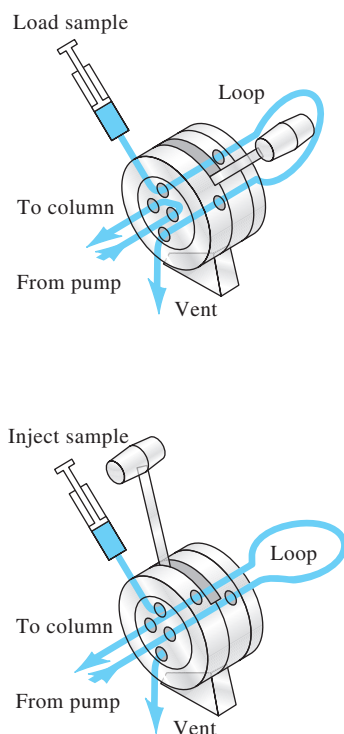


FIGURE 28-6 A sampling loop for LC. With the valve handle as shown on the left, the loop is filled from the syringe, and the mobile phase flows from pump to column. When the valve is placed in the position on the right, the loop is inserted between the pump and the column so that the mobile phase sweeps the sample onto the column. (Courtesy of Beckman-Coulter, Inc.)

Most chromatographs today are sold with autoinjectors. Such units are capable of injecting samples into the LC from vials on a sample carousel or from microtiter plates. They usually contain sampling loops and a syringe pump for injection volumes from less than 1 μL to more than 1 mL. Some have controlled-temperature environments that allow for sample storage and for carrying out derivatization reactions prior to injection. Most are programmable to allow for unattended injections into the LC system. Commercial UHPLC systems have injection volumes of 0.1–50 μL .⁵

28C-4 Columns for HPLC

Liquid-chromatographic columns are usually constructed from smooth-bore stainless steel tubing. HPLC columns are sometimes made from heavy-walled glass tubing and polymer tubing, such as polyetheretherketone (PEEK). In addition, stainless steel columns lined with glass or PEEK are also available. Hundreds of packed columns differing in size and packing are available. The cost of standard-sized, nonspecialty columns ranges from \$200 to more than \$500. Specialized columns, such as chiral columns, can cost more than \$1000.

Analytical Columns

Most liquid-chromatographic columns range from 5 to 25 cm long. Straight columns are invariably used. Sometimes length is added by coupling two or more columns. The inside diameter of analytical columns is often 3–5 mm; the most common particle size of packings is 3 or 5 μm . The most common columns are 10 or 15 cm long, 4.6 mm in inside diameter, and packed with 5- μm particles. Columns of this type generate 40,000–70,000 plates/meter (typically about 10,000 plates/column).

In the 1980s microcolumns became available with inside diameters of 1–4.6 mm and lengths of 3–7.5 cm. These columns, which are packed with 3- or 5- μm particles, achieve as many as 100,000 plates/m and have the advantage of speed and minimal solvent consumption. This latter property is of considerable importance because the high-purity solvents required for LC are expensive to purchase and to dispose of after use. Figure 28-7 illustrates the speed with which a separation can be performed on a microbore column. In this example, eight diverse components are separated in about 15 s. The column was 4 cm long and had an inside diameter of 4 mm; it was packed with 3- μm particles.

Precolumns

There are two types of precolumns. A precolumn between the mobile phase reservoir and the injector is used for mobile-phase conditioning and is termed a *scavenger column*. The solvent partially dissolves the silica packing and ensures that the mobile phase is saturated with silicic acid prior to entering the analytical column. This saturation minimizes losses of the stationary phase from the analytical column.

⁵J. De Vos, K. Broeckhoven, and S. Eeltink, *Anal. Chem.*, **2016**, *88*, 262, DOI: 10.1021/acs.analchem.5b04381.

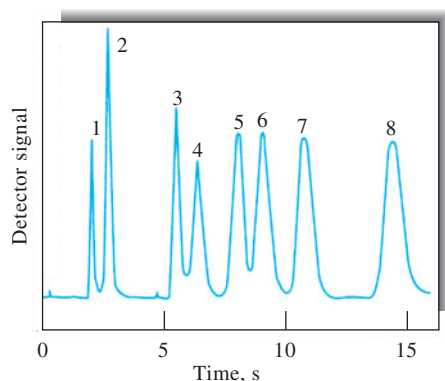


FIGURE 28-7 High-speed isocratic separation. Column dimensions: 4 cm long, 0.4 cm inside diameter; packing: 3- μ m spherisorb; mobile phase: 4.1% ethyl acetate in *n*-hexane. Compounds: (1) *p*-xylene, (2) anisole, (3) benzyl acetate, (4) dioctyl phthalate, (5) dipentyl phthalate, (6) dibutyl phthalate, (7) dipropyl phthalate, (8) diethyl phthalate. (Adapted from R. P. W. Scott, *Small Bore Liquid Chromatography Columns: Their Properties and Uses*, New York: Wiley, 1984, p. 156.)

A second type of precolumn is a *guard column*, positioned between the injector and the analytical column. A guard column is a short column packed with a similar stationary phase as the analytical column. The purpose of the guard column is to prevent impurities, such as highly retained compounds and particulate matter, from reaching and contaminating the analytical column. The guard column is replaced regularly and serves to increase the lifetime of the analytical column.

Column Temperature Control

For some applications, close control of column temperature is not necessary and columns are operated at room temperature. Often, however, better, more reproducible chromatograms are obtained by maintaining constant column temperature. Most modern commercial instruments are now equipped with heaters that control column temperatures to a few tenths of a degree from near ambient to 150°C. Columns may also be fitted with water jackets fed from a constant-temperature bath to give precise temperature control. Many chromatographers consider temperature control to be essential for reproducible separations.

Temperature control is important in UHPLC for the same reasons that it is important in ordinary HPLC, but frictional heating resulting from ultra-high pressures creates temperature gradients in the column. These gradients may be minimized by good temperature control, which leads to column stability and high-efficiency separations. Temperatures above 100°C can lead to column degradation, so operating temperatures in the range of 60–100°C strike a good compromise for UHPLC.⁶

28C-5 Types of Column Packings

Two basic types of packings have been used in LC, *pellicular* and *porous particle*. The original pellicular particles were spherical, nonporous, glass or polymer beads with typical diameters of 30–40 μ m. A thin, porous layer of silica, alumina, a polystyrene-divinylbenzene synthetic resin, or an ion-exchange resin was deposited on the surface of these beads. Small porous microparticles have completely replaced these large pellicular particles. In recent years, small ($\sim$ 5 μ m) pellicular packings have been reintroduced for separation of proteins and large biomolecules.

The typical porous particle packing for LC consists of porous microparticles having diameters ranging from 3 to 10 μ m; for a given size particle, a very narrow particle size distribution is desirable. The particles are composed of silica, alumina, the synthetic resin polystyrene-divinylbenzene, or an ion-exchange resin. Silica is by far the most common packing in LC. Silica particles are prepared by agglomerating submicron silica particles under conditions that lead to larger particles having highly uniform diameters. The resulting particles are often coated with thin organic films, which are chemically or physically bonded to the surface. Column packings for specific chromatographic modes are discussed in later sections of this chapter.

28C-6 Detectors

No detectors for LC are as universally applicable as the flame ionization and thermal conductivity detectors for GC described in Section 27B-4. GC detectors were specifically developed to measure small concentrations of analytes in flowing gas streams. On the other hand, LC detectors have often been traditional analytical instruments adapted with flow cells to measure low concentrations of solutes in liquid streams. A major challenge in the development of LC has been in adapting and improving such devices.⁷

Characteristics of the Ideal Detector

The ideal detector for LC should have all of the properties listed in Section 27B-4 for GC detectors with the exception that an LC detector need not be responsive over as great a temperature range. In addition, an HPLC detector should have minimal internal volume to reduce zone broadening (Section 28B-2) and should be compatible with liquid flow.

Types of Detectors

Liquid chromatographic detectors are of two basic types. *Bulk-property detectors* respond to a mobile-phase bulk property, such as refractive index, dielectric constant, or density, that is modulated by the presence of solutes. In contrast, *solute-property*

⁶S. Fekete et al., *Trends Anal. Chem.*, **2014**, 63, 2, DOI: 10.1016/j.trac.2014.08.007.

⁷For more detailed discussions, see *Liquid Chromatography: Fundamentals and Instrumentation*, F. Salvatore, P. R. Haddad, C. Poole, and D. K. Lloyd, eds., Boston: Elsevier, 2013; R. P. W. Scott, *Chromatographic Detectors*, New York: Dekker, 1996; R. P. W. Scott, *Liquid Chromatography Detectors*, 2nd ed., Amsterdam: Elsevier, 1986.

TABLE 28-1 Performance of HPLC Detectors

HPLC Detector	Commercially Available	Mass LOD* (typical)	Linear Range [†] (decades)
Absorbance	Yes	10 pg	3–4
Fluorescence	Yes	10 fg	5
Electrochemical	Yes	100 pg	4–5
Refractive index	Yes	1 ng	3
Conductivity	Yes	100 pg–1 ng	5
Mass spectrometry	Yes	<1 pg	5
FTIR	Yes	1 µg	3
Light scattering	Yes	1 µg	5
Optical activity	No	1 ng	4
Element selective	Yes	1 ng	4–5
Photoionization	Yes	<1 pg	4

Sources: From manufacturer's literature; *Handbook of Instrumental Techniques for Analytical Chemistry*, F. Settle, ed., Upper Saddle River, NJ: Prentice-Hall, 1997; E. S. Yeung and R. E. Synovec, *Anal. Chem.*, **1986**, 58, 1237A, DOI: 10.1021/ac00125a797.

*Mass LODs (limits of detection) depend on compound, instrument, and HPLC conditions, but those given are typical values with commercial systems when available.

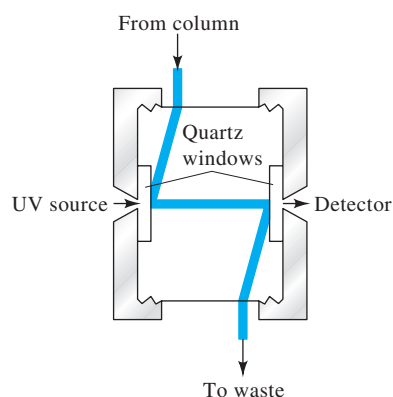
[†]Typical values from the preceding sources.

detectors respond to some property of solutes, such as UV absorbance, fluorescence, or diffusion current, that is not possessed by the mobile phase.

Table 28-1 lists the most common detectors for HPLC and some of their most important properties. The most widely used detectors for LC are based on absorption of UV or visible radiation (see Figure 28-8). Fluorescence, refractive-index, and electrochemical detectors are also widely used. Mass spectrometry (MS) detectors are currently quite popular. Such LC/MS systems can greatly aid in identifying the analytes exiting from the HPLC column as discussed later in this section.

UV-Visible Absorption Detectors

Figure 28-8 is a schematic of a typical, Z-shaped, flow-through cell for absorption measurements on eluents from a chromatographic column. To minimize extracolumn band broadening, the volume of such a cell should be kept as small as possible. Typical cell volumes are 1–10 µL with cell path lengths of 2–10 mm.


FIGURE 28-8 A UV-visible absorption cell for HPLC.

Many absorption detectors are double-beam devices in which one beam passes through the eluent cell and the other beam is a reference beam. Matched photoelectric detectors are then used to compare the intensities of the two beams. Alternatively, a chopped beam system similar to that shown in Figure 9-13b (see also Figure 13-13b) is used in conjunction with a single phototransducer. In either case, the chromatogram consists of a plot of the absorbance (logarithm of the ratio of the two transduced signals) as a function of time. Single-beam instruments are also encountered. Here, intensity measurements of the solvent system are stored in a computer memory and ultimately recalled for the calculation of absorbance.

UV Absorption Detectors with Filters. The earliest absorption detectors were filter photometers with a mercury lamp as the source. Most commonly, the intense line at 254 nm was isolated by filters; with some instruments, lines at 250, 313, 334, and 365 nm could also be used by substitution of filters. This type of detector is restricted to solutes that absorb at one of these wavelengths. As shown in Section 14B, several organic functional groups and a number of inorganic species exhibit broad absorption bands that encompass one or more of these UV wavelengths.

Deuterium or tungsten filament sources with interference filters were also used to provide a simple means of detecting absorbing species as they eluted from a column. Some instruments were dual wavelength devices, or they were equipped with filter wheels containing several filters that could be rapidly switched to detect various species as they eluted. Today, filter instruments have been almost entirely replaced by scanning and diode array spectrometers.

Absorption Detectors with Scanning Capabilities. Most HPLC manufacturers offer detectors that consist of a scanning

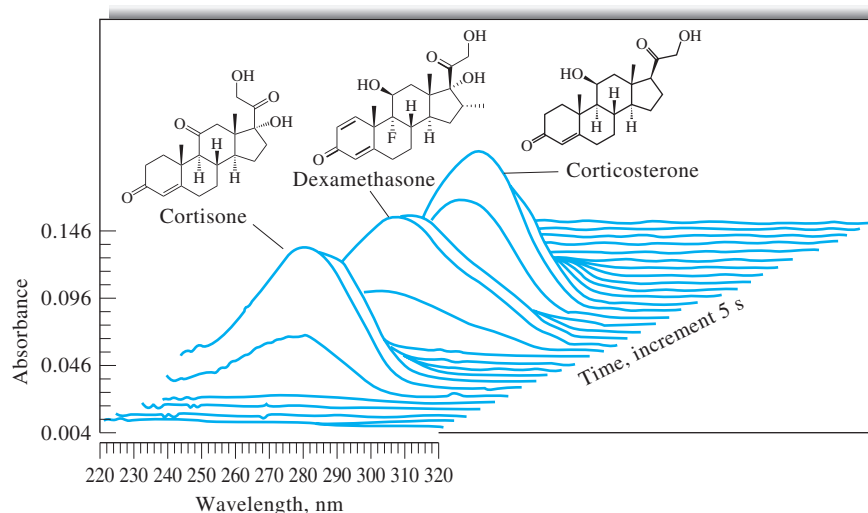


FIGURE 28-9 Absorption spectra of the eluent from a mixture of three steroids taken at 5-second intervals. (Source: Agilent Technologies)

spectrophotometer with grating optics. Some are limited to UV radiation; others encompass both UV and visible radiation. Several operational modes can be chosen. For example, the entire chromatogram can be obtained at a single wavelength; alternatively, when eluent peaks are sufficiently separated in time, different wavelengths can be chosen for each peak. Here again, computer control is often used to select the best wavelength for each eluent. Where entire spectra are desired for identification purposes, the flow of eluent can be stopped for a sufficient period to permit scanning the wavelength region of interest.

The most powerful UV spectrophotometric detectors are array-based instruments as described in Section 7E-3 and Figure 13-25.⁸ Several manufacturers offer such instruments, which permit collection of an entire spectrum in approximately 1 second. Thus, spectral data for each chromatographic peak can be collected and stored as it appears at the end of the column. One form of presentation of the spectral data, which is helpful in identification of species and for choosing conditions for quantitative determination, is a three-dimensional plot such as that shown in Figure 28-9. Here, spectra were obtained at successive 5-second intervals. The appearance and disappearance of each of the three steroids in the eluent can be seen.

Infrared Absorption Detectors. Two types of infrared detectors have been offered commercially. The first is a filter instrument similar in design to that shown in Figure 16-13. The second, and more sophisticated, type of infrared detector is based on Fourier transform instruments similar to those discussed in Section 16B-1. Several of the manufacturers of Fourier

transform infrared (FTIR) instruments offer accessories that permit their use as HPLC detectors. Infrared detector cells are similar in construction to those used with UV radiation except that windows are constructed of sodium chloride or calcium fluoride. Cell path lengths range from 0.2 to 1.0 mm and volumes from 1.5 to 10 μL .

A major limitation to the use of IR detectors lies in the low transparency of many useful solvents. For example, the broad infrared absorption bands for water and the alcohols largely preclude the use of this detector for many applications. Also, the use of aqueous mobile phases can lead to rapid deterioration of cell materials unless special windows are used. The poor limits of detection for many solutes has likewise limited the use of IR detectors. The introduction of mass spectrometric detectors for LC (discussed later in this section) has led to a dramatic decline in applications of IR detection.

Fluorescence Detectors

Fluorescence detectors for HPLC are similar in design to the fluorometers and spectrofluorometers described in Section 15B-2. In most, fluorescence is observed by a photoelectric transducer located at 90° to the excitation beam. The simplest detectors use a mercury excitation source and one or more filters to isolate a band of emitted radiation. More sophisticated instruments are based on a xenon source and use a grating monochromator to isolate the fluorescence radiation. Laser-induced fluorescence is also used because of its sensitivity and selectivity.

As was pointed out in Chapter 15, an inherent advantage of fluorescence methods is their high sensitivity, which is typically greater by more than an order of magnitude than most absorption procedures. This advantage has been exploited in LC for the separation and determination of the components of samples that fluoresce. As was discussed in Section 15C-2, fluorescent

⁸See R. P. W. Scott, *Chromatographic Detectors*, New York: Dekker, 1996; *Diode Array Detection in HPLC*, L. Huber and S. A. George, eds., New York: Dekker, 1993.

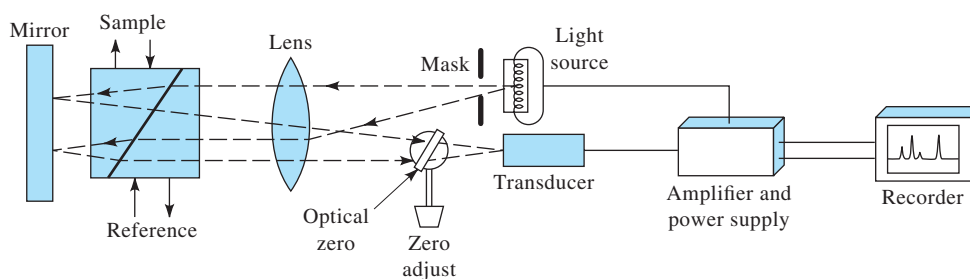


FIGURE 28-10 Schematic diagram of a differential refractive-index detector. (Courtesy of Waters Associates, Inc., Milford, MA.)

compounds are frequently encountered in the analysis of such materials as pharmaceuticals, natural products, clinical samples, and petroleum products. Often, the number of fluorescing species can be enlarged by preliminary treatment of samples with reagents that form fluorescent derivatives. For example, dansyl-chloride (5-dimethylamino-naphthalene-1-sulphonyl chloride), which reacts with primary and secondary amines, amino acids, and phenols to give fluorescent compounds, has been widely used for the detection of amino acids in protein hydrolyzates.

Refractive-Index Detectors

Figure 28-10 shows a schematic diagram of a differential refractive-index detector in which the solvent passes through one half of the cell on its way to the column; the eluate then flows through the other chamber. The two compartments are separated by a glass plate mounted at an angle such that bending of the incident beam occurs if the two solutions differ in refractive index. The resulting displacement of the beam with respect to the photosensitive surface of a detector causes variation in the output signal, which, when amplified and recorded, provides the chromatogram.

Refractive-index detectors have the significant advantage of responding to nearly all solutes. That is, they are general detectors analogous to flame or thermal conductivity detectors in GC. In addition, they are reliable and unaffected by flow rate. They are, however, highly temperature dependent and must be maintained at a constant temperature to a few thousandths of a degree centigrade. Furthermore, they are not as sensitive as many other types of detectors and are not compatible with gradient elution methods. Refractive-index detectors find application in the determination of some analytes such as sugars.

Evaporative Light Scattering Detector

One of the newer types of detectors for HPLC is the *evaporative light scattering detector* (ELSD). In this detector, the column effluent is passed into a nebulizer where it is converted into a fine mist by a flow of nitrogen or air. The fine droplets are then carried through a controlled-temperature drift tube where evaporation of the mobile phase occurs, leading to formation of fine particles of the analyte. The cloud of analyte particles then passes through a laser beam. The scattered radiation is detected at right angles to the flow by a silicon photodiode.

A major advantage of this type of detector is that its response is approximately the same for all nonvolatile solutes. In addition it is significantly more sensitive than the refractive-index detector, with detection limits of 0.2 ng/ μ L. A disadvantage is that mobile-phase compositions are limited to only volatile components.

Electrochemical Detectors

Electrochemical detectors of several types are currently available from instrument manufacturers.⁹ These devices are based on amperometry, voltammetry, coulometry, and conductometry. The first three of these methods are discussed in Chapters 24 and 25.

Although electroanalytical procedures have as yet not been exploited to the extent of optical detectors, they appear to offer advantages, in many instances, of high sensitivity, simplicity, convenience, and widespread applicability. This last property is illustrated in Figure 28-11, which depicts the potential ranges at which oxidation or reduction of sixteen organic functional groups occurs. In principle, species containing any of these groups could be detected by amperometric, voltammetric, or coulometric procedures. A major limitation of electrochemical detectors, however, is that they are not compatible with gradient elution.

Several electrochemical detector cells for HPLC applications have been described in the literature, and several are available from commercial sources. Figure 28-12 is an example of a thin-layer flow-through cell for amperometric detection (see also Figure 25-17). Here, the electrode surface is part of a channel wall formed by sandwiching a 50- μ m Teflon gasket between two machined blocks of Kel-F plastic. The indicator electrode is platinum, gold, glassy carbon, or carbon paste. A reference electrode, and often a counter electrode, is located downstream from the indicator electrode block. The cell volume is 1–5 μ L. A useful modification of this cell, which is available commercially, includes two working electrodes, which can be operated in series or in parallel.¹⁰ The former configuration, in which the eluent

⁹For brief description of commercially available detectors, see B. E. Erickson, *Anal. Chem.*, **2000**, 72, 353A, DOI: 10.1021/ac002813b.

¹⁰D. A. Roston, R. E. Shoup, and P. T. Kissinger, *Anal. Chem.*, **1982**, 54, 1417A, DOI: 10.1021/ac00250a767.

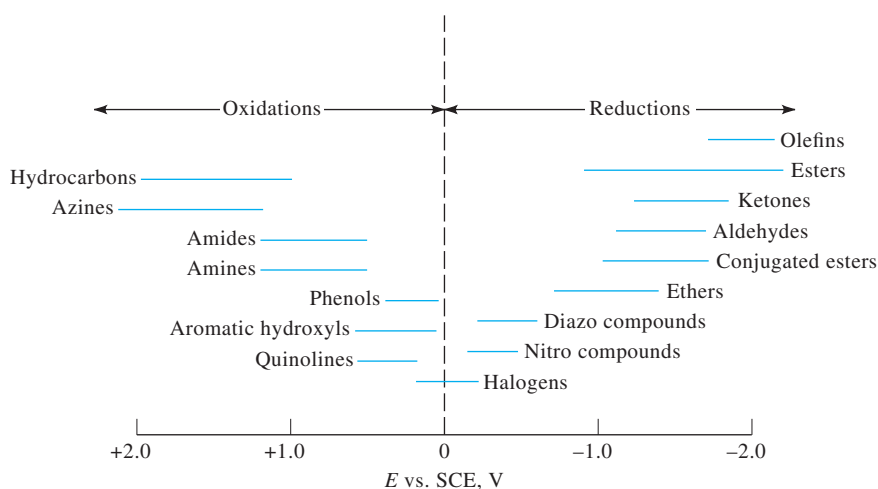
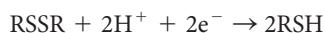


FIGURE 28-11 Potential amperometric detection of organic functional groups. The horizontal lines show the range of oxidation or reduction potentials for electroactive compounds containing the indicated functional groups. SCE = saturated calomel electrode.

flows first over one electrode and then over the second, requires that the analyte undergo a reversible oxidation (or reduction) at the upstream electrode. The second electrode then operates as a cathode (or an anode) to determine the oxidation (or reduction) product. This arrangement enhances the selectivity of the detection system. An interesting application of this system is for the detection and determination of the components in mixtures containing both thiols and disulfides. Here, the upstream mercury electrode reduces the disulfides at about -1.0 V. That is,



A downstream mercury electrode is then oxidized in the presence of the thiols from the original sample as well as those formed at the upstream electrode. That is,



In the parallel configuration, the two electrodes are rotated so that the axis between them is at 90° to the stream flow. The

two can then be operated at different potentials (relative to a downstream reference electrode), which often gives an indication of peak purity. Alternatively, one electrode can be operated as a cathode and the other as an anode, thus making possible simultaneous detection of both oxidants and reductants.

Voltammetric, conductometric, and coulometric detectors are also available commercially. Conductometric detectors are discussed further in Section 28F-3.

Mass Spectrometric Detectors

The combination of LC and mass spectrometry would seem to be an ideal merger of separation and detection.¹¹ Just as in GC, a mass spectrometer can greatly aid in identifying species as they elute from the chromatographic column. There are major problems, however, in the coupling of these two techniques. A gas-phase sample is needed for mass spectrometry, and the output of the LC column is a solute dissolved in a solvent. As a first step, the solvent must be vaporized. When vaporized, however, the LC solvent produces a gas volume that is 10–1000 times greater than the carrier gas in GC. Hence, most of the solvent must also be removed.

There have been several devices developed to solve the problems of solvent removal and LC column interfacing. Today, the most popular approaches use a low-flow-rate atmospheric pressure ionization technique. A block diagram of a typical LC/MS system is shown in Figure 28-13. The HPLC system is often a nanoscale capillary LC system with flow rates in the $\mu\text{L}/\text{min}$ range. Alternatively, some interfaces allow flow rates as high as $1\text{--}2\text{ mL}/\text{min}$, which is typical of conventional HPLC conditions. The most common ionization sources are electrospray ionization and atmospheric pressure chemical ionization (see

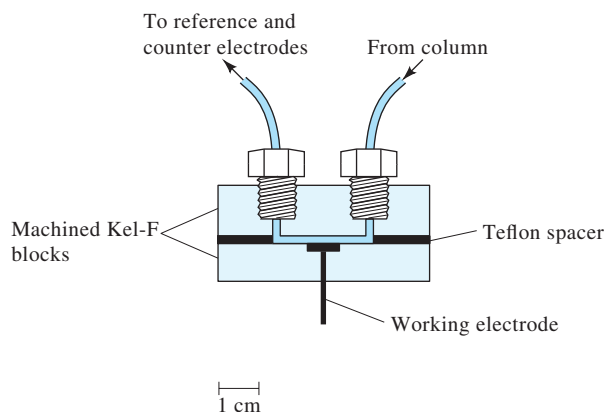


FIGURE 28-12 Amperometric thin-layer detector cell for HPLC.

¹¹W. M. A. Niessen, *Liquid Chromatography-Mass Spectrometry*, 3rd ed., Boca Raton, FL: CRC Press, 2006; R. E. Ardrey, *Liquid Chromatography-Mass Spectrometry: An Introduction*, Chichester, UK: Wiley, 2003.

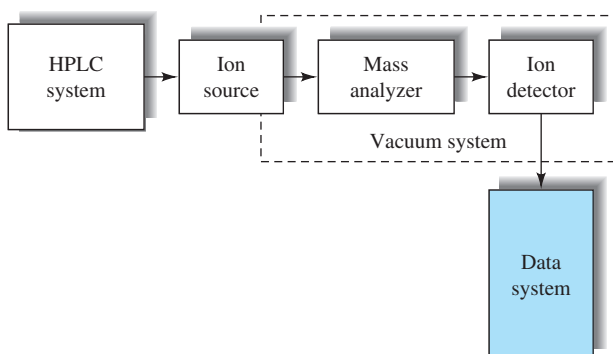


FIGURE 28-13 Block diagram of an LC/MS system. The effluent from the LC column is introduced to an atmospheric pressure ionization source, such as an electrospray or a chemical ionization source. The ions produced are sorted by the mass analyzer and detected by the ion detector.

Section 20B). The combination of HPLC and mass spectrometry gives high selectivity because unresolved peaks can be isolated by monitoring only a selected mass. The LC/MS technique can provide fingerprinting of a particular eluate instead of relying on retention time as in conventional HPLC. The combination also can give molecular mass, structural information, and accurate quantitative analysis.¹²

For some complex mixtures, the combination of LC and MS does not provide enough resolution. In recent years it has become feasible to couple two or more mass analyzers to form tandem mass spectrometers (see Section 20C-5). When combined with LC, the tandem mass spectrometry system is called an LC/MS/MS instrument. Tandem mass spectrometers are usually triple quadrupole systems or quadrupole ion-trap spectrometers. To attain higher resolution than can be achieved with a quadrupole, the final mass analyzer in a tandem MS system can be a time-of-flight mass spectrometer. Sector mass spectrometers can also be combined to give tandem systems. Ion cyclotron resonance and ion-trap mass spectrometers can be operated in such a way as to provide not only two stages of mass analysis but n stages. Such MS ^{n} systems (see Section 20C-5) provide the analysis steps sequentially within a single mass analyzer. These have been combined with LC systems in LC/MS ^{n} instruments.

With state-of-the-art, computer-controlled LC/MS systems, both real-time and computer-reconstructed chromatograms and spectra of the eluted peaks can be obtained.

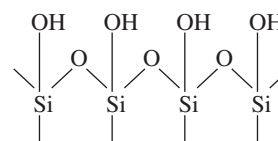
28D PARTITION CHROMATOGRAPHY

The most widely used type of HPLC is *partition chromatography* in which the stationary phase is a second liquid that is immiscible with the liquid mobile phase. Partition chromatography can be

subdivided into *liquid-liquid* and *liquid-bonded-phase* chromatography. The difference between the two lies in the way that the stationary phase is held on the support particles of the packing. The liquid is held in place by physical adsorption in liquid-liquid chromatography, while it is attached by chemical bonding in bonded-phase chromatography. Early partition chromatography was exclusively liquid-liquid; now, however, bonded-phase methods predominate because of their greater stability and compatibility with gradient elution. Liquid-liquid packings are today relegated to certain special applications. We restrict our discussion in this section to bonded-phase partition chromatography.¹³

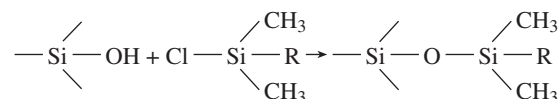
28D-1 Bonded-Phase Packings

The supports for most bonded-phase packings for partition chromatography are prepared from rigid silica, or silica-based, compositions. These solids are formed as uniform, porous, mechanically sturdy particles commonly having diameters of 1.5–10 μm , with 3- and 5- μm particles being most common. The surface of fully hydrolyzed silica (hydrolyzed by heating with 0.1 M HCl for a day or two) is made up of chemically reactive silanol groups. That is,



Typical silica surfaces contain about 8 $\mu\text{mol}/\text{m}^2$ of OH groups and have surface areas of 100–300 m^2/g .

The most useful bonded-phase coatings are siloxanes formed by reaction of the hydrolyzed surface with an organochlorosilane. For example,



where R is an alkyl group or a substituted alkyl group.

Surface coverage by silanization is limited to 4 $\mu\text{mol}/\text{m}^2$ or less because of steric effects. The unreacted SiOH groups, unfortunately, impart an undesirable polarity to the surface, which may lead to tailing of chromatographic peaks, particularly for basic solutes. To lessen this effect, siloxane packings are frequently *capped* by further reaction with chlorotrimethylsilane that, because of its smaller size, can bond some of the unreacted silanol groups.



Tutorial: Learn more about **partition chromatography** at www.tinyurl.com/skoogpia7

¹²For a review of commercial LC/MS systems, see B. E. Erickson, *Anal. Chem.*, **2000**, 72, 711A, DOI: 10.1021/ac0029758.

¹³For a report on retention mechanisms in bonded-phase chromatography, see J. G. Dorsey and W. T. Cooper, *Anal. Chem.*, **1994**, 66, 857A, DOI: 10.1021/ac00089a002.

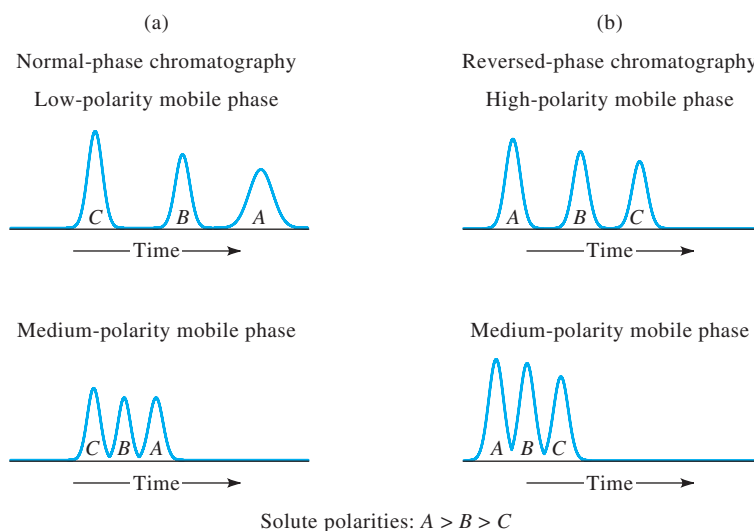


FIGURE 28-14 Relationship between polarity and elution times for normal-phase and reversed-phase chromatography.

Normal- and Reversed-Phase Packings

Two types of partition chromatography are distinguishable based on the relative polarities of the mobile and stationary phases. Early work in LC was based on highly polar stationary phases such as triethylene glycol or water; a relatively nonpolar solvent such as hexane or *i*-propyl ether then served as the mobile phase. For historic reasons, this type of chromatography is now called *normal-phase chromatography*. In *reversed-phase chromatography*, the stationary phase is nonpolar, often a hydrocarbon, and the mobile phase is a relatively polar solvent (such as water, methanol, acetonitrile, or tetrahydrofuran).¹⁴

In normal-phase chromatography, the *least* polar component is eluted first; *increasing* the polarity of the mobile phase then *decreases* the elution time. With reversed-phase chromatography, however, the *most* polar component elutes first, and *increasing* the mobile-phase polarity *increases* the elution time. These relationships are illustrated in Figure 28-14. As shown in Figure 28-1, normal-phase partition chromatography and adsorption chromatography overlap considerably. In fact, retention in most types of normal-phase chromatography appears to be governed by adsorption-displacement processes.¹⁵

Bonded-phase packings are classified as reversed phase when the bonded coating is nonpolar in character and as normal phase when the coating contains polar functional groups. It has been estimated that more than three quarters of all HPLC separations are currently performed in columns with reversed-

phase packings. The major advantage of reversed-phase separations is that water can be used as the mobile phase. Water is an inexpensive, nontoxic, UV-transparent solvent compatible with biological solutes. Also, mass transfer is rapid with nonpolar stationary phases, as is solvent equilibration after gradient elution. Most commonly, the R group of the siloxane in these coatings is a C₈ chain (*n*-octyl) or a C₁₈ chain (*n*-octyldecyl).

The mechanism by which these surfaces retain solute molecules is not entirely clear. The retention mechanism appears to be quite complex and very different from bulk phase partitioning because the nonpolar stationary phase is anchored at one end.¹⁶ Regardless of the detailed mechanism of retention, a bonded coating can usually be treated as if it were a conventional, physically retained liquid.

Figure 28-15 illustrates the effect of chain length of the alkyl group on performance. As expected, longer chains produce packings that are more retentive. In addition, longer chain lengths permit the use of larger samples. For example, the maximum sample size for a C₁₈ packing is roughly double that for a C₄ preparation under similar conditions.

In most applications of reversed-phase chromatography, elution is carried out with a highly polar mobile phase, for example, an aqueous solution containing various concentrations of such solvents as methanol, acetonitrile, or tetrahydrofuran. In this mode, care must be taken to avoid pH values greater than about 7.5 because the silica can form soluble silicate species, causing dissolution of the stationary phase. Also, hydrolysis of the siloxane can occur in alkaline solutions, which leads to

¹⁴For further discussion of reversed-phase HPLC, see L. R. Snyder, J. J. Kirkland, and J. L. Glajch, *Practical HPLC Method Development*, 2nd ed., New York: Wiley, 1997; A. M. Krstulovic and P. R. Brown, *Reversed-Phase High Performance Liquid Chromatography*, New York: Wiley, 1982.

¹⁵See J. G. Dorsey et al., *Anal. Chem.*, **1994**, 66, 500R, DOI: 10.1021/ac00084a019.

¹⁶L. A. Cole and J. G. Dorsey, *Anal. Chem.*, **1992**, 64, 1317, DOI: 10.1021/ac00037a004; L. A. Cole, J. G. Dorsey, and K. A. Dill, *Anal. Chem.*, **1992**, 64, 1324, DOI: 10.1021/ac00037a005.

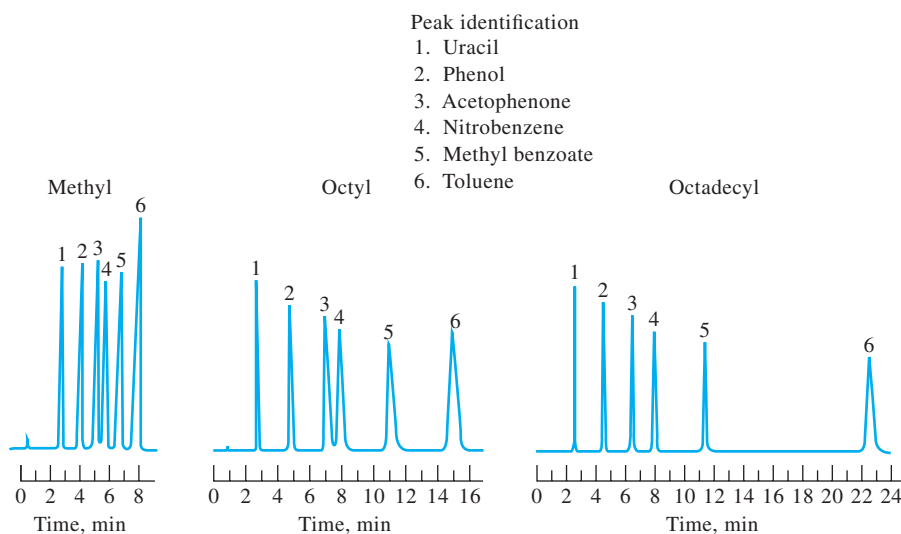


FIGURE 28-15 Effect of chain length on performance of reversed-phase siloxane columns packed with 5- μ m particles. Mobile phase: 50:50 methanol-water. Flow rate: 1.0 mL/min.

degradation or destruction of the packing. Acid hydrolysis of the siloxane can limit the pH to about 2.5 in acidic solutions.

In commercial normal-phase bonded packings, the R in the siloxane structure is a polar functional group such as cyano, $-\text{C}_2\text{H}_4\text{CN}$; diol, $-\text{C}_3\text{H}_6\text{OCH}_2\text{CHOHCH}_2\text{OH}$; amino, $-\text{C}_3\text{H}_6\text{NH}_2$; and dimethylamino, $-\text{C}_3\text{H}_6\text{N}(\text{CH}_3)_2$. The polarities of these packing materials vary over a considerable range, with the cyano type being the least polar and the amino types the most. Diol packings are intermediate in polarity. With normal-phase packings, elution is carried out with relatively non-polar solvents, such as ethyl ether, chloroform, and *n*-hexane.

28D-2 Method Development in Partition Chromatography

Method development tends to be more complex in LC than in GC because in a liquid mobile phase the sample components interact with both the stationary phase and the mobile phase.¹⁷ In contrast, in GC, the mobile phase simply carries the sample components through the stationary phase and makes no contribution to the separation process. That is, in GC separations are not significantly affected by whether the mobile phase is helium, nitrogen, or hydrogen. In marked contrast, the success of a partition chromatographic separation is often critically dependent on whether the mobile phase is, say, acetonitrile, hexane, or dioxane.

Column Selection in Partition Chromatographic Separations

Successful chromatography with interactive mobile phases requires a proper balance of intermolecular forces among the

three active participants in the separation process—the solute, the mobile phase, and the stationary phase. These intermolecular forces are described qualitatively in terms of the relative polarity of each of the three reactants. The polarities of the various analyte functional groups increase in the following order: hydrocarbons < ethers < esters < ketones < aldehydes < amides < amines < alcohols. Water is more polar than compounds containing any of the preceding functional groups.

Often, in choosing a column for a partition chromatographic separation, the polarity of the stationary phase is matched roughly with that of the analytes; a mobile phase of considerably different polarity is then used for elution. This procedure is generally more successful than one in which the polarities of the solute and mobile phase are matched but different from that of the stationary phase. Here, the stationary phase often cannot compete successfully for the sample components; retention times then become too short for practical application. At the other extreme, of course, is the situation where the polarity of the solute and stationary phases are too much alike and totally different from that of the mobile phase. Here, retention times become inordinately long.

In summary, then, polarities for solute, mobile phase, and stationary phase must be carefully blended if good partition chromatographic separations are to be realized in a reasonable time. Unfortunately, theories of mobile-phase and stationary-phase interactions with any given set of sample components are imperfect, and at best, the user can only narrow the choice of stationary phase to a general type. Having made this choice, some trial-and-error experiments must be performed in which chromatograms are obtained with various mobile phases until a satisfactory separation is realized. The high degree of correlation between mobile-phase composition and retention factors, however, allows prediction of approximate retention times from only

¹⁷For more information, see L. R. Snyder, J. J. Kirkland, and J. L. Glajch, *Practical HPLC Method Development*, 2nd ed., New York: Wiley, 1997.

TABLE 28-2 Properties of Common Chromatographic Mobile Phases

Solvent	Refractive Index ^a	Viscosity, cP ^b	Boiling Point, °C	Polarity Index, <i>P'</i>	Eluent Strength, ^c ϵ^0
Fluoroalkanes ^d	1.27–1.29	0.4–2.6	50–174	<–2	–0.25
Cyclohexane	1.423	0.90	81	0.04	–0.2
<i>n</i> -Hexane	1.372	0.30	69	0.1	0.01
1-Chlorobutane	1.400	0.42	78	1.0	0.26
Carbon tetrachloride	1.457	0.90	77	1.6	0.18
<i>i</i> -Propyl ether	1.365	0.38	68	2.4	0.28
Toluene	1.494	0.55	110	2.4	0.29
Diethyl ether	1.350	0.24	35	2.8	0.38
Tetrahydrofuran	1.405	0.46	66	4.0	0.57
Chloroform	1.443	0.53	61	4.1	0.40
Ethanol	1.359	1.08	78	4.3	0.88
Ethyl acetate	1.370	0.43	77	4.4	0.58
Dioxane	1.420	1.2	101	4.8	0.56
Methanol	1.326	0.54	65	5.1	0.95
Acetonitrile	1.341	0.34	82	5.8	0.65
Nitromethane	1.380	0.61	101	6.0	0.64
Ethylene glycol	1.431	16.5	182	6.9	1.11
Water	1.333	0.89	100	10.2	Large

^aAt 25°C.^bThe centipoise is a common unit of viscosity; in SI units, 1 cP = 1 mN · s · m^{–2}.^cOn Al₂O₃. Multiplication by 0.8 gives ϵ^0 on SiO₂.^dProperties depend on molecular mass range of data given.

a few trial-and-error experiments. If resolution of all of the components of a mixture proves to be impossible, a different type of column may have to be chosen.

Mobile-Phase Selection in Partition Chromatography

In Section 26D-4, three methods were described for improving the resolution of a chromatographic column; each is based on varying one of the three parameters (*N*, *k*, and α) contained in Equation 26-26.¹⁸ In LC, the retention factor *k* is experimentally the most easily manipulated of the three because of the strong dependence of this constant on the composition of the mobile phase. As noted earlier, for optimal performance, *k* should be in the ideal range between 2 and 10; for complex mixtures, however, this range must often be expanded to perhaps 0.5–20 to provide time for peaks for all of the components to appear.

Sometimes, adjustment of *k* alone is not sufficient to produce individual peaks with no overlap. In this case we must resort to variations in the selectivity factor α . Here again, the

simplest way of bringing about changes in α is by altering the mobile-phase composition, taking care, however, to keep *k* within a reasonable range. Alternatively, α can be changed by choosing a different column packing.

Effect of Solvent Strength on Retention Factors. Solvents that interact strongly with solutes are often termed “strong” solvents. Strong solvents are often, but not always, polar solvents. Solvent strength depends on the nature of the analyte and stationary phase. Several indexes have been developed for quantitatively describing the polarity of solvents. The most useful of these for partition chromatography is the polarity index *P'*, which was developed by Snyder.¹⁹ This parameter is based on solubility measurements for the substance in question in three solvents: dioxane (a low dipole proton acceptor), nitromethane (a high dipole proton acceptor), and ethyl alcohol (a high dipole proton donor). The polarity index is a numerical measure of the relative polarity of various solvents. Table 28-2 lists polarity indexes (and other properties) for a number of solvents used in partition chromatography. Note

¹⁸For more information on selecting stationary and mobile phases, see L. R. Snyder, J. J. Kirkland, and J. L. Glajch, *Practical HPLC Method Development*, 2nd ed., New York: Wiley, 1997.

¹⁹L. R. Snyder, *J. Chromatogr. Sci.*, **1978**, 16, 223, DOI: 10.1093/chromsci/16.6.223.

that the polarity index varies from 10.2 for highly polar water to -2 for the highly nonpolar fluoroalkanes. Any desired polarity index between these limits can be achieved by mixing two appropriate solvents. Thus, the polarity index P'_{AB} of a mixture of solvents A and B is given by

$$P'_{AB} = \phi_A P'_A + \phi_B P'_B \quad (28-2)$$

where P'_A and P'_B are the polarity indexes of the two solvents and ϕ_A and ϕ_B are the volume fractions of solvents A and B.

In Section 26D-4, it was pointed out that the easiest way to improve the chromatographic resolution of two species is by manipulating the retention factor k , which can in turn be varied by changing the polarity index of the solvent. Here, adjustment of P' is easily accomplished by the use of mobile phases that consist of a mixture of two solvents. Typically, a two-unit change in P' results (very roughly) in a ten-fold change in k . That is, for a normal-phase separation

$$\frac{k_2}{k_1} = 10^{(P'_2 - P'_1)/2} \quad (28-3)$$

where k_1 and k_2 are the initial and final values of k for a solute and P'_1 and P'_2 are the corresponding polarity indexes. For a reversed-phase column

$$\frac{k_2}{k_1} = 10^{(P'_1 - P'_2)/2} \quad (28-4)$$

It should be emphasized that these equations apply only approximately. Nonetheless, as shown in Example 28-1, they can be useful.

EXAMPLE 28-1

In a reversed-phase column, a solute was found to have a retention time of 31.3 min, and an unretained species required 0.48 min for elution when the mobile phase was 30% (by volume) methanol and 70% water. Calculate (a) k and (b) a water-methanol composition that should bring k to a value of about 5.

Solution

(a) Application of Equation 26-12 yields

$$k = (31.3 - 0.48)/0.48 = 64$$

(b) To obtain P' for the mobile phase we substitute polarity indexes for methanol and water from Table 28-2 into Equation 28-2 to give

$$P' = (0.30 \times 5.1) + (0.70 \times 10.2) = 8.7$$

Substitution of this result into Equation 28-4 gives

$$\frac{5}{64} = 10^{(P'_2 - 8.7)/2}$$

If we take the logarithm of both sides of this equation, we can solve for P'_2

$$\begin{aligned} -1.11 &= \frac{P'_2 - 8.7}{2} = 0.5P'_2 - 4.35 \\ P'_2 &= 6.5 \end{aligned}$$

Letting x be the volume fraction of methanol in the new solvent mixture and substituting again into Equation 28-2, we find

$$\begin{aligned} 6.5 &= (x \times 5.1) + (1 - x) \times 10.2 \\ x &= 0.73 \text{ or } 73\% \end{aligned}$$

Often, in reversed-phase separations, we use a solvent mixture containing water and a polar organic solvent. The retention factor can then be manipulated by varying the water concentration as shown by Example 28-1. The effect of such manipulations is shown by the chromatograms in Figure 28-16a and b where the sample was a mixture of six steroids. With a mixture of 41% acetonitrile and 59% water, k had a value of 5 and all of the analytes were eluted in such a short time (~ 2 min) that the separation was quite incomplete. By increasing the concentration of water to 70%, elution took place over 7 min, which doubled the value of k . Now the total elution time was sufficient to achieve a separation, but the α value for compounds 1 and 3 was not great enough to resolve them.

Effect of Mobile Phase on Selectivities. In many cases, adjusting k to a suitable level is all that is needed to give a satisfactory separation. When two bands still overlap, however, as in Figure 28-16a and b, the selectivity factor α for the two species must be made larger. Such a change can be brought about most conveniently by changing the chemical nature of the mobile phase while holding the predetermined value of k *more or less the same*. The most widely accepted approach for this type of optimization is called the *solvent selectivity triangle*.²⁰ Here, the mobile-phase effects contributing to selectivity are considered to be the result of proton donor, proton acceptor, and dipolar interactions. Solvents incorporating these interactions are then chosen for statistically designed experiments to find the optimal solvent mixture.²¹

For reversed-phase chromatography, the three solvent modifiers are methanol, acetonitrile, and tetrahydrofuran. Water is then used to adjust the solvent strength of the mixtures and yield a suitable value of k . Three binary solvent compositions (water plus modifier) define the three vertices of the

²⁰For more information, see J. C. Berridge, *Techniques for the Automated Optimization of HPLC Separations*, Chichester, UK: Wiley, 1985.

²¹See J. L. Glajch, J. J. Kirkland, K. M. Squire, and J. M. Minor, *J. Chromatogr. A*, **1980**, 199, 57, DOI: 10.1016/S0021-9673(01)91361-5; L. R. Snyder, *J. Chromatogr. Sci.*, **1978**, 16, 223–234, DOI: 10.1093/chromsci/16.6.223.

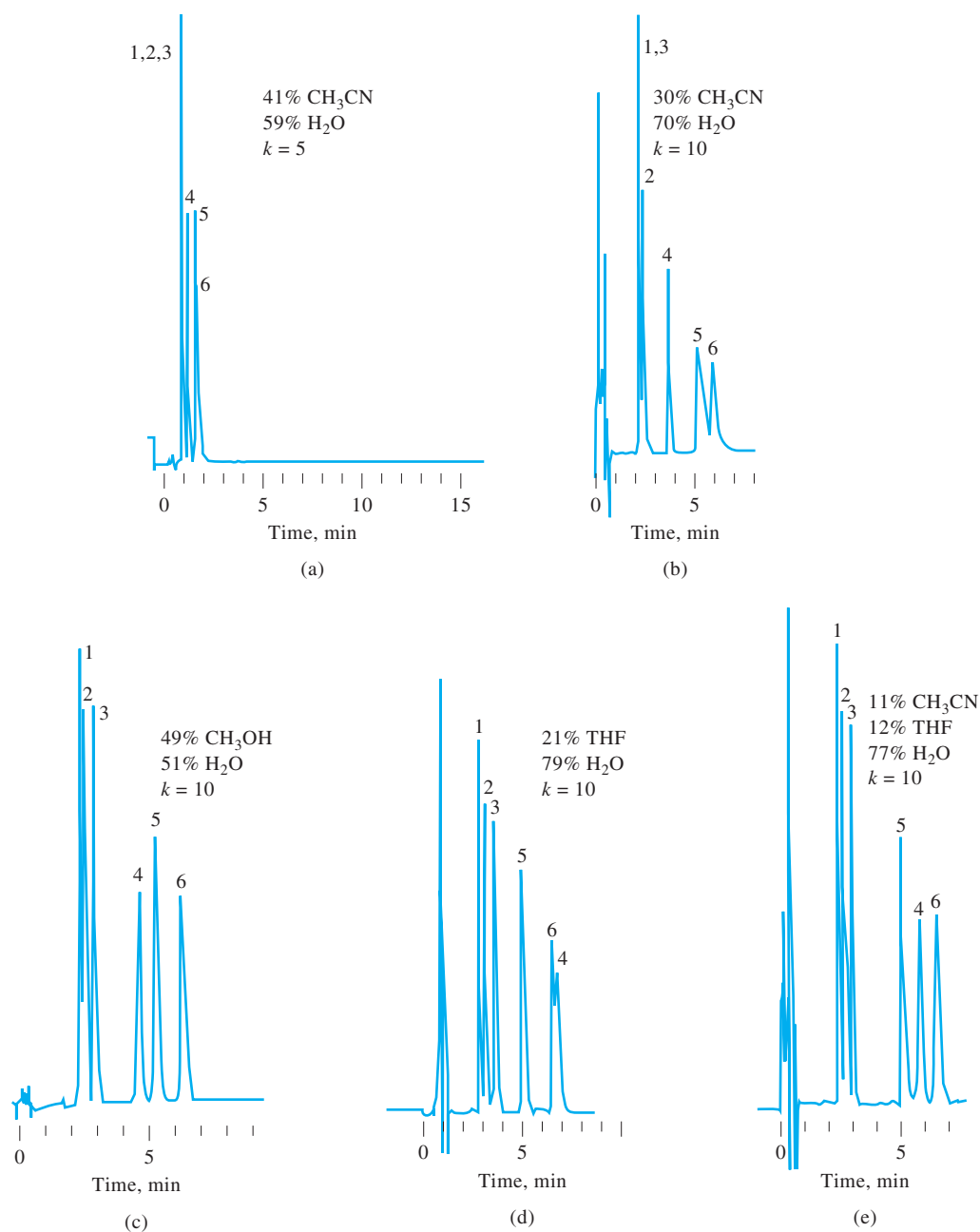


FIGURE 28-16 Systematic approach to the separation of six steroids. The use of water to adjust k is shown in (a) and (b). The effects of varying α at constant k are shown in (b), (c), (d), and (e). Column: 0.4×150 mm packed with $5 \mu\text{m}$ C₈ bonded, reversed-phase particles. Temperature: 50°C . Flow rate: $3.0 \text{ cm}^3/\text{min}$. Detector: UV 254 nm. THF = tetrahydrofuran. CH₃CN = acetonitrile. Compounds: (1) prednisone, (2) cortisone, (3) hydrocortisone, (4) dexamethasone, (5) corticosterone, (6) cortodoxolone.

solvent triangle as shown in Figure 28-17. Usually, seven to ten experiments are enough to define a solvent composition that will produce the best selectivity for a suitable k range. Several other statistical design schemes, such as full factorial designs, have also been used, but the solvent-triangle approach requires fewer experiments. Software is also available to aid in automated schemes for solvent optimization. In addition, simulation

software, such as the popular DryLab,²² is widely used for method development and optimization.

Figure 28-16 illustrates the systematic, solvent-triangle approach to the development of a separation of six steroids by

²²DryLab is now part of Molnár-Institute, <http://molnar-institute.com/drylab/>. It was developed by LC Resources, Inc., Walnut Creek, CA.

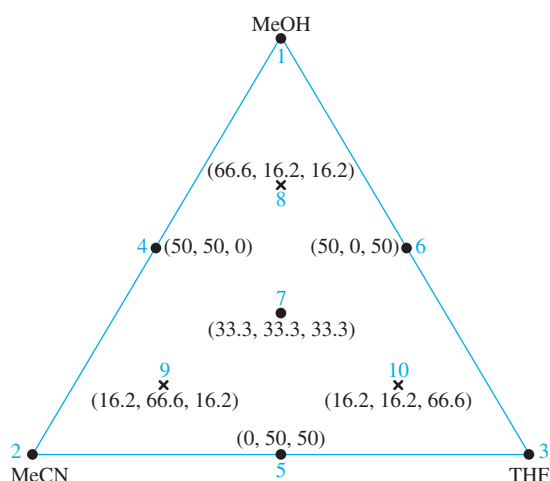


FIGURE 28-17 Solvent selectivity triangle for optimization of reversed-phase separations. Ten mixtures of the three organic solvents (methanol, acetonitrile, and tetrahydrofuran) are shown with the relative proportions indicated in parentheses (MeOH, MeCN, and THF). Water is used to maintain solvent strength and keep the k value within an appropriate range.

reversed-phase chromatography. The first two chromatograms show the results from initial experiments to determine the minimal value of k required. Here, the last eluting component is used to calculate k . With k for component 6 equal to 10, room exists on the time scale for discrete peaks; here, however, the α value for components 1 and 3 (and to a lesser extent 5 and 6) is not great enough for satisfactory resolution. Further experiments were then performed to find better α values; in each case, the water concentration was adjusted to a level that yielded $k = 10$ for the last eluting component. The results of experiments with methanol-water and tetrahydrofuran-water mixtures are shown in Figure 28-16c and d. Several additional experiments involving the systematic variation of pairs of the three organic solvents were performed (in each case adjusting k to 10 with water). Finally, the mixture shown in Figure 28-16e was chosen as the best mobile phase for the separation of the particular group of compounds.

For normal-phase separations, a similar solvent triangle is used in which the selectivity solvents are ethyl ether, methylene chloride, and chloroform;²³ the solvent strength adjustment is then made with n -hexane. With these solvent systems, optimization can be accomplished with a minimal number of experiments.

28D-3 Applications of Partition Chromatography

Reversed-phase bonded packings, when used in conjunction with highly polar solvents (often aqueous), approach the ideal, universal system for LC. Because of their wide range of

TABLE 28-3 Typical Applications of Partition Chromatography

Field	Typical Mixtures
Pharmaceuticals	Antibiotics, sedatives, steroids, analgesics
Biochemical	Amino acids, proteins, carbohydrates, lipids
Food products	Artificial sweeteners, antioxidants, aflatoxins, additives
Industrial chemicals	Condensed aromatics, surfactants, propellants, dyes
Pollutants	Pesticides, herbicides, phenols, polychlorinated biphenyls
Forensic science	Drugs, poisons, blood alcohol, narcotics
Clinical chemistry	Bile acids, drug metabolites, urine extracts, estrogens

applicability, their convenience, and the ease with which k and α can be altered by manipulation of aqueous mobile phases, these packings are frequently applied before all others for exploratory separations with new types of samples.

Table 28-3 lists a few typical examples of the multitude of uses of partition chromatography in various fields. A more complete picture of the many applications of this technique can be obtained by consulting review articles and books.²⁴ Figure 28-18 illustrates two of many thousands of applications of bonded-phase partition chromatography to the analysis of consumer and industrial materials.

Derivative Formation

In some instances, it is useful to convert the components of a sample to a derivative before, or sometimes after, chromatographic separation. Such treatment may be desirable (1) to reduce the polarity of the species so that partition rather than adsorption or ion-exchange columns can be used; (2) to increase the detector response, and thus sensitivity, for all of the sample components; and (3) to selectively enhance the detector response to certain components of the sample.

Figure 28-19 illustrates the use of derivatives to reduce polarity and enhance sensitivity. The sample was made up of thirty amino acids of physiological importance. Often, such a separation would be performed on an ion-exchange column with photometric detection based on postcolumn reaction of the amino acids with a colorimetric reagent such as ninhydrin. The

²³Note that chlorinated solvents are now subject to strict environmental regulations.

²⁴See, for example, L. R. Snyder, *Anal. Chem.*, **2000**, 72, 412A, DOI: 10.1021/ac002846r; J. G. Dorsey et al., *Anal. Chem.*, **1998**, 70, 591R, DOI: 10.1021/a1980022h; *HPLC: Practical and Industrial Applications*, J. K. Swadesh, ed., 2nd ed., Boca Raton, FL: CRC Press, 2001; *Handbook of Pharmaceutical Analysis by HPLC*, S. Ahuja and M. W. Dong, eds., San Diego, CA: Elsevier, 2005.

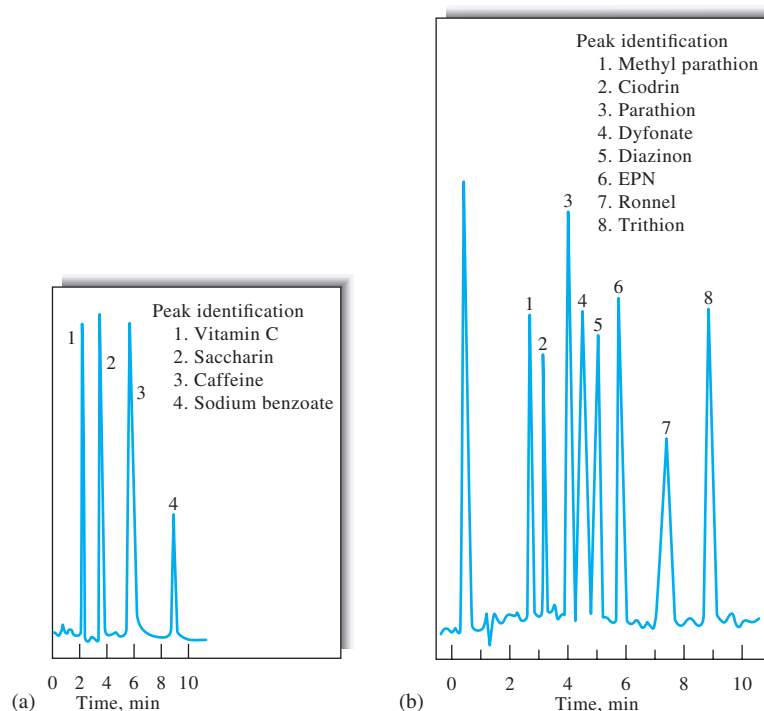


FIGURE 28-18 Typical applications of bonded-phase chromatography. (a) Soft-drink additives. Column: 4.6×250 mm packed with polar (nitrile) bonded-phase packing. Isocratic elution with 6% HOAc to 94% H₂O. Flow rate: 1.0 mL/min. (Courtesy of BTR Separations, a DuPont ConAgra affiliate.) (b) Organophosphate insecticides. Column 4.5×250 mm packed with $5 \mu\text{m}$ C₈ bonded-phase particles. Gradient elution: 67% CH₃OH to 33% H₂O, to 80% CH₃OH to 20% H₂O. Flow rate: 2 mL/min. Both used 254-nm-UV detectors.

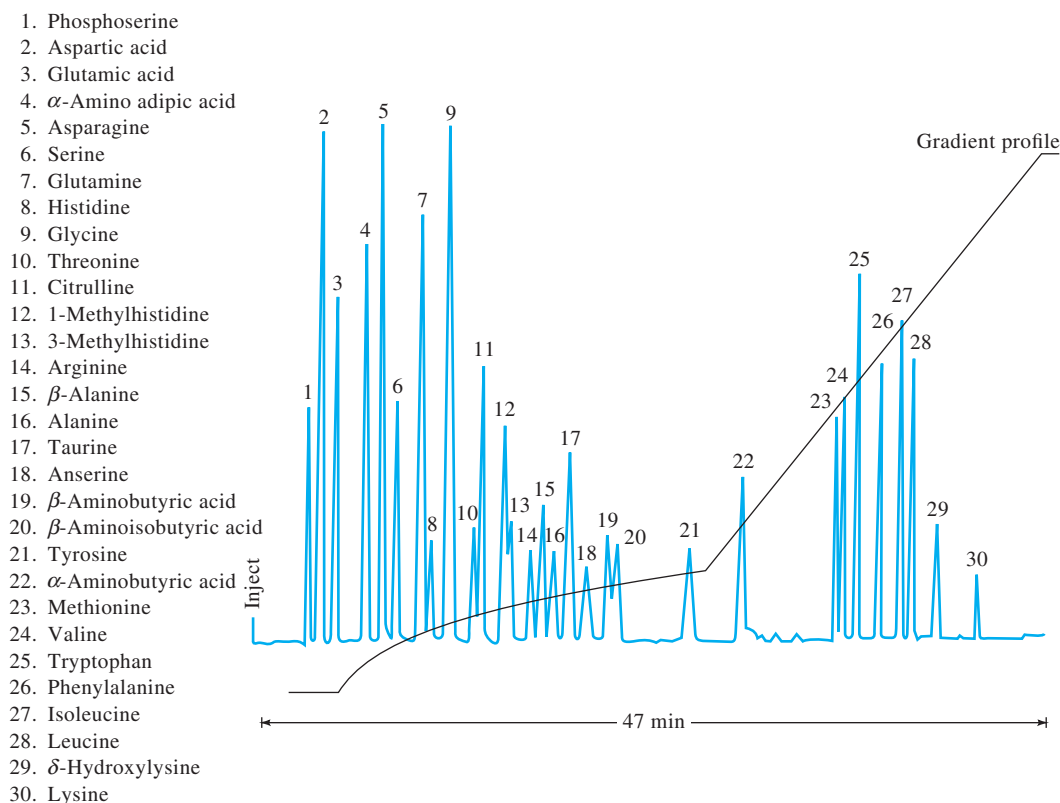


FIGURE 28-19 Chromatogram of orthophthalaldehyde derivatives of 30 amino acids of physiological importance. Column: $5 \mu\text{m}$ C₁₈, reversed-phase. Solvent A: 0.05 M Na₂HPO₄, pH 7.4; 96:2:2 CH₃OH, THF, H₂O. Fluorescence detector: excitation 334 nm; emission 425 nm. (Reprinted with permission from R. Pfeifer et al., *Amer. Lab.*, **1983**, 15 (3), 86. Copyright 1983 by International Scientific Communications, Inc.)

chromatogram shown in Figure 28-19 was obtained by automatic, precolumn derivative formation with orthophthalaldehyde. The substituted isoindoles²⁵ formed by the reaction exhibit intense fluorescence at 425 nm, which permits detection down to a few picomoles (10^{-12} mol). Furthermore, the polarity of the derivatives is such that separation on a C_{18} reversed-phase packing becomes feasible. The advantages of this newer procedure are speed and smaller sample size. Derivatives are frequently used with selective detectors such as fluorescence, electrochemical, and mass spectrometric detectors.

Ion-Pair Chromatography

Ion-pair chromatography,²⁶ sometimes called *paired-ion chromatography*, is a subset of reversed-phase chromatography in which easily ionizable species are separated on reversed-phase columns. In this type of chromatography, an organic salt containing a large organic counterion, such as a quaternary ammonium

ion or alkyl sulfonate, is added to the mobile phase as an ion-pairing reagent. Two mechanisms for separation are postulated. In the first, the counterion forms an uncharged ion pair with a solute ion of opposite charge in the mobile phase. This ion pair then partitions into the nonpolar stationary phase, giving differential retention of solutes based on the affinity of the ion pair for the two phases. Alternatively, the counterion is retained strongly by the normally neutral stationary phase and imparts a charge to this phase. Separation of organic solute ions of the opposite charge then occurs by formation of reversible ion-pair complexes with the more strongly retained solutes forming the strongest complexes with the stationary phase. Some unique separations of both ionic and nonionic compounds in the same sample can be accomplished by this form of partition chromatography.

Figure 28-20 illustrates the separation of ionic and nonionic compounds using alkyl sulfonates of various chain lengths as ion-pairing agents. Note that a mixture of C_5 - and C_7 -alkyl sulfonates gives the best separation results.

Applications of ion-pair chromatography frequently overlap those of ion-exchange chromatography, which are discussed in Section 28F. For the separation of small inorganic and organic ions, ion exchange is usually preferred unless selectivity is a problem.

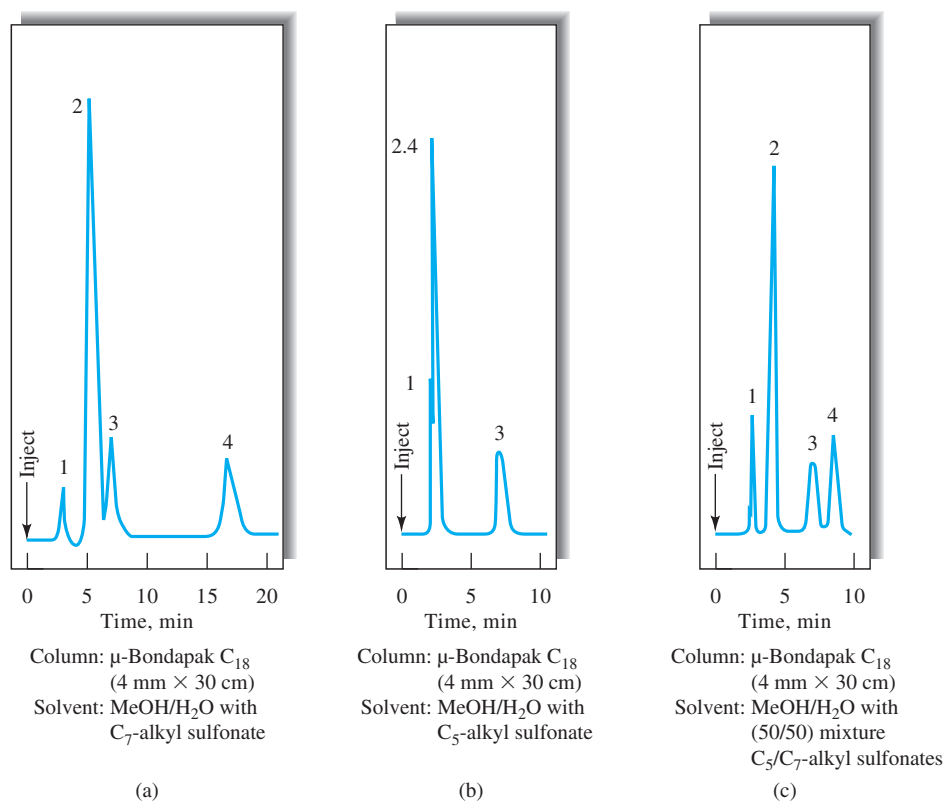


FIGURE 28-20 Chromatograms illustrating separations of mixtures of ionic and nonionic compounds by ion-pair chromatography. Compounds: (1) niacinamide, (2) pyridoxine, (3) riboflavin, (4) thiamine. At pH 3.5, niacinamide is strongly ionized, and riboflavin is nonionic. Pyridoxine and thiamine are weakly ionized. Column: μ -Bondapak, C_{18} , 4 mm $\times$ 30 cm. Mobile phase: (a) MeOH and H₂O with C_7 -alkyl sulfonate; (b) MeOH and H₂O with C_5 -alkyl sulfonate; (c) MeOH and H₂O with 1:1 mixture of C_5 - and C_7 -alkyl sulfonates. (Courtesy of Waters Associates, Inc.)

²⁵For the structure of these compounds, see S. S. Simons Jr. and D. E. Johnson, *J. Amer. Chem. Soc.*, **1976**, 98, 7098, DOI: 10.1021/ja00438a070.

²⁶See T. Cecchi, *Ion-Pair Chromatography and Related Techniques*, Boca Raton, FL: CRC Press, 2010.

Chiral Chromatography

Tremendous advances have been made in recent years in separating compounds that are nonsuperimposable mirror images of each other called *chiral compounds*.²⁷ Such mirror images are called *enantiomers*. Either chiral mobile-phase additives or chiral stationary phases are required for these separations. Preferential complexation between the *chiral resolving agent* (additive or stationary phase) and one of the isomers results in a separation of the enantiomers. The chiral resolving agent must have chiral character itself to recognize the chiral nature of the solute.

Chiral stationary phases have received the most attention.²⁸ Here, a chiral agent is immobilized on the surface of a solid support. Chiral stationary phases are available from a number of manufacturers. Several different modes of interaction can occur between the chiral resolving agent and the solute.²⁹ In one type, the interactions are due to attractive forces such as those between π bonds, hydrogen bonds, or dipoles. In another type, the solute can fit into chiral cavities in the stationary phase to form inclusion complexes. No matter what the mode, the ability to separate these very closely related compounds is of extreme importance in many fields. Figure 28-21 shows the separation of

a racemic mixture of an ester on a chiral stationary phase. Note the excellent resolution of the *R* and *S* enantiomers.

Chiral separations are increasingly being accomplished by supercritical fluid chromatography as described in Chapter 29.

28E ADSORPTION CHROMATOGRAPHY

Adsorption, or liquid-solid, chromatography is the classic form of LC first introduced by Tswett at the beginning of the last century. Because of the strong overlap between normal-phase partition chromatography and adsorption chromatography, many of the principles and techniques used for the former apply to adsorption chromatography. In fact, in many normal-phase separations, adsorption-displacement processes govern retention.

Finely divided silica and alumina are the only stationary phases that find use for adsorption chromatography. Silica is preferred for most applications because of its higher sample capacity. The adsorption characteristics of the two substances parallel one another. For both, retention times become longer as the polarity of the analyte increases.

It has been found that the polarity index P' , which was described in Section 28D-2, can also serve as a rough guide to the strengths of solvents for adsorption chromatography. A much better index, however, is the *eluent strength* ϵ^0 , which is the adsorption energy per unit area of solvent.³⁰ The eluent strength depends on the adsorbent, with ϵ^0 values for silica being about 0.8 of alumina. The values for ϵ^0 in the last column of Table 28-2 are for alumina. Note that solvent-to-solvent differences in ϵ^0 roughly parallel those for P' .

Because of the versatility and ready availability of bonded stationary phases, use of traditional adsorption chromatography with solid stationary phases has decreased in recent years in favor of normal-phase chromatography.

28F ION CHROMATOGRAPHY

Ion chromatography refers to modern, efficient methods for separating and determining ions on columns with relatively low ion-exchange capacity. Although ion-exchange separations have been around since ion-exchange resins were developed in the mid-1930s, ion chromatography as it is practiced today was first developed in the mid-1970s when it was shown that anion or cation mixtures can be resolved on HPLC columns packed with anion-exchange or cation-exchange resins. At that time, detection was generally performed with conductivity measurements, which were not ideal because of high

²⁷For further information, see *Chiral Separation Techniques: A Practical Approach*, 3rd ed., G. Subramanian, ed., New York: Wiley, 2006; *Chiral Separations for Pharmaceutical and Biotechnological Products*, S. Ahuja, ed., Hoboken, NJ: Wiley, 2011; S. Ahuja, *Chiral Separations by Chromatography*, New York: Oxford University Press, 2000.

²⁸For a review on chiral stationary phases, see D. W. Armstrong and B. Zhang, *Anal. Chem.*, **2001**, 73, 557A, DOI: 10.1021/ac012526n.

²⁹For a review on chiral interactions, see M. C. Ringo and C. E. Evans, *Anal. Chem.*, **1998**, 70, 315A, DOI: 10.1021/ac9818428.

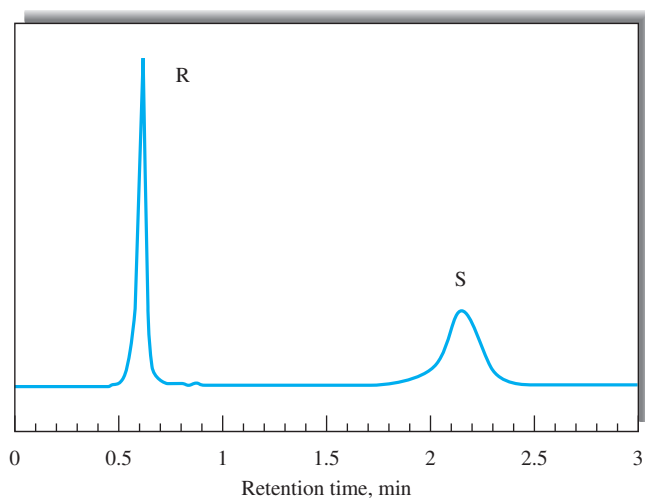


FIGURE 28-21 Chromatogram of a racemic mixture of *N*-(1-naphthyl)leucine ester on a dinitrobenzene-leucine chiral stationary phase. The *R* and *S* enantiomers are seen to be well separated. Column: 4.6×50 mm; mobile phase, 20% 2-propanol in hexane; flow rate 1.2 mL/min; UV detector at 254 nm. (From L. H. Bluhm, Y. Wang, and T. Li, *Anal. Chem.*, **2000**, 72, 5201, DOI: 10.1021/ac000568q. With permission of American Chemical Society.)

³⁰See L. R. Snyder, *Principles of Adsorption Chromatography*, Chap. 8, New York: Dekker, 1968.

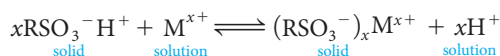
electrolyte concentrations in the mobile phase. The development of low-exchange-capacity columns allowed the use of low-ionic-strength mobile phases that could be further deionized (ionization suppressed) to allow high-sensitivity conductivity detection. Currently, several other detector types are available for ion chromatography, including spectrophotometric and electrochemical.³¹

Ion chromatography was an outgrowth of ion-exchange chromatography, which during the Manhattan project³² was developed for the separation of closely related rare earth cations with cation-exchange resins. This monumental work, which laid the theoretical groundwork for ion-exchange separations, was extended after World War II to many other types of materials; ultimately, it led to automated methods for the separation and detection of amino acids and other ionic species in complex mixtures. The development of modern HPLC began in the late 1960s, but its application to separation of ionic species was delayed by the lack of a sensitive general method of detecting such eluted ionic species as alkali and alkaline earth cations and halide, acetate, and nitrate anions. This situation was remedied in 1975 by the development by workers at Dow Chemical Company of an eluent suppressor technique, which made possible the conductometric detection of eluted ions.³³ This technique is described in Section 28F-3.

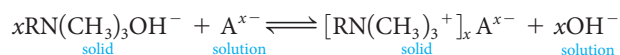
28F-1 Ion-Exchange Equilibria

Ion-exchange processes are based on exchange equilibria between ions in solution and ions of like sign on the surface of an essentially insoluble, high-molecular-mass solid. Natural ion-exchangers, such as clays and zeolites, have been recognized and used for several decades. Synthetic ion-exchange resins were first produced in the mid-1930s for water softening, water deionization, and solution purification. The most common active sites for cation-exchange resins are the sulfonic acid group $\text{—SO}_3^-\text{H}^+$, a strong acid, and the carboxylic acid group $\text{—COO}^-\text{H}^+$, a weak acid. Anionic exchangers contain strongly basic tertiary amine groups $\text{—N(CH}_3)_3^+\text{OH}^-$ or weakly basic primary amine groups $\text{—NH}_3^+\text{OH}^-$.

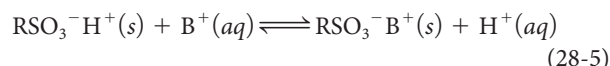
When a sulfonic acid ion-exchanger is brought in contact with an aqueous solvent containing a cation M^{x+} , an exchange equilibrium is set up that can be described by



where RSO_3^-H^+ represents one of many sulfonic acid groups attached to a large polymer molecule. Similarly, a strong-base exchanger interacts with the anion A^{x-} as shown by the reaction



As an example of the application of the mass-action law to ion-exchange equilibria, we will consider the reaction between a singly charged ion B^+ with a sulfonic acid resin held in a chromatographic column. From a neutral solution, initial retention of B^+ ions at the head of the column occurs because of the reaction



Elution with a dilute solution of hydrochloric acid shifts the equilibrium in Equation 28-5 to the left, causing part of the B^+ ions in the stationary phase to be transferred to the mobile phase. These ions then move down the column in a series of transfers between the stationary and mobile phases.

The equilibrium constant K_{ex} for the exchange reaction shown in Equation 28-5 takes the form

$$K_{\text{ex}} = \frac{[\text{RSO}_3^-\text{B}^+]_{\text{s}}[\text{H}^+]_{\text{aq}}}{[\text{RSO}_3^-\text{H}^+]_{\text{s}}[\text{B}^+]_{\text{aq}}} \quad (28-6)$$

Here, $[\text{RSO}_3^-\text{B}^+]_{\text{s}}$ and $[\text{RSO}_3^-\text{H}^+]_{\text{s}}$ are concentrations (strictly activities) of B^+ and H^+ in the solid phase. Rearranging yields

$$\frac{[\text{RSO}_3^-\text{B}^+]_{\text{s}}}{[\text{B}^+]_{\text{aq}}} = K_{\text{ex}} \frac{[\text{RSO}_3^-\text{H}^+]_{\text{s}}}{[\text{H}^+]_{\text{aq}}} \quad (28-7)$$

During the elution, the aqueous concentration of hydrogen ions is much larger than the concentration of the singly charged B^+ ions in the mobile phase. Also, the exchanger has an enormous number of exchange sites relative to the number of B^+ ions being retained. Thus, the overall concentrations $[\text{H}^+]_{\text{aq}}$ and $[\text{RSO}_3^-\text{H}^+]_{\text{s}}$ are not affected significantly by shifts in the equilibrium 28-5. Therefore, when $[\text{RSO}_3^-\text{H}^+]_{\text{s}} \gg [\text{RSO}_3^-\text{B}^+]_{\text{s}}$ and $[\text{H}^+]_{\text{aq}} \gg [\text{B}^+]_{\text{aq}}$, the right-hand side of Equation 28-7 is substantially constant, and we can write

$$\frac{[\text{RSO}_3^-\text{B}^+]_{\text{s}}}{[\text{B}^+]_{\text{aq}}} = K = \frac{c_{\text{s}}}{c_{\text{M}}} \quad (28-8)$$

where K is a constant that corresponds to the distribution constant as defined by Equations 26-1 and 26-2. All of the equations in Table 26-5 (Section 26E) can then be applied to ion-exchange chromatography in the same way as to the other types, which have already been considered.

Note that K_{ex} in Equation 28-6 represents the affinity of the resin for the ion B^+ relative to another ion (here, H^+). Where K_{ex} is large, a strong tendency exists for the solid phase to retain B^+ ; where K_{ex} is small, the reverse is true. By selecting a common reference ion such as H^+ , distribution ratios for different ions on a given type of resin can be experimentally compared. Such experiments reveal that polyvalent ions are much more strongly held than singly charged species. Within a given charge group,

³¹For a brief review of ion chromatography, see J. S. Fritz, *Anal. Chem.*, **1987**, 59, 335A, DOI: 10.1021/ac00131a737; P. R. Haddad, *Anal. Chem.*, **2001**, 73, 266A, DOI: 10.1021/ac012440u. For monographs on the subject, see J. S. Fritz and D. T. Gjerde, *Ion Chromatography*, 4th ed., Weinheim, Germany: Wiley-VCH, 2009; H. Small, *Ion Chromatography*, New York: Plenum Press, 1989.

³²The Manhattan project was the U.S. effort, with help from Canada and the United Kingdom, during World War II to develop nuclear weapons.

³³H. Small, T. S. Stevens, and W. C. Bauman, *Anal. Chem.*, **1975**, 47, 1801, DOI: 10.1021/ac60361a017.

however, differences appear that are related to the size of the hydrated ion as well as to other properties. Thus, for a typical sulfonated cation-exchange resin, values for K_{ex} decrease in the order $\text{Ti}^+ > \text{Ag}^+ > \text{Cs}^+ > \text{Rb}^+ > \text{K}^+ > \text{NH}_4^+ > \text{Na}^+ > \text{H}^+ > \text{Li}^+$. For divalent cations, the order is $\text{Ba}^{2+} > \text{Pb}^{2+} > \text{Sr}^{2+} > \text{Ca}^{2+} > \text{Ni}^{2+} > \text{Cd}^{2+} > \text{Cu}^{2+} > \text{Co}^{2+} > \text{Zn}^{2+} > \text{Mg}^{2+} > \text{UO}_2^{2+}$.

For anions, K_{ex} for a strong-base resin decreases in the order $\text{SO}_4^{2-} > \text{C}_2\text{O}_4^{2-} > \text{I}^- > \text{NO}_3^- > \text{Br}^- > \text{Cl}^- > \text{HCO}_2^- > \text{CH}_3\text{CO}_2^- > \text{OH}^- > \text{F}^-$. This sequence somewhat depends on the type of resin and reaction conditions and should thus be considered only approximate.

28F-2 Ion-Exchange Packings

Historically, ion-exchange chromatography was performed on small, porous beads formed during emulsion copolymerization of styrene and divinylbenzene. The presence of divinylbenzene (usually $\sim 8\%$) results in cross-linking, which makes the beads mechanically stable. To make the polymer active toward ions, acidic or basic functional groups are bonded chemically to the structure. The most common groups are sulfonic acid and quaternary amines.

Figure 28-22 shows the structure of a strong acid resin. Note the cross-linking that holds the linear polystyrene molecules together. The other types of resins have similar structures except for the active functional group.

Porous polymeric particles are not entirely satisfactory for chromatographic packings because of the slow rate of diffusion of analyte molecules through the micropores of the polymer matrix and because of the compressibility of the matrix. To overcome this problem, two newer types of packings have been developed and are in more general use than the porous polymer type. One is a polymeric bead packing in which the bead surface is coated with a synthetic ion-exchange resin. A second

type of packing is prepared by coating porous microparticles of silica, such as those used in adsorption chromatography, with a thin film of the exchanger. Particle diameters are typically $3\text{--}10\ \mu\text{m}$. With either type, faster diffusion in the polymer film leads to enhanced efficiency. Polymer-based packings have higher capacity than silica-based packings and can be used over a broad pH range. Silica-based ion exchangers give higher efficiencies but suffer from a limited pH range of stability and an incompatibility with suppressor-based detection (see next section).

28F-3 Inorganic-Ion Chromatography

The mobile phase in ion-exchange chromatography must have the same general properties required for other types of chromatography. That is, it must dissolve the sample, have a solvent strength that leads to reasonable retention times (appropriate k values), and interact with solutes in such a way as to lead to selectivity (suitable α values). The mobile phases in ion-exchange chromatography are aqueous solutions that may contain moderate amounts of methanol or other water-miscible organic solvents; these mobile phases also contain ionic species, often in the form of a buffer. Solvent strength and selectivity are determined by the kind and concentration of these added ingredients. In general, the ions of the mobile phase compete with analyte ions for the active sites on the ion-exchange packing.

Two types of ion chromatography are currently in use: *suppressor-based* and *single-column*. They differ in the method used to prevent the conductivity of the eluting electrolyte from interfering with the measurement of analyte conductivities.

Ion Chromatography Based on Suppressors

As noted earlier, the widespread application of ion chromatography for the determination of inorganic species was inhibited by the lack of a good general detector, which would permit quantitative determination of ions on the basis of chromatographic

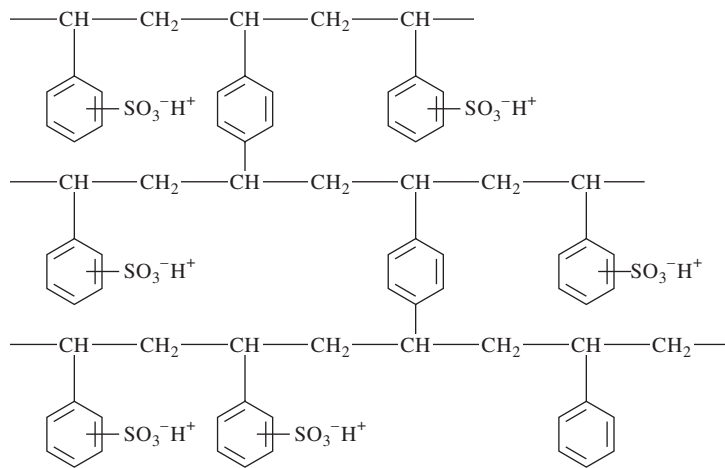


FIGURE 28-22 Structure of a cross-linked polystyrene ion-exchange resin. Similar resins are used in which the $-\text{SO}_3^- \text{H}^+$ group is replaced by $-\text{COO}^- \text{H}^+$, $-\text{NH}_3^+ \text{OH}^-$, and $\text{N}(\text{CH}_3)_3^+ \text{OH}^-$ groups.

peak areas. Conductivity detectors are an excellent choice for this task. They can be highly sensitive, they are universal for charged species, and as a general rule, they respond in a predictable way to concentration changes. Furthermore, such detectors are simple, inexpensive to construct and maintain, easy to miniaturize, and ordinarily give prolonged, trouble-free service. The only limitation to conductivity detectors proved to be a serious one, which delayed their general use. This limitation arises from the high electrolyte concentration required to elute most analyte ions in a reasonable time. As a result, the conductivity from the mobile-phase components tends to swamp that from analyte ions, which greatly reduces the detector sensitivity.

In 1975 the problem created by the high conductance of eluents was solved by the introduction of an *eluent suppressor column* immediately following the ion-exchange column.³⁴ The suppressor column is packed with a second ion-exchange resin that effectively converts the ions of the eluting solvent to a molecular species of limited ionization without affecting the conductivity due to analyte ions. For example, when cations are being separated and determined, hydrochloric acid is chosen as the eluting reagent, and the suppressor column is an anion-exchange resin in the hydroxide form. The product of the reaction in the suppressor is water. That is



The analyte cations are not retained by this second column.

For anion separations, the suppressor packing is the acid form of a cation-exchange resin and sodium bicarbonate or carbonate is the eluting agent. The reaction in the suppressor is



The largely undissociated carbonic acid does not contribute significantly to the conductivity.

An inconvenience associated with the original suppressor columns was the need to regenerate them periodically (typically, every 8 to 10 h) to convert the packing back to the original acid or base form. In the 1980s, however, micromembrane suppressors that operate continuously became available. Figure 28-23 illustrates a typical micromembrane suppressor. Here, the eluent and the suppressor solutions flow in opposite directions on either side of the permeable ion-exchange membrane *M*. For the analysis of anions, the membranes are cation-exchange resins; for cations, they are anion exchangers. When, for example, sodium ions are to be removed from the eluent, acid for regeneration flows continuously in the suppressor stream. Sodium ions from the eluent exchange with hydrogen ions from the membrane and then migrate across the membrane where they exchange with hydrogen ions of the regeneration reagent. The device has a remarkably high exchange rate. For example, it

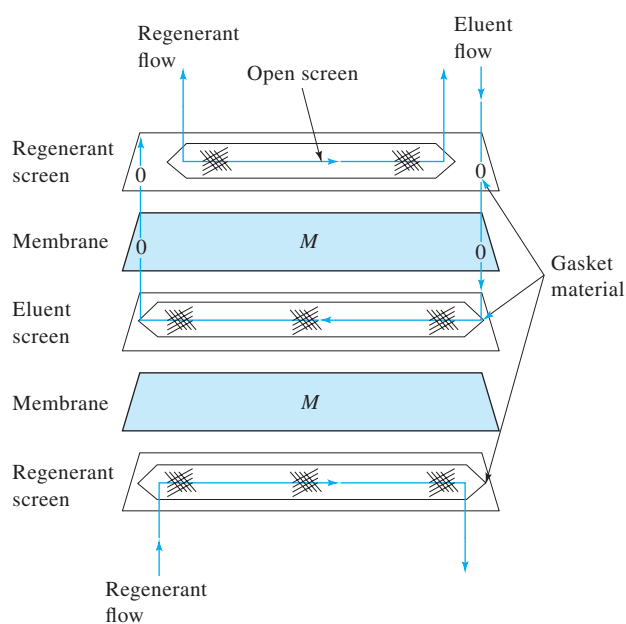


FIGURE 28-23 A micromembrane suppressor. Eluent flows through a narrow channel that contains a plastic screen that reduces the void volume and appears to increase mass-transfer rates. The eluent is separated from the suppressor solution by 50- μm exchange membranes. Regenerant flow is in the direction opposite to eluent flow. (Courtesy of Dionex Corporation, Sunnyvale, CA.)

is capable of removing essentially all of the sodium ions from a 0.1 M solution of sodium hydroxide when the eluent flow rate is 2 mL/min.

In recently designed commercial instruments, regeneration of suppressor solutions is performed automatically with electro-generated hydrogen or hydroxyl ions so that interruptions in the use of the instruments for regeneration are not required. Figure 28-24 shows two applications of ion chromatography based on a suppressor column and conductometric detection. In each case, the ions were present in the parts-per-million range, and the sample sizes were 50 μL in one case and 100 μL in the other. The method is particularly important for anion analysis because no other rapid and convenient method for handling mixtures of this type now exists.

Single-Column Ion Chromatography

Commercial ion chromatography instrumentation that requires no suppressor column is also available. This approach depends on the small differences in conductivity between sample ions and the prevailing eluent ions. To amplify these differences, low-capacity exchangers are used that permit elution with solutions with low electrolyte concentrations. Furthermore, eluents of low conductivity are chosen.

Single-column ion chromatography offers the advantage of not requiring special equipment for suppression. However, it is a somewhat less sensitive method for determining anions than suppressor column methods.

³⁴H. Small, T. S. Stevens, and W. C. Bauman, *Anal. Chem.*, **1975**, *47*, 1801, DOI: 10.1021/ac60361a017.

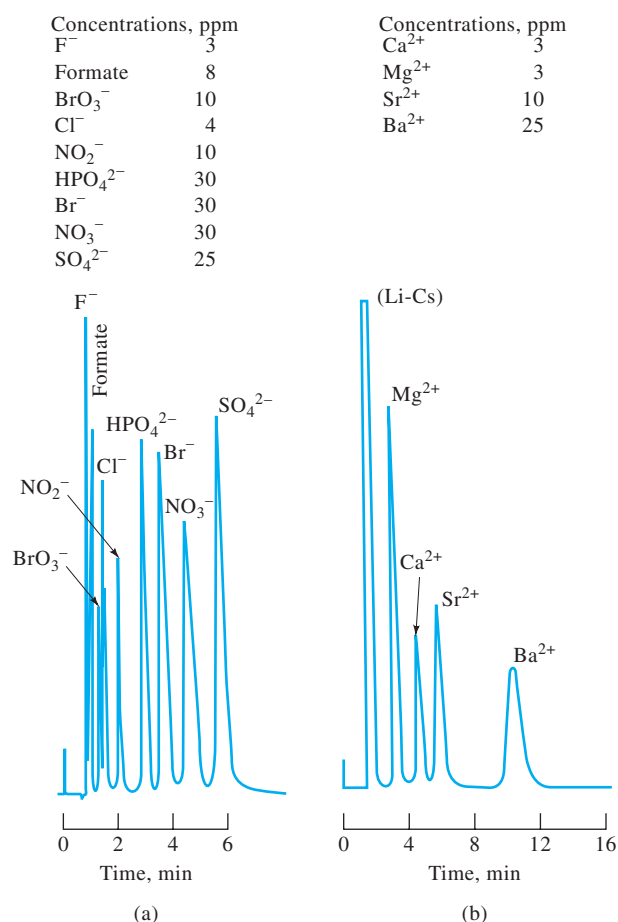


FIGURE 28-24 Typical applications of ion chromatography. (a) Separation of anions on an anion-exchange column. Eluent: 0.0028 M NaHCO₃ to 0.0023 M Na₂CO₃. Sample size: 50 μ L. (b) Separation of alkaline earth ions on a cation-exchange column. Eluent: 0.025 M phenylenediamine dihydrochloride to 0.0025 M HCl. Sample size: 100 μ L. (Courtesy of Dionex Corp., Sunnyvale, CA.)

Capillary Ion Chromatography

Chromatographic systems in general and ion chromatography equipment in particular have been part of the miniaturization trend.³⁵ Capillary ion chromatography utilizes capillary columns of ~ 0.4 mm diameter, which in turn provides scaled down volumes of reagents and flow rates of about a factor of 100. For example, if a standard analytical ion chromatography system consumes 50 L of eluent per month, a capillary system would consume 0.5 L per month. Further, manufacturers offer electrolytic eluent generation equipment to allow regeneration of suppressor solutions by adding acid or base at the column head as mentioned in Section 28F-3, thus eliminating eluent preparation steps in the analytical process.³⁶

Nowhere have the advantages of miniaturization and automation been more important than in the scientific packages aboard the rovers of the Mars exploration program. A small, lightweight, low-power capillary ion chromatography system is under development for possible inclusion in a Mars mission. The system features low eluent consumption, gas pressure pumping (no heavy mechanical pump) at ~ 200 psi at a flow rate of 1.0 μ L/min, and the capability for determining a large number of ions likely to be found in the Martian surface at concentrations of 0.5–100 μ M.³⁷

28F-4 Organic and Biochemical Applications of Ion-Exchange Chromatography

Ion-exchange chromatography has been applied to a variety of organic and biochemical systems, including drugs and their metabolites, serums, food preservatives, vitamin mixtures, sugars, and pharmaceutical preparations. An example of one of these applications is shown in Figure 28-25 in which 1×10^{-8} mol each of seventeen amino acids was separated on a cation-exchange column.

28F-5 Ion-Exclusion Chromatography

Ion-exclusion chromatography is not a form of ion chromatography because neutral species rather than ions are being separated. Nevertheless, it is convenient to discuss it here because ion-exclusion chromatography, like ion chromatography, uses ion-exchange columns to achieve separations.

The theory and applications of ion-exclusion chromatography are conveniently illustrated by the separation of simple carboxylic acids shown by the chromatogram in Figure 28-26. Here, the stationary phase was a cation-exchange resin in its acidic form, and elution was accomplished with a dilute solution of hydrochloric acid. The analytical column was followed by a suppressor column packed with a cation-exchange resin in the silver form. The hydrogen ions of the eluent were exchanged for silver ions, which then precipitated the chloride ions, thus removing the ions contributed by the eluent. The undissociated form of the analyte acids was distributed between the mobile phase in the column and the immobilized liquid held in the pores of the packing. The distribution constants for the various acids are primarily related to the inverse of their dissociation constants, although other factors also play a part in the extent to which various species are distributed between the two phases.

Ion-exclusion chromatography finds numerous applications for identification and determination of acidic species in milk, coffee, wine, and many other products of commerce. Salts

³⁵P. Kuban and P. K. Dasgupta, *J. Sep. Sci.*, **2004**, 27, 1441, DOI: 10.1002/jssc.200401824.

³⁶P. Bodsky and F. Hoefler, *Fundamentals of Capillary Ion Chromatography*, Sunnyvale, CA: Thermo Fisher Scientific, 2012.

³⁷C. P. Shelor et al., *Astrobiology*, **2014**, 14, 577, DOI: 10.1089/ast.2013.1131.

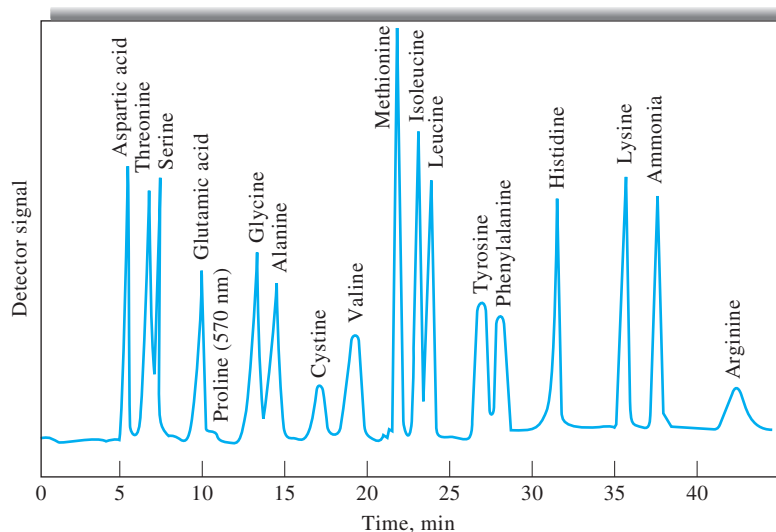


FIGURE 28-25 Separation of amino acids on an ion-exchange column. Packing: cation exchange with particle size of 8 μm . Pressure: 2700 psi. (Reprinted with permission from J. R. Benson, *Amer. Lab.*, **1972**, 4 (10), 60. Copyright 1972 by International Scientific Communications, Inc.)

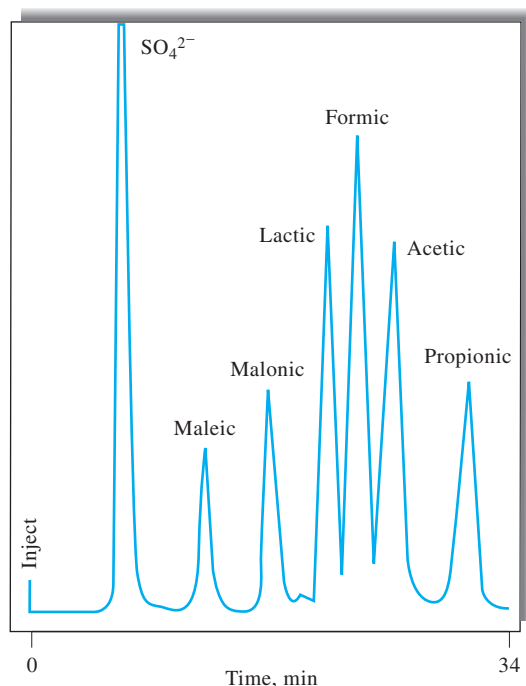


FIGURE 28-26 An ion-exclusion chromatogram for a mixture of six weak acids.

28G SIZE-EXCLUSION CHROMATOGRAPHY

Size-exclusion, or gel, chromatography is a powerful technique that is particularly applicable to high-molecular-mass species.³⁸ Packings for size-exclusion chromatography consist of small ($\sim 10 \mu\text{m}$) silica or polymer particles containing a network of uniform pores into which solute and solvent molecules can diffuse. While in the pores, molecules are effectively trapped and removed from the flow of the mobile phase. The average residence time in the pores depends on the effective size of the analyte molecules. Molecules larger than the average pore size of the packing are excluded and thus suffer essentially no retention; such species are the first to be eluted. Molecules having diameters significantly smaller than the pores can penetrate or permeate throughout the pore maze and are thus entrapped for the greatest time; these are last to be eluted. Between these two extremes are intermediate-size molecules whose average penetration into the pores of the packing depends on their diameters. Within this group, fractionation occurs, which is directly related to molecular size and to some extent molecular shape. Note that size-exclusion separations differ from the other procedures we have been considering in that no chemical or physical

of weak acids can be analyzed because they are converted to the corresponding acid by the hydrogen ions in the exchanger. Weak bases and their salts can also be determined by ion-exclusion chromatography. In these applications an anion-exchange column in the hydroxide form is used.

³⁸For monographs on this subject, see A. Striegel, W. W. Yau, J. J. Kirkland, and D. D. Bly, *Modern Size-Exclusion Chromatography: Practice of Gel Permeation and Gel Filtration Chromatography*, 2nd ed., Hoboken, NJ: Wiley, 2009; C. S. Wu, ed., *Handbook of Size Exclusion Chromatography*, 2nd ed., New York: Dekker, 2004; C. S. Wu, ed., *Column Handbook for Size Exclusion Chromatography*, San Diego: Academic Press, 1999.

interaction between analytes and the stationary phase are involved. Indeed, such interactions are avoided because they lead to lower column efficiencies. Also note that, unlike other forms of chromatography, there is an upper limit to retention time because no analyte species is retained longer than those that totally permeate the stationary phase.

28G-1 Column Packings

Two types of packing for size-exclusion chromatography are encountered: polymer beads and silica-based particles, both of which have diameters of 5–10 μm . Silica particles have the advantages of greater rigidity, which leads to easier packing and permits the use of higher pressures; greater stability, which permits the use of a wider range of solvents, including water; more rapid equilibration with new solvents; and stability at higher temperatures. The disadvantages of silica-based particles include their tendency to retain solutes by adsorption and their potential for catalyzing the degradation of solute molecules.

Most early size-exclusion chromatography was carried out on cross-linked styrene-divinylbenzene copolymers similar in structure (except that the sulfonic acid groups are absent) to that shown in Figure 28-22. The pore size of these polymers is controlled by the extent of cross-linking and hence the relative amount of divinylbenzene present during manufacture. Polymeric packings having several different average pore sizes are thus marketed. Originally, styrene-divinylbenzene gels were hydrophobic and thus could be used only with nonaqueous mobile phases. Now, however, hydrophilic gels are available, making possible the use of aqueous solvents for the separation of large, water-soluble molecules such as sugars. These hydrophilic gels are sulfonated divinylbenzenes or polyacrylamides. Chromatography based on the hydrophilic packings was once called *gel filtration*, and techniques based on hydrophobic packings were termed *gel permeation*. Today, both techniques are described as size-exclusion methods. With both types of packings many pore diameters are available. Ordinarily, a given

packing will accommodate a 2-to-2.5-decade range of molecular mass. The average molecular mass suitable for a given packing may be as small as a few hundred or as large as several million.

Table 28-4 lists the properties of some typical commercial size-exclusion packings.

28G-2 Theory of Size-Exclusion Chromatography

The total volume V_t of a column packed with a porous polymer or silica gel is given by

$$V_t = V_g + V_i + V_o \quad (28-9)$$

where V_g is the volume occupied by the solid matrix of the polymer or gel, V_i is the volume of solvent held in its pores, and V_o is the free volume outside the particles. Assuming no mixing or diffusion, V_o also represents the theoretical volume of solvent required to transport through the column those components too large to enter the pores. In fact, however, some mixing and diffusion will occur, and as a result the unretained components will appear in a Gaussian-shape band with a concentration maximum at V_o . For components small enough to enter freely into the pores of the polymer or gel, band maxima will appear at the end of the column at an eluent volume corresponding to $(V_i + V_o)$. Generally, V_p , V_o , and V_g are of the same order of magnitude; thus, a size-exclusion column permits separation of the large molecules of a sample from the small molecules with a minimal volume of eluate.

Molecules of intermediate size are able to transfer to some fraction K of the solvent held in the pores; the elution volume V_e for these retained molecules is

$$V_e = V_o + KV_i \quad (28-10)$$

Equation 28-10 applies to all of the solutes on the column. For molecules too large to enter the pores, $K = 0$ and $V_e = V_o$; for molecules that can enter the pores unhindered, $K = 1$ and $V_e = (V_o + V_i)$. In deriving Equation 28-10, the assumption is made

TABLE 28-4 Properties of Typical Packings for Size-Exclusion Chromatography

Type	Particle Size, μm	Average Pore Size, $\text{\AA}$	Molecular Mass Exclusion Limit*
Polystyrene-divinylbenzene	10	100	700
		1000	$(0.1 \text{ to } 20) \times 10^4$
		10^4	$(1 \text{ to } 20) \times 10^4$
		10^5	$(1 \text{ to } 20) \times 10^5$
		10^6	$(5 \text{ to } >10) \times 10^6$
Silica	10	125	$(0.2 \text{ to } 5) \times 10^4$
		300	$(0.03 \text{ to } 1) \times 10^5$
		500	$(0.05 \text{ to } 5) \times 10^5$
		1000	$(5 \text{ to } 20) \times 10^5$

*Molecular mass above which no retention occurs.

that no interaction, such as adsorption, occurs between the solute molecules and the polymer or gel surfaces. With adsorption, the amount of interstitially held solute will increase; with small molecules, K will then be greater than unity.

Equation 28-10 rearranges to

$$K = (V_e - V_o)/V_i = c_s/c_M \quad (28-11)$$

where K is the distribution constant for the solute (see Equations 26-1 and 26-2). Values of K range from zero for totally excluded large molecules to unity for small molecules. The distribution constant is a valuable parameter for comparing data from different packings. In addition, it makes possible the application of all of the equations in Table 26-5 to exclusion chromatography.

The useful molecular mass range for a size-exclusion packing is conveniently illustrated by means of a calibration curve

such as that shown in Figure 28-27a. Here, molecular mass, which is directly related to the size of solute molecules, is plotted against retention volume V_R , where V_R is the product of the retention time and the volumetric flow rate. Note that the ordinate scale is logarithmic. The *exclusion limit* is the maximum molecular mass of a species that will penetrate the pores. All species having greater molecular mass than the exclusion limit are so large that they are not retained and elute together to give peak A in the chromatogram shown in Figure 28-27b. Below the *permeation limit*, the solute molecules can penetrate into the pores completely. All molecules below this molecular mass are so small that they elute as the single band labeled D. As molecular masses decrease from the exclusion limit, solute molecules spend more and more time, on the average, in the particle pores and thus move progressively more slowly. It is in the selective permeation region that fractionation occurs, yielding individual solute peaks such as B and C in the chromatogram.

Experimental calibration curves, similar in appearance to the hypothetical one in Figure 28-27a, are usually obtained by



Simulation: Learn more about **size-exclusion chromatography** at www.tinyurl.com/skoogpia7

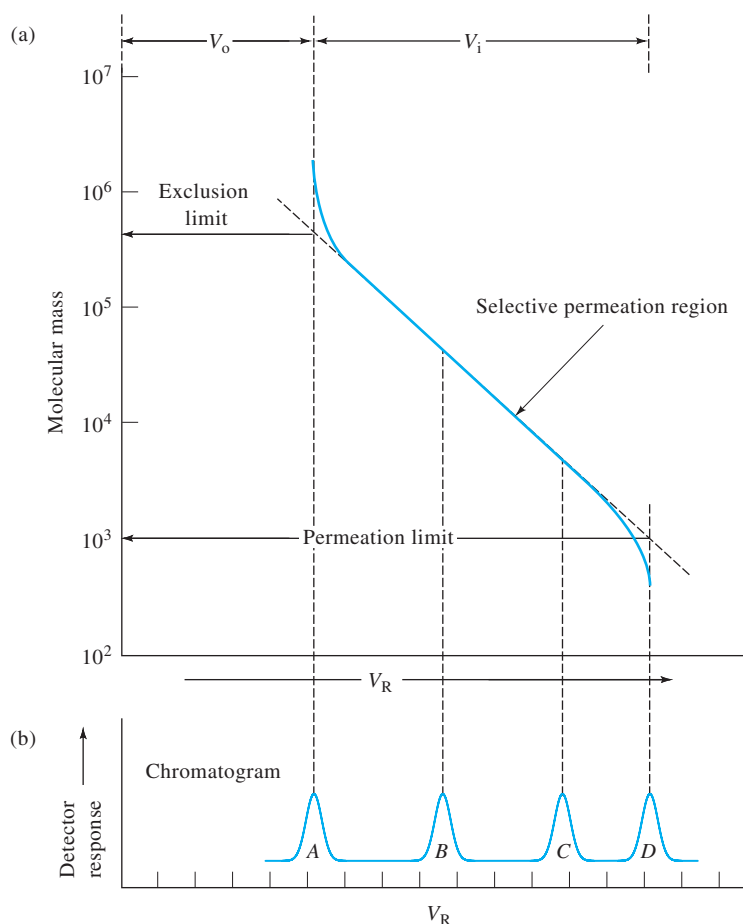


FIGURE 28-27 (a) Calibration curve for a size-exclusion column. (b) Chromatogram showing peak A containing all compounds with molecular masses greater than the exclusion limit, peaks B and C consisting of compounds within the selective permeation region, and peak D containing all compounds smaller than the permeation limit.

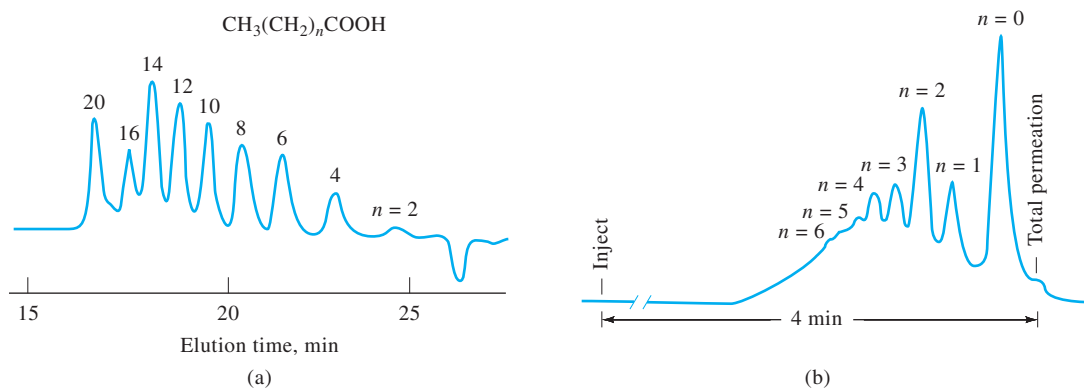


FIGURE 28-28 Applications of size-exclusion chromatography. (a) Separation of fatty acids. Column: polystyrene based, 7.5×600 nm, with exclusion limit of 1×10^3 . Mobile phase: tetrahydrofuran. Flow rate: 1.2 mL/min. Detector: refractive index. (b) Analysis of a commercial epoxy resin (n = number of monomeric units in the polymer). Column: porous silica 6.2×250 mm. Mobile phase: tetrahydrofuran. Flow rate: 1.3 mL/min. Detector: UV absorption. (Courtesy of BTR Separations.)

means of standards. Often, such curves are supplied by manufacturers of size-exclusion columns. Molecular-mass calibration standards should be as similar as possible in chemical functionality to the sample components.

28G-3 Applications of Size-Exclusion Chromatography

Gel filtration and gel permeation methods are complementary in that gel filtration is applied to water-soluble samples and gel permeation is used for substances in less-polar organic solvents. One useful application of the size-exclusion procedure is the separation of high-molecular-mass, natural-product molecules from low-molecular-mass species and from salts. For example, a gel with an exclusion limit of several thousand can separate proteins from amino acids and low-molecular-mass peptides.

Another useful application of gel permeation chromatography is the separation of homologs and oligomers. These applications are illustrated by the two examples shown in Figure 28-28. The first shows the separation of a series of fatty acids ranging in molecular mass $\mathcal{M}$ from 116 to 344 on a polystyrene-based packing with an exclusion limit of 1000. The second is a chromatogram of a commercial epoxy resin, again on a polystyrene packing. Here, n refers to the number of monomeric units ($\mathcal{M} = 264$) in the molecules.

Another important application of size-exclusion chromatography is the rapid determination of the molecular mass or the molecular mass distribution of large polymers or natural products. The key to such determinations is an accurate molecular mass calibration. Calibrations can be accomplished by means of standards of known molecular mass (see Figure 28-27) or by the universal calibration method. The latter method relies on the principle that the product of the intrinsic molecular viscosity η and molecular mass $\mathcal{M}$ is proportional to hydrodynamic volume

(effective volume including solvation sheath). Ideally, molecules are separated in size-exclusion chromatography according to hydrodynamic volume. Hence, a universal calibration curve can be obtained by plotting $\log(\eta\mathcal{M})$ versus the retention volume V_R . Alternatively, absolute calibration can be achieved by using a molar mass-sensitive detector such as a low-angle, light-scattering detector.

The most important advantages of size-exclusion procedures include (1) short and well-defined separation times (all solutes leave the column between V_o and $V_o + V_i$ in Equation 28-9 and Figure 28-27); (2) narrow bands, which lead to good sensitivity; (3) freedom from sample loss because solutes do not interact with the stationary phase; and (4) absence of column deactivation brought about by interaction of solute with the packing.

The disadvantages are (1) only a limited number of bands can be accommodated because the time scale of the chromatogram is short and (2) inapplicability to samples of similar size, such as isomers. Generally, a 10% difference in molecular mass is required for reasonable resolution.

Some of separations traditionally done by size-exclusion chromatography are now being accomplished by field-flow fractionation as discussed in Chapter 30.

28H AFFINITY CHROMATOGRAPHY

Affinity chromatography involves covalently bonding a reagent, called an *affinity ligand*, to a solid support.³⁹ Typical affinity ligands are antibodies, enzyme inhibitors, or other molecules that

³⁹For details on affinity chromatography, see M. Zachariou, ed., *Affinity Chromatography: Methods and Protocols*, 2nd ed., Totowa, NJ: Humana Press, 2007; D. S. Hage ed., *Handbook of Affinity Chromatography*, 2nd ed., Boca Raton, FL: CRC Press, 2006.

reversibly and selectively bind to analyte molecules in the sample. When the sample passes through the column, only the molecules that selectively bind to the affinity ligand are retained. Molecules that do not bind pass through the column with the mobile phase. After the undesired molecules are removed, the retained analytes can be eluted by changing the mobile-phase conditions.

The stationary phase for affinity chromatography is a solid such as agarose or a porous glass bead to which the affinity ligand is immobilized. The mobile phase in affinity chromatography has two distinct roles to play. First, it must support the strong binding of the analyte molecules to the ligand. Second, once the undesired species are removed, the mobile phase must weaken or eliminate the analyte-ligand interaction so that the analyte can be eluted. Often, changes in pH or ionic strength are used to change the elution conditions during the two stages of the process.

Affinity chromatography has the major advantage of extraordinary specificity. The primary use is in the rapid isolation of biomolecules during preparative work.

28I THIN-LAYER CHROMATOGRAPHY

Planar chromatographic methods include *thin-layer chromatography* (TLC) and *paper chromatography* (PC). Each makes use of a flat, relatively thin layer of material that is either self-supporting or is coated on a glass, plastic, or metal surface. The mobile phase moves through the stationary phase by capillary action, sometimes assisted by gravity or an electrical potential.

Currently, most planar chromatography is based on the thin-layer technique, which is faster, has better resolution, and is more sensitive than its paper chromatography equivalent. This section is devoted to thin-layer methods.

28I-1 The Scope of Thin-Layer Chromatography

In terms of theory, types of stationary and mobile phases, and applications, thin-layer and LC are remarkably similar. TLC techniques in fact have been used to develop conditions for HPLC separations. At one time TLC methods were widely used in the pharmaceutical industry. Today, such techniques have largely been replaced by LC methods, which are readily automated and faster. Thin-layer chromatography has found widespread use in clinical laboratories and is the backbone of many biochemical and biological studies. It also finds extensive use in industrial laboratories.⁴⁰ Because of these many areas of application, TLC remains a very important technique.

⁴⁰Books devoted to the principles and applications of thin-layer chromatography include *Handbook of Thin-Layer Chromatography*, 3rd ed., J. Sherma and B. Fried, eds., New York: Dekker, 2003; B. Fried and J. Sherma, *Thin Layer Chromatography*, 4th ed., New York: Dekker, 1999. For reviews of planar chromatography, see J. Sherma, *Anal. Chem.*, **2010**, 82, 4895, DOI: 10.1021/ac902643v; J. Anal. Chem., **2008**, 80, 4253, DOI: 10.1021/ac702341; *Anal. Chem.*, **2006**, 78, 3841, DOI: 10.1021/ac060098l.

28I-2 Principles of Thin-Layer Chromatography

Typical thin-layer separations are performed on a glass plate coated with a thin and adherent layer of finely divided particles; this layer constitutes the stationary phase. The particles are similar to those described in the discussion of adsorption, normal- and reversed-phase partition, ion-exchange, and size-exclusion-column chromatography. Mobile phases are also similar to those found in HPLC.

Preparation of Thin-Layer Plates

A thin-layer plate is prepared by spreading an aqueous slurry of the finely ground solid on the clean surface of a glass or plastic plate or microscope slide. Often a binder is incorporated into the slurry to enhance adhesion of the solid particles to the glass and to one another. The plate is then allowed to stand until the layer has set and adheres tightly to the surface; for some purposes, it may be heated in an oven for several hours. Several chemical supply houses offer precoated plates of various kinds. Costs are a few dollars per plate. The common plate sizes in centimeters are 5×20 , 10×20 , and 20×20 .

Commercial plates come in two categories, conventional and high-performance. Conventional plates have thicker layers (200–250 μm) of particles having particle sizes of 20 μm or greater. High-performance plates usually have film thicknesses of 100 μm and particle diameters of 5 μm or less. High-performance plates provide sharper separations in shorter times. They suffer, however, from having a significantly smaller sample capacity than conventional plates.

Sample Application

Sample application is perhaps the most critical aspect of thin-layer chromatography, particularly for quantitative measurements. Usually, the sample, as a 0.01–0.1% solution, is applied as a spot 1–2 cm from the edge of the plate. For best separation efficiency, the spot should have a minimal diameter—about 5 mm for qualitative work and smaller for quantitative analysis. For dilute solutions, three or four repetitive applications are used, with drying between.

Manual application of samples is performed by touching a capillary tube containing the sample to the plate or by use of a hypodermic syringe. A number of mechanical dispensers, which increase the precision and accuracy of sample application, are now offered commercially.

Plate Development

Plate development is the process in which a sample is carried through the stationary phase by a mobile phase; it is analogous to elution in LC. The most common way of developing a plate is to place a drop of the sample near one edge of the plate and mark its position with a pencil. After the sample solvent has evaporated, the plate is placed in a closed container saturated with vapors of the developing solvent. One end of the plate is immersed in the developing solvent, with care being taken to

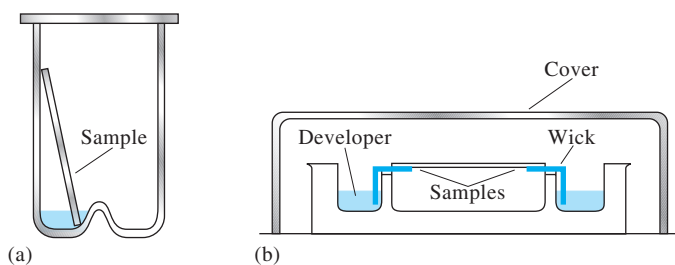


FIGURE 28-29 (a) Ascending-flow developing chamber. (b) Horizontal-flow developing chamber, in which samples are placed on both ends of the plate and developed toward the middle, thus doubling the number of samples that can be accommodated.

avoid direct contact between the sample and the developer (Figure 28-29). After the developer has traversed one half or two thirds of the length of the plate, the plate is removed from the container and dried. The positions of the components are then determined in any of several ways.

Locating Analytes on the Plate

Several methods can be used to locate sample components after separation. Two common methods, which can be applied to most organic mixtures, involve spraying with a solution of sulfuric acid or placing the plate in a chamber containing a few crystals of iodine. Both of these reagents react with organic compounds on the plate to yield dark products. Several specific reagents (such as ninhydrin) are also useful for locating separated species.

Another method of detection is based on incorporating a fluorescent material into the stationary phase. After development, the plate is examined under UV light. The sample components quench the fluorescence of the material so that all of the plate fluoresces except where the nonfluorescing sample components are located.

Figure 28-30 is an idealized drawing showing the appearance of a plate after development and the corresponding chromatogram. Sample 1 contained two components, whereas sample 2 contained one. Frequently, the spots on a real plate exhibit tailing, giving signals that are not symmetric as are those in the figure.

28I-3 Performance Characteristics of Thin-Layer Plates

Most of the terms and relationships developed for column chromatography in Section 26B can, with slight modification, be applied to thin-layer chromatography. One new term, the *retardation factor* or R_F , is required.

The Retardation Factor

The developed thin-layer plate for a single solute is shown in Figure 28-30a for sample 2. The retardation factor for this solute is given by

$$R_F = \frac{d_R}{d_M} \quad (28-12)$$

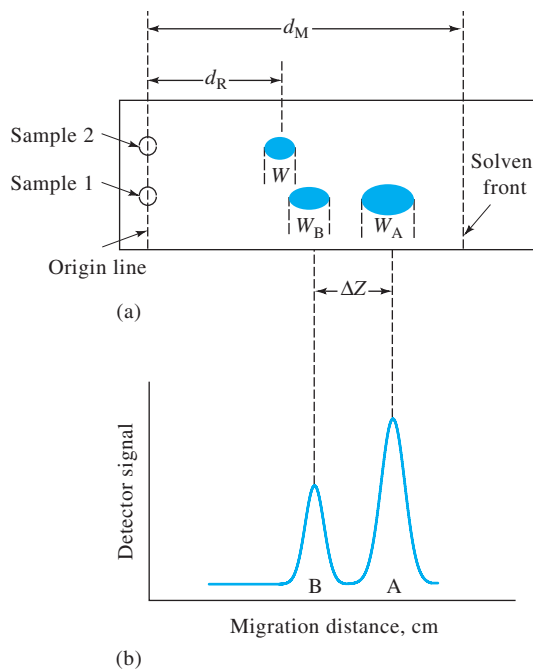


FIGURE 28-30 (a) Thin-layer plate after development. (b) Thin-layer chromatogram for sample 1.

where d_R and d_M are linear distances measured from the origin line. Values for R_F can vary from 1 for solutes that are not retarded to a value that approaches 0. Note that if the spots are not symmetric, as they are in Figure 28-30a, the measurement of d_R is based on the position of maximum intensity.

The Retention Factor

All of the equations in Table 26-5 can be adapted to thin-layer chromatography. To apply these equations, it is only necessary to relate d_R and d_M as defined in Figure 28-30a to t_R and t_M , which are defined in Figure 26-4. To arrive at these relationships, consider the single solute that appears in sample 2 in Figure 28-30a. Here, t_M and t_R correspond to times required for the mobile phase and the solute to travel a fixed distance—in this case, d_R . For the mobile phase, this time is equal to the distance divided by its linear velocity u , or

$$t_M = d_R/u \quad (28-13)$$

The solute does not reach this same point, however, until the mobile phase has traveled the distance d_M . Therefore,

$$t_R = d_M/u \quad (28-14)$$

Substitution of Equations 28-13 and 28-14 into Equation 26-12 yields

$$k = \frac{d_M - d_R}{d_R} \quad (28-15)$$

The retention factor k can also be expressed in terms of the retardation factor by rewriting Equation 28-15 in the form

$$k = \frac{1 - d_R/d_M}{d_R/d_M} = \frac{1 - R_F}{R_F} \quad (28-16)$$

Retention factors derived in this way can be used for method development in column chromatography as described in Section 28D-2. Obtaining retention factors by thin-layer chromatography is, however, usually simpler and more rapid than obtaining the data from experiments on a column.

Plate Heights

Approximate plate heights can also be determined for a given type of packing by thin-layer chromatographic measurements. Thus, for sample 2 in Figure 28-30a, the plate count is given by the equation

$$N = 16 \left(\frac{d_R}{W} \right)^2 \quad (28-17)$$

where d_R and W are defined in the figure. The plate height is then given by

$$H = d_R/N \quad (28-18)$$

28I-4 Applications of Thin-Layer Chromatography

The thin-layer method can be applied both to qualitative identification of compounds and to quantitative analysis.

Qualitative TLC

The data from a single chromatogram usually do not provide sufficient information to permit identification of the various species present in a mixture because of the variability of R_F values with sample size, the thin-layer plate, and the development conditions. In addition, there is always the possibility that two quite different solutes may exhibit identical or nearly identical R_F values under a given set of conditions.

Identification Methods. Sometimes, tentative identification can be made by applying to the plate solutions of purified species suspected to be present in the unknown. A match in R_F values provides evidence as to the identity of one of the components. In other cases, the identity of separated species can be confirmed by a scraping-and-dissolution technique. Here, the area containing the analyte is scraped from the plate with

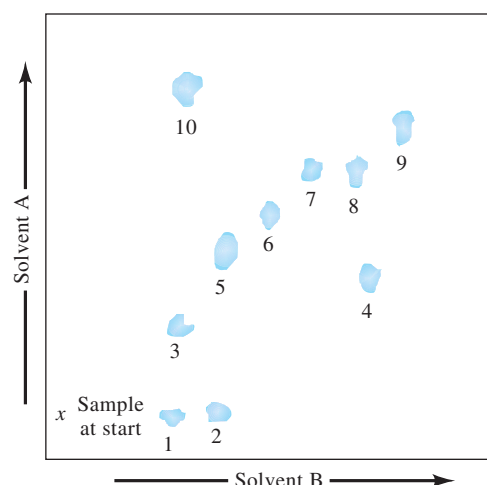


FIGURE 28-31 Two-dimensional thin-layer chromatogram (silica gel) of some amino acids. Solvent A: toluene, 2-chloroethanol, pyridine. Solvent B: chloroform, benzyl alcohol, acetic acid. Amino acids: (1) aspartic acid, (2) glutamic acid, (3) serine, (4) β -alanine, (5) glycine, (6) alanine, (7) methionine, (8) valine, (9) isoleucine, (10) cysteine.

a razor blade or spatula and the contents collected on a piece of glazed paper. After transfer to a suitable container, the analyte is dissolved in an appropriate solvent and separated from the stationary phase by centrifugation or filtration. Spectroscopic techniques, such as fluorescence, UV-visible absorption, nuclear magnetic resonance, or FTIR, are then used to identify the species. Thin-layer chromatography has also been on-line coupled to a number of spectrometric detection techniques. The coupling to TLC to mass spectrometers is an active area of research.⁴¹

Two-Dimensional Planar Chromatography. Figure 28-31 illustrates the separation of amino acids in a mixture by development in two dimensions. The sample was placed in one corner of a square plate and development was performed in the ascending direction with solvent A. This solvent was then removed by evaporation, and the plate was rotated 90°, following which an ascending development with solvent B was performed. After solvent removal, the positions of the amino acids were determined by spraying with ninhydrin, a reagent that forms a pink to purple product with amino acids. The spots were identified by comparison of their positions with those of standards. Imaging techniques have been used for quantitative analysis of two-dimensional separations.⁴²

⁴¹P. Bernard-Savary and C. F. Poole, *J. Chromatogr. A*, **2015**, 1421, 184, DOI: 10.1016/j.chroma.2015.08.002.

⁴²M. Medic-Saric et al., *J. Planar. Chromatogr.-Mod. TLC*, **2004**, 17, 459, DOI: 10.1556/JPC.17.2004.6.12.

Quantitative Analysis

A semiquantitative estimate of the amount of a component present can be obtained by comparing the area of a spot with that of a standard. More accurate results can be obtained by scraping the spot

from the plate, extracting the analyte from the stationary-phase solid, and measuring the analyte by a suitable physical or chemical method. In a third method, a scanning densitometer can be used to measure fluorescence or absorption of the spot.

>> QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

28-1 List the types of substances to which each of the following chromatographic methods is most applicable:

- (a) gas-liquid
- (b) liquid adsorption
- (c) liquid-liquid partition
- (d) reversed-phase partition
- (e) ion exchange
- (f) gel permeation
- (g) gas-solid
- (h) gel filtration
- (i) ion-pair

28-2 Describe three general methods for improving resolution in partition chromatography.

28-3 Describe a way to manipulate the retention factor of a solute in partition chromatography.

28-4 How can the selectivity factor be manipulated in (a) GC and (b) LC?

28-5 In preparing a hexane-acetone gradient for an alumina HPLC column, is it desirable to increase or decrease the proportion of hexane as the column is eluted?

28-6 What is meant by the linear-response range of a detector?

28-7 Define

- (a) isocratic elution.
- (b) gradient elution.
- (c) stop-flow injection.
- (d) reversed-phase packing.
- (e) normal-phase packing.
- (f) ion-pairing chromatography.
- (g) ion chromatography.
- (h) bulk property detector.
- (i) solute property detector.
- (j) sparging.

28-8 What is a guard column in partition chromatography?

28-9 In what way are normal-phase partition chromatography and adsorption chromatography similar?

*** 28-10** What is the order in which the following compounds would be eluted from an HPLC column containing a reversed-phase packing?

- (a) benzene, diethyl ether, *n*-hexane
- (b) acetone, dichloroethane, acetamide

*** 28-11** What is the order of elution of the following compounds from a normal-phase packed HPLC column?

- (a) ethyl acetate, acetic acid, dimethylamine
- (b) propylene, hexane, benzene, dichlorobenzene

28-12 Describe the fundamental difference between adsorption and partition chromatography.

28-13 Describe the fundamental difference between ion-exchange and size-exclusion chromatography.

*** 28-14** What types of species can be separated by HPLC but not by GC?

28-15 Describe the various kinds of pumps used in HPLC. What are the advantages and disadvantages of each?

28-16 Describe the differences between single-column and suppressor-column ion chromatography.

28-17 Mass spectrometry is an extremely versatile detection system for GC. However, interfacing an HPLC system to a mass spectrometer is a much more difficult task. Describe the major reasons why it is more difficult to combine HPLC with mass spectrometry than it is to combine GC with mass spectrometry.

28-18 Which of the GC detectors in Table 27-1 are suitable for HPLC? Why are some of the detectors listed unsuitable for HPLC?

28-19 Although temperature does not have nearly the effect on HPLC separations that it has on GC separations, it nonetheless can play an important role. Discuss how and why temperature might or might not influence the following separations:

- (a) a reversed-phase chromatographic separation of a steroid mixture.
- (b) an adsorption chromatographic separation of a mixture of closely related isomers.



*** 28-20** Two components in an HPLC separation have retention times that differ by 15 s. The first peak elutes in 9.0 min and the peak widths are approximately equal. The dead time t_M was 65 s. Use a spreadsheet to find the minimal number of theoretical plates needed to achieve the following resolution R_s values: 0.50, 0.75, 0.90, 1.0, 1.10, 1.25, 1.50, 1.75, 2.0, 2.5. How would the results change if peak 2 were twice as broad as peak 1?



*** 28-21** An HPLC method was developed for the separation and determination of ibuprofen in rat plasma samples as part of a study of the time course of the drug in laboratory animals. Several standards were chromatographed and the following results obtained:

Ibuprofen Concentration, μg/mL	Relative Peak Area
0.5	5.0
1.0	10.1
2.0	17.2
3.0	19.8
6.0	39.7
8.0	57.3
10.0	66.9
15.0	95.3

>> QUESTIONS AND PROBLEMS (continued)

Next, a 10 mg/kg sample of ibuprofen was administered orally to a laboratory rat. Blood samples were drawn at various times after administration of the drug and subjected to HPLC analysis. The following results were obtained:

Time, h	Peak Area
0	0
0.5	91.3
1.0	80.2
1.5	52.1
2.0	38.5
3.0	24.2
4.0	21.2
6.0	18.5
8.0	15.2

Find the concentration of ibuprofen in the blood plasma for each of the times given above and plot the concentration versus time. On a percentage basis, during what half-hour period (first, second, third, etc.) is most of the ibuprofen lost?

- * 28-22 In a normal-phase partition column, a solute was found to have a retention time of 29.1 min, and an unretained sample had a retention time of 1.05 min when the mobile phase was 50% by volume chloroform and 50% *n*-hexane. Calculate (a) k for the solute and (b) a solvent composition that would bring k down to a value of about 10.

Challenge Problem

28-23 Assume for simplicity that the HPLC plate height, H , can be given by Equation 26-23 as

$$H = \frac{B}{u} + C_S u + C_M u = \frac{B}{u} + C u$$

where $C = C_S + C_M$.

- (a) By using calculus to find the minimum H , show that the optimal velocity u_{opt} can be expressed as

$$u_{\text{opt}} = \sqrt{\frac{B}{C}}$$

- (b) Show that this leads to a minimum plate height H_{min} given by

$$H_{\text{min}} = 2\sqrt{BC}$$

- (c) Under some conditions for chromatography, C_S is negligible compared to C_M . For packed HPLC columns, C_M is given by

$$C_M = \frac{\omega d_p^2}{D_M}$$

where ω is a dimensionless constant, d_p is the particle size of the column packing, and D_M is the diffusion coefficient in the mobile phase. The B coefficient can be expressed as

$$B = 2\gamma D_M$$

where γ is also a dimensionless constant. Express u_{opt} and H_{min} in terms of D_M , d_p , and the dimensionless constants γ and ω .

- (d) If the dimensionless constants are on the order of unity, show that u_{opt} and H_{min} can be expressed as

$$u_{\text{opt}} \approx \frac{D_M}{d_p} \quad \text{and} \quad H_{\text{min}} \approx d_p$$

- (e) Under the preceding conditions, how could the plate height be reduced by one-third? What would happen to the optimal velocity under these conditions? What would happen to the number of theoretical plates N for the same length column?
- (f) For the conditions in part (e), how could you maintain the same number of theoretical plates while reducing the plate height by one-third?
- (g) The preceding discussion assumes that all band broadening occurs within the column. Name two sources of extracolumn band broadening that might also contribute to the overall width of HPLC peaks.

Supercritical Fluid Chromatography and Extraction

Two techniques based on the use of supercritical fluids were developed during the 1970s and early 1980s and today play an important role in the analysis of environmental, biomedical, and food samples. These methods are **supercritical fluid chromatography (SFC)** and **supercritical fluid extraction (SFE)**. Commercial instrumentation for both of these techniques became available in the mid-1980s, and their use has grown rapidly in the analytical community. In this chapter we describe the theory, instrumentation, and applications of both SFC and SFE.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

29A PROPERTIES OF SUPERCRITICAL FLUIDS

A *supercritical fluid* is formed whenever a substance is heated above its critical temperature. The *critical temperature* of a substance is the temperature above which a distinct liquid phase cannot exist, regardless of pressure. The vapor pressure of a substance at its critical temperature is its *critical pressure*. At temperatures and pressures above its critical temperature and pressure (its critical point), a substance is called a *supercritical fluid*. Supercritical fluids have densities, viscosities, and other properties that are intermediate between those of the substance in its gaseous and liquid states. Table 29-1 compares some properties of supercritical fluids to those of typical gases and liquids. These properties are important in gas, liquid, and supercritical fluid chromatography and extractions.

Table 29-2 lists properties of four out of perhaps two dozen compounds that have been used as mobile phases in SFC. Note that their critical temperatures, and the pressures at these temperatures, are well within the operating conditions of ordinary high-performance liquid chromatography (HPLC).

An important property of supercritical fluids, and one that is related to their high densities ($0.2\text{--}0.5\text{ g/cm}^3$), is their ability to dissolve large nonvolatile molecules. For example, supercritical carbon dioxide readily dissolves *n*-alkanes containing from five to twenty-two carbon atoms, di-*n*-alkylphthalates in which the alkyl groups contain four to sixteen carbon atoms, and various polycyclic aromatic hydrocarbons consisting of several rings. Certain important industrial processes are based on the high solubility of organic species in supercritical carbon dioxide. For example, this medium has been used for extracting caffeine from coffee beans to give decaffeinated coffee and for extracting nicotine from cigarette tobacco.

A second important property of many supercritical fluids is that analytes dissolved in them can be easily recovered by simply allowing the solutions to equilibrate with the atmosphere at relatively low temperatures. Thus, an analyte dissolved in supercritical carbon dioxide, the most commonly used solvent, can be recovered by simply reducing the pressure and allowing the fluid to evaporate under ambient laboratory conditions. This

TABLE 29-1 Comparison of Properties of Supercritical Fluids with Liquids and Gases

Property	Gas (STP)	Supercritical Fluid	Liquid
Density, g/cm ³	$(0.6\text{--}2) \times 10^{-3}$	0.2–0.5	0.6–2
Diffusion coefficient, cm ² /s	$(1\text{--}4) \times 10^{-1}$	$10^{-3}\text{--}10^{-4}$	$(0.2\text{--}2) \times 10^{-5}$
Viscosity, g cm ⁻¹ s ⁻¹	$(1\text{--}3) \times 10^{-4}$	$(1\text{--}3) \times 10^{-4}$	$(0.2\text{--}3) \times 10^{-2}$

Note: All data are to an order of magnitude only.

TABLE 29-2 Properties of Some Supercritical Fluids

Fluid	Critical Temperature, °C	Critical Pressure, atm	Critical Point Density, g/mL	Density at 400 atm, g/mL
CO ₂	31.3	72.9	0.47	0.96
N ₂ O	36.5	71.7	0.45	0.94
NH ₃	132.5	112.5	0.24	0.40
<i>n</i> -Butane	152.0	37.5	0.23	0.50

Source: From M. L. Lee and K. E. Markides, *Science*, **1987**, 235, 1342, DOI: 10.1126/science.235.4794.1342, with permission. Data taken from *Matheson Gas Data Book* and *CRC Handbook of Chemistry and Physics*.

property is particularly important with thermally unstable analytes. Another advantage of many supercritical fluids is that they are inexpensive, innocuous, and generally nontoxic substances. For these reasons, SFC has been touted as a “green” alternative to HPLC. Supercritical fluid carbon dioxide is especially attractive for extractions and chromatography. Finally, supercritical fluids have the advantage that solute diffusivities are an order of magnitude higher than in liquid solvents and viscosities are an order of magnitude lower. These last two advantages are important in both chromatography and extractions with supercritical fluids.

29B SUPERCRITICAL FLUID CHROMATOGRAPHY

SFC, in which the mobile phase is a supercritical fluid, is a hybrid of gas chromatography (GC) and liquid chromatography (LC) that combines some of the best features of each. For certain applications, it is superior to both GC and HPLC.¹ In 1985 several instrument manufacturers began to offer equipment specifically designed for SFC, and the use of such equipment has expanded rapidly. The SFC technique has become an important tool in many industrial, regulatory, and academic laboratories. Both packed and open tubular columns are used in SFC.

¹For additional information on SFC, see *Supercritical Fluid Chromatography: Advances and Applications in Pharmaceutical Analysis*, G. K. Webster, ed., Boca Raton, FL: CRC Press, 2014; *Practical Supercritical Fluid Chromatography and Extraction*, M. Caude and D. Thiebaut, eds., Amsterdam: Harwood, 2000; L. Taylor, *J. Supercrit. Fluids*, **2009**, 47, 566, DOI: 10.1016/j.supflu.2008.09.012; G. Guichon and A. Tarafder, *J. Chromatogr. A*, **2011**, 1218, 1037, DOI: 10.1016/j.chroma.2010.12.047; E. Lesellier and C. West, *J. Chromatogr. A*, **2015**, 1382, 2, DOI: 10.1016/j.chroma.2014.12.083.

SFC is important because it permits the separation and determination of a group of compounds not conveniently handled by either GC or LC. These compounds (1) are nonvolatile or thermally unstable so that GC procedures are inapplicable and (2) contain no functional groups that make possible detection by the spectroscopic or electrochemical techniques used in HPLC. In 1986, Chester² estimated that up to 25% of all separation problems faced by scientists involve mixtures containing such intractable species.

29B-1 Instrumentation and Operating Variables

As mentioned earlier, the pressures and temperatures required for creating supercritical fluids derived from several common gases and liquids lie well within the operating limits of ordinary HPLC equipment. Thus, as shown in Figure 29-1, instruments for SFC are similar in most aspects to the instruments for HPLC described in Section 28C.³ There are two important differences between the two techniques, however. First, a thermostatted column oven, similar to that used in GC (Section 27B-3), is required to provide precise temperature control of the mobile phase; second, a restrictor, or back-pressure device, is used to maintain the desired pressure in the column. In the case of open tubular columns, which were popular in the 1980s, the restrictor was also used to convert the eluent from a supercritical fluid to a gas for transfer to the detector, typically a flame ionization detector (FID).

²T. L. Chester, *J. Chromatogr. Sci.*, **1986**, 24, 226, DOI: 10.1093/chromsci/24.6.226.

³For an historical perspective on instrumental developments in SFC, see M. Saito, *J. Biosci. Bioeng.*, **2013**, 115, 590, DOI: 10.1016/j.jbiosc.2012.12.008.

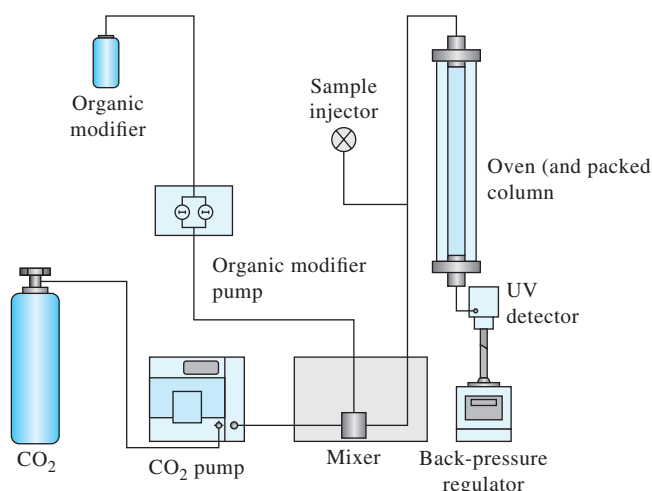


FIGURE 29-1 Block diagram of an instrument for packed-column SFC.

Beginning in the late 1980s, packed column SFC began receiving a good deal of interest and today packed columns are widely used. Figure 29-1 shows a schematic diagram of a typical packed column SFC instrument. With the regulator, pressure control is independent of the mobile phase flow rate. With packed columns, polar modifiers, which are not compatible with FIDs, can be pumped together with the main fluid with binary pump designs, and mixed to increase the solvating tendency of the mobile phase. Also, the same types of detectors used in HPLC, particularly UV-visible detectors, are useful with packed columns.

Effects of Pressure

In SFC, retention of analytes is influenced by mobile phase density, which is a function of temperature, pressure, and mobile phase composition. In open tubular SFC, pressure changes have a pronounced effect on the retention factor k and thus the retention time t_R . The density of a supercritical fluid increases rapidly and nonlinearly with increases in pressure. Such density increases cause a rise in the solvent power of the mobile phase, which in turn shortens elution time. For example, the elution time for hexadecane is reported to decrease from 25 to 5 min as the pressure of carbon dioxide is raised from 70 to 90 atm. An effect similar to that of temperature programming in GC and gradient elution in HPLC can be achieved by linearly increasing the column pressure or by regulating the pressure to create linear density increases. Figure 29-2 illustrates the improvement in chromatograms realized by pressure programming. The decompression of fluids as they travel through the column can cause temperature changes that can affect separations and thermodynamic measurements. The most common pressure profiles used in SFC are often constant (isobaric) for a given length of time followed by a linear or asymptotic increase to a final pressure. In addition to pressure programming, temperature programming and mobile-phase gradients can be used.

With packed column SFC and the use of modifiers, mobile phase composition has become more important than pressure and density, which are used mostly for fine-tuning purposes.⁴

Stationary Phases

Packed columns can provide more theoretical plates and handle larger sample volumes than open tubular columns. Because of the low viscosity of supercritical media, columns can be much

⁴L. Nováková et al., *Anal. Chim. Acta*, **2014**, 824, 18, DOI: 10.1016/j.aca.2014.03.034.

Sample: 1. cholesteryl octanoate
2. cholesteryl decylate
3. cholesteryl laurate
4. cholesteryl myristate
5. cholesteryl palmitate
6. cholesteryl stearate

Column: DB-1
Mobile phase: CO₂
Temperature: 90°C
Detector: FID Flame ionization

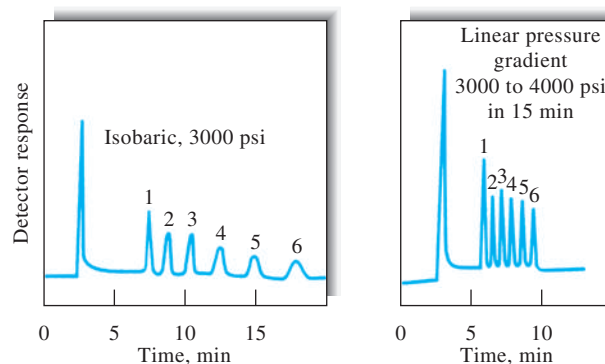


FIGURE 29-2 Effect of pressure programming in open tubular SFC. Note the shorter time for the pressure-gradient chromatogram on the right compared with the constant pressure (isobaric) chromatogram on the left. (Courtesy of Brownlee Labs, Santa Clara, CA.)



Simulation: Learn more about **supercritical fluid chromatography** at www.tinyurl.com/skoogpia7

longer than those used in LC, and column lengths of 10–20 m and inside diameters of 50 or 100 μm are common with open tubular columns. For difficult separations, columns 60 m or longer have been used. Open tubular columns are similar to the fused-silica wall-coated (FSWC) open tubular columns described in Section 27C-1. Packed columns are usually made of stainless steel, 10–25 cm long. Typical columns are 3.0–4.6 mm i.d. packed with particles ranging in size from 2–5 μm . More than 100,000 plates have been achieved with packed columns.

When packed columns first became popular for SFC, separations were compared to normal phase HPLC since CO_2 is relatively nonpolar. As a result, only polar stationary phases were considered. However, with the increasing use of modifiers to vary the mobile phase polarity, a variety of stationary phases ranging from polar (silica gel) to nonpolar (C18 bonded silica) have proven to be useful. Retention mechanisms can thus be altered by varying the stationary phase and mobile phase composition.⁴ Many of the stationary phases used in HPLC have been applied to SFC.

Mobile Phases

The most widely used mobile phase for SFC is carbon dioxide. It is a good solvent for a variety of nonpolar organic molecules. In addition, it transmits in the UV region and is odorless, nontoxic, readily available, and remarkably inexpensive compared to other chromatographic solvents. Its critical temperature of 31°C and its critical pressure of 72.9 atm permit a wide selection of temperatures and pressures without exceeding the operating limits of modern HPLC equipment. In most applications of modern, packed-column SFC, organic modifiers are added in order to improve analyte solubility, avoid precipitation, and vary analyte retention. Typically, for an isocratic elution, a modifier concentration of 5% or less is used. Polar modifiers, such as methanol, ethanol, and acetonitrile have been used as well as nonpolar compounds like hexane. For samples with a wide range of polarities, gradient elution with 5–50% modifier is the most common separation technique.⁵

Ethane, pentane, dichlorodifluoromethane, diethyl ether, ammonia, and tetrahydrofuran have also served as mobile phases in SFC.

Detectors

With open tubular columns, the flame ionization detector of GC can be used. This can be an advantage as discussed in Section 27B-4, since the FID is a universal-type detector and is largely trouble free. With packed columns, many of the detectors used in HPLC are applicable including UV-visible absorption, fluorescence, and mass spectrometry. Mass spectrometry has become quite popular because of the ease of interfacing atmospheric pressure chemical ionization and electrospray ionization sources.⁶

⁵E. Lesellier and C. West, *J. Chromatogr. A*, **2015**, 1382, 2, DOI: 10.1016/j.chroma.2014.12.083.

⁶A. Grand-Guillaume Perrenoud, J.-L. Veuthey, and D. Guillarme, *J. Chromatogr. A*, **2014**, 1339, 174, DOI: 10.1016/j.chroma.2014.03.006; J. D. Pinkston, W. Wen, K. L. Morand, D. A. Tirey, D. T. Stanton, *Anal. Chem.*, **2006**, 78, 7467, DOI: 10.1021/ac061033l.

29B-2 Comparison to Other Types of Chromatography

The data in Tables 29-1 and 29-2 reveal that several physical properties of supercritical fluids are intermediate between gases and liquids. Hence, this type of chromatography combines some of the characteristics of both GC and LC. At first, particularly when open tubular columns were popular, SFC was thought to be a complement to GC. The relatively nonpolar mobile phase was used with nonpolar analytes and a relatively nonpolar stationary phase. Because CO_2 had more solvating power than GC carrier gases, SFC began to be thought of as a way to increase the range of samples that could be analyzed by GC. At present, SFC is considered to be more like HPLC because it uses nearly the same hardware and software. Mobile phases can be binary or ternary mixtures with CO_2 as the major constituent. Gradient elution is performed with relatively polar stationary phases. SFC has several advantages over reverse phase HPLC.⁷ The lower viscosity of supercritical CO_2 compared to HPLC mobile phases means that higher flow rates can be used, which leads to faster separations. The operating costs of SFC are lower than HPLC because solvent consumption is lower. Finally, SFC is more environmentally friendly than HPLC.

Figures 29-3 and 29-4 compare the performance characteristics of a packed column when elution is performed with supercritical carbon dioxide and a conventional liquid mobile phase.

⁷L. Taylor, *Anal. Chem.*, **2010**, 82, 4925, DOI: 10.1021/ac101194x.

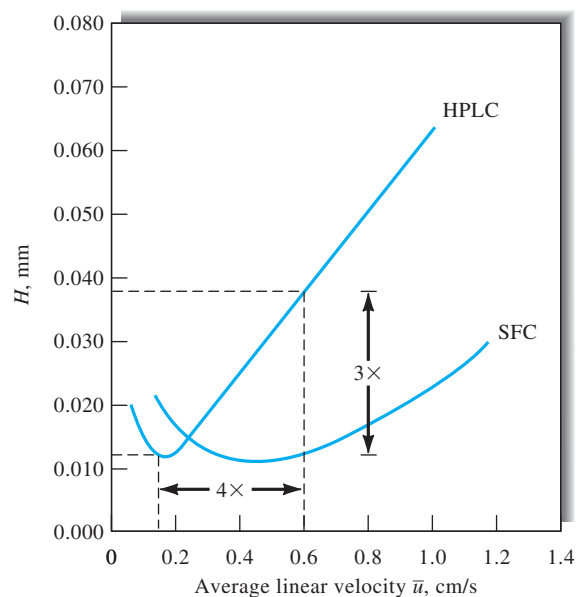


FIGURE 29-3 Performance characteristics of a 5- μm octadecyl (C_{18}) bonded silica column when elution is carried out with a conventional mobile phase (HPLC) and supercritical carbon dioxide SFC. (Adapted from D. R. Gere, *Science*, **1983**, 222, 253, DOI: 10.1126/science.6414083.)

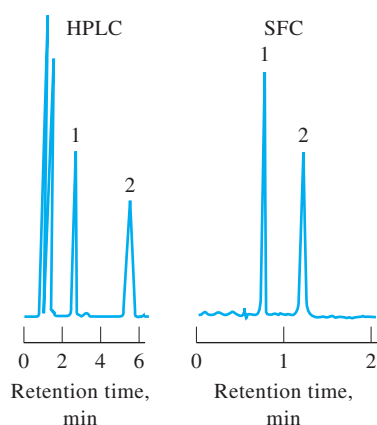


FIGURE 29-4 Comparison of chromatograms obtained by conventional partition chromatography (HPLC) and SFC. Column: 20 cm $\times$ 4.6 mm with 10- μ m reversed-phase bonded packing. Analytes: (1) biphenyl and (2) terphenyl. For HPLC: mobile phase, 65% CH₃OH to 35% H₂O; flow rate, 4 mL/min; linear velocity, 0.55 cm/s; sample size, 10 μ L. For SFC: mobile phase, CO₂; flow rate, 5.4 mL/min; linear velocity, 0.76/s; sample size, 3 μ L. (Adapted from D. R. Gere, *Science*, **1983**, 222, 253.)

Figure 29-3 shows that at a linear mobile-phase velocity of 0.6 cm/s, the supercritical column yields a plate height of 0.013 mm, whereas the plate height with a liquid eluent is three times as large or 0.039 mm. Thus, a reduction in peak width by a factor of $\sqrt{3}$ should be realized with SFC. Alternatively, the linear velocity can be increased by a factor of 4 at the plate height corresponding to the minimum in the HPLC curve; this increase would result in a reduction of analysis time by a factor of 4. These advantages are reflected in the two chromatograms shown in Figure 29-4.

The mobile phase plays different roles in GC, LC, and SFC. Ordinarily, in GC the mobile phase serves but one purpose—zone movement. As we have seen in Chapter 28, in LC the mobile phase provides not only transport of solute molecules but also interactions with solutes that influence selectivity factors (α values). When a molecule dissolves in a supercritical medium, the process resembles volatilization but at a much lower temperature than would normally be used in GC. Thus, at a given temperature, the vapor pressure for a large molecule in a supercritical fluid may be 10^{10} times greater than in the absence of the fluid. Because of this, high-molecular-mass compounds, thermally unstable species, polymers, and large biological molecules can be eluted from a column at relatively low temperatures. Interactions between solute molecules and the molecules of a supercritical fluid must occur to account for their solubility in these media. The solvent power is thus a function of the chemical composition and the density of the fluid. Therefore, in contrast to GC, it is possible to vary α values by changing the mobile phase.

Figure 29-5 compares the application range of SFC with GC, LC, and size-exclusion chromatography (SEC). Note that LC and SFC are applicable over molecular mass ranges that are several orders of magnitude greater than GC. As we have noted earlier, SEC can be applied to even larger molecules.

29B-3 Applications

The SFC technique has been applied to a wide variety of materials, including natural products, drugs, foods, pesticides and herbicides, surfactants, polymers and polymer additives, fossil fuels, and explosives and propellants. One of the most important applications of SFC is to chiral separations, particularly in the pharmaceutical industry.⁸ The increased diffusion rates of SFC over LC eluents lead to higher resolution and sharper peaks to enable accurate measurements of enantiomeric purity. Figure 29-6 shows the separation of the enantiomers of metoprolol, a beta-blocking drug, by both HPLC and SFC. Although both methods achieve complete resolution of the enantiomers, the separation requires only 6 min with SFC compared to 22 min with HPLC. Note also the higher resolution and more symmetric peaks obtained by SFC. In addition to higher resolution, method development time can also be decreased with SFC compared to HPLC methods because the desired resolution for enantiomers can often be achieved without investigating as many stationary phases and mobile phases as in HPLC. In cases where a single column is insufficient to produce the desired separation, the lower pressure drop in SFC makes it easier to couple

⁸V. Desfontaine, D. Guilleme, E. Francotte, and L. Nováková, *J. Pharm. Biomed. Anal.*, **2015**, 113, 56, DOI: 10.1016/j.jpba.2015.03.007; K. Kalíková, T. Šlechtová, J. Vozka, E. Tesařová, *Anal. Chim. Acta*, **2014**, 821, 1, DOI: 10.1016/j.aca.2014.02.036.

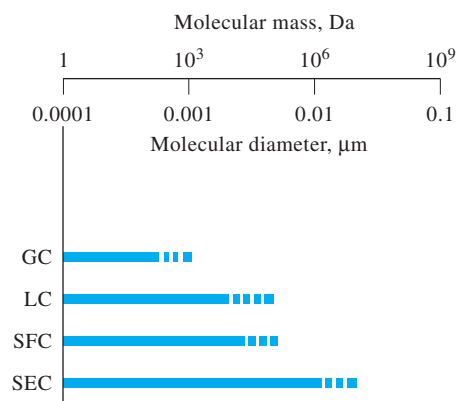


FIGURE 29-5 Range of molecular masses and sizes over which column chromatographic techniques can be applied. (From M. L. Lee and K. E. Markides, *Science*, **1987**, 235, 1342, DOI: 10.1126/science.235.4794.1342. With permission.)

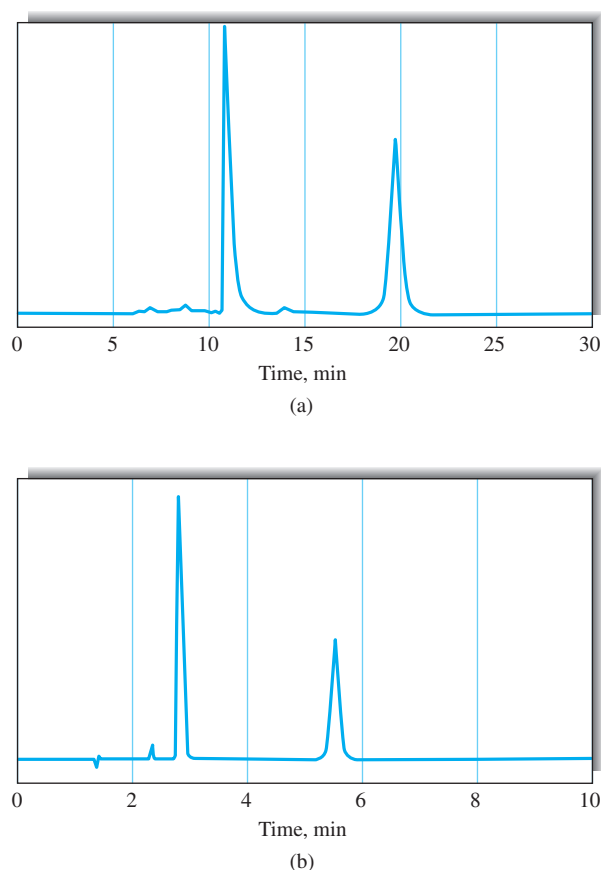


FIGURE 29-6 Separation of enantiomers of metoprolol by HPLC (a) and SFC (b) on a Chiralcel OD stationary phase. In (a), the mobile phase was 20% 2-propanol in hexane with 1% diethylamine; selectivity $\alpha = 2.67$ and resolution $R_s = 4.8$. In (b), CO_2 was used containing 20% methanol with 0.5% isopropylamine; $\alpha = 2.77$ and $R_s = 12.7$. (From K. W. Phinney, *Anal. Chem.*, **2000**, 72, 204A, DOI: 10.1021/ac0027542; adapted from M. S. Villeneuve and R. J. Anderegg, *J. Chromatogr. A*, **1998**, 826, 217, DOI: 10.1016/S0021-9673(98)00696-7.)

columns with different stationary phases. Preparative chiral separations by SFC are very attractive because the mobile phase can be readily removed by simply venting the CO_2 .

SFC is also very useful for achiral separations. Figures 29-7, 29-8, and 29-9 illustrate three typical and diverse applications of SFC. Figure 29-7 shows the separation of a series of dimethylpolysiloxane oligomers ranging in molecular mass from 400 to 700 Da. This chromatogram was obtained by using a 10 m $\times$ 100 μm inside-diameter fused-silica capillary coated with a 0.25- μm film of 5% phenylpolysiloxane. The mobile phase was CO_2 at 140°C, and the following pressure program was used: 80 atm for 20 min, then a linear gradient from 80 to 280 atm at 5 atm/min. A FID was used.

Figure 29-8 shows chromatograms of nine different drugs using packed column SFC with gradient elution. Six different stationary phases are shown. The stationary phase can be seen

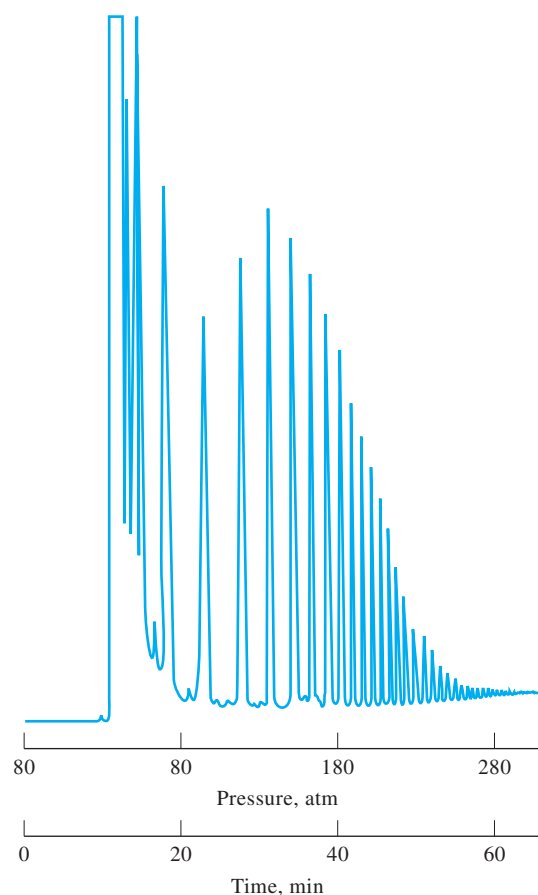


FIGURE 29-7 Separation of oligomers of dimethylpolysiloxane by SFC. (From C. M. White and R. K. Houck, *HRC&CC*, **1986**, 9, 4, DOI: 10.1002/jhrc.1240090103. With permission.)

to affect the retention of the analytes and the elution order. The acidic compounds (1, 2, 3, and 4) eluted rapidly on bare silica (a) and hybrid silica (b). Neutral compounds (5 and 6) did not elute at all on the amide stationary phase (f). The basic compounds (7, 8, and 9) did not elute on bare silica, but all eluted on the 2-EP phase (d). Several of the bases showed broadened peaks and significant tailing (b, c, and d).

Figure 29-9 illustrates the determination of triglycerides in vegetable oil by high efficiency, low-pressure SFC using “superficially porous particles.” These particles are made from nonporous silica surrounded by a porous shell layer with properties similar to the porous materials used in HPLC. Triglycerides are chains of three fatty acids attached to a glycerol backbone. Note how retention increases in changing from 10% acetonitrile as a modifier to 30%. This was attributed to a decrease in solubility of the triglycerides when the proportion of polar modifier was increased. Column length was also increased by using four columns in series instead of the three columns used in Figure 29-9. The triglycerides were found to be very soluble in the CO_2 mobile phase, which was judged to be a good replacement for the organic solvents used in nonaqueous reversed phase HPLC.

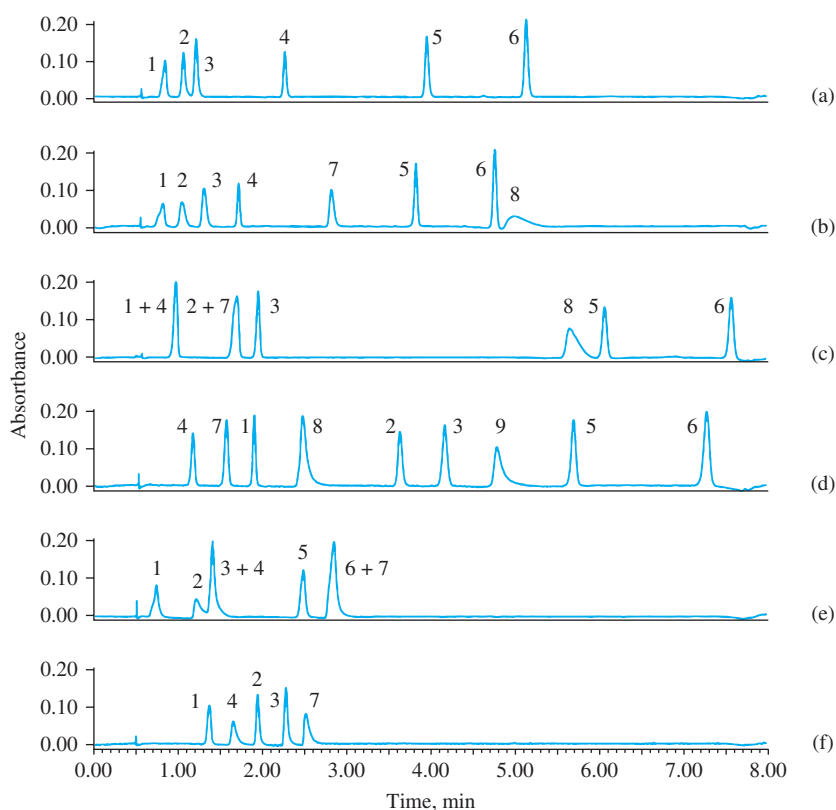


FIGURE 29-8 Separation of nine drugs by SFC on different stationary phases. Columns: 150 mm $\times$ 4.6 mm with 5- μ m packing. (a) Waters Viridis silica; (b) Waters Viridis BEH; (c) Phenomenex Luna Hilic diol; (d) Phenomenex PrincetonSFC 2-EP; (e) Waters Viridis fluorophenyl; (f) Waters X-bridge amide. Analytes: (1) ibuprofen; (2) mefenamic acid; (3) doclofenac; (4) caffeine; (5) chlorthalidane; (6) hydrochlorothiazide; (7) midazolam; (8) propranolol; (9) dopamine. SFC 150 bar pressure at 40°C. Gradient of 5–40% methanol over 9 min with CO₂. UV detection at 220 nm. (Adapted from A. Grand-Guillaume Perrenoud, J. Boccard, J. L. Veuthey, and D. Guilleme, *J. Chromatogr. A*, **2012**, 1262, 205, DOI: 10.1016/j.chroma.2012.08.091.)

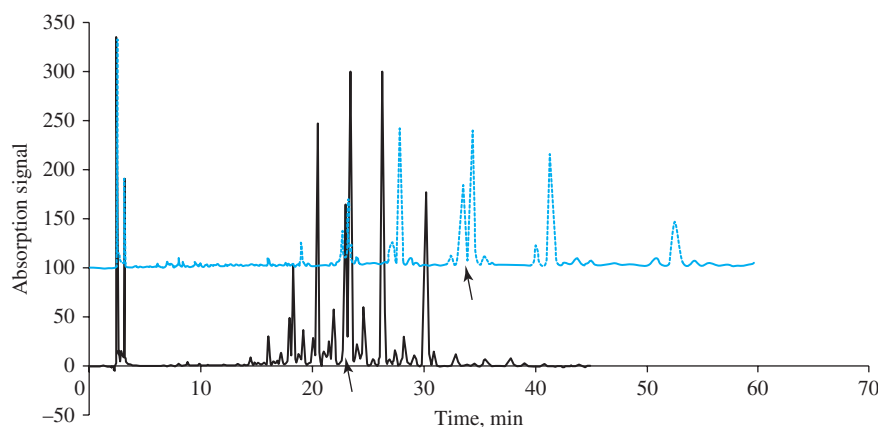


FIGURE 29-9 Chromatogram of rapeseed oil showing various triglycerides. Bottom solid line, 10% acetonitrile, upper dashed line, 30% acetonitrile. Arrows show chains PLL/LLL bottom, and POLn/OLL, where fatty acids are P = palmitic, L = linoleic, PO = palmitoleic, Ln = linolinic, and O = oleic. Column, three Phenomenex Kinetex C18 columns in series, 3 $\times$ 150 mm $\times$ 4.6 mm with 2.6- μ m packing. SFC, pressure 10 Mpa, 15°C. Mobile phase CO₂ with acetonitrile, UV detection at 210 nm. (Adapted from E. Lesellier, A. Latos, and A. L. Oliveira, *J. Chromatogr. A*, **2014**, 1327, 141, DOI: 10.1016/j.chroma.2013.12.046.)

29C SUPERCRITICAL FLUID EXTRACTION

Often, the analysis of complex materials requires as a preliminary step separation of the analyte or analytes from a sample matrix. Ideally, an analytical separation method should be rapid, simple, and inexpensive; should give quantitative recovery of analytes without loss or degradation; should yield a solution of the analyte that is sufficiently concentrated to permit the final measurement to be made without the need for concentration; and should generate little or no laboratory wastes.

For many years, one of the most common methods for performing analytical separations on complex environmental, pharmaceutical, food, and petroleum samples was based on extraction of bulk samples with hydrocarbon or chlorinated organic solvents using a Soxhlet extractor. Unfortunately, liquid extractions frequently fail to meet several of the ideal criteria listed in the previous paragraph. They usually require several hours or more to achieve satisfactory recoveries of analytes, and sometimes never do. The solvent costs are often high. The solution of the recovered analyte is often so dilute that a concentration step must follow the extraction. Analyte degradation or loss as well as atmospheric pollution may accompany this concentration step.

Solid-phase extraction, or liquid-solid extraction, can overcome several of these problems.⁹ Solid-phase extraction techniques use membranes or small, disposable, syringe-barrel columns or cartridges. A hydrophobic organic compound is coated or chemically bonded to powdered silica to form the solid extracting phase. The compounds can be nonpolar, moderately polar, or polar. For example, an octadecyl (C_{18}) bonded silica is a common packing. The functional groups bonded to the packing attract hydrophobic compounds in the sample by van der Waals interactions and extract them from the aqueous solution.

SFE is another way to avoid many of the problems that plague organic liquid extractants. Beginning in the mid-1980s, chemists began to investigate the use of supercritical fluids for separating analytes. Such extractions were shown to have many advantages as discussed in the following sections.¹⁰

⁹For more information see D. A. Skoog, D. M. West, F. J. Holler, and S. R. Crouch, *Fundamentals of Analytical Chemistry*, 9th ed., Belmont, CA: Brooks/Cole, 2014, pp. 856–857.

¹⁰See *Supercritical Fluid Extraction: Technology, Applications and Limitations*, J. Lindy, ed., Hauppauge, NY: Nova Science Publishers, 2014; *Supercritical Fluids as Solvents and Reaction Media*, G. Brunner, ed., Amsterdam: Elsevier, 2004, Chap. 4; *Practical Supercritical Fluid Chromatography and Extraction*, M. Caude and D. Thiebaut, eds., Amsterdam: Harwood, 2000; *Supercritical Fluids in Chromatography and Extraction*, R. M. Smith and S. B. Hawthorne, eds., Amsterdam: Elsevier, 1997; L. Taylor, in *Handbook of Instrumental Techniques for Analytical Chemistry*, F. Settle, ed., Chap. 11, Upper Saddle River, NJ: Prentice-Hall, 1997.

29C-1 Advantages of SFE

Some of the advantages of SFE include the following:

1. SFE is generally fast. The rate of mass transfer between a sample matrix and an extraction fluid is determined by the rate of diffusion of a species in the fluid and the viscosity of the fluid—the greater the diffusion rate and the lower the viscosity, the greater the rate of mass transfer. As we have noted earlier, both of these variables are more favorable for supercritical fluids than for typical liquid solvents. Because of this, SFE can generally be completed in 10–60 minutes, whereas extractions with an organic liquid may require several hours or even days.
2. The solvent strength of a supercritical fluid can be varied by changes in the pressure and to a lesser extent in the temperature. In contrast, the solvent strength of an organic liquid is essentially constant regardless of conditions. This property allows the conditions for extraction with a supercritical fluid to be optimized for a given class of analytes.
3. Many supercritical fluids are gases at ambient conditions. Thus, recovery of analytes is simple compared to organic liquids, which must be vaporized by heating, leading to possible decomposition of thermally unstable analytes or loss of volatile analytes. In contrast, a supercritical fluid can be separated from the analyte by simply releasing pressure. Alternatively, the analyte stream can be bubbled through a small vial containing a good solvent for the analyte, which will dissolve in the small volume of solvent.
4. Some supercritical fluids are cheap, inert, and non-toxic. Thus, they are readily disposed of after an extraction is completed by allowing them to evaporate into the atmosphere.

29C-2 Instrumentation

The instrumentation for SFE can be relatively simple as shown in Figure 29-10. Instrument components include a fluid source, most commonly a tank of carbon dioxide; a syringe pump having a pressure rating of at least 400 atm and a flow rate for the pressurized fluid of at least 2 mL/min; a valve to control the flow of the critical fluid into a heated extraction cell having a capacity of a few milliliters; and an exit valve leading to a flow restrictor that depressurizes the fluid and transfers it into a collection device. In the simplest instruments, the flow restrictor is 10–50 cm of capillary tubing. In modern sophisticated commercial instruments, the restrictors are variable and controlled manually or automatically. Commercial SFE systems are available from several instrument manufacturers.

An SFE system can be operated in one of two ways. In the dynamic extraction mode, the valve between the extraction cell and the restrictor remains open so that the sample is continually supplied with fresh supercritical fluid and the extracted

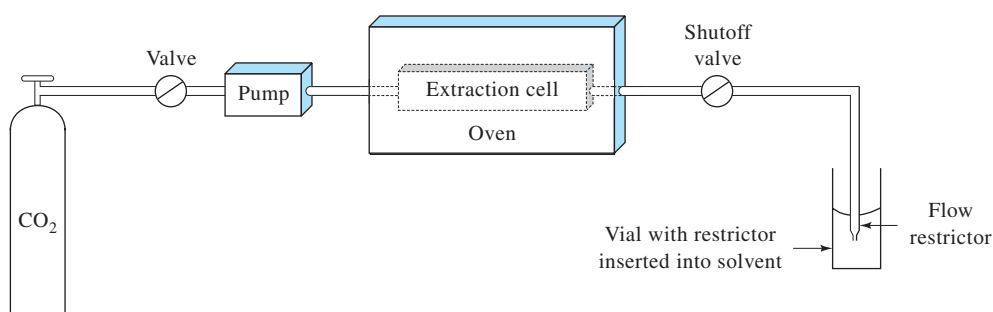


FIGURE 29-10 A typical arrangement for off-line SFE. The shutoff valve is required for static SFE but not dynamic SFE.

material flows into the collection vessel where depressurization occurs. In the static extraction mode, the valve between the extraction cell and the restrictor is closed and the extraction cell is pressurized under static conditions. After a suitable period, the exit valve is opened and the cell contents are transferred through the restrictor by a dynamic flow of fluid from the pump. The dynamic mode is more widely used than the static mode.

29C-3 Supercritical Fluid Choice

Two dozen or more supercritical fluids have been described as extraction media, but as in SFC, by far the most widely used substance is carbon dioxide alone or containing an organic modifier. The best choice of fluid is determined by a number of variables, including polarity and solubility of the analytes and the matrix components, physical nature of the matrix, concentration of the analytes, moisture content of the sample, and kinetic considerations.¹¹ Unfortunately, the current theory of SFE is not well developed, and final conditions in most cases must be determined empirically. Carbon dioxide has been the fluid of choice in most studies. It is an excellent solvent for nonpolar species, such as alkanes and terpenes, and a decent extraction medium for moderately polar species, such as polycyclic aromatic hydrocarbons, polychlorinated biphenyls, aldehydes, esters, alcohols, organic chloro pesticides, and fats. It is generally not a good extraction medium for highly polar compounds unless strongly polar modifiers such as methanol are added. Modifiers can be introduced into the extraction system either by a second pump or by injection into the sample prior to extraction.

A variety of modifiers have been used to enhance the polarity of supercritical fluid carbon dioxide, including several of the lower-molecular-mass alcohols, propylene carbonate, 2-methoxyethanol, methylene chloride, and certain organic acids. The most common modifier is methanol. Figure 29-11 demonstrates the improved efficiency achieved by the presence

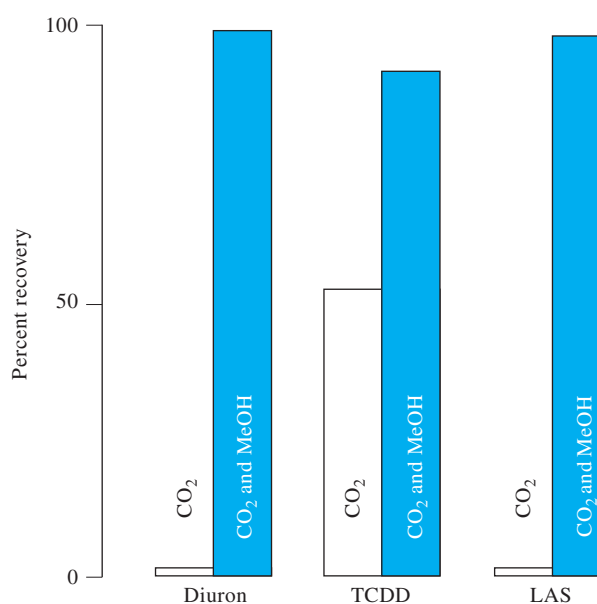


FIGURE 29-11 Comparison of extraction efficiencies obtained by using CO₂ and CO₂ modified with methanol. A soil sample was used. All extractions were for 30 min. Diuron is a common herbicide that is an aromatic substituted derivative of urea. TCDD is 2,3,7, 8-tetrachlorodibenzo-*p*-dioxin. LAS is linear alkylbenzenesulfonate detergent.

of a small amount of methanol in the extraction of various materials from soil samples.

29C-4 Off-Line and On-Line Extractions

Two types of methods have been used to collect analytes after extraction: *off-line* and *on-line*. In off-line collection, which is the simpler of the two, the analytes are collected by immersing the restrictor in a few milliliters of solvent and allowing the gaseous supercritical fluid to escape into the atmosphere (see Figure 29-10). Analytes have also been collected on adsorbents, such as silica. The adsorbed analytes are then eluted with a small volume of a liquid solvent. In either case the separated analytes are then

¹¹For a review of the effects of many of these variables, see M. E. P. McNally, *Anal. Chem.*, **1995**, 67, 308A, DOI: 10.1021/ac00105a723.

TABLE 29-3 Some Typical Analytical Applications of SFE

Material	Analyte*	Supercritical Fluid	Extraction Time, min	Off-line (1), On-line (2)
Soils	Pesticides	CO ₂	20	1
River sediments	PAHs	CO ₂ /5% MeOH	120	1
Smoke, urban dust	PAHs	CO ₂	15	2
Railroad bed soil	PCBs, PAHs	CO ₂ /MeOH	45	1
Foods	Fats	CO ₂ /MeOH	12	1
Spices, bubble gum	Aromas and fragrances	CO ₂	10	2
Serum	Cholesterol	CO ₂	30	1
Coal, fly ash	PCBs, dioxins	CO ₂	15	2
Polymers	Additives and oligomers	CO ₂	15	2
Animal tissue	Drug residues	CO ₂	9	1

*PAHs = polycyclic aromatic hydrocarbons; PCBs = polychlorinated biphenyls.

identified by any of several optical, electrochemical, or chromatographic methods.

In the on-line method, the effluent from the restrictor, after depressurization, is transferred directly to a chromatograph system. In most cases the latter is a GC or an SFC instrument, although occasionally an HPLC system has been used. The principal advantages of an on-line system are the elimination of sample handling between the extraction and the measurement and the potential for enhanced sensitivity because no dilution of the analyte occurs.

29C-5 Typical Applications of SFE

Hundreds of applications of SFE have appeared in the literature. Many applications are for the analysis of environmental samples. Others have been for the analysis of foods, plants, biomedical samples, and industrial materials. Table 29-3 provides a few typical applications of off-line and on-line SFE. In addition to the analytical uses of SFE, there are many preparative and isolation uses in the pharmaceutical and polymer industries.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

29-1 Define

- (a) critical temperature and critical pressure of a gas.
- (b) supercritical fluid.

29-2 What properties of a supercritical fluid are important in chromatography?

29-3 How do instruments for SFC differ from those for (a) HPLC and (b) GC?

29-4 Describe the effect of pressure on supercritical fluid chromatograms.

29-5 List some of the advantageous properties of supercritical CO₂ as a mobile phase for chromatographic separations compared to HPLC mobile phases.

29-6 Compare SFC with other column chromatographic methods.

>> QUESTIONS AND PROBLEMS (continued)

- 29-7** For supercritical carbon dioxide, predict the effect that the following changes will have on the elution time in an SFC experiment:
- (a) Increasing the flow rate (at constant temperature and pressure)
 - (b) Increasing the pressure (at constant temperature and flow rate)
 - (c) Increasing the temperature (at constant pressure and flow rate)
- 29-8** For SFE, differentiate between
- (a) on-line and off-line processes.
 - (b) static and dynamic extractions.
- 29-9** List the advantages and any disadvantages of SFE compared to liquid-liquid extractions.
- 29-10** How are analytes usually recovered after an SFE?

Challenge Problem

- 29-11** In an interesting paper, Zheng and coworkers (J. Zheng, L. T. Taylor, J. D. Pinkston, and M. L. Mangels, *J. Chromatogr. A*, **2005**, 1082, 220, DOI: 10.1016/j.chroma.2005.04.086) discuss the elution of polar and ionic compounds in SFC.
- (a) Why are highly polar or ionic compounds usually not eluted in SFC?
 - (b) What types of mobile-phase additives have been used to improve the elution of highly polar or ionic compounds?
 - (c) Why is ion-pairing SFC not often used?
 - (d) Why are ammonium salts sometimes added as mobile-phase modifiers in SFC?
 - (e) The authors describe an SFC system that uses mass spectrometry (MS) as a detector. Discuss the interfacing of an SFC unit to a mass spectrometer. Compare the compatibility of SFC with MS to that of HPLC and GC with MS.
 - (f) The authors studied the effect of column outlet pressure on the elution of sodium 4-dodecylbenzene sulfonate on three different stationary phases with five mobile-phase additives. What effect was observed, and what was the explanation for the effect?
 - (g) What elution mechanisms were considered by the authors?
 - (h) Which mobile-phase additive gave the fastest elution of the sulfonate salts? Which provided the longest retention times?
 - (i) Did a silica column give results similar to or different from a cyano bonded-phase column?

Gas Chromatography

In **gas chromatography**, the components of a vaporized sample are separated as a result of differential distribution between a mobile gaseous phase and a liquid or a solid stationary phase held in a column. In performing a gas chromatographic separation, the sample is vaporized and injected onto the head of a chromatographic column. Elution is brought about by the flow of an inert gaseous mobile phase. In contrast to most other types of chromatography, the mobile phase does not interact with molecules of the analyte; its only function is to transport the analyte through the column.

There are two types of gas chromatography: *gas-liquid chromatography* (GLC) and *gas-solid chromatography* (GSC). GLC is used throughout all fields of science; its name is usually shortened to *gas chromatography* (GC).¹ GSC is based on a solid stationary phase in which retention of analytes occurs because of physical adsorption. The application of GSC is limited because of semipermanent retention of active or polar molecules and severe tailing of elution peaks. Tailing is a result of the non-linear nature of the adsorption process. Thus, this technique is not widely used except for the separation of certain low-molecular-mass gaseous species; we discuss the method briefly in Section 27F.

In GLC the analyte is partitioned between a gaseous mobile phase and a liquid phase immobilized on the surface of an inert solid packing or on the walls of a capillary tubing. The concept of GLC was first suggested in 1941 by Martin and Synge, who were also responsible for the development of liquid-liquid partition chromatography. More than a decade was to elapse, however, before the value of GLC was demonstrated experimentally² and this technique began to be used as a routine laboratory tool. In 1955 the first commercial apparatus for GLC appeared on the market. Since that time, the growth in applications of this technique has been phenomenal. Currently, nearly a million gas chromatographs are in use throughout the world.

27A PRINCIPLES OF GLC

The general principles of chromatography, which were developed in Chapter 26, and the mathematical relationships summarized in Section 26E are applicable to GC with only minor modifications that arise from the compressibility of gaseous mobile phases.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

¹For detailed treatment of GC, see C. Poole, ed., *Gas Chromatography*, Amsterdam: Elsevier, 2012; H. M. McNair and J. M. Miller, *Basic Gas Chromatography*, 2nd ed., Hoboken, NJ: Wiley, 2009; R. L. Grob and E. F. Barry, eds., *Modern Practice of Gas Chromatography*, 4th ed., New York: Wiley-Interscience, 2004; R. P. W. Scott, *Introduction to Analytical Gas Chromatography*, 2nd ed., New York: Marcel Dekker, 1997; W. Jennings, E. Mittlefehldt, and P. Strempel, *Analytical Gas Chromatography*, 2nd ed., Orlando, FL: Academic Press, 1997.

²A. T. James and A. J. P. Martin, *Analyst*, **1952**, 77, 915–932, DOI: 10.1039/AN9527700915.

27A-1 Retention Volumes

To take into account the effects of pressure and temperature in GC, it is often useful to use retention volumes rather than the retention times that were discussed in Section 26B. The relationship between the two is given in Equations 27-1 and 27-2

$$V_R = t_R F \quad (27-1)$$

$$V_M = t_M F \quad (27-2)$$

where F is the average volumetric flow rate within the column; V and t are retention volumes and times, respectively; and the subscripts R and M refer to species that are retained and not retained on the column. The flow rate within the column is not directly measurable. Instead, the rate of gas flow as it exits the column is determined experimentally with a flow meter, which is discussed in Section 27B. For popular soap-bubble-type flow meters, where the gas is saturated with water, the average flow rate F is related to the measured flow rate F_m by

$$F = F_m \times \frac{T_c}{T} \times \frac{(P - P_{H_2O})}{P} \quad (27-3)$$

where T_c is the column temperature in kelvins, T is the temperature at the flow meter, and P is the gas pressure at the end of the column. Usually P and T are the ambient pressure and temperature. The term involving the vapor pressure of water, P_{H_2O} , is a correction for the pressure used when the gas is saturated with water.

Both V_R and V_M depend on the average pressure within the column—a quantity that lies intermediate between the inlet pressure P_i and the outlet pressure P (atmospheric pressure). The *pressure drop correction factor* j , also known as the *compressibility factor*, accounts for the pressure within the column being a nonlinear function of the P_i/P ratio. Corrected retention volumes V_R^0 and V_M^0 , which correspond to volumes at the average column pressure, are obtained from the relationships

$$V_R^0 = j t_R F \quad \text{and} \quad V_M^0 = j t_M F \quad (27-4)$$

where j can be calculated from the relationship

$$j = \frac{3[(P_i/P)^2 - 1]}{2[(P_i/P)^3 - 1]} \quad (27-5)$$

The *specific retention volume* V_g is then defined as

$$V_g = \frac{V_R^0 - V_M^0}{m_s} \times \frac{273}{T_c} = \frac{jF(t_R - t_M)}{m_s} \times \frac{273}{T_c} \quad (27-6)$$

where m_s is the mass of the stationary phase, a quantity determined at the time of column preparation.

27A-2 Relationship between V_g and K

The specific retention volume V_g can be related to the distribution constant K_c . To do so, we substitute the expression relating t_R and t_M to k (Equation 26-12) into Equation 27-6, which gives

$$V_g = \frac{jF t_M k}{m_s} \times \frac{273}{T_c}$$

Combining this expression with Equation 27-4 yields

$$V_g = \frac{V_M^0 k}{m_s} \times \frac{273}{T_c}$$

Substituting Equation 26-9 for k gives (here, V_M^0 and V_M are identical)

$$V_g = \frac{K V_s}{m_s} \times \frac{273}{T_c}$$

The density of the liquid on the stationary phase ρ_s is given by

$$\rho_s = \frac{m_s}{V_s}$$

where V_s is the stationary-phase volume. Thus,

$$V_g = \frac{K}{\rho_s} \times \frac{273}{T_c} \quad (27-7)$$

Note that V_g at a given temperature depends only on the distribution constant of the solute and the density of the liquid making up the stationary phase.

27A-3 Effect of Mobile-Phase Flow Rate

Equation 26-23 and the relationships shown in Table 26-3 are fully applicable to GC. The longitudinal diffusion term (B/u) is more important in GLC, however, than in other chromatographic processes because of the much larger diffusion rates in gases (10^4 – 10^5 times greater than liquids). As a result, the minima in curves relating plate height H to flow rate (van Deemter plots) are usually considerably broadened in GC (see Figure 26-8).

27B INSTRUMENTS FOR GLC

Many changes and improvements in gas chromatographic instruments have appeared in the marketplace since their commercial introduction. In the 1970s, electronic integrators and computer-based data-processing equipment became common. The 1980s saw computers being used for automatic control of most instrument variables, such as column temperature, flow rates, and sample injection. In addition, very high-performance instruments became available at moderate costs, and perhaps most important, open tubular columns were developed capable of separating components of complex mixtures in relatively short times. Today, more than 50 instrument manufacturers offer several hundred different models of gas chromatographic equipment at costs that vary from about \$1000 to more than \$50,000.

The basic components of a typical instrument for performing GC are shown in Figure 27-1. A description of each component follows.



Exercise: Learn more about **gas chromatography** at www.tinyurl.com/skoopia7

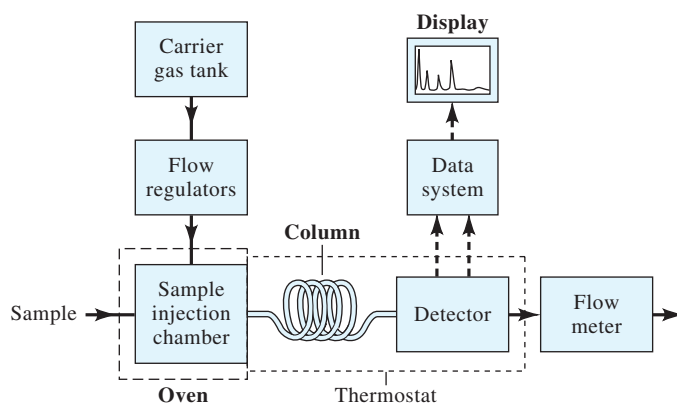


FIGURE 27-1 Block diagram of a typical gas chromatograph.

27B-1 Carrier Gas System

The mobile-phase gas in GC is called the *carrier gas* and must be chemically inert. Helium is the most common mobile-phase gas used, although argon, nitrogen, and hydrogen are also used. These gases are available in pressurized tanks. Pressure regulators, gauges, and flow meters are required to control the flow rate of the gas. In addition, the carrier gas system often contains a molecular sieve to remove impurities and water.

Flow rates are normally controlled by a two-stage pressure regulator at the gas cylinder and some sort of pressure regulator or flow regulator mounted in the chromatograph. Inlet pressures usually range from 10 to 50 psi (lb/in.²) above room pressure, which lead to flow rates of 25 to 150 mL/min with packed columns and 1 to 25 mL/min for open tubular capillary columns. Generally, it is assumed that flow rates are constant if the inlet pressure remains constant. Flow rates can be established by a rotometer at the column head; this device, however, is not as accurate as the simple soap-bubble meter shown in Figure 27-2. Usually, the flow meter is located at the end of the column as shown in Figure 27-1. A soap film is formed in the path of the gas when a rubber bulb containing an aqueous solution of soap or detergent is squeezed; the time required for this film to move between two graduations on the buret is measured and converted to volumetric flow rate (see Figure 27-2). Note that volumetric flow rates and linear flow velocities are related by Equation 26-6 or 26-7. Most modern computer-controlled gas chromatographs are equipped with electronic flow meters that can be regulated to maintain the flow rate at the desired level.

27B-2 Sample Injection Systems

To achieve high column efficiency, the sample must be of a suitable size and introduced as a “plug” of vapor; slow injection or oversize samples cause band spreading and poor resolution. Calibrated microsyringes, such as those shown in Figure 27-3, are often used to inject liquid samples through a rubber or silicone diaphragm, or septum, into a heated sample port located at the



FIGURE 27-2 A soap-bubble flow meter. (Courtesy Agilent Technologies.)

head of the column. The sample port (Figure 27-4) is ordinarily about 50°C above the boiling point of the least volatile component of the sample. For ordinary packed analytical columns, sample sizes range from a few tenths of a microliter to 20 μ L. Capillary columns require samples that are smaller by a factor of 100 or more. With these columns, a *sample splitter* is often needed to deliver a small known fraction (1:50 to 1:500) of the injected sample, with the remainder going to waste. Commercial gas chromatographs intended for use with capillary columns incorporate such splitters, and they also allow for *splitless injection* to improve sensitivity or for use with packed columns. With splitless inlets, the purge valve closes at injection and stays closed for 30–60 seconds. During this time the sample vapor can go



FIGURE 27-3 A set of microsyringes for sample injection. (Courtesy Hamilton Company.)

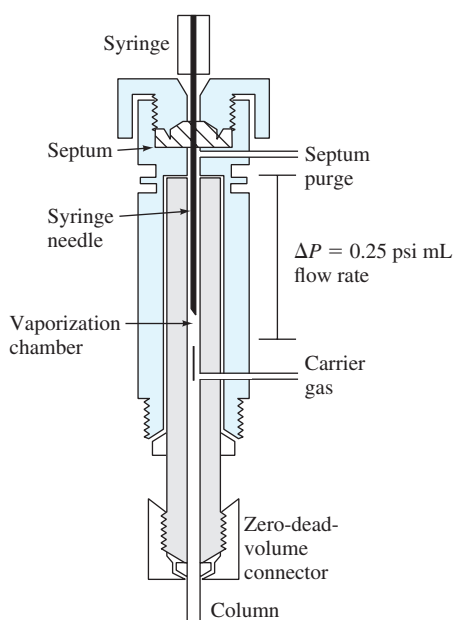


FIGURE 27-4 Cross-sectional view of a microflash vaporizer direct injector.

only onto the column. When the purge valve opens, any remaining vapor is rapidly vented. *On-column inlets* are also available for capillary GC. With this type of inlet, the entire sample is injected onto the column as a liquid, which is later vaporized by temperature programming of the column or inlet. With on-column inlets, the analyte is separated from the solvent by thermal and solvent effects.³

For the most reproducible sample injection, newer gas chromatographs use autoinjectors and autosamplers, such as the system shown in Figure 27-5. With such autoinjectors, syringes are filled, and the sample injected into the chromatograph auto-

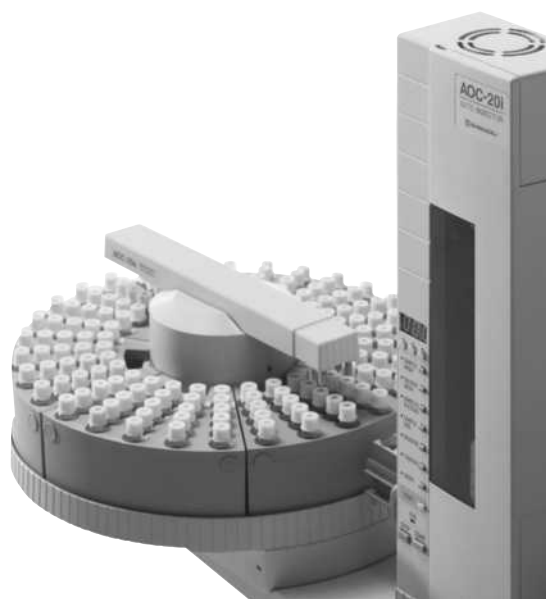


FIGURE 27-5 An autoinjection system with autosampler for gas chromatography. (Shimadzu Corp.)

matically. In the autosampler, samples are contained in vials on a sample turntable. The autoinjector syringe picks up the sample through a septum on the vial and injects the sample through a septum on the chromatograph. With the unit shown, up to 150 sample vials can be placed on the turntable. Injection volumes can vary from 0.1 μL with a 10- μL syringe to 200 μL with a 200- μL syringe. The reproducibility of autoinjection systems is quite good with standard deviations of replicate samples often as low as 0.3%.

For introducing gases, a sample valve, such as that shown in Figure 27-6, is often used instead of a syringe. With such devices, sample sizes can be reproduced to better than 0.5% relative. Liquid samples can also be introduced through a sampling valve. Solid samples are introduced as solutions or alternatively are sealed into thin-walled vials that can be inserted at the head of the column and punctured or crushed from the outside.

27B-3 Column Configurations and Column Ovens

Two general types of columns are used in GC, *packed columns* and *capillary columns*. In the past, the vast majority of gas chromatographic analyses used packed columns. For most current applications, packed columns have been replaced by the more efficient capillary columns.

Chromatographic columns vary in length from less than 2 to 60 m or more. They are constructed of stainless steel, glass, fused silica, or Teflon. In order to fit into an oven for thermostating, they are usually formed as coils having diameters of 10–30 cm (see Figure 27-7). A detailed discussion of columns, column packings, and stationary phases is found in Section 27C.

³N. H. Snow, in *Modern Practice of Gas Chromatography*, R. L. Grob and E. F. Barry, eds., 4th ed., Chap. 9, New York: Wiley-Interscience, 2004.

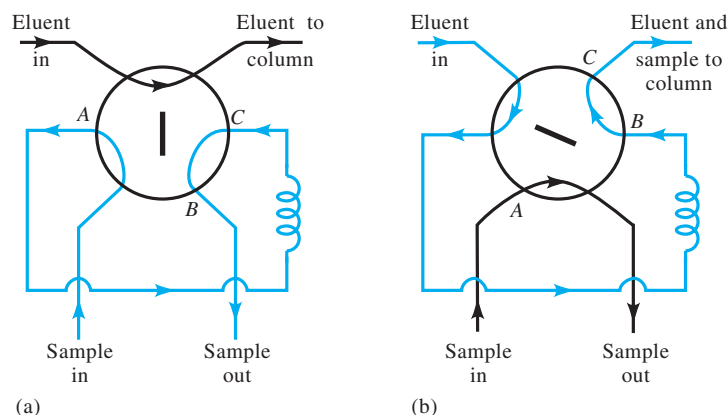


FIGURE 27-6 A rotary sample valve: valve position (a) is for filling the sample loop ACB; position (b) is for introduction of sample into column.



FIGURE 27-7 Fused-silica capillary columns. (Courtesy of Restek Corp., Bellefonte, PA.)

Column temperature is an important variable that must be controlled to a few tenths of a degree for precise work. Thus, the column is ordinarily housed in a thermostatted oven. The optimal column temperature depends on the boiling point of the sample and the degree of separation required. Roughly, a temperature equal to or slightly above the average boiling point of a sample results in a reasonable elution time (2 to 30 min). For samples with a broad boiling range, it is often desirable to take advantage of *temperature programming*, in which the column temperature is increased either continuously or in steps as the

separation proceeds. Figure 27-8 shows the improvement in a chromatogram brought about by temperature programming.

In general, optimal resolution is associated with minimal temperature; the cost of lowered temperature, however, is an increase in elution time and therefore the time required to complete an analysis. Figures 27-8a and 27-8b illustrate this principle.

Analytes of limited volatility can sometimes be determined by forming derivatives that are more volatile. Likewise, derivatization is sometimes used to enhance detection or chromatographic performance.

27B-4 Gas Chromatographic Detectors

Dozens of detectors have been investigated and used with gas chromatographic separations.⁴ We first describe the characteristics that are most desirable in a gas chromatographic detector and then discuss the most widely used devices. In some cases, gas chromatographs are coupled to spectroscopic instruments such as mass and infrared spectrometers. With such systems, the spectral device not only detects the appearance of the analytes as they elute from the column but also helps to identify them.

Characteristics of the Ideal Detector

The ideal detector for GC has the following characteristics:

1. Adequate sensitivity. Just what constitutes adequate sensitivity cannot be described in quantitative terms. For example, the sensitivities of the detectors described in this section vary by a factor of 10^7 . Yet all are widely used and are adequate for certain tasks; the least sensitive are not, however, satisfactory for certain applications. In general, the sensitivities of present-day detectors lie in the range of 10^{-8} to 10^{-15} g solute/s.

⁴See L. A. Colon and L. J. Baird, in *Modern Practice of Gas Chromatography*, R. L. Grob and E. F. Barry, eds., 4th ed., Chap. 6, New York: Wiley-Interscience, 2004.

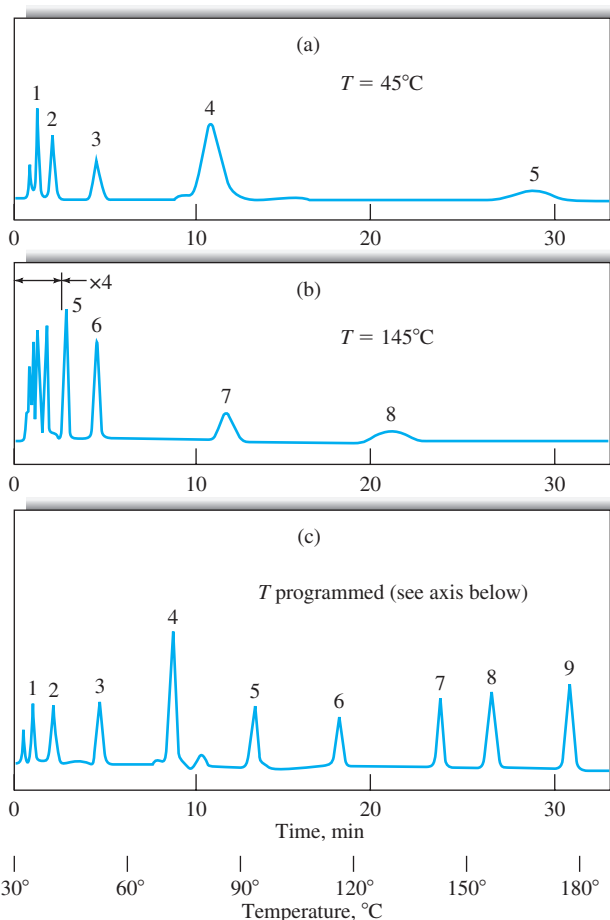


FIGURE 27-8 Effect of temperature on gas chromatograms: (a) isothermal at 45°C, (b) isothermal at 145°C, and (c) programmed at 30–180°C. (From W. E. Harris and H. W. Habgood, *Programmed Temperature Gas Chromatography*, New York: Wiley, 1966, p. 10. Reprinted with permission.)

2. Good stability and reproducibility.
3. A linear response to solutes that extends over several orders of magnitude.
4. A temperature range from room temperature to at least 400°C.
5. A short response time independent of flow rate.
6. High reliability and ease of use. The detector should be foolproof in the hands of inexperienced operators, if possible.
7. Similarity in response toward all solutes or alternatively a highly predictable and selective response toward one or more classes of solutes.
8. The detector should be nondestructive.

Needless to say, no detector exhibits all of these characteristics. Some of the most common detectors are listed in Table 27-1. Several of the most widely used detectors are described in the paragraphs that follow.

Flame Ionization Detectors

The flame ionization detector (FID) is the most widely used and generally applicable detector for GC. With an FID such as the one shown in Figure 27-9, effluent from the column is directed into a small air-hydrogen flame. Most organic compounds produce ions and electrons when pyrolyzed at the temperature of an air-hydrogen flame. Detection involves monitoring the current produced by collecting these charge carriers. A potential difference of a few hundred volts applied between the burner tip and a collector electrode located above the flame causes the ions and electrons to move toward the collector. The resulting current ($\sim 10^{-12}$ A) is then measured with a sensitive picoammeter.

The ionization of carbon compounds in the FID is not fully understood, although the number of ions produced is roughly

TABLE 27-1 Typical Gas Chromatographic Detectors

Type	Applicable Samples	Typical Detection Limit
Flame ionization	Hydrocarbons	1 pg/s
Thermal conductivity	Universal detector	500 pg/mL
Electron capture	Halogenated compounds	5 fg/s
Mass spectrometer (MS)	Tunable for any species	0.25–100 pg
Thermionic	Nitrogen and phosphorous compounds	0.1 pg/s (P), 1 pg/s (N)
Electrolytic conductivity (Hall)	Compounds containing halogens, sulfur, or nitrogen	0.5 pg Cl/s, 2 pg S/s, 4 pg N/s
Photoionization	Compounds ionized by UV radiation	2 pg C/s
Fourier transform IR (FTIR)	Organic compounds	0.2–40 ng

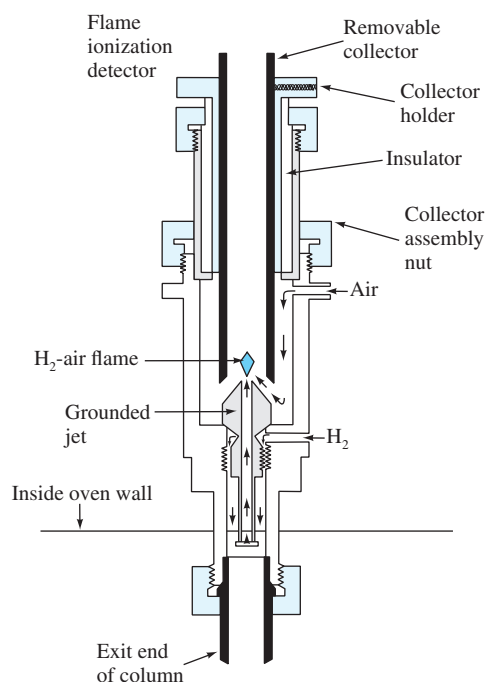


FIGURE 27-9 A typical flame ionization detector. (Courtesy of Agilent Technologies.)

proportional to the number of *reduced* carbon atoms in the flame. Because the FID responds to the number of carbon atoms entering the detector per unit of time, it is a *mass-sensitive* rather than a *concentration-sensitive* device. As such, this detector has the advantage that changes in flow rate of the mobile phase have little effect on detector response.

Functional groups, such as carbonyl, alcohol, halogen, and amine, yield fewer ions or none at all in a flame. In addition, the detector is insensitive toward noncombustible gases such as H_2O , CO_2 , SO_2 , CO , noble gases, and NO_x . These properties make the FID a most useful general detector for the analysis of most organic samples, including those contaminated with water and the oxides of nitrogen and sulfur.

The FID exhibits a high sensitivity ($\sim 10^{-13}$ g/s), large linear response range ($\sim 10^7$), and low noise. It is generally rugged and easy to use. Disadvantages of the FID are that it destroys the sample during the combustion step and requires additional gases and controllers.

Thermal Conductivity Detectors

The *thermal conductivity detector* (TCD), which was one of the earliest detectors for GC, is still widely used. This device contains an electrically heated source whose temperature at constant electrical power depends on the thermal conductivity of the surrounding gas. The heated element may be a fine platinum, gold, or tungsten wire or, alternatively, a small thermistor. The electrical resistance of this element depends on the thermal conductivity of the gas. Figure 27-10a shows a cross-sectional view of one of the temperature-sensitive elements in a TCD.

Twin detectors are usually used, one being located ahead of the sample-injection chamber and the other immediately beyond the column. The detector elements are labeled *sample* and *reference* in Figure 27-10b. Alternatively, the gas stream can be split. The detectors are incorporated in two arms of a bridge circuit. The bridge circuit is arranged so that the thermal conductivity of the carrier gas is canceled. In addition, the effects of variations in temperature, pressure, and electrical power are minimized.

Modulated single-filament TCDs are also available. Here, the analytical and reference gases are passed alternately over a tiny filament held in a low-volume ($\sim 5\text{-}\mu\text{L}$) cell. The gases are switched at a frequency of 10 Hz. The output is thus a 10-Hz signal whose amplitude is proportional to the difference in thermal conductivity of the analytical and reference gases. Because the amplifier responds only to a 10-Hz signal, thermal noise in the system is largely eliminated.

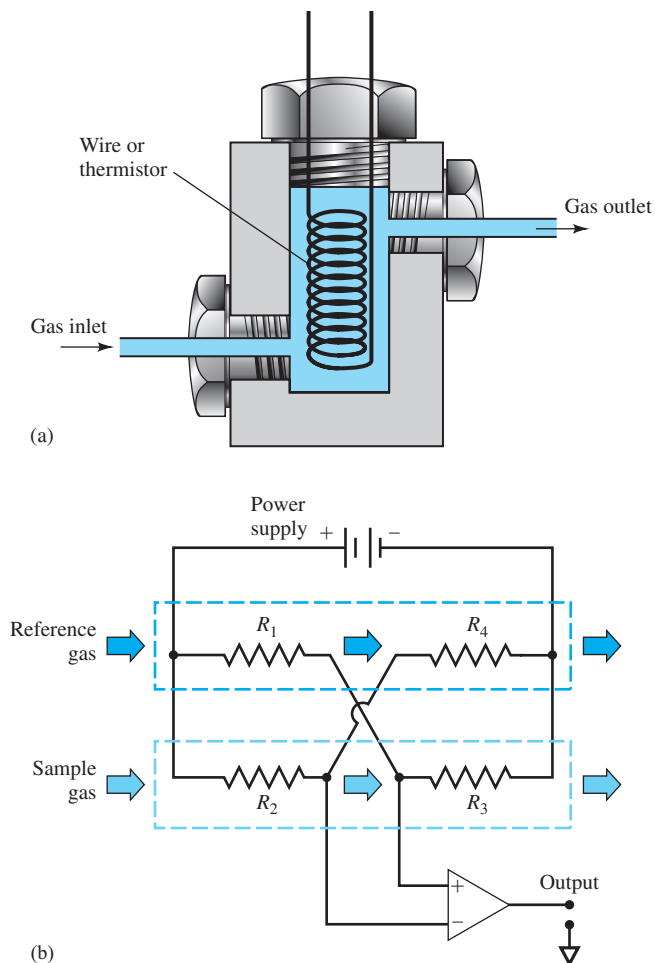


FIGURE 27-10 Schematic of (a) a thermal conductivity detector cell and (b) an arrangement of two sample detector cells (R_2 and R_3) and two reference detector cells (R_1 and R_4). (Adapted from F. Rastrelloa, P. Placidi, A. Scorzonia, E. Cozzanib, M. Messinab, I. Elmib, S. Zampolli, and G. C. Cardinali, *Sensors and Actuators A*, **2012**, 178, 49, DOI: 10.1016/j.sna.2012.02.008.)

The thermal conductivities of helium and hydrogen are roughly six to ten times greater than those of most organic compounds. Thus, even small amounts of organic species cause relatively large decreases in the thermal conductivity of the column effluent, which results in a marked rise in the temperature of the detector. Detection by thermal conductivity is less satisfactory with carrier gases whose conductivities closely resemble those of most sample components.

The advantages of the TCD are its simplicity, its large linear dynamic range ($\sim 10^5$), its general response to both organic and inorganic species, and its nondestructive character, which permits collection of solutes after detection. Its chief limitation is its relatively low sensitivity ($\sim 10^{-8}$ g solute/mL carrier gas). Other detectors exceed this sensitivity by factors of 10^4 to 10^7 . It should be noted that the low sensitivity of TCDs often precludes their use with capillary columns where sample amounts are very small.

Electron-Capture Detectors

The electron-capture detector (ECD) has become one of the most widely used detectors for environmental samples because it selectively responds to halogen-containing organic compounds, such as pesticides and polychlorinated biphenyls. As shown in Figure 27-11, the sample eluate from a column is passed over a radioactive β emitter, usually nickel-63. An electron from the emitter causes ionization of the carrier gas (often nitrogen) and the production of a burst of electrons. In the absence of organic species, a constant standing current between a pair of electrodes results from this ionization process. The current decreases significantly, however, in the presence of organic molecules containing electronegative functional groups that tend to capture electrons.

The ECD is selective in its response. Compounds such as halogens, peroxides, quinones, and nitro groups are detected with high sensitivity. The detector is insensitive to functional groups such as amines, alcohols, and hydrocarbons. An important

application of the ECD is for the detection and quantitative determination of chlorinated insecticides.

ECDs are highly sensitive and have the advantage of not altering the sample significantly (in contrast to the FID, which consumes the sample). The linear response of the detector, however, is limited to about two orders of magnitude.

Thermionic Detectors

The thermionic detector is selective toward organic compounds containing phosphorus and nitrogen. Its response to a phosphorus atom is approximately 10 times greater than to a nitrogen atom and 10^4 to 10^6 times larger than to a carbon atom. Compared with the FID, the thermionic detector is approximately 500 times more sensitive to phosphorus-containing compounds and 50 times more sensitive to nitrogen-bearing species. These properties make thermionic detection particularly useful for sensing and determining the many phosphorus-containing pesticides.

A thermionic detector is similar in structure to the FID shown in Figure 27-9. The column effluent is mixed with hydrogen, passes through the flame tip assembly, and is ignited. The hot gas then flows around an electrically heated rubidium silicate bead, which is maintained at about 180 V with respect to the collector. The heated bead forms a plasma having a temperature of 600–800°C. Exactly what occurs in the plasma to produce unusually large numbers of ions from phosphorus- or nitrogen-containing molecules is not fully understood; but large ion currents result, which are useful for determining compounds containing these two elements.

Electrolytic Conductivity Detectors

In the Hall electrolytic conductivity detector, compounds containing halogens, sulfur, or nitrogen are mixed with a reaction gas in a small reactor tube, usually made of nickel. The reaction tube is kept at 850–1000°C. The products are then dissolved in a liquid, which produces a conductive solution. The

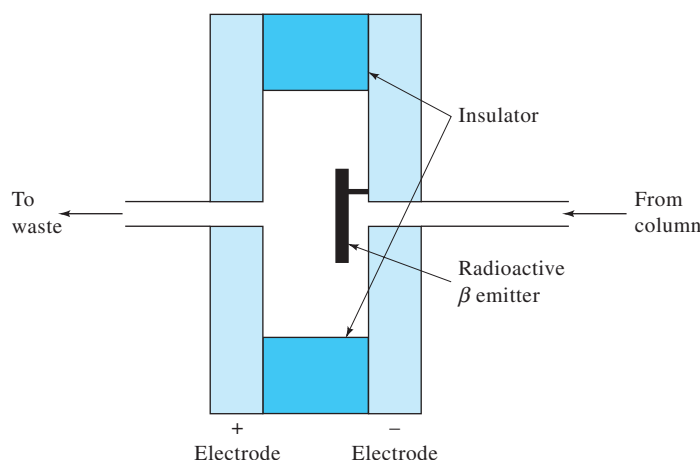


FIGURE 27-11 Schematic diagram of an ECD.

change in conductivity as a result of the ionic species in the conductance cell is then measured. A typical detector is illustrated in Figure 27-12.

In the halogen mode, hydrogen is used as the reaction gas. Halogen-containing compounds are converted to HX and dissolved in *n*-propyl alcohol as the conductivity solvent. In this mode, sulfur-containing compounds are converted to H₂S and nitrogen-containing compounds to NH₃, which do not give significant responses because both are poorly ionized in the solvent. The limit of detection is ~ 0.5 pg Cl/s, and the linear range is 10^6 .

In the sulfur mode, the reaction gas is air, which converts the sulfur-containing compounds to SO₂. The conductivity solvent is methyl alcohol with a small amount of water. The SO₂ in the presence of water is converted to sulfite and sulfate ions. Nitrogen-containing compounds are converted to N₂ and nitrogen oxides and show little or no response. Halogen-containing compounds are converted to HX and must be removed with a postreaction scrubber prior to detection. In the sulfur mode, approximately 2 pg S/s can be detected with a linear range of three orders of magnitude.

In the nitrogen mode, hydrogen is used as the reaction gas, as in the halogen mode. However, here, water containing a small amount of an organic solvent is used as the conductivity

solvent. In this solvent, the NH₃ produced is converted to NH₄⁺. The HX and H₂S produced from halogen- and sulfur-containing compounds must be removed with a postreaction scrubber. The limit of detection is ~ 4 pg N/s with a linear range of three orders of magnitude.

Dry electrolytic conductivity detectors are also available. These differ from the conventional detectors in that they do not use a solvent but instead detect the product ions in the gas phase. The dry detector is responsive to chlorine- and bromine-containing compounds. It can be used in series with an FID.

Photoionization Detector

In the photoionization detector, molecules eluting from the GC column are photoionized by ultraviolet radiation from a 10.2 eV hydrogen or a 11.7 eV argon lamp. This source ionizes species with an ionization potential below the lamp energy. Compounds with a higher ionization potential do not absorb the energy and thus are not detected. The ions and electrons produced by photoionization are then collected at a pair of biased electrodes. The detector is most sensitive for aromatic hydrocarbons and organosulfur or organophosphorus compounds that are easily photoionized. The linear range is as high as seven orders of magnitude.

Atomic Emission Detectors

In the atomic emission detector (AED), the effluent from the GC column is introduced into a microwave-induced plasma (MIP), an inductively coupled plasma (ICP), or a direct current plasma (DCP). The MIP has been most widely used and is available commercially. The MIP is used in conjunction with a diode array or charge-coupled-device atomic emission spectrometer as shown in Figure 27-13. The plasma is sufficiently energetic to atomize all of the elements in a sample and to excite their characteristic atomic emission spectra. Hence, the AED is an *element-selective detector*. As shown on the right of the figure, the positionable diode array is capable of monitoring simultaneously several elements at any given setting.

Figure 27-14 illustrates the power of element-selective detection. The sample in this case consisted of a gasoline containing a small concentration of methyl tertiary butyl ether (MTBE), an antiknock agent, as well as several aliphatic alcohols in low concentrations. The upper chromatogram, obtained by monitoring the carbon emission line at 198 nm, consists of a myriad of peaks that would be very difficult to sort out and identify. In contrast, when the oxygen line at 777 nm is used to record the chromatogram (Figure 27-14b), peaks for the alcohols and for MTBE are evident and readily identifiable.

Flame Photometric Detector

The *flame photometric detector* (FPD) has been widely applied to the analysis of air and water pollutants, pesticides, and coal hydrogenation products. It is a selective detector that is primarily responsive to compounds containing sulfur and phosphorus. In this detector, the eluent is passed into a low-temperature hydrogen-air flame, which converts some of the phosphorus to

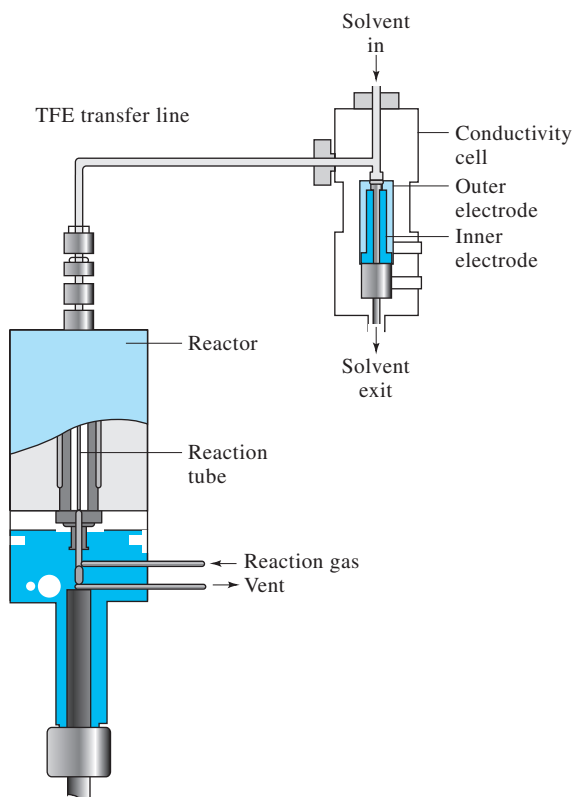


FIGURE 27-12 Diagram of a Hall electrolytic conductivity detector. (Courtesy of ThermoElectron Corp.)

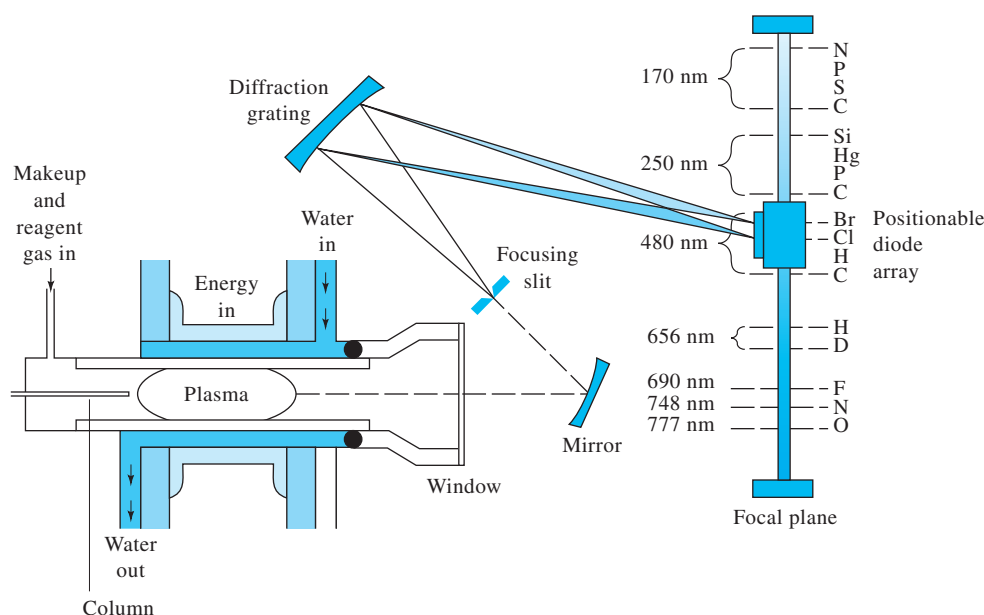
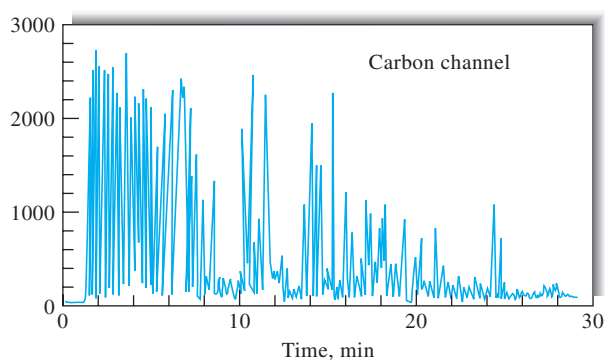
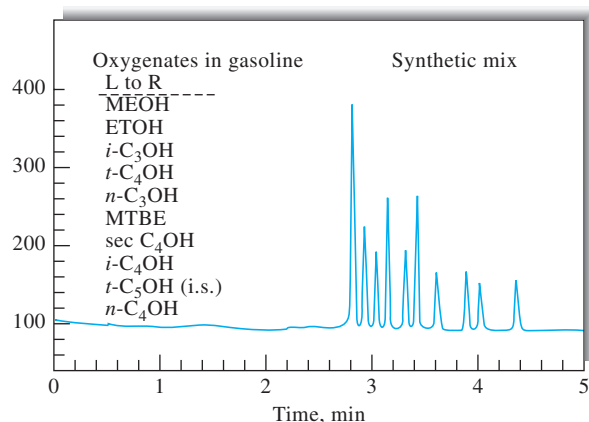


FIGURE 27-13 An AED for GC. (Courtesy of Agilent Technologies.)



(a)



(b)

FIGURE 27-14 Chromatogram for a gasoline sample containing a small amount of MTBE and several aliphatic alcohols. (a) Monitoring a carbon emission line and (b) monitoring an oxygen emission line. (Courtesy of Agilent Technologies.)

an HPO species that emits bands of radiation centered at about 510 and 526 nm. Sulfur in the sample is simultaneously converted to S_2 , which emits a band centered at 394 nm. However, the sulfur chemiluminescence detector discussed later in this section provides lower detection limits and wider linear working range than does the FPD. Suitable filters are used to isolate the appropriate bands, and their intensity is recorded photometrically. Other elements that have been detected by flame photometry include the halogens, nitrogen, and several metals, such as tin, chromium, selenium, and germanium.

Mass Spectrometry Detectors

One of the most powerful detectors for GC is the mass spectrometer. Mass spectrometers and applications of mass spectrometry are discussed in Chapters 11 and 20. The combination of GC with mass spectrometry is known as *GC/MS*.⁵ As discussed in Chapter 11, a mass spectrometer measures the mass-to-charge ratio (m/z) of ions that have been produced from the sample. Most of the ions produced are singly charged ($z = 1$) so that mass spectrometrists often speak of measuring the mass of ions when mass-to-charge ratio is actually measured. Currently, nearly fifty instrument companies offer GC/MS equipment. The flow rate from capillary columns is generally low enough that the column output can be fed directly into the ionization chamber of the mass spectrometer. A schematic of a typical system is shown in Figure 27-15. Prior to the advent of capillary columns in GC,

⁵For additional information, see O. D. Sparkman, Z. E. Penton, and F. G. Kitson, *Gas Chromatography and Mass Spectrometry*, 2nd ed., Amsterdam: Elsevier, 2011; M. C. McMaster, *GC/MS: A Practical User's Guide*, 2nd ed., New York: Wiley, 2008.

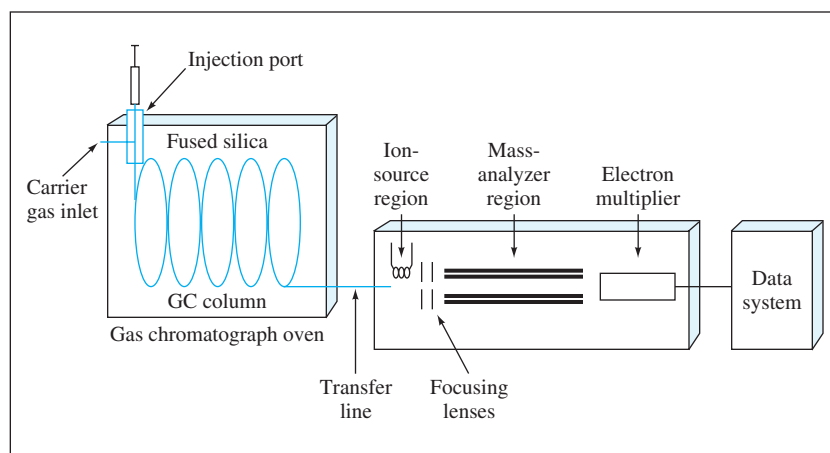


FIGURE 27-15 Schematic of a typical capillary GC/MS system. The effluent from the GC is passed into the inlet of the mass spectrometer, where the molecules in the gas are fragmented, ionized, analyzed, and detected.

when packed columns were used, it was necessary to minimize the large volumes of carrier gas eluting from the GC. Various jet, membrane, and effusion separators were used for this purpose. However, in many cases such devices also removed a significant amount of the analyte and were thus quite inefficient. Currently, capillary columns are invariably used in GC/MS instruments, and such separators are no longer needed.

Thermal degradation of components can be a difficulty in GC/MS. Not only can the GC injection port and GC column cause degradation but also the heated metal surfaces in the mass spectrometer ion source may cause problems. Lowering the temperature can minimize degradation. Often, however, the mass spectrometer can be used to identify decomposition products, which can lead to chromatographic modifications that solve the degradation problem.

The most common ion sources used in GC/MS are electron ionization and chemical ionization. Ion sources for mass spectrometry are discussed in detail in Section 20B. The most common mass analyzers are quadrupole and ion-trap analyzers. These analyzers are described in Sections 11B-2 and 20C-3. Time-of-flight mass analyzers are also used, but not as frequently as quadrupoles and ion traps.

In GC/MS, the mass spectrometer scans the masses repetitively during a chromatographic experiment. If the chromatographic run is 10 minutes, for example, and a scan is taken each second, 600 mass spectra are recorded. The data can be analyzed by the data system in several different ways. First, the ion abundances in each spectrum can be summed and plotted as a function of time to give a *total-ion chromatogram*. This plot is similar to a conventional chromatogram. One can also display the mass spectrum at a particular time during the chromatogram to identify the species eluting at that time. Finally, a single mass-to-charge (m/z) value can be selected and monitored throughout the chromatographic experiment, a technique known as *selected-ion*

monitoring. Mass spectra of selected ions obtained during a chromatography experiment are known as *mass chromatograms*.

GC/MS instruments have been used for the identification of thousands of components that are present in natural and biological systems. For example, these procedures have permitted characterization of the odor and flavor components of foods, identification of water pollutants, medical diagnosis based on breath components, and studies of drug metabolites.

An example of one application of GC/MS is shown in Figure 27-16. The upper figure is the total-ion chromatogram of a five-component mixture. Also shown are mass chromatograms at $m/z = 74$ and $m/z = 93$. From these, the identities of components 1, 2, and 5 can be determined.

Mass spectrometry can also be used to obtain information about incompletely separated components. For example, the mass spectrum of the front edge of a GC peak may be different from that of the middle part of the peak or the trailing edge if the peak is due to more than one component. With mass spectrometry, we can not only determine that a peak is due to more than one species but also identify the various unresolved components. GC has also been coupled to tandem mass spectrometers or to Fourier transform mass spectrometers to give GC/MS/MS or GC/MSⁿ systems. These are extremely powerful tools for identifying components in mixtures.

GC Coupled with Spectroscopic Detection

GC is often coupled with the selective techniques of spectroscopy and electrochemistry to provide powerful tools for separating and identifying the components of complex mixtures. Combinations of GC with mass spectrometry (GC/MS), Fourier



Animation: Learn more about GC/MS at www.tinyurl.com/skoogpia7

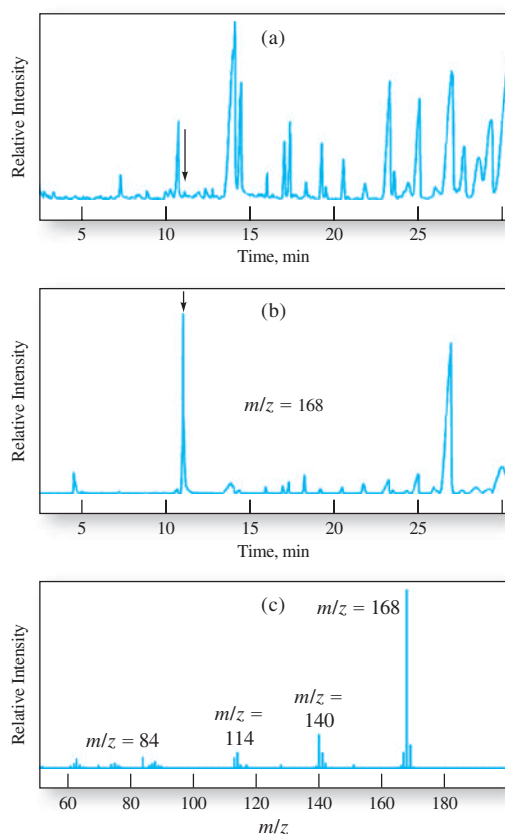


FIGURE 27-16 Typical outputs for a GC/MS system. In (a) the total ion chromatogram of an extract from a termite sample is shown. In (b) the ion at $m/z = 168$ was monitored during the chromatogram. In (c), the complete mass spectrum of the compound eluting at $t = 10.46$ minutes is presented, allowing it to be identified as β -carboline norharmane, an important alkaloid. (Adapted from S. Itakura, S. Kawabata, H. Tanaka, and A. Enoki, *J. Insect Sci.*, **2008**, 8, 13.)

transform infrared spectroscopy (GC/FTIR), nuclear magnetic resonance spectroscopy, and electroanalytical methods are sometimes termed *hyphenated methods*.⁶

In early systems, the eluates from the GC column were collected as separate fractions in a cold trap, and a nondestructive, nonselective detector was used to indicate their appearance. The composition of each fraction was then investigated by nuclear magnetic resonance spectrometry, infrared spectroscopy, or by electroanalytical measurements. A serious limitation to this approach was the very small (usually micromolar) quantities of solute contained in a fraction.

Most modern hyphenated methods monitor the effluent from the chromatographic column continuously by spectroscopic methods. The combination of two techniques based on different principles can achieve tremendous selectivity. Today's

computer-based GC instruments incorporate large databases for comparing spectra and identifying compounds.

Other Types of Detectors

Several other types of GC detectors are useful for specific applications. The *sulfur chemiluminescence detector* is based on the reaction between certain sulfur compounds and ozone. The resulting luminescence intensity is proportional to the concentration of sulfur. This detector has proven particularly useful for the determination of pollutants such as mercaptans. In the sulfur chemiluminescence detector, the eluent is mixed with hydrogen and air, and combustion takes place as in the FID. The resulting gases are then mixed with ozone, and the intensity of the emission produced is measured. The linear range is about five orders of magnitude, and the limit of detection for sulfur is approximately 0.5 pg/s. The sulfur chemiluminescence detector has also been adapted to supercritical fluid chromatography.

The nitrogen-specific chemiluminescence detector is quite similar to the sulfur detector. The nitrous oxide combustion product reacts with ozone to produce chemiluminescence. The detector responds linearly to nitrogen over about four orders of magnitude. The limit of detection for nitrogen is about 5 pg/s. The detector can be used for organic nitrogen compounds and for inorganic compounds such as ammonia, hydrazine, HCN, and nitrogen oxides.

27C GAS CHROMATOGRAPHIC COLUMNS AND STATIONARY PHASES

The pioneering gas-liquid chromatographic studies in the early 1950s were carried out on packed columns in which the stationary phase was a thin film of liquid retained by adsorption on the surface of a finely divided, inert solid support. From theoretical studies made during this early period, it became apparent that unpacked columns having inside diameters of a few tenths of a millimeter could provide separations superior to those on packed columns in both speed and column efficiency.⁷ In such *capillary columns*, the stationary phase was a film of liquid a few tenths of a micrometer thick that uniformly coated the interior of a capillary tubing. In the late 1950s such *open tubular columns* were constructed and the predicted performance characteristics were confirmed experimentally in several laboratories, with open tubular columns having 300,000 plates or more being described.⁸ Today, open tubular columns predominate in GC

⁷For a thorough discussion of packed and capillary column technology, see E. F. Barry and R. L. Grob, *Columns for Gas Chromatography*, Hoboken, NJ: Wiley-Interscience, 2007.

⁸In 1987 a world record for length of an open tubular column and number of theoretical plates was set, as attested in the *Guinness Book of Records*, by Chrompack International Corporation of the Netherlands. The column was a fused-silica column drawn in one piece and having an internal diameter of 0.32 mm and a length of 2.1 km, or 1.3 miles. The column was coated with a 0.1 μ m film of polydimethylsiloxane. A 1300 m section of this column contained more than 2 million plates.

⁶For reviews on hyphenated methods, see C. L. Wilkins, *Science*, **1983**, 222, 291, DOI: 10.1126/science.6353577; C. L. Wilkins, *Anal. Chem.*, **1989**, 59, 571A, DOI: 10.1021/ac00135a001.

because, with no packing, columns can be narrower and longer, leading to higher efficiencies than with packed columns.

Despite such spectacular performance characteristics, capillary columns did not gain widespread use until more than two decades after their invention. The reasons for the delay were several, including small sample capacities; fragility of columns; mechanical problems associated with sample introduction and connection of the column to the detector; difficulties in coating the column reproducibly; short lifetimes of poorly prepared columns; tendencies of columns to clog; and patents, which limited commercial development to a single manufacturer (the original patent expired in 1977). The most significant development in capillary GC occurred in 1979 when fused-silica capillaries were introduced. Since then an impressive list of commercially available capillary columns for various applications has appeared. As a result, the majority of applications that have appeared in the past few years use capillary columns.⁹

27C-1 Capillary Columns

Capillary columns are also called open tubular columns because of the open flow path through them. They are of two basic types: *wall-coated open tubular* (WCOT) and *support-coated open tubular* (SCOT) columns.¹⁰ Wall-coated columns are simply capillary tubes coated with a thin layer of the stationary phase. In SCOT columns, the inner surface of the capillary is lined with a thin film ($\sim 30\ \mu\text{m}$) of a support material, such as diatomaceous earth. This type of column holds several times as much stationary phase as does a wall-coated column and thus has a greater sample capacity. Generally, the efficiency of a SCOT column is less than that of a WCOT column but significantly greater than that of a packed column.

Early WCOT columns were constructed of stainless steel, aluminum, copper, or plastic. Later, glass columns began to be used. Often, the glass was etched with gaseous hydrochloric acid, strong aqueous hydrochloric acid, or potassium hydrogen fluoride to give a rough surface, which bonded the stationary phase more tightly. The most widely used capillary columns are *fused-silica wall-coated* (FSWC) *open tubular columns*. Fused-silica capillaries are drawn from specially purified silica that contains minimal amounts of metal oxides. These capillaries have much thinner walls than glass columns. The tubes are given added strength by an outside protective polyimide coating, which is applied as the capillary tubing is drawn. The resulting columns are quite flexible and can be bent into coils with diameters of a few inches. Figure 27-7 shows fused-silica open tubular columns. Fused-silica capillary columns are available commercially and offer several important advantages such as physical

strength, much lower reactivity toward sample components, and flexibility. For most applications, they have replaced the older-type WCOT glass columns.

The most widely used silica open tubular columns have inside diameters of 0.32 and 0.25 mm. Higher-resolution columns are also available with diameters of 0.20 and 0.15 mm. Such columns are more troublesome to use and are more demanding on the injection and detection systems. Thus, a sample splitter must be used to reduce the size of the sample injected onto the column and a more sensitive detector system with a rapid response time is required.

Larger 530- μm capillaries, sometimes called *megabore columns*, are also available commercially. These columns will tolerate sample sizes that are similar to those for packed columns. The performance characteristics of megabore open tubular columns are not as good as those of smaller-diameter columns but are significantly better than those of packed columns.

Table 27-2 compares the performance characteristics of fused-silica capillary columns with other types of wall-coated columns as well as with support-coated and packed columns.

27C-2 Packed Columns

Modern packed columns are fabricated from glass or metal tubing; they are typically 2–3 m long and have inside diameters of 2–4 mm. These tubes are densely packed with a uniform, finely divided packing material, or solid support, coated with a thin layer (0.05–1 μm) of the stationary liquid phase. Columns are usually formed as coils with diameters of roughly 15 cm to permit convenient thermostating in an oven.

Solid Support Materials

The packing, or solid support in a packed column, holds the liquid stationary phase in place so that as large a surface area as possible is exposed to the mobile phase. The ideal support consists of small, uniform, spherical particles with good mechanical strength and a specific surface area of at least 1 m^2/g . In addition, the material should be inert at elevated temperatures and be uniformly wetted by the liquid phase. No material is yet available that meets all of these criteria perfectly.

The earliest, and still the most widely used, packings for GC were prepared from naturally occurring diatomaceous earth, which consists of the skeletons of thousands of species of single-celled plants that once inhabited ancient lakes and seas. Figure 27-17 is an enlarged photo of a diatom obtained with a scanning electron microscope. Such plants received their nutrients and disposed of their wastes via molecular diffusion through their pores. As a result, their remains are well-suited as support materials because GC is also based on the same kind of molecular diffusion.

Particle Size of Supports

As shown in Figure 26-11, the efficiency of a gas chromatographic column increases rapidly with decreasing particle

⁹For more information on columns in GC, see E. F. Barry, in *Modern Practice of Gas Chromatography*, R. L. Grob and E. F. Barry, eds., 4th ed., Chap. 3, New York: Wiley-Interscience, 2004.

¹⁰For a detailed description of open tubular columns, see M. L. Lee, F. J. Yang, and K. D. Bartle, *Open Tubular Column Gas Chromatography: Theory and Practice*, New York: Wiley, 1984.

TABLE 27-2 Properties and Characteristics of Typical GC Columns

	Type of Column			
	FSWC*	WCOT†	SCOT‡	Packed
Length, m	10–100	10–100	10–100	1–6
Inside diameter, mm	0.1–0.3	0.25–0.75	0.5	2–4
Efficiency, plates/m	2000–4000	1000–4000	600–1200	500–1000
Sample size, ng	10–75	10–1000	10–1000	10–10 ⁶
Relative pressure	Low	Low	Low	High
Relative speed	Fast	Fast	Fast	Slow
Flexibility?	Yes	No	No	No
Chemical inertness	Best			
		→ Poorest		

*Fused silica, wall-coated open tubular column.

†Wall-coated open tubular metal, plastic, or glass columns.

‡Support-coated open tubular column (also called porous-layer open tubular, or PLOT).

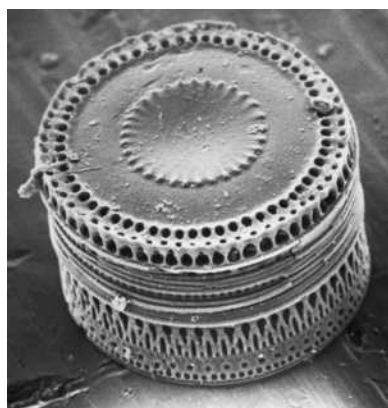


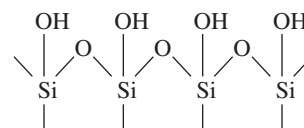
FIGURE 27-17 A photomicrograph of a diatom. Magnification 5000×. (BIOPHOTO ASSOCIATES/Getty Images)

diameter of the packing. The pressure difference required to maintain an acceptable flow rate of carrier gas, however, varies inversely as the square of the particle diameter; the latter relationship has placed lower limits on the size of particles used in GC because it is not convenient to use pressure differences that are greater than about 50 psi. As a result, the usual support particles are 60–80 mesh (250–170 μm) or 80–100 mesh (170–149 μm).

27C-3 Adsorption on Column Packings or Capillary Walls

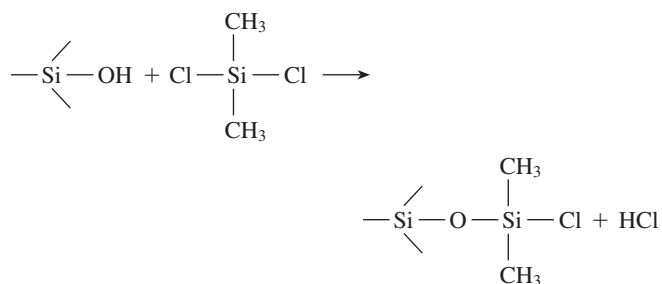
A problem that has plagued GC from its beginning has been the physical adsorption of polar or polarizable analyte species, such as alcohols or aromatic hydrocarbons, on the silicate surfaces of column packings or capillary walls. Adsorption causes distorted peaks, which are broadened and often exhibit a tail (recall Figure 26-5). Adsorption occurs with silanol groups that form on

the surface of silicates by reaction with moisture. Thus, a fully hydrolyzed silicate surface has the structure

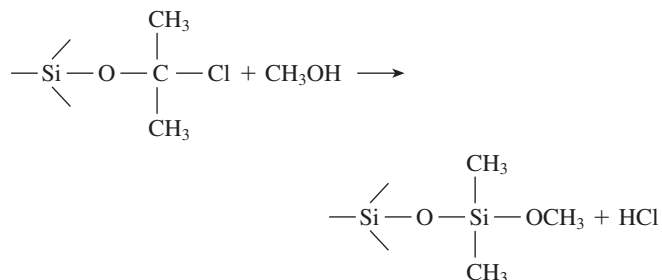


The SiOH groups on the support surface have a strong affinity for polar organic molecules and tend to retain them by adsorption.

Support materials can be deactivated by silanization with dimethylchlorosilane (DMCS). The reaction is



When the support is washed with methanol, the second chloride is replaced by a methoxy group. That is,



Silanized surfaces of column packings may still show a residual adsorption, which apparently occurs with metal oxide impurities in the diatomaceous earth. Acid washing prior to silanization removes these impurities. Fused silica used for manufacturing open tubular columns is largely free of this type of impurity. Because of this, fewer problems with adsorption arise with fused-silica columns.

27C-4 The Stationary Phase

Desirable properties for the immobilized liquid phase in a gas-liquid chromatographic column include (1) *low volatility* (ideally, the boiling point of the liquid should be at least 100°C higher than the maximum operating temperature for the column); (2) *thermal stability*; (3) *chemical inertness*; and (4) *solvent characteristics* such that k and α (Sections 26B-5 and 26B-6, respectively) values for the solutes to be resolved fall within a suitable range.

Many liquids have been proposed as stationary phases in the development of GLC. Currently, fewer than a dozen are commonly used. The proper choice of stationary phase is often crucial to the success of a separation. Qualitative guidelines for stationary-phase selection can be based on a literature review, an Internet search, prior experience, or advice from a vendor of chromatographic equipment and supplies.

The retention time for an analyte on a column depends on its distribution constant, which in turn is related to the chemical nature of the liquid stationary phase. To separate various sample components, their distribution constants must be sufficiently different to accomplish a clean separation. At the same time, these constants must not be extremely large or extremely small because large distribution constants lead to prohibitively long retention times and small constants produce such short retention times that separations are incomplete.

To have a reasonable residence time in the column, an analyte must show some degree of compatibility (solubility) with the stationary phase. Here, the principle of “like dissolves like” applies, where “like” refers to the polarities of the analyte and the immobilized liquid. The polarity of a molecule, as indicated by its dipole moment, is a measure of the electric field produced by separation of charge within the molecule. Polar stationary phases contain functional groups such as —CN, —CO, and —OH. Hydrocarbon-type stationary phases and dialkyl siloxanes are nonpolar, but polyester phases are highly polar. Polar analytes include alcohols, acids, and amines; solutes of medium polarity include ethers, ketones, and aldehydes. Saturated hydrocarbons are nonpolar. Generally, the polarity of the stationary phase should match that of the sample components. When the match is good, the order of elution is determined by the boiling point of the eluents.

Classification of Stationary Phases

Many different schemes have been reported to classify stationary phases and thereby simplify stationary-phase selection. Most of these are based on solute probes that test specific interactions between the solute and the liquid phase by measuring solute retention characteristics. Two of the most important classifications are based on the work of Rohrschneider and McReynolds.¹¹ The result was the production of lists of stationary phases and the compound classes that can be separated by each phase (see Table 27-3). Likewise, numerical values, known as McReynolds constants, are available that can guide the user in selecting a stationary phase to separate analytes having different functional groups, such as alcohols from aldehydes or ketones.¹²

¹¹L. Rohrschneider, *J. Chromatogr.*, **1966**, 22, 6, DOI: 10.1016/S0021-9673(01)97064-5; W. O. McReynolds, *J. Chromatogr. Sci.*, **1970**, 8, 685, DOI: 10.1093/chromsci/8.12.685.

¹²J. A. Dean, *Analytical Chemistry Handbook*, pp. 4.34–4.37, New York: McGraw-Hill, 1995.

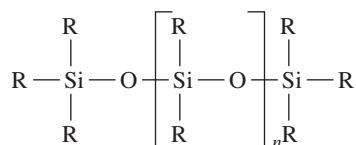
TABLE 27-3 Some Common Liquid Stationary Phases for Gas-Liquid Chromatography

Stationary Phase	Common Trade Name	Maximum Temperature, °C	Common Applications
Polydimethyl siloxane	OV-1, SE-30	350	General-purpose nonpolar phase, hydrocarbons, polynuclear aromatics, steroids, PCBs
5% Phenyl-polydimethyl siloxane	OV-3, SE-52	350	Fatty acid methyl esters, alkaloids, drugs, halogenated compounds
50% Phenyl-polydimethyl siloxane	OV-17	250	Drugs, steroids, pesticides, glycols
50% Trifluoropropyl-polydimethyl siloxane	OV-210	200	Chlorinated aromatics, nitroaromatics, alkyl-substituted benzenes
Polyethylene glycol	Carbowax 20M	250	Free acids, alcohols, ethers, essential oils, glycols
50% Cyanopropyl-polydimethyl siloxane	OV-275	240	Polyunsaturated fatty acids, rosin acids, free acids, alcohols

Some Widely Used Stationary Phases

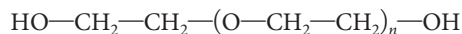
Table 27-3 lists the most widely used stationary phases for both packed and open tubular column GC in order of increasing polarity. These six liquids can probably provide satisfactory separations for 90% or more of samples encountered.

Five of the liquids listed in Table 27-3 are polydimethyl siloxanes that have the general structure



In the first of these, polydimethyl siloxane, the —R groups are all —CH₃, giving a liquid that is relatively nonpolar. In the other polysiloxanes shown in the table, a fraction of the methyl groups are replaced by functional groups such as phenyl (—C₆H₅), cyanopropyl (—C₃H₆CN), and trifluoropropyl (—C₃H₆CF₃). The percentage description in each case gives the amount of substitution of the named group for methyl groups on the polysiloxane backbone. Thus, for example, 5% phenyl-polydimethyl siloxane has a phenyl ring bonded to 5% (by number) of the silicon atoms in the polymer. These substitutions increase the polarity of the liquids to various degrees.

The fifth entry in Table 27-3 is a polyethylene glycol with the structure



This compound is used widely for separating polar species. Figure 27-18 illustrates applications of the phases listed in Table 27-3 for open tubular columns.

Bonded and Cross-Linked Stationary Phases

Commercial columns are advertised as having bonded or cross-linked stationary phases. The purpose of bonding and cross-linking is to provide a longer-lasting stationary phase that is not disrupted at elevated temperatures or during temperature programming. With use, untreated columns slowly lose their stationary phase because of “bleeding,” in which a small amount of immobilized liquid is carried out of the column during the elution process. Such columns are also recommended for on-column injection where a large solvent volume is used. Indeed, cross-linked or bonded columns may be backflushed to remove contaminants without significant loss of stationary phase.

Bonding involves attaching a monomolecular layer of the stationary phase to the silica surface of the column by a chemical reaction. For commercial columns, the nature of the reaction is usually proprietary.

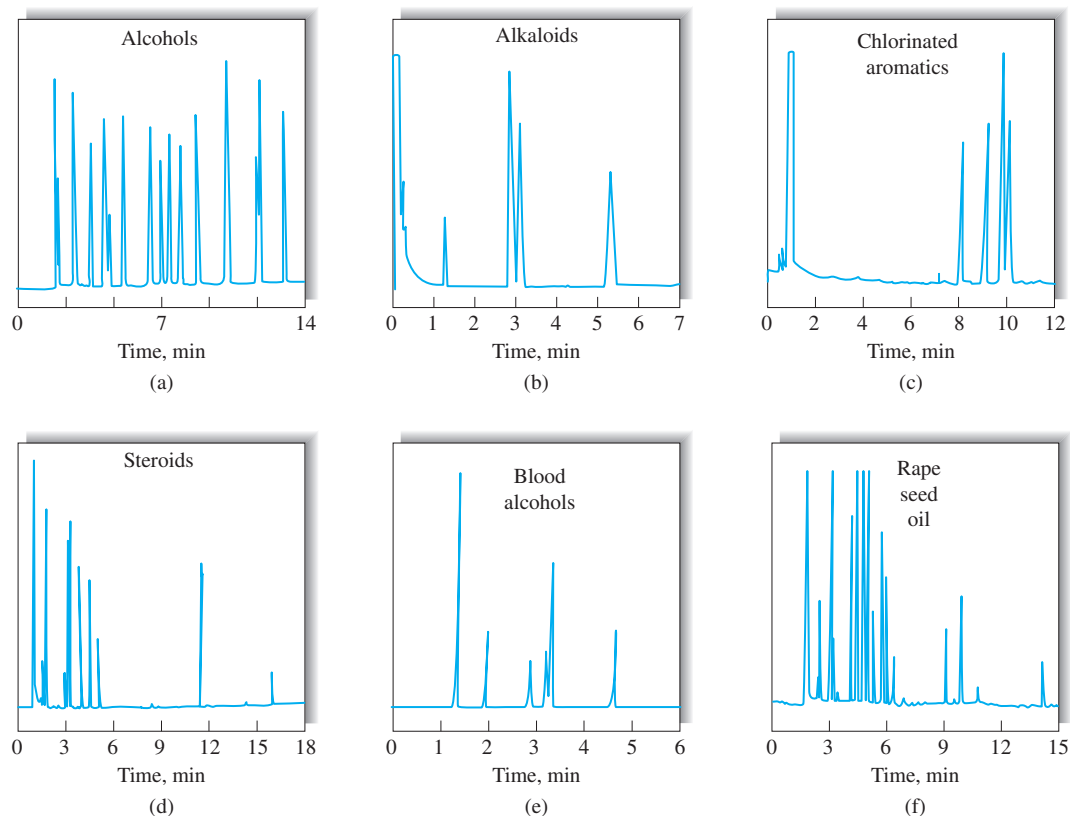


FIGURE 27-18 Typical chromatograms from open tubular columns coated with (a) polydimethyl siloxane, (b) 5% (phenyl methyl)dimethyl siloxane, (c) 50% (phenyl methyl)dimethyl siloxane, (d) 50% poly(trifluoropropyl-dimethyl) siloxane, (e) polyethylene glycol, and (f) 50% poly(cyanopropyl-dimethyl) siloxane. (Source: Agilent Technologies.)

Cross-linking is carried out *in situ* after a column is coated with one of the polymers listed in Table 27-3. One way of cross-linking is to incorporate a peroxide into the original liquid. When the film is heated, reaction between the methyl groups in the polymer chains is initiated by a free radical mechanism. The polymer molecules are then cross-linked through carbon-to-carbon bonds. The resulting films are less extractable and have considerably greater thermal stability than do untreated films. Cross-linking has also been initiated by exposing the coated columns to gamma radiation.

Film Thickness

Commercial columns are available having stationary phases that vary in thickness from 0.1 to 5 μm . Film thickness primarily affects the retentive character and the capacity of a column as discussed in Section 26C-3. Thick films are used with highly volatile analytes because such films retain solutes for a longer time, thus providing a greater time for separation to take place. Thin films are useful for separating species of low volatility in a reasonable length of time. For most applications with 0.25- or 0.32-mm columns, a film thickness of 0.25 μm is recommended. With megabore columns, 1- to 1.5- μm films are often used. Today, columns with 8- μm films are marketed.

Chiral Stationary Phases

In recent years, much effort has been devoted to developing methods for the separation of enantiomers by gas or liquid chromatography.¹³ Two approaches have been used. One is based on forming derivatives of the analyte with an optically active reagent that forms a pair of diastereomers that can be separated on an achiral column. The alternative method is to use a chiral liquid as the stationary phase. A number of amino acid-derived chiral phases have been developed for this purpose, and others are becoming available commercially.

27D APPLICATIONS OF GC

To evaluate the importance of GC, we must distinguish between the two roles the method plays. First, GC is a tool for performing separations. In this role, GC methods are unsurpassed when applied to complex organic, metal-organic, and biochemical systems made up of volatile species or species that can be derivatized to yield volatile substances. The second role that GC plays is in the completion of an analysis. In this role, retention times or volumes are used for qualitative identification, and peak heights or peak areas provide quantitative information. For qualitative purposes, GC is much more limited than most of the spectroscopic methods considered in earlier chapters. Thus, an important trend in the field has been in the direction of combining the remarkable separation capabilities of GC with the superior identification properties of such instruments as mass, infrared, and nuclear magnetic resonance spectrometers (see Section 27B-4).

27D-1 Qualitative Analysis

Gas chromatograms are widely used to establish the purity of organic compounds. The appearance of additional peaks reveals any contaminants present, and the areas under these peaks provide estimates of the extent of contamination. Such areas are only estimates because different components may have widely differing detector response factors. Gas chromatographic techniques are also useful for evaluating the effectiveness of purification procedures.

In theory, GC retention times should be useful for identifying components in mixtures. In fact, however, the applicability of such data is limited by the number of variables that must be controlled to obtain reproducible results. Nevertheless, GC provides an excellent means of confirming the presence or absence of a suspected compound in a mixture, provided that an authentic sample of the substance is available. No new peaks in the chromatogram of the mixture should appear on addition of the known compound, and enhancement of an existing peak should be observed. The evidence is particularly convincing if the effect can be duplicated on different columns and at different temperatures.

On the other hand, because a chromatogram provides only a single piece of information about each species in a mixture (the retention time), the application of the technique to the qualitative analysis of complex samples of unknown composition is limited. This limitation has been largely overcome by linking chromatographic columns directly with ultraviolet, infrared, and mass spectrometers to produce hyphenated instruments as discussed elsewhere.

Selectivity Factors

We have seen (Section 26B-6) that the selectivity factor α for compounds A and B is given by the relationship

$$\alpha = \frac{K_B}{K_A} = \frac{(t_R)_B - t_M}{(t_R)_A - t_M} = \frac{(t'_R)_B}{(t'_R)_A}$$

where $(t'_R)_A = [(t_R)_A - t_M]$ is the adjusted retention time for species A. If a standard substance is chosen as compound B, then α can provide an index for identification of compound A, which is largely independent of column variables other than temperature. Numerical tabulations of selectivity factors for pure compounds relative to a common standard can be prepared and then used for the characterization of solutes. Unfortunately, finding a universal standard that yields selectivity factors of reasonable magnitude for all types of analytes is impossible. Thus, the amount of selectivity factor data available in the literature is presently limited.

The Retention Index

The retention index I was first proposed by E. Kovats in 1958 for identifying solutes from chromatograms.¹⁴ The retention index for any given solute can be calculated from a chromatogram of a mixture of that solute with at least two normal alkanes having retention times that bracket that of the solute. The retention index scale is based on normal alkanes. By definition, the retention index for a normal alkane is equal to 100 times the number of carbons in the compound *regardless of the column packing, the*

¹³For a review of chiral stationary-phase separations by GC, see J. V. Hinshaw, *LC-GC*, **1993**, 11, 644; E. F. Barry, in *Modern Practice of Gas Chromatography*, R. L. Grob and E. F. Barry, eds., 4th ed., Chap. 3, New York: Wiley-Interscience, 2004.

¹⁴E. Kovats, *Helv. Chim. Acta*, **1958**, 41, 1915, DOI: 10.1002/hlca.19580410703.

temperature, or other chromatographic conditions. The retention indexes for all compounds other than normal alkanes vary, often by several hundred retention index units, with column variables.

It has long been known that within a homologous series, a plot of the logarithm of adjusted retention time ($t'_R = t_R - t_M$) versus the number of carbon atoms is linear, provided the lowest member of the series is excluded. Such a plot for C_4 to C_9 normal alkane standards is shown in Figure 27-19. Also indicated on the ordinate are log-adjusted retention times for three compounds on the same column and at the same temperature. Their retention indexes are then obtained by multiplying the corresponding abscissa values by 100. Thus, the retention index for toluene is 749, and for benzene it is 644.

Normally, a graphical procedure is not required to determine retention indexes. Instead, adjusted retention data are calculated by interpolation from a chromatogram of a mixture of the solute of interest and two or more alkane standards.

It is important to reiterate that the retention index for a normal alkane is independent of temperature and column packing. Thus, I for heptane, by definition, is always 700. In contrast, retention indexes of all other solutes may, and often do, vary widely from one column to another. For example, the retention index for acenaphthene on a cross-linked polydimethyl siloxane stationary phase at 140°C is 1460. With 5% phenyl-polydimethyl siloxane as the stationary phase, it is 1500 at the same temperature, and with polyethylene glycol as the stationary phase, the retention index is 2084.

The retention index system has the advantage of being based on readily available reference materials that cover a wide boiling range. In addition, the temperature dependence of retention indexes is relatively small.

The use of retention data on two or more GC columns can improve the chances of correctly identifying an unknown compound. The columns can be used in separate experiments or sometimes they can be used in tandem. The use of two or more columns in series is termed *multidimensional chromatography*.¹⁵ Likewise, the responses of two or more GC detectors can greatly aid in qualitative identification.

The combination of GC with various spectroscopic detectors, particularly with mass spectrometry, can greatly aid in identifying components. In fact, GC/MS is now the premier technique for separating and identifying species in mixtures.

The National Institute for Standards and Technology maintains NIST Standard Reference Database 1A, which contains mass spectral data on nearly 250,000 compounds and retention indices on 70,000 of those. Along with the database, the agency has developed a sophisticated software application to search and match experimental spectra and retention indices to those of known compounds.¹⁶

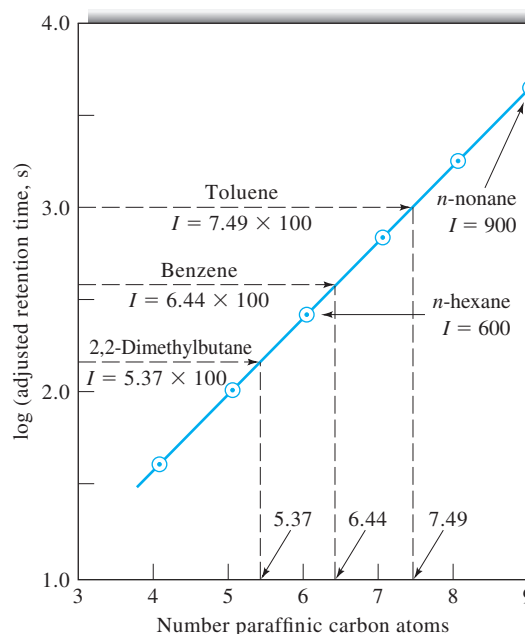


FIGURE 27-19 Graphical illustration of the method for determining retention indexes for three compounds. Stationary phase: squalane. Temperature: 60°C. Retention indexes for normal alkane standards nonane and hexane are indicated.

27D-2 Quantitative Analysis

The peak height or peak area of an eluate from a GC column has been widely used for quantitative and semiquantitative analyses. An accuracy of 1% relative is attainable under carefully controlled conditions using either the external or the internal standard method. As with most analytical tools, reliability is directly related to the control of variables; the nature of the sample also plays a part in determining the potential accuracy. The general discussion of quantitative chromatographic analysis given in Section 26F-2 applies to GC as well as to other types; therefore, no further consideration of this topic is given here.

27E ADVANCES IN GC

Although GC is quite a mature technique, there have been many developments in recent years in theory, instrumentation, columns, and practical applications. We discuss here some developments in high-speed GC and in miniaturized GC systems.

27E-1 High-Speed GC¹⁷

Researchers in GC have often focused on achieving ever higher resolution to separate more and more complex mixtures. In most separations, conditions are varied to separate the most difficult-to-separate pair of components, the so-called *critical pair*. Many of the components of interest, under these conditions, are highly overseparated. The basic idea of high-speed

¹⁵For reviews of two-dimensional GC, see M. Adahchour, J. Beens, R. J. J. Vreuls, U. A. Th. Brinkman, *Trends Anal. Chem. (TRAC)*, **2006**, 25, 438, DOI: 10.1016/j.trac.2006.03.002; **2006**, 25, 540, DOI: 10.1016/j.trac.2006.04.004; **2006**, 25, 726, DOI: 10.1016/j.trac.2006.03.005.

¹⁶NIST Standard Reference Database 1A, Gaithersburg, MD: National Institute of Standards and Technology, 2014, <http://tinyurl.com/ncozs4c>.

¹⁷For more information, see R. D. Sacks, in *Modern Practice of Gas Chromatography*, R. L. Grob and E. F. Barry, eds., 4th ed., Ch. 5, New York: Wiley-Interscience, 2004.

GC is that, for many separations of interest, higher speed can be achieved albeit at the expense of some selectivity and resolution.

The principles of high-speed separations can be demonstrated by substituting Equation 26-5 into Equation 26-11

$$\frac{L}{t_R} = u \times \frac{1}{1 + k_n} \quad (27-8)$$

where k_n is the retention factor for the last component of interest in the chromatogram. If we rearrange Equation 27-8 and solve for the retention time of the last component of interest, we obtain

$$t_R = \frac{L}{u} \times (1 + k_n) \quad (27-9)$$

Equation 27-9 tells us that we can achieve faster separations by using short columns, higher-than-usual carrier gas velocities and small retention factors. For example, if we reduce the column length L by a factor of 4 and increase the carrier-gas velocity u by a factor of 5, the analysis time t_R is reduced by a factor of 20. The price paid is reduced resolving power caused by increased band broadening and reduced peak capacity (the number of peaks that will fit in the chromatogram).

Research workers in the field have been designing instrumentation and chromatographic conditions to optimize separation speed at the lowest cost in terms of resolution and peak capacity.¹⁸ They have designed systems to achieve tunable columns and high-speed temperature programming. A tunable column is a series combination of a polar and a nonpolar column.

Figure 27-20 shows the separation of twelve compounds prior to initiating a programmed temperature ramp and nineteen compounds after the temperature program was begun. The total time required was 140 s. These workers have also been using high-speed GC with mass spectrometry detection including time-of-flight detection.¹⁹

27E-2 Miniaturized GC Systems

For many years there has been a desire to miniaturize GC systems to the microchip level. Miniature GC systems are useful in space exploration, in portable instruments for field use, and in environmental monitoring. Early work reported on GC columns etched onto a microchip.²⁰ However, the relatively poor chromatographic performance of such devices led to removal of the column from the chip in the commercial systems that were produced.

Microfabricated columns have been designed using substrates of silicon, several metals, and polymers.²¹ Relatively deep, narrow channels are etched into the substrate. These channels have low dead volume to reduce band broadening and high surface area to increase stationary-phase volume. A complete microfabricated GC system has been described.²² Figure 27-21 shows the microfabricated GC column and

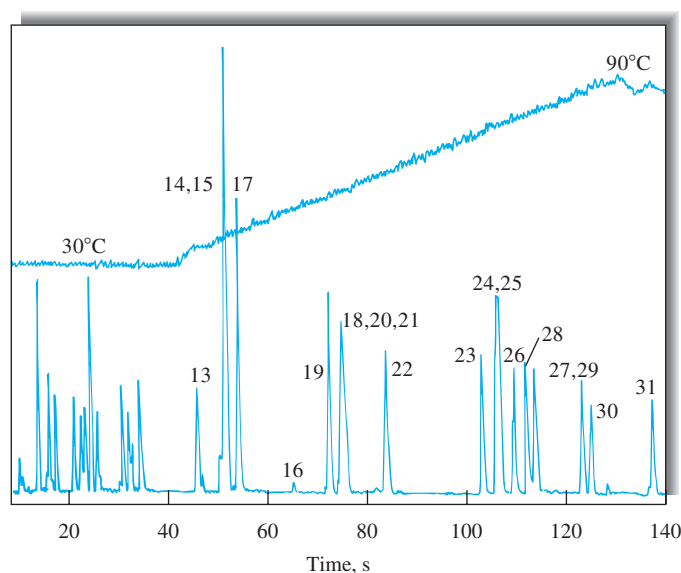


FIGURE 27-20 High-speed chromatogram obtained with isothermal operation (30°C) for 37 s followed by a 35°C/min temperature ramp to 90°C. (From H. Smith and R. D. Sacks, *Anal. Chem.*, **1998**, 70, 4960. Copyright 1998 American Chemical Society.)

¹⁹C. Leonard and R. Sacks, *Anal. Chem.*, **1999**, 71, 5177, DOI: 10.1021/ac990631f.

²⁰See S. C. Terry et al., *IEEE Trans. Electron Devices*, **1979**, 26, 1880, DOI: 10.1109/T-ED.1979.19791; J. B. Angell et al., *Sci. Am.*, **1983**, 248 (4), 44, DOI: 10.1038/scientificamerican0483-44.

²¹G. Lambertus et al., *Anal. Chem.*, **2004**, 76, 2629, DOI: 10.1021/ac030367x.

²²D. Gaddes et al., *J. Chromatograph. A*, **2014**, 1349, 96, DOI: 10.1016/j.chroma.2014.04.087.

¹⁸H. Smith and R. D. Sacks, *Anal. Chem.*, **1998**, 70, 4960, DOI: 10.1021/ac980463b.

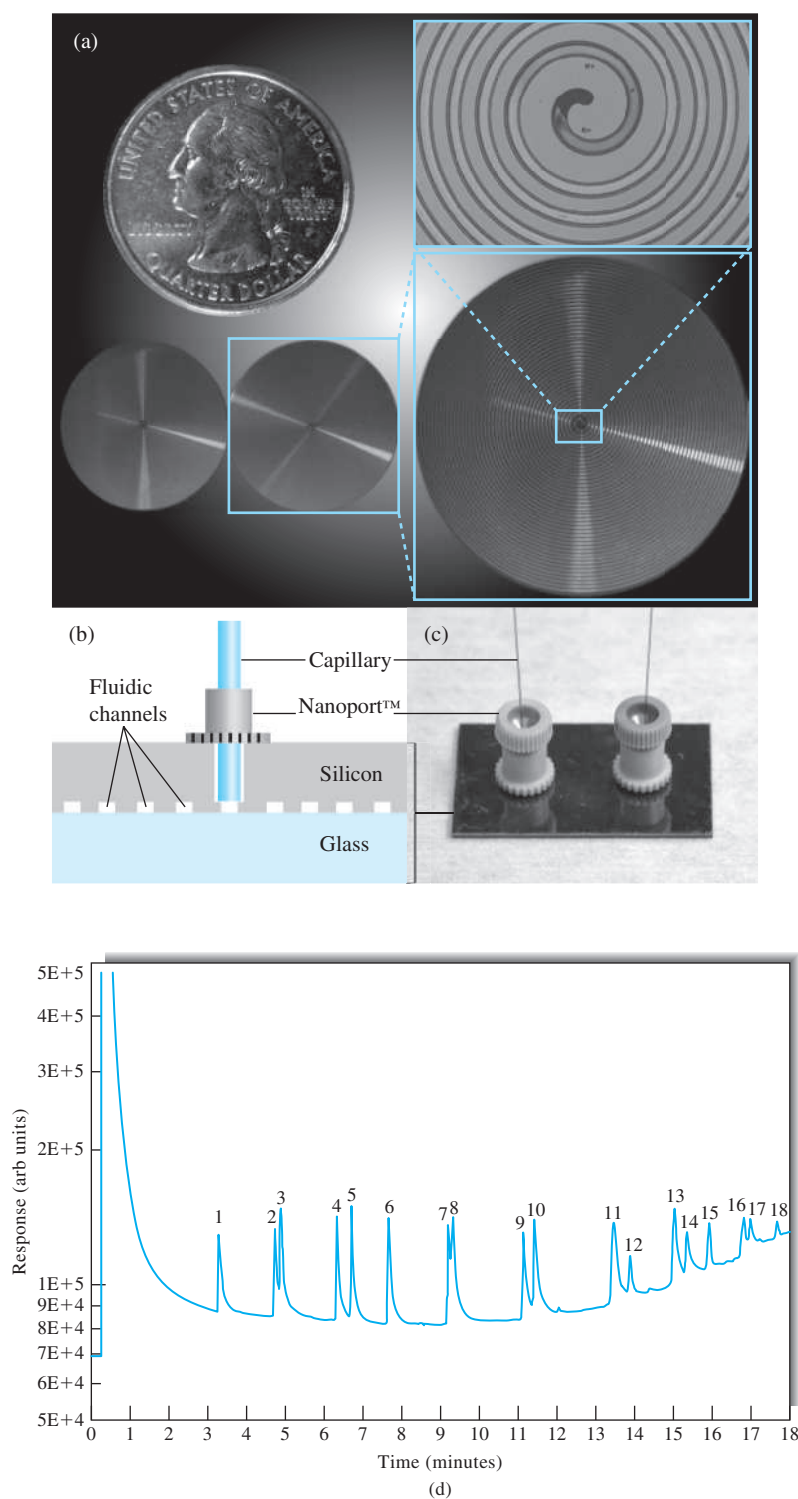


FIGURE 27-21 (a) Photo of a 2-m microfabricated GC column in relation to a U.S. quarter dollar. The two insets on the right show the column at two different zoom levels. The lighter regions in the images are the channel tracks and the darker regions are bonded silicon wall regions. (b) Schematic illustration in cross-sectional view of the capillary tubing to GC column fluidic channel connection. (c) Photo of the assembled GC column. The column with the fluid connections are carefully aligned and attached at the inlet and outlet. The silica tubing connects the injection port to the column. (d) Separation of a mixture of polycyclic aromatic hydrocarbons on the column shown in (a)–(c). The separation was started at 45°C and was ramped to 325°C at a rate of 15°C/min. All 18 components of the mixture were successfully separated. (D. Gaddes et al., *J. Chromatograph. A*, **2014**, 1349, 96. With permission.)

various connections to tubing and an injection port (a, b, and c). A chromatogram of a mixture of polycyclic aromatic hydrocarbons is also shown (d).

27F GAS-SOLID CHROMATOGRAPHY

GSC is based on adsorption of gaseous substances on solid surfaces. Distribution constants are generally much larger than those for GLC. As a result, GSC is useful for the separation of species that are not retained by gas-liquid columns, such as the components of air, hydrogen sulfide, carbon disulfide, nitrogen oxides, carbon monoxide, carbon dioxide, and the rare gases.

GSC is performed with both packed and open tubular columns. For the latter, a thin layer of the adsorbent is affixed to the inner walls of the capillary. Such columns are sometimes called *porous-layer open tubular*, or PLOT, columns.

27F-1 Molecular Sieves

Molecular sieves are aluminum silicate ion exchangers, whose pore size depends on the kind of cation present. Commercial preparations of these materials are available in particle sizes of 40–60 mesh to 100–120 mesh. The sieves are classified according to the maximum diameter of molecules that can enter the pores. Commercial molecular sieves come in pore sizes of 4, 5, 10, and 13 Å. Molecules smaller than these dimensions penetrate into the interior of the particles where adsorption takes place. For such molecules, the surface area is enormous when

compared with the area available to larger molecules. Thus, molecular sieves can be used to separate small molecules from large. For example, a 6-ft, 5-Å packing at room temperature will easily separate a mixture of helium, oxygen, nitrogen, methane, and carbon monoxide in the order given.

Figure 27-22a shows a typical molecular sieve chromatogram. In this application two packed columns were used, one an ordinary gas-liquid column and the other a molecular sieve column. The former retains only the carbon dioxide and passes the remaining gases at rates corresponding to the carrier rate. When the carbon dioxide elutes from the first column, a switch briefly directs the flow around the second column to avoid permanent adsorption of the carbon dioxide on the molecular sieve. After the carbon dioxide signal has returned to zero, the flow is switched back through the second column, thereby permitting separation and elution of the remainder of the sample components.

27F-2 Porous Polymers

Porous polymer beads of uniform size are manufactured from styrene cross-linked with divinylbenzene (Section 28F-2). The pore size of these beads is uniform and is controlled by the amount of cross-linking. Porous polymers have found considerable use in the separation of gaseous polar species such as hydrogen sulfide, oxides of nitrogen, water, carbon dioxide, methanol, and vinyl chloride. A typical application of an open tubular column lined with a porous polymer (PLOT column) is shown in Figure 27-22b.

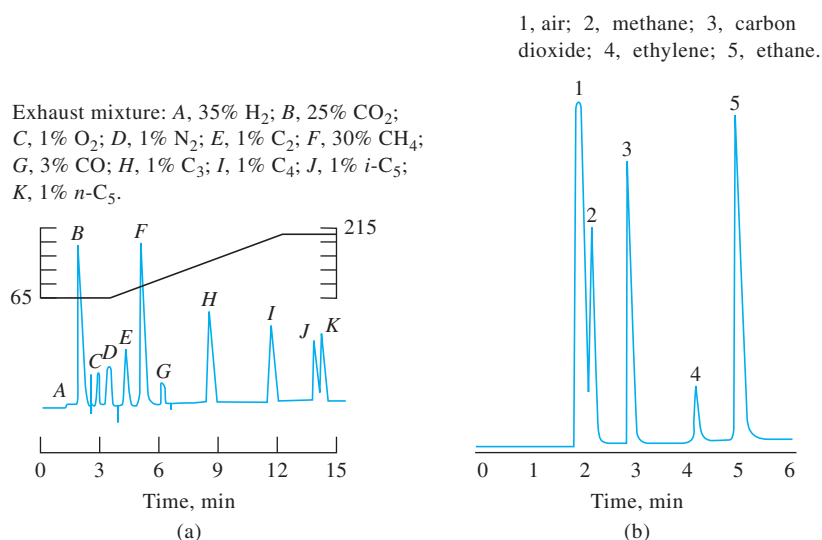


FIGURE 27-22 Typical gas-solid chromatographic separations: (a) a 5 ft. $\times$ 1/8 in. molecular sieve column; (b) a 30 m $\times$ 0.53 mm PLOT column. C_n = hydrocarbon with *n* carbons.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

- 27-1** How do gas-liquid and gas-solid chromatography differ?
- 27-2** How does a soap-bubble flow meter work?
- 27-3** What is meant by temperature programming in GC? Why is it frequently used?
- 27-4** Define (a) retention volume, (b) corrected retention volume, (c) specific retention volume.
- 27-5** What is the difference between a concentration-sensitive and a mass-sensitive detector? Are the following detectors mass or concentration sensitive? (a) thermal conductivity, (b) atomic emission, (c) thermionic, (d) electron captive, (e) flame photometric, (f) flame ionization.
- 27-6** Describe the principles of operation for the detectors listed in Question 27-5.
- 27-7** What are the major advantages and the principal limitations of each of the detectors listed in Question 27-5?
- * **27-8** What is the difference between a total-ion chromatogram and a mass chromatogram?
- 27-9** Discuss why the combination of GC and mass spectrometry is so powerful.
- 27-10** What are *hyphenated* GC methods? Briefly describe two hyphenated methods.
- * **27-11** What is the packing material used in most packed GC columns?
- 27-12** How do the following open tubular columns differ?
- (a) PLOT columns (b) WCOT columns (c) SCOT columns
- 27-13** What are megabore open tubular columns? Why are they used?
- * **27-14** What are the advantages of fused-silica capillary columns compared with glass or metal columns?
- 27-15** What properties should the stationary-phase liquid for GC possess?
- * **27-16** What is the effect of stationary-phase film thickness on gas chromatograms?
- 27-17** Why are gas chromatographic stationary phases often bonded and cross-linked? What do these terms mean?
- 27-18** List the variables that lead to (a) band broadening and (b) band separation in GLC.
- 27-19** What are retention indexes? Describe how they are determined.
- 27-20** The same polar compound is gas chromatographed on an SE-30 (very nonpolar) column and then on a Carbowax 20M (very polar column). How will $K = c_s/c_M$ vary between the two columns?
- * **27-21** Use the retention data given in the following table to calculate the retention index of 1-hexene.

Sample	Retention Time, min
Air	0.571
<i>n</i> -pentane	2.16
<i>n</i> -hexane	4.23
1-hexene	3.15

>> QUESTIONS AND PROBLEMS (continued)



* 27-22 A GC column was operated under the following conditions:

column: 1.10 m $\times$ 2.0 mm, packed with Chromosorb P; mass of stationary liquid added, 1.40 g; density of liquid, 1.02 g/mL

pressures: inlet, 26.1 psi above room; room, 748 torr

measured outlet flow rate: 25.3 mL/min

temperature: room, 21.2°C; column, 102.0°C

retention times: air, 18.0 s; methyl acetate, 1.98 min; methyl propionate, 4.16 min; methyl *n*-butyrate, 7.93 min

peak widths of esters at base: 0.19, 0.39, and 0.79, respectively

Calculate

- the average flow rate in the column.
- the corrected retention volumes for air and the three esters.
- the specific retention volumes for the three components.
- the distribution constants for each of the esters.
- a corrected retention volume and retention time for methyl *n*-hexanoate.



* 27-23 From the data in Problem 27-22, calculate

- the retention factor k for each component.
- selectivity factor α for each adjacent pair of compounds.
- the average number of theoretical plates and plate height for the column.
- the resolution for each adjacent pair of compounds.

27-24 The stationary-phase liquid in the column described in Problem 27-23 was didecylphthalate, a solvent of intermediate polarity. If a nonpolar solvent such as a silicone oil had been used instead, would the retention times for the three compounds be larger or smaller? Why?



27-25 One method for quantitative determination of the concentration of constituents in a sample analyzed by GC is the area-normalization method. Here, complete elution of all of the sample constituents is necessary. The area of each peak is then measured and corrected for differences in detector response to the different eluates. This correction involves dividing the area by an empirically determined correction factor. The concentration of the analyte is found from the ratio of its corrected area to the total corrected area of all peaks. For a chromatogram containing three peaks, the relative areas were found to be 16.4, 45.2, and 30.2 in the order of increasing retention time. Calculate the percentage of each compound if the relative detector responses were 0.60, 0.78, and 0.88, respectively.



27-26 Determine the concentration of species in a sample using the peak areas and relative detector responses for the five gas chromatographic peaks given in the following table. Use the area-normalization method described in Problem 27-25.

Also shown are the relative responses of the detector. Calculate the percentage of each component in the mixture.

Compound	Relative Peak Area	Relative Detector Response
A	32.5	0.70
B	20.7	0.72
C	60.1	0.75
D	30.2	0.73
E	18.3	0.78

27-27 What would be the effect of the following on the plate height of a column? Explain.

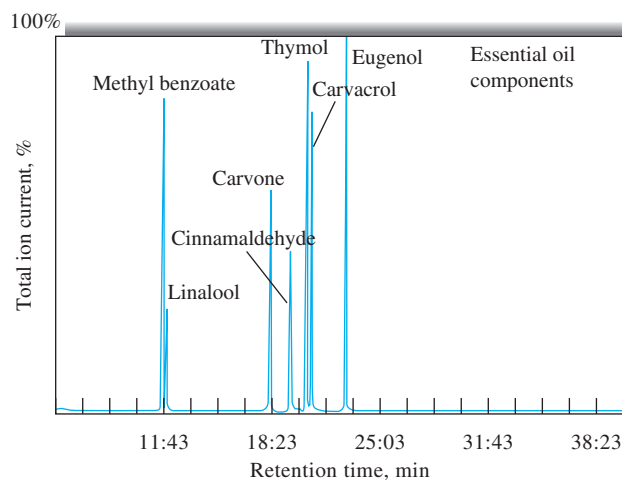
- Increasing the mass of the stationary phase relative to the packing mass.
- Decreasing the rate of sample injection.
- Increasing the injection port temperature.
- Increasing the flow rate.
- Reducing the particle size of the packing.
- Decreasing the column temperature.

*** 27-28** What kinds of mixtures are separated by GSC?

27-29 Why is GSC not used nearly as extensively as GLC?

Challenge Problem

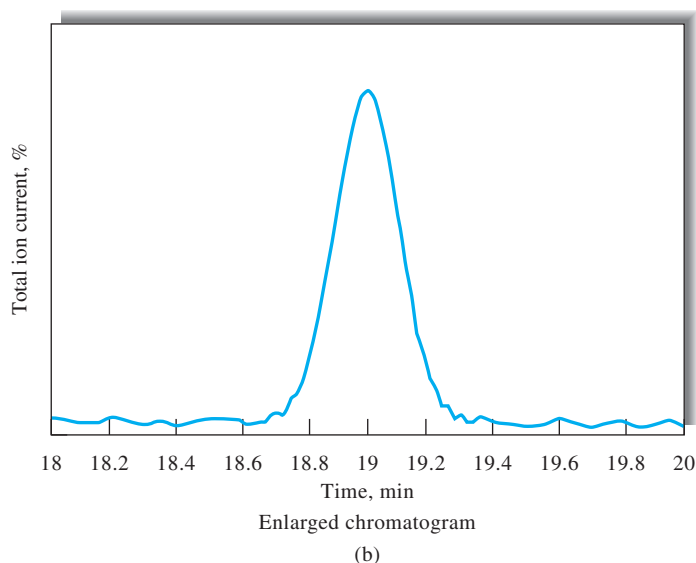
27-30 Cinnamaldehyde is the component responsible for cinnamon flavor. It is also a potent antimicrobial compound present in essential oils (see M. Friedman, N. Kozukue, and L. A. Harden, *J. Agric. Food Chem.*, **2000**, 48, 5702, DOI: 10.1021/jf000585g). The GC response of an artificial mixture containing six essential oil components and methyl benzoate as an internal standard is shown in part (a) of the figure.



(a)

» QUESTIONS AND PROBLEMS (continued)

- (a) An idealized enlargement of the region near the cinnamaldehyde peak is given in part (b) of the figure. Determine the retention time for cinnamaldehyde.



- (b) From part (b) of the figure, determine the number of theoretical plates for the column.
- (c) The fused-silica column was $0.25 \text{ mm} \times 30 \text{ cm}$ with a $0.25\text{-}\mu\text{m}$ film. Determine the height equivalent to a theoretical plate from the data in parts (a) and (b).
- (d) Quantitative data were obtained by using methyl benzoate as the internal standard. The following results were found for calibration curves of cinnamaldehyde, eugenol, and thymol. The values under each component represent the peak area of the component divided by the peak area of the internal standard.

Concentration, mg sample/ 200 μL	Cinnamaldehyde	Eugenol	Thymol
0.50	—	0.4	—
0.65	—	—	1.8
0.75	1.0	0.8	—
1.10	—	1.2	—
1.25	2.0	—	—
1.30	—	—	3.0
1.50	—	1.5	—
1.90	3.1	2.0	4.6
2.50	4.0	—	5.8

Determine the calibration curve equations for each component. Include the R^2 values.

- (e) From the data in part (d), determine which of the components has the highest calibration curve sensitivity? Which has the lowest?
- (f) A sample containing the three essential oils in part (d) gave the peak areas relative to the internal standard area: cinnamaldehyde, 2.6; eugenol, 0.9; thymol, 3.8. Determine the concentrations of each of the oils in the sample and the standard deviations in concentration.

- (g) A study was made of the decomposition of cinnamaldehyde in cinnamon oil. The oil was heated for various times at different temperatures. The following data were obtained:

Temp, °C	Time, min	Cinnamaldehyde, %
25, initial		90.9
40	20	87.7
—	40	88.2
—	60	87.9
60	20	72.2
—	40	63.1
—	60	69.1
100	20	66.1
—	40	57.6
—	60	63.1
140	20	64.4
—	40	53.7
—	60	57.1
180	20	62.3
—	40	63.1
—	60	52.2
200	20	63.1
—	40	64.5
—	60	63.3
210	20	74.9
—	40	73.4
—	60	77.4

Determine whether temperature has a statistical effect on the decomposition of cinnamaldehyde using analysis of variance (ANOVA). (For how to perform ANOVA, see S. R. Crouch and F. J. Holler, *Applications of Microsoft® Excel in Analytical Chemistry*, 3rd ed., Chap. 3, Belmont, CA: Cengage Learning, 2017.) In the same way, determine if time of heating has an effect.

- (h) Using the data in part (g), assume that decomposition begins at 60°C and test the hypothesis that there is no effect of temperature or time.

Capillary Electrophoresis, Electrochromatography, and Field-Flow Fractionation

This chapter deals with three relatively new separation methods. We first consider the principles of electrophoretic separations, with particular emphasis on **capillary electrophoresis**, and the applications of this versatile technique to various types of analytical problems. Capillary zone electrophoresis, capillary gel electrophoresis, capillary isotachopheresis, capillary isoelectric focusing, and micellar electrokinetic chromatography are described. A brief discussion of **electrochromatography** is then given. The chapter concludes with a discussion of the principles and applications of **field-flow fractionation** techniques, which are used in the separation of polymers, colloids, and other macromolecules.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

In capillary electrophoresis and electrochromatography, separations occur in a buffer-filled capillary tube under the influence of an electric field as seen in the schematic of Figure 30-1. Separations in field-flow fractionation, on the other hand, occur in a thin ribbon-like flow channel under the influence of a sedimentation, electrical, or thermal field applied perpendicular to the flow direction.

30A AN OVERVIEW OF ELECTROPHORESIS

Electrophoresis is a separation method based on the differential rate of migration of charged species in an applied dc electric field. This separation technique was first developed by the Swedish chemist Arne Tiselius in the 1930s for the study of serum proteins; he was awarded the 1948 Nobel Prize in Chemistry for this work.

Electrophoresis on a macro scale has been applied to a variety of difficult analytical separation problems: inorganic anions and cations, amino acids, catecholamines, drugs, vitamins, carbohydrates, peptides, proteins, nucleic acids, nucleotides, polynucleotides, and numerous other species.

A particular strength of electrophoresis is its unique ability to separate charged macromolecules of interest in biotechnology and in research in the biological sciences. For many years, electrophoresis has been the powerhouse method for separating proteins (enzymes, hormones, and antibodies) and nucleic acids (DNA and RNA) with unparalleled resolution. For example, to sequence DNA it is necessary to distinguish between long-chain polynucleotides that have as many as perhaps 200–500 bases and that differ by only a single nucleotide. Only electrophoresis has sufficient resolving power to handle this problem. Without electrophoresis, for example, the Human Genome Project would have been nearly impossible because human DNA contains some three billion nucleotides.

An electrophoretic separation is performed by injecting a small band of the sample into an aqueous buffer solution contained in a narrow tube or on a flat porous support medium

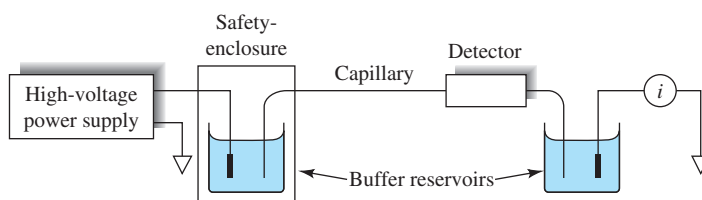


FIGURE 30-1 Schematic of a capillary electrophoresis system.

such as paper or a semisolid gel. A high voltage is applied across the length of the buffer by means of a pair of electrodes located at each end of the buffer. This field causes ions of the sample to migrate toward one of the electrodes. The rate of migration of a given species depends on its charge and its size. Separations are then based on differences in charge-to-size ratios for the various analytes in a sample. The larger this ratio, the faster an ion migrates in the electric field.

30A-1 Types of Electrophoresis

Electrophoretic separations are currently performed in two quite different formats: one is called *slab electrophoresis* and the other *capillary electrophoresis*. The first is the classical method that has been used for many years to separate complex, high-molecular-mass species of biological and biochemical interest. Slab separations are carried out on a thin flat layer or slab of a porous semisolid gel containing an aqueous buffer solution within its pores. This slab has dimensions of a few centimeters on a side and, like a chromatographic thin-layer plate, is capable of separating several samples simultaneously. Samples are introduced as spots or bands on the slab, and a dc electric field is applied across the slab for a fixed period. When the separations are complete, the field is discontinued and the separated species are visualized by staining in much the same way as was described for thin-layer chromatography in Section 28I-2.

Slab electrophoresis is now the most widely used separation tool in biochemistry and biology. Monographs, textbooks, and journals in the life sciences contain hundreds of photographs of developed electrophoretic slabs. Capillary electrophoresis, which is an instrumental version of electrophoresis, was developed in the mid to late 1980s. It has become an important tool for a wide variety of analytical separation problems. In many cases, this new method of performing electrophoretic separations is a substitute for slab electrophoresis with several important advantages that are described later in this chapter.

30A-2 The Basis for Electrophoretic Separations

The migration rate v of an ion (cm/s) in an electric field is equal to the product of the field strength E (V cm⁻¹) and the electrophoretic mobility μ_e (cm² V⁻¹ s⁻¹). That is,

$$v = \mu_e E \quad (30-1)$$

The electrophoretic mobility is in turn proportional to the ionic charge on the analyte and inversely proportional to frictional retarding factors. The electric field acts on only ions. If two species differ either in charge or in the frictional forces they experience while moving through the buffer, they will be separated from each other. Neutral species are not affected by the field and are thus not separated. The frictional retarding force on an analyte ion is determined by the size and shape of the ion and the viscosity of the migration medium. For ions of the same size, the greater the charge, the greater the driving force, and the faster the rate of migration. For ions of the same charge, the smaller the ion, the smaller the frictional forces, and the faster the rate of migration. The ion's *charge-to-size ratio* combines these two effects. Note that in contrast to chromatography, only one phase is involved in an electrophoretic separation.

30B CAPILLARY ELECTROPHORESIS

As useful as conventional slab electrophoresis is, this type of electrophoretic separation is typically slow, labor intensive, and difficult to automate. Slab electrophoresis does not yield very precise quantitative information. During the mid to late 1980s, there was explosive growth in research and application of electrophoresis performed in capillary tubes, and several commercial instruments appeared. *Capillary electrophoresis* (CE) yields high-speed, high-resolution separations on exceptionally small sample volumes (0.1–10 nL in contrast to slab electrophoresis, which requires samples in the μ L range). Additionally, the separated species are eluted from one end of the capillary, so quantitative detectors, similar to those found in high-performance liquid chromatography (HPLC), can be used instead of the cumbersome staining techniques of slab electrophoresis.¹

30B-1 Migration Rates in CE

As Equation 30-1 shows, the migration rate of an ion v depends on the electric field strength. The electric field in turn is proportional

¹For additional information on CE, see *Capillary Electrophoresis and Microchip Capillary Electrophoresis*, C. D. Garcia, K. Y. Chumbimuni-Torres, and E. Carrilho, eds., Hoboken, NJ: Wiley, 2013; *Analysis and Detection by Capillary Electrophoresis*, M. L. Marina, A. Rios, and M. Valcarcel, eds., Vol. 45 of *Comprehensive Analytical Chemistry*, D. Barcelo, ed., Amsterdam: Elsevier, 2005; R. Weinberger, *Practical Capillary Electrophoresis*, 2nd ed., New York: Academic Press, 2000; *High Performance Capillary Electrophoresis*, M. G. Khaleli, ed., New York: Wiley, 1998.

to the magnitude of the applied voltage V and inversely proportional to the length L over which it is applied. Thus

$$v = \mu_e \times \frac{V}{L} \quad (30-2)$$

This relationship indicates that high applied voltages are desirable to achieve rapid ion migration and fast separations. It is desirable to have rapid separations, but it is even more important to achieve high-resolution separations. So we must examine the factors that determine resolution in electrophoresis.

30B-2 Plate Heights in CE

In chromatography, both longitudinal diffusion and mass-transfer resistance contribute to band broadening. However, because only a single phase is used in electrophoresis, in theory only longitudinal diffusion needs to be considered. In practice, however, Joule heating can add variance as can the injection process. Although CE is not a chromatographic process, separations are often described in a manner similar to chromatography. For example, in electrophoresis, we calculate the plate count N by

$$N = \frac{\mu_e V}{2D} \quad (30-3)$$

where D is the diffusion coefficient of the solute ($\text{cm}^2 \text{s}^{-1}$). Because resolution increases as the plate count increases, it is desirable to use high applied voltages to achieve high-resolution separations. Note that for electrophoresis, contrary to the situation in chromatography, the plate count does not increase with the length of the column.

With gel slab electrophoresis, joule heating limits the magnitude of the applied voltage to about 500 V. Here, one of the strengths of the capillary format compared with the slab format is realized. Because the capillary is quite long and has a small cross-sectional area, the solution resistance through the capillary is exceptionally high. Because power dissipation is inversely

proportional to resistance ($P = I^2/R$), much higher voltages can be applied to capillaries than to slabs for the same amount of heating. Additionally, the high surface-to-volume ratio of the capillary provides efficient cooling. As a result of these two factors, band broadening due to thermally driven convective mixing does not occur to a significant extent in capillaries. Electric fields of 100–400 V/cm are typically used. High-voltage power supplies of 10–25 kV are normal. The high fields lead to corresponding improvements in speed and resolution over those seen in the slab format. CE peak widths often approach the theoretical limit set by longitudinal diffusion. CE normally yields plate counts in the range of 100,000–200,000, compared to the 5,000–20,000 plates typical for HPLC. Plate counts of nearly 3,000,000 have been reported for capillary zone electrophoresis of dansylated amino acids,² and plate counts of more than 10,000,000 have been reported for capillary gel electrophoresis of polynucleotides.³

30B-3 Electroosmotic Flow

A unique feature of CE is *electroosmotic flow*. When a high voltage is applied across a fused-silica capillary tube containing a buffer solution, electroosmotic flow usually occurs, in which the bulk liquid migrates toward the cathode. The rate of migration can be substantial. For example, a 50 mM pH 8 buffer flows through a 50-cm capillary toward the cathode at approximately 5 cm/min with an applied voltage of 25 kV.⁴

As shown in Figure 30-2, the cause of electroosmotic flow is the electric double layer that develops at the silica-solution interface. At pH values higher than 3, the inside wall of a silica

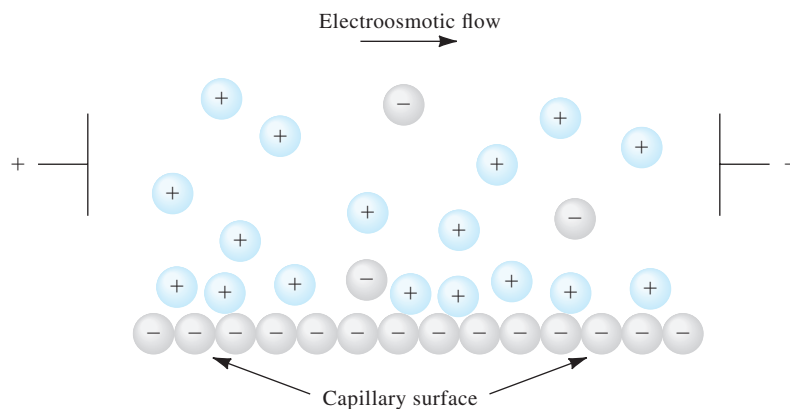


FIGURE 30-2 Charge distribution at a silica-capillary interface and resulting electroosmotic flow. (From A. G. Ewing, R. A. Wallingford, and T. M. Olefirowicz, *Anal. Chem.*, **1989**, 61, 292A, DOI: 10.1021/ac00179a722. Copyright 1989 American Chemical Society.)

²R. D. Smith, J. A. Olivares, N. T. Nguyen, and H. R. Udseth, *Anal. Chem.*, **1988**, 60, 436, DOI: 10.1021/ac00156a013.

³A. Guttman, A. S. Cohen, D. N. Heiger, and B. L. Karger, *Anal. Chem.*, **1990**, 62, 137, DOI: 10.1021/ac00201a010.

⁴J. D. Olechno, J. M. Y. Tso, J. Thayer, and A. Wainright, *Amer. Lab.*, **1990**, 22 (17), 51.

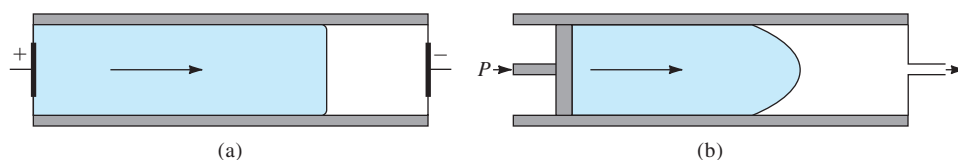


FIGURE 30-3 Flow profiles for liquids under (a) electroosmotic flow and (b) pressure-induced flow.

capillary is negatively charged because of ionization of the surface silanol groups ($\text{Si}-\text{OH}$). Buffer cations congregate in the electrical double layer adjacent to the negative surface of the silica capillary. The cations in the diffuse outer layer of the double layer are attracted toward the cathode, or negative electrode, and because they are solvated, they drag the bulk solvent along with them. As shown in Figure 30-3a, electroosmosis leads to bulk solution flow that has a flat profile across the tube because flow originates at the walls of the tubing. This profile is in contrast to the laminar (parabolic) profile observed with the pressure-driven flow encountered in HPLC (Figure 30-3b). Because the profile is essentially flat, electroosmotic flow does not contribute significantly to band broadening the way pressure-driven flow does in liquid chromatography.

The rate of electroosmotic flow is generally greater than the electrophoretic migration velocities of the individual ions and effectively becomes the mobile-phase pump of CE. Even though analytes migrate according to their charges within the capillary, the electroosmotic flow rate is usually sufficient to sweep all positive, neutral, and even negative species toward the same end of the capillary, so all can be detected as they pass by a common point (see Figure 30-4). The resulting *electropherogram* looks like a chromatogram but with narrower peaks.

The electroosmotic flow velocity v is given by an equation similar to Equation 30-1. That is,

$$v = \mu_{\text{eo}}E \quad (30-4)$$

In the presence of electroosmosis, the velocity of an ion is the sum of its electrophoretic migration velocity and the electroosmotic flow velocity. Thus,

$$v = (\mu_e + \mu_{\text{eo}})E \quad (30-5)$$

As a result of electroosmosis, the order of elution in a typical electrophoretic separation is, first, the fastest cation followed by successively slower cations, then all the neutrals in a single zone, and finally the slowest anion followed by successively faster anions (see Figure 30-4). In some instances, the rate of electroosmotic flow may not be great enough to surpass the rate at which some of the anions move toward the anode, in which case these species move in that direction instead of toward the cathode.

The migration time t_m in CE is the time it takes for a solute to migrate from the point of introduction to the detector. If a capillary of total length L is used and the length to the detector is l , the migration time is

$$t_m = \frac{l}{(\mu_e + \mu_{\text{eo}})E} = \frac{lL}{(\mu_e + \mu_{\text{eo}})V} \quad (30-6)$$

The number of theoretical plates in the presence of electroosmotic flow can be found from an expression analogous to Equation 26-21:

$$N = 16 \left(\frac{t_m}{W} \right)^2 \quad (30-7)$$

where W , as in chromatography, is the peak width measured at the base of the peak.

It is possible to reverse the direction of the normal electroosmotic flow by adding a cationic surfactant to the buffer. The surfactant adsorbs on the capillary wall and makes the wall positively charged. Now buffer anions congregate near the wall and are swept toward the positive electrode. This ploy is often used to speed up the separation of anions.



Simulation: Learn more about **capillary electrophoresis** at www.tinyurl.com/skoogpia7

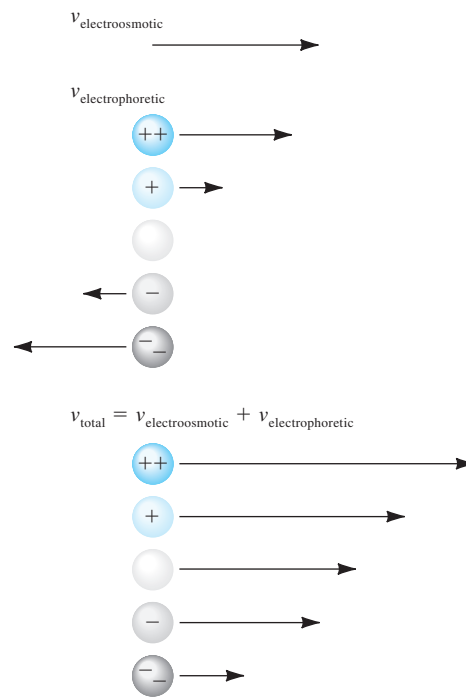


FIGURE 30-4 Velocities in the presence of electroosmotic flow. The length of the arrow next to an ion indicates the magnitude of its velocity; the direction of the arrow indicates the direction of motion. The negative electrode is to the right and the positive electrode to the left of this section of solution.

Electroosmosis is often desirable in certain types of CE, but in other types it is not. Electroosmotic flow can be minimized by modifying the inside capillary walls with a reagent like trimethylchlorosilane that bonds to the surface and reduces the number of surface silanol groups (see Section 28D-1).

30B-4 Instrumentation for CE

As shown in Figure 30-1, the instrumentation for CE is relatively simple. Although CE instruments can be readily constructed in the laboratory, commercial instruments are often desirable for safety reasons and for ease of connecting detectors. CE instruments are available from a number of instrument companies. A buffer-filled fused-silica capillary, typically 10–100 μm in internal diameter and 30–100 cm long, extends between two buffer reservoirs that also hold platinum electrodes. Like the capillary tubes used in gas chromatography (GC), the outside walls of the fused-silica capillary are typically coated with polyimide for durability, flexibility, and stability. The sample is introduced at one end and detection occurs at the other. A voltage of 5–30 kV dc is applied across the two electrodes. The polarity of this high voltage can be as indicated in Figure 30-1 or can be reversed to allow rapid separation of anions. High-voltage electrophoresis compartments are usually safety interlocked to protect the user.

Although the instrumentation is conceptually simple, significant experimental difficulties in sample introduction and detection arise due to the very small volumes involved. Because the volume of a normal capillary is 4–5 μL , injection and detection volumes must be on the order of a few nanoliters or less.

Sample Introduction

The most common sample-introduction methods are *electrokinetic injection* and *pressure injection*. With electrokinetic injection, one end of the capillary and its electrode are removed from their buffer compartment and placed in a small cup containing the sample. A voltage is then applied for a measured time, causing the sample to enter the capillary by a combination of ionic migration and electroosmotic flow. The capillary end and electrode are then returned to the regular buffer solution for the duration of the separation. This injection technique discriminates by injecting larger amounts of the more mobile ions relative to the slower-moving ions.

In pressure injection, the sample-introduction end of the capillary is also placed in a small cup containing the sample, but here a pressure difference drives the sample solution into the capillary. The pressure difference can be produced by applying a vacuum at the detector end, by pressurizing the sample, or by elevating the sample end (hydrodynamic injection). Pressure injection does not discriminate because of ion mobility, but it cannot be used in gel-filled capillaries.

For both electrokinetic injection and pressure injection, the volume injected is controlled by the duration of the injection. Injections of 5–50 nL are common, and volumes below 100 pL have been reported. For a buffer with density and viscosity near

TABLE 30-1 Detectors for CE

Type of Detector	Representative Detection Limit* (attomoles detected)
Spectrometry	
Absorption [†]	1–1000
Fluorescence	1–0.01
Thermal lens [†]	10
Raman [†]	1000
Chemiluminescence [†]	1–0.0001
Mass spectrometry	1–0.01
Electrochemical	
Conductivity [†]	100
Potentiometry [†]	1
Amperometry	0.1

Sources: B. Huang, J. J. Li, L. Zhang, J. K. Cheng, *Anal. Chem.*, **1996**, 68, 2366, DOI: 10.1021/ac9511253; S. C. Beale, *Anal. Chem.*, **1998**, 70, 279R, DOI: 10.1021/a19800141; S. N. Krylov and N. J. Dovichi, *Anal. Chem.*, **2000**, 72, 111R, DOI: 10.1021/a1000014c; S. Hu and N. J. Dovichi, *Anal. Chem.*, **2002**, 74, 2833, DOI: 10.1021/ac0202379.

*Detection limits quoted have been determined with injection volumes ranging from 18 pL to 10 nL.

[†]Mass detection limit converted from concentration detection limit using a 1-nL injection volume.

the values for water, a height differential of 5 cm for 10 s hydrodynamically injects about 6 nL with a 75- μm inside-diameter capillary.

Microinjection tips constructed from capillaries drawn to very small diameters allow sampling from picoliter environments such as single cells or substructures within single cells. This technique has been used to study amino acids and neurotransmitters from single cells. Other novel injection techniques have been described in the literature.⁵ Commercial CE systems are available with thermostatted multiposition carousels for automated sampling.

Detection

Because the separated analytes move past a common point in most types of CE, detectors are similar in design and function to those described for HPLC. Table 30-1 lists several of the detection methods that have been reported for CE. The second column of the table shows representative detection limits for these detectors.

Absorption Methods. Both fluorescence and absorption detectors are widely used in CE, although absorption methods are more common because they are more generally applicable. To keep the detection volume on the nanoliter scale or smaller, detection is performed on-column. In this case a small section

⁵For examples, see C.-C. Chen et al., *Anal. Chem.*, **2007**, 79, 195, DOI: 10.1021/ac061377b; S. I. Cho et al., *J. Chromatogr. A*, **2005**, 1063, 253, DOI: 10.1016/j.chroma.2004.11.088; L. M. Ponton and C. E. Evans, *Anal. Chem.*, **2001**, 73, 1974, DOI: 10.1021/ac001145r.

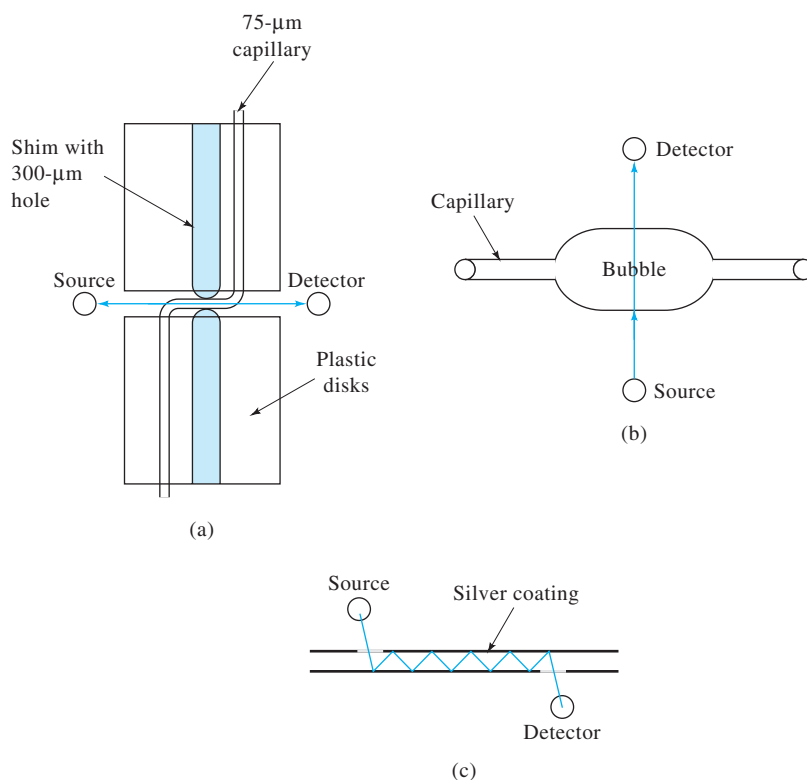


FIGURE 30-5 Three types of cells for improving the sensitivity of absorption measurements in CE: (a) a 3-mm Z cell, (b) a 150- μm bubble cell, (c) a multireflection cell.

of the protective polyimide coating is removed from the exterior of the capillary by burning or etching. That section of the capillary then serves as the detector cell. Unfortunately, the path length for such measurements is no more than 50–100 μm , which restricts detection limits in concentration terms; because such small volumes are involved, however, mass detection limits are equal to or better than those for HPLC.

Several cell designs have been used for increasing the measurement path length to improve the sensitivity of absorption methods. Three of these are shown in Figure 30-5. In the commercial detector shown in Figure 30-5a, the end of the capillary is bent into a Z shape, which produces a path length as long as ten times the capillary diameter. Increases in path length can lead to decreases in peak efficiency and thus resolution. In some cases, special lenses, such as spherical ball lenses, are inserted between the source and the Z cell and between the cell and the detector.⁶ Such lenses improve sensitivity by focusing the light into the cell and onto the detector.

Figure 30-5b shows a second way to increase the absorption path length. In this example, a bubble is formed near the end of the capillary. In the commercial version of this technique, the

bubble for a 50- μm capillary has an inside diameter of 150 μm , which gives a threefold increase in path length.

A third method for increasing the path length of radiation by reflection is shown in Figure 30-5c. In this technique, a reflective coating of silver is deposited on the end of the capillary. The source radiation then undergoes multiple reflections during its transit through the capillary, which significantly increase the path length.

Commercial CE systems are available with diode array and charge-coupled device (CCD) detectors that allow spectra to be collected over the UV-visible range in less than 1 s.

Indirect Detection. Indirect absorbance detection has been used for species of low molar absorptivity that are difficult to detect without derivatization. An ionic chromophore is placed in the electrophoresis buffer. The detector then receives a constant signal due to the presence of this substance. The analyte displaces some of these ions, just as in ion-exchange chromatography, so that the detector signal decreases during the passage of an analyte band through the detector. The analyte is then determined from the decrease in absorbance. The electropherogram in Figure 30-6 was generated by using indirect absorbance detection with 4-mM chromate ion as the chromophore; this ion absorbs radiation strongly at 254 nm in the buffer. Although the peaks obtained are

⁶D. M. Spence, A. M. Sekelsky, and S. R. Crouch, *Instrum. Sci. Technol.*, **1996**, 24, 103, DOI: 10.1080/10739149608000472.

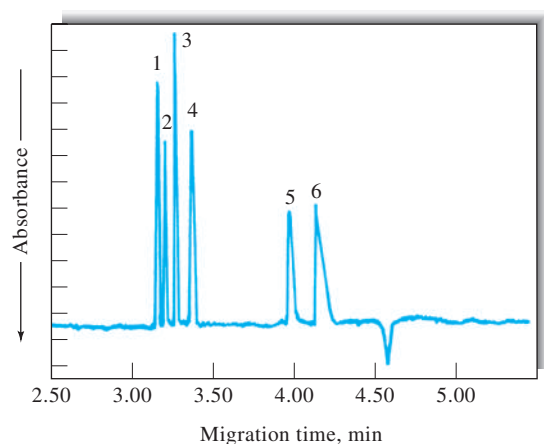


FIGURE 30-6 Electropherogram of a six-anion mixture by indirect detection with 4-mM chromate ion at 254 nm. Peak: (1) bromide (4 ppm), (2) chloride (2 ppm), (3) sulfate (4 ppm), (4) nitrate (4 ppm), (5) fluoride (1 ppm), (6) phosphate (6 ppm).

negative (decreasing A) peaks, they appear as positive peaks in Figure 30-6 because the detector polarity was reversed.

Fluorescence Detection. Just as in HPLC, fluorescence detection yields increased sensitivity and selectivity for fluorescent analytes or fluorescent derivatives. Laser-based instrumentation is preferred to focus the excitation radiation on the small capillary and to achieve the low detection limits available from intense sources. Laser-induced-fluorescence attachments are available that couple with commercial CE instruments. Laser-induced fluorescence has allowed detection of as little as 10 zeptomoles or 6000 molecules.⁷

⁷S. Wu and N. Dovichi, *J. Chromatogr.*, **1989**, 480, 141, DOI: 10.1016/S0021-9673(01)84284-9.

Electrochemical Detection. Two types of electrochemical detection have been used with CE: conductivity and amperometry. One of the problems with electrochemical detection has been that of isolating the detector electrodes from the high voltage required for the separation. One method for isolation involves inserting a porous glass or graphite joint between the end of the capillary and a second capillary containing the detector electrodes.

Mass Spectrometric Detection. The very small volumetric flow rates of less than 1 $\mu\text{L}/\text{min}$ from electrophoresis capillaries make it feasible to couple the effluent directly to the ionization source of a mass spectrometer. The most common sample-introduction and ionization interface for this purpose is electrospray (Section 20B-4), although fast atom bombardment, matrix-assisted laser desorption-ionization (MALDI) spectrometry, and inductively coupled plasma mass spectrometry (ICPMS) have also been used. Because the liquid sample must be vaporized before entering the mass spectrometry (MS) system, it is important that volatile buffers be used. Capillary electrophoresis-mass spectrometry (CE/MS) systems have become quite important in the life sciences for determining large biomolecules that occur in nature, such as proteins, DNA fragments, and peptides.⁸

Figure 30-7 is a schematic of a typical electrospray interface coupled to a quadrupole mass spectrometer. Note that the capillary is positioned between the isolated high-voltage region and the electrospray source. The high-voltage end of the capillary was at 30–50 kV with respect to common. The low-voltage end was maintained at 3–5 kV and charged the droplets.

⁸For more information on mass spectrometric detection, see J. A. Olivares, N. T. Nguyen, C. R. Yonker, and R. D. Smith, *Anal. Chem.*, **1987**, 59, 1230, DOI: 10.1021/ac00135a034; D. C. Simpson and R. D. Smith, *Electrophoresis*, **2005**, 26, 1291, DOI: 10.1002/elps.200410132; K. Vuignier, J.-L. Veuthey, P.-A. Carrupt, and J. Schappler, *Electrophoresis*, **2012**, 33, 3306, DOI: 10.1002/elps.201200116.

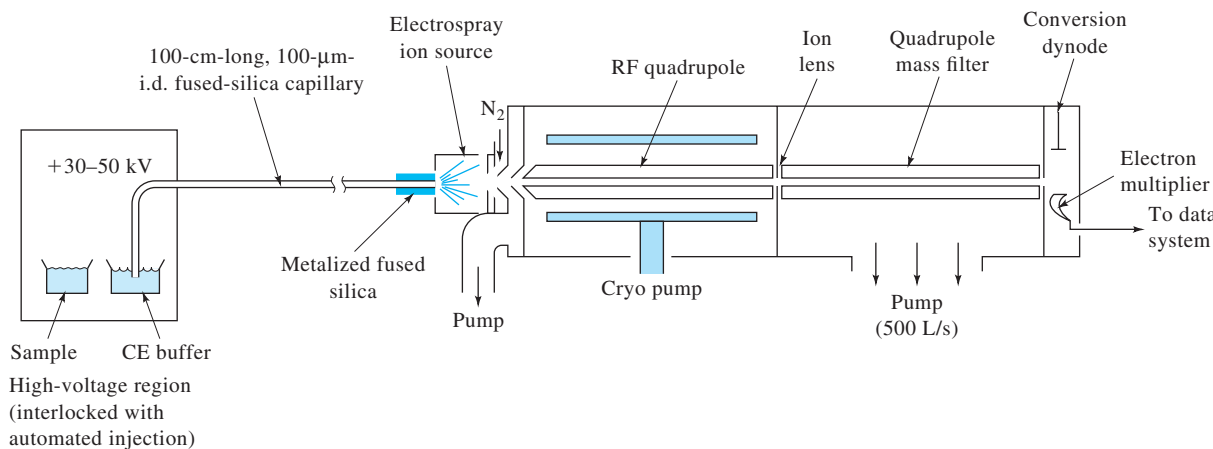


FIGURE 30-7 An instrument for CE/MS. The high-voltage (anode) end was maintained at 30–50 kV in an electrically isolated, interlocked box. Electrical contact at the low-voltage (cathode) end was made by silver deposited on the capillary and a stainless steel sheath. This electrical contact was at 3–5 kV with respect to common, which also charged the electrospray. The flow of nitrogen at $\sim 70^\circ\text{C}$ for desolvation was 3–6 L/min. (From R. D. Smith, J. A. Olivares, N. T. Nguyen, and H. R. Udseth, *Anal. Chem.*, **1988**, 60, 436, DOI: 10.1021/ac00156a013. With permission.)

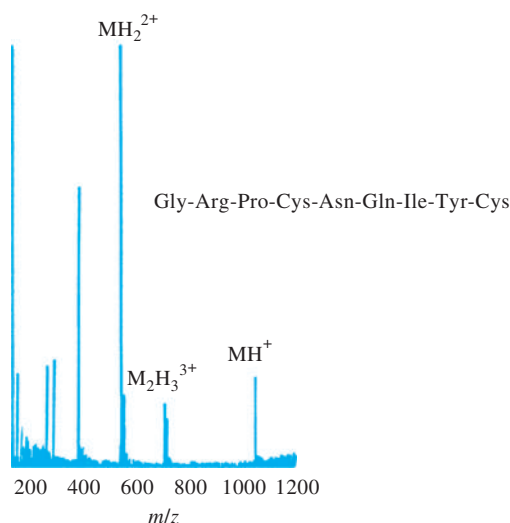


FIGURE 30-8 Electrospray ionization mass spectrum for vasotocin. (From R. D. Smith, J. A. Olivares, N. T. Nguyen, and H. R. Udseth, *Anal. Chem.*, **1988**, 60, 436. With permission.)

Similar electrospray instruments are available commercially coupled with either quadrupole or ion-trap mass spectrometers.⁹ Ion-trap mass spectrometers can allow CE/MS/MS or CE/MSⁿ operation.

Figure 30-8 shows the electrospray mass spectrum obtained for vasotocin, a polypeptide having a molecular mass of 1050. Note the presence of doubly and triply charged species. With higher-molecular-mass species, ions are often observed with charges of +12 or more. Ions with such large charges make it possible to detect high-molecular-mass analytes with a quadrupole instrument with a relatively modest mass range. Typical detection limits for CE/MS are of a few tens of femtomoles for molecules with molecular masses of 100,000 or more.

Commercial CE Systems

Complete CE systems are sold by several instrument companies. More than two dozen companies offer supplies and accessories for CE. The initial cost of equipment and the expense of maintenance for CE are generally significantly lower than those for ion chromatographic and atomic spectroscopic instruments. Thus, commercial CE instruments with standard absorption or fluorescence detectors cost in the range of \$10,000 to \$75,000. Addition of mass spectrometric detection can raise the cost significantly.

30C APPLICATIONS OF CE

Capillary electrophoretic separations can be performed in several different modes.¹⁰ These include capillary zone electrophoresis, capillary gel electrophoresis, capillary isoelectric

focusing, capillary isotachopheresis, and micellar electrokinetic chromatography. The sections that follow illustrate typical applications of each of these techniques.

30C-1 Capillary Zone Electrophoresis

In capillary zone electrophoresis (CZE), the buffer composition is constant throughout the region of the separation. The applied field causes each of the different ionic components of the mixture to migrate according to its own mobility and to separate into zones that may be completely resolved or that may be partially overlapped. Completely resolved zones have regions of buffer between them as illustrated in Figure 30-9a. The situation is analogous to elution column chromatography, where regions of mobile phase are located between zones containing separated analytes.

Separation of Small Ions

For most electrophoretic separations of small ions, the smallest analysis time occurs when the analyte ions move in the same direction as the electroosmotic flow. Thus, for cation separations, the walls of the capillary are untreated and the electroosmotic flow and the cation movement are toward the cathode. For the separation of anions, on the other hand, the electroosmotic flow is usually reversed by treating the walls of the capillary with an alkyl ammonium salt, such as cetyl trimethylammonium bromide. The positively charged ammonium ions become attached to the silica surface, yielding a positively charged, immobile surface layer. This, in turn, creates a negatively charged, mobile solution layer, which is attracted toward the anode, reversing the electroosmotic flow.

In the past, the most common method for analysis of small anions has been ion chromatography. For cations, the preferred techniques have been atomic absorption spectroscopy and ICPMS. Figure 30-10 illustrates the speed and resolution of electrophoretic separations of small anions. Here, thirty anions were separated cleanly in just more than 3 minutes. Typically, an ion-exchange separation of only three or four anions can be accomplished in this brief period. Figure 30-11 (page 802) further illustrates the speed at which separations can be carried out. In this example, nineteen cations were separated in less than 2 minutes. CE methods were once predicted to replace the more established methods because of lower equipment costs, smaller-sample-size requirements, and shorter analysis times. However, because variations in electroosmotic flow rates make reproducing CE separations difficult, LC methods and atomic spectrometric methods are still widely used for small inorganic ions.

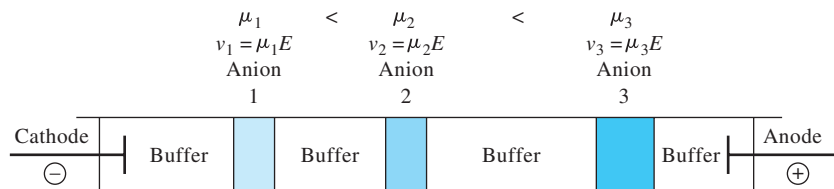
Separation of Molecular Species

A variety of synthetic herbicides, pesticides, and pharmaceuticals that are ions or can be derivatized to yield ions have been separated and analyzed by CZE. Figure 30-12 (page 802) illustrates this type of application, in which three anti-inflammatory drugs with ionizable carboxyl groups are separated in less than 15 min.

⁹Agilent Technologies, Santa Clara, CA.

¹⁰For reviews, see V. A. Galievsky, A. S. Stasheuski, S. N. Krylov, *Anal. Chem.*, **2015**, 87, 157, DOI: 10.1021/ac504219r; M. Geiger, A. L. Hogerton, and M. T. Bowser, *Anal. Chem.*, **2012**, 84, 577, DOI: 10.1021/ac203205a; N. W. Frost, M. Jing, and M. T. Bowser, *Anal. Chem.*, **2010**, 82, 4682, DOI: 10.1021/ac101151k.

(a) Zone electrophoresis



(b) Isotachopheresis



(c) Isoelectric focusing

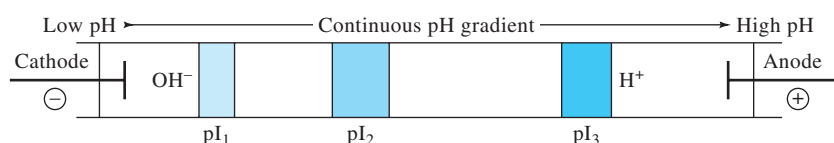


FIGURE 30-9 Three modes of separation by electrophoresis. In zone electrophoresis (a), the ions separate into zones 1, 2, and 3. The zones shown are completely resolved with buffer between each zone. In isotachopheresis (b), the sample is injected between a leading, high-mobility buffer and a trailing, low-mobility buffer. In isoelectric focusing (c), a continuous pH gradient exists along the length of the capillary. Analyte ions migrate to the pH corresponding to their isoelectric points.

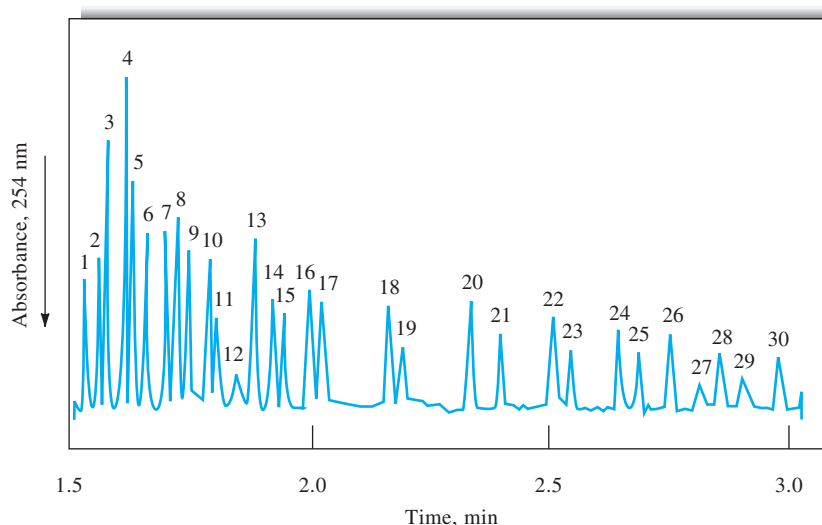


FIGURE 30-10 Electropherogram showing the separation of thirty anions. Capillary internal diameter 50 μm (fused silica). Detection: indirect UV, 254 nm. Peaks 1 = thio-sulfate (4 ppm), 2 = bromide (4 ppm), 3 = chloride (2 ppm), 4 = sulfate (4 ppm), 5 = nitrite (4 ppm), 6 = nitrate (4 ppm), 7 = molybdate (10 ppm), 8 = azide (4 ppm), 9 = tungstate (10 ppm), 10 = monofluorophosphate (4 ppm), 11 = chlorate (4 ppm), 12 = citrate (2 ppm), 13 = fluoride (1 ppm), 14 = formate (2 ppm), 15 = phosphate (4 ppm), 16 = phosphite (4 ppm), 17 = chlorite (4 ppm), 18 = galactarate (5 ppm), 19 = carbonate (4 ppm), 20 = acetate (4 ppm), 21 = ethanesulfonate (4 ppm), 22 = propionate (5 ppm), 23 = propanesulfonate (4 ppm), 24 = butyrate (5 ppm), 25 = butanesulfonate (4 ppm), 26 = valerate (5 ppm), 27 = benzoate (4 ppm), 28 = *l*-glutamate (5 ppm), 29 = pentanesulfonate (4 ppm), 30 = *d*-gluconate (5 ppm). (From W. R. Jones and P. Jandik, *J. Chromatogr. A*, **1991**, 546, 445, DOI: 10.1016/s0021-9673(01)93043-2. With permission.)

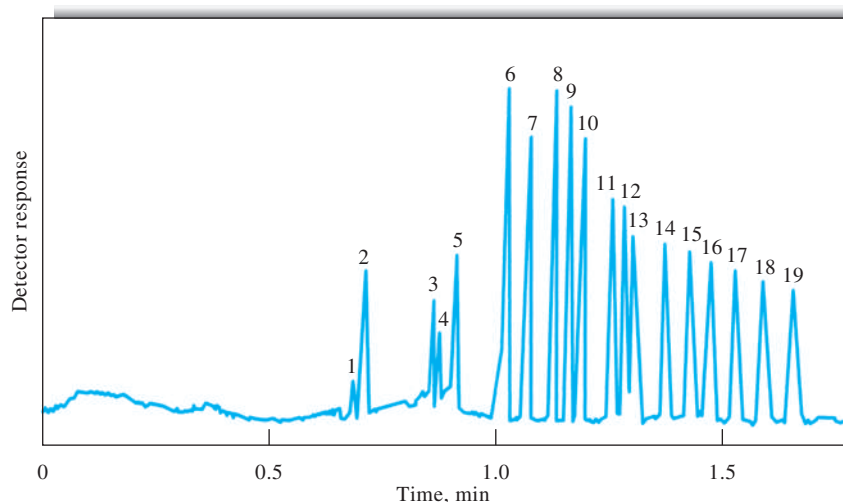


FIGURE 30-11 Separation of alkali, alkaline earths, and lanthanides. Capillary: 36.5 cm $\times$ 75 μ m fused silica, +30 kV. Injection: hydrostatic, 20 s at 10 cm. Detection: indirect UV, 214 nm. Peaks 1 = rubidium (2 ppm), 2 = potassium (5 ppm), 3 = calcium (2 ppm), 4 = sodium (1 ppm), 5 = magnesium (1 ppm), 6 = lithium (1 ppm), 7 = lanthanum (5 ppm), 8 = cerium (5 ppm), 9 = praseodymium (5 ppm), 10 = neodymium (5 ppm), 11 = samarium (5 ppm), 12 = europium (5 ppm), 13 = gadolinium (5 ppm), 14 = terbium (5 ppm), 15 = dysprosium (5 ppm), 16 = holmium (5 ppm), 17 = erbium (5 ppm), 18 = thulium (5 ppm), 19 = ytterbium (5 ppm). (From A. Weston, P. R. Brown, P. Jandik, W. R. Jones, and A. L. Heckenberg, *J. Chromatogr. A*, **1992**, 593, 289, DOI: 10.1016/0021-9673(92)80297-8. With permission.)

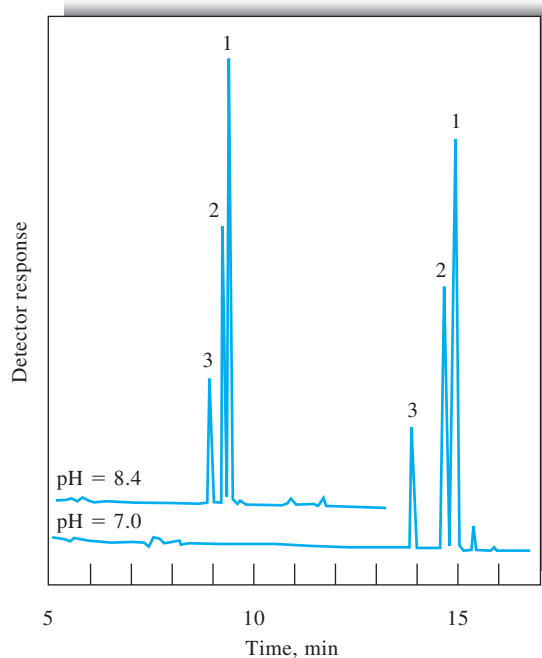


FIGURE 30-12 Separation of anti-inflammatory drugs by CZE. Detection: UV at 215 nm. Analytes: (1) naproxen, (2) ibuprofen, and (3) tolmetin. (Reprinted with permission from A. Wainright, *J. Microcolumn. Sep.*, **1990**, 2, 166, DOI: 10.1002/mcs.1220020403.)

Proteins, amino acids, and carbohydrates have all been separated in minimum times by CZE. In the case of neutral carbohydrates, the separations are preceded by formation of negatively charged borate complexes. CE has become a major technique in the rapidly expanding field of proteomics. Analyses of proteins in urine samples are being used to diagnose chronic kidney and coronary artery disease. Fingerprinting of various proteins by CZE is also becoming important. Figure 30-13 shows an example of CZE with laser-induced fluorescence detection for identification of *Staphylococcus* species. Capillary gel electrophoresis is widely used in DNA sequencing as discussed in the next section.

30C-2 Capillary Gel Electrophoresis

Capillary gel electrophoresis (CGE) is generally performed in a porous gel polymer matrix with a buffer mixture that fills the pores of the gel. In early slab electrophoresis studies, the primary purpose of the polymeric medium was to reduce analyte dispersion by convection and diffusion and to provide a convenient medium for detection and scanning. This type of medium provided a molecular sieving action that retarded the migration of analyte species to various extents depending on the pore size of

the polymer and the size of the analyte ions. This sieving action is particularly helpful in separating macromolecules such as proteins, DNA fragments, and oligonucleotides that have similar charges but differ in size. Most macroscale electrophoresis separations are now carried out on a gel slab, but some capillary electrophoretic separations of species that differ in size are also performed in gels contained in capillary tubes.

Types of Gels

The most common type of gel used in electrophoresis is a polyacrylamide polymer formed by polymerizing acrylamide ($\text{CH}_2=\text{CH}-\text{CO}-\text{NH}_2$) in the presence of a cross-linking agent. The pore size of the polymer depends on the ratio of monomer to cross-linking agent. Increases in the amount of the cross-linking agent produce smaller pore size. Other gels that have been used for CGE include agarose (a polysaccharide extracted from a marine alga), methyl cellulose, various derivatives of cellulose, polyvinyl alcohol, polyethyleneoxide, dextran, polydimethylacrylamide, and polyethylene glycol. Because of the difficulty of filling capillaries with rigid gels, entangled polymer gel networks are often used. These allow easy filling and flushing of the capillary. A typical separation of

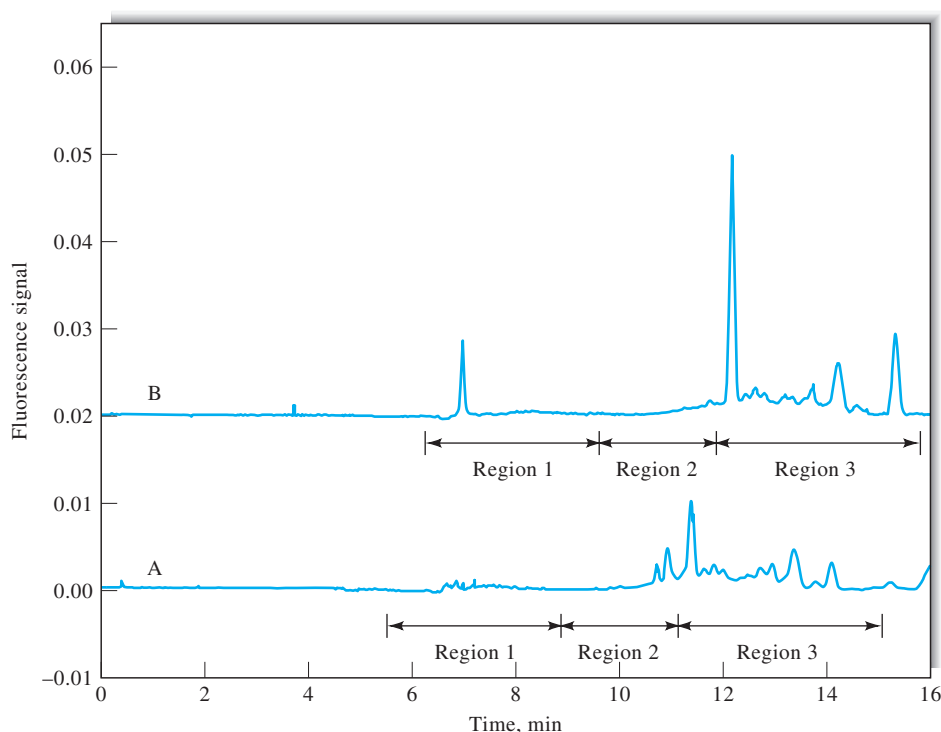


FIGURE 30-13 Fingerprinting proteins by CZE. The electropherograms show samples of *Staphylococcus* (*S. aureus*) grown in two different culture media (A and B). The CE peaks were clustered in the three regions identified. Migration times were used to identify culture media and to differentiate *S. aureus* from *S. epidermidis*. Capillary: uncoated 50 μm fused silica. Detection: laser-induced fluorescence. Buffer: pH 7.0 phosphate. (Adapted from M. T. Valedo, C. Pelaez-Lorenzo, R. Gonzalez, M. de Frutos, J. C. Diez-Masa, *Anal. Chim. Acta*, **2010**, 658, 81, DOI: 10.1016/j.aca.2009.10.053.)

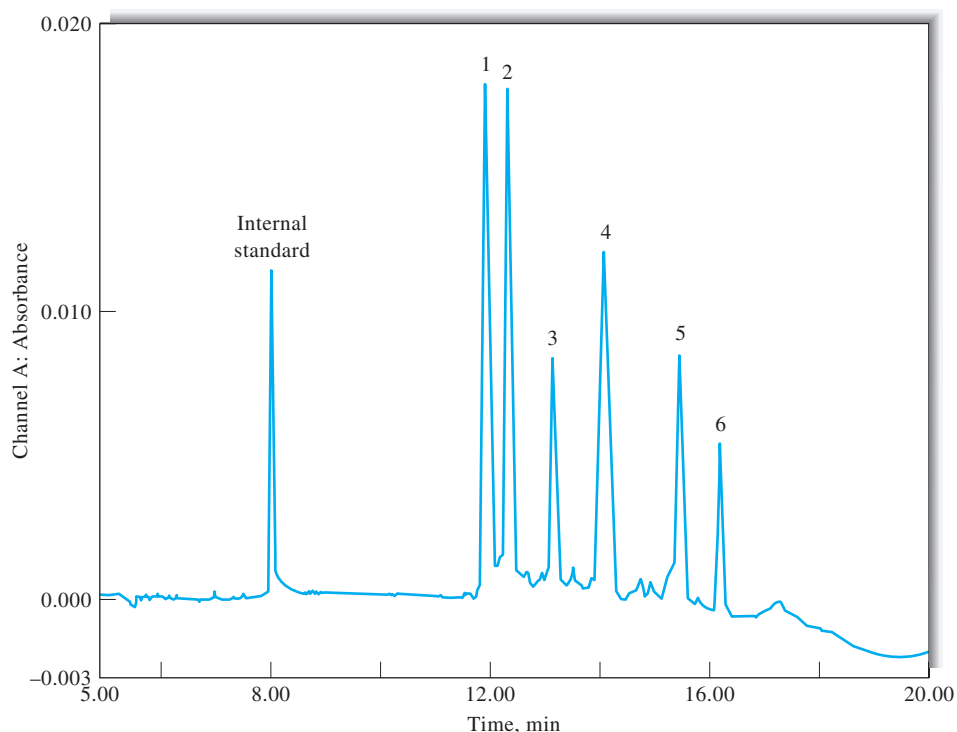


FIGURE 30-14 Capillary gel separation of SDS denatured proteins in a polyethylene glycol column. UV absorption detection at 214 nm. Proteins: (1) α -lactalbumin, (2) soybean trypsin inhibitor, (3) carbonic anhydrase, (4) ovalbumin, (5) bovine serum albumin, and (6) phosphorylase B. (Reprinted from K. Ganzler et al., *Anal. Chem.*, **1992**, 64, 2665. Copyright 1992 American Chemical Society.)

a standard protein mixture in a polyethylene glycol gel is illustrated in Figure 30-14.

CE in DNA Sequencing

A major goal of the Human Genome Project was to determine the order of occurrence of the four bases, adenine (A), cytosine (C), guanine (G), and thymine (T), in DNA molecules. The sequence defines an individual's genetic code. The need for sequencing DNA has spawned the development of several new analytical instruments. Among the most attractive of these approaches is capillary array electrophoresis.¹¹ In this technique, as many as ninety-six capillaries are operated in parallel. The capillaries are filled with a separation matrix, normally a linear polyacrylamide gel. The capillaries have inner diameters of 35–75 μm and are 30–60 cm long.

In sequencing, DNA extracted from cells is fragmented by various approaches. Depending on the terminal base in the fragment, one of four fluorescent dyes is attached to the various fragments. The sample contains many different-size fragments, each with a fluorescent label. Under the influence of the electrophoretic field, lower-molecular-mass fragments move faster and arrive at the detector sooner than higher-molecular-mass fragments. The DNA sequence is determined by the dye color

sequence of the eluting fragments. Lasers are used to excite the dye fluorescence. Several different techniques have been described for detecting the fluorescence. One method uses a scanning system in which the capillary bundle is moved relative to the excitation laser and a four-wavelength detection system. In the detection system illustrated in Figure 30-15, a laser beam is focused onto the capillary array by a lens. The region illuminated by the laser is imaged onto a CCD detector (see Section 7E-3). Filters allow wavelength selection to detect the four colors. Simultaneous separation of eleven DNA fragments in 100 capillaries has been reported.¹² Other designs include sheath-flow detector systems and a detector that uses two diode lasers for excitation. Commercial instrumentation is available with prices ranging from \$90,000 to more than \$1,000,000.¹³ DNA sequencers and other genetic analyzers have been miniaturized using lab-on-a-chip technology.¹⁴ CE played a major role in identifying human remains in the debris of the World Trade Center disaster.

¹²K. Ueno and E. S. Yeung, *Anal. Chem.*, **1994**, 66, 1424, DOI: 10.1021/ac00081a010.

¹³For reviews of DNA sequencers, see X. Li et al., *PLoS One*, **2013**, 8, e67744, DOI: 10.1371/journal.pone.0067744; E. Ban, et al., *Electrophoresis*, **2012**, 33, 2, DOI: 10.1002/elps.201100344; T. C. Glenn, *Mol. Ecol. Resour.*, **2011**, 11, 759, DOI: 10.1111/j.1755-0998.2011.03024.x; H. Kambara, *Chem. Rec.*, **2010**, 10, 8, DOI: 10.1002/tcr.200900010.

¹⁴C. A. Emrich, H. Tian, I. L. Medintz, and R. A. Mathies, *Anal. Chem.*, **2002**, 74, 5076, DOI: 10.1021/ac020236g; Agilent Technologies, Santa Clara, CA.

¹¹For a review, see I. Kheterpal and R. A. Mathies, *Anal. Chem.*, **1999**, 71, 31A, DOI: 10.1021/ac990099w.

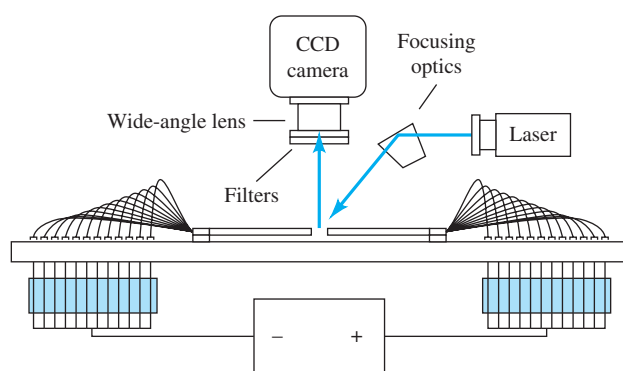


FIGURE 30-15 On-column laser fluorescence detection system for capillary array electrophoresis. A laser is focused as a line onto the array of capillaries at a 45° angle. The fluorescence is filtered and detected by a CCD camera through a wide-angle lens. (From K. Ueno and E. S. Yeung, *Anal. Chem.*, **1994**, *66*, 1424. Copyright 1994 American Chemical Society.)

30C-3 Capillary Isotachopheresis

In capillary isotachopheresis all analyte bands ultimately migrate at the same velocity; hence, the name from *iso* for same and *tach* for speed. In any particular application, either cations or anions can be separated, but not both at the same time. In an isotachopheresis separation, the sample is injected between two buffers: a leading one containing ions of a higher mobility than any of the analyte ions and a terminating one with ions of a lower mobility than the sample ions (Figure 30-9b). For example, in separating anions, chloride ions might be contained in the leading buffer and slow-moving heptanoate ions in the terminating buffer. For a separation of anions, the leading electrolyte solution is connected to the anode and the terminating one to the cathode.

When the electric field is first applied in an isotachopheretic separation, analyte ions migrate as in zone electrophoresis, each ion with its unique velocity given by the product $\mu_e E$. This difference in migration rates results in the separation of the various analyte species into adjacent bands, with fastest species located in a band directly adjacent to the leading buffer and the slowest just ahead of the terminal buffer. After the bands have formed, they then move at the same velocity. The reason that the bands have the same velocity is that the electric field becomes smaller for the more mobile bands and greater for the slower bands, so that the current is the same, as it must be, in all parts of the buffer. The ionic current that results from the flow of ions in the buffer is analogous to the dc current in a circuit consisting of several resistors connected in series to a battery where the current must be identical in all of the resistors. Hence, the potential across each of the resistors must vary in such a way that Ohm's law is obeyed.

When equilibrium is reached in an isotachopheretic experiment, each sample component is migrating in a band sandwiched

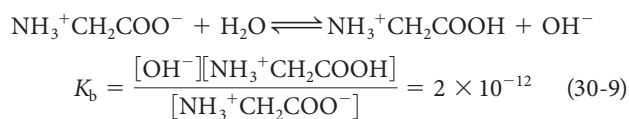
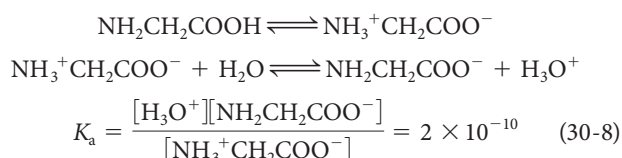
between a band that contains the next-slower-moving ions and the next-faster-moving band, as shown in Figure 30-9b. The boundary between bands is sharp. If a solute species starts to move into the next-faster band, it encounters a lower field, which reduces its velocity until it drops back into its original band. Note in Figure 30-9b that, in contrast to zone electrophoresis or elution chromatography, the analyte bands are immediately adjacent to one another and are not separated by bands of the buffer.

30C-4 Capillary Isoelectric Focusing

Capillary isoelectric focusing is used to separate amphiprotic species, such as amino acids and proteins that contain a weak carboxylic acid group and a weak-base amine group.

Properties of Amphiprotic Compounds

An *amphiprotic* compound is a species that in solution is capable of both donating and accepting a proton. A typical amino acid, such as glycine, is an amphiprotic compound. When glycine is dissolved in water, three important equilibria operate:

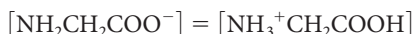


The first reaction constitutes a kind of internal acid-base reaction and is analogous to the reaction one would observe between a simple carboxylic acid and an amine. The typical aliphatic amine has a base dissociation constant of 10^{-4} to 10^{-5} , and many carboxylic acids have acid dissociation constants of about the same magnitude. The result is that the first reaction proceeds far to the right, with the product or products being the predominant species in the solution.

The amino acid product in the first reaction, bearing both a positive and a negative charge, is called a *zwitterion*. As shown by the equilibrium constants for the second and third reactions (Equations 30-8 and 30-9), the zwitterion of glycine is stronger as an acid than as a base. Thus, an aqueous solution of glycine is somewhat acidic.

The zwitterion of an amino acid, containing as it does a positive and a negative charge, has no tendency to migrate to an electric field, but the singly charged anionic and cationic species are attracted to oppositely charged electrodes. No net migration of the amino acid occurs in an electric field when the pH of the solvent is such that the concentrations of the anionic and cationic forms are identical. The pH at which no net migration occurs is called the *isoelectric point* (pI) and is an important physical constant for characterizing amino acids. The isoelectric

point is readily related to the ionization constants for the species. Thus, for glycine, at the isoelectric point,



If we divide K_a by K_b and substitute this equality, we obtain at the isoelectric point

$$\frac{K_a}{K_b} = \frac{[\text{H}_3\text{O}^+]_{\text{iso}} [\text{NH}_2\text{CH}_2\text{COO}^-]}{[\text{OH}^-]_{\text{iso}} [\text{NH}_3^+\text{CH}_2\text{COOH}]} = \frac{[\text{H}_3\text{O}^+]_{\text{iso}}}{[\text{OH}^-]_{\text{iso}}} \quad (30-10)$$

Substituting $K_w/[\text{H}_3\text{O}^+]_{\text{iso}}$ for $[\text{OH}^-]_{\text{iso}}$ in Equation 30-10 and rearranging yields

$$[\text{H}_3\text{O}^+]_{\text{iso}} = \sqrt{\frac{K_a K_w}{K_b}} \quad (30-11)$$

We can convert Equation 30-11 into the isoelectric point pH (pI) by taking the negative logarithms of both sides. Thus, pI can be expressed as

$$\begin{aligned} \text{pI} = -\log[\text{H}_3\text{O}^+]_{\text{iso}} &= \frac{(-\log K_a - \log K_w + \log K_b)}{2} \\ \text{pI} &= \frac{(\text{p}K_a + \text{p}K_w - \text{p}K_b)}{2} \end{aligned} \quad (30-12)$$

For glycine, $\text{p}K_a = -\log(2 \times 10^{-10}) = 9.7$, $\text{p}K_b = 11.7$ and $\text{p}K_w = 14.0$. Thus,

$$\text{pI} = (9.7 + 14.0 - 11.7)/2 = 6.0$$

Hence, the isoelectric point pI for glycine occurs at a pH of 6.0.

Separation of Amphiprotic Species

In isoelectric separation of amphiprotic species, the separation is performed in a buffer mixture that continuously varies in pH along its length. This pH gradient is prepared from a mixture of several different *ampholytes* in an aqueous solution. Ampholytes are amphoteric compounds usually containing carboxylic and amino groups. Ampholyte mixtures having different pH ranges can be prepared or are available from several commercial sources.

To perform an isoelectric focusing experiment in a capillary tube, the analyte mixture is dissolved in a dilute solution of the ampholytes, which is then transferred to the tube. One end of the capillary is then inserted in a solution of strong base, such as sodium hydroxide, that also holds the cathode. The other end of the tube is immersed in a solution of a strong acid, such as phosphoric, that also holds the anode. When the electric field is applied, hydrogen ions begin to migrate from the anode compartment toward the cathode. Hydroxide ions from the cathode begin to move in the opposite direction. If a component of the ampholyte or the analyte has a net negative charge, it migrates toward the positive anode. As it migrates it passes into continuously lower pH regions, where progressive protonation of the species occurs, which lowers its negative charge. Ultimately, it reaches the pH where its net charge is zero (its isoelectric point). Migration of the species then ceases. This process goes on for each ampholyte species and ultimately provides a continuous pH gradient throughout the tube. Analyte ions also migrate until they reach

their isoelectric points. These processes then result in the separation of each analyte into a narrow band that is located at the pH of its isoelectric point (see Figure 30-9c). Very sharp focusing is realized in such systems. Note that isoelectric focusing separations are based on differences in equilibrium properties of the analytes (K_a , K_b) rather than on differences in rates of migration. Once each analyte has migrated to a region where it is neutral, the positions of bands become constant and no longer change with time.

Mobilization of Focused Bands

To detect the focused bands in a capillary isoelectric focusing separation, it is necessary to move, or mobilize, the contents of the capillary so that the bands pass the detector located at one end. This mobilization can be accomplished by applying a pressure difference, just as for sample loading, or by simply changing the solution in the electrode compartment. During the focusing step, equal numbers of H^+ and OH^- ions enter opposite ends of the capillary, so the pH gradient remains stable. Suppose that sodium chloride is added to the sodium hydroxide solution after focusing is finished. Now both Cl^- and OH^- migrate into the cathode end of the column, and the sum of these two concentrations is balanced by H^+ entering the opposite end. That means that there is now less OH^- than H^+ flowing into the capillary. The pH decreases at the cathode end. The pH gradient is no longer stable. It moves toward the cathode end, and along with it go the focused bands. The bands that pass the detector first are the ones corresponding to proteins with the most alkaline isoelectric points. Figure 30-16 shows electropherograms for a set of low molecular mass pI markers (A) and a mixture of three proteins (B). In this case, mobilization was achieved by electroosmosis.

30C-5 Micellar Electrokinetic Chromatography

In 1984, Terabe and collaborators¹⁵ described a modification of CE that permitted the separation of low-molecular-mass aromatic phenols and nitro compounds with equipment such as shown in Figure 30-1. *Micellar electrokinetic chromatography* (MEKC) is a type of CE that offers several unique features.¹⁶ Like CE, MEKC provides highly efficient separations on microvolumes of sample solution without the need for the high-pressure pumping system required for HPLC. Unlike normal CE, however, MEKC allows uncharged species to be separated as well as charged species. In MEKC a mobile phase is transported across a stationary phase by electroosmotic flow. As shown in Figure 30-3, electroosmotic pumping leads to a flat-plug profile rather than the parabolic profile of pressure-induced flow. The flat profile of osmotic pumping leads to narrow bands and thus high separation efficiencies.

¹⁵S. Terabe, K. Otsuka, K. Ichikawa, A. Tsuchiya, and T. Ando, *Anal. Chem.*, **1984**, 56, 111, DOI: 10.1021/ac00265a031; S. Terabe, K. Otsuka, and T. Ando, *Anal. Chem.*, **1985**, 57, 834, DOI: 10.1021/ac00281a014.

¹⁶For a brief review, see S. Terabe, *Anal. Chem.*, **2004**, 76, 240A, DOI: 10.1021/ac0415859.

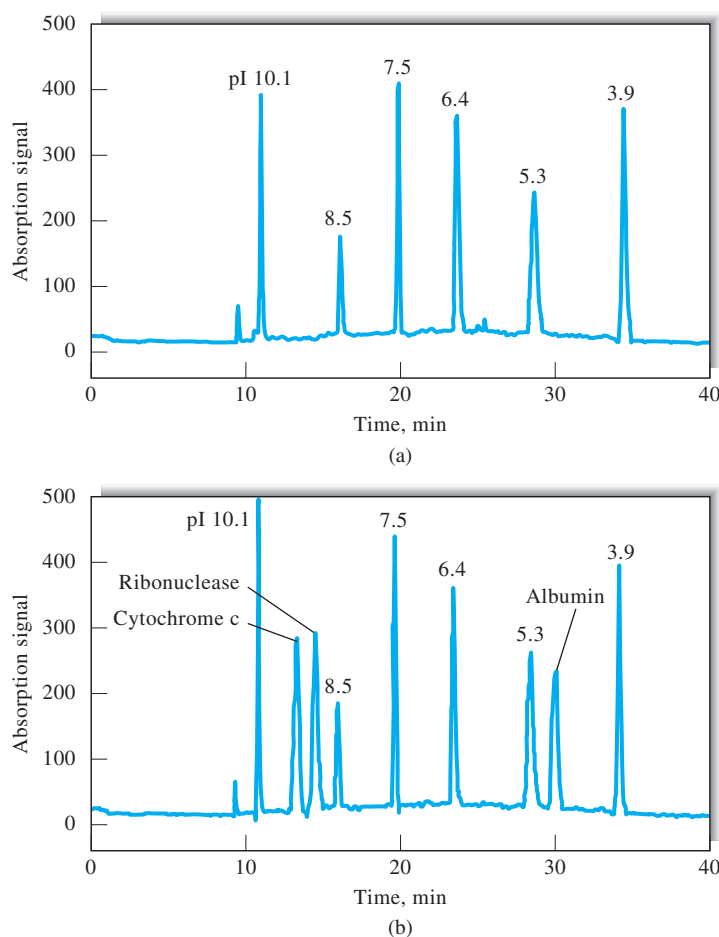


FIGURE 30-16 Capillary isoelectric focusing of proteins. Isoelectric pI values listed above peaks. In (a), a set of low molecular mass markers of known pI is shown. In (b), the proteins ribonuclease, cytochrome c, and albumin are shown along with the markers. Detection was by UV absorption. (From M. Horká, F. Růžicka, J. Horký, V. Holá, and K. Šlais, *J. Chromatogr. B*, **2006**, *841*, 152, DOI: 10.1016/j.jchromb.2006.05.013, with permission.)

Micelles form in aqueous solutions when the concentration of an ionic species having a long-chain hydrocarbon tail is increased above a certain level called the *critical micelle concentration* (CMC). At this point the surfactant begins to form spherical aggregates made up of 40–100 ions with their hydrocarbon tails in the interior of the aggregate and their charged ends exposed to water on the outside. Micelles constitute a stable second phase that can incorporate nonpolar compounds in the hydrocarbon interior of the particles, thus *solubilizing* the nonpolar species. Solubilization is commonly encountered when a greasy material or surface is washed with a detergent solution.

In MEKC, surfactants are added to the operating buffer in amounts that exceed the CMC. For most applications to date, the surfactant has been sodium dodecyl sulfate (SDS). The surface of an ionic micelle of this type has a large negative charge, which gives it a large electrophoretic mobility. Most buffers, however, exhibit such a high electroosmotic flow rate toward the

negative electrode that the anionic micelles are carried toward that electrode also but at a much reduced rate. Thus, during an experiment, the buffer mixture consists of a faster-moving aqueous phase and a slower-moving micellar phase. When a sample is introduced into this system, the components distribute themselves between the aqueous phase and the hydrocarbon phase in the interior of the micelles. The positions of the resulting equilibria depend on the polarity of the solutes. With polar solutes the aqueous solution is favored; with nonpolar compounds, the hydrocarbon environment is preferred.

The phenomena just described are quite similar to what occurs in an LC column except that the “stationary phase” moves along the length of the column but at a much slower rate than the mobile phase. The mechanism of separation is identical in the two cases and depends on differences in distribution constants for analytes between the mobile aqueous phase and the hydrocarbon *pseudostationary phase*. The process is thus

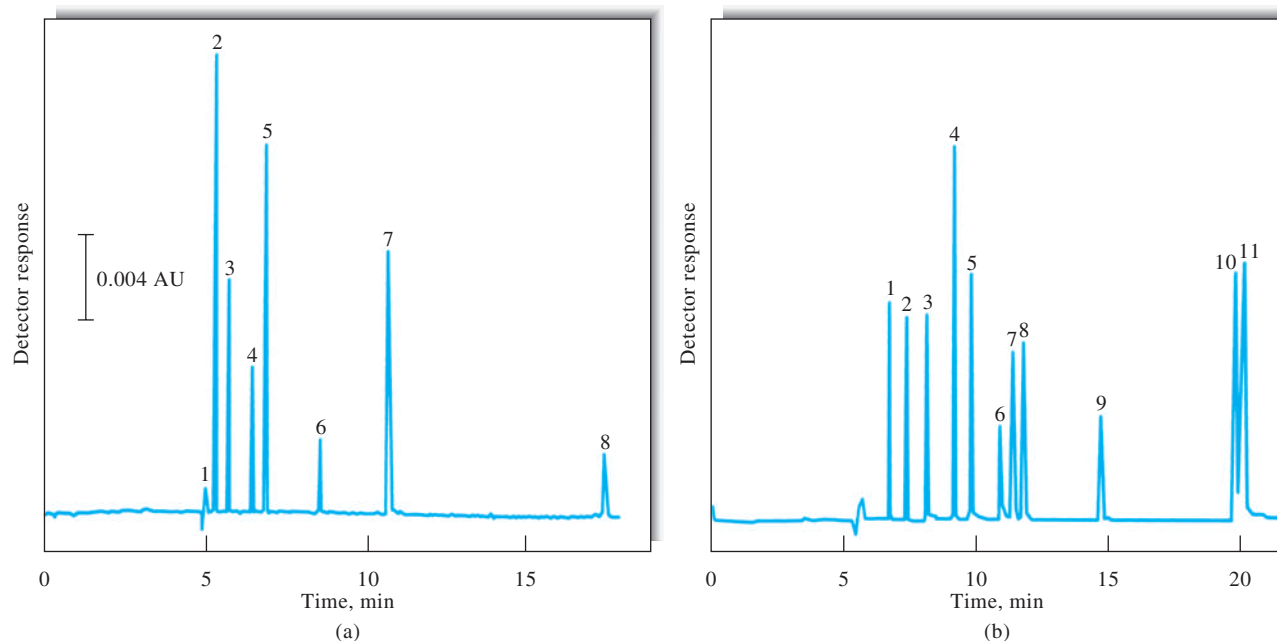


FIGURE 30-17 Typical separations by MEKC. (a) Some test compounds: 1 = methanol, 2 = resorcinol, 3 = phenol, 4 = *p*-nitroaniline, 5 = nitrobenzene, 6 = toluene, 7 = 2-naphthol, 8 = Sudan III; capillary, 50- μ m inside diameter, 500 mm to the detector; applied voltage, $\sim$ 15 kV; detection UV absorption at 210 nm. (b) Analysis of a cold medicine: 1 = acetaminophen, 2 = caffeine, 3 = sulpyrine, 4 = naproxen, 5 = guaiphenesin, 6 = noscapine, 7 = chlorpheniramine, 8 = tipepidine, 9 = noscapine, 10 = noscapine, 11 = chlorpheniramine and tipepidine; applied voltage, 20 kV; capillary, as in (a); detection UV absorption at 220 nm. (From S. Terabe, *Trends Anal. Chem.*, **1989**, 8, 129, DOI: 10.1016/0165-9936(89)85022-8. With permission.)

true chromatography; hence, the name “micellar electrokinetic chromatography.” Figure 30-17 illustrates two typical separations by MEKC.

MEKC has become important for chiral separations.¹⁷ Here, chiral resolving agents are used as in HPLC to preferentially complex one of the isomers. Either a chiral resolving agent with detergent properties, such as a bile acid, is used to form the micelles or a resolving agent, such as a cyclodextrin, is added to a detergent that is itself achiral. In many aspects, chiral separations are easier to develop by MEKC than by LC, although LC is still the dominant technique in industry because of its familiarity.

One advantage that MEKC has over HPLC is much higher column efficiencies (100,000 plates or more). In addition, changing the second phase in MEKC is accomplished by changing only the micellar composition of the buffer. In contrast, in HPLC, the second phase can be altered only by changing the type of column packing or column. The MEKC technique is useful for separating small molecules that are impossible to separate by traditional gel electrophoresis. On-line preconcentration techniques have been reported.¹⁸ Mass spectrometric detection is possible, although the surfactants can cause difficulties, particularly when electrospray ionization is used.

30D PACKED COLUMN ELECTROCHROMATOGRAPHY

Electrochromatography is a hybrid of HPLC and CE that offers some of the best features of the two methods.¹⁹ Like HPLC and MEKC, it is applicable to the separation of neutral species or charged species. Electrochromatography with packed columns is, however, the least mature of the various electroseparation techniques. In this method, a polar solvent is usually driven by electroosmotic flow through a capillary packed with a reversed-phase HPLC packing. Separations depend on the distribution of the analyte species between the mobile phase and the liquid stationary phase held on the packing. Figure 30-18 shows a typical electrochromatogram for the separation of sixteen polyaromatic hydrocarbons (PAHs) in a 33-cm-long capillary having an inside diameter of 75 μ m. The mobile phase consisted of acetonitrile in a 4-mM sodium borate solution. The stationary phase consisted of 3- μ m octadecylsilica particles.

30E FIELD-FLOW FRACTIONATION

Field-flow fractionation (FFF) describes a group of analytical techniques that are quite useful in the separation and characterization of dissolved or suspended materials such as polymers,

¹⁷R. Vaspalec and P. Boček, *Chem. Rev.*, **2000**, 100, 3715, DOI: 10.1021/cr9411583.

¹⁸S. Terabe, *Anal. Chem.*, **2004**, 76, 240A.

¹⁹L. A. Colón, Y. Guo, and A. Fermier, *Anal. Chem.*, **1997**, 69, 461A, DOI: 10.1021/ac9717245.

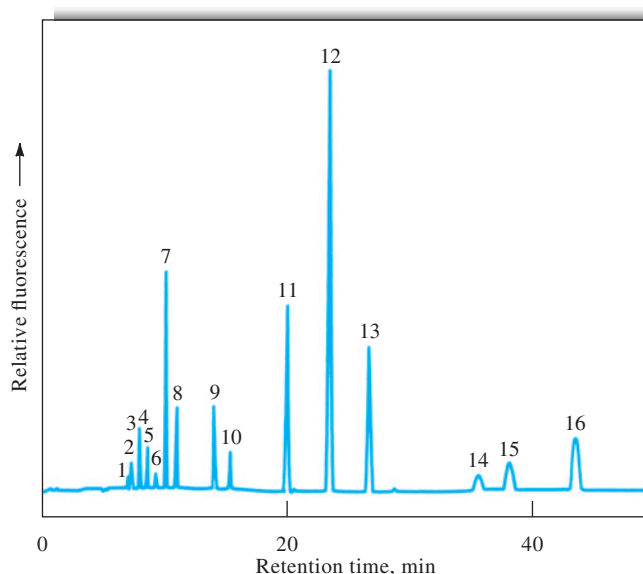


FIGURE 30-18 Electrochromatogram showing the electrochromatographic separation of 16 PAHs ($\sim 10^{-6}$ to 10^{-8} M of each compound). The peaks are identified as follows: (1) naphthalene, (2) acenaphthylene, (3) acenaphthene, (4) fluorene, (5) phenanthrene, (6) anthracene, (7) fluoranthene, (8) pyrene, (9) benz[a]anthracene, (10) chrysene, (11) benzo[b]fluoranthene, (12) benzo[k]fluoranthene, (13) benzo[a]pyrene, (14) dibenz[a,h]anthracene, (15) benzo[ghi]perylene, and (16) indeo[1,2,3-cd]pyrene. (From C. Yan, R. Dadoo, H. Zhao, R. N. Zare, and D. J. Rakestraw, *Anal. Chem.*, **1995**, 67, 2026, DOI: 10.1021/ac00109a020. With permission.)

large particles, and colloids. Although the FFF concept was first described by Giddings in 1966,²⁰ only in the last twenty years have practical applications and advantages over other methods been shown.²¹

²⁰J. C. Giddings, *Sep. Sci.*, **1966**, 1, 123, DOI: 10.1080/01496396608049439.

²¹For a review of FFF methods, see J. C. Giddings, *Anal. Chem.*, **1995**, 67, 592A, DOI: 10.1021/ac00115a001.

30E-1 Separation Mechanisms²²

Separations in FFF occur in a thin ribbon-like flow channel such as that shown in Figure 30-19. The channel is typically 25–100 cm long and 1–3 cm wide. The thickness of the ribbon-like structure is

²²M. E. Schimpf, K. Caldwell, and J. C. Giddings, eds., *Field-Flow Fractionation Handbook*, New York: Wiley, 2000, Part I.

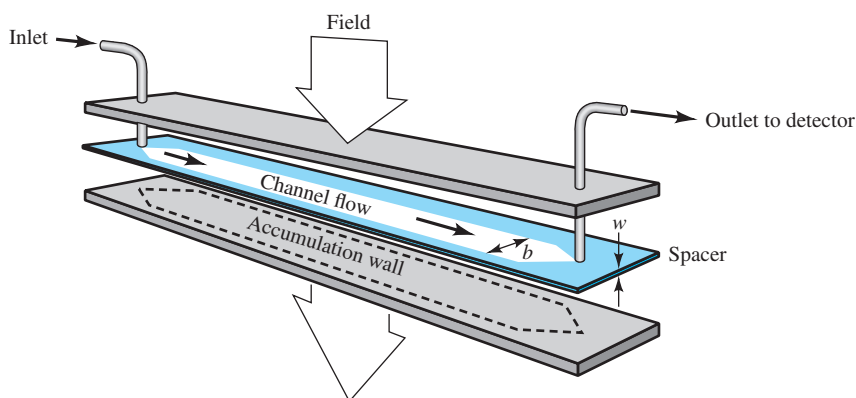


FIGURE 30-19 Schematic diagram of FFF flow channel sandwiched between two walls. An external field (electrical, thermal, or centrifugal) is applied perpendicular to the flow direction.

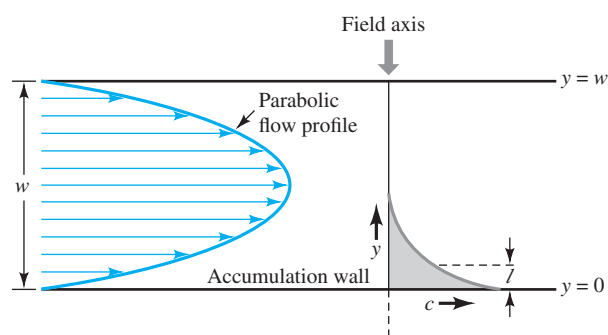


FIGURE 30-20 When the field is applied in FFF, components migrate to the accumulation wall where an exponential concentration profile exists as seen on the right. Components extend a distance y into the channel. The average thickness of the layer is l , which differs for each component. The main channel flow of the eluting solvent is then turned on, resulting in the parabolic flow profile shown on the left.

usually 50–500 μm . The channel is usually cut from a thin spacer and sandwiched between two walls. An electrical, thermal, or centrifugal field is applied perpendicular to the flow direction. Alternatively, a cross flow perpendicular to the main flow direction can be used.

In practice, the sample is injected at the inlet to the channel. The external field is next applied across the face of the channel as illustrated in Figure 30-19. In the presence of the field, sample components migrate toward the *accumulation wall* at a velocity determined by the strength of the interaction of the constituent with the field. Sample components rapidly reach a steady-state concentration distribution near the accumulation wall as shown in Figure 30-20. The mean thickness of the component layer l is related to the diffusion coefficient of the molecule D and to the field-induced velocity U toward the wall. The faster the component moves in the field, the thinner the layer near the wall. The larger the diffusion coefficient, the thicker the layer. Because the sample components have different values of D and U , the mean layer thickness will vary among components.

Once components have reached their steady-state profiles near the accumulation wall, the channel flow is begun. The flow

is laminar, resulting in the parabolic profile shown on the left in Figure 30-20. The main carrier flow has its highest velocity in the center of the channel and its lowest velocity near the walls. Components that interact strongly with the field are compressed very near the wall as shown by component A in Figure 30-21. Here, they are eluted by slow-moving solvent. Components B and C protrude more into the channel and experience a higher solvent velocity. The elution order is thus C, then B, and then A. Components separated by FFF flow through a UV-visible absorption, refractive index, or fluorescence detector located at the end of the flow channel and similar to those used in HPLC separations. The separation results are revealed by a plot of detector response versus time, called a *fractogram*, similar to a chromatogram in chromatography.

30E-2 FFF Methods

Different FFF subtechniques result from the application of different types of fields or gradients.²³ To date, the methods that have been most widely reported are *sedimentation FFF*, *electrical FFF*, *thermal FFF*, and *flow FFF*. *Acoustic*, *magnetic*, and *dielectrophoretic FFF* have also been reported.²⁴

Sedimentation FFF

Sedimentation FFF is the most widely used form. In this technique, the channel is coiled and made to fit inside a centrifuge basket as illustrated in Figure 30-22. Components with the highest mass and density are driven to the wall by the sedimentation (centrifugation) force and elute last. Low-mass species are eluted first. There is relatively high selectivity between particles of different



Simulation: Learn more about **field-flow fractionation** at www.tinyurl.com/skoogpia7

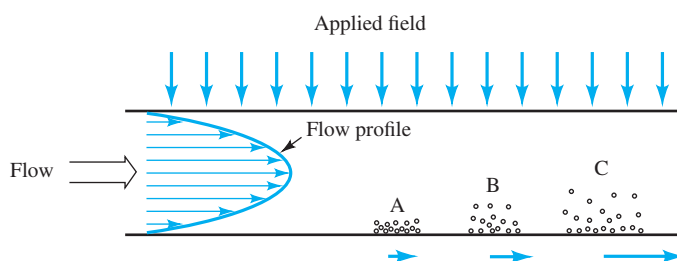


FIGURE 30-21 Three components A, B, and C are shown compressed against the accumulation wall in FFF to different amounts because of different interactions with the external field. When the flow is begun, component A experiences the lowest solvent velocity because it is the closest to the wall. Component B protrudes more into the channel where it experiences a higher flow velocity. Component C, which interacts the least with the field, experiences the highest solvent-flow velocity and thus is displaced the most rapidly by the flow.

²³For a discussion of the various FFF methods, see J. C. Giddings, *Unified Separation Science*, New York: Wiley, 1991, Chap. 9; M. E. Schimpf, K. Caldwell, and J. C. Giddings, eds., *Field-Flow Fractionation Handbook*, New York: Wiley, 2000.

²⁴See F. A. Messaud et al., *Prog. Polym. Sci.*, **2009**, 34, 351, DOI: 10.1016/j.progpolymsci.2008.11.001.



FIGURE 30-22 Sedimentation FFF apparatus. (Courtesy of Postnova Analytics, Salt Lake City, UT.)

size in sedimentation FFF. A separation of polystyrene beads of various diameters by sedimentation FFF is shown in Figure 30-23.

Because the centrifugation forces are relatively weak for small molecules, sedimentation FFF is most applicable for molecules with molecular masses exceeding 10^6 . Such systems as polymers, biological macromolecules, natural and industrial colloids, emulsions, and subcellular particles appear to be amenable to separation by sedimentation FFF.

Electrical FFF

In electrical FFF, an electric field is applied perpendicular to the flow direction. Retention and separation occur based on electrical charge. Species with the highest charge are driven most effectively toward the accumulation wall. Species of lower charge are not as compacted and protrude more into the higher-flow region. Hence, species of the lowest charge are eluted first, with highly charged species retained the most.

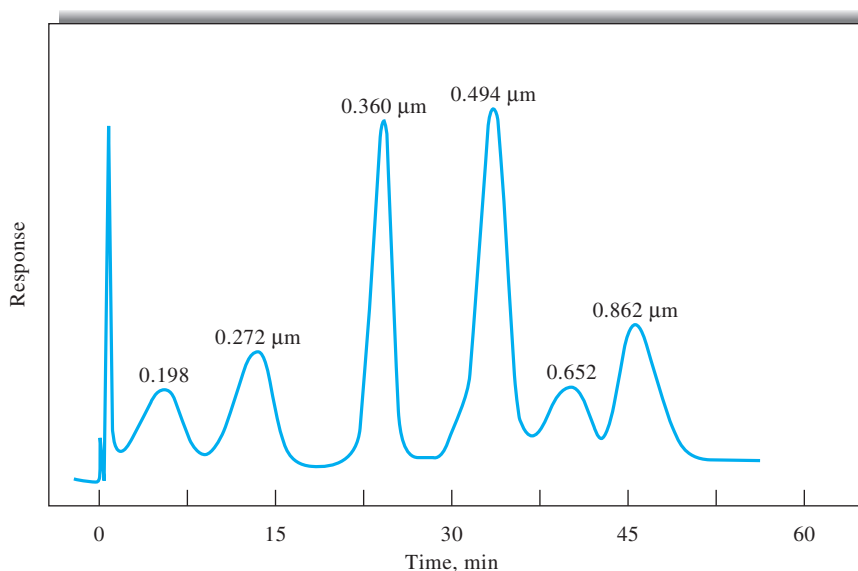


FIGURE 30-23 Fractogram illustrating separation of polystyrene beads of various diameters by sedimentation FFF. The channel flow rate was 2 mL/min. (Courtesy of Postnova Analytics, Salt Lake City, UT.)

Because electric fields are quite powerful, even small ions are amenable to separation by electrical FFF. However, electrolysis effects have limited the application of this method to the separation of mixtures of proteins and other large molecules.

Thermal FFF

In thermal FFF, a thermal field is applied perpendicular to the flow direction by forming a temperature gradient across the FFF channel. The temperature difference induces thermal diffusion in which the velocity of movement is related to the thermal diffusion coefficient of the species.

Thermal FFF is particularly well suited for the separation of synthetic polymers with molecular masses in the range of 10^3 to 10^7 . The technique has significant advantages over size-exclusion chromatography for high-molecular-mass polymers. On the other hand, low-molecular-mass polymers appear to be better separated by size-exclusion methods. In addition to polymers, particles and colloids have been separated by thermal FFF.²⁵

Flow FFF

Perhaps the most versatile of all the FFF subtechniques is flow FFF, in which the external field is replaced by a slow cross flow

of the carrier liquid. The perpendicular flow transports material to the accumulation wall in a nonselective manner. However, steady-state layer thicknesses are different for various components because they depend not only on the transport rate but also on molecular diffusion. Exponential distributions of differing thicknesses are formed as in normal FFF.

Two types of flow FFF have been described: symmetric flow FFF and asymmetric flow FFF. In symmetric flow FFF, the cross flow is introduced through a porous frit on the top wall. This flow exits the channel through an ultrafiltration membrane and porous frit on the accumulation wall. In asymmetric flow FFF, only the bottom wall is permeable. The upper wall is a solid wall impermeable to the carrier fluid. A single inlet flow is split into two flows, the channel flow and the cross flow. The cross flow exits as in symmetric flow FFF. Both medium- and high-temperature asymmetric flow FFF have been developed. Asymmetric flow FFF offers a simpler structure, and thus easier construction, compared to symmetric flow FFF.²⁶

Flow FFF has been applied to the separation of proteins, synthetic polymers, and a variety of colloidal particles. Figure 30-24 illustrates the separation of three proteins by flow FFF. The reproducibility is illustrated by the fractograms for the three injections.

²⁵P. M. Shiundu, G. Liu, and J. C. Giddings, *Anal. Chem.*, **1995**, 67, 2705, DOI: 10.1021/ac00111a032.

²⁶See reference 24.

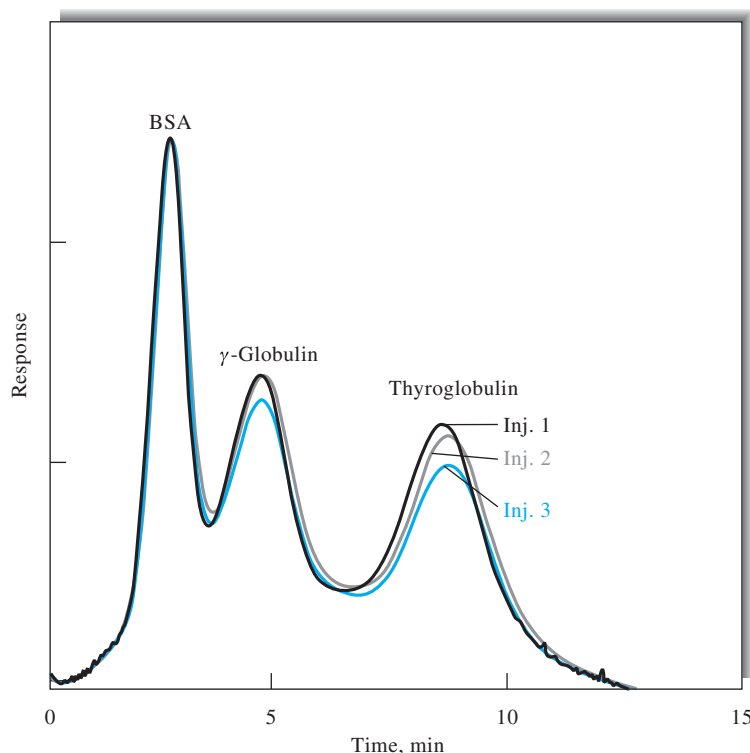


FIGURE 30-24 Separation of three proteins by flow FFF. Three separate injections are shown. In the experiment shown, the sample was concentrated at the head of the channel by means of an opposing flow. BSA = bovine serum albumin. (From H. Lee, S. K. R. Williams, and J. C. Giddings, *Anal. Chem.*, **1998**, 70, 2495, DOI: 10.1021/ac9710792. Copyright 1998 American Chemical Society.)

30E-3 Advantages of FFF over Chromatographic Methods

FFF has several advantages over ordinary chromatographic methods for some applications. First, no packing material or stationary phase is needed for separation to occur. In some chromatographic systems, there may be undesirable interactions between the packing material or stationary phase and the sample constituents. Some solvents or sample materials adsorb or react with the stationary phase or its support. Shear forces in chromatography can lead to degradation of some samples. Macromolecules and particles are particularly prone to these adverse interactions.

The geometry and flow profiles involved in FFF are well characterized. Likewise, the effects of most external fields can be readily modeled. As a result, fairly exact theoretical predictions

of retention and plate height can be made in FFF. Chromatographic predictions are still rather inexact in comparison.

Finally, the external field governs FFF retention. With electrical, sedimentation, and flow FFF, the perpendicular forces can be varied rapidly and in a time-programmed fashion. This gives FFF a certain versatility in adapting to different types of samples. Methods can also be readily optimized for resolution and separation speed.

Although FFF is a fairly recent addition to the repertoire of analytical separation methods, it has been shown to be highly complementary to chromatography. The FFF methods are best suited for macromolecules and particles that are for the most part beyond the molecular mass range of chromatographic methods. On the other hand, chromatographic methods are superior for low-molecular-mass substances.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

- 30-1** What is electroosmotic flow? Why does it occur?
- 30-2** Suggest how electroosmotic flow might be suppressed.
- 30-3** Why does pH affect separation of amino acids by electrophoresis?
- 30-4** What is the principle of separation by CZE?
- * **30-5** A certain inorganic cation has an electrophoretic mobility of $5.27 \times 10^{-4} \text{ cm}^2 \text{ s}^{-1} \text{ V}^{-1}$. This same ion has a diffusion coefficient of $9.5 \times 10^{-6} \text{ cm}^2 \text{ s}^{-1}$. If this ion is separated from other cations by CZE with a 50.0-cm capillary, what is the expected plate count N at applied voltages of
 - (a) 10.0 kV?
 - (b) 20.0 kV?
 - (c) 30.0 kV?
- * **30-6** The cationic analyte of Problem 30-5 was separated by CZE in a 50.0-cm capillary at 10.0 kV. Under the separation conditions, the electroosmotic flow rate was 0.85 mm s^{-1} toward the cathode. If the detector was 45.0 cm from the injection end of the capillary, how long would it take in minutes for the analyte cation to reach the detector after the field is applied?
- 30-7** What is the principle of micellar electrokinetic capillary chromatography? How does it differ from CZE?
- 30-8** Describe a major advantage of micellar electrokinetic capillary chromatography over conventional liquid chromatography.
- 30-9** Three large proteins are ionized at the pH at which an electrical FFF separation is carried out. If the ions are designated A^{2+} , B^{+} , and C^{3+} , predict the order of elution.
- 30-10** What determines the elution order in sedimentation FFF?
- 30-11** List the major advantages and limitations of FFF compared to chromatographic methods.

» QUESTIONS AND PROBLEMS (continued)

Challenge Problem



30-12 Doxorubicin (DOX) is a widely used anthracycline that has been effective in treatments of leukemia and breast cancer in humans (A. B. Anderson, C. M. Ciriaks, K. M. Fuller, and E. A. Ariaga, *Anal. Chem.*, **2003**, 75, 8, DOI: 10.1021/ac020426r). Unfortunately, side effects, such as liver toxicity and drug resistance, have been reported. In a recent study, Anderson et al. used laser-induced fluorescence (LIF) as a detection mode for CE to investigate metabolites of DOX in single cells and subcellular fractions. The following are results similar to those obtained by Anderson et al. for quantifying doxorubicin by LIF. The CE peak areas were measured as a function of the DOX concentration to construct a calibration curve.

DOX Concentration, nM	Peak Area
0.10	0.10
1.00	0.80
5.00	4.52
10.00	8.32
20.00	15.7
30.00	26.2
50.00	41.5

- Find the equation for the calibration curve and the standard deviations of the slope and intercept. Find the R^2 value.
- Rearrange the equation found in part (a) to express concentration in terms of the measured area.
- The limit of detection (LOD) for DOX was found to be 3×10^{-11} M. If the injection volume was 100 pL, what was the LOD in moles?
- Two samples of unknown DOX concentration were injected and peak areas of 11.3 and 6.97 obtained. What were the concentrations and their standard deviations?
- Under certain conditions, the DOX peak required 300 s to reach the LIF detector. What time would be required if the applied voltage were doubled? What time would be required if the capillary length were doubled at the same applied voltage?
- The capillary used in part (e) under normal conditions had a plate count of 100,000. What would N be if the capillary length were doubled at the same applied voltage? What would N be if the applied voltage were doubled at the original capillary length?
- For a 40.6-cm-long capillary of inside diameter 50 μm , what would the plate height be for a capillary with $N = 100,000$?
- For the same capillary as in part (g), what is the variance σ^2 of a typical peak?

An Introduction to Infrared Spectrometry

The infrared (IR) region of the spectrum encompasses radiation with wavenumbers ranging from about 12,800 to 10 cm^{-1} or wavelengths from 0.78 to 1000 μm . Because of similar applications and instrumentation, the IR spectrum is usually subdivided into three regions, the near-IR, the mid-IR, and the far-IR. The techniques and the applications of methods based on the three IR spectral regions differ considerably as discussed in this chapter.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

Table 16-1 gives the rough limits of each of the three IR regions. Measurements in the near-IR region are often made with photometers and spectrophotometers similar in design and components to the instruments described in earlier chapters for ultraviolet (UV)-visible spectrometry. The most important applications of this spectral region have been to the quantitative analysis of industrial and agricultural materials and for process control. Applications of near-IR spectrometry are discussed in Section 17D.

Until the early 1980s, instruments for the mid-IR region were largely of the dispersive type and used diffraction gratings. Since that time, however, mid-IR instrumentation has dramatically changed so that now almost all new instruments are of the Fourier transform (FT) type. Photometers based on interference filters also find use for measuring the composition of gases and atmospheric contaminants.

The appearance of relatively inexpensive FT spectrometers in the last decade has markedly increased the number and type of applications of mid-IR radiation. This increase has come about because interferometric instruments can produce improvements of an order of magnitude, or more, in signal-to-noise ratios and detection limits over dispersive instruments. Before the appearance of these instruments, the mid-IR spectral region was used largely for qualitative organic analysis and structure determinations based on absorption spectra. Now, in contrast, mid-IR spectrometry is beginning to be used in addition for quantitative analysis of complex samples by both absorption and emission spectrometry. Applications of this spectral region are also beginning to appear for microscopic studies of surfaces, analysis of solids by attenuated total reflectance and diffuse reflectance, photoacoustic measurements, and other uses. Several of these applications are described in Chapter 17.

In the past the far-IR region of the spectrum, although potentially quite valuable, had limited use because of experimental difficulties. The few sources of this type of radiation are notoriously weak and are further attenuated by the need for order-sorting filters that prevent radiation of higher grating orders from reaching the detector. FT spectrometers, with their much higher throughput, largely alleviate this problem and make the far-IR spectral region much more accessible to scientists. Several applications of far-IR spectroscopy are described in Section 17E.

TABLE 16-1 IR Spectral Regions

Region	Wavelengths (λ), μm	Wavenumbers ($\bar{\nu}$), cm^{-1}	Frequencies (ν), Hz
Near	0.78 to 2.5	12800 to 4000	3.8×10^{14} to 1.2×10^{14}
Middle	2.5 to 50	4000 to 200	1.2×10^{14} to 6.0×10^{12}
Far	50 to 1000	200 to 10	6.0×10^{12} to 3.0×10^{11}
Most used	2.5 to 15	4000 to 670	1.2×10^{14} to 2.0×10^{13}

In this chapter, we first deal with the mechanisms of the absorption, emission, and reflection of IR radiation using absorption spectroscopy as the basis for this discussion. We follow this discussion with a description of the components of IR instruments and how these are arranged in dispersive and non-dispersive instruments as well as in FT spectrometers.¹

16A THEORY OF IR ABSORPTION SPECTROMETRY

IR absorption, emission, and reflection spectra for molecular species can be rationalized by assuming that all arise from various changes in energy brought about by transitions of molecules from one vibrational or rotational energy state to another. In this section we use molecular absorption to illustrate the nature of these transitions.

16A-1 Introduction

Figure 16-1 shows a typical output from a commercial IR spectrophotometer. Although the y -axis is shown as linear in transmittance, modern computer-based spectrophotometers can also produce spectra that are linear in absorbance. The abscissa in this spectrum is linear in wavenumbers with units of reciprocal centimeters. A wavelength scale is also shown at the top of the plot. Computer-based spectrophotometers can produce a variety of other spectral formats such as linear in wavelength, baseline corrected, and derivative and smoothed spectra.

A linear wavenumber scale is usually preferred in IR spectroscopy because of the direct proportionality between this quantity and both energy and frequency. The frequency of the absorbed radiation is, in turn, the molecular vibrational frequency actually responsible for the absorption process. Frequency, however, is seldom if ever used

as the abscissa because of the inconvenient size of the unit; that is, a frequency scale of the plot in Figure 16-1 would extend from 1.2×10^{14} to 2.0×10^{13} Hz. Although the axis in terms of wavenumbers is often referred to as a frequency axis, keep in mind that this terminology is not strictly correct because the wavenumber $\bar{\nu}$ is only proportional to frequency ν . The relationships are given in Equation 16-1.

$$\bar{\nu} (\text{cm}^{-1}) = \frac{1}{\lambda (\mu\text{m})} \times 10^4 (\mu\text{m}/\text{cm}) = \frac{\nu (\text{Hz})}{c (\text{cm/s})} \quad (16-1)$$

Finally, note that the horizontal scale of Figure 16-1 changes at 2000 cm^{-1} , with the units at higher wavenumbers being represented by half the linear distance of those at lower wavenumbers. The expanded scale in the region from 2000 to 650 cm^{-1} permits easier identification of the numerous IR bands that appear in this region.

Dipole Moment Changes during Vibrations and Rotations

IR radiation is not energetic enough to bring about the kinds of electronic transitions that we have encountered in our discussions of UV and visible radiation. Absorption of IR radiation is thus confined largely to molecular species that have small energy differences between various vibrational and rotational states.

To absorb IR radiation, a molecule must undergo a net change in dipole moment as it vibrates or rotates. Only under these circumstances can the alternating electric field of the radiation interact with the molecule and cause changes in the amplitude of one of its motions. For example, the charge distribution around a molecule such as hydrogen chloride is not symmetric because the chlorine has a higher electron density than the hydrogen. Thus, hydrogen chloride has a significant dipole moment and is said to be polar. The dipole moment is determined by the magnitude of the charge difference and the distance between the two centers of charge. As a hydrogen chloride molecule vibrates, a regular fluctuation in its dipole moment occurs, and a field is established that can interact with the electric field associated with radiation. If the frequency of the radiation exactly matches a natural vibrational frequency of the molecule, absorption of the radiation takes place that produces a change in the amplitude of the molecular vibration. Similarly, the rotation of asymmetric molecules around their centers of mass results in periodic dipole moment fluctuations that allow interaction with the radiation field.

¹For detailed treatments of IR spectrometry, see *Introduction to Experimental Infrared Spectroscopy: Fundamentals and Practical Methods*, M. Tasumi, ed., Chichester, UK: Wiley, 2015; P. Larkin, *Infrared and Raman Spectroscopy: Principles and Spectral Interpretation*, Waltham, MA: Elsevier, 2011; N. B. Colthup, L. H. Daly, and S. E. Wiberley, *Introduction to Infrared and Raman Spectroscopy*, 3rd ed., San Diego: Academic Press, 1990; C.-P. S. Hsu, in *Handbook of Instrumental Techniques for Analytical Chemistry*, F. Settle, ed., Upper Saddle River, NJ: Prentice Hall, 1997, Chap. 15.

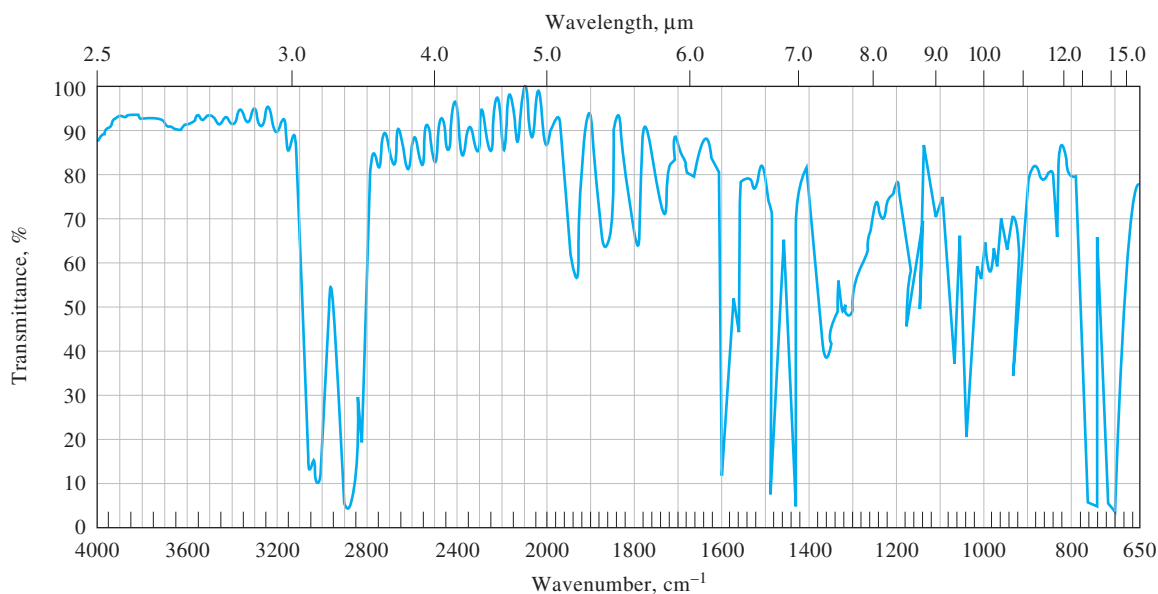


FIGURE 16-1 IR absorption spectrum of a thin polystyrene film. Note the scale change on the x-axis at 2000 cm^{-1} .

No net change in dipole moment occurs during the vibration or rotation of homonuclear species such as O_2 , N_2 , or Cl_2 . As a result, such compounds cannot absorb IR radiation. With the exception of a few compounds of this type, all other molecular species absorb IR radiation.

Rotational Transitions

The energy required to cause a change in rotational level is quite small and corresponds to radiation of $\bar{\nu} \leq 100\text{ cm}^{-1}$ ($\lambda > 100\text{ }\mu\text{m}$). Because rotational levels are quantized, absorption by gases in this far-IR region is characterized by discrete, well-defined lines. In liquids or solids, intramolecular collisions and interactions cause broadening of the lines into a continuum.

Vibrational-Rotational Transitions

Vibrational energy levels are also quantized, and for most molecules the energy differences between quantum states correspond to the mid-IR region. The IR spectrum of a gas usually consists of a series of closely spaced lines, because there are several rotational energy levels for each vibrational level. On the other hand, rotation is highly restricted in liquids and solids; in such samples, discrete vibrational-rotational lines disappear, leaving only somewhat broadened vibrational bands.

Types of Molecular Vibrations

The relative positions of atoms in a molecule are not fixed but instead fluctuate continuously because of a multitude of different types of vibrations and rotations about the bonds in the molecule. For a simple diatomic or triatomic molecule, it is easy to

define the number and nature of such vibrations and relate these to energies of absorption. An analysis of this kind becomes difficult if not impossible for molecules made up of many atoms. Not only do large molecules have a large number of vibrating centers, but also interactions among several centers can occur and must be taken into account for a complete analysis.

Vibrations fall into the basic categories of stretching and bending. A stretching vibration involves a continuous change in the interatomic distance along the axis of the bond between two atoms. Bending vibrations are characterized by a change in the angle between two bonds and are of four types: scissoring, rocking, wagging, and twisting. These are shown schematically in Figure 16-2.

All of the vibration types shown in Figure 16-2 may be possible in a molecule containing more than two atoms. In addition, interaction or coupling of vibrations can occur if the vibrations involve bonds to a single central atom. The result of coupling is a change in the characteristics of the vibrations involved.

In the treatment that follows, we first consider isolated vibrations represented by a simple harmonic oscillator model. Modifications to the theory of the harmonic oscillator, which are needed to describe a molecular system, are taken up next. Finally, the effects of vibrational interactions in molecular systems are discussed.

16A-2 Mechanical Model of a Stretching Vibration in a Diatomic Molecule

The characteristics of an atomic stretching vibration can be approximated by a mechanical model consisting of two masses connected by a spring. A disturbance of one of these masses



Tutorial: Learn more about **IR absorption** at www.tinyurl.com/skoogpia7

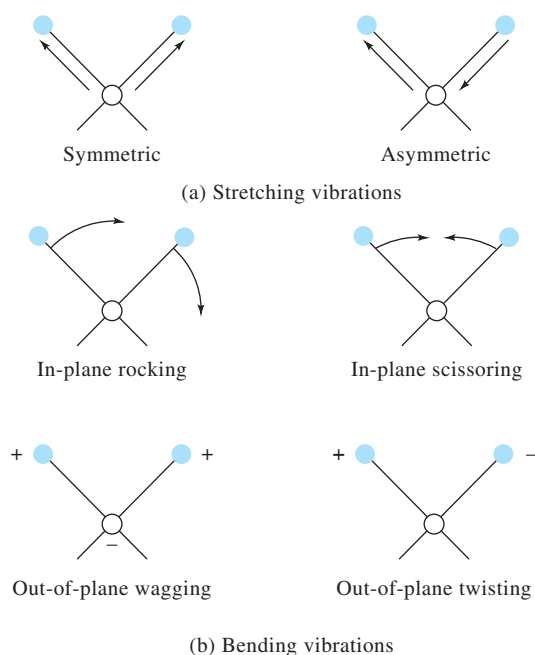


FIGURE 16-2 Types of molecular vibrations. Note that + indicates motion from the page toward the reader and – indicates motion away from the reader.

along the axis of the spring results in a vibration called a *simple harmonic motion*.

Let us first consider the vibration of a single mass attached to a spring that is hung from an immovable object (see Figure 16-3a). If the mass is displaced a distance y from its equilibrium position by application of a force along the axis of the spring, the

restoring force F is proportional to the displacement (Hooke's law). That is,

$$F = -ky \quad (16-2)$$

where k is the force constant, which depends on the stiffness of the spring. The negative sign indicates that F is a restoring force. This means that the direction of the force is opposite the direction of the displacement. Thus, the force tends to restore the mass to its original position.

Potential Energy of a Harmonic Oscillator

The potential energy E of the mass and spring can be arbitrarily assigned a value of zero when the mass is in its rest, or equilibrium, position. As the spring is compressed or stretched, however, the potential energy of this system increases by an amount equal to the work required to displace the mass. If, for example, the mass is moved from some position y to $y + dy$, the work and hence the change in potential energy dE is equal to the force F times the distance dy . Thus,

$$dE = -F dy \quad (16-3)$$

Combining Equations 16-3 and 16-2 yields

$$dE = ky dy$$

Integrating between the equilibrium position $y = 0$ and y gives

$$\int_0^E dE = \int_0^y ky dy$$

$$E = \frac{1}{2}ky^2 \quad (16-4)$$

The potential-energy curve for a simple harmonic oscillation, derived from Equation 16-4, is a parabola, as depicted in

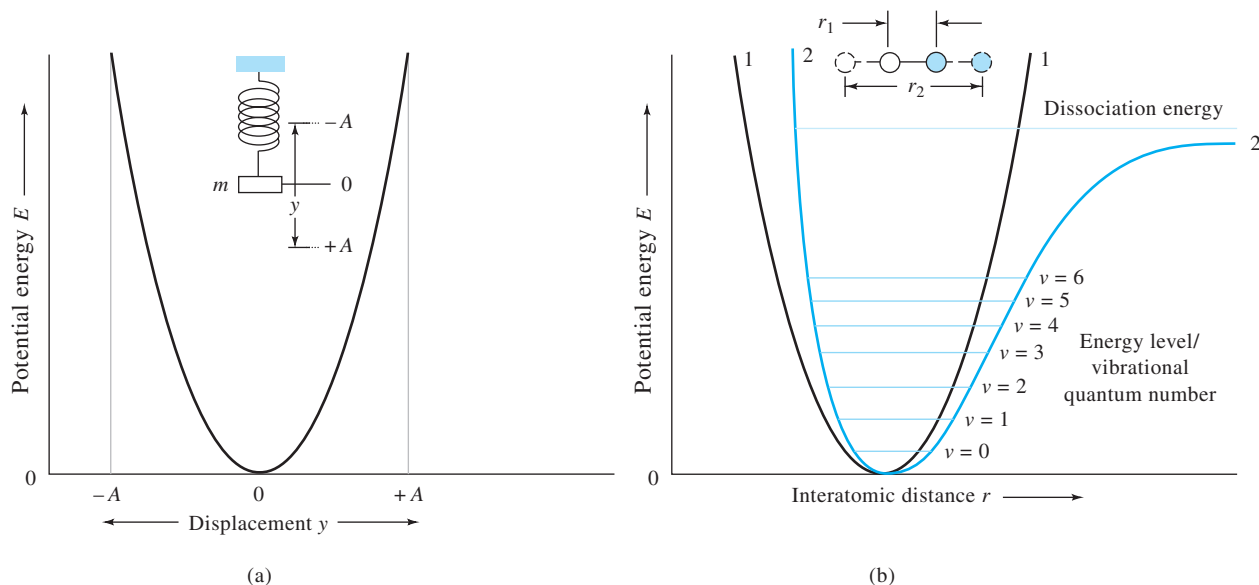


FIGURE 16-3 Potential-energy diagrams. (a) Harmonic oscillator. (b) Curve 1, harmonic oscillator; curve 2, anharmonic motion.

Figure 16-3a. Notice that the potential energy is a maximum when the spring is stretched or compressed to its maximum amplitude A , and it decreases to zero at the equilibrium position.

Vibrational Frequency

The motion of the mass as a function of time t can be deduced from classical mechanics as follows. Newton's second law states that

$$F = ma$$

where m is the mass and a is its acceleration. But acceleration is the second derivative of distance with respect to time. Thus,

$$a = \frac{d^2y}{dt^2}$$

Substituting these expressions into Equation 16-2 gives

$$m \frac{d^2y}{dt^2} = -ky \quad (16-5)$$

A solution to this equation must be a periodic function such that its second derivative is equal to the original function times $-k/m$. A suitable cosine relationship meets this requirement. Thus, the instantaneous displacement of the mass at time t can be written as

$$y = A \cos 2\pi\nu_m t \quad (16-6)$$

where ν_m is the natural vibrational frequency and A is the maximum amplitude of the motion. The second derivative of Equation 16-6 with respect to time is

$$\frac{d^2y}{dt^2} = -4\pi^2\nu_m^2 A \cos 2\pi\nu_m t \quad (16-7)$$

Substitution of Equations 16-6 and 16-7 into Equation 16-5 gives

$$A \cos 2\pi\nu_m t = \frac{4\pi^2\nu_m^2 m}{k} A \cos 2\pi\nu_m t$$

The natural frequency of the oscillation is then

$$\nu_m = \frac{1}{2\pi} \sqrt{\frac{k}{m}} \quad (16-8)$$

where ν_m is the natural frequency of the mechanical oscillator. Although it depends on the force constant of the spring and the mass of the attached body, the natural frequency is independent of the energy imparted to the system; changes in energy merely result in a change in the amplitude A of the vibration.

The equation just developed may be modified to describe the behavior of a system consisting of two masses m_1 and m_2 connected by a spring. This is accomplished by substituting the reduced mass μ for the single mass m where

$$\mu = \frac{m_1 m_2}{m_1 + m_2} \quad (16-9)$$

Thus, the vibrational frequency for such a system is given by

$$\nu_m = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} = \frac{1}{2\pi} \sqrt{\frac{k(m_1 + m_2)}{m_1 m_2}} \quad (16-10)$$

Molecular Vibrations

The approximation is ordinarily made that the behavior of a molecular vibration is analogous to the mechanical model just described. Thus, the frequency of the molecular vibration is calculated from Equation 16-10 after substituting the masses of the two atoms for m_1 and m_2 . The quantity k is the force constant of the chemical bond, which is a measure of its stiffness.

16A-3 Quantum Treatment of Vibrations

The equations of ordinary mechanics that we have used thus far do not completely describe the behavior of particles of atomic dimensions. For example, the quantized nature of molecular vibrational energies, and of other atomic and molecular energies as well, does not appear in these equations. We may, however, invoke the concept of the simple harmonic oscillator to develop the wave equations of quantum mechanics. Solutions of these equations for potential energies have the form

$$E = \left(\nu + \frac{1}{2} \right) \frac{h}{2\pi} \sqrt{\frac{k}{\mu}} \quad (16-11)$$

where h is Planck's constant, and ν is the *vibrational quantum number*, which can take only positive integer values (including zero). Thus, in contrast to ordinary mechanics where vibrators can assume any potential energy, quantum mechanical vibrators can take on only certain discrete energies.

It is interesting to note that the factor $\sqrt{k/\mu}/2\pi$ appears in both the classical and the quantum equations; by substituting Equation 16-10 into 16-11, we find

$$E = \left(\nu + \frac{1}{2} \right) h\nu_m \quad (16-12)$$

where ν_m is the vibrational frequency of the classical model.²

We now assume that transitions in vibrational energy levels can be brought about by absorption of radiation, provided the energy of the radiation exactly matches the difference in energy levels ΔE between the vibrational quantum states and provided also the vibration causes a change in dipole moment. This difference is identical between any pair of adjacent levels, because ν in Equations 16-11 and 16-12 can assume only whole numbers; that is,

$$\Delta E = h\nu_m = \frac{h}{2\pi} \sqrt{\frac{k}{\mu}} \quad (16-13)$$

²Unfortunately, the generally accepted symbol for the vibrational quantum number ν is similar in appearance to the Greek nu, ν , which symbolizes frequency. Thus, constant care must be exercised to avoid confusing the two in equations such as Equation 16-12.

At room temperature, the majority of molecules are in the ground state $v = 0$; thus, from Equation 16-12,

$$E_0 = \frac{1}{2}h\nu_m$$

Promotion to the first excited state $v = 1$ with energy

$$E_1 = \frac{3}{2}h\nu_m$$

requires radiation of energy

$$\left(\frac{3}{2}h\nu_m - \frac{1}{2}h\nu_m\right) = h\nu_m$$

The frequency of radiation ν that will bring about this change is identical to the classical vibrational frequency of the bond ν_m . That is,

$$E_{\text{radiation}} = h\nu = \Delta E = h\nu_m = \frac{h}{2\pi} \sqrt{\frac{k}{\mu}}$$

or

$$\nu = \nu_m = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}} \quad (16-14)$$

If we wish to express the radiation in wavenumbers, we substitute Equation 6-3 and rearrange:

$$\bar{\nu} = \frac{1}{2\pi c} \sqrt{\frac{k}{\mu}} = 5.3 \times 10^{-12} \sqrt{\frac{k}{\mu}} \quad (16-15)$$

where $\bar{\nu}$ is the wavenumber of an absorption maximum (cm^{-1}), k is the force constant for the bond in newtons per meter (N/m), c is the velocity of light (cm s^{-1}), and μ is the reduced mass (kg) defined by Equation 16-9.³

IR measurements in conjunction with Equation 16-14 or 16-15 permit the evaluation of the force constants for various types of chemical bonds. Generally, k has been found to lie in the range between 3×10^2 and 8×10^2 N/m for most single bonds, with 5×10^2 serving as a reasonable average value. Double and triple bonds are found by this same means to have force constants of about two and three times this value (1×10^3 and 1.5×10^3 N/m, respectively). With these average experimental values, Equation 16-15 can be used to estimate the wavenumber of the fundamental absorption band, or the absorption due to the transition from the ground state to the first excited state, for a variety of bond types. The following example demonstrates such a calculation.

EXAMPLE 16-1

Calculate the approximate wavenumber and wavelength of the fundamental absorption due to the stretching vibration of a carbonyl group C=O.

Solution

The mass of the carbon atom in kilograms is given by

$$\begin{aligned} m_1 &= \frac{12 \times 10^{-3} \text{ kg/mol}}{6.0 \times 10^{23} \text{ atom/mol}} \times 1 \text{ atom} \\ &= 2.0 \times 10^{-26} \text{ kg} \end{aligned}$$

Similarly, for oxygen,

$$m_2 = (16 \times 10^{-3}) / (6.0 \times 10^{23}) = 2.7 \times 10^{-26} \text{ kg}$$

and the reduced mass μ is given by (Equation 16-9)

$$\begin{aligned} \mu &= \frac{2.0 \times 10^{-26} \text{ kg} \times 2.7 \times 10^{-26} \text{ kg}}{(2.0 + 2.7) \times 10^{-26} \text{ kg}} \\ &= 1.1 \times 10^{-26} \text{ kg} \end{aligned}$$

As noted earlier, the force constant for the typical double bond is about 1×10^3 N/m. Substituting this value and μ into Equation 16-15 gives

$$\begin{aligned} \bar{\nu} &= 5.3 \times 10^{-12} \text{ s/cm} \sqrt{\frac{1 \times 10^3 \text{ N/m}}{1.1 \times 10^{-26} \text{ kg}}} \\ &= 1.6 \times 10^3 \text{ cm}^{-1} \end{aligned}$$

The carbonyl stretching band is found experimentally to be in the region of 1600 to 1800 cm^{-1} (6.3 to $5.6 \mu\text{m}$).

Selection Rules

As given by Equations 16-12 and 16-13, the energy for a transition from energy level 1 to 2 or from level 2 to 3 should be identical to that for the 0 to 1 transition. Furthermore, quantum theory indicates that the only transitions that can take place are those in which the vibrational quantum number changes by unity; that is, the *selection rule* states that $\Delta v = \pm 1$. Because the vibrational levels are equally spaced for a harmonic oscillator, only a single absorption peak should be observed for a given molecular vibration. In addition to the $\Delta v = \pm 1$ selection rule, there must be a change in dipole moment during the vibration.

Anharmonic Oscillator

Thus far, we have considered the classical and quantum mechanical treatments of the harmonic oscillator. The potential energy of such a vibrator changes periodically as the distance between the masses fluctuates (Figure 16-3a). From qualitative considerations, however, this description of a molecular vibration appears imperfect. For example, as the two atoms approach one

³By definition, the newton has the units of $\text{N} = \text{kg m/s}^2$. Thus, $\sqrt{k/\mu}$ has units of s^{-1} .

another, coulombic repulsion between the two nuclei produces a force that acts in the same direction as the restoring force of the bond. For this reason, the potential energy can be expected to rise more rapidly than the harmonic approximation predicts. At the other extreme of oscillation, a decrease in the restoring force, and thus the potential energy, occurs as the interatomic distance approaches that at which dissociation of the atoms takes place.

In theory, the wave equations of quantum mechanics permit the derivation of more nearly correct potential-energy curves for molecular vibrations. Unfortunately, however, the mathematical complexity of these equations prevents their quantitative application to all but simple diatomic systems. Qualitatively, the curves take the anharmonic form shown as curve 2 in Figure 16-3b. Such curves depart from harmonic behavior by varying degrees, depending on the nature of the bond and the atoms involved. Note, however, that the harmonic and anharmonic curves are nearly alike at low potential energies.

Anharmonicity leads to deviations of two kinds. At higher quantum numbers, ΔE becomes smaller (see curve 2 in Figure 16-3b), and the selection rule is not rigorously followed. As a result, weaker transitions called *overtone*s are sometimes observed. These transitions correspond to $\Delta v = \pm 2$ or ± 3 . The frequencies of such overtone transitions are approximately two or three times that of the fundamental frequency, and the intensities are lower than that of the fundamental.

Vibrational spectra are further complicated by two different vibrations in a molecule interacting to give absorptions at frequencies that are approximately the sums or differences of their fundamental frequencies. Again, the intensities of these sum and difference bands are generally low.

16A-4 Vibrational Modes

It is ordinarily possible to deduce the number and kinds of vibrations in simple diatomic and triatomic molecules and whether these vibrations will lead to absorption. Complex molecules may contain several types of atoms as well as bonds. For these molecules, the many types of possible vibrations give rise to IR spectra that are much more difficult to analyze or predict.

The number of possible vibrations in a polyatomic molecule can be calculated as follows. Three coordinates are needed to locate a point in space. To fix N points requires a set of three coordinates for each point, for a total of $3N$. Each coordinate corresponds to one degree of freedom for one of the atoms in a polyatomic molecule. For this reason, a molecule containing N atoms is said to have $3N$ *degrees of freedom*.

In defining the motion of a molecule, we need to consider (1) the motion of the entire molecule through space (that is, the translational motion of its center of gravity); (2) the rotational motion of the entire molecule around its center of gravity; and (3) the motion of each of its atoms relative to the other atoms, or in other words, its individual vibrations. Because all atoms in the molecule move in concert through space, definition of

translational motion requires three coordinates and thus this common motion requires three of the $3N$ degrees of freedom. Another three degrees of freedom are needed to describe the rotation of the molecule as a whole. The remaining $3N - 6$ degrees of freedom involve interatomic motion and hence represent the number of possible vibrations within the molecule. A linear molecule is a special case because, by definition, all of the atoms lie on a single, straight line. Rotation about the bond axis is not possible, and two degrees of freedom suffice to describe rotational motion. Thus, the number of vibrations for a linear molecule is given by $3N - 5$. Each of the $3N - 6$ or $3N - 5$ vibrations is called a *normal mode*.

For each normal mode of vibration, there is a potential-energy relationship such as that shown by the solid lines in Figure 16-3b. The same selection rules discussed earlier apply for each of these relationships. In addition, to the extent that a vibration approximates harmonic behavior, the differences between the energy levels of a given vibration are the same; that is, a single absorption peak appears for each vibration having a change in dipole moment.

Four factors tend to produce fewer experimental bands than would be expected from the theoretical number of normal modes. Fewer absorption bands are found when (1) the symmetry of the molecules is such that no change in dipole moment results from a particular vibration; (2) the energies of two or more vibrations are identical or nearly identical; (3) the absorption intensity is so low as to be undetectable by ordinary means; or (4) the vibrational energy is in a wavelength region beyond the range of the instrument used to measure absorption.

Occasionally, more absorption bands are found than are expected based on the number of normal modes. We have already mentioned the overtone bands that occur at two or three times the frequency of the fundamental. In addition, *combination bands* are sometimes encountered when a photon excites two vibrational modes simultaneously. The frequency of the combination band is approximately the sum or difference of the two fundamental frequencies. This phenomenon occurs when a quantum of energy is absorbed by two bonds rather than one.

16A-5 Vibrational Coupling

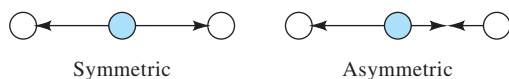
The energy of a vibration, and thus the wavelength of the corresponding absorption maximum, may be influenced by (or coupled with) other vibrators in the molecule. A number of factors influence the extent of such coupling.

1. Strong coupling between stretching vibrations occurs only when there is an atom common to the two vibrations.
2. Interaction between bending vibrations requires a common bond between the vibrating groups.
3. Coupling between a stretching and a bending vibration can occur if the stretching bond forms one side of the angle that varies in the bending vibration.

- Interaction is greatest when the coupled groups have individual energies that are nearly equal.
- Little or no interaction is observed between groups separated by two or more bonds.
- Coupling requires that the vibrations be of the same symmetry species.⁴

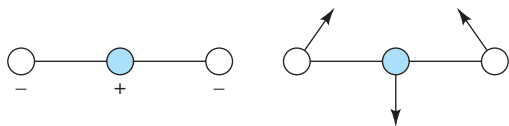
As an example of coupling effects, let us consider the IR spectrum of carbon dioxide. If no coupling occurred between the two C=O bonds, an absorption band would be expected at the same wavenumber as that for the C=O stretching vibration in an aliphatic ketone (about 1700 cm^{-1} , or $6\text{ }\mu\text{m}$; see Example 16-1). Experimentally, carbon dioxide exhibits two absorption maxima, one at 2350 cm^{-1} ($4.3\text{ }\mu\text{m}$) and the other at 667 cm^{-1} ($15\text{ }\mu\text{m}$).

Carbon dioxide is a linear molecule and thus has $(3 \times 3) - 5 = 4$ normal modes. Two stretching vibrations are possible; furthermore, interaction between the two can occur because the bonds involved are associated with a common carbon atom. As can be seen, one of the coupled vibrations is symmetric and the other is asymmetric.



The symmetric vibration causes no change in dipole moment, because the two oxygen atoms simultaneously move away from or toward the central carbon atom. Thus, the symmetric vibration is IR inactive. In the asymmetric vibration, one oxygen moves away from the carbon atom as the carbon atom moves toward the other oxygen. As a consequence, a net change in charge distribution occurs periodically, producing a change in dipole moment, so absorption at 2350 cm^{-1} results.

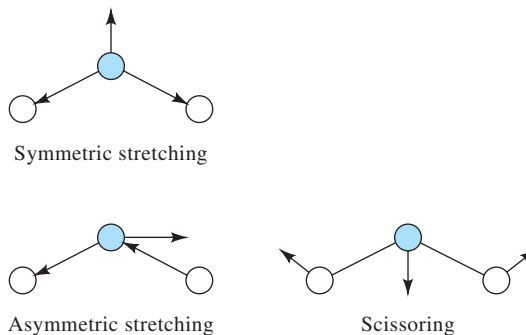
The remaining two vibrational modes of carbon dioxide involve scissoring, as shown here.



These two bending vibrations are the resolved components at 90° to one another of the bending motion in all possible planes around the bond axis. The two vibrations are identical in energy and thus produce a single absorption band at 667 cm^{-1} . Quantum states that are identical, as these are, are said to be *degenerate*.

It is of interest to compare the spectrum of carbon dioxide with that of a nonlinear, triatomic molecule such as water, sulfur dioxide, or nitrogen dioxide. These molecules have

$(3 \times 3) - 6 = 3$ vibrational modes that take the following forms:



Because the central atom is not in line with the other two, the symmetric stretching vibration produces a change in dipole moment and is thus IR active. For example, stretching peaks at 3657 and 3766 cm^{-1} (2.74 and $2.66\text{ }\mu\text{m}$) appear in the IR spectrum for the symmetric and asymmetric stretching vibrations of the water molecule. There is only one component to the scissoring vibration for this nonlinear molecule because motion in the plane of the molecule constitutes a rotational degree of freedom. For water, the bending vibration causes absorption at 1595 cm^{-1} ($6.27\text{ }\mu\text{m}$). The difference in behavior of linear and nonlinear triatomic molecules with two and three absorption bands, respectively, illustrates how IR absorption spectroscopy can sometimes be used to deduce molecular shapes.

Coupling of vibrations is a common phenomenon. As a result, the position of an absorption band corresponding to a given organic functional group cannot be specified exactly. For example, the C—O stretching frequency in methanol is 1034 cm^{-1} ($9.67\text{ }\mu\text{m}$), in ethanol it is 1053 cm^{-1} ($9.50\text{ }\mu\text{m}$), and in 2-butanol it is 1105 cm^{-1} ($9.05\text{ }\mu\text{m}$). These variations result from a coupling of the C—O stretching with adjacent C—C stretching or C—H vibrations.

Although interaction effects may lead to uncertainties in the identification of functional groups contained in a compound, it is this very effect that provides the unique features of an IR absorption spectrum that are so important for the positive identification of a specific compound.

16B IR INSTRUMENTATION

Three types of instruments for IR absorption measurements are commonly available: (1) dispersive spectrophotometers with a grating monochromator, (2) FT spectrometers employing an interferometer (Section 7I), and (3) nondispersive photometers using a filter or an absorbing gas that are used for analysis of atmospheric gases at specific wavelengths.

Until the 1980s, the most widely used instruments for IR measurements were dispersive spectrophotometers. Now, however, this type of instrument has been largely displaced for

⁴For a discussion of symmetry operations and symmetry species, see F. A. Cotton, *Chemical Applications of Group Theory*, 3rd ed., New York: Wiley 1990; R. S. Drago, *Physical Methods for Chemists*, 2nd ed., Philadelphia: Saunders, 1992; A. Vincent, *Molecular Symmetry and Group Theory*, Chichester: Wiley, 2001.

mid- and far-IR measurements by FT spectrometers because of their speed, reliability, signal-to-noise advantage, and convenience. Dispersive spectrometers are still used in the near-IR where they are often extensions of UV-visible instruments, but many dedicated near-IR instruments are of the Fourier transform-IR (FTIR) type.

16B-1 FT Spectrometers

The theoretical basis and the inherent advantages of FT and other multiplex instruments were discussed in some detail in Section 7I, and the reader may find it worthwhile to review that section before proceeding further. Two types of multiplex instruments have been described for the IR region. In the FT spectrometer, coding is accomplished by splitting the source into two beams whose path lengths can be varied periodically to give interference patterns. The FT is then used for data processing.⁵ The second is the Hadamard transform spectrometer, which is a dispersive instrument that uses a moving mask at the focal plane of a monochromator for encoding the spectral data. Hadamard transform IR instruments have not been widely adopted and will, therefore, not be discussed further in this book.⁶

When FTIR spectrometers first appeared in the marketplace, they were bulky, expensive (>\$100,000), and required frequent mechanical adjustments. For these reasons, their use was limited to special applications where their unique characteristics (speed, high resolution, sensitivity, and unparalleled wavelength precision and accuracy) were essential. FT instruments have now been reduced to benchtop size and have become reliable and easy to maintain. Furthermore, the price of simpler models has been reduced to the point where they are competitive with all but the simplest dispersive instruments (~\$15,000 and more). For these reasons, FT instruments have largely displaced dispersive instruments in the laboratory.⁷

Components of FT Instruments

The majority of commercially available FTIR instruments are based on the Michelson interferometer, although other types of optical systems are also encountered. We shall consider the Michelson design only, which is illustrated in Figure 7-43.⁸

Drive Mechanism. Requirements for satisfactory interferograms (and thus satisfactory spectra) are that the moving mirror have constant velocity and that its position be precisely known at many points along its travel. The planarity of the mirror must also remain constant during its entire sweep of 10 cm or more.

In the far-IR region, where wavelengths range from 50 to 1000 μm (200 to 10 cm^{-1}), displacement of the mirror by a fraction of a wavelength, and accurate measurement of its position, can be accomplished by means of a motor-driven micrometer screw. A more precise and sophisticated mechanism is required for the mid- and near-IR regions, however. Here, the mirror mount is generally floated on an air bearing held within close-fitting stainless steel sleeves (see Figure 16-4). The mount is driven by a linear drive motor and an electromagnetic coil similar to the voice coil in a loudspeaker; an increasing current in the coil drives the mirror at constant velocity. After reaching its terminus, the mirror is returned rapidly to the starting point for the next sweep by a rapid reversal of the current. The length of travel varies from 1 to about 20 cm; the scan rates range from 0.01 to 10 cm/s.

Two additional features of the mirror system are necessary for successful operation. The first is a means of sampling the interferogram at precisely spaced retardation intervals. The second is a method for determining exactly the zero retardation point to permit signal averaging. If this point is not known precisely, the signals from repetitive sweeps would not be in phase, and averaging would then tend to degrade rather than improve the signal.

The problem of precise signal sampling and signal averaging can be accomplished by using two or three interferometers rather than one, with a single mirror mount holding the movable mirrors. Figure 16-4 is a schematic showing such an arrangement. The components and radiation paths for each of the three interferometer systems are indicated by the subscripts 1, 2, and 3, respectively. System 1 is the IR system that ultimately provides an interferogram similar to that shown as curve A in Figure 16-5. System 2 is a so-called laser-fringe reference system, which provides sampling-interval information. It consists of a He-Ne laser S_2 , an interferometric system including mirrors MM_2 and M_2 , a beamsplitter B_2 , and a transducer T_2 . The output from this system is a cosine wave, as shown in C of Figure 16-5. This signal is converted electronically to the square-wave form shown in D; sampling begins or terminates at each successive zero crossing. The laser-fringe reference system gives a highly reproducible and regularly spaced sampling interval. In most instruments, the laser signal is also used to control the speed of the mirror-drive system at a constant level.

The third interferometer system, sometimes called the white-light system, uses a tungsten source S_3 and transducer T_3 sensitive to visible radiation. Its mirror system is fixed to give a zero retardation displaced to the left from that for the analytical signal (see interferogram B, Figure 16-5). Because the source is polychromatic, its power at zero retardation is much larger than

⁵For detailed discussions of FTIR spectroscopy, see B. C. Smith, *Fundamentals of Fourier Transform Infrared Spectroscopy*, 2nd ed., Boca Raton, FL: CRC Press, 2011; P. R. Griffiths and J. A. deHaseth, *Fourier Transform Infrared Spectroscopy*, 2nd ed., New York: Wiley, 2007.

⁶For a description of the Hadamard transform and Hadamard transform spectroscopy, see D. K. Graff, *J. Chem. Educ.*, **1995**, 72, 304, DOI: 10.1021/ed072p304; *Fourier, Hadamard, and Hilbert Transforms in Chemistry*, A. G. Marshall, ed., New York: Plenum Press, 1982.

⁷For reviews of commercial FTIR spectrometers, see J. P. Smith and V. Hinson-Smith, *Anal. Chem.*, **2003**, 75, 37A, DOI: 10.1021/ac031224d; R. Mukhopadhyay, *Anal. Chem.*, **2004**, 76, 369A, DOI: 10.1021/ac041652z.

⁸The Michelson interferometer was designed and built in 1891 by A. A. Michelson. He was awarded the 1907 Nobel Prize in physics for the invention of interferometry.

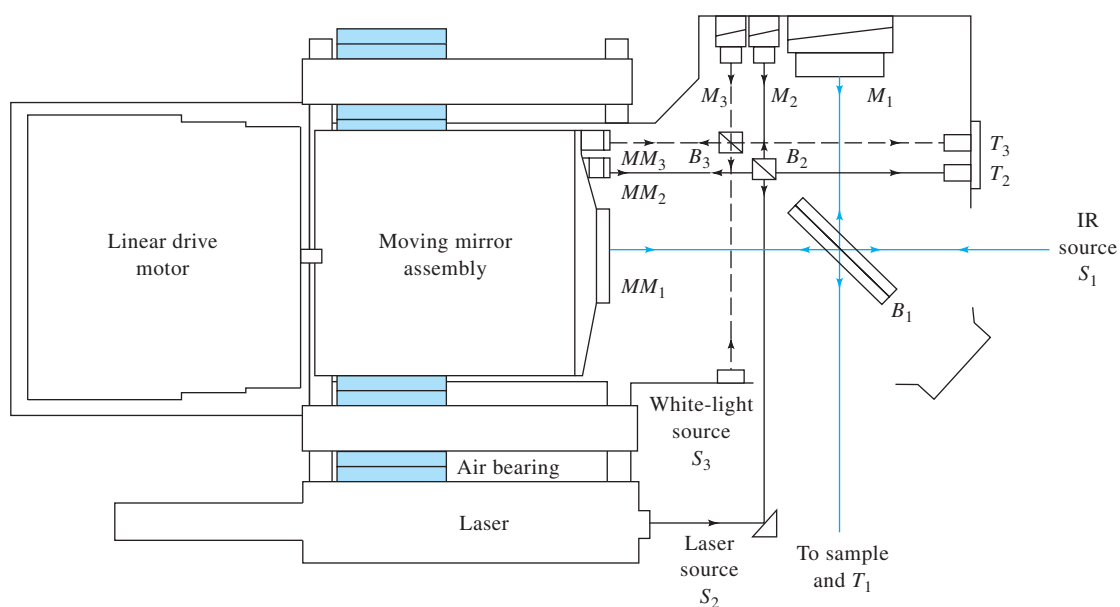


FIGURE 16-4 Interferometers in an FTIR spectrometer. Subscript 1 defines the radiation path in the IR interferometer. Subscripts 2 and 3 refer to the laser and white-light interferometers, respectively. (Courtesy of Thermo Fisher Scientific Inc., Waltham, MA.)

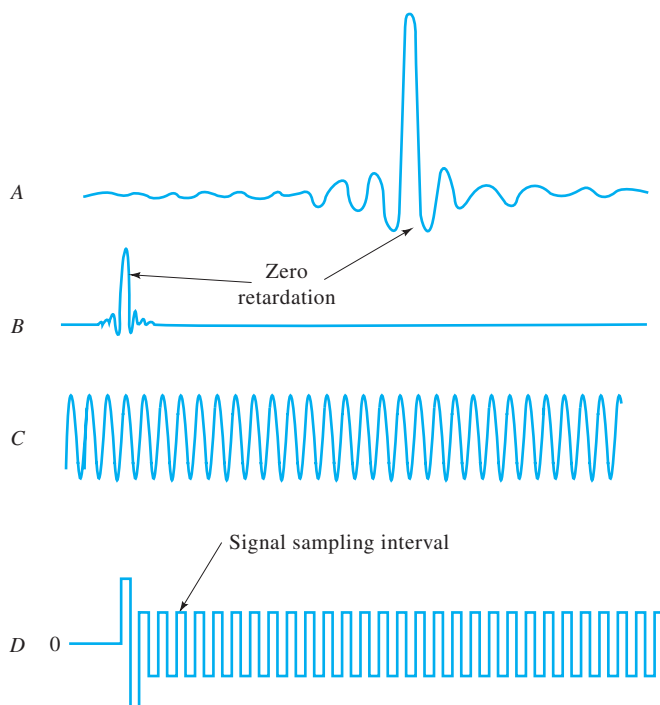


FIGURE 16-5 Time-domain signals for the three interferometers contained in many FTIR instruments. Curve A, IR signal; curve B, white-light signal; curve C, laser-fringe reference signal; curve D, square-wave electrical signal formed from the laser signal. (Adapted from P. R. Griffiths, *Anal. Chem.*, **1974**, *46*, 645A, DOI: 10.1021/ac60343a756. Copyright 1974 American Chemical Society.)



FIGURE 16-6 Photo of a basic, benchtop FTIR spectrometer suitable for student use. Spectra are recorded in a few seconds and displayed on a laptop computer. Software is available for controlling the instrument, collecting and processing data, and analyzing the results. An IR database is included for analysis and spectral interpretation. (Courtesy of Thermo Electron Corp., Madison, WI.)

any signal before and after that point. Thus, this maximum is used to trigger the start of data sampling for each sweep at a highly reproducible point.

The triple interferometer system just described leads to remarkable precision in determining spectral frequencies, which significantly exceeds that realizable with conventional grating instruments. This high reproducibility is particularly important when many scans are to be averaged. Contemporary instruments, such as the benchtop unit pictured in Figure 16-6, are able to achieve excellent frequency precision with one or two interferometers. In the instrument diagrammed in Figure 16-7, the interferometer is actually two parallel interferometers, one to modulate the IR radiation from the source before it passes through the sample and the second to modulate the red light from the He-Ne laser to provide the reference signal for acquiring data from the IR detector. No white-light source is used, and the IR interferogram is used to establish zero retardation. The maximum in the IR interferogram is an excellent reference because this is the only point at which all wavelengths interfere constructively.

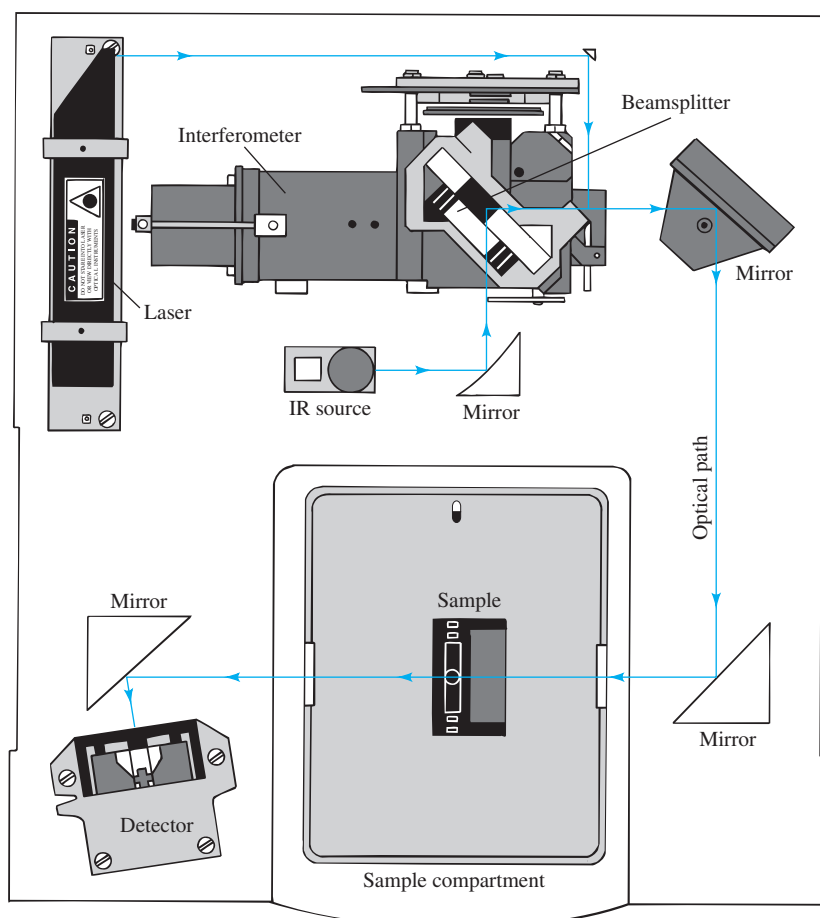


FIGURE 16-7 Diagram of a basic FTIR spectrometer. Radiation of all frequencies from the IR source is reflected into the interferometer where it is modulated by the moving mirror on the left. The modulated radiation is then reflected from the two mirrors on the right through the sample in the compartment at the bottom. After passing through the sample, the radiation falls on the transducer. A data-acquisition system attached to the transducer records the signal and stores it in the memory of a computer as an interferogram. (Courtesy of Thermo Fisher Scientific, Waltham, MA.)

The benchtop system shown in Figure 16-6 is capable of providing spectra with a resolution of better than 1 cm^{-1} . An educational kit is available to aid in using the instrument in an academic environment. The kit includes information on IR theory, FTIR instrumentation, and several experiments.

Instrument Designs

FTIR spectrometers can be single-beam or double-beam instruments. Figure 16-8 shows the optics of a basic single-beam spectrometer, which sells in the range of \$15,000 to \$25,000. A typical procedure for determining transmittance or absorbance with this type of instrument is to first obtain a reference interferogram by scanning a reference (usually air) twenty or thirty times, coadding the data, and storing the results in the memory of the instrument computer (usually after transforming it to the spectrum). A sample is then inserted in the radiation path and the process repeated. The ratio of sample and reference spectral data is then computed to give the transmittance at various frequencies. From this ratio the absorbance is calculated as a function of wavenumber. Ordinarily, modern IR sources and detectors are sufficiently stable so that reference spectra need to be obtained only occasionally.

A double-beam spectrometer is illustrated in Figure 16-9. The mirrors directing the interferometer beam through the sample and reference cells are oscillated rapidly compared to the movement of the interferometer mirror so that sample and reference information can be obtained at each mirror position. The double-beam design compensates for source and detector drifts.

Performance Characteristics of Commercial Instruments

A number of instrument manufacturers offer several models of FTIR instruments. The least expensive has a range of 7800 to 350 cm^{-1} (1.3 to $29\text{ }\mu\text{m}$) with a resolution of a few wavenumbers. This performance can be obtained with a scan time as brief as 1 second. More expensive instruments with interchangeable beamsplitters, sources, and transducers offer expanded frequency ranges and higher resolutions. For example, one instrument is reported to produce spectra from the far-IR (10 cm^{-1} or $1000\text{ }\mu\text{m}$) through the visible region to $50,000\text{ cm}^{-1}$, or 200 nm .



Simulation: Learn more about IR instrumentation at www.tinyurl.com/skoogpia7

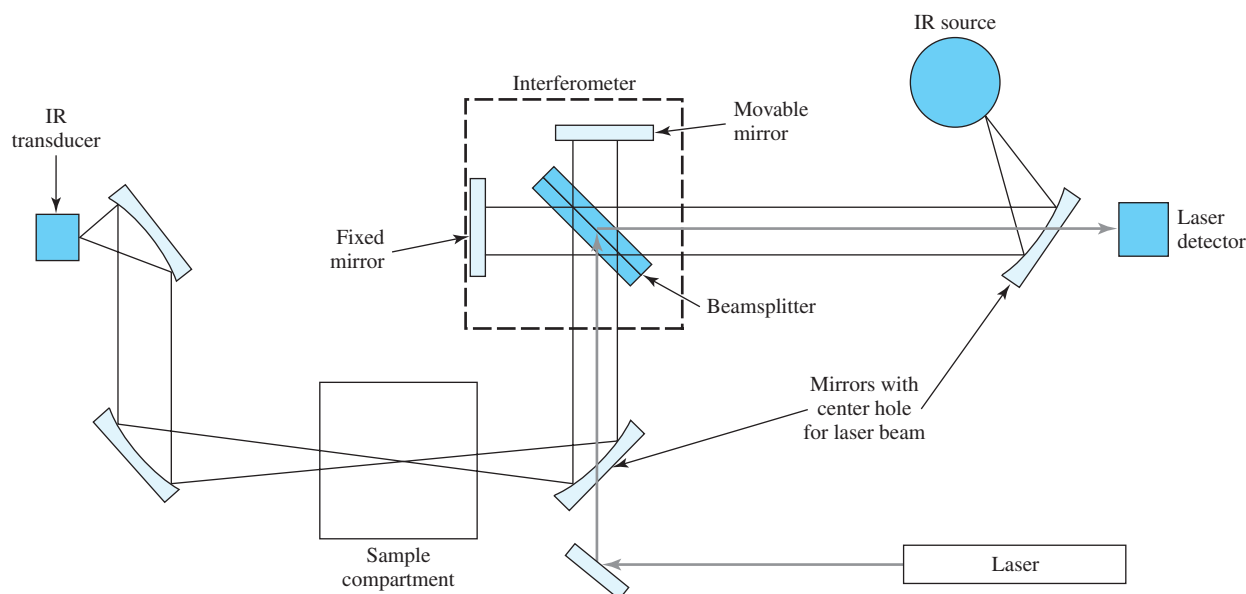


FIGURE 16-8 Single-beam FTIR spectrometer. In one arm of the interferometer, the IR source radiation travels through the beamsplitter to the fixed mirror, back to the beamsplitter, and through the sample to the IR transducer. In the other arm, the IR source radiation travels to the beamsplitter, is reflected to the movable mirror, and travels back through the beamsplitter to the sample and to the transducer. When the two beams meet again at the beamsplitter, they can interfere with each other if the phase difference (path difference) is appropriate. A plot of the signal versus mirror displacement is the interferogram. The interferogram contains information about all the frequencies present. The spectrum, intensity versus wavenumber, is the FT of the interferogram. It can be calculated with a computer from the signal versus mirror displacement. An empty sample compartment allows the reference spectrum to be calculated. Next, the sample is placed in the sample compartment and the sample spectrum is obtained. The absorbance is then calculated at each wavenumber from the ratio of the sample intensity to the reference intensity.

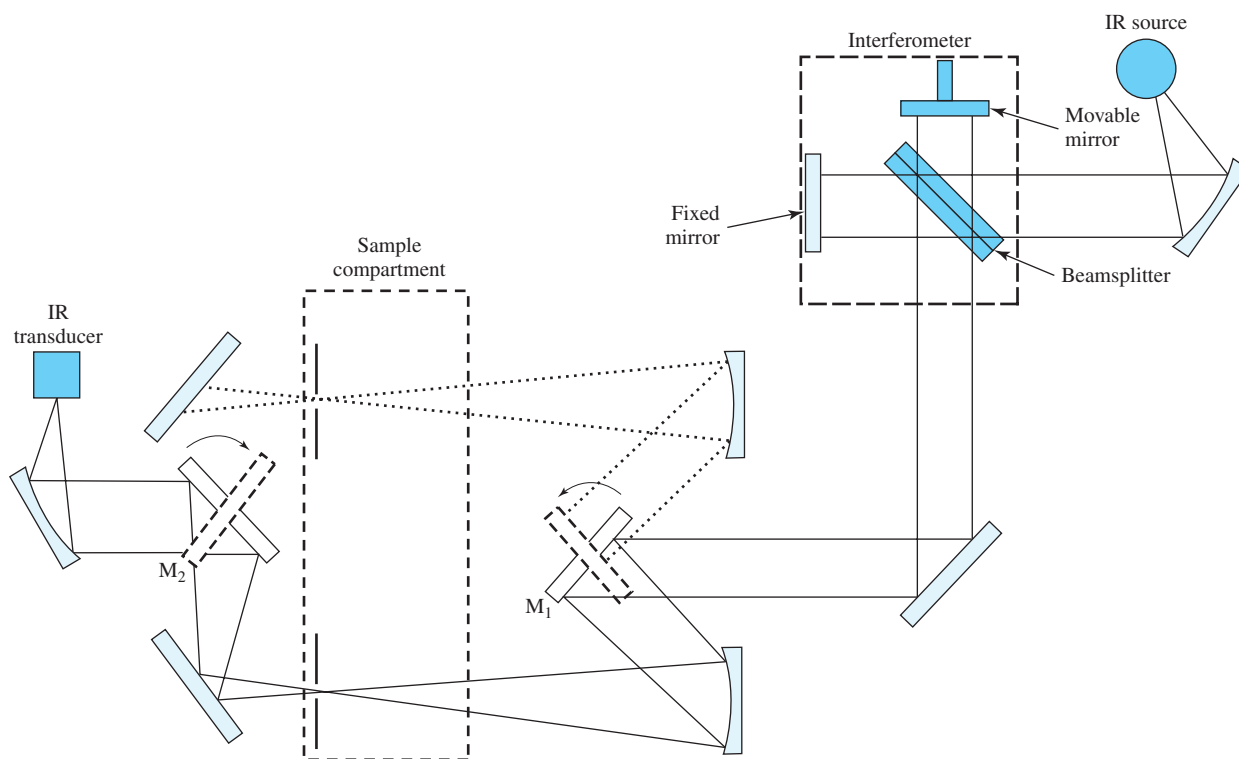


FIGURE 16-9 Double-beam FTIR spectrometer. The beam emerging from the interferometer strikes mirror M_1 , which in one position directs the beam through the reference cell and in the other position directs it through the sample cell. Mirror M_2 , which is synchronized to M_1 , alternately directs the reference beam and the sample beam to the transducer.

Resolutions for commercial instruments vary from 8 to less than 0.01 cm^{-1} . Several minutes are required to obtain a complete spectrum at the highest resolution.

Advantages of FT Spectrometers

Over most of the mid-IR spectral range, FT instruments have signal-to-noise ratios that are better than those of a good-quality dispersive instrument, usually by more than an order of magnitude. The enhanced signal-to-noise ratio can, of course, be traded for rapid scanning, with good spectra being attainable in a few seconds in most cases. Interferometric instruments are also characterized by high resolutions ($<0.1 \text{ cm}^{-1}$) and highly accurate and reproducible frequency determinations. The latter property is particularly helpful when spectra are to be subtracted for background correction.

Another advantage of FT instruments is that their optics provide a much larger energy throughput (one to two orders of magnitude) than do dispersive instruments, which are limited in throughput by the need for narrow slit widths. The potential gain in throughput, however, may be partially offset by the lower sensitivity of the fast-response detector required for the interferometric measurements. Finally, it should be noted that the interferometer is free from the problem of stray radiation because each IR frequency is, in effect, chopped at a different frequency.

FTIR instruments have advantages over dispersive instruments in all areas of chemistry, particularly for (1) very high-resolution work that is encountered with gaseous mixtures having complex spectra resulting from the superposition of vibrational and rotational bands, (2) the study of samples with high absorbances, (3) the study of substances with weak absorption bands (for example, the study of compounds that are chemisorbed on catalyst surfaces), (4) investigations requiring fast scanning such as kinetic studies or detection of chromatographic effluents, (5) collecting IR data from very small samples, (6) obtaining reflection spectra, and (7) IR emission studies.

16B-2 Dispersive Instruments

Although nearly all instruments produced today are FT systems, many dispersive spectrophotometers are still found in laboratories. Dispersive IR spectrophotometers are generally double-beam, recording instruments, which use reflection gratings for dispersing radiation. As was pointed out in Section 13D-2, the double-beam design is less demanding with respect to the performance of sources and detectors—an important characteristic because of the relatively low intensity of IR sources, the low sensitivity of IR transducers, and the resulting requirement for large signal amplifications (see Section 16C).

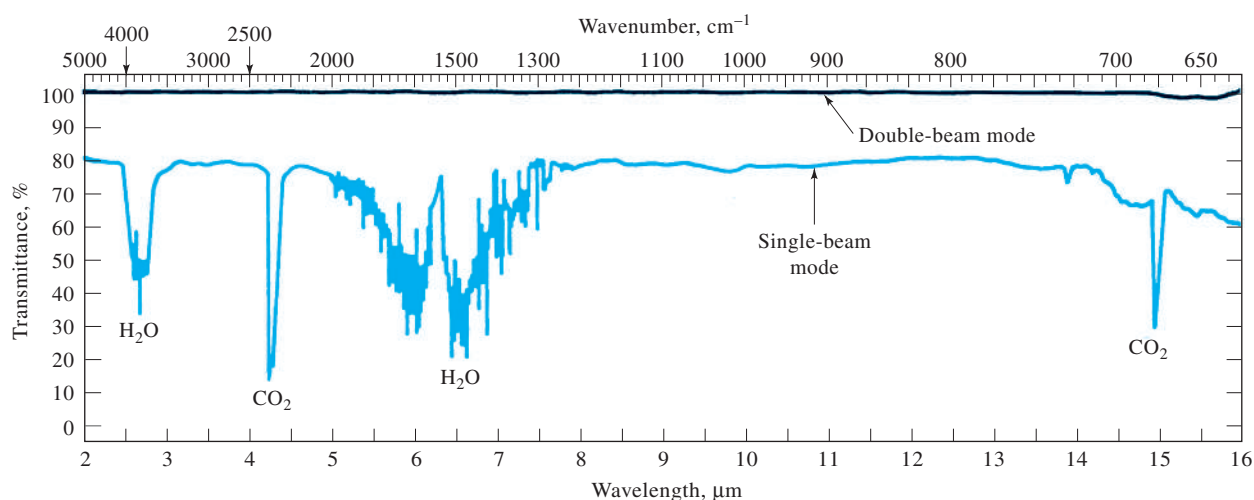


FIGURE 16-10 Single- and double-beam spectra of atmospheric water vapor and CO_2 . In the lower, single-beam trace, the absorption of atmospheric gases is apparent. The top, double-beam spectrum shows that the reference beam compensates very well for this absorption and allows for a stable 100% T baseline to be obtained. (From J. D. Ingle Jr. and S. R. Crouch, *Spectrochemical Analysis*, p. 409, Upper Saddle River, NJ: Prentice-Hall, 1988. With permission.)

An additional reason for the general use of double-beam instruments in the IR region is shown in Figure 16-10. The lower curve reveals that atmospheric water and carbon dioxide absorb radiation in some important spectral regions and can cause serious interference problems. The upper curve shows that the reference beam compensates nearly perfectly for absorption by both compounds. A stable 100% T baseline results.

Generally, dispersive IR spectrophotometers incorporate a low-frequency chopper (five to thirty cycles per second) that permits the detector to discriminate between the signal from the source and signals from extraneous radiation, such as IR emission from various bodies surrounding the transducer. Low chopping rates are demanded by the slow response times of the IR transducers used in most dispersive instruments. In general, the optical designs of dispersive instruments do not differ greatly from the double-beam UV-visible spectrophotometers discussed in the previous chapter except that the sample and reference compartments are always located between the source and the monochromator in IR instruments. This arrangement is possible because IR radiation, in contrast to UV-visible, is not sufficiently energetic to cause photochemical decomposition of the sample. Placing the sample and reference before the monochromator, however, has the advantage that most scattered radiation and IR emission, generated within the cell compartment, is effectively removed by the monochromator and thus does not reach the transducer.

Figure 16-11 shows schematically the arrangement of components in a typical IR spectrophotometer. Like many inexpensive dispersive IR instruments, it is an optical null type,

in which the radiant power of the reference beam is reduced, or attenuated, to match that of the beam passing through the sample. Attenuation is accomplished by imposing a device that removes a continuously variable fraction of the reference beam. The attenuator commonly takes the form of a comb, the teeth of which are tapered so that there is a linear relationship between the lateral movement of the comb and the decrease in power of the beam. Movement of the comb occurs when a difference in power of the two beams is sensed by the detection system. Most dispersive IR instruments are older instruments that use mechanical recorders instead of computers. For such instruments, the comb movement is synchronized with the recorder pen so that its position gives a measure of the relative power of the two beams and thus the transmittance of the sample.

Note that three types of systems link the components of the instrument in Figure 16-11: (1) a radiation linkage indicated by dashed lines, (2) a mechanical linkage shown by thick dark lines, and (3) an electrical connection shown by narrow solid lines.

Radiation from the source is split into two beams, half passing into the sample-cell compartment and the other half into the reference area. The reference beam then passes through the attenuator and on to a chopper. The chopper consists of a motor-driven disk that alternately reflects the reference or transmits the sample beam into the monochromator. After dispersion by a prism or grating, the alternating beams fall on the transducer and are converted to an electrical signal. The signal is amplified and passed to the synchronous rectifier, a device that is mechanically or electrically coupled to the chopper to cause

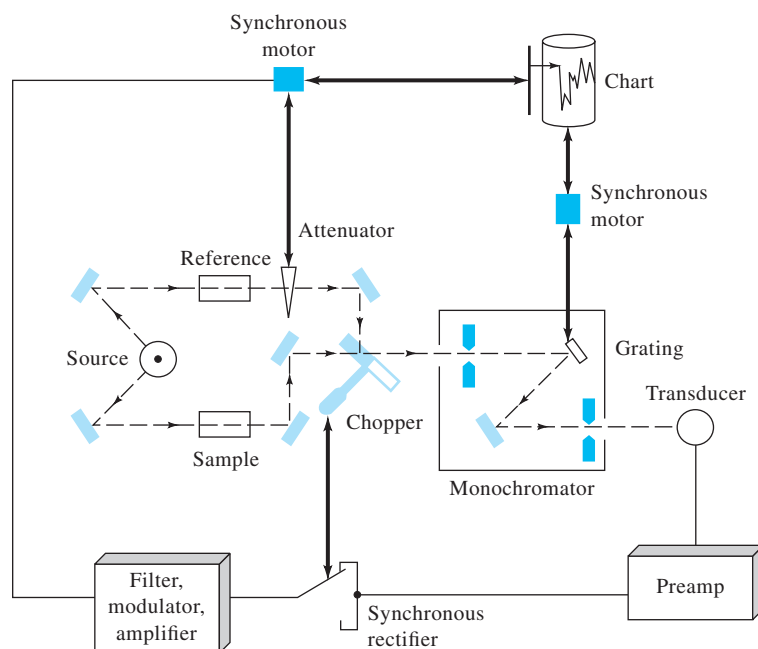


FIGURE 16-11 Schematic diagram of a double-beam, dispersive IR spectrophotometer. The heavy black lines indicate mechanical linkages, and the light lines indicate electrical connections. The radiation path is designated by dashed lines.

the rectifier switch and the beam leaving the chopper to change simultaneously. If the two beams are identical in power, the signal from the rectifier is constant direct current. If, on the other hand, the two beams differ in power, a fluctuating, or ac, current is produced, the phase of which is determined by which beam is the more intense. The current from the rectifier is filtered and further amplified to drive a synchronous motor in one direction or the other, depending on the phase of the input current. The synchronous motor is mechanically linked to both the attenuator and the pen drive of the recorder and causes both to move until a null balance is achieved. A second synchronous motor simultaneously drives the chart and varies the wavelength. Frequently, a mechanical linkage between the wavelength and slit drives varies the slit width to keep the radiant power reaching the detector approximately constant.

The reference-beam attenuator system, such as the one just described, creates three limitations to the performance of dispersive IR instruments. First, the response of the attenuator system always lags behind the transmittance changes, particularly in scanning regions where the signal is changing most rapidly. Second, the momentum associated with both the mechanical attenuator and the recorder system may result in the pen drive overshooting the true transmittance. Third, in regions where the transmittance approaches zero, almost no radiation reaches the transducer, and the exact null position cannot be established accurately. The result is sluggish transducer response and rounded peaks. Figure 16-12

illustrates transmittance overshoot and rounded peaks in regions of low transmittance (1700 and 3000 cm^{-1}).

16B-3 Nondispersive Instruments

Several simple, rugged instruments have been designed for quantitative IR analysis. Some are simple filter photometers, whereas others use filter wedges in lieu of a dispersing element to provide entire spectra. Finally, some gas analyzers use no wavelength-selection device at all. Generally, these instruments are less complex, more rugged, easier to maintain, and less expensive than the instruments we have previously described in this chapter.

Filter Photometers

Figure 16-13 shows a schematic diagram of a portable, IR filter photometer designed for quantitative analysis of various substances in the atmosphere.⁹ Different models are available that are factory calibrated for 1, 5, 30, or more than 100 gases. The instrument is computer controlled. For many compounds, fixed bandpass filters of 1.8 , 3.3 , 3.6 , 4.0 , 4.2 , 4.5 , 4.7 , 8 , 11 , and $14\text{ }\mu\text{m}$ can be used. A continuously variable filter, which transmits in the range between about 7.7 and $14.1\text{ }\mu\text{m}$

⁹See P. A. Wilks, *Amer. Lab.*, **1994** (12), 44; *Proc. Control Qual.*, **1992**, 3, 283.

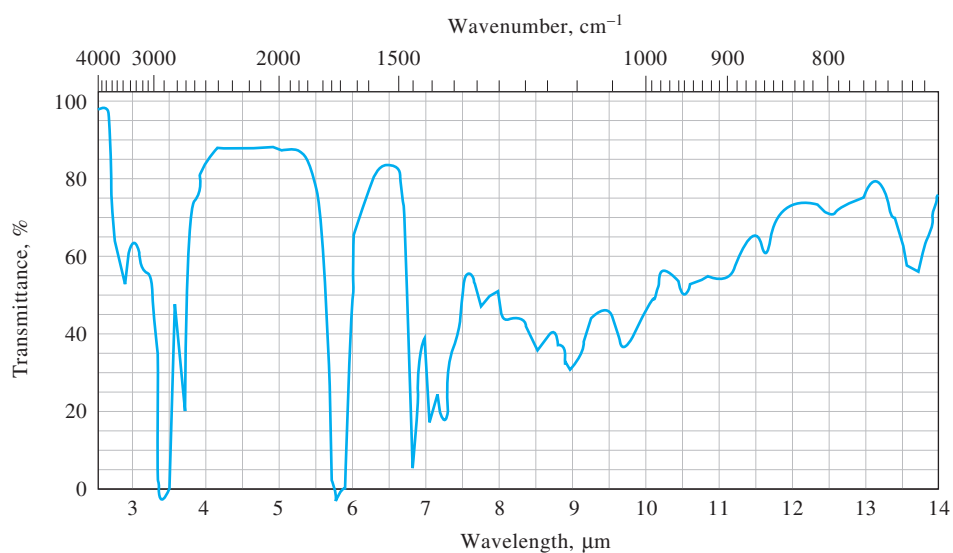


FIGURE 16-12 IR spectrum of *n*-hexanal illustrating overshoot at low %*T*.

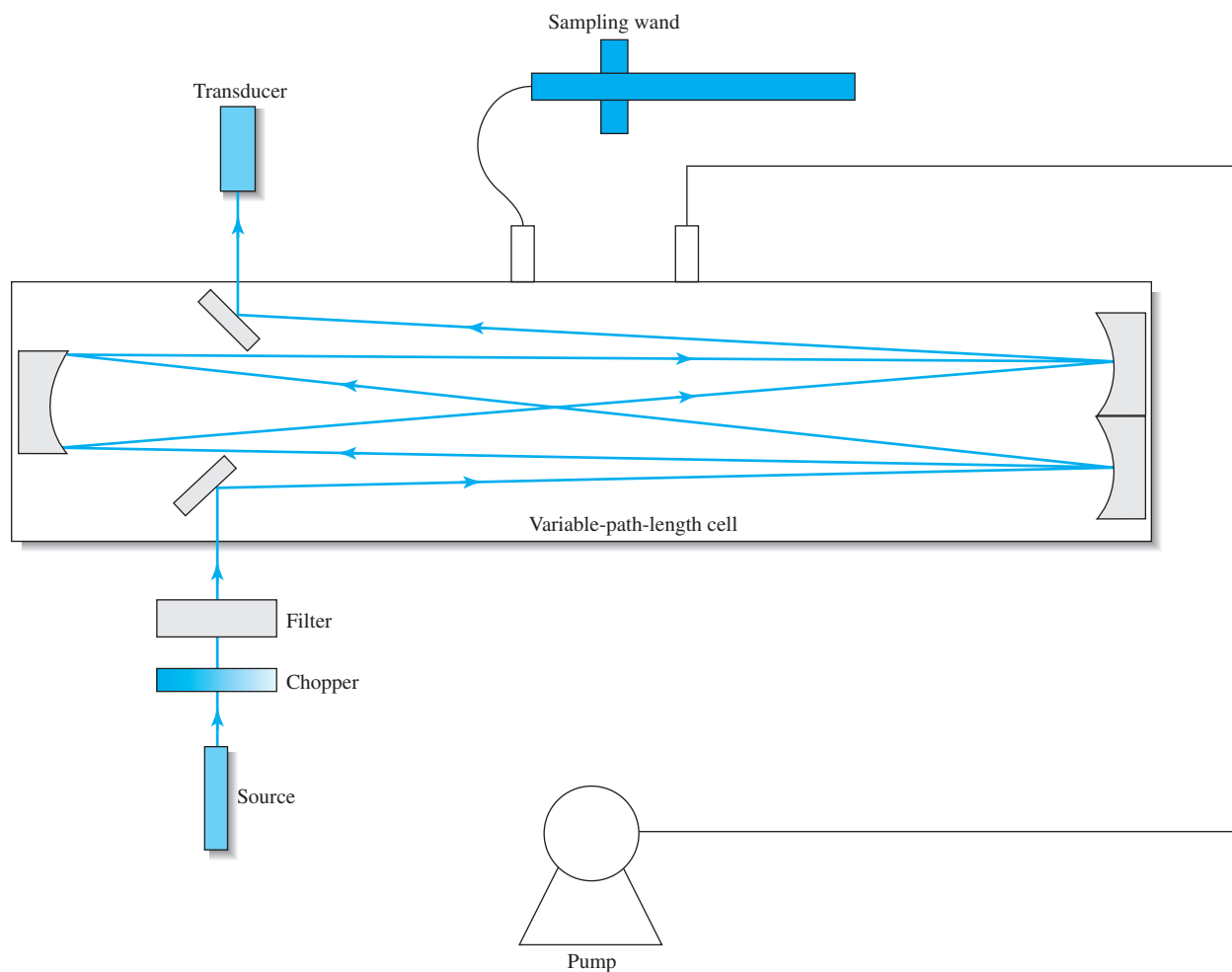


FIGURE 16-13 A portable IR photometer designed for gas analysis. The sample is introduced into the cell by means of a pump. The path length can be changed by altering the number of mirror reflections. (Courtesy of Thermo Fisher Scientific, Waltham, MA.)

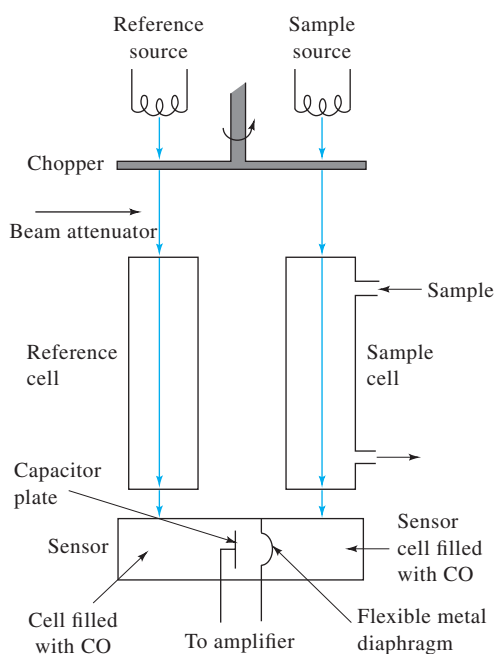


FIGURE 16-14 A nondispersive IR photometer for monitoring CO.

(1300 to 710 cm^{-1}), can also be used for selecting alternative wavelengths or for spectral scanning applications. The source is a nichrome-wire filament, and the transducer is a pyroelectric device (see Section 16C for descriptions of sources and transducers).

The gaseous sample is brought into the cell by means of a battery-operated pump at a rate of 20 L/min. In the cell, three gold-plated mirrors are used in a folded-path-length design. Path lengths of 0.5 m and 12.5 m may be selected. Detection of many gases at sub-parts-per-million levels, particularly with the long-path-length setting, have been reported with this photometer.

A recently designed filter IR instrument has been reported for determination of CO_2 in industrial settings. The estimated limit of detection was 1 ppm.¹⁰

Photometers without Filters

Photometers, which have no wavelength-restricting device, are widely used to monitor gas streams for a single component.¹¹ Figure 16-14 shows a typical nondispersive instrument designed to determine carbon monoxide (CO) in a gaseous

mixture. The reference cell is a sealed container filled with a nonabsorbing gas; as shown in the figure, the sample flows through a second cell that is of similar length. The chopper blade is so arranged that the beams from identical sources are chopped simultaneously at the rate of about five times per second. Selectivity is obtained by filling both compartments of the sensor cell with the gas being analyzed, CO in this example. The two chambers of the detector are separated by a thin, flexible, metal diaphragm that serves as one plate of a capacitor; the second plate is contained in the sensor compartment on the left.

In the absence of CO in the sample cell, the two sensor chambers are heated equally by IR radiation from the two sources. If the sample contains CO, however, the right-hand beam is attenuated somewhat and the corresponding sensor chamber becomes cooler with respect to its reference counterpart. As a result, the diaphragm moves to the right and the capacitance of the capacitor changes. This change in capacitance is sensed by the amplifier system. The amplifier output drives a servomotor that moves the beam attenuator into the reference beam until the two compartments are again at the same temperature. The instrument thus operates as a null balance device. The chopper serves to provide an ac signal, which is less sensitive to drift and $1/f$ noise.

This type of instrument is highly selective because heating of the sensor gas occurs only from that narrow portion of the spectrum of radiation absorbed by the CO in the sample. The device can be adapted to the determination of any IR-absorbing gas (e.g., CO_2 , NO, or H_2S).¹²

Filter Correlation Analyzers

As shown in Figure 16-15, filter correlation analyzers use a rotating gas filter through which the IR beam passes. The filter has two compartments, one for the gas of interest and the other for a nonabsorbing gas such as nitrogen. When the gas of interest is in the beam, it selectively attenuates the IR source to produce a reference beam. The sample beam is produced by the transparent gas. Typically, the IR source is chopped at a fairly high frequency (360 Hz) whereas the filter rotates at a fairly low frequency (30 Hz). A modulated signal is produced that is related to the concentration of the analyte gas.

Filter correlation analyzers are available for such gases as CO_2 and CO. They can be set up to detect trace levels (<0.1 ppm) or higher amounts. The analyzers are calibrated by constructing a working curve by dilution of a gas standard.

¹⁰See J. Hodgkinson et al., *Sens. Actuators B*, **2013**, 186, 580, DOI: 10.1016/j.snb.2013.06.006.

¹¹For the determination of CO in cigarette smoke, see *Appl. Spectrosc.*, **2006**, 60, 272, DOI: 10.1366/000370206776342616.

¹²For CO_2 in seawater, see *Marine Chemistry*, **2006**, 100, 24, DOI: 10.1016/j.marchem.2005.10.009; for NO in cigarette smoke, see reference 11; for H_2S in acidic alkylamine gas streams, see *Hydrocarbon Processing*, **2009**, 88 (February, issue 2), 73.

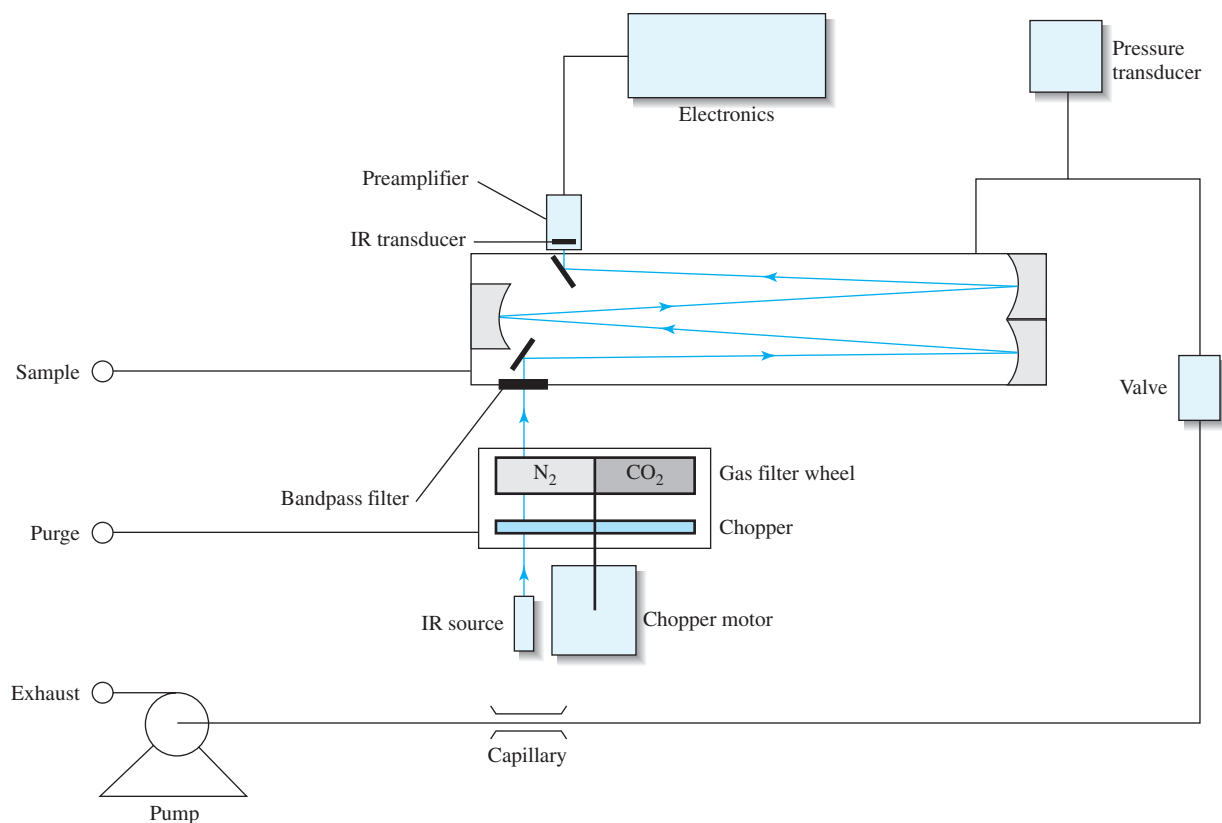


FIGURE 16-15 An IR filter correlation analyzer for determining CO_2 . The sample is drawn into the sample cell by a pump. The chopped IR source radiation (360 Hz) alternates between the N_2 and CO_2 sides of the filter wheel, which rotates at 30 Hz. The CO_2 side provides a reference beam that cannot be further attenuated by CO_2 in the sample cell. The N_2 side produces the sample beam by allowing the IR radiation to pass through to the cell where it can be attenuated by CO_2 in the sample. The modulated amplitude of the chopped detector signal is related to the CO_2 concentration in the sample. Other gases do not modulate the detector signal because they absorb the reference and sample beams equally. (Courtesy of Thermo Fisher Scientific Inc., Waltham, MA.)

16C IR SOURCES AND TRANSDUCERS

Instruments for measuring IR absorption all require a source of continuous IR radiation and an IR transducer. The desirable characteristics of these instrument components were listed in Sections 7B and 7E. In this section we describe sources and transducers that are found in modern IR instruments.

16C-1 Sources

IR sources consist of an inert solid that is heated electrically to a temperature between 1500 and 2200 K. These sources produce continuum radiation approximating that of a blackbody (see Figure 6-22). The maximum radiant intensity at these temperatures occurs between 5000 and 5900 cm^{-1} (2 and $1.7\text{ }\mu\text{m}$). At longer wavelengths, the intensity falls off smoothly until it is about 1% of the maximum at 670 cm^{-1} ($15\text{ }\mu\text{m}$). On the short wavelength side, the decrease is much more rapid, and a similar reduction in intensity is observed at about $10,000\text{ cm}^{-1}$ ($1\text{ }\mu\text{m}$).

The Nernst Glower

The Nernst glower is composed of rare earth oxides formed into a cylinder having a diameter of 1 to 3 mm and a length of 2 to 5 cm. Platinum leads are sealed to the ends of the cylinder to permit electrical connections to what amounts to a resistive heating element. As current passes through the device, its temperature increases to between 1200 K and 2200 K. The Nernst glower has a large negative temperature coefficient of electrical resistance, and it must be heated externally to a dull red heat before the current is large enough to maintain the desired temperature. Because the resistance decreases with increasing temperature, the source circuit must be designed to limit the current or the glower rapidly becomes so hot that it is destroyed.

Figure 16-16 shows the spectral output of a Nernst glower operated at approximately 2200 K. Note that the overall curve shape is similar to that of a blackbody. The small peaks and depressions are a function of the chemical composition of the device.

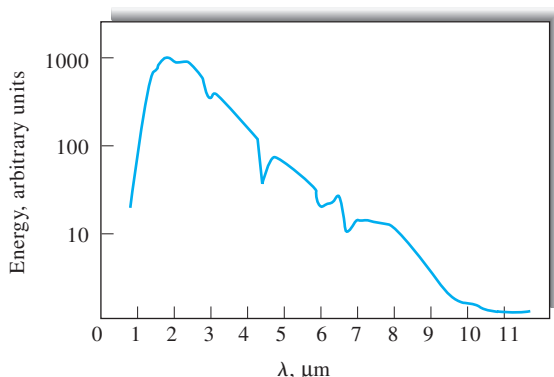


FIGURE 16-16 Spectral distribution of energy from a Nernst glower operated at approximately 2200 K.

The Globar

A Globar is a ceramic rod, usually silicon carbide or a composite material, electrically heated to more than 1000 K. Older dispersive instruments used Globars that were heated as high as 1500 K to provide sufficient intensity. The sensitivity of modern FTIR instruments has meant that lower source temperatures (1000 to 1300 K) can be used, resulting in longer source lifetimes.

Globar sources for FTIR must be temperature controlled for high stability. Some simple sources in older instruments used a fan to provide air cooling, but water cooling is more effective. The Globar has the advantage of a positive temperature coefficient of resistance that makes it possible to control temperature electronically. The most stable sources, therefore, use electronic temperature control, which eliminates the inconvenience and temperature fluctuations of circulating water. Electronic temperature control is also said to increase the source lifetime. Spectral energies of the Globar and Nernst glower are similar, except in the region below 5 μm , where the Globar is significantly more intense. Spectral energies of the Globar and the Nernst glower are comparable except in the region below 5 μm , where the Globar provides a significantly greater output.

Incandescent Wire Source

A source with somewhat lower intensity but with longer life than the Globar or Nernst glower is a tightly wound spiral of nichrome wire heated to about 1100 K by an electrical current. A rhodium wire heater sealed in a ceramic cylinder has similar properties, although it is more expensive. Nichrome wire sources are less intense than many IR sources. However, the incandescent wire source requires no cooling and is nearly maintenance free. For this reason, the nichrome wire source is often used where reliability is paramount, such as in process analyzers.

The Mercury Arc

For the far-IR region of the spectrum ($\lambda > 50 \mu\text{m}$), none of the thermal sources just described provides sufficient radiant power for convenient detection. In this region, a high-pressure mercury arc is often used. This device consists of a quartz-jacketed tube containing mercury vapor at a pressure greater than

1 atmosphere. When an electric current passes through the vapor, an internal plasma source is formed that provides continuum radiation in the far-IR region.

The Tungsten Filament Lamp

An ordinary tungsten filament lamp is a convenient source for the near-IR region of 4000 to 12,800 cm^{-1} (2.5 to 0.78 μm).

The Carbon Dioxide Laser Source

A tunable carbon dioxide laser is used as an IR source for monitoring the concentrations of certain atmospheric pollutants and for determining absorbing species in aqueous solutions.¹³ A carbon dioxide laser produces a band of radiation in the range of 900 to 1100 cm^{-1} (11 to 9 μm), which consists of about 100 closely spaced discrete lines. As described in Section 7B-3, any one of these lines can be chosen by tuning the laser. Although the range of wavelengths available is limited, the region from 900 to 1100 cm^{-1} is one particularly rich in absorption bands arising from the interactive stretching modes of CO_2 . Thus, this source is useful for quantitative determination of a number of important species such as ammonia, butadiene, benzene, ethanol, nitrogen dioxide, and trichloroethylene. An important property of the laser source is the radiant power available in each line, which is several orders of magnitude greater than that of blackbody sources.

Carbon dioxide lasers are widely used in remote-sensing applications such as light detection and ranging (lidar). The operating principle of lidar is similar to that of radar. The lidar system transmits radiation out to a target where it interacts with and is altered by the target. Some of the radiation is then reflected back to the lidar instrument where it is analyzed and used to obtain information about the target. By means of lidar, distance, speed, rotation, chemical composition, and concentration of remote targets can be obtained.

Semiconductor IR Laser Sources

Tunable semiconductor lasers have also been used as sources (see Section 7B-3 for a description of these devices) in the near- and mid-IR regions. As an example, the Sample Analysis at Mars (SAM) experiment on the Mars rover Curiosity's Science Laboratory uses a tunable diode laser spectrometer in the 2.78 μm region to detect carbon, hydrogen, and oxygen in the Martian environment and to determine their isotopic ratios. A quantum cascade tunable laser is used to detect methane by scanning across several of its rotational lines in the 3.3 μm region. Both lasers are used in Curiosity's two-channel tunable laser spectrometer (TLS) that can achieve a spectral resolution of 0.00002 cm^{-1} . Methane is determined by the difference in

¹³See A. A. Demidov, in *Introduction to Laser Spectroscopy*, 2nd ed., D. L. Andrews and A. A. Demidov, eds., New York: Kluwer Academic/Plenum Press, 2002; Z. Zelinger, M. Strizik, P. Kubat, and S. Civis, *Anal. Chim. Acta*, **2000**, 422, 179, DOI: 10.1016/S0003-2670(00)01069-2; P. L. Meyer, M. W. Sigrist, *Rev. Sci. Instrum.*, **1990**, 61, 1779, DOI: 10.1063/1.1141097.

the absorption signals obtained from a closed sample cell filled with the Mars atmosphere and the same cell evacuated by means of a turbo-molecular pump. How much of the methane detected arises from a chemical source in the Martian atmosphere and how much comes from the rover itself remains an open question.

16C-2 IR Transducers

IR transducers are of three general types: (1) pyroelectric transducers, (2) photoconducting transducers, and (3) thermal transducers. The first is found in photometers, many FTIR spectrometers, and dispersive spectrophotometers. Photoconducting transducers are also found in FTIR instruments. Thermal detectors are found in older dispersive instruments but are too slow to be used in FTIR spectrometers.

Pyroelectric Transducers

Pyroelectric transducers are constructed from single crystalline wafers of pyroelectric materials, which are insulators (dielectric materials) with very special thermal and electrical properties. Triglycine sulfate ($\text{NH}_2\text{CH}_2\text{COOH}$)₃ · H₂SO₄ (usually deuterated, DTGS, and often with a fraction of the glycines replaced with alanine, DLaTGS), is the most important pyroelectric material used for IR detection systems.

When an electric field is applied across any dielectric material, polarization takes place, with the magnitude of the polarization being a function of the dielectric constant of the material. For most dielectrics, this induced polarization rapidly decays to zero when the external field is removed. Pyroelectric substances, in contrast, retain a strong temperature-dependent polarization after removal of the field. Thus, by sandwiching the pyroelectric crystal between two electrodes, one of which is IR transparent, a temperature-dependent capacitor is produced. Changing its temperature by irradiating it with IR radiation alters the charge distribution across the crystal, which can be detected as a current in an external electrical circuit connecting the two sides of the capacitor. The magnitude of this current is proportional to the surface area of the crystal and the rate of change of polarization with temperature. Pyroelectric crystals lose their residual polarization when they are heated to a temperature called the *Curie point*. For triglycine sulfate, the Curie point is 47°C.

Pyroelectric transducers exhibit response times that are fast enough to allow them to track the changes in the time-domain signal from an interferometer. For this reason, many FTIR spectrometers for the mid-IR region use this type of transducer.

Photoconducting Transducers

IR photoconducting transducers consist of a thin film of a semiconductor material, such as lead sulfide, mercury telluride-cadmium telluride (MCT), or indium antimonide, deposited on a nonconducting glass surface and sealed in an evacuated envelope

to protect the semiconductor from the atmosphere. Absorption of radiation by these materials promotes nonconducting valence electrons to a higher energy-conducting state, thus decreasing the electrical resistance of the semiconductor. Typically, a photoconductor is placed in series with a voltage source and load resistor, and the voltage drop across the load resistor serves as a measure of the power of the beam of radiation.

A lead sulfide photoconductor is the most widely used transducer for the near-IR region of the spectrum from 10,000 to 333 cm⁻¹ (1 to 3 μm). It can be operated at room temperature. For mid- and far-IR radiation, MCT photoconductor transducers are used. They must be cooled with liquid nitrogen (77 K) to minimize thermal noise. The long-wavelength cutoff, and many of the other properties of these transducers, depend on the ratio of the mercury telluride to cadmium telluride, which can be varied continuously.

The MCT transducer is faster and more sensitive than the DTGS transducer discussed in the previous section. For this reason, the MCT transducer also finds widespread use in FTIR spectrometers, particularly those requiring fast response times, such as spectrometers interfaced to gas chromatographs and those intended for kinetic measurements.

Thermal Transducers

Thermal transducers, whose responses depend on the heating effect of radiation, are found in older dispersive spectrometers for detection of all but the shortest IR wavelengths. With these devices, the radiation is absorbed by a small blackbody and the resultant temperature rise is measured. The radiant power level from a spectrophotometer beam is minute (10⁻⁷ to 10⁻⁹ W), so that the heat capacity of the absorbing element must be as small as possible if a detectable temperature change is to be produced. Under the best of circumstances, temperature changes are confined to a few thousandths of a kelvin.

The problem of measuring IR radiation by thermal means is compounded by thermal noise from the surroundings. For this reason, thermal transducers are housed in a vacuum and carefully shielded from thermal radiation emitted by other nearby objects. To further minimize the effects of extraneous heat sources, the beam from the source is always chopped. In this way, the analyte signal, after transduction, has the frequency of the chopper and can be separated electronically from extraneous noise signals, which are ordinarily broad band or vary only slowly with time.

Thermocouples. In its simplest form, a thermocouple consists of a pair of junctions formed when two pieces of a metal such as bismuth are fused to each end of a dissimilar metal such as antimony. A potential difference between the two junctions varies with their difference in temperature.

The transducer junction for IR radiation is formed from very fine wires or alternatively by evaporating the metals onto a nonconducting support. In either case, the junction is usually

blackened (to improve its heat-absorbing capacity) and sealed in an evacuated chamber with a window that is transparent to IR radiation. The response time is typically about 30 ms.

The reference junction, which is usually housed in the same chamber as the active junction, is designed to have a relatively large heat capacity and is carefully shielded from the incident radiation. Because the analyte signal is chopped, only the difference in temperature between the two junctions is important; therefore, the reference junction does not need to be maintained at constant temperature. To enhance sensitivity, several thermocouples may be connected in series to produce a *thermopile*.

Bolometers. A bolometer is a type of resistance thermometer constructed of strips of metals, such as platinum or nickel, or from a semiconductor. This type of semiconductor device is sometimes called a *thermistor*. Semiconductor materials exhibit a relatively large change in resistance as a function of temperature. The responsive element in a bolometer is kept small and blackened to absorb the radiant heat. Bolometers are not so extensively used as other IR transducers for the mid-IR region. However, a germanium bolometer, operated at 1.5 K, is an excellent transducer for radiation in the range of 5 to 400 cm^{-1} (2000 to 25 μm). The response time is a few milliseconds.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



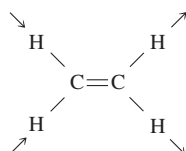
Problems with this icon are best solved using spreadsheets.

- * **16-1** The IR spectrum of CO shows a vibrational absorption band centered at 2170 cm^{-1} .
 - (a) What is the force constant for the CO bond?
 - (b) At what wavenumber would the corresponding peak for ^{14}CO occur?
- * **16-2** Gaseous HCl exhibits an IR absorption at 2890 cm^{-1} due to the hydrogen-chlorine stretching vibration.
 - (a) Calculate the force constant for the bond.
 - (b) Calculate the wavenumber of the absorption band for DCl assuming the force constant is the same as that calculated in part (a).
- 16-3** Calculate the absorption frequency corresponding to the —C—H stretching vibration treating the group as a simple diatomic C—H molecule with a force constant of $k = 5.0 \times 10^2 \text{ N/m}$. Compare the calculated value with the range found in correlation charts (such as the one shown in Figure 17-6). Repeat the calculation for the deuterated bond.
- 16-4** The force constant of the NO bond is approximately $1.53 \times 10^3 \text{ N/m}$. Treating the NO group as a simple diatomic molecule, calculate the absorption frequency in Hz and the wavelength and wavenumber of the fundamental absorption. How does this compare to the experimental value found for NO and NO dimers by Varetto, E. L.; Pimentel, G. C., *J. Chem. Phys.*, **1971**, 55, 3813, DOI: 10.1063/1.1676666.
- * **16-5** The wavelength of the fundamental O—H stretching vibration is about 3.0 μm . What is the approximate wavenumber and wavelength of the first overtone band for the O—H stretch?
- * **16-6** The wavelength of the first overtone of the N—H stretching vibration is about 1.5 μm . What is the approximate wavenumber and wavelength of the fundamental band for the N—H stretch?
- * **16-7** Sulfur dioxide is a nonlinear molecule. How many vibrational modes will this compound have? How many IR absorption bands would sulfur dioxide be expected to have?

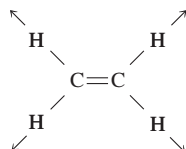
» QUESTIONS AND PROBLEMS (continued)

*** 16-8** Indicate whether the following vibrations are active or inactive in the IR spectrum.

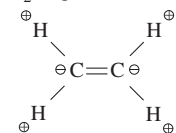
Molecule	Motion
(a) CH_3-CH_3	C—C stretching
(b) CH_3-CCl_3	C—C stretching
(c) SO_2	Symmetric stretching
(d) $\text{CH}_2=\text{CH}_2$	C—H stretching:



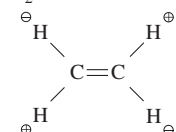
(e) $\text{CH}_2=\text{CH}_2$	C—H stretching:
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(f) $\text{CH}_2=\text{CH}_2$	CH_2 wag:
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(g) $\text{CH}_2=\text{CH}_2$	CH_2 twist:
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16-9 What are the major advantages of an FTIR spectrometer compared with a dispersive instrument?



16-10 What length of mirror drive in an FTIR spectrometer would be required to provide a resolution of

(a) 0.010 cm^{-1} , (b) 0.50 cm^{-1} , and (c) 2.0 cm^{-1} ?

*** 16-11** It was stated that at room temperature (25°C) the majority of molecules are in the ground vibrational energy level ($\nu = 0$).

(a) Use the Boltzmann equation (Equation 8-1) to calculate the excited-state and ground-state population ratios for HCl: $N(\nu = 1)/N(\nu = 0)$. The fundamental vibrational frequency of HCl occurs at 2885 cm^{-1} .

(b) Use the results of part (a) to find $N(\nu = 2)/N(\nu = 0)$

16-12 Why are nondispersive IR instruments often used for the determination of gases rather than dispersive IR spectrometers?

16-13 The first FTIR instruments used three different interferometer systems. Briefly, describe how it has been possible to simplify the optical systems in more contemporary instruments.

*** 16-14** In a particular trace analysis via FTIR, a set of sixteen interferograms were collected. The signal-to-noise ratio (S/N) associated with a particular spectral peak was approximately 4:1. How many interferograms would have to be collected and averaged if the goal is to obtain an $S/N = 20:1$?



- * **16-15** Radiation leaves a source at a natural frequency of (a) 9.10×10^{13} Hz. What is the wavenumber corresponding to this frequency? If a Michelson interferometer has a mirror velocity of 1.00 cm/s, what will the frequency be at the transducer. Find the wavenumber at the source and frequency at the transducer for source radiation of (b) 9.20×10^{13} Hz and (c) 9.30×10^{13} Hz.

Challenge Problem

- 16-16 (a)** The IR spectrum of gaseous N_2O shows three strong absorption bands at 2224 cm^{-1} , 1285 cm^{-1} , and 2089 cm^{-1} . In addition two quite weak bands are observed at 2563 cm^{-1} and 2798 cm^{-1} . It is known that N_2O is a linear molecule, but assume it is not known whether the structure is $\text{N}-\text{N}-\text{O}$ or $\text{N}-\text{O}-\text{N}$. Use the IR data to decide between the two structures. What vibrations can be assigned to the strong absorption bands? What are possible causes of the weak absorptions?
- (b)** The IR spectrum of HCN shows three strong absorption bands at 3312 cm^{-1} , 2089 cm^{-1} , and 712 cm^{-1} . From this information alone, can you deduce whether HCN is linear or nonlinear? Assuming that HCN is linear, assign vibrations to the three absorption bands.
- (c)** How many fundamental vibrational modes are expected for BF_3 ? Which of these are expected to be IR active? Why? Sketch the vibrations.
- (d)** How many fundamental vibrational modes would you predict for (1) methane, (2) benzene, (3) toluene, (4) ethylene, and (5) carbon tetrachloride?

Applications of Infrared Spectrometry

Infrared (IR) absorption spectrometry is a versatile tool that is applied to the qualitative and quantitative determination of molecular species of all types. In this chapter we first focus on the uses of mid-IR absorption and reflection spectrometry for structural investigations of molecular compounds, particularly organic compounds and species of interest in biochemistry. We then examine several of the other applications of IR spectroscopy, including photoacoustic IR, near- and far-IR spectroscopy, IR emission, and IR microscopy.

As shown in Table 17-1, the applications of IR spectrometry fall into three major categories based on the three IR spectral regions. The most widely used region is the mid-IR, which extends from about 670 to 4000 cm^{-1} (2.5 to $14.9\text{ }\mu\text{m}$). In this region, absorption, reflection, and emission spectra are used for both qualitative and quantitative analysis. The near-IR region, from 4000 to $14,000\text{ cm}^{-1}$ (0.75 to $2.5\text{ }\mu\text{m}$), also finds considerable use for the routine quantitative determination of certain species, such as water, carbon dioxide, sulfur, low-molecular-mass hydrocarbons, amine nitrogen, and other compounds of interest in agriculture and in industry. These determinations are often based on diffuse-reflectance measurements of untreated solid or liquid samples or absorption studies of gases. The primary use of the far-IR region (15 to $1000\text{ }\mu\text{m}$) has been for the determination of the structures of inorganic and metal-organic species based on absorption measurements.

17A MID-IR ABSORPTION SPECTROMETRY

Mid-IR absorption and reflection spectrometry are major tools for determining the structure of organic and biochemical species. In this section we examine mid-IR absorption applications. Section 17B is devoted to mid-IR reflectance measurements.¹

17A-1 Sample Handling

As we have seen in earlier chapters, ultraviolet (UV) and visible molecular spectra are obtained most conveniently from dilute solutions of the analyte. Absorbance measurements in the optimal range are obtained by suitably adjusting either the



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

¹For further reading see R. M. Silverstein, F. X. Webster, D. Kiemle, and D. L. Bryce, *Spectrometric Identification of Organic Compounds*, 8th ed., Chap. 2, New York: Wiley, 2015; *Introduction to Experimental Infrared Spectroscopy: Fundamentals and Practical Methods*, M. Tasumi, ed., Chichester, UK: Wiley, 2015; P. Larkin, *Infrared and Raman Spectroscopy: Principles and Spectral Interpretation*, Waltham, MA: Elsevier, 2011; B. Schrader, *Infrared and Raman Spectroscopy*, New York: VCH, 1995; N. B. Colthup, L. H. Daly, and S. E. Wiberley, *Introduction to Infrared and Raman Spectroscopy*, 3rd ed., San Diego: Academic Press, 1990.

TABLE 17-1 Major Applications of IR Spectrometry

Spectral Regions	Measurement Type	Kind of Analysis	Applicable Samples
Near-IR	Diffuse reflectance	Quantitative	Solid or liquid commercial materials
	Absorption	Quantitative	Gaseous mixtures
Mid-IR	Absorption	Qualitative	Pure solid, liquid, or gases
		Quantitative	Complex liquid, solid, or gaseous mixtures
		Chromatographic	Complex liquid, solid, or gaseous mixtures
Far-IR	Reflectance	Qualitative	Pure solids or liquids
	Emission	Quantitative	Atmospheric samples
	Absorption	Qualitative	Pure inorganic or organometallic species

concentration or the cell length. Unfortunately, this approach is often not applicable for IR spectroscopy because no good solvents are transparent throughout the region of interest. As a result, sample handling is frequently the most difficult and time-consuming part of an IR spectrometric analysis.² In this section we outline some of the common techniques for preparation of samples for IR absorption measurements.

Gases

The spectrum of a low-boiling-point liquid or gas can be obtained by permitting the sample to expand into an evacuated cylindrical cell equipped with suitable windows. For this purpose, a variety of cylindrical cells are available with path lengths that range from a few centimeters to 10 m or more. The longer path lengths are obtained in compact cells by providing reflecting internal surfaces, so that the beam makes numerous passes through the sample before exiting the cell (see Figure 16-13).

²See *Handbook of Spectroscopy*, G. Gauglitz and D. S. Moore, eds, Vol. 1, Section 1, Weinheim, Germany: Wiley-VCH, 2014; *Practical Sampling Techniques for Infrared Analysis*, P. B. Coleman, ed., Boca Raton, FL: CRC Press, 1993; T. J. Porro and S. C. Pattacini, *Spectroscopy*, **1993**, 8 (7), 40; *Ibid.*, 8 (8), 39; A. L. Smith, *Applied Infrared Spectroscopy*, New York: Wiley, 1979, Chap. 4.

Solutions

When feasible, a convenient way of obtaining IR spectra is on solutions prepared to contain a known concentration of sample, as is generally done in UV-visible spectrometry. This technique is somewhat limited in its applications, however, by the availability of solvents transparent over significant regions in the IR.

Solvents. Figure 17-1 lists several common solvents used for IR studies of organic compounds. This figure illustrates that no single solvent is transparent throughout the entire mid-IR region.

Water and the alcohols are difficult to use as solvents in IR spectrometry. Water shows several strong absorption bands in the IR region, as can be seen in Figure 17-2. The spectrum of water is shown along with the spectrum of an aqueous solution of aspirin. The computer-calculated difference spectrum reveals the spectrum of the water-soluble aspirin. Water and alcohols also attack alkali-metal halides, the most common materials used for cell windows. Hence, water-insoluble window materials, such as barium fluoride, must be used with such solvents. Care must also be taken to dry the solvents shown in Figure 17-1 before use with typical cells.

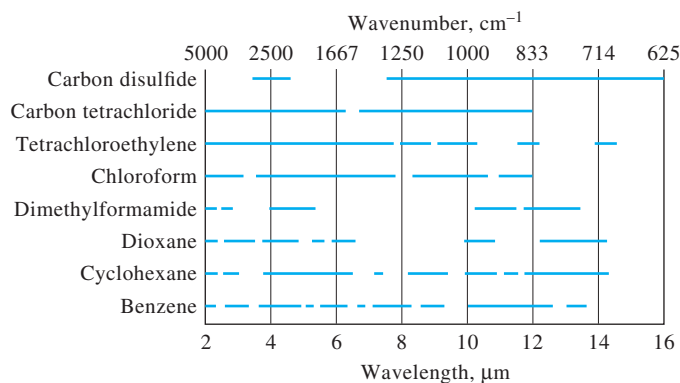


FIGURE 17-1 Solvents for IR spectroscopy. Horizontal lines indicate useful regions.

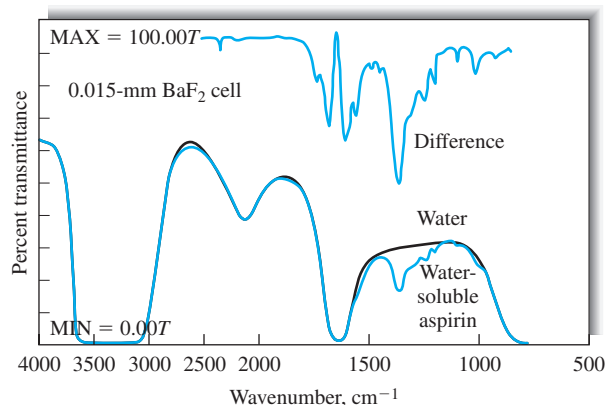


FIGURE 17-2 IR spectra of water and of an aqueous solution containing water-soluble aspirin. The top spectrum shows the computer-generated difference spectrum obtained by subtracting the water spectrum from that of the solution. (From *Practical Sampling Techniques for Infrared Analysis*, P. B. Coleman, ed., Boca Raton, FL: CRC Press, 1993. With permission.)

Cells. Because of the tendency for solvents to absorb IR radiation, IR liquid cells are ordinarily much narrower (0.01 to 1 mm) than those used in the UV and visible regions. Often, relatively high sample concentrations (from 0.1% to 10%) are required because of the short path lengths and the low molar absorptivities of IR bands. Liquid cells are frequently designed for easy disassembly and use Teflon spacers to allow variation in path length (see Figure 17-3). Fixed-path-length cells can be filled or emptied with a hypodermic syringe.

A variety of window materials are available as listed in Table 17-2. Window materials are often chosen on the basis of cost, range of transparency, solubility in the solvent, and reactivity with the sample or solvent. Sodium chloride and potassium bromide windows are most commonly used and are the least

expensive. Even with care, however, their surfaces eventually become fogged because of absorption of moisture. Polishing with a buffing powder can, in many cases, return them to their original condition.

For samples that are wet or aqueous, calcium and barium fluoride are suitable, although neither transmits well throughout the entire mid-IR region. Silver bromide is often used, although

TABLE 17-2 Common IR Window Materials

Window Material	Applicable Range, cm^{-1}	Water Solubility, g/100 g H_2O , 20°C
Sodium chloride	40,000–625	36.0
Potassium bromide	40,000–385	65.2
Potassium chloride	40,000–500	34.7
Cesium iodide	40,000–200	160.0
Fused silica	50,000–2,500	Insoluble
Calcium fluoride	50,000–1,100	1.51×10^{-3}
Barium fluoride	50,000–770	0.12 (25°C)
Thallium bromide iodide, KRS-5	16,600–250	<0.0476
Silver bromide	20,000–285	1.2×10^{-7}
Zinc sulfide, Irtran-2	10,000–715	Insoluble
Zinc selenide, Irtran-4	10,000–515	Insoluble
Polyethylene	625–30	Insoluble

From *Practical Sampling Techniques for Infrared Analysis*, P. B. Coleman, ed., Boca Raton, FL: CRC Press, 1993.

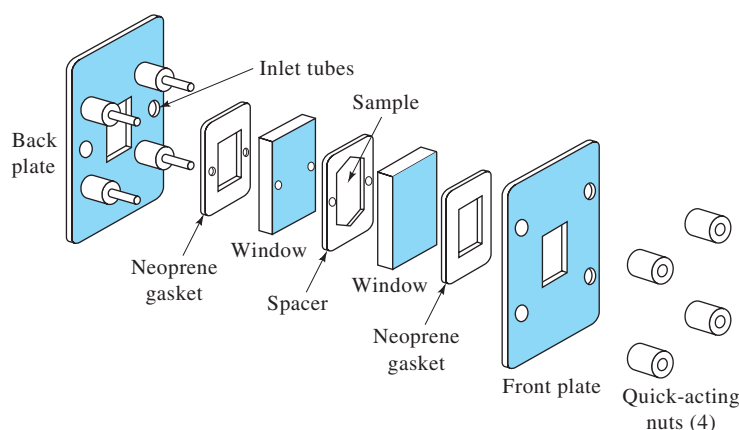


FIGURE 17-3 Expanded view of a demountable IR cell for liquid samples. Teflon spacers ranging from 0.015 to 1 mm thick are available. (Courtesy of Perkin-Elmer Corp., Norwalk, CT.)

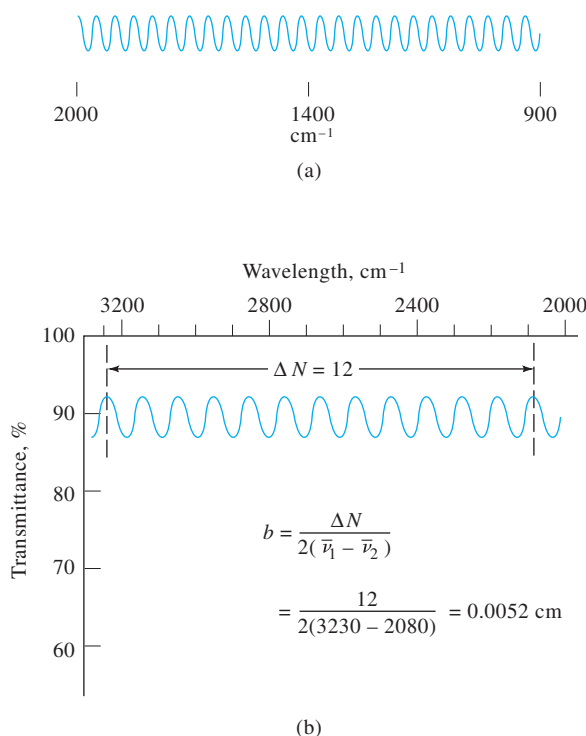


FIGURE 17-4 Determination of the path length of IR cells from interference fringes. Fringes are observed with an empty cell in the light path (a). In (b) an illustration of the calculation of the path length is given.

it is more expensive, is photosensitive, and is easily scratched. The Irtran materials are also resistant to chemical attack and are water insoluble. They are somewhat expensive and have high indexes of refraction (2.2 to 2.4), which can lead to interference fringes because of mismatches with the sample index of refraction (often around 1.4 to 1.6).

Figure 17-4 shows how the thickness b of very narrow IR cells can be determined by the interference fringe method. The interference pattern of Figure 17-4a is obtained by inserting an empty cell in the light path of the sample beam and recording the transmittance against air in the reference beam. The reflected radiation from the cell walls interferes with the transmitted beam to produce the interference maxima and minima. The maxima occur when the radiation reflected from the two internal surfaces of the cell has traveled a distance that is an integral multiple N of the wavelength of the radiation transmitted without reflection. Constructive interference then occurs whenever the wavelength is equal to $2b/N$. That is,

$$\frac{2b}{N} = \lambda \quad (17-1)$$

As shown in Figure 17-4b, the number of interference fringes ΔN between two known wavelengths λ_1 and λ_2 are counted and introduced into the equation

$$\Delta N = \frac{2b}{\lambda_1} - \frac{2b}{\lambda_2} = 2b\bar{\nu}_1 - 2b\bar{\nu}_2$$

or

$$b = \frac{\Delta N}{2(\bar{\nu}_1 - \bar{\nu}_2)} \quad (17-2)$$

It should be noted that interference fringes are ordinarily not seen when a cell is filled with liquid because the refractive index of most liquids approaches that of the window material and reflection is minimized (Equation 6-15). On the other hand, interference can be observed between 2800 and 2000 cm^{-1} in the spectrum shown in Figure 16-1. The sample is a sheet of polystyrene, which has a refractive index considerably different from that of air. For this reason, significant reflection occurs at the two interfaces of the sheet. Equation 17-2 is often used to calculate the thickness of thin polymer films.

Liquids

When the amount of a liquid sample is small or when a suitable solvent is unavailable, it is common practice to obtain spectra on the pure (neat) liquid. The thickness of a very thin film of liquid presents a sufficiently short path length to produce satisfactory spectra. Commonly, a drop of the neat liquid is squeezed between two rock-salt plates to give a layer 0.015 mm thick or less. The two plates, held together by capillary action, are then placed in the beam path. This technique does not give particularly reproducible transmittance data, but the resulting spectra are usually satisfactory for qualitative investigations.

Solids

Most organic compounds exhibit numerous absorption bands throughout the mid-IR region, and finding a solvent that does not have overlapping bands is often impossible. Because of this, spectra are often obtained on dispersions of the solid in a liquid or solid matrix. Generally, in these techniques, the solid sample must be ground until its particle size is much less than the wavelength of the radiation to avoid the effects of scattered radiation.

Pelleting. One of the most popular techniques for handling solid samples has been KBr pelleting (other alkali-metal halides have also been used). Halide salts have the property of cold flow, in which they have glasslike transparent or translucent properties when sufficient pressure is applied to the finely powdered materials. In using this technique, a milligram or less of the finely ground sample is intimately mixed with about 100 mg of

dried, spectroscopic grade, potassium bromide powder. Mixing can be carried out with a mortar and pestle or, better, in a small ball mill. The mixture is then pressed in a special die at 10,000 to 15,000 pounds per square inch to yield a transparent disk. Best results are obtained if the disk is formed in a vacuum to eliminate occluded air. The disk is then held in the instrument beam for spectroscopic examination. The resulting spectra frequently exhibit bands at 3450 and 1640 cm^{-1} (2.9 and $6.1\text{ }\mu\text{m}$) due to absorbed moisture.

A good pellet spectrum should have a relatively flat (non-sloping) baseline with the intensities of the strongest bands in the $5\text{--}15\%$ T ($0.7\text{--}1.3\text{ A}$) range. If the strongest bands show smaller absorbances, this can be corrected by repeating the formation of the pellet with a smaller amount of KBr. If the absorbances of the strongest bands are too high (e.g., totally absorbing), the pellet can be broken into smaller fragments, and one of these can be repressed with more KBr.

With many compounds, KBr pelleting produces excellent spectra that appear in many spectral libraries. Being ionic, KBr transmits throughout most of the IR region with a lower cutoff of about 400 cm^{-1} . Ion exchange can occur with some samples such as amine hydrochlorides or inorganic salts. With the former, bands of the amine hydrobromide are often found. Polymorphism can also occur because of the forces involved in grinding and pressing the pellet. These can convert one polymorph into another. Although KBr is the most frequently used pelleting salt, materials such as CsI and CsBr are sometimes used. Cesium iodide has greater transparency at low frequencies than KBr.

Mulls. IR spectra of solids that are not soluble in an IR-transparent solvent or are not conveniently pelleted in KBr are often obtained by dispersing the analyte in a mineral oil or a fluorinated hydrocarbon *mull*. To obtain good mull spectra, the sample must be ground to an average particle size that is lower than about one-fifth of the shortest wavelength to be used. For many mid-IR spectra, the starting point of the scan is $4000\text{--}5000\text{ cm}^{-1}$. This corresponds to wavelengths of $2\text{--}2.5\text{ }\mu\text{m}$. Hence, grinding to a particle size lower than $0.4\text{--}0.5\text{ }\mu\text{m}$ is needed. Larger particles scatter radiation in proportion to the fourth power of the radiation frequency and give rise to sloping baselines. Often $20\text{--}30\text{ mg}$ samples are used for grinding, but samples as small as $5\text{--}10\text{ mg}$ can be used. Grinding should result in an even, smooth, glossy sample in the mortar. Mulls are then formed by adding a small drop of a heavy hydrocarbon (usually Nujol[™]) and continuing the grinding process until a smooth paste, similar in appearance to toothpaste or cold cream, results. If hydrocarbon bands are likely to interfere, Fluorolube[™], a halogenated polymer, can be used instead. In either case, the resulting mull is then examined as a film between flat salt plates.

Other Methods for Solids. The IR behavior of solids can also be obtained by casting a thin film of the sample as is done with many polymers. For some samples, spectra can be acquired by reflectance techniques and by the photoacoustic method. These spectra are often similar to absorption spectra and provide

the same kinds of information. Reflectance and photoacoustic methods are discussed in Sections 17B and 17C.

17A-2 Qualitative Analysis

The general use of mid-IR spectroscopy by chemists for identifying organic compounds began in the late 1950s with the appearance of inexpensive and easy-to-use double-beam recording spectrophotometers that produce spectra in the region of 5000 to 670 cm^{-1} (2 to $15\text{ }\mu\text{m}$). The appearance of this type of instrument (as well as nuclear magnetic resonance and mass spectrometers) revolutionized the way chemists identify organic, inorganic, and biological species. The introduction of these tools dramatically reduced the time required to perform a qualitative analysis or structural determination.

Figure 17-5 shows four typical spectra obtained with an inexpensive double-beam instrument. Identification of an organic compound from a spectrum of this kind is a two-step process. The first step involves determining the functional groups present by examining the *group frequency region*, which encompasses radiation from about 3600 cm^{-1} to approximately 1250 cm^{-1} (see Figure 17-5). The second step involves a detailed comparison of the spectrum of the unknown with the spectra of pure compounds that contain all of the functional groups found in the first step. The fingerprint region, from 1200 to 600 cm^{-1} (Figure 17-5) is particularly useful here because small differences in the structure and constitution of a molecule result in significant changes in the appearance and distribution of absorption bands in this region. Consequently, a close match between two spectra in the fingerprint region (as well as others) constitutes almost certain evidence that the two compounds are identical.

Group Frequencies

We have noted that the approximate frequency (or wavenumber) at which an organic functional group, such as $\text{C}=\text{O}$, $\text{C}=\text{C}$, $\text{C}-\text{H}$, $\text{C}\equiv\text{C}$, or $\text{O}-\text{H}$, absorbs IR radiation can be calculated from the masses of the atoms and the force constant of the bond between them (Equations 16-14 and 16-15). These frequencies, called *group frequencies*, are seldom totally invariant because of interactions with other vibrations associated with one or both of the atoms composing the group. On the other hand, such interaction effects are ordinarily small; as a result, a range of frequencies can be assigned within which it is highly probable that the absorption maximum for a given functional group will be found. Table 17-3 lists group frequencies for several common functional groups. A more detailed presentation of group frequencies is found in the *correlation chart* shown in Figure 17-6.³ Note that, although most group frequencies fall in the range of 3600 to 1250 cm^{-1} , a few fall in the fingerprint region. These include the $\text{C}-\text{O}-\text{C}$ stretching vibration at about 1200 cm^{-1} and the $\text{C}-\text{Cl}$ stretching vibration at 700 to 800 cm^{-1} .

³Other correlation charts can be found in R. M. Silverstein, F. X. Webster, D. Kiemle, and D. L. Bryce, *Spectrometric Identification of Organic Compounds*, 8th ed., Chap. 2, New York: Wiley, 2015.

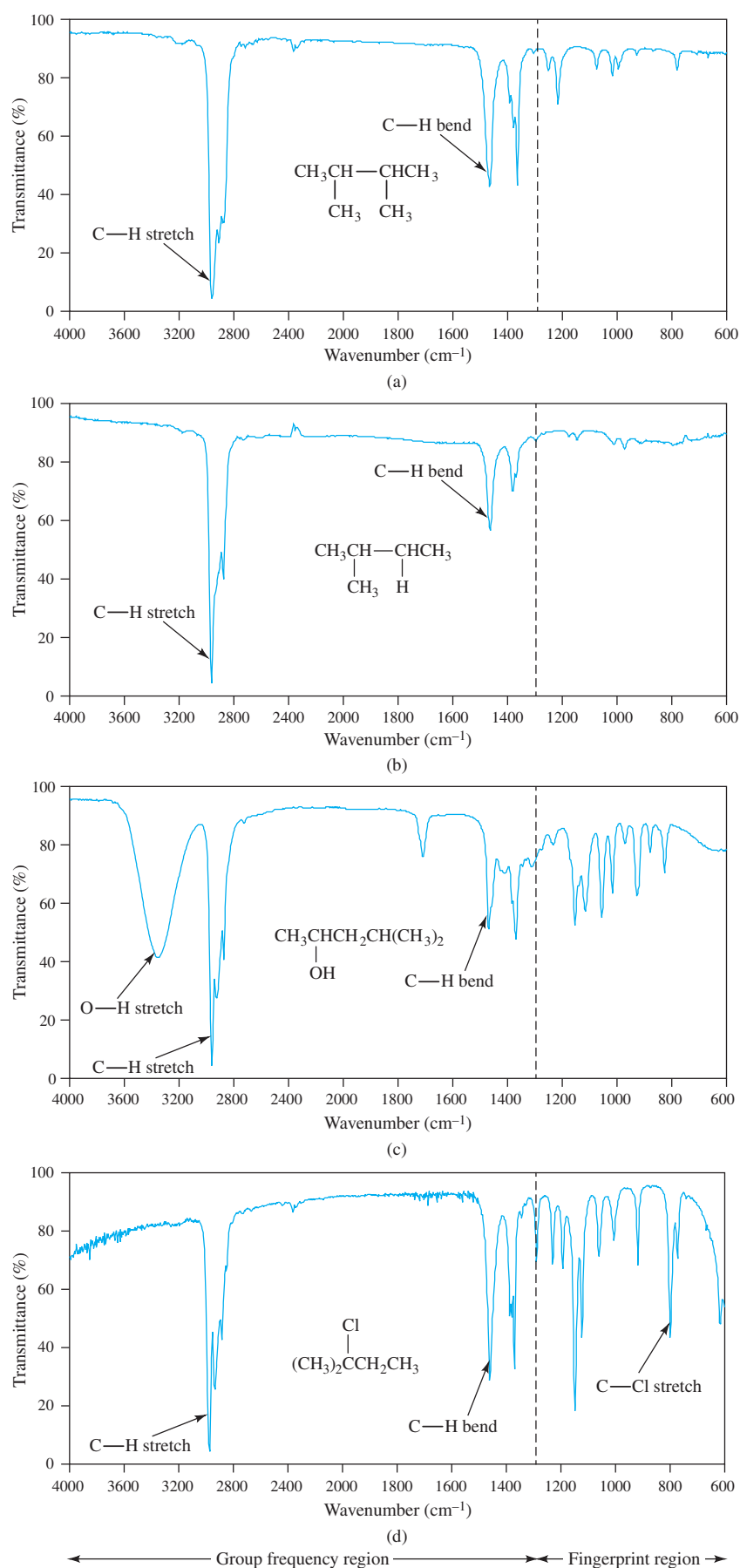


FIGURE 17-5 The group frequency region of the mid-IR ($\sim 3600\text{--}1250\text{ cm}^{-1}$) is used to identify common functional groups. The fingerprint region ($\sim 1200\text{--}600\text{ cm}^{-1}$) is used to identify compounds. (Adapted from J. C. Gilbert and S. F. Martin, *Experimental Organic Chemistry: A Miniscale and Microscale Approach*, 6th ed., Figures 8.11 (a) and (b), 10.11 (c), and 14.7 (d), Boston: Cengage Learning, 2016.)

TABLE 17-3 Abbreviated Table of Group Frequencies for Organic Functional Groups

Bond	Type of Compound	Frequency Range, cm^{-1}	Intensity
C—H	Alkanes	2850–2970	Strong
C—H	Alkenes (>C=C<H)	1340–1470	Strong
		3010–3095	Medium
C—H	Alkynes ($\text{—C}\equiv\text{C—H}$)	675–995	Strong
		3300	Strong
C—H	Aromatic rings	3010–3100	Medium
		690–900	Strong
O—H	Monomeric alcohols, phenols	3590–3650	Variable
	Hydrogen-bonded alcohols, phenols	3200–3600	Variable, sometimes broad
	Monomeric carboxylic acids	3500–3650	Medium
	Hydrogen-bonded carboxylic acids	2500–2700	Broad
N—H	Amines, amides	3300–3500	Medium
C=C	Alkenes	1610–1680	Variable
C=C	Aromatic rings	1500–1600	Variable
C $\equiv$ C	Alkynes	2100–2260	Variable
C—N	Amines, amides	1180–1360	Strong
C $\equiv$ N	Nitriles	2210–2280	Strong
C—O	Alcohols, ethers, carboxylic acids, esters	1050–1300	Strong
C=O	Aldehydes, ketones, carboxylic acids, esters	1690–1760	Strong
NO ₂	Nitro compounds	1500–1570	Strong
		1300–1370	Strong

Several group frequencies are identified in the four spectra in Figure 17-5. All four spectra contain an absorption band at 2900 to 3000 cm^{-1} , which corresponds to a C—H stretching vibration and generally indicates the presence of one or more alkane groups (see Table 17-3). The two bands at about 1375 cm^{-1} and 1450 cm^{-1} are also characteristic group frequencies for C—H groups and result from bending vibrations in the molecule. The spectrum in Figure 17-5c illustrates the group frequency for an O—H stretching vibration at about 3200 cm^{-1} as well as the alkane group frequencies (these bands are also present in the spectrum for *n*-hexanal shown in Figure 16-12).

Finally, the characteristic group frequency for a C—Cl stretching vibration is shown at about 800 cm^{-1} in Figure 17-5d.

Group frequencies and correlation charts permit intelligent guesses as to what functional groups are likely to be present or absent in a molecule. Ordinarily, it is impossible to identify unambiguously either the sources of all of the bands in a given spectrum or the exact identity of the molecule. Instead, group frequencies and correlation charts serve as a starting point in the identification process.

The Fingerprint Region

Small differences in the structure and constitution of a molecule result in significant changes in the distribution of absorption maxima in this region of the spectrum that extends from about 1200 to 600 cm^{-1} (8 to 14 μm). The fingerprint region is thus well suited for identifying compounds based on spectral comparisons. Most single bonds give rise to absorption bands at these frequencies; because their energies are about the same, strong interaction occurs between neighboring bonds. The absorption bands are thus composites of these various interactions and depend on the overall skeletal structure of the molecule. Exact interpretation of spectra in this region is seldom possible because of the complexity of the spectra. However, for identification purposes, the richness of spectral features is a distinct advantage. Many compounds show unique absorption bands in this region, which is very useful for final identification.

Figure 17-5a and b illustrates the unique character of IR spectra, particularly in the fingerprint region. The two molecules shown differ by just one methyl group, yet the two spectra differ dramatically in the fingerprint region.

As shown in Figure 17-6 (pages 420–421), a number of inorganic groups such as sulfate, phosphate, nitrate, and carbonate absorb IR radiation in the fingerprint region ($<1200\text{ cm}^{-1}$ or $>8.3\text{ }\mu\text{m}$).

Limitations to the Use of Correlation Charts

The unambiguous establishment of the identity or the structure of a compound is seldom possible from correlation charts alone. Uncertainties frequently arise from overlapping group frequencies, spectral variations as a function of the physical state of the sample (that is, whether it is a solution, a mull, in a pelleted form, and so forth), and instrumental limitations.

In using group frequencies, it is essential that the entire spectrum, rather than a small isolated portion, be considered and interrelated. Interpretation based on one part of the spectrum should be confirmed or rejected by studying other regions.

To summarize, correlation charts serve only as a guide for further and more careful study. Several excellent monographs describe the absorption characteristics of functional groups in detail.⁴ A study of these characteristics, as well as the other physical properties of the sample, may permit unambiguous identification. IR spectroscopy, when used in conjunction with mass spectrometry, nuclear magnetic resonance, and elemental analysis, usually leads to positive identification of a species.

Collections of Spectra

As just noted, correlation charts seldom suffice for the positive identification of an organic compound from its IR spectrum. However, several catalogs of IR spectra are available that assist in qualitative identification by providing comparison spectra for a large number of pure compounds.⁵ Manually searching large catalogs of spectra is slow and tedious. For this reason, computer-based search systems are widely used.

Computer Search Systems

Virtually all IR instrument manufacturers now offer computer search systems to aid in identifying compounds from stored IR spectral data.⁶ The position and relative magnitudes of peaks in the spectrum of the analyte are determined and stored in memory to give a peak profile, which can be compared with profiles of pure compounds stored on disk, CD-ROM, or a website. The computer then matches profiles and prints a list of compounds having spectra similar to that of the analyte. Usually, the spectrum of the ana-

lyte and that of each potential match can be shown simultaneously on the computer display for comparison, as shown in Figure 17-7.

In 1980 the Sadtler Standard IR Collection and the Sadtler Commercial IR Collection became available as software packages. Currently, this library contains more than 220,000 spectra.⁷ Included are vapor-phase spectra and condensed-phase spectra of pure compounds. Individual databases in application areas such as polymers, industrial products, pure organic compounds, forensic sciences, and the environment are available. Several manufacturers of Fourier transform IR (FTIR) instruments have now incorporated these packages into their instrument computers, thus creating instantly available IR libraries of hundreds of thousands of compounds.

The Sadtler algorithm consists of a search system in which the spectrum of the unknown compound is first coded according to the location of its strongest absorption peak; then each additional strong band ($\%T < 60\%$) in ten regions 200 cm^{-1} wide from 4000 to 2000 cm^{-1} are coded by their location. Finally, the strong bands in seventeen regions 100 cm^{-1} wide from 2100 to 400 cm^{-1} are coded in a similar way. The compounds in the library are coded in this same way. The data are organized by the location of the strongest band with only those compounds having the same strongest band being considered in any sample identification. This procedure is rapid and produces a list of potential matches within a short period. For example, searching a base library of 50,000 compounds by this procedure requires only a few seconds.

In addition to search systems, artificial intelligence-based computer programs seek to determine the structure or substructure from spectral profiles. Some of these programs use data from different types of instruments (nuclear magnetic resonance, mass, and IR spectrometers, etc.) to determine structures.⁸

17A-3 Quantitative Applications

Quantitative IR absorption methods differ somewhat from UV-visible molecular spectroscopic methods because of the greater complexity of the spectra, the narrowness of the absorption bands, and the instrumental limitations of IR instruments. Quantitative data obtained with older dispersive IR instruments were generally significantly inferior in quality to data obtained with UV-visible spectrophotometers. Modern FTIR instruments, however, achieve markedly better precision and accuracy than their dispersive predecessors. Meticulous attention to detail is, however, essential for obtaining good-quality results.⁹



Tutorial: Learn more about IR spectral interpretation and identification at www.tinyurl.com/skoogpia7

⁴R. M. Silverstein, F. X. Webster, D. Kiemle, and D. L. Bryce, *Spectrometric Identification of Organic Compounds*, 8th ed., New York: Wiley, 2015; B. Schrader, *Infrared and Raman Spectroscopy*, New York: VCH, 1995; N. B. Colthup, L. H. Daly, and S. E. Wiberley, *Introduction to Infrared and Raman Spectroscopy*, 3rd ed., San Diego: Academic Press, 1990.

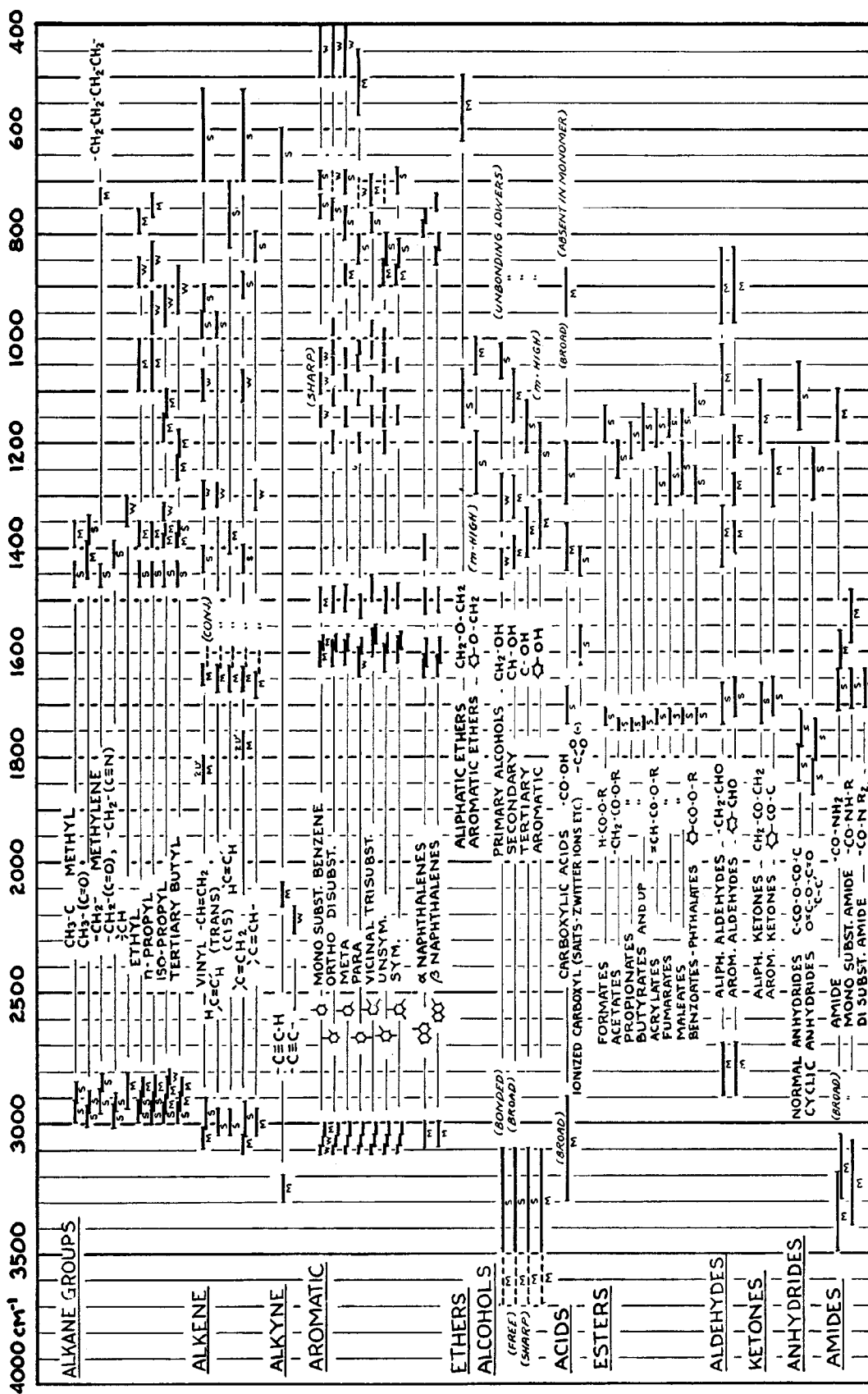
⁵See Informatics/Sadtler Group, Bio-Rad Laboratories, Inc., Philadelphia, PA; C. J. Pouchert, *The Aldrich Library of FT-IR Spectra*, Milwaukee, WI: Aldrich Chemical Co., 1997; Thermo Fisher Scientific-Spectral Libraries (<http://www.thermoscientific.com/en/products/spectral-libraries.html>).

⁶See E. Pretsch, G. Toth, M. E. Monk, and M. Badertscher, *Computer Aided Structure Elucidation: Spectra Interpretation and Structure Generation*, New York: VCH-Wiley, 2003; B. C. Smith, *Infrared Spectral Interpretation: A Systematic Approach*, Boca Raton, FL: CRC Press, 1998, Chap. 9; W. O. George and H. Willis, *Computer Methods in UV, Visible and IR Spectroscopy*, New York: Springer-Verlag, 1990.

⁷See note 5.

⁸See, for example, M. E. Munk, *J. Chem. Inf. Comput. Sci.*, **1998**, 38, 997, DOI: 10.1021/ci980083r.

⁹For a discussion of quantitative FTIR spectroscopy, see B. C. Smith, *Fundamentals of Fourier Transform Infrared Spectroscopy*, 2nd ed., Boca Raton, FL: CRC Press, 2011; P. R. Griffiths and J. A. deHasech, *Fourier Transform Infrared Spectrometry*, 2nd ed., New York: Wiley, 2007.



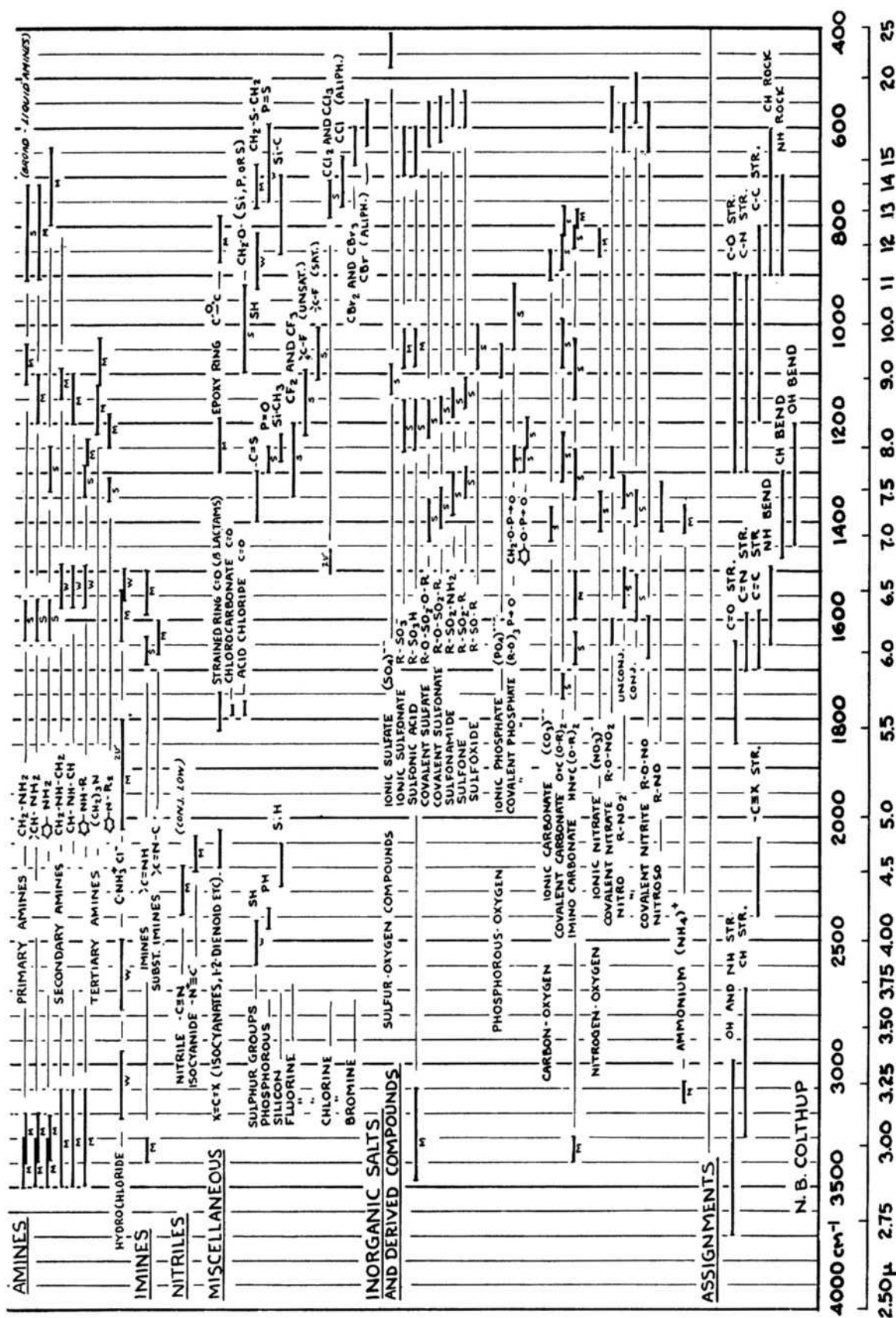


FIGURE 17-6 Correlation chart. (From N. Colthup, *J. Opt. Soc. Am.*, **1950**, *40*, 397.)

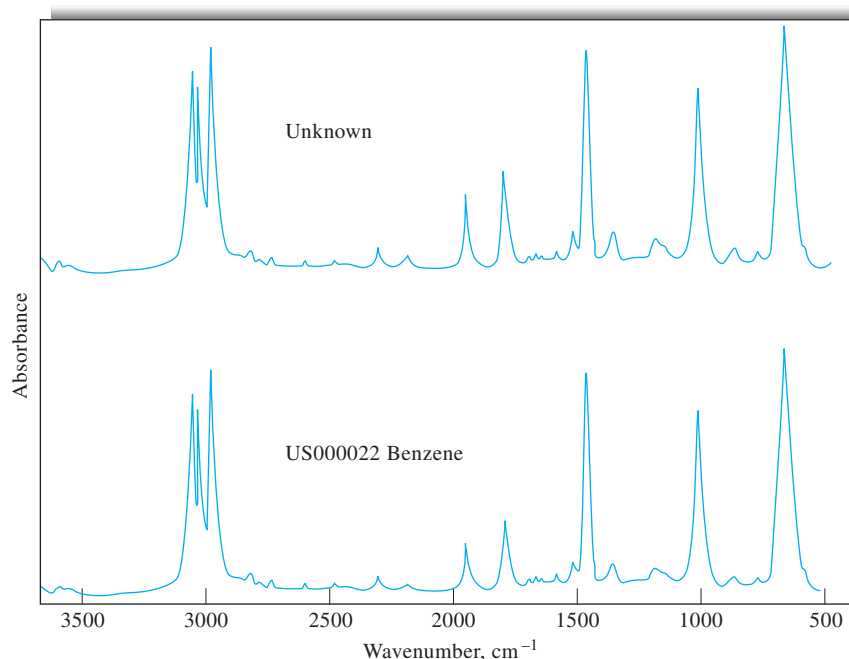


FIGURE 17-7 Plot of an unknown spectrum and the best match from a computer search report. (Courtesy of Informatics/Sadtler Group, Bio-Rad Laboratories, Inc., Philadelphia, PA.)

Deviations from Beer's Law

With IR radiation, instrumental deviations from Beer's law are more common than with UV and visible wavelengths because IR absorption bands are relatively narrow. Furthermore, with dispersive instruments, the low intensity of sources and low sensitivities of transducers in this region require the use of relatively wide monochromator slit widths; thus, the bandwidths used are frequently of the same order of magnitude as the widths of absorption peaks. We have pointed out (Section 13B-2) that this combination of circumstances usually leads to a nonlinear relationship between absorbance and concentration. As discussed in Section 16B-1, FTIR instruments have better performance characteristics. Hence, Beer's law deviations are not quite as serious as with dispersive instruments. In either case, quantitative analysis usually requires that empirical calibration curves be used.

Absorbance Measurement

Matched absorption cells for solvent and solution are ordinarily used in the UV and visible regions, and the measured absorbance is then found from the relation

$$A = \log \frac{P_{\text{solvent}}}{P_{\text{solution}}}$$

The use of the solvent in a matched cell as a reference absorber has the advantage of largely canceling out the effects of radiation losses due to reflection at the various interfaces, scattering and absorption by the solvent, and absorption by the container windows. This technique is seldom practical for

measurements in the IR region because of the difficulty in obtaining cells whose transmission characteristics are identical. Most IR cells have very short path lengths, which are difficult to duplicate exactly. In addition, the cell windows are readily attacked by contaminants in the atmosphere and the solvent. Because of this, their transmission characteristics change continually with use. For these reasons, a reference absorber is often dispensed with entirely in IR work, and the intensity of the radiation passing through the sample is simply compared with that of the unobstructed beam. In either case, the resulting transmittance is ordinarily less than 100%, even in regions of the spectrum where no absorption by the sample occurs (see Figure 17-5).

For quantitative work, it is necessary to correct for the scattering and absorption by the solvent and the cell. Two methods are used. In the so-called cell-in-cell-out procedure, spectra of the pure solvent and the analyte solution are obtained successively with respect to the unobstructed reference beam. The same cell is used for both measurements. The transmittance of each solution versus the reference beam is then determined at an absorption maximum of the analyte. These transmittances can be written as

$$T_0 = P_0/P_r$$

and

$$T_s = P/P_r$$

where P_r is the power of the unobstructed beam and T_0 and T_s are the transmittances of the solvent and analyte solution,

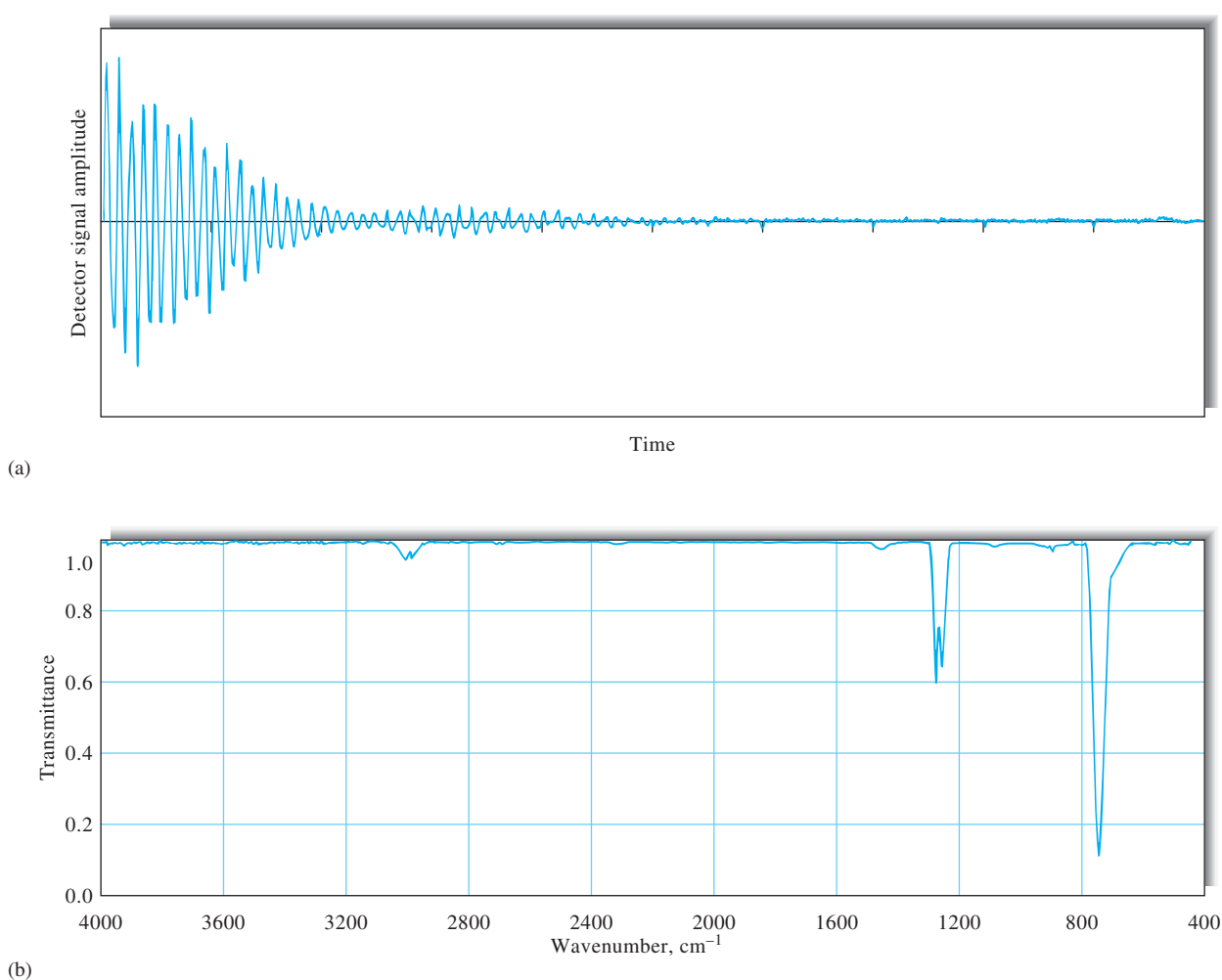


FIGURE 17-8 (a) Interferogram obtained from a typical FTIR spectrometer for methylene chloride. The plot shows detector signal output as a function of time or displacement of the moving mirror of the interferometer. (b) IR spectrum of methylene chloride produced by the Fourier transformation of the data in (a). Note that the Fourier transform takes signal intensity collected as a function of time and produces transmittance as a function of frequency after subtraction of a background interferogram and proper scaling.

respectively, against this reference. If P_r remains constant during the two measurements, then the transmittance of the sample with respect to the solvent can be obtained by division of the two equations. That is,

$$T = T_s/T_0 = P/P_0$$

With modern FTIR spectrometers, the reference interferogram is obtained with no sample in the sample cell. Then, the sample is placed in the cell, and a second interferogram obtained. Figure 17-8a shows an interferogram collected using an FTIR spectrometer with methylene chloride, CH_2Cl_2 , in the sample cell. The Fourier transform is then applied to the two interferograms to compute the IR spectra of the reference and the sample. The ratio of the two spectra can then be computed to produce an IR spectrum of the analyte such as the one illustrated in Figure 17-8b.

An alternative way of obtaining P_0 and T for a single absorption band is the baseline method. For an instrument that displays transmittance, the solvent transmittance is assumed to be constant or at least to change linearly between the shoulders of the absorption peak, as illustrated in Figure 17-9. The quantities T_0 and T_s are then obtained as shown in the figure. For direct absorbance readout, as is shown in Figure 17-7, the absorbance is assumed to be constant or to change linearly under the absorption band. The peak absorbance is then obtained by subtracting the baseline absorbance.

Typical Applications

With the exception of homonuclear molecules, all organic and inorganic molecular species absorb radiation in the IR region. IR spectrophotometry thus offers the potential for determining

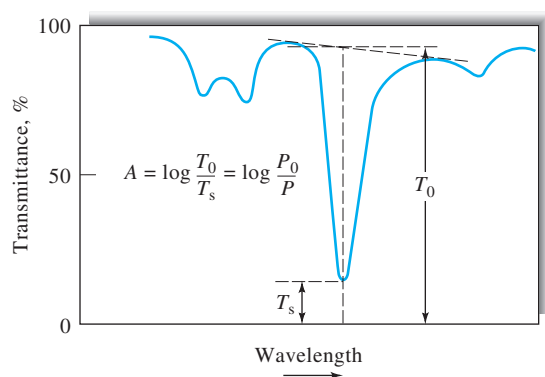


FIGURE 17-9 The baseline method for determining the absorbance of an absorption maximum.

an unusually large number of substances. Moreover, the uniqueness of an IR spectrum leads to a degree of specificity that is matched or exceeded by relatively few other analytical methods. This specificity has found particular application to analysis of mixtures of closely related organic compounds. Two examples that typify these applications follow.

Analysis of a Mixture of Aromatic Hydrocarbons. A typical application of quantitative IR spectroscopy involves the resolution of C_8H_{10} isomers in a mixture that includes *o*-xylene, *m*-xylene, *p*-xylene, and ethylbenzene. The IR absorption spectra of the individual components in the range of 12 to 15 μm in

cyclohexane solvent is shown in Figure 17-10. Useful absorption bands for determination of the individual compounds occur at 13.47, 13.01, 12.58, and 14.36 μm , respectively. Unfortunately, however, the absorbance of a mixture at any one of these wavelengths is not entirely determined by the concentration of just one component because of overlapping absorption bands. Thus, molar absorptivities for each of the four compounds must be determined at the four wavelengths. Then, four simultaneous equations can be written that permit the calculation of the concentration of each species from four absorbance measurements (see Section 14D-2). Alternatively, chemometric techniques, such as factor analytical methods,¹⁰ can use entire spectral regions to determine the individual components. Such methods can be used even when the relationship between absorbance and concentration is nonlinear (as frequently occurs in the IR region).

Determination of Air Contaminants. The recent proliferation of government regulations with respect to atmospheric contaminants has demanded the development of sensitive, rapid, and highly specific methods for a variety of chemical compounds. IR absorption procedures appear to meet this need better than any other single analytical tool.

Table 17-4 demonstrates the potential of IR spectroscopy for the analysis of mixtures of gases. The standard sample of

¹⁰E. R. Malinowski, *Factor Analysis in Chemistry*, 3rd ed., New York: Wiley, 2002.

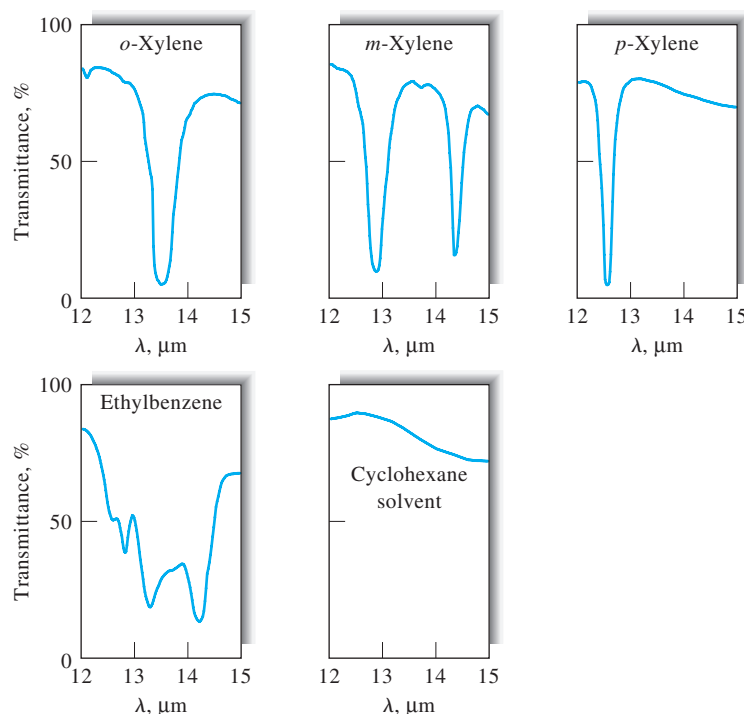


FIGURE 17-10 Spectra of C_8H_{10} isomers in cyclohexane.

TABLE 17-4 An Example of IR Determinations of Air Contaminants

Contaminant	Concn, ppm	Found, ppm	Relative Error, %
Carbon monoxide	50.0	49.1	1.8
Methyl ethyl ketone	100.0	98.3	1.7
Methanol	100.0	99.0	1.0
Ethylene oxide	50.0	49.9	0.2
Chloroform	100.0	99.5	0.5

Courtesy of Thermo Fisher Scientific, Inc.

air containing five species in known concentration was analyzed with a computerized version of the instrument shown in Figure 16-13; a 20-m gas cell was used. The data were printed out within a minute or two after sample introduction.

Table 17-5 shows potential applications of IR filter photometers (such as that shown in Figure 16-13) for the quantitative determination of various chemicals in the atmosphere for the purpose of assuring compliance with Occupational Safety and Health Administration (OSHA) regulations.

Of the more than 400 chemicals for which maximum tolerable limits have been set by OSHA, more than half appear to have absorption characteristics suitable for determination by means of IR filter photometers or spectrophotometers. With

such a large number of absorbing compounds, we expect overlapping absorption bands. However, the method can provide a moderately high degree of selectivity.

Disadvantages and Limitations to Quantitative IR Methods

There are several disadvantages to quantitative analysis by IR spectrometry. Among these are the frequent nonadherence to Beer's law and the complexity of spectra. The richness of spectral features enhances the probability of overlapping absorption bands. In addition, with older dispersive instruments, the narrowness of bands and the effects of stray radiation make absorbance measurements critically dependent on the slit width and the wavelength setting. Finally, the narrow-path-length cells required for many analyses are inconvenient to use and may lead to significant analytical uncertainties. For these reasons, the analytical errors associated with a quantitative IR analysis often cannot be reduced to the level associated with UV and visible methods, even with considerable care and effort.

17B MID-IR REFLECTION SPECTROMETRY

IR reflection spectrometry has found a number of applications, particularly for dealing with solid samples that are difficult to handle, such as polymer films and fibers, foods, rubber-based

TABLE 17-5 IR Determination of Gases for OSHA Compliance

Compound	Permissible Exposure Limit, ppm*	Wavelength, μm	Minimum Detectable Concentration, ppm [†]
Carbon disulfide	2	4.54	0.5
β -Chloroprene	10	11.4	4
Diborane	0.1	3.9	0.05
Ethylenediamine	10	13.0	0.4
Hydrogen cyanide	4.7 [‡]	3.04	0.4
Methyl mercaptan	0.5	3.38	0.4
Nitrobenzene	1	11.8	0.2
Pyridine	5	14.2	0.2
Sulfur dioxide	2	8.6	0.5
Vinyl chloride	1	10.9	0.3

Courtesy of Thermo Fisher Scientific, Inc.

*Cal/OSHA exposure limits for 8-h weighted average as of 26-April-2013. See <http://tinyurl.com/ksbnhbm>.

[†]For a 20.25-m cell.

[‡]Short-term exposure limit: 15-min time-weighted average that shall not be exceeded at any time during the work day.

materials, agriculture products, and many others.¹¹ Mid-IR reflection spectra, although not identical to the corresponding absorption spectra, are similar in general appearance and provide the same information as do their absorption counterparts. Reflectance spectra can be used for both qualitative and quantitative analysis. Most instrument manufacturers now offer adapters that fit into the cell compartments of IR absorption instruments and make it possible to obtain reflection spectra readily.

17B-1 Types of Reflection

Reflected radiation is of four types: *specular reflection*, *diffuse reflection*, *internal reflection*, and *attenuated total reflectance* (ATR). Specular reflection occurs when the reflecting medium is a smooth polished surface. Here, the angle of reflection is identical to the incident angle of the radiation. If the surface is made up of an IR absorber, the relative intensity of reflection is less for wavelengths that are absorbed than for wavelengths that are not. Thus, a plot of reflectance R , which is the fraction of the incident radiant energy reflected, versus wavelength or wavenumber provides a spectrum for a compound similar in general appearance to a transmission spectrum for the species. Specular reflection spectra find some use for examining and characterizing the smooth surfaces of solids and coated solids but are not as widely used as diffuse- and total-reflection spectra. We will, therefore, focus on the latter two types of spectra.

17B-2 Diffuse-Reflectance Spectrometry

Diffuse-reflectance IR Fourier transform spectrometry (DRIFTS) is an effective way of directly obtaining IR spectra on powdered samples with a minimum of sample preparation.¹² In addition to saving time in sample preparation, it permits conventional IR spectral data to be gathered on samples not appreciably altered from their original state. The widespread use of diffuse-reflectance measurements had to await the general availability of FTIR instruments in the mid-1970s because the intensity of radiation reflected from powders is too low to be measured at medium resolution and adequate signal-to-noise ratios with dispersive instruments.

Diffuse reflection is a complex process that occurs when a beam of radiation strikes the surface of a finely divided powder. With this type of sample, specular reflection occurs at each plane surface. However, because there are many of these surfaces

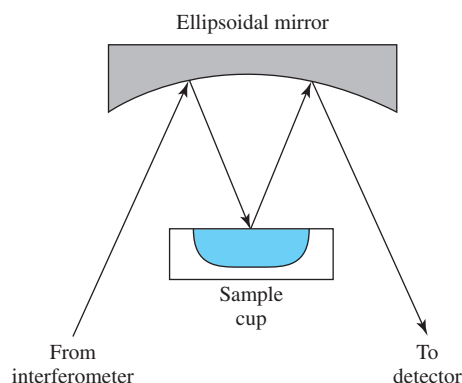


FIGURE 17-11 A diffuse-reflectance attachment for an FTIR spectrometer.

and they are randomly oriented, radiation is reflected in all directions. The intensity of the reflected radiation is roughly independent of the viewing angle.

A number of models have been developed to describe in quantitative terms the intensity of diffuse reflected radiation. The most widely used of these models was developed by Kubelka and Munk.¹³ Fuller and Griffiths in their discussion of this model show that the relative reflectance intensity for a powder $f(R'_\infty)$ is given by¹⁴

$$f(R'_\infty) = \frac{(1 - R'_\infty)^2}{2R'_\infty} = \frac{k}{s}$$

where R'_∞ is the ratio of the reflected intensity of the sample to that of a nonabsorbing standard, such as finely ground potassium chloride. The quantity k is the molar absorption coefficient of the analyte, and s is a scattering coefficient. For a diluted sample, k is related to the molar absorptivity ϵ and the molar concentration of the analyte c by the relationship

$$k = 2.303\epsilon c$$

Reflectance spectra then consist of a plot of $f(R'_\infty)$ versus wavenumber (see Figure 17-12b).

Instrumentation

Currently, most manufacturers of FTIR instruments offer adapters that fit in cell compartments and permit diffuse-reflectance measurements. Figure 17-11 illustrates one type of adapter. The collimated beam from the interferometer is directed to an ellipsoidal mirror and then to the sample. The sample is usually ground and mixed with KBr or KCl as a diluent. The mixture is then placed in a sample cup 3–4 mm deep and about 10–15 mm in diameter. A complex combination of reflection, absorption, and scattering occurs before the beam is directed to the detector.

¹¹See F. M. Mirabella, ed., *Modern Techniques in Applied Molecular Spectroscopy*, New York: Wiley, 1998; G. Kortum, *Reflectance Spectroscopy*, New York: Springer, 1969; N. J. Harrick, *Internal Reflection Spectroscopy*, New York: Wiley, 1967; for the theory of reflectance spectroscopy, see B. Hapke, *Theory of Reflectance and Emittance Spectroscopy*, 2nd ed., Cambridge: Cambridge University Press, 2012.

¹²For additional information, see A. Drochner and G. H. Vogel, in *Methods in Physical Chemistry*, 2nd ed., R. Schäfer and P. C. Schmidt, eds., Ch. 14, Somerset, NJ: Wiley, 2012; M. Milosevic, S. L. Berets, *Appl. Spectrosc. Rev.*, **2002**, 37, 347, DOI: 10.1081/asr-120016081; P. R. Griffiths and M. P. Fuller in *Advances in Infrared and Raman Spectroscopy*, R. J. H. Clark and R. E. Hester, eds., Vol. 9, Chap. 2, London: Heydon and Sons, 1982.

¹³P. Kubelka and E. Munk, *Tech. Phys.*, **1931**, 12, 593; P. Kubelka, *J. Opt. Soc. Am.*, **1948**, 38, 448, DOI: 10.1364/josa.38.000448.

¹⁴M. P. Fuller and P. R. Griffiths, *Anal. Chem.*, **1978**, 50, 1906, DOI: 10.1021/ac50035a045.

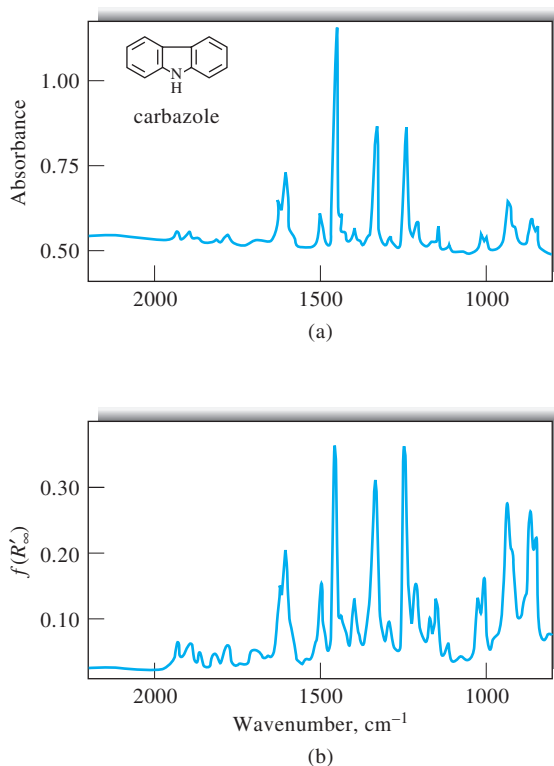


FIGURE 17-12 Comparison of the absorption spectrum (a) for carbazole with its diffuse-reflectance spectrum (b).

To obtain a spectrum with a single-beam instrument, the signal for the sample is first stored. A reference signal with a good reflector, such as finely ground KBr or KCl, is then recorded in place of the sample. The ratio of these signals is then taken to give the reflectance.

Comparison of Absorption and Reflection Spectra

Figure 17-12 compares the conventional IR absorption spectrum for carbazole obtained by means of a KBr pellet with the diffuse-reflectance spectrum of a 5% finely ground mixture of carbazole in potassium chloride. Note that the peak locations are the same in the two spectra but that the relative peak heights differ considerably. The differences are typical, with minor peaks generally appearing larger in reflection spectra.

17B-3 ATR Spectrometry

Internal-reflection spectroscopy is a technique for obtaining IR spectra of samples that are difficult to deal with, such as solids of limited solubility, films, threads, pastes, adhesives, and powders.¹⁵

Principles

When a beam of radiation passes from a more dense to a less dense medium, reflection occurs. The fraction of the incident

beam reflected increases as the angle of incidence becomes larger; beyond a certain critical angle, reflection is complete. It has been shown both theoretically and experimentally that during the reflection process the beam penetrates a small distance into the less dense medium before reflection occurs. The depth of penetration, which varies from a fraction of a wavelength up to several wavelengths, depends on the wavelength, the index of refraction of the two materials, and the angle of the beam with respect to the interface. The penetrating radiation is called the *evanescent wave*. At wavelengths where the less dense medium absorbs the evanescent radiation, attenuation of the beam occurs, which is known as *attenuated total reflectance*, or ATR. The resulting ATR spectrum resembles that of a conventional IR spectrum with some differences.

Instrumentation

Figure 17-13 shows an apparatus for ATR measurements. As can be seen from the upper figure, the sample (here, a solid) is placed on opposite sides of a transparent crystalline material of high refractive index. By proper adjustment of the incident angle, the radiation undergoes multiple internal reflections before passing from the crystal to the detector. Absorption and attenuation take place at each of these reflections.

Figure 17-13b is an optical diagram of an adapter that fits into the cell area of most IR spectrometers and permits ATR measurements. Cells for liquid samples are also available.

ATR Spectra

ATR spectra are similar but not identical to ordinary absorption spectra. In general, although the same bands are observed, their relative intensities differ. With ATR spectra, the absorbances, although dependent on the angle of incidence, are independent of sample thickness, because the radiation penetrates only a few micrometers into the sample. The *effective penetration depth* d_p depends on the wavelength of the beam, the refractive indexes of the crystal and the sample, and the beam angle. The penetration depth can be calculated from

$$d_p = \frac{\lambda_c}{2\pi[\sin^2\theta - (n_s/n_c)^2]^{1/2}}$$

where λ_c is the wavelength in the crystal (λ/n_c), θ is the angle of incidence, and n_s and n_c are the refractive indexes of the sample and crystal, respectively. Note that the effective penetration depth can be changed by changing the crystal material, the angle of incidence, or both. It is possible to obtain a depth profile of a surface using ATR spectroscopy. In practice, a multi-reflection crystal with a 45° angle can accommodate most routine samples.

One of the major advantages of ATR spectroscopy is that absorption spectra are readily obtainable on a wide variety of sample types with a minimum of preparation. Threads, yarns, fabrics, and fibers can be studied by pressing the samples against the dense crystal. Pastes, powders, or suspensions can be handled

¹⁵See F. M. Mirabella, ed., *Modern Techniques in Applied Molecular Spectroscopy*, New York: Wiley, 1998; G. Kortum, *Reflectance Spectroscopy*, New York: Springer, 1969; N. J. Harrick, *Internal Reflection Spectroscopy*, New York: Wiley, 1967.

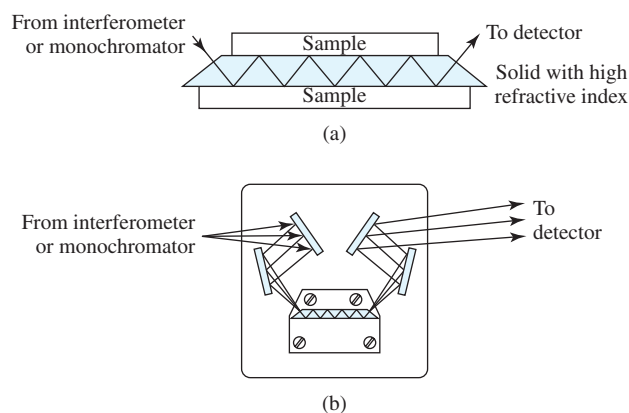


FIGURE 17-13 ATR apparatus. In (a) a solid sample is shown mounted on an internal reflection crystal of high refractive index. The materials used as ATR crystals include KRS-5, AgCl, Ge, Si, and the Irtran materials. Solid samples can be pressed against the crystal to obtain optical contact. In (b), a typical attachment for ATR is shown. With many attachments, the internal reflection plate can be positioned in the holder to provide several incident angles.

in a similar way. Aqueous solutions can also be accommodated provided the crystal is not water soluble. There are even ATR flow cells available. ATR spectroscopy has been applied to many substances, such as polymers, rubbers, and other solids. It is of interest that the resulting spectra are free from the interference fringes mentioned previously.

The spectra obtained with ATR methods can differ from IR absorption spectra because of distortions that occur near strong absorption bands where the sample refractive index may change rapidly. Also, the orientation of the sample on the ATR crystal can influence band shapes and relative intensities. However, the ATR band intensity is usually proportional to concentration so that quantitative measurements can be made.

17C PHOTOACOUSTIC IR SPECTROSCOPY

Photoacoustic spectroscopy (PAS) provides a way to obtain UV, visible, and IR absorption spectra of solids, semisolids, or turbid liquids.¹⁶ Acquisition of spectra for these materials by ordinary methods is usually difficult at best and often impossible because of light scattering and reflection.

17C-1 The Photoacoustic Effect

PAS is based on a light absorption effect that was first investigated in the 1880s by Alexander Graham Bell and others. This effect is observed when a gas in a closed cell is irradiated with a chopped beam of radiation of a wavelength that is absorbed by the gas. The absorbed radiation causes periodic heating of the

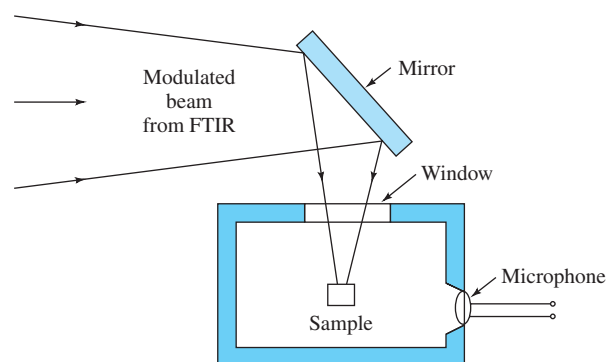


FIGURE 17-14 Diagram of a photoacoustic attachment for an FTIR spectrometer.

gas, which in turn results in regular pressure fluctuations within the chamber. If the chopping rate lies in the acoustical frequency range, these pulses of pressure can be detected by a sensitive microphone. The photoacoustic effect has been used since the turn of the century for the analysis of absorbing gases and has recently taken on new importance for this purpose with the advent of tunable IR lasers as sources. Of greater importance, however, has been the use of the photoacoustic effect for obtaining absorption spectra of solids and turbid liquids.

17C-2 Photoacoustic IR Spectra

Photoacoustic IR spectroscopy saw limited application before the advent of FTIR instruments. Now, several manufacturers make photoacoustic accessories for FTIR instruments. In photoacoustic measurements, the sample is placed in a small sample cup within the photoacoustic attachment as illustrated in Figure 17-14. The photoacoustic spectrometer chamber is filled with a high-thermal-conductivity gas such as helium or nitrogen and placed in the FTIR sample compartment. The mirror shown deflects the modulated beam onto the sample. Absorption of the IR beam by the sample can result in nonradiative decay of the excited vibrational states of the sample molecules. This can transfer heat to the surface of the sample and result in the generation of a modulated acoustic wave in the gas inside the chamber. A very sensitive microphone then detects the acoustic wave.

Photoacoustic spectra are normally plotted in a format similar to absorption spectra as shown in Figure 17-15. The spectra are usually plotted as a ratio to a background scan of a totally absorbing material such as carbon black. Although the frequencies of transitions are the same as in absorption spectra, relative intensities depend on wavelength and modulation frequency.

Photoacoustic spectra can be obtained on samples with essentially no sample preparation. The only requirement is that the sample must fit within the sample cup. In addition to solids and turbid liquids, photoacoustic methods have been used for detecting the components of mixtures separated by thin-layer and high-performance liquid chromatography and for monitoring the concentrations of gaseous pollutants in the atmosphere.

¹⁶See K. H. Michaelian, *Photoacoustic Infrared Spectroscopy*, 2nd ed., Weinheim, Germany: Wiley-VCH, 2010.

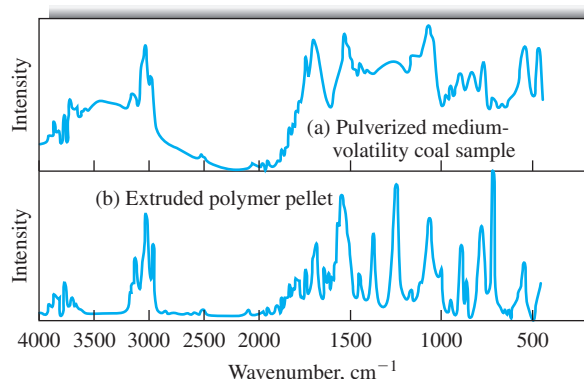


FIGURE 17-15 Photoacoustic IR spectra of (a) a pulverized coal sample and (b) an extruded polymer pellet. (From *Practical Sampling Techniques for Infrared Analysis*, P. B. Coleman, ed., Boca Raton, FL: CRC Press, 1993.)

17D NEAR-IR SPECTROSCOPY

The near-IR (NIR) region of the spectrum extends from the upper wavelength end of the visible region at about 770 nm to 2500 nm (13,000 to 4000 cm^{-1}).¹⁷ Absorption bands in this region are overtones or combinations (Section 16A-4) of fundamental stretching vibrations that occur in the region of 3000 to 1700 cm^{-1} . The bonds involved are usually C—H, N—H, and O—H. Because the bands are overtones or combinations, their molar absorptivities are low and detection limits are on the order of 0.1%.¹⁸

In contrast to mid-IR spectrometry, the most important uses of NIR radiation are for the routine *quantitative* determination of species, such as water, proteins, low-molecular-mass hydrocarbons, and fats, in products of the agricultural, food, petroleum, and chemical industries. Both diffuse-reflection and transmission measurements are used, although diffuse reflectance is by far the more widely used.

17D-1 Instrumentation

Four different types of instruments are available for the NIR region. Grating instruments are similar to those used for UV-visible absorption spectroscopy. There are also discrete filter instruments usually containing filter wheels for selecting different wavelengths. These are less flexible than other instruments but useful for fixed, well-characterized samples. In addition, acousto-optic tunable filter (AOTF) instruments are available. Acousto-optic filters and filter-based instruments are discussed

in Section 7C-3. Speed and ruggedness are the main advantages of AOTF devices. Fourier transform spectrometers are also available commercially for NIR spectrometry. The advantages normally associated with FT instruments in the mid-IR region, such as high throughput and resolution, are less applicable in the NIR region. However, the wavelength reproducibility and signal-to-noise-ratio characteristics are still major advantages of FT systems.

Most NIR spectrometers use tungsten-halogen lamps with quartz windows. Cells for absorption measurements are usually quartz or fused silica transparent up to about 3000 nm. Cell lengths vary from 0.1 to 10 cm. Detectors range from PbS and PbSe photoconductors to InSb and InAs photodiodes. Array detectors, such as InGaAs detectors, have also become available for the region. Several commercial UV-visible spectrophotometers are designed to operate from 180 to 2500 nm and can thus be used to obtain NIR spectra.

Several solvents are used for NIR studies. Some of these are listed in Figure 17-16. Note that only carbon tetrachloride and carbon disulfide are transparent throughout the entire NIR region.

17D-2 Data Processing in NIR Spectrometry

NIR spectral bands are normally broad and often overlapping. There are rarely clean spectral bands that allow simple correlation with analyte concentration. Instead, multivariate calibration techniques are used.¹⁹ Most commonly, partial least squares, principal components regression, and artificial neural networks are used. Such calibration involves development of a calibration model through obtaining results on a “training set” that includes as many of the conditions encountered in the samples as possible. Problems and pitfalls in developing the models have been discussed.²⁰

The manufacturers of NIR instruments include software packages for developing calibration models. In addition, third-party software for multivariate calibration is readily available.

17D-3 Applications of NIR Absorption Spectrometry

In contrast to mid-IR spectroscopy, NIR absorption spectra are less useful for identification and more useful for quantitative analysis of compounds containing functional groups made up of hydrogen bonded to carbon, nitrogen, and oxygen.

¹⁷In the literature of NIR spectrometry, the abscissa is usually *wavelength* in nanometers or micrometers in contrast to mid-IR spectra in which the abscissa is *wavenumber* in units of $(\text{cm})^{-1}$.

¹⁸For a general reference on NIR, see *Handbook of Near-Infrared Analysis*, 3rd ed., D. A. Burns and E. W. Ciurczak, eds., Boca Raton, FL: Taylor & Francis, 2008.

¹⁹For a discussion of multivariate techniques, see K. R. Beebe, R. J. Pell, and M. B. Seasholtz, *Chemometrics: A Practical Guide*, Chap. 5, New York: Wiley, 1998; H. Martens and T. Naes, *Multivariate Calibration*, New York: Wiley, 1989; R. G. Brereton, *Applied Chemometrics for Scientists*, Chap. 6, Chichester, UK: Wiley, 2007; K. Varmuza and P. Fitzmoser, *Introduction to Multivariate Statistics in Chemometrics*, Chap. 4, Boca Raton, FL: CRC Press, 2009.

²⁰For example, see M. A. Arnold, J. J. Burmeister, and G. W. Small, *Anal. Chem.*, **1998**, *70*, 1773, DOI: 10.1021/ac9710801; L. Zhang, G. W. Small, and M. A. Arnold, *Anal. Chem.*, **2003**, *75*, 5905, DOI: 10.1021/ac034495x.

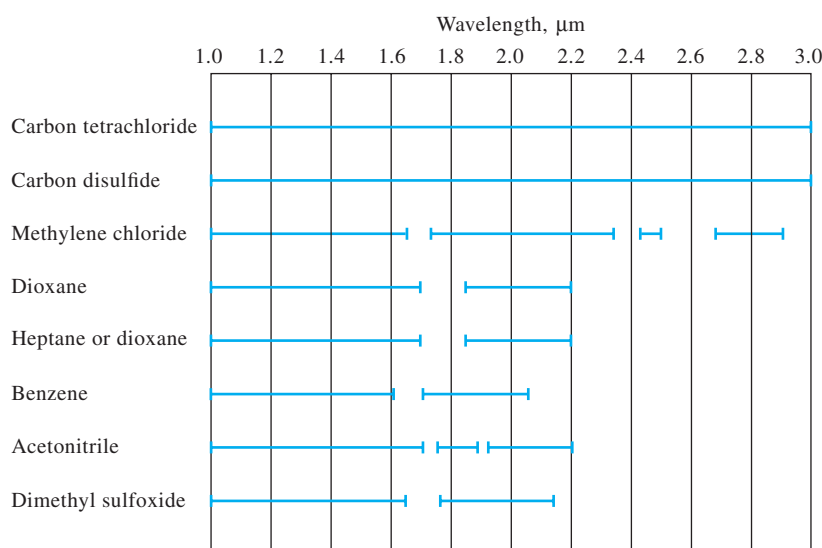


FIGURE 17-16 Some useful solvents for NIR spectroscopy. Solid lines indicate satisfactory transparency for use with 1-cm cells.

Such compounds can often be determined with accuracies and precisions equivalent to UV-visible spectroscopy, which are usually much better than those of mid-IR spectroscopy. Some applications include the determination of water in a variety of samples, including glycerol, hydrazine, organic films, and fuming nitric acid; the quantitative determination of phenols, alcohols, organic acids, and hydroperoxides based on the first overtone of the O—H stretching vibration that absorbs radiation at about 7100 cm^{-1} ($1.4\text{ }\mu\text{m}$); and the determination of esters, ketones, and carboxylic acids based on their absorption in the region of $3300\text{ to }3600\text{ cm}^{-1}$ ($2.8\text{ to }3.0\text{ }\mu\text{m}$). The absorption in this last case is the first overtone of the carbonyl stretching vibration.

NIR spectrophotometry is also a valuable tool for identification and determination of primary and secondary amines in the presence of tertiary amines in mixtures. The analyses are generally carried out in carbon tetrachloride solutions in 10-cm cells. Primary amines are determined directly by measurement of the absorbance of a combination N—H stretching band at about 5000 cm^{-1} ($2.0\text{ }\mu\text{m}$); neither secondary nor tertiary amines absorb radiation in this region. Primary and secondary amines have several overlapping absorption bands in the $3300\text{ to }10,000\text{ cm}^{-1}$ ($1\text{ to }3\text{ }\mu\text{m}$) region due to various N—H stretching vibrations and their overtones. On the other hand, tertiary amines can have no such bands. Thus, one of these bands gives the secondary amine concentration after correction for the absorption by the primary amine.

17D-4 Applications of NIR Reflectance Spectrometry

NIR reflectance spectroscopy has become a most important tool for the routine quantitative determination of constituents

in finely ground solids.²¹ The most widespread use of this technique has been for the determination of protein, moisture, starch, oil, lipids, and cellulose in agricultural products such as grains and oilseeds.²² For example, Canada sells all its wheat based on guaranteed protein content, so to certify the grain, the Canadian Grain Commission formerly ran more than 600,000 Kjeldahl protein determinations annually. It was estimated in 1984 that 80% to 90% of all Canadian grain was analyzed for protein by NIR reflectance spectroscopy at a savings of more than \$500,000 a year in analytical costs.²³

In NIR reflectance spectroscopy, the finely ground solid sample is irradiated with one or more narrow bands of radiation ranging in wavelength from $1\text{ to }2.5\text{ }\mu\text{m}$ or $10,000\text{ to }4000\text{ cm}^{-1}$. Diffuse reflectance occurs in which the radiation penetrates the surface layer of the particles, excites vibrational modes of the analyte molecule, and is then scattered in all directions. A reflectance spectrum is thus produced that is dependent on the composition of the sample. A typical reflectance spectrum for a sample of wheat is shown in Figure 17-17. The ordinate in this case is the logarithm of the reciprocal of reflectance R , where R is the ratio of the intensity of radiation reflected from the sample to the reflectance from a standard reflector, such as finely ground barium sulfate or magnesium oxide. The reflectance

²¹See, J. M. Soriano-Disla, L. J. Janik, R. A. Viscarra Rossel, L. M. Macdonald, M. J. McLaughlin, *Appl. Spectrosc. Rev.*, **2014**, 49, 139, DOI: 10.1080/05704928.2013.811081.

²²H. Cen and Y. He., *Trends Food Sci. Technol.*, **2007**, 18, 72, DOI: 10.1016/j.tifs.2006.09.003.

²³S. A. Borman, *Anal. Chem.*, **1984**, 56, 933A, DOI: 10.1021/ac00272a727.

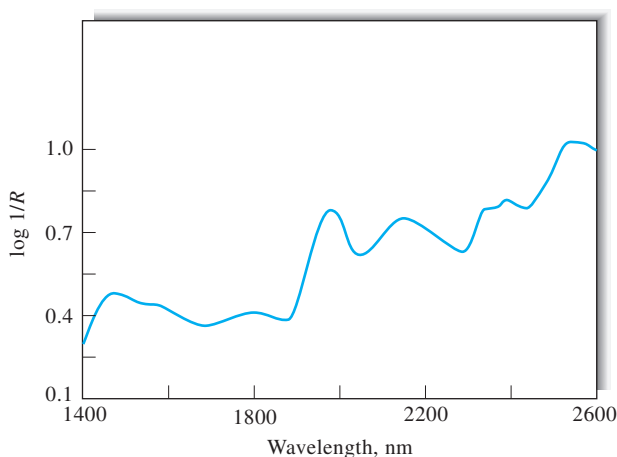


FIGURE 17-17 Diffuse-reflectance NIR spectrum for a sample of wheat.

band at 1940 nm is a water band used for moisture determinations. The band at about 2100 nm is, in fact, two overlapping bands, one for starch and the other for protein. By making measurements at several wavelengths in this region, the concentrations of each of these components can be determined.

Instruments for diffuse-reflectance measurements are available from commercial sources. Some of these use several interference filters to provide narrow bands of radiation. Others are equipped with grating monochromators. Ordinarily, reflectance measurements are made at two or more wavelengths for each analyte species being determined.

The great advantage of NIR reflectance methods is their speed and the simplicity of sample preparation. Once method development has been completed, analysis of solid samples for several species can be completed in a few minutes. Accuracies and precisions of 1% to 2% relative are regularly reported.

17E FAR-IR SPECTROSCOPY

The far-IR region is particularly useful for inorganic studies because absorption due to stretching and bending vibrations of bonds between metal atoms and both inorganic and organic ligands generally occurs at frequencies lower than 650 cm^{-1} ($>15\text{ }\mu\text{m}$). For example, heavy-metal iodides generally absorb radiation in the region below 100 cm^{-1} , and the bromides and chlorides have bands at higher frequencies. Absorption frequencies for metal-organic bonds ordinarily depend on both the metal atom and the organic portion of the species.

Far-IR studies of inorganic solids have also provided useful information about lattice energies of crystals and transition energies of semiconducting materials.

Molecules composed only of light atoms absorb far-IR radiation if they have skeletal bending modes that involve more than two atoms other than hydrogen. Important examples are

substituted benzene derivatives, which generally show several absorption bands. The spectra are frequently quite specific and useful for identifying a particular compound. There are also characteristic group frequencies in the far-IR region.

Pure rotational absorption by gases is observed in the far-IR region, provided the molecules have permanent dipole moments. Examples include H_2O , O_3 , HCl , and AsH_3 . Absorption by water is troublesome; elimination of its interference requires evacuation or at least purging of the spectrometer.

Prior to the advent of Fourier transform spectrometers, experimental difficulties made it difficult to obtain good far-IR spectra. The sources available in this region are quite low in intensities. Order-sorting filters are needed with grating instruments in this region to minimize radiation diffracted from higher grating orders. These reduced the already low throughputs of dispersive spectrometers. It is no surprise that Fourier transform spectrometry was first applied in this spectral region. The high throughput of interferometers and the relatively low mechanical tolerances required combine to make it fairly simple to obtain FTIR spectra in the far-IR region with good signal-to-noise ratios.

17F IR EMISSION SPECTROSCOPY

Molecules that absorb IR radiation are also capable of emitting characteristic IR wavelengths when they are heated. The principal deterrent to the analytical application of this phenomenon has been the poor signal-to-noise characteristics of the IR emission signal, particularly when the sample is at a temperature only slightly higher than its surroundings. With interferometry, however, interesting and useful applications have appeared in the literature.

An early example of the application of IR emission spectroscopy is found in a paper that describes the use of a Fourier transform spectrometer for the identification of microgram quantities of pesticides.²⁴ Samples were prepared by dissolving them in a suitable solvent followed by evaporation on a NaCl or KBr plate. The plate was then heated electrically near the spectrometer entrance. Pesticides such as DDT, malathion, and diel-drin were identified in amounts as low as 1 to $10\text{ }\mu\text{g}$.

Equally interesting has been the use of interferometry for the remote detection of components emitted from industrial smoke stacks. In one of these applications, an interferometer was mounted on an 8-inch reflecting telescope.²⁵ With the telescope focused on the plume from an industrial plant, CO_2 and SO_2 were readily detected at a distance of several hundred feet.

²⁴J. Coleman and M. J. D. Low, *Spectrochim. Acta*, **1966**, 22, 1293, DOI: 10.1016/0371-1951(66)80034-6.

²⁵M. J. D. Low and F. K. Clancy, *Env. Sci. Technol.*, **1967**, 1, 73, DOI: 10.1021/es60001a007.

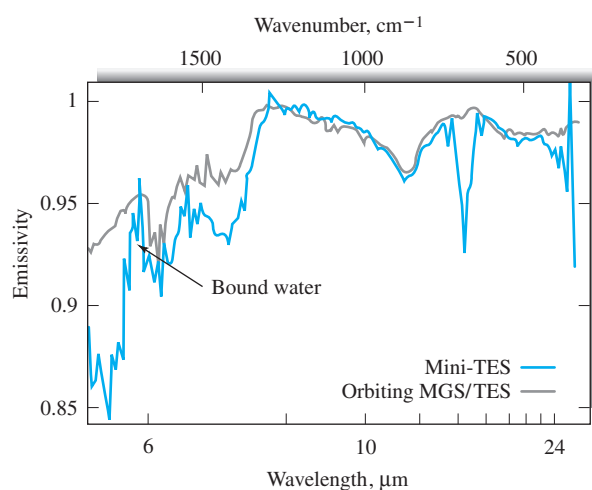


FIGURE 17-18 Spectra from Mars rover *Spirit* showing an unidentified mineral containing bound water in its crystal structure. Minerals such as zeolites and gypsum are possible candidates. Mini-TES = mini thermal-emission spectrometer, MGS = Mars Global Surveyor. (Courtesy of NASA/JPL/Arizona State University.)

The Mars Global Surveyor launched from Cape Canaveral in November 1996. It reached Mars orbit in 1997. One of five instruments on board was an IR emission spectrometer, called a thermal emission spectrometer. The spacecraft completed its mapping mission in 2001, providing measurements of the Martian surface and atmosphere. The Mars rover *Spirit* had a mini thermal-emission spectrometer on board capable of indicating the composition of nearby soils and rocks. Figure 17-18 shows an IR emission spectrum taken in early 2004 that showed evidence of an unidentified water-containing mineral. Thermal emission spectra have also indicated the presence of carbonates and other hydrated minerals. These may have been produced by long-standing bodies of water. *Spirit* stopped communicating in March 2010; efforts to resume transmissions were concluded in May 2011.

The Rover Environmental Monitoring Station (REMS) aboard the Mars Rover Curiosity contains a suite of IR sensors located on one of its booms. These measure the IR radiation emitted by the ground and atmosphere, which are used to calculate temperatures.

17G IR MICROSCOPY AND IMAGING

IR microspectrometry is used for obtaining absorption or reflection spectra of species in samples having physical dimensions in the range of 10 to 500 μm .²⁶ IR microscopes, which were first introduced by several instrument manufacturers in the 1980s, generally consist of two microscopes, one an ordinary optical microscope and the other an IR device with reflection optics

that reduce the size of the IR beam to about that of the sample. The optical microscope is used to visually locate the sample particle or spot to be studied with the IR beam. The IR source is an ordinary Fourier transform spectrometer and not a grating instrument because of the much greater sensitivity of FTIR instruments. The detector is usually a liquid-nitrogen-cooled mercury telluride–cadmium telluride photoconductive device that is more sensitive than other types of IR detectors.

IR microscopy has been applied to identification of polymer contaminants, imperfections in polymer films, and examination of layers of laminated polymer sheets. The IR microscope has proven quite useful in forensic science.²⁷ Forensic applications include identification of micro-samples of fibers, paint, and explosives. IR microscopy has also been applied in the textile industry to characterize single fibers.

In a general sense, IR microspectroscopy can be divided into sample *mapping* or sample *imaging* applications.²⁸ In sample mapping, the spectrum of a region of sample is taken, and then another region is moved into the measurement zone and a new spectrum obtained. This sequential acquisition of spectra is repeated until the entire area of interest of the sample has been mapped. In contrast, imaging is accomplished by focusing an image of the sample on an array detector and taking a digital picture. The intensity of the radiation passing through each sample region is measured simultaneously by the array detector. The mapping/imaging distinction can be somewhat blurry since some commercial instruments use an array of detectors to acquire a line map prior to repositioning the sample and repeating the process several times.

In *hyperspectral imaging*, images at multiple wavelength regions are acquired simultaneously with a two-dimensional array detector. Both the mid-IR and the near-IR regions have been used, although the instrumentation is different in the two spectral regions. Raman spectroscopy (see Chapter 18) is also widely used in hyperspectral imaging applications. These imaging techniques are often referred to as *chemical imaging* methods.

Biomedical applications of IR imaging and mapping include examination of tissue samples and imaging of individual cells.²⁹ IR microspectroscopy has been applied to the examination of agricultural, plant, and food materials as well as to polymers and pharmaceuticals.³⁰ Although many of these techniques are fairly new, we will undoubtedly see many more applications in the near future.

²⁷See *Infrared and Raman Spectroscopy in Forensic Science*, J. M. Chalmers, H. G. M. Edwards, and M. D. Hairgreaves, eds., Chichester, UK: Wiley, 2012.

²⁸P. R. Griffiths and E. V. Miseso in *Infrared and Raman Imaging*, 2nd ed., Chap. 1, R. Salzer and H. W. Siesler, eds., Weinheim, Germany: Wiley-VCH, 2014.

²⁹C. Krafft and J. Popp, Chap. 3, S. Gamsjaeger, R. Mendelsohn, K. Klaushofer, and E. P. Paschalis, Chap. 4, M. Diem et al., Chap. 5, in *Infrared and Raman Imaging*, 2nd ed., R. Salzer and H. W. Siesler, eds., Weinheim, Germany: Wiley-VCH, 2014.

³⁰H. Schulz, A. Krahmer, A. Naumann, and G. Gudi, Chap. 6, V. Bellon-Maurel and N. Gorretta, Chap. 7, E. Grotheer et al. in *Infrared and Raman Spectroscopic Imaging*, 2nd ed., R. Salzer and H. W. Siesler, eds., Weinheim, Germany: Wiley-VCH, 2014.

²⁶See *Practical Guide to Infrared Microspectroscopy*, H. J. Humecki, ed., New York: Marcel Dekker, 1995; J. Katon, *Infrared Microspectroscopy in Modern Techniques in Molecular Spectroscopy*, F. Mirabella, ed., New York: Wiley, 1998.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

- 17-1** Cyclohexanone exhibits its strongest IR absorption band at $5.86\ \mu\text{m}$, and at this wavelength there is a linear relationship between absorbance and concentration.
- Identify the part of the molecule responsible for the absorbance at this wavelength.
 - Suggest a solvent that would be suitable for a quantitative analysis of cyclohexanone at this wavelength.
 - A solution of cyclohexanone ($4.0\ \text{mg/mL}$) in the solvent selected in part (b) exhibits a blank-corrected absorbance of 0.800 in a cell with a path length of $0.025\ \text{mm}$. What is the detection limit for this compound under these conditions if the noise associated with the spectrum of the solvent is 0.001 absorbance units?
- * 17-2** The spectrum in Figure 17-19 was obtained for a liquid with the empirical formula of $\text{C}_3\text{H}_6\text{O}$. Identify the compound.
- 17-3** The spectrum in Figure 17-20 is that of a high-boiling-point liquid having the empirical formula $\text{C}_9\text{H}_{10}\text{O}$. Identify the compound as closely as possible.
- * 17-4** The spectrum in Figure 17-21 is for an acrid-smelling liquid that boils at 50°C and has a molecular mass of about 56 . What is the compound? What impurity is clearly present?
- * 17-5** The spectrum in Figure 17-22 is that of a nitrogen-containing substance that boils at 97°C . What is the compound?
- 17-6** Why are quantitative analytical methods based on NIR radiation often more precise and accurate than methods based on mid-IR radiation?
- 17-7** Discuss the origins of photoacoustic IR spectra. What are the usual types of samples for photoacoustic measurements?
- 17-8** Why do IR spectra seldom show regions at which the transmittance is 100% ?
- * 17-9** A quantitative analysis of methyl iso-butyl ketone (MIBK) was performed by measuring absorbances at $1745\ \text{cm}^{-1}$. Solutions of differing MIBK concentration gave the absorbances shown in the table.



Concentration of MIBK, %	A
1.00	0.12
3.00	0.28
5.00	0.42
7.00	0.56
9.00	0.68

Samples 1, 2, and 3 containing an unknown amount of MIBK gave the absorbances that follow:

Sample	A
1	0.49
2	0.74
3	0.34

Determine the equation of the calibration curve and the concentrations of the three samples.

- * 17-10** An empty cell showed 20 interference fringes in the wavelength range of 6.0 to $12.2\ \mu\text{m}$. Calculate the path length of the cell.
- * 17-11** An empty cell exhibited 17.5 interference fringes in the region of 1050 to $1580\ \text{cm}^{-1}$. What was the path length of the cell?
- * 17-12** Estimate the thickness of the polystyrene film that yielded the spectrum shown in Figure 16-1.

» QUESTIONS AND PROBLEMS (continued)

Challenge Problem

- 17-13 (a)** Consider an ATR experiment with a sample having a refractive index of 1.03 at 2000 cm^{-1} . The ATR crystal was AgCl with a refractive index of 2.00 at this wavelength. For an angle of incidence of 45° , what is the effective penetration depth of the evanescent wave? How would the penetration depth change if the angle were changed to 60° ?
- (b)** For the same experiment in part (a) and a 60° incidence angle, find the penetration depths for sample refractive indexes varying from 1.00 to 1.70 in steps of 0.10. Plot penetration depth as a function of refractive index. Determine the refractive index for which the penetration depth becomes zero. What happens at this point?
- (c)** For a sample with a refractive index of 1.37 at 2000 cm^{-1} and an incidence angle of 45° , plot the penetration depth versus the ATR crystal refractive index. Vary the crystal refractive index over the range of 2.00 to 4.00 in steps of 0.25. Which crystal, AgCl ($n_c = 2.00$) or Ge ($n_c = 4.00$), gives the smaller penetration depth? Why?
- (d)** An aqueous solution with a refractive index of 1.003 is measured with an ATR crystal with a refractive index of 2.8. The incidence angle is 45° . What is the effective penetration depth at 3000 cm^{-1} , 2000 cm^{-1} , and 1000 cm^{-1} ? Is absorption by the aqueous solvent as much of a problem in ATR as in normal IR absorption measurements? Why or why not?
- (e)** Work by S. Ekgasit and H. Ishida (S. Ekgasit and H. Ishida, *Appl. Spectrosc.*, **1996**, *50*, 1187; *Appl. Spectrosc.*, **1997**, *51*, 461) describes a new method to obtain a depth profile of a sample surface using ATR spectroscopy. Describe the principles of this new approach. Why are spectra taken with different degrees of polarization? What is the complex index of refraction?

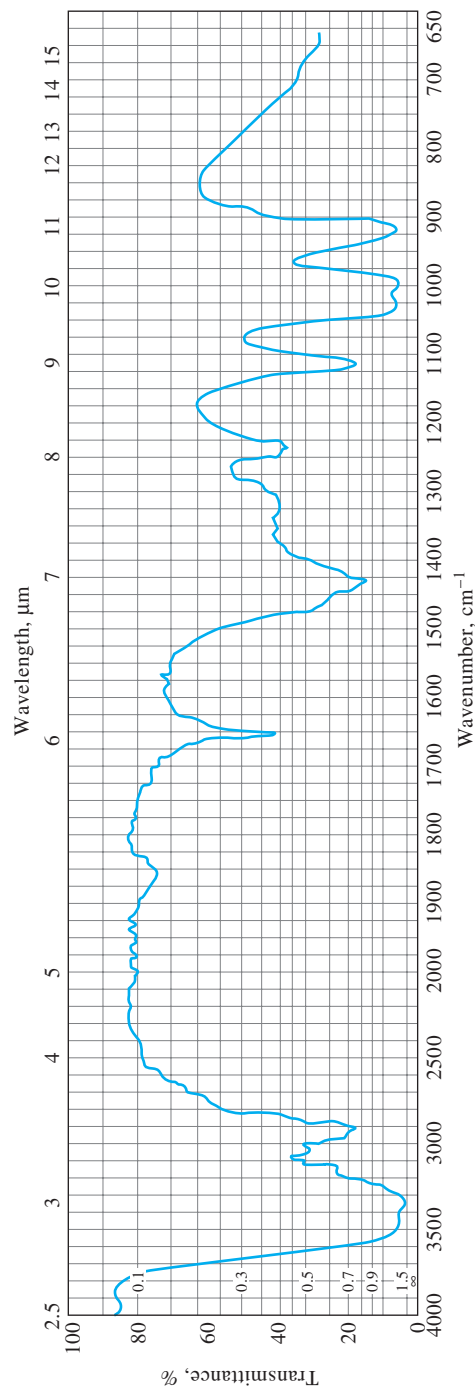


FIGURE 17-19 See Problem 17-2. (Spectrum courtesy of Thermodynamics Research Center Data Project, Texas A&M University, College Station, Texas.)

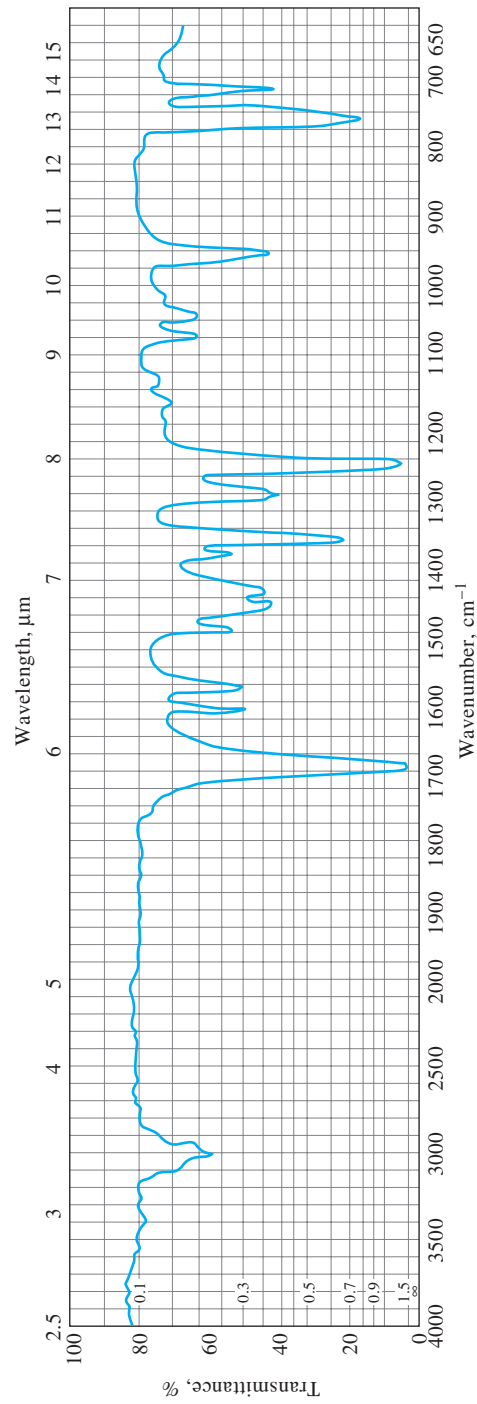


FIGURE 17-20 See Problem 17-3. (Spectrum courtesy of Thermodynamics Research Center Data Project, Texas A&M University, College Station, Texas.)

QUESTIONS AND PROBLEMS (continued)

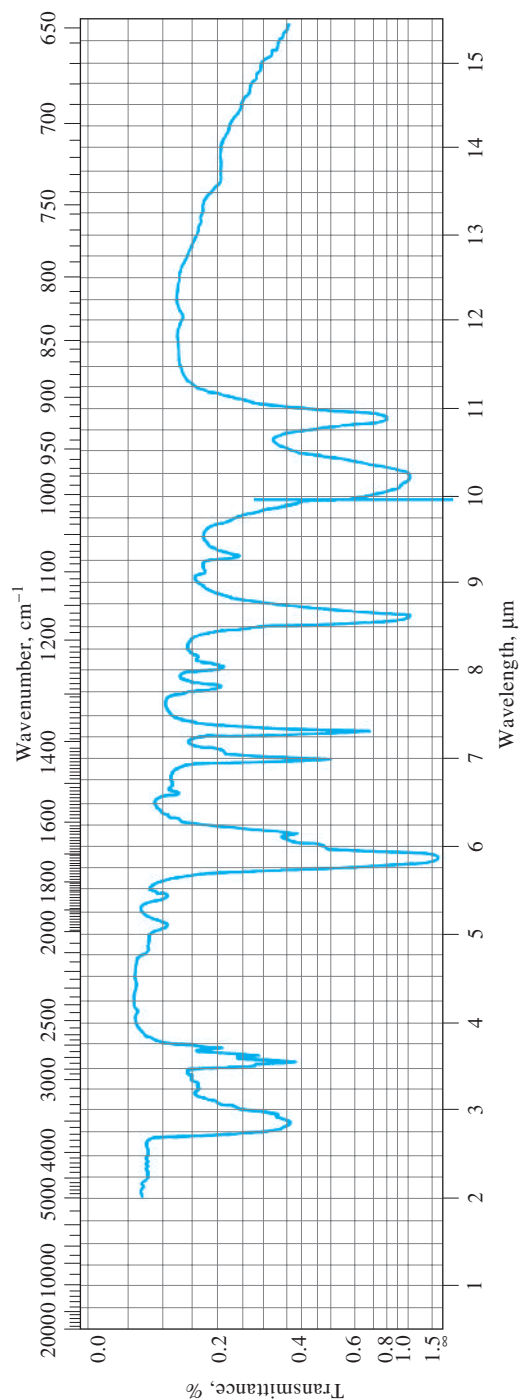


FIGURE 17-21 See Problem 17-4. (Spectrum courtesy of Thermodynamics Research Center Data Project, Texas A&M University, College Station, Texas.)

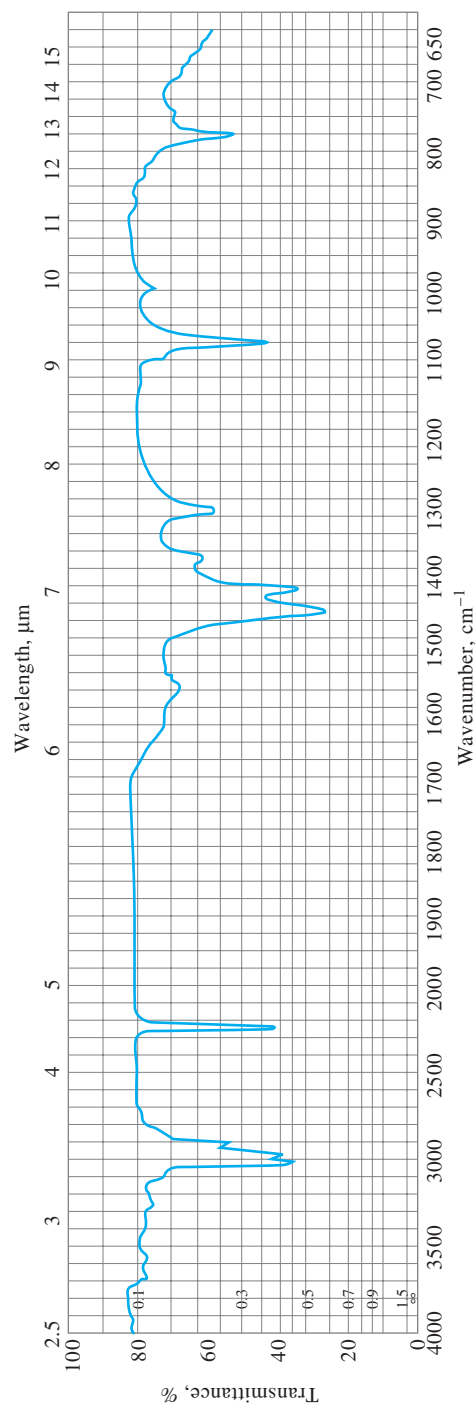


FIGURE 17-22 See Problem 17-5. (Spectrum courtesy of Thermodynamics Research Center Data Project, Texas A&M University, College Station, Texas.)

Raman Spectroscopy

When radiation passes through a transparent medium, the species present scatter a fraction of the beam in all directions (Section 6B-10). Raman scattering results from the same type of quantized vibrational changes associated with infrared (IR) absorption. Thus, the difference in wavelength between the incident and scattered visible radiation corresponds to wavelengths in the mid-IR region. Indeed, the Raman scattering spectrum and IR absorption spectrum for a given species often resemble one another quite closely. Although IR spectroscopy is still the more widely used vibrational spectroscopic tool, for several problems, **Raman spectroscopy** can provide more useful and selective information.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

In 1928 the Indian physicist C. V. Raman discovered that the visible wavelength of a small fraction of the radiation scattered by certain molecules differs from that of the incident beam and furthermore that the shifts in wavelength depend on the chemical structure of the molecules responsible for the scattering. Raman was awarded the 1931 Nobel Prize in physics for this discovery and for his systematic exploration of it.¹

Although there can be striking similarities between Raman spectra and IR spectra, enough differences remain between the kinds of groups that are IR active and Raman active to make the techniques complementary rather than competitive. Water is quite useful as a solvent in Raman spectroscopy, which is a tremendous advantage of Raman over IR. In addition, because Raman scattering is usually measured in the visible or near-IR region, glass or quartz cells can be used, avoiding the inconvenience of working with sodium chloride or other atmospherically unstable window materials. Despite these advantages, Raman spectroscopy was not widely used by chemists until laser sources in the 1960s made Raman spectra a good deal easier to obtain. In recent years, Raman spectroscopy has become a routine tool thanks to the laser, the array detector, and the availability of commercial instrumentation at moderate cost.

18A THEORY OF RAMAN SPECTROSCOPY

Raman spectra are acquired by irradiating a sample with a nearly monochromatic laser source in the visible or near-IR spectral region. During irradiation, the spectrum of the scattered radiation is measured at some angle (often 90°) with a suitable spectrometer. With the exception of Resonance Raman scattering, the excitation wavelength is usually well removed from an absorption band to minimize fluorescence. The Raman experiment was illustrated previously in Figure 6-18. At the very most, the intensities of Raman lines are 0.001% of the intensity of the

¹For discussions of the theory and practice of Raman spectroscopy, see P. Vandenabeele, *Practical Raman Spectroscopy: An Introduction*, Chichester, UK: Wiley, 2013; E. Smith and G. Dent, *Modern Raman Spectroscopy: A Practical Approach*, Chichester, UK: Wiley, 2005; J. R. Ferraro, K. Nakamoto, and C. W. Brown, *Introductory Raman Spectroscopy*, 2nd ed., San Diego: Academic Press, 2003; R. L. McCreery, *Raman Spectroscopy for Chemical Analysis*, New York: Wiley, 2000.

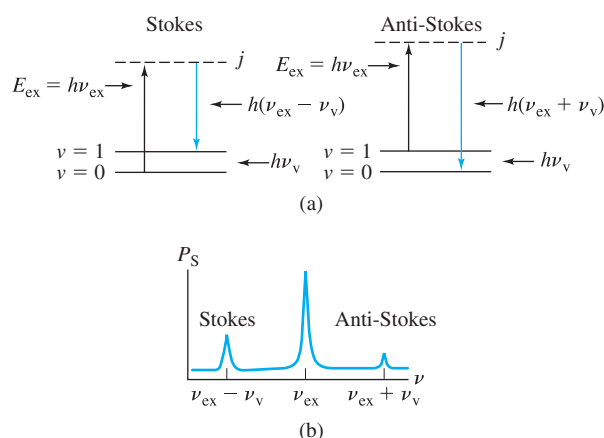


FIGURE 18-1 The origin of Raman spectra. In (a) radiation from a source that is incident on the sample produces scattering at all angles. The incident radiation causes excitation (a) to a virtual level j and subsequent reemission of a photon of lower (left) or higher (right) energy. The Raman spectrum (b) consists of lower-frequency emissions called Stokes scattering and higher-frequency emissions termed anti-Stokes scattering. Usually, the ground vibrational level ($\nu = 0$) is more highly populated than the excited vibration levels so that the Stokes lines are more intense than the anti-Stokes lines. Elastically scattered radiation is of the same frequency as the excitation beam and is called Rayleigh scattering.

source. Because of this, it might seem more difficult to detect and measure Raman bands than IR vibrational bands. However, the Raman scattered radiation is in the visible or near-IR regions for which more sensitive detectors are available. Hence, today, obtaining Raman spectra is nearly as easy as obtaining IR spectra.

18A-1 Excitation of Raman Spectra

In Figure 18-1, the sample is irradiated by a monochromatic beam of energy $h\nu_{ex}$. Because the excitation wavelength is usually well away from an absorption band, excitation can be considered to involve a *virtual state* of energy level j , indicated by the dashed line in Figure 18-1a.² A molecule in the ground vibrational level ($\nu = 0$) can absorb a photon of energy $h\nu_{ex}$ and reemit a photon of energy $h(\nu_{ex} - \nu_v)$, as shown on the left side of Figure 18-1a. When the scattered radiation is of a lower frequency than the excitation radiation, it is called *Stokes scattering*. Molecules in a vibrationally excited state ($\nu = 1$) can also scatter radiation inelastically and produce a Raman signal of energy $h(\nu_{ex} + \nu_v)$. Scattered radiation of a higher frequency than the source radiation is called *anti-Stokes scattering*. Elastic scattering can also occur with emission of a photon of the same energy as the excitation photon, $h\nu_{ex}$. Scattered radiation of the same frequency as the source is termed *Rayleigh scattering*. Note that the frequency shifts of the inelastically scattered radiation

$(\nu_{ex} + \nu_v) - \nu_{ex} = \nu_v$ and $(\nu_{ex} - \nu_v) - \nu_{ex} = -\nu_v$ correspond to the vibrational frequency, ν_v . The simplified Raman spectrum corresponding to the transitions shown is given in Figure 18-1b.

Figure 18-2 depicts a portion of the Raman spectrum of carbon tetrachloride that was obtained by using an argon-ion laser having a wavelength of 488.0 nm as the source. As is usually the case for Raman spectra, the abscissa of Figure 18-2 is the wavenumber shift $\Delta\bar{\nu}$, which is defined as the difference in wavenumbers (cm^{-1}) between the observed radiation and that of the source. Note that three Raman lines are found on both sides of the Rayleigh lines and that the pattern of shifts on each side is identical. That is, Stokes lines are found at wavenumbers that are 218, 314, and 459 cm^{-1} smaller than the Rayleigh lines, and anti-Stokes lines occur at 218, 314, and 459 cm^{-1} greater than the wavenumber of the source. It should also be noted that additional lines can be found at ± 762 and $\pm 790 \text{ cm}^{-1}$. Because the anti-Stokes lines are appreciably less intense than the corresponding Stokes lines, only the Stokes part of a spectrum is generally used. Furthermore, the abscissa of the plot is often labeled simply “wavenumber $\bar{\nu}$, cm^{-1} ” rather than “wavenumber or Raman shift $\Delta\bar{\nu}$.” It is noteworthy that fluorescence may interfere seriously with the observation of Stokes shifts but not with anti-Stokes. With fluorescing samples, anti-Stokes signals may, therefore, be more useful despite their lower intensities.

It is important to appreciate that the magnitude of Raman shifts is *independent of the wavelength of excitation*. Thus, Raman shifts identical to those shown in Figure 18-2 are observed for carbon tetrachloride regardless of whether excitation was carried out with an argon-ion laser (488.0) or a helium-neon laser (632.8 nm).

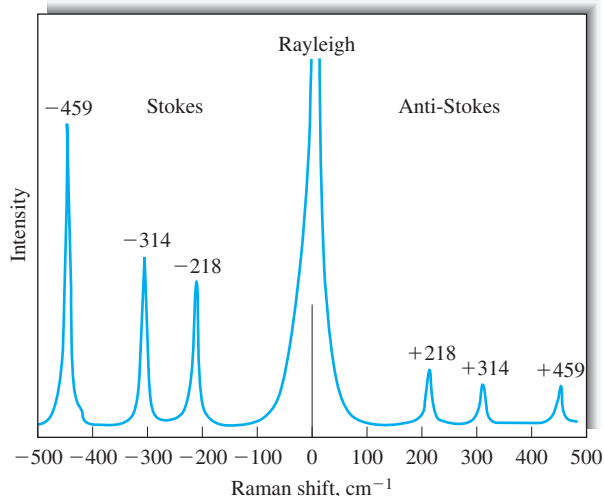


FIGURE 18-2 Raman spectrum of CCl_4 , excited by laser radiation of $\lambda_{ex} = 488 \text{ nm}$ ($\bar{\nu}_{ex} = 20,492 \text{ cm}^{-1}$). The number above the Raman lines is the Raman shift, $\Delta\nu = \nu_{ex} \pm \nu_v$, in cm^{-1} . Stokes-shifted lines are often given positive values rather than negative values as shown. (From J. R. Ferraro, K. Nakamoto, and C. W. Brown, *Introductory Raman Spectroscopy*, 2nd ed., San Diego: Academic Press, 2003. Reprinted with permission.)

²It should be noted that the virtual state is not a real state, but merely a mental construct to help in visualizing the scattering process.

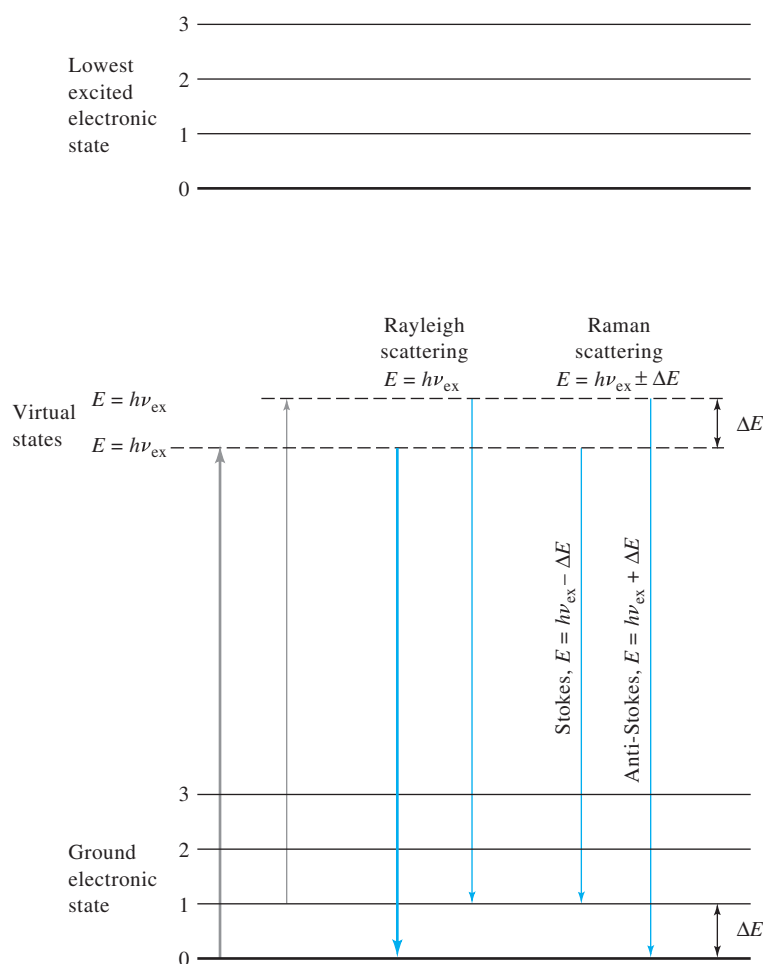


FIGURE 18-3 Origins of Rayleigh and Raman scattering.

Superficially, the Raman Stokes shifts to lower energies (longer wavelengths) are analogous to the Stokes shifts found in molecular fluorescence (see Section 6C-6). We note, however, that Raman and fluorescence spectra arise from fundamentally different processes.

18A-2 Mechanisms of Raman and Rayleigh Scattering

In normal Raman spectroscopy, spectral excitation is normally carried out by radiation having a wavelength that is well away from any absorption bands of the analyte. (Resonance Raman [see Section 18D-1] is an exception where excitation occurs near or within an absorption band.) The energy-level diagram of Figure 18-1a is expanded in Figure 18-3 and provides a qualitative picture of the sources of Raman and Rayleigh scattering. The heavy arrow on the far left depicts the energy change in the molecule when it interacts with a photon from the source. The increase in energy is equal to the energy of the photon $h\nu_{\text{ex}}$. It is important to appreciate that the process shown is *not quantized*; thus, depending on the frequency of the radiation from the

source, the energy of the molecule can assume any of an infinite number of values, or *virtual states*, between the ground state and the lowest (first) electronic excited state shown in the upper part of the diagram. The second and lighter arrow on the left shows the type of change that would occur if the molecule encountered by the photon happened to be in the first vibrational level of the electronic ground state. At room temperature, the fraction of the molecules in this state is small. Thus, as indicated by the thickness of the arrows, the probability of this process occurring is much smaller.

The middle set of arrows depicts the changes that produce Rayleigh scattering. Again the more probable change is shown by the wider arrow. Note that no energy is lost in Rayleigh scattering, and because of this the collisions between the photon and the molecule are said to be *elastic*.

Finally, the energy changes that produce Stokes and anti-Stokes emission are depicted on the right. The two differ from the Rayleigh radiation by frequencies corresponding to $\pm\Delta E$, the energy of the first vibrational level of the ground state, $h\nu_v$. Note that if the bond were IR active, the energy of its absorption

would also be ΔE . Thus, the Raman *frequency shift* and the IR *absorption frequency* are identical.

Note also that the relative populations of the two upper energy states are such that Stokes emission is much favored over anti-Stokes. In addition, Rayleigh scattering has a considerably higher probability of occurring than Raman scattering because the most probable event is the energy transfer to molecules in the ground state and reemission by the return of these molecules to the ground state. Finally, it should be noted that the ratio of anti-Stokes to Stokes intensities increases with temperature because a larger fraction of the molecules is in the first vibrationally excited state under these circumstances.

18A-3 Wave Model of Raman and Rayleigh Scattering

Let us assume that a beam of radiation having a frequency ν_{ex} is incident on a solution of an analyte. The electric field E of this radiation can be described by the equation

$$E = E_0 \cos(2\pi\nu_{\text{ex}}t) \quad (18-1)$$

where E_0 is the amplitude of the wave. When the electric field of the radiation interacts with an electron cloud of an analyte bond, it induces a dipole moment m in the bond that is given by

$$m = \alpha E = \alpha E_0 \cos(2\pi\nu_{\text{ex}}t) \quad (18-2)$$

where α is a proportionality constant called the *polarizability* of the bond. This constant is a measure of the deformability of the bond in an electric field.

The polarizability α varies as a function of the distance between nuclei according to the equation

$$\alpha = \alpha_0 + (r - r_{\text{eq}}) \left(\frac{\partial \alpha}{\partial r} \right) \quad (18-3)$$

where α_0 is the polarizability of the bond at the equilibrium internuclear distance r_{eq} and r is the internuclear separation at any instant. The change in internuclear separation varies with the frequency of the vibration ν_v as given by

$$r - r_{\text{eq}} = r_m \cos(2\pi\nu_v t) \quad (18-4)$$

where r_m is the maximum internuclear separation relative to the equilibrium position.

Substituting Equation 18-4 into 18-3 gives

$$\alpha = \alpha_0 + \left(\frac{\partial \alpha}{\partial r} \right) r_m \cos(2\pi\nu_v t) \quad (18-5)$$

We can then obtain an expression for the induced dipole moment m by substituting Equation 18-5 into Equation 18-2. Thus,

$$m = \alpha_0 E_0 \cos(2\pi\nu_{\text{ex}}t) + E_0 r_m \left(\frac{\partial \alpha}{\partial r} \right) \cos(2\pi\nu_v t) \cos(2\pi\nu_{\text{ex}}t) \quad (18-6)$$

If we use the trigonometric identity for the product of two cosines

$$\cos x \cos y = [\cos(x + y) + \cos(x - y)]/2$$

we obtain from Equation 18-6

$$m = \alpha_0 E_0 \cos(2\pi\nu_{\text{ex}}t) + \frac{E_0}{2} r_m \left(\frac{\partial \alpha}{\partial r} \right) \cos[2\pi(\nu_{\text{ex}} - \nu_v)t] + \frac{E_0}{2} r_m \left(\frac{\partial \alpha}{\partial r} \right) \cos[2\pi(\nu_{\text{ex}} + \nu_v)t] \quad (18-7)$$

The first term in this equation represents Rayleigh scattering, which occurs at the excitation frequency ν_{ex} . The second and third terms in Equation 18-7 correspond, respectively, to the Stokes and anti-Stokes frequencies of $\nu_{\text{ex}} - \nu_v$ and $\nu_{\text{ex}} + \nu_v$. In essence then, the excitation frequency is modulated by the vibrational frequency of the bond. It is important to note that the selection rules for Raman scattering require that there be a change in polarizability during the vibration—that is, $\partial\alpha/\partial r$ in Equation 18-7 must be greater than zero for Raman lines to appear. The selection rules also predict that Raman lines corresponding to fundamental modes of vibration occur with $\Delta\nu = \pm 1$. Just as with IR spectroscopy, much weaker overtone transitions appear at $\Delta\nu = \pm 2$.

We have noted that, for a given bond, the energy *shifts* observed in a Raman experiment should be identical to the *energies* of its IR absorption bands, provided the vibrational modes involved are both IR and Raman active. Figure 18-4 illustrates the similarity of the two types of spectra; it is seen that there are several bands with identical $\bar{\nu}$ and $\Delta\bar{\nu}$ values for the two compounds. We should also note, however, that the relative intensities of the corresponding bands are frequently quite different. Moreover, certain peaks that occur in one spectrum are absent in the other.

The differences between a Raman spectrum and an IR spectrum are not surprising when it is considered that the basic mechanisms, although dependent on the same vibrational modes, arise from processes that are mechanistically different. IR absorption requires that there be a change in dipole moment or charge distribution during the vibration. Only then can radiation of the same frequency interact with the molecule and promote it to an excited vibrational state. In contrast, scattering involves a momentary distortion of the electrons distributed around a bond in a molecule, followed by reemission of the radiation as the bond returns to its normal state. In its distorted form, the molecule is temporarily polarized; that is, it develops momentarily an induced dipole that disappears on relaxation and reemission. Because of this fundamental difference in mechanism, the Raman activity of a given vibrational mode may



Tutorial: Learn more about **Raman and IR spectra** at www.tinyurl.com/skoogpia7

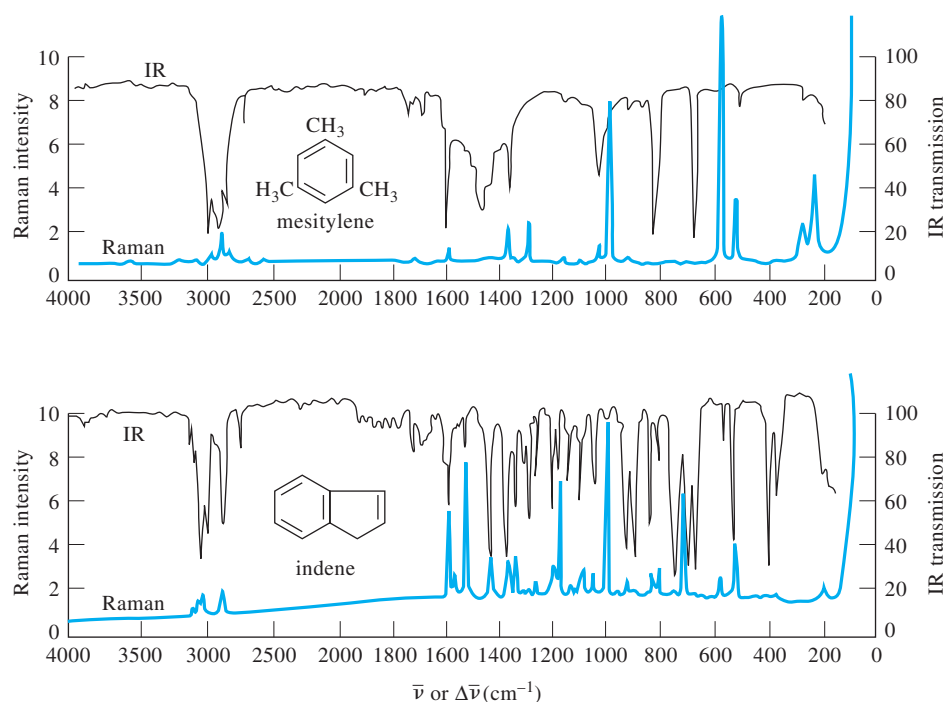


FIGURE 18-4 Comparison of Raman and IR spectra for mesitylene and indene. (Courtesy of Perkin-Elmer Corp., Norwalk, CT.)

differ markedly from its IR activity. For example, a homonuclear molecule such as nitrogen, chlorine, or hydrogen has no dipole moment either in its equilibrium position or when a stretching vibration causes a change in the distance between the two nuclei. Thus, absorption of radiation (IR) of the vibrational frequency cannot occur. On the other hand, the polarizability of the bond between the two atoms of such a molecule varies periodically in phase with the stretching vibrations, reaching a maximum at the greatest separation and a minimum at the closest approach. A Raman shift corresponding in frequency to that of the vibrational mode results.

It is of interest to compare the IR and the Raman activities of coupled vibrational modes such as those described earlier (page 396) for the carbon dioxide molecule. In the symmetric mode, no change in the dipole moment occurs as the two oxygen atoms move away from or toward the central carbon atom; thus, this mode is IR inactive. The polarizability, however, fluctuates in phase with the vibration because distortion of bonds becomes easier as they lengthen and more difficult as they shorten. Raman activity is associated with this mode.

In contrast, the dipole moment of carbon dioxide fluctuates in phase with the asymmetric vibrational mode. Thus, an IR absorption band arises from this mode. On the other hand, as the polarizability of one of the bonds increases as it lengthens, the polarizability of the other decreases, resulting in no net change in the molecular polarizability. Thus, the asymmetric stretching vibration is Raman inactive. For molecules with a center of symmetry, such as CO_2 , no IR active transitions are in

common with Raman active transitions. This is often called the *mutual exclusion principle*.

Often, as in the foregoing examples, parts of Raman and IR spectra are complementary, each being associated with a different set of vibrational modes within a molecule. For noncentrosymmetric molecules, many vibrational modes may be both Raman and IR active. For example, all of the vibrational modes of sulfur dioxide yield both Raman and IR bands. The intensities of the bands differ, however, because the probabilities for the transitions are different for the two mechanisms. Raman spectra are often simpler than IR spectra because the occurrence of overtone and combination bands is less common in Raman spectra.

18A-4 Intensity of Normal Raman Bands

The intensity or radiant power of a normal Raman band depends in a complex way on the polarizability of the molecule, the intensity of the source, and the concentration of the active group, as well as other factors. In the absence of absorption, the power of Raman emission increases with the fourth power of the frequency of the source. However, advantage can seldom be taken of this relationship because of the likelihood that ultraviolet (UV) irradiation will cause photodecomposition of the analyte or fluorescence of a sample constituent.

Raman intensities are usually directly proportional to the concentration of the active species. In this regard, Raman spectroscopy more closely resembles fluorescence than absorption with its logarithmic concentration-intensity relationship.

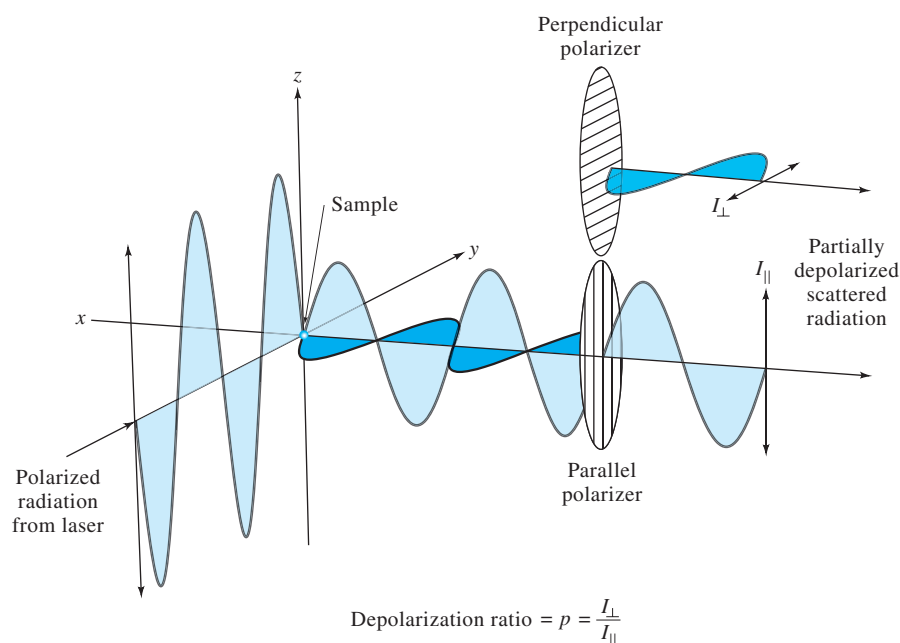


FIGURE 18-5 Depolarization resulting from Raman scattering.

18A-5 Raman Depolarization Ratios

In addition to intensity and frequency information, Raman measurements provide an additional variable that can be useful in determining the structure of molecules: the *depolarization ratio*.³ In this discussion, it is important to distinguish carefully between the terms *polarizability* and *polarization*. The former term describes a *molecular* property having to do with the deformability of a bond. Polarization, in contrast, is a property of a beam of radiation and describes the plane in which the radiation vibrates.

When Raman spectra are excited by plane-polarized radiation, as they are when a laser source is used, the scattered radiation is found to be polarized to various degrees depending on the type of vibration responsible for the scattering. The nature of this effect is illustrated in Figure 18-5, where radiation from a laser source is shown as being polarized in the yz plane. Part of the resulting scattered radiation is shown as being polarized parallel to the original beam, that is, in the xz plane; the intensity of this radiation is symbolized by the subscript $\parallel$. The remainder of the scattered beam is polarized in the xy plane, which is perpendicular to the polarization of the original beam; the intensity of this perpendicularly polarized radiation is shown by the subscript $\perp$. The depolarization ratio p is defined as

$$p = \frac{I_{\perp}}{I_{\parallel}} \quad (18-8)$$

Experimentally, the depolarization ratio may be obtained by inserting a Polaroid sheet or other polarizer between the sample

and the monochromator. Spectra are then obtained with the axis of the sheet oriented parallel with first the xz and then the xy plane shown in Figure 18-5.

The depolarization ratio depends on the symmetry of the vibrations responsible for the scattering. For example, the band for carbon tetrachloride at 459 cm^{-1} (Figure 18-2) arises from a totally symmetric “breathing” vibration involving the simultaneous movement of the four tetrahedrally arranged chlorine atoms toward and away from the central carbon atom. The depolarization ratio is 0.005, indicating minimal depolarization; the 459-cm^{-1} line is thus said to be polarized. In contrast, the carbon tetrachloride bands at 218 and 314 cm^{-1} , which arise from nonsymmetrical vibrations, have depolarization ratios of about 0.75. From scattering theory it is possible to demonstrate that the maximum depolarization for nonsymmetric vibrations is $6/7$, and for symmetric vibrations the ratio is always less than this number. The depolarization ratio is thus useful in correlating Raman lines with modes of vibration.

18B INSTRUMENTATION

Instrumentation for modern Raman spectroscopy consists of a laser source, a sample illumination system, and a suitable spectrometer as illustrated in Figure 18-6.⁴ The performance requirements for these components are more stringent than for the molecular spectrometers we have already described, however,

³D. P. Strommen, *J. Chem. Educ.*, **1992**, 69, 803, DOI: 10.1021/ed069p803.

⁴For a description of Raman instrumentation, see P. Vandenabeele, *Practical Raman Spectroscopy: An Introduction*, Chichester, UK: Wiley, 2013, Chap. 4.

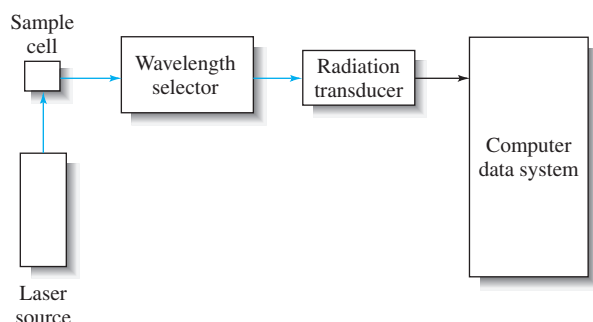


FIGURE 18-6 Block diagram of a Raman spectrometer. The laser radiation is directed into a sample cell. The Raman scattering is usually measured at right angles to avoid viewing the source radiation. A wavelength selector isolates the desired spectral region. The transducer converts the Raman signal into a proportional electrical signal that is processed by the computer data system.

because of the inherent weakness of the Raman scattering signal compared with the signal produced by the Rayleigh scattering.

18B-1 Sources

The sources used in modern Raman spectrometry are nearly always lasers because their high intensity is necessary to produce Raman scattering of sufficient intensity to be measured with a reasonable signal-to-noise ratio. Five of the most common lasers used for Raman spectroscopy are listed in Table 18-1. Because the intensity of Raman scattering varies as the fourth power of the frequency, argon and krypton ion sources that emit in the blue and green region of the spectrum have an advantage over the other sources shown in the table. For example, the argon ion line at 488 nm provides Raman lines that are nearly three times as intense as those excited by the He-Ne source, given the same input power. However, these short-wavelength sources can produce significant fluorescence and cause photodecomposition of the sample.

The last two sources in the table, which emit near-IR radiation, are finding more and more use as excitation sources. Near-IR sources have two major advantages over shorter-wavelength

lasers. The first is that they can be operated at much higher power (up to 50 W) without causing photodecomposition of the sample. The second is that they are not energetic enough to populate a significant number of fluorescence-producing excited electronic energy states in most molecules. As a result, fluorescence is generally much less intense or nonexistent with these lasers. The Nd-YAG laser, used in Fourier transform Raman (FT-Raman) spectrometers is particularly effective in eliminating fluorescence. The two lines of the diode laser at 785 and 830 nm also markedly reduce fluorescence in most cases.

Figure 18-7 illustrates an example where the Nd-YAG source completely eliminates background fluorescence. The upper curve was obtained with conventional Raman equipment using the 514.5-nm line from an argon-ion laser for excitation. The sample was anthracene, and most of the recorded signal arises from the fluorescence of that compound. The lower curve in blue is for the same sample recorded with a Fourier transform spectrometer equipped with a Nd-YAG laser that emitted at 1064 nm. Note the total absence of fluorescence background signal.

The excitation wavelength in Raman spectrometry must be carefully chosen. Not only is photodecomposition and fluorescence a problem but colored samples and some solvents can absorb the incident radiation or the Raman-scattered radiation. Thus, there is a need for more than one source or for multiple wavelength sources.

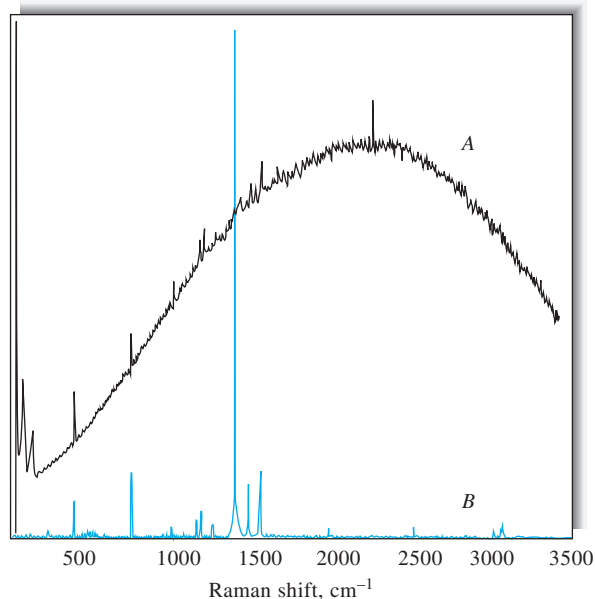


FIGURE 18-7 Spectra of anthracene taken with a conventional Raman instrument with an argon-ion laser source at 514.5 nm (A) and with an FT-Raman instrument with a Nd-YAG source at 1064 nm (B). (From B. Chase, *Anal. Chem.*, **1987**, 59, 881A, DOI: 10.1021/ac00141a714. Copyright 1987 American Chemical Society.)

TABLE 18-1 Some Common Laser Sources for Raman Spectroscopy

Laser Type	Wavelength, nm
Argon ion	488.0 or 514.5
Krypton ion	413.1, 530.9, 647.1
Helium-neon	632.8
Diode	660–880
Nd-YAG	1064

18B-2 Sample-Illumination System

Sample handling for Raman spectroscopic measurements is simpler than for IR spectroscopy because glass can be used for windows, lenses, and other optical components instead of the more fragile and atmospherically less stable crystalline halides. In addition, the laser source is easily focused on a small sample area and the emitted radiation efficiently transported to the slit or entrance aperture of a spectrometer. As a result, very small samples can be investigated. In fact, a common sample holder for nonabsorbing liquid samples is an ordinary glass-melting-point capillary.

Gas Samples

Gases are normally contained in glass tubes 1–2 cm in diameter and about 1 mm thick. Gases can also be sealed in small capillary tubes. For weak scatterers, an external multiple-pass setup with mirrors can be used as shown in Figure 18-8a. The resulting Raman scattering perpendicular to the sample tube and to the excitation laser beam is then focused on the entrance slit of the spectrometer by a large lens (L2 in the figure).

Liquid Samples

Liquids can be sealed in ampoules, glass tubes, or capillaries. Figure 18-8b and c show two of many systems for illuminating liquids. In Figure 18-8b a capillary cell is shown. Capillaries can be as small as 0.5–0.1 mm bore and 1 mm long. The spectra of nanoliter volumes of sample can be obtained with capillary cells. A large cylindrical cell, such as that illustrated in Figure 18-8c, can be used to reduce local heating, particularly for absorbing samples. The laser beam is focused to an area near the wall to minimize absorption of the incident beam. Further reduction of localized heating is often achieved by rotating the cell with an attached motor.

A major advantage of sample handling in Raman spectroscopy compared with IR arises because water is a weak Raman scatterer but a strong absorber of IR radiation. Thus, aqueous solutions can be studied by Raman spectroscopy but only with difficulty by IR. This advantage is particularly important for biological and inorganic systems and in studies dealing with water pollution.

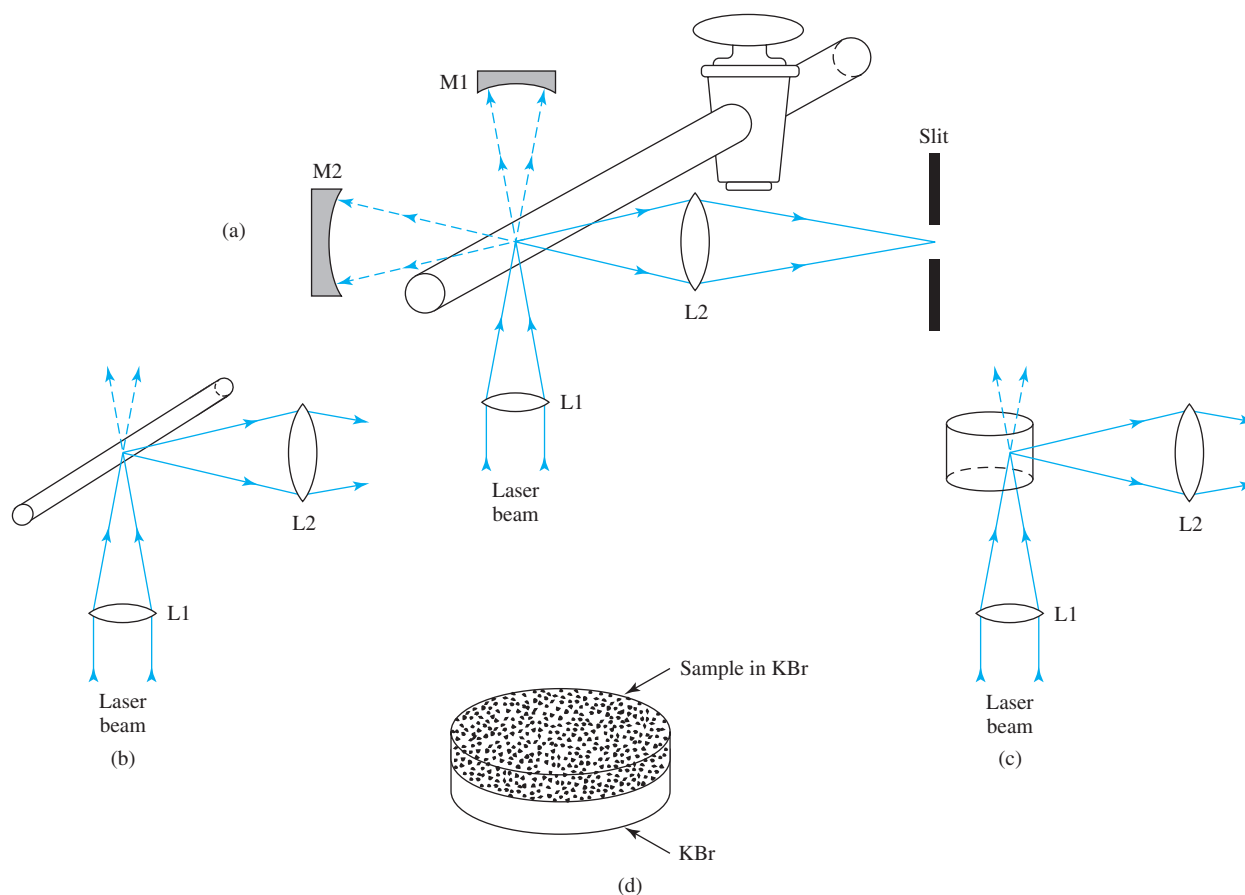


FIGURE 18-8 Sample illumination systems for Raman spectroscopy. In (a), a gas cell is shown with external mirrors for passing the laser beam through the sample multiple times. Liquid cells can be capillaries (b) or cylindrical cells (c). Solids can be determined as powders packed in capillaries or as KBr pellets (d). (Adapted from J. R. Ferraro, K. Nakamoto, and C. W. Brown, *Introductory Raman Spectroscopy*, 2nd ed., San Diego: Academic Press, 2003. Reprinted with permission.)

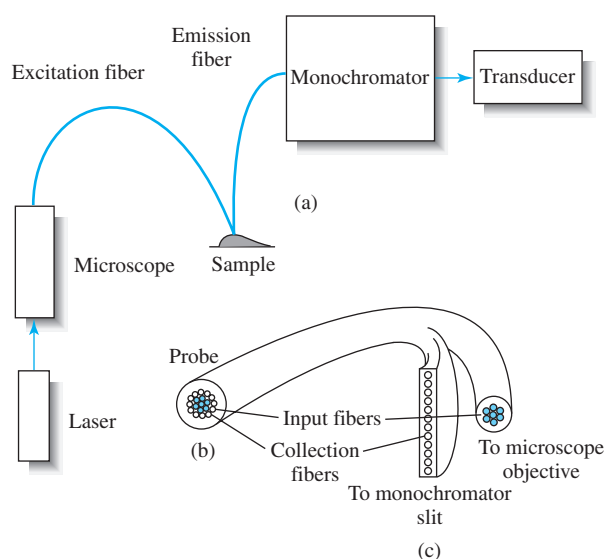


FIGURE 18-9 Raman spectrometer with fiber-optic probe. In (a) a microscope objective focuses the laser radiation onto excitation fibers that transport the beam to the sample. The Raman scattering is collected by emission fibers and carried to the entrance slit of a monochromator or to the entrance of an interferometer. A radiation transducer, such as a photomultiplier tube, converts the scattered light intensity to a proportional current or pulse rate, (b) end view of the probe, and (c) end view of collection fibers at entrance slit of monochromator. The colored circles represent the input fiber and the uncolored circles the collection fibers. (Adapted from R. L. McCreery, M. Fleischmann, and P. Hendra, *Anal. Chem.*, **1983**, 55, 146, DOI: 10.1021/ac00252a039. Copyright 1983 American Chemical Society.)

Solid Samples

Raman spectra of solid samples are often acquired by filling a small cavity or capillary with the sample after it has been ground to a fine powder. Polymers can usually be examined directly with no sample pretreatment. In some cases, KBr pellets similar to those used in IR spectroscopy are used as shown in Figure 18-8d. Dilution with KBr can reduce decomposition of the sample produced by local heating.

Fiber-Optic Sampling

One of the significant advantages of Raman spectrometry is that it is based on visible or near-IR radiation that can be transmitted for a considerable distance (as much as 100 m or more) through optical fibers. Figure 18-9 shows the arrangement of a typical Raman instrument that uses a fiber-optic probe. In this experiment, a microscope objective lens is used to focus the laser excitation beam on one end of an excitation fiber of a fiber bundle. These fibers bring the excitation radiation to the sample. Fibers can be immersed in liquid samples or used to illuminate solids. A second fiber or fiber bundle collects the Raman scattering and transports it to the entrance slit of the spectrometer. Several commercial instruments are now available with such probes.

The Raman spectrum shown in Figure 18-10 illustrates how a fiber-optic probe can be used to monitor chemical processes. In this case a fiber-optic probe was used to monitor the hanging drop crystallization of aprotinin (a serine protease inhibitor) and $(\text{NH}_4)_2\text{SO}_4$ in aqueous solution. Raman bands were attributed to both the protein and the salt. By using chemometric techniques, changes in the spectrum during crystallization were correlated with depletion of both the protein and the salt. The authors were able to determine accurately supersaturation of aprotinin using this technique.

Fiber-optic probes are proving very useful for obtaining Raman spectra in locations remote from the sample site. Examples include hostile environments, such as hazardous reactors or molten salts; biological samples, such as tissues and arterial walls; and environmental samples, such as groundwater and seawater.

Raman Microprobe

A popular accessory for Raman spectrometers is the Raman microprobe. The first developments in Raman microscopy occurred in the 1970s. Today, several instrument companies make microprobe attachments. With these, the sample is placed on the stage of a microscope where it is illuminated by visible light. After selecting the area to be viewed and adjusting the focus, the illumination lamp is turned off and the exciting laser beam is directed to the sample. With modern optics, the Raman microprobe can obtain high-quality Raman spectra without sample preparation on picogram amounts of sample with 1- μm spatial resolution.

Raman Mapping and Imaging

Raman spectroscopy can be used for mapping and imaging applications similar to those described in Section 17-G for IR spectroscopy. In mapping, a full spectrum is acquired before the sample or fiber optic probe is repositioned for another spectrum. The process is repeated until the desired two-dimensional resolution is achieved. Acquiring a multi-position map in this

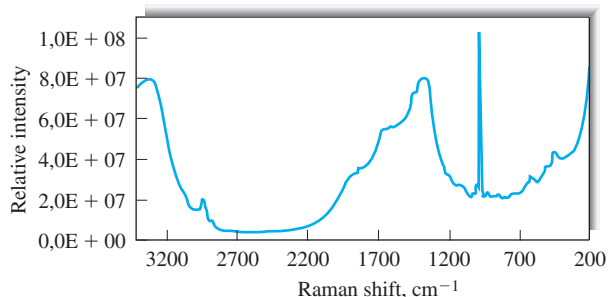


FIGURE 18-10 Raman spectrum of an aqueous solution containing aprotinin (100 mg/mL) and $(\text{NH}_4)_2\text{SO}_4$ (1.0 M) in 50 mM sodium acetate buffer at pH 4.5 and 24°C. A diode laser source at 785 nm was used with a CCD detector. (From R. E. Tamagawa, E. A. Miranda, and K. A. Berglund, *Cryst. Growth Des.*, **2002**, 2, 511, DOI: 10.1021/cg025544m. Copyright 2002 American Chemical Society.)

manner with a spectrum at each position can be very time-consuming. Fast mapping techniques have been described to speed up the process.

In imaging applications, a larger area is illuminated with a partially defocused laser beam. By adjusting the wavelength selector (filters or monochromator), the user attempts to isolate a specific Raman band. The intensity of this band is then viewed as a function of position in the resulting image. In practice, the bandwidth isolated is usually larger than that of a single Raman band and hence a range of wavelengths is imaged.

Since the introduction of Raman mapping and imaging in the 1990s, many different applications have been described.⁵ These include such diverse areas as the examination of meteorites in the search for extraterrestrial life, the identification of materials useful in forensic science, the examination of cancerous cells in human breast tissue, the identification of contaminants in pharmaceutical powders, and the characterization of such new materials as carbon nanotubes and graphene ribbons. One of the major advantages of Raman spectroscopy in biomedical applications is that molecular information can be acquired without the use of radioactive or fluorescent labels, which can introduce contaminants and perturb the sample. The valuable information provided by Raman spectroscopy ensures that we will very likely see an increasing number of these applications in the future.

18B-3 Raman Spectrometers

Until the early 1980s, Raman spectrometers were similar in design and used the same type of components as the classical ultraviolet-visible dispersing instruments described in Section 13D-3. Most spectrometers used double-grating systems to minimize the amount of stray and Rayleigh-scattered radiation reaching the transducer. Photomultipliers served as transducers. Now, however, most Raman spectrometers being marketed are either Fourier transform instruments equipped with cooled germanium transducers or multichannel instruments based on charge-coupled devices (CCDs).

Wavelength-Selection Devices and Transducers

A high-quality wavelength-selection device is required in Raman spectroscopy to separate the relatively weak Raman lines from the intense Rayleigh-scattered radiation. Traditional dispersive Raman spectrometers used double- or even triple-grating monochromators for this purpose. In recent years, holographic interference filters, called *notch filters*, and holographic gratings have improved to the extent that they have virtually eliminated the need for multiple-grating monochromators. In fact, the combination of a notch filter and a high-quality grating

monochromator is now found in most commercial dispersive instruments.

Instruments with monochromators invariably use photomultiplier tubes as transducers because of the weak signals being measured. Many spectrometers also use photon-counting systems to measure the Raman intensity. Because photon counting is inherently a digital technique, such systems are readily interfaced to modern computer data systems.

Many newer Raman instruments have replaced the single-wavelength output monochromator with a spectrograph and an array detector. The photodiode array was the first array detector to be used. It allows the simultaneous collection of entire Raman spectra. Photodiode arrays are typically used in conjunction with an image intensifier to amplify the weak Raman signal.

More recently, charge-transfer devices, such as CCDs and charge-injection devices (CIDs), have been used in Raman spectrometers. Figure 18-11 shows a fiber-optic Raman spectrometer that uses a CCD as a multichannel detector. This instrument contains high-quality bandpass and band-rejection (notch) filters to provide good stray light rejection. The CCD array can be a two-dimensional array or in some cases a linear array.

Fourier Transform Raman Spectrometers

The Fourier transform Raman (FT-Raman) instrument uses a Michelson interferometer, similar to that used in FTIR spectrometers, and a continuous-wave (CW) Nd-YAG laser as shown in Figure 18-12. The use of a 1064-nm (1.064- μm) source virtually eliminates fluorescence and photodecomposition of samples. Hence, dyes and other fluorescing compounds can be investigated with FT-Raman instruments. The FT-Raman instrument also provides superior frequency precision relative to conventional instruments, which enable spectral subtractions and high-resolution measurements.

One disadvantage of the FT-Raman spectrometer is that water absorbs in the 1000-nm region, which can cancel the Raman advantage of being able to use aqueous solutions. Also, optical filtering, as shown in Figure 18-12, is a necessity. The stray light from the exciting laser must be eliminated because it can saturate many transducers. The Rayleigh-scattered line is often six orders of magnitude greater than the Stokes-shifted Raman lines, and the intensity of this line must be minimized before striking the transducer. Holographic notch filters and other filter types are used for this purpose. Because the Raman scattering from a Nd-YAG laser can occur at wavelengths as long as 1700 nm, photomultipliers and many array detectors are not used. Most FT-Raman instruments instead use InGaAs, Ge, and other photoconductive devices as transducers. These devices are usually operated at cryogenic temperatures.

⁵See for example *Infrared and Raman Spectroscopic Imaging*, R. Salzer and H. W. Siesler, eds., 2nd ed., Weinheim, Germany: Wiley-VCH, 2014; *Infrared and Raman Spectroscopy in Forensic Science*, J. M. Chalmers, H. G. M. Edwards, and M. D. Hairgreaves, eds., Chichester, UK: Wiley, 2012.



Tutorial: Learn more about **Raman instrumentation** at www.tinyurl.com/skoogpia7

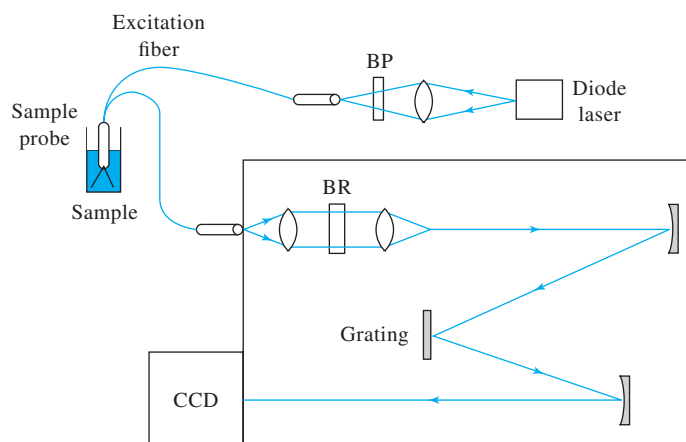


FIGURE 18-11 Fiber-optic Raman spectrometer with spectrograph and CCD detector. The bandpass filter (BP) is used to isolate a single laser line. The band-rejection filter (BR) minimizes the Rayleigh-scattered radiation.

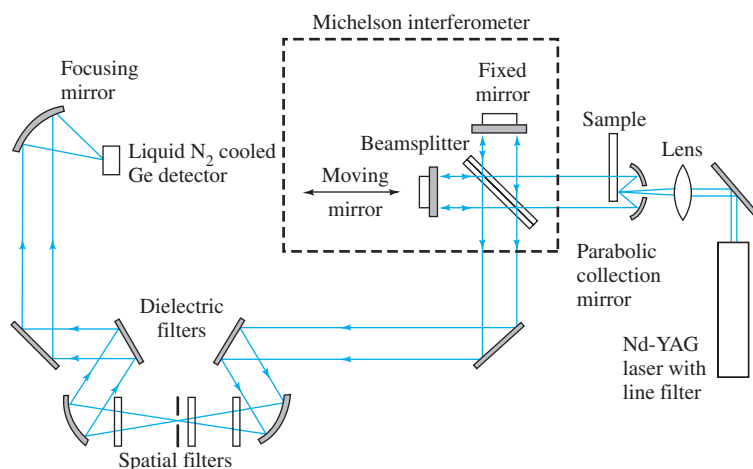


FIGURE 18-12 Optical diagram of an FT-Raman instrument. The laser radiation passes through the sample and then into the interferometer, consisting of the beamsplitter and the fixed and movable mirrors. The output of the interferometer is then extensively filtered to remove stray laser radiation and Rayleigh scattering. After passing through the filters, the radiation is focused onto a cooled Ge detector.

The FT-Raman spectrometer has a number of unique advantages for Raman spectrometry. However, the limitations previously noted mean that dispersive Raman instruments will still be used for some time.

Portable Raman Spectrometers

The availability of inexpensive diode lasers, narrow bandwidth filters, and CCD detectors has fostered the development of portable and handheld Raman instruments. Several instrument companies have introduced these devices. They are finding applications in identifying, verifying, and certifying materials for industry, homeland security, and forensics.

18C APPLICATIONS OF RAMAN SPECTROSCOPY

Raman spectroscopy has been applied to the qualitative and quantitative analysis of inorganic, organic, and biological systems.⁶

⁶See *Analytical Raman Spectroscopy*, J. G. Grasselli and B. J. Bulkin, eds., New York: Wiley, 1991.

18C-1 Raman Spectra of Inorganic Species

The Raman technique is often superior to IR spectroscopy for investigating inorganic systems because aqueous solutions can usually be used.⁷ In addition, the vibrational energies of metal-ligand bonds are generally in the range of 100 to 700 cm^{-1} , a region of the IR that is experimentally difficult to study. These vibrations are frequently Raman active, however, and lines with $\Delta\bar{\nu}$ values in this range are readily observed. Raman studies are potentially useful sources of information concerning the composition, structure, and stability of coordination compounds. For example, numerous halogen and halogenoid complexes produce Raman spectra and thus are amenable to investigation by this means. Metal-oxygen bonds are also Raman active. Spectra for such species as VO_3^{4-} , $\text{Al}(\text{OH})_4^-$, $\text{Si}(\text{OH})_6^{2-}$, and $\text{Sn}(\text{OH})_6^{2-}$ have been obtained. Raman studies have been useful in determining the probable structures of these and similar species. For example, in perchloric acid solutions, vanadium(IV) appears to be present as $\text{VO}^{2+}(\text{aq})$ rather than as $\text{V}(\text{OH})_2^{2+}(\text{aq})$. Studies of boric acid solutions show that the anion formed by acid dissociation is the tetrahedral $\text{B}(\text{OH})_4^-$ rather than H_2BO_3^- . Dissociation constants for strong acids such as H_2SO_4 , HNO_3 , H_2SeO_4 , and H_5IO_6 have been calculated from Raman measurements. It seems probable that the future will see even wider use of Raman spectroscopy for theoretical verification and structural studies of inorganic systems.

18C-2 Raman Spectra of Organic Species

Raman spectra are similar to IR spectra in that they have regions useful for functional group detection and fingerprint regions that permit the identification of specific compounds. Daimay et al. have published a comprehensive treatment of Raman functional group frequencies.⁸

Raman spectra yield more information about certain types of organic compounds than do their IR counterparts. For example, the double-bond stretching vibration for olefins results in weak and sometimes undetected IR absorption. On the other hand, the Raman band (which like the IR band, occurs at about 1600 cm^{-1}) is intense, and its position is sensitive to the nature of substituents as well as to their geometry. Thus, Raman studies are likely to yield useful information about the olefinic functional group that may not be revealed by IR spectra. This statement applies to cycloparaffin derivatives as well; these compounds have a characteristic Raman band in the region of 700 to 1200 cm^{-1} . This band has been attributed to a breathing vibration in which the nuclei move in and out symmetrically with respect to the center of the ring. The position of the band decreases continuously from 1190 cm^{-1} for cyclopropane to

700 cm^{-1} for cyclooctane. Raman spectroscopy thus appears to be an excellent diagnostic tool for the estimation of ring size in paraffins. The IR band associated with this vibration is weak or nonexistent.

18C-3 Biological and Forensic Applications of Raman Spectroscopy

Raman spectroscopy has been applied widely for the study of biological systems.⁹ The advantages of this technique include the small sample requirement, the minimal sensitivity to water, the spectral detail, and the conformational and environmental sensitivity.

Raman spectroscopy has become a useful tool in forensic science.¹⁰ Raman methods, often implemented with portable or handheld instruments, have been used in the analysis of body fluids, gunshot residues, and trace evidence. In forensic applications, Raman spectroscopy is generally considered confirmatory to results obtained by other instrumental methods.

18C-4 Quantitative Applications

Raman spectra tend to be less cluttered with bands than IR spectra. Because of this, peak overlap in mixtures is less likely, and quantitative measurements are simpler. In addition, Raman sampling devices are not subject to attack by moisture, and small amounts of water in a sample do not interfere. Despite these advantages, Raman spectroscopy has only recently been exploited widely for quantitative analysis. The increasing use of Raman spectroscopy is due to the availability of inexpensive, routine Raman instrumentation.

There are, however, some drawbacks to the use of Raman spectroscopy for quantitative analysis. First, there are matrix effects that can occur in Raman measurements. Absorption of the Raman signal by concomitants in the sample can sometimes lead to errors. The method of standard additions is often used to compensate for matrix effects. There can also be effects from sample inhomogeneity. Another drawback occurs because of instrument instability. Fluctuations in laser intensity between samples and standards or samples and samples with added analyte can influence results. Changes can also occur in the position of Raman bands due to changes in temperature or instrument conditions. Also, sample positioning in the laser beam can change measurement results. Quantitative results attained with FT-Raman instruments are often superior to those

⁷See K. Nakamoto, *Infrared and Raman Spectra of Inorganic and Coordination Compounds*, 5th ed., New York: Wiley, 1996.

⁸L. Daimay, N. B. Colthup, W. G. Fatley, and J. G. Grasselli, *The Handbook of Infrared and Raman Characteristic Frequencies of Organic Molecules*, New York: Academic Press, 1991.

⁹See J. R. Ferraro, K. Nakamoto, and C. W. Brown, *Introductory Raman Spectroscopy*, 2nd ed., San Diego: Academic Press, 2003, Chap. 6; *Infrared and Raman Spectroscopy of Biological Materials*, H. U. Gremlich and B. Yan, eds., New York: Marcel Dekker, 2001; *Biological Applications of Raman Spectroscopy*, T. G. Spiro, ed., Vols. 1–3, New York: Wiley, 1987–88.

¹⁰For a review of Raman applications in forensic science, see K. C. Doty et al., *J. Raman Spectrosc.*, **2016**, 47, 39, DOI: 10.1002/jrs.4826; see also *Infrared and Raman Spectroscopy in Forensic Science*, J. M. Chalmers, H. G. M. Edwards, and M. D. Hairgreaves, eds., Chichester, UK: Wiley, 2012.

with dispersive systems due to the higher stability of FT-Raman systems and the larger aperture of the spectrometer.

Because laser beams can be precisely focused, it becomes possible to perform quantitative analyses on very small samples. The Raman microprobe has been used to determine analytes in single bacterial cells, components in individual particles of smoke and fly ash, and species in microscopic inclusions in minerals. Surfaces have been examined by tuning the instrument to a given vibrational mode. This results in an image of regions on a surface where a particular bond or functional group is present as discussed previously.

The Raman microprobe has played a critical role in the authentication of some presumed ancient documents such as the Vinland map (see the Instrumental Analysis in Action feature at the end of Section 3). In the case of the map, the presence of TiO_2 in the ink was shown conclusively by Raman microscopy.

18D OTHER TYPES OF RAMAN SPECTROSCOPY

Advancements in tunable lasers led to several new Raman spectroscopic methods in the early 1970s. A brief discussion of the applications of some of these techniques follows.

18D-1 Resonance Raman Spectroscopy

Resonance Raman scattering refers to a phenomenon in which Raman line intensities are greatly enhanced by excitation with wavelengths that closely approach that of an electronic absorption band of an analyte.¹¹ Under this circumstance, the magnitudes of Raman lines associated with the most symmetric vibrations are enhanced by a factor of 10^2 to 10^6 . As a result, resonance Raman spectra have been obtained at analyte concentrations as low as 10^{-8} M. This level of sensitivity is in contrast to normal Raman studies, which are ordinarily limited to concentrations greater than 0.1%. Furthermore, because resonance enhancement is restricted to the Raman bands associated with the chromophore, resonance Raman spectra are usually quite selective.

Figure 18-13a illustrates the energy changes responsible for resonance Raman scattering. This figure differs from the energy diagram for normal Raman scattering (Figure 18-3) in that the electron is promoted into an excited electronic state followed by an immediate relaxation to a vibrational level of the electronic ground state. As shown in the figure, resonance Raman scattering differs from fluorescence (Figure 18-13b) in that relaxation to the ground state is not preceded by prior relaxation to the lowest vibrational level of the excited electronic state. The

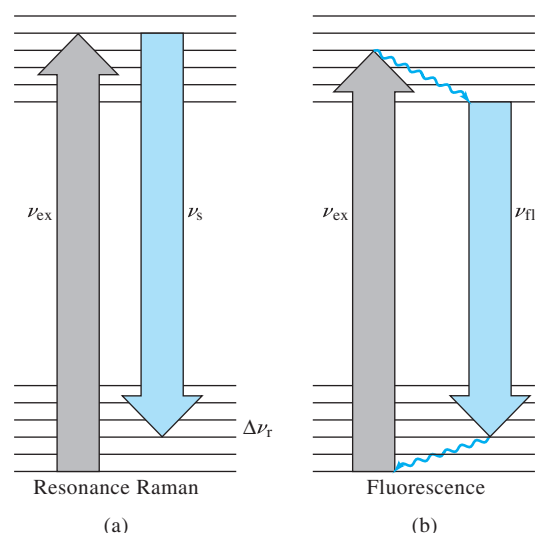


FIGURE 18-13 Energy diagram for (a) resonance Raman scattering and (b) fluorescence emission. Radiationless relaxation is shown as wavy arrows. In the resonance Raman case, the excited electron immediately relaxes into a vibrational level of the ground electronic state giving up a Stokes photon ν_s . In fluorescence, relaxation to the lowest vibrational level of the excited electronic state occurs prior to emission. Resonance Raman scattering is nearly instantaneous, and the spectral bands are very narrow. Fluorescence emission usually takes place on the nanosecond time scale. Fluorescence spectra are usually broad because of the many vibrational states.

time scales for the two phenomena are also quite different, with Raman relaxation occurring in less than 10^{-14} s compared with the 10^{-6} to 10^{-10} s for fluorescence emission.

Line intensities in a resonance Raman experiment increase rapidly as the excitation wavelength approaches the wavelength of the electronic absorption band. Thus, to achieve the greatest signal enhancement for a broad range of absorption maxima, a tunable laser is required. With intense laser radiation, sample decomposition can become a major problem because electronic absorption bands often occur in the UV region. To circumvent this problem, it is common practice to circulate the sample past the focused beam of the laser. Circulation is normally accomplished in one of two ways: by pumping a solution or liquid through a capillary mounted in the sample position or by rotating a cylindrical cell containing the sample through the laser beam. Thus, only a small fraction of the sample is irradiated at any instant, and heating and sample decomposition are minimized.

Perhaps the most important application of resonance Raman spectroscopy has been to the study of biological molecules under physiologically significant conditions; that is, in the presence of water and at low to moderate concentration levels. As an example, the technique has been used to determine the oxidation state and spin of iron atoms in hemoglobin and cytochrome *c*. In these molecules, the resonance Raman bands are

¹¹For brief reviews, see T. G. Spiro and R. S. Czernuszewicz in *Physical Methods in Bioinorganic Chemistry*, L. Que, ed., Sausalito, CA: University Science Books, 2000; S. A. Asher, *Anal. Chem.*, **1993**, 65, 59A, DOI: 10.1021/ac00050a001.

due solely to vibrational modes of the tetrapyrrole chromophore. None of the other bands associated with the protein is enhanced, and at the concentrations normally used, these bands do not interfere.

Time-resolved resonance Raman spectrometry is a technique that allows collection of Raman spectra of excited state molecules. It has been used to study intermediates in enzyme reactions, the spectra of carotenoid excited states, ultrafast electron transfer steps, and a variety of other biological and bioinorganic processes.¹² Time-discrimination methods have been used to overcome a major limitation of resonance Raman spectroscopy, namely, fluorescence interference either by the analyte itself or by other species present in the sample.

18D-2 Surface-Enhanced Raman Spectroscopy

In surface-enhanced Raman spectroscopy (SERS),¹³ Raman spectra are acquired in the usual way on samples that are adsorbed on the surface of colloidal metal particles (usually silver, gold, or copper) or on roughened surfaces of pieces of these metals. For reasons that are finally becoming understood, at least semiquantitatively, the Raman lines of the adsorbed molecule are often enhanced by a factor of 10^3 to 10^6 .

Surface enhancement is thought to arise from two factors. First, there is an enhancement due to the electromagnetic field. This occurs because the incident electromagnetic wave interacts with the metal surface to excite localized surface plasmons, which amplifies the field near the surface. Because Raman scattering scales as the fourth power of the field, this effect can enhance signals by as much as a factor of 10,000. The second factor is a chemical enhancement that occurs because of the adsorbed molecule interacting with the surface. This effect can enhance the Raman signal by as much as 100 times. The net enhancement is the product of the two effects, which can be approximately 1 million.¹⁴ When surface enhancement is combined with the resonance enhancement technique discussed in the previous section, the net

increase in signal intensity is roughly the product of the intensity produced by each of the techniques. Consequently, detection limits in the range of 10^{-9} to 10^{-12} M have been observed.

Several sample-handling techniques are used for SERS. In one technique, colloidal silver or gold particles are suspended in a dilute solution (usually aqueous) of the sample. The solution is then held or flowed through a narrow glass tube while it is excited by a laser beam. In another method, a thin film of colloidal metal particles is deposited on a glass slide and a drop or two of the sample solution spotted on the film. The Raman spectrum is then obtained in the usual manner. Alternatively, the sample may be deposited electrolytically on a roughened metal electrode, which is then removed from the solution and exposed to the laser excitation source.

18D-3 Nonlinear Raman Spectroscopy

In Section 7B-3, we pointed out that many lasers are intense enough to produce significant amounts of nonlinear radiation. Throughout the 1970s and 1980s, many Raman techniques were developed that depend on polarization induced by second-order and higher field strengths. These techniques are termed *nonlinear Raman methods*.¹⁵ Included in these methods are *stimulated Raman scattering*, *the hyper-Raman effect*, *stimulated Raman gain*, *inverse Raman spectroscopy*, *coherent anti-Stokes Raman spectroscopy*, and *coherent Stokes Raman spectroscopy*. The most widely used of these methods is coherent anti-Stokes Raman spectroscopy, or CARS.

Nonlinear techniques have been used to overcome some of the drawbacks of conventional Raman spectroscopy, particularly its low efficiency, its limitation to the visible and near-UV regions, and its susceptibility to interference from fluorescence. A major disadvantage of nonlinear methods is that they tend to be analyte specific and often require several different tunable lasers to be applicable to diverse species. To date, none of the nonlinear methods has found widespread application among nonspecialists. However, many of these methods have shown considerable promise. As less expensive and more routinely useful lasers become available, nonlinear Raman methods, particularly CARS, should become more widely used.

¹²J. R. Kincaid and K. Czarniecki, in *Comprehensive Coordination Chemistry II*, J. A. McCleverty and T. J. Meyer, eds., Oxford: Elsevier, 2004.

¹³See M. J. Weaver, S. Zou, and H. Y. Chan, *Anal. Chem.*, **2000**, *72*, 38A, DOI: 10.1021/ac0027136. American Chemical Society.

¹⁴See, P. L. Stiles, J. A. Dieringer, N. C. Shah, and R. P. Van Duyne, *Ann. Rev. Anal. Chem.*, **2008**, *1*, 601, DOI: 10.1146/annurev.anchem.1.031207.112814.

¹⁵See J. R. Ferraro, K. Nakamoto, and C. W. Brown, *Introductory Raman Spectroscopy*, 2nd ed., San Diego: Academic Press, 2003, pp. 194–202.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.

 Problems with this icon are best solved using spreadsheets.

18-1 What is a virtual state?

18-2 Why does the ratio of anti-Stokes to Stokes intensities increase with sample temperature?



* **18-3** An antihistamine shows sharp peaks at Raman shifts of $\Delta\bar{\nu} = 488, 725, 875, 925,$ and 1350 cm^{-1} . At what wavelengths in nanometers would the Stokes and anti-Stokes lines for the antihistamine appear if the source were

(a) a helium-neon laser (632.8 nm)?

(b) an argon-ion laser (488.0 nm)?

* **18-4** Assume the excitation sources in Problem 18-3 have the same power. (a) Compare the relative intensities of the Raman lines of the antihistamine for each of the two excitation sources. (b) If the intensities were recorded with a typical monochromator-photomultiplier system, why would the measured intensity ratios differ from the ratio calculated in part (a)?

* **18-5** For vibrational states, the Boltzmann equation can be written as

$$\frac{N_1}{N_0} = \exp(-\Delta E/kT)$$

where N_0 and N_1 are the populations of the lower and higher energy states, respectively, ΔE is the energy difference between the states, k is Boltzmann's constant, and T is the temperature in kelvins.

For temperatures of 20°C and 40°C , calculate the ratios of the intensities of the anti-Stokes and Stokes lines for CCl_4 at (a) 218 cm^{-1} , (b) 459 cm^{-1} , (c) 790 cm^{-1} .

For each temperature and Raman shift, calculate how much more intense the Stokes line is compared to the anti-Stokes line.

18-6 The following questions deal with laser sources in Raman spectroscopy.

(a) Under what circumstances would a helium-neon laser be preferable to an argon-ion laser?

(b) Under what circumstances would a diode laser be preferable to an argon-ion or helium-neon laser?

(c) Why are UV emitting sources avoided?

* **18-7** The following Raman data were obtained for CHCl_3 with the polarizer of the spectrometer set (1) parallel to the plane of polarization of the laser and (2) at 90° to the plane of the source.

	Relative Intensities		
	$\Delta\bar{\nu}, \text{cm}^{-1}$	(1) $I_{\parallel}$	(2) $I_{\perp}$
(a)	760	0.60	0.46
(b)	660	8.4	0.1
(c)	357	7.9	0.6
(d)	258	4.2	3.2

Calculate the depolarization ratio and indicate which Raman lines are polarized.

18-8 Discuss the advantages and disadvantages of FT-Raman spectrometers compared to conventional dispersive Raman instruments.

>> QUESTIONS AND PROBLEMS (continued)

Challenge Problem

18-9 The following questions all deal with the similarities and differences between IR spectrometry and Raman spectrometry.

- (a) What are the requirements for a vibrational mode in a molecule to show IR absorption? What are the requirements for a vibrational mode to be Raman active? Why do these requirements differ? Under what circumstances will vibrational modes be both Raman and IR active? Under what circumstances will vibrational modes be Raman active but not IR active and vice versa?
- (b) Consider the molecule chloroacetonitrile (ClCH_2CN). How many vibrational modes should this molecule have? Why might one observe fewer Raman bands than expected?
- (c) Chloroacetonitrile shows a strong Raman band at 2200 cm^{-1} due to the $\text{C}\text{--}\text{N}$ stretching mode. The corresponding IR absorption is very weak or absent. By comparing spectra in the 2200 cm^{-1} region, what can you conclude about the $\text{C}\text{--}\text{N}$ stretching mode in chloroacetonitrile?
- (d) Compare and contrast IR and Raman spectrometry with respect to optics, cell materials, sample handling, solvent compatibility, and applicability to various sample types.
- (e) Compare and contrast the sources and transducers used in Raman spectrometers to those used in FTIR instruments. Consider both FT-Raman and dispersive Raman spectrometers in your comparison.
- (f) Compare and contrast IR and Raman spectrometry with respect to qualitative usefulness, detection limits, quantitative analysis, and instrumental complexity.

Molecular Mass Spectrometry

Mass spectrometry is perhaps the most widely applicable of all the analytical tools available because the technique can provide information about (1) the elemental composition of samples of matter; (2) the structures of inorganic, organic, and biological molecules; (3) the qualitative and quantitative composition of complex mixtures; (4) the structure and composition of solid surfaces; and (5) isotopic ratios of atoms in samples.

We have already discussed in Chapter 11 how mass spectrometry is used by chemists for the identification and quantitative determination of one or more elements in a sample of matter. This chapter describes how mass spectrometry is used to collect the type of information listed in items (2) and (3) in the previous paragraph. Chapter 21 describes how mass spectrometry is used for elucidating the structure and composition of surfaces. Finally, in Section 20D, we discuss the use of isotopic ratios determined by mass spectrometry.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

The first general application of molecular mass spectrometry for routine chemical analysis occurred in the early 1940s, when the techniques began to be adopted by the petroleum industry for the quantitative analysis of hydrocarbon mixtures produced in catalytic cracking units. Before this time, analyses of mixtures of this type, which often contained as many as nine hydrocarbon components, were carried out by fractional distillation followed by refractive-index measurements of the separated components. Typically, 200 hours or more of operator time was required to complete an analysis. It was found that similar information could be obtained in a few hours or less with a mass spectrometer. This improved efficiency led to the appearance and rapid improvement of commercial mass spectrometers. Beginning in the 1950s, these commercial instruments began to be adapted by chemists for the identification and structural elucidation of a wide variety of organic compounds. This use of the mass spectrometer combined with the invention of nuclear magnetic resonance and the development of infrared spectrometry revolutionized the way organic chemists identify and determine the structure of molecules. This application of mass spectrometry is still extremely important.

Applications of molecular mass spectrometry dramatically changed in the decade of the 1980s, as a result of the development of new methods for producing ions from nonvolatile or thermally labile molecules, such as those frequently encountered by biochemists and biologists. Since about 1990, explosive growth in the area of biological mass spectrometry has occurred because of these new ionization methods. Now, mass spectrometry is being applied to the determination of the structure of polypeptides, proteins, and other high-molecular-mass biopolymers.

In this chapter, we first describe the nature of molecular mass spectra and define some terms used in molecular mass spectrometry. We next consider the various techniques used to form ions from analyte molecules in mass spectrometers and the types of spectra produced by these techniques. We then describe in some detail the various types of mass spectrometers used in molecular mass spectrometry (other than the quadrupole and time-of-flight instruments, which received detailed treatment

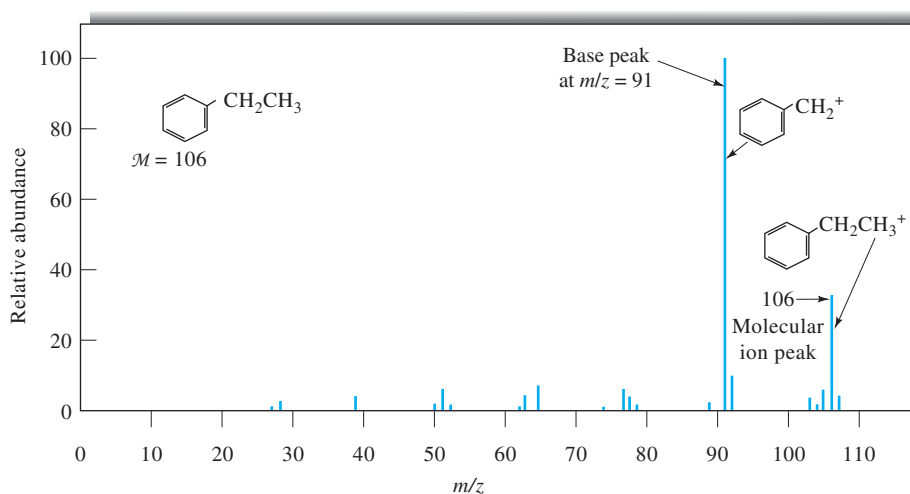
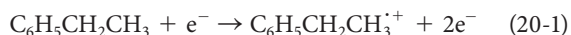


FIGURE 20-1 Mass spectrum of ethyl benzene.

in Section 11B). Finally, we describe several of the current applications of molecular mass spectrometry.¹

20A MOLECULAR MASS SPECTRA

Figure 20-1 illustrates how mass spectral data are usually presented. The analyte was ethyl benzene, which has a nominal molecular mass of 106 daltons (Da). To obtain this spectrum, ethyl benzene vapor was bombarded with a stream of electrons that led to the loss of an electron by the analyte and formation of the ion $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3^+$ as shown by the reaction



The charged species $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_3^+$ is the *molecular ion*. As indicated by the dot, the molecular ion is a radical ion that has the same molecular mass as the molecule.

The collision between energetic electrons and analyte molecules usually imparts enough energy to the molecules to leave them in an excited state. Relaxation then often occurs by fragmentation of part of the molecular ions to produce ions of lower masses. For example, a major product in the case of ethyl benzene is $\text{C}_6\text{H}_5\text{CH}_2^+$, which results from the loss of a CH_3 group. Other smaller positively charged fragments are also formed in lesser amounts.

The positive ions produced in electron ionization (EI) are attracted through the slit of a mass spectrometer where they are

sorted according to their mass-to-charge ratios and displayed in the form of a mass spectrum. Note that the plot shown in Figure 20-1 is in the form of a bar graph that relates the relative intensity of mass peaks to their mass-to-charge ratio. The largest peak in a spectrum, termed the *base peak*, is arbitrarily assigned a value of 100. The heights of the remaining peaks are then computed as a percentage of the base-peak height. Modern mass spectrometers are programmed to automatically recognize the base peak. They then normalize the remaining peaks in the spectrum relative to the base peak.

20B ION SOURCES

The starting point for a mass spectrometric analysis is the formation of gaseous analyte ions, and the scope and the utility of a mass spectrometric method is dictated by the ionization process. The appearance of mass spectra for a given molecular species strongly depends on the method used for ion formation. Table 20-1 lists many of the ion sources that have been used in molecular mass spectrometry.² Note that these methods fall into three major categories: *gas-phase sources*, *desorption sources*, and *ambient desorption sources*. With a gas-phase source, which includes the first three sources in the table, the sample is first vaporized and then ionized. With a desorption source, however, the solid- or liquid-state sample is converted directly into gaseous ions. An advantage of desorption sources is that they

¹For detailed discussions of mass spectrometry, see J. H. Gross, *Mass Spectrometry: A Textbook*, Heidelberg, Germany: Springer, 2011; J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., Chichester, UK: Wiley, 2007; R. M. Smith, *Understanding Mass Spectra: A Basic Approach*, 2nd ed., New York: Wiley, 2004; E. de Hoffman and V. Stroobant, *Mass Spectrometry: Principles and Applications*, 2nd ed., Chichester, UK: Wiley, 2002.

²For more information about ion sources, see J. Greaves and J. Roboz, *Mass Spectrometry for the Novice*, Boca Raton, FL: CRC Press, 2014; J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., Chichester, UK: Wiley, 2007; E. de Hoffman and V. Stroobant, *Mass Spectrometry: Principles and Applications*, 2nd ed., Chichester, UK: Wiley, 2002; for an historical perspective, see F. W. McLafferty, *Ann. Rev. Anal. Chem.*, **2011**, 4, 1, DOI: 10.1146/annurev-anchem-061010-114018.

TABLE 20-1 Ion Sources for Molecular Mass Spectrometry

Basic Type	Name and Acronym	Ionizing Agent
Gas phase	Electron ionization (EI)	Energetic electrons
	Chemical ionization (CI)	Reagent gaseous ions
	Field ionization (FI)	High-potential electrode
Desorption	Field desorption (FD)	High-potential electrode
	Electrospray ionization (ESI)	High electrical field
	Matrix-assisted laser desorption/ionization (MALDI)	Laser beam
	Plasma desorption (PD)	Fission fragments from ^{252}Cf
	Fast atom bombardment (FAB)	Energetic atomic beam
	Secondary-ion mass spectrometry (SIMS)	Energetic beam of ions
	Thermospray ionization (TS)	High temperature
Ambient desorption	Desorption electrospray ionization (DESI)	Charged droplet spray
	Direct analysis in real time (DART)	Excited atoms or molecules

are applicable to nonvolatile and thermally unstable samples. Ambient sources allow desorption ionization with minimal sample pretreatment and without the enclosures of typical ionization sources.³ Currently, commercial mass spectrometers are equipped with accessories that permit interchangeable use of several ionization sources.

Gas-phase sources are generally restricted to the ionization of thermally stable compounds that have boiling points less than about 500°C. In most cases, this requirement limits gaseous sources to compounds with molecular masses less than roughly 10³ Da. Desorption and ambient sources, which do not require volatilization of analyte molecules, are applicable to analytes having molecular masses as large as 10⁵ Da.

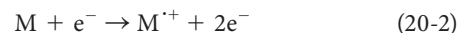
Ion sources are also classified as being *hard sources* or *soft sources*. Hard ionization sources impart enough energy to analyte molecules to leave them in a highly excited energy state. Relaxation then involves rupture of bonds, producing fragment ions that have mass-to-charge ratios less than that of the molecular ion. Soft ionization sources cause little fragmentation. Thus, the mass spectrum from a soft ionization source often consists of the molecular ion peak and only a few, if any, other peaks. Figure 20-2 illustrates the difference in spectra obtained from a hard ionization source and a soft ionization source.

Both hard- and soft-source spectra are useful for analysis. The many peaks in a hard-source spectrum provide useful information about the kinds of functional groups and thus structural information about analytes. Soft-source spectra are useful because they supply accurate information about the molecular mass of the analyte molecule or molecules.

20B-1 Electron Ionization

Historically, ions for mass analysis were produced by *electron ionization*, formerly called electron-impact ionization.⁴ In this process, the sample is brought to a temperature high enough to produce a molecular vapor, which is then ionized by bombarding the resulting molecules with a beam of energetic electrons. Despite certain disadvantages, this technique is still of major importance and is the one on which many libraries of mass spectral data are based.

Figure 20-3 is a schematic of a basic EI source. Electrons are emitted from a heated tungsten or rhenium filament and accelerated by applying approximately 70 V between the filament and the anode. As shown in the figure, the paths of the electrons and molecules are at right angles and intersect near the center of the source, where collision and ionization occur. The primary product is singly charged positive ions formed when the energetic electrons approach molecules closely enough to cause them to lose electrons by electrostatic repulsion. Electron ionization is not very efficient, and only about one molecule in a million undergoes the primary reaction



Here, M represents the analyte molecule, and M⁺ is its molecular ion. The positive ions produced by electron ionization are attracted through the slit in the first accelerating plate by a small potential difference (typically 5 V) that is applied between this plate and the repellers shown in Figure 20-3. With magnetic sector instruments, high voltages (10³ to 10⁴ V) are applied to the accelerator plates, which give the ions their final velocities before

³See G. A. Harris, A. S. Galhena, and F. M. Fernandez, *Anal. Chem.*, **2011**, 83, 4508, DOI: 10.1021/ac200918u.

⁴The term *electron impact* is no longer in favor because it gives an erroneous idea of the processes that occur in this method of ionization.

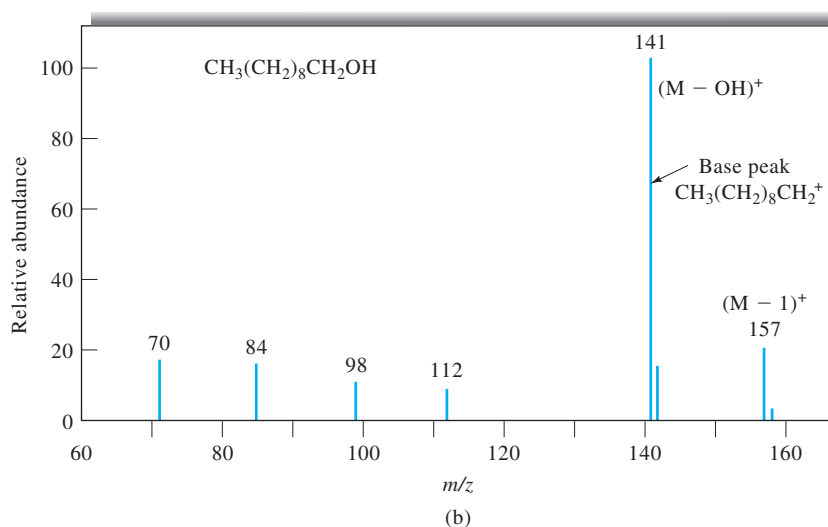
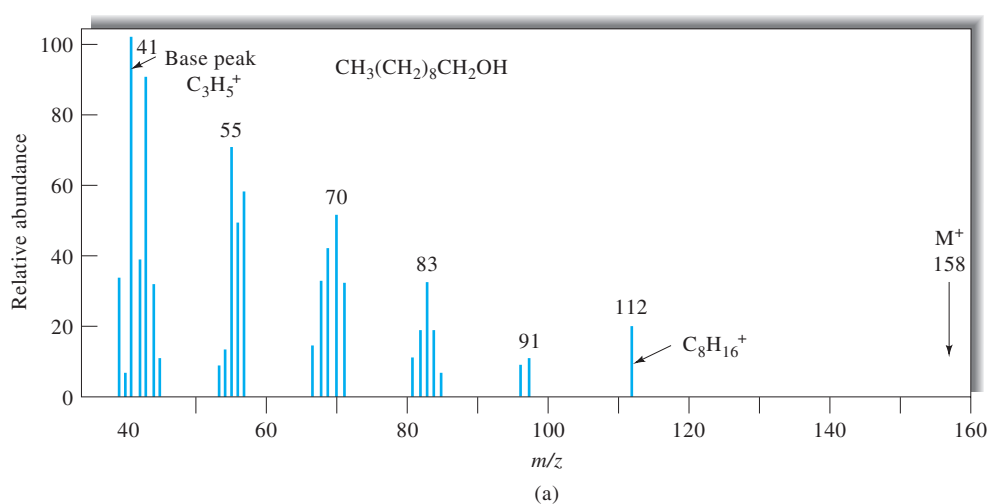


FIGURE 20-2 Mass spectrum of 1-decanol from (a) a hard ionization source (EI) and (b) a soft ionization source (chemical ionization [CI]).

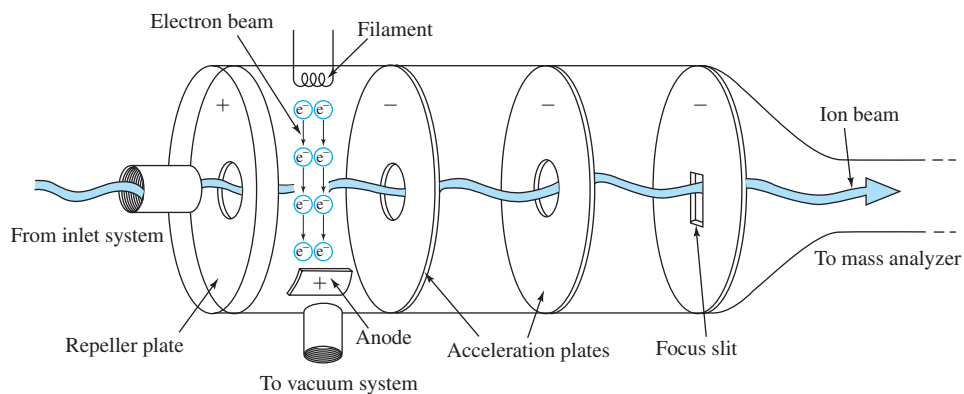


FIGURE 20-3 Diagram of an EI source. (Adapted from D. L. Pavia, G. M. Lampman, G. S. Kriz, and J. R. Vyvyanh, *Introduction to Spectroscopy*, 5th ed., Stamford, CT: Cengage Learning, 2015, p. 110.)

they enter the mass analyzer. Commercial EI sources are more complex than that shown in Figure 20-3 and may use additional electrostatic or magnetic fields to manipulate the electron or ion beam. A typical kinetic energy produced in an EI source is calculated in Example 20-1.

EXAMPLE 20-1

- (a) Calculate the kinetic energy that a singly charged ion ($z = 1$) will acquire if it is accelerated through a potential of 10^3 V in an EI source. (b) Does the kinetic energy of the ion depend on its mass? (c) Does the velocity of the ion depend on its mass?

Solution

- (a) The kinetic energy (KE) added to the ion is due to the accelerating potential V and is given by the equation

$$KE = qV = zeV$$

where e is the electronic charge (1.6×10^{-19} coulombs). Thus, for $z = 1$

$$KE = 1 \times 1.6 \times 10^{-19} \text{ C} \times 10^3 \text{ V} = 1.6 \times 10^{-16} \text{ J}$$

- (b) The kinetic energy that an ion acquires in the source is independent of its mass and depends only on its charge and the accelerating potential.
- (c) The translational component of the kinetic energy of an ion is a function of the ion mass m and its velocity v as given by the equation

$$KE = (1/2)mv^2 \quad \text{or} \quad v = (2 KE/m)^{1/2}$$

Thus, if all ions acquire the same amount of kinetic energy, those ions with largest mass must have the smallest velocity.

EI Spectra

To form a significant number of gaseous ions at a reproducible rate, electrons from the filament in the source must be

accelerated by a voltage of greater than about 50 V. The low mass and high kinetic energy of the resulting electrons cause little increase in the translational energy of impacted molecules. Instead, the molecules are left in highly excited vibrational and rotational states. Subsequent relaxation then usually takes place by extensive fragmentation, giving a large number of positive ions of various masses that are less than (and occasionally, because of collisions, greater than) that of the molecular ion. These lower-mass ions are called *product* or *fragment ions*. Table 20-2 shows some typical fragmentation reactions that follow formation of a *molecular ion* from a hypothetical molecule ABCD. In Example 20-2, the kinetic energy of electrons accelerated through a potential difference of 70 V is calculated.

EXAMPLE 20-2

- (a) Calculate the energy (in J/mol) that electrons acquire as a result of being accelerated through a potential of 70 V. (b) How does this energy compare to that of a typical chemical bond?

Solution

- (a) The kinetic energy KE of an individual electron is equal to the product of the charge on the electron e times the potential V through which it has been accelerated. Multiplying the kinetic energy of a single electron by Avogadro's number, N , gives the energy per mole:

$$\begin{aligned} KE &= eVN \\ &= (1.60 \times 10^{-19} \text{ C/e}^-)(70 \text{ V})N \\ &= (1.12 \times 10^{-17} \text{ CV/e}^-)(6.02 \times 10^{23} \text{ e}^-/\text{mol}) \\ &= 6.7 \times 10^6 \text{ J/mol} \quad \text{or} \quad 6.7 \times 10^3 \text{ kJ/mol} \end{aligned}$$

- (b) Typical bond energies fall in the 200- to 600-kJ/mol range. Therefore, an electron that has been accelerated through 70 V generally has considerably more energy than that required to break a chemical bond.

TABLE 20-2 Some Typical Reactions in an EI Source

Molecular ion formation	$ABCD + e^- \rightarrow ABCD^{\bullet+} + 2e^-$
Fragmentation	$ABCD^{\bullet+} \rightarrow A^+ + BCD^{\bullet}$ $\quad \quad \quad \rightarrow A^{\bullet} + BCD^+ \rightarrow BC^+ + D$ $\quad \quad \quad \rightarrow CD^{\bullet} + AB^+ \rightarrow B + A^+$ $\quad \quad \quad \rightarrow AB^{\bullet} + CD^+ \rightarrow D + C^+$ $\quad \quad \quad \quad \quad \rightarrow C + D^+$
Rearrangement followed by fragmentation	$ABCD^{\bullet+} \rightarrow ADBC^{\bullet+} \rightarrow BC^{\bullet} + AD^+$ $\quad \quad \quad \quad \quad \rightarrow AD^{\bullet} + BC^+$
Collision followed by fragmentation	$ABCD^{\bullet+} + ABCD \rightarrow (ABCD)_2^{\bullet+} \rightarrow BCD^{\bullet} + ABCDA^+$

The complex mass spectra that result from electron ionization are useful for compound identification. On the other hand, with certain types of molecules, fragmentation is so complete that the molecular ion is not detected. Without a molecular ion, important information for determining the molecular mass of the analyte is lost. Figure 20-4 shows typical EI spectra for two simple organic molecules: methylene chloride and 1-pentanol. Note that in each of the spectra, the base peak corresponds to a fragment of the molecule, which has a mass significantly less than the molecular mass of the original compound. For methylene chloride, the base peak occurs at a mass-to-charge ratio m/z of 49, which corresponds to the loss of one Cl atom. For 1-pentanol, the base peak is found at an m/z of 44, which is that of the product ion CH_2CHOH^+ . Quite frequently, the base peaks in EI spectra correspond to fragments and not to the molecular ion.

The molecular ion peak occurs at a mass corresponding to the molecular mass of the analyte. Thus, molecular ion peaks appear at $m/z = 84$ for methylene chloride and at $m/z = 88$ for 1-pentanol. The molecular ion peak is, of course, very important in structural determinations, because its m/z value provides the

molecular mass of the compound. Unfortunately, it is not always possible to identify the molecular ion peak. Indeed, electron ionization of certain molecules yields no molecular ion peak at all (see Figure 20-2a).

Isotope Peaks

It is interesting to note in the spectra shown in Figures 20-1 and 20-4 that peaks occur at mass-to-charge ratios greater than that of the molecular ion. These peaks are attributable to ions having the same chemical formula, but different isotopic compositions. For example, for methylene chloride, the more important isotopic species are $^{12}\text{C}^1\text{H}_2^{35}\text{Cl}_2$ ($\mathcal{M} = 84$), $^{13}\text{C}^1\text{H}_2^{35}\text{Cl}_2$ ($\mathcal{M} = 85$), $^{12}\text{C}^1\text{H}_2^{35}\text{Cl}^{37}\text{Cl}$ ($\mathcal{M} = 86$), $^{13}\text{C}^1\text{H}_2^{35}\text{Cl}^{37}\text{Cl}$ ($\mathcal{M} = 87$), and $^{12}\text{C}^1\text{H}_2^{37}\text{Cl}_2$ ($\mathcal{M} = 88$), where $\mathcal{M}$ is molecular mass. Peaks for each of these species can be seen in Figure 20-4a. The size of the various peaks depends on the relative



Tutorial: Learn more about sources for MS at www.tinyurl.com/skoogpia7

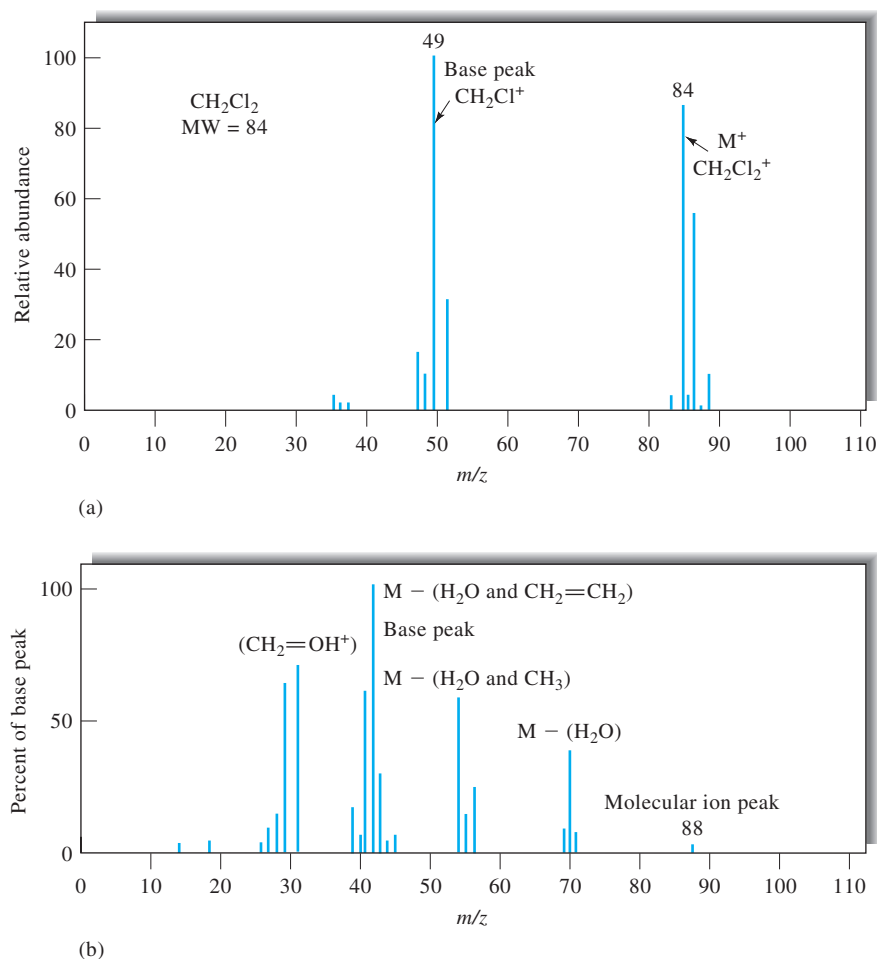


FIGURE 20-4 EI mass spectra of (a) methylene chloride and (b) 1-pentanol.

TABLE 20-3 Natural Abundance of Isotopes of Some Common Elements

Element ^a	Most Abundant Isotope	Abundance of Other Isotopes Relative to 100 Parts of the Most Abundant ^b	
Hydrogen	¹ H	² H	0.015
Carbon	¹² C	¹³ C	1.08
Nitrogen	¹⁴ N	¹⁵ N	0.37
Oxygen	¹⁶ O	¹⁷ O	0.04
		¹⁸ O	0.20
Sulfur	³² S	³³ S	0.80
		³⁴ S	4.40
Chlorine	³⁵ Cl	³⁷ Cl	32.5
Bromine	⁷⁹ Br	⁸¹ Br	98.0
Silicon	²⁸ Si	²⁹ Si	5.1
		³⁰ Si	3.4

^aFluorine (¹⁹F), phosphorus (³¹P), sodium (²³Na), and iodine (¹²⁷I) have no additional naturally occurring isotopes.

^bThe numerical entries indicate the average number of isotopic atoms present for each 100 atoms of the most abundant isotope; thus, for every 100 ¹²C atoms there will be an average of 1.08 ¹³C atoms.

natural abundance of the isotopes. Table 20-3 lists the most common isotopes for atoms that occur widely in organic compounds. Note that fluorine, phosphorus, iodine, and sodium occur only as single isotopes.

The small peak for ethyl benzene at m/z 107 in Figure 20-1 is due to the presence of ¹³C in the molecules. The intensities of peaks due to incorporation of two or more ¹³C atoms in ethyl benzene can be predicted with good precision but are normally so small as to be undetectable because of the low probability of there being more than one ¹³C atom in a small molecule. As will be shown in Section 20D-1, isotope peaks sometimes provide a useful means for determining the formula for a compound.

Collision Product Peaks

Ion-molecule collisions, such as that shown by the last reaction in Table 20-2, can produce peaks at higher mass numbers than that of the molecular ion. At ordinary sample pressures, however, the only important reaction of this type is one in which the collision transfers a hydrogen atom to the ion to give a protonated molecular ion. The result is an enhanced $(M + 1)^+$ peak. This transfer is a second-order reaction, and the amount of product depends strongly on the reactant concentration. Consequently, the height of an $(M + 1)^+$ peak due to this reaction increases much more rapidly with increases in sample pressure than do the heights of other peaks. This phenomenon usually makes it possible to detect this atom-transfer reaction.

Advantages and Disadvantages of EI Sources

EI sources are convenient to use. They produce high ion currents, which lead to good sensitivities. The extensive fragmentation and resulting large number of peaks is also an advantage

because it often makes unambiguous identification of analytes possible. Extensive fragmentation can also be a disadvantage, however, when it results in the disappearance of the molecular ion peak so that the molecular mass of analytes cannot be easily established. Another limitation of the EI source is the need to volatilize the sample, which may result in thermal degradation of some analytes before ionization can occur. The effects of thermal decomposition can sometimes be minimized by carrying out the volatilization from a heated probe located close to the entrance slit of the spectrometer. At the lower pressure of the source area, volatilization occurs at a lower temperature. Furthermore, less time is allowed for thermal decomposition to take place. As mentioned earlier, EI sources are applicable only to analytes having molecular masses smaller than about 10^3 Da.

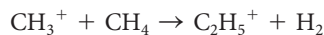
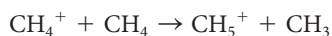
20B-2 Chemical Ionization Sources and Spectra

Most modern mass spectrometers are designed so that electron ionization and chemical ionization can be carried out interchangeably. Such sources are called EI-CI sources. In CI, gaseous atoms of the sample (from either a batch inlet or a heated probe) are ionized by collision with ions produced by electron bombardment of an excess of a reagent gas. Usually, positive ions are used, but negative ion CI is occasionally used with analytes that contain very electronegative atoms. Chemical ionization is probably the second-most common procedure for producing ions for mass spectrometry.⁵

⁵For a more detailed discussion of chemical ionization, see J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., Chap. 7, Chichester, UK: Wiley, 2007.

To carry out CI experiments, it is necessary to modify the electron beam ionization area shown in Figure 20-3 by adding vacuum pump capacity and by reducing the width of the slit to the mass analyzer. These measures allow a reagent pressure of about 100 Pa (0.75 torr) to be maintained in the ionization area while maintaining the pressure in the analyzer below 10^{-7} Pa. With these changes, a gaseous reagent is introduced into the ionization region in an amount such that the concentration ratio of reagent to sample is 10^3 to 10^4 . Because of this large concentration difference, the electron beam reacts nearly exclusively with reagent molecules.

One of the most common reagents for CI is methane, which reacts with high-energy electrons to give several ions such as CH_4^+ , CH_3^+ , and CH_2^+ . The first two predominate and represent about 90% of the reaction products. These ions react rapidly with additional methane molecules as follows:



Generally, collisions between the analyte molecule M and CH_5^+ or C_2H_5^+ are highly reactive and involve proton or hydride transfer. For example,



Note that proton-transfer reactions give the $(\text{M} + \text{H})^+$ ion whereas the hydride transfer produces an ion with a mass one less than the analyte, or the $(\text{MH} - \text{H})^+$ ion. With some compounds, an $(\text{MH} + 29)^+$ peak is also produced from transfer of a C_2H_5^+ ion to the analyte. A variety of other reagents, including propane, isobutane, and ammonia, are used for chemical ionization. Each produces a somewhat different spectrum with a given analyte.

Figure 20-2 contrasts the CI and EI spectra for 1-decanol. The EI spectrum (Figure 20-2a) shows evidence for rapid and extensive fragmentation of the molecular ion. Thus, no detectable peaks are observed above mass 112, which corresponds to the ion $\text{C}_8\text{H}_{16}^+$.

The base peak is provided by the ion C_3H_5^+ at mass 41. Other peaks for C_3 species are grouped around the base peak. A similar series of peaks, found at 14, 28, and 42 mass units greater, correspond to ions with one, two, and three additional CH_2 groups.

Relative to the EI spectrum, the CI spectrum shown in Figure 20-2b is simple indeed, consisting of the $(\text{MH} - \text{H})^+$ peak, a base peak corresponding to a molecular ion that has lost an OH group, and a series of peaks differing from one another by 14 mass units. As in the EI spectrum, these peaks arise from ions formed by cleavage of adjacent carbon-carbon bonds. As we have just noted, chemical ionization spectra generally

contain well-defined $(\text{M} + \text{H})^+$ or $(\text{MH} - \text{H})^+$ peaks resulting from the addition or abstraction of a proton in the presence of the reagent ion.

20B-3 Field Ionization Sources and Spectra

In *field ionization* sources, ions are formed under the influence of a large electric field (10^8 V/cm). Such fields are produced by applying high voltages (10 to 20 kV) to specially formed emitters consisting of numerous fine tips having diameters of less than 1 μm . The emitter often takes the form of a fine tungsten wire ($\sim 10 \mu\text{m}$ diameter) on which microscopic carbon dendrites, or whiskers, have been grown by the pyrolysis of benzonitrile in a high electric field. The result of this treatment is a growth of many hundreds of carbon microtips projecting from the surface of the wire as can be seen in the photomicrograph of Figure 20-5.

Field ionization emitters are mounted 0.5 to 2 mm from the cathode, which often also serves as a slit. The gaseous sample from a batch inlet system is allowed to diffuse into the high-field area around the microtips of the anode. The electric field is concentrated at the emitter tips, and ionization occurs via a quantum mechanical tunneling mechanism in which electrons from the analyte are extracted by the microtips of the anode. Little vibrational or rotational energy is imparted to the analyte, and little fragmentation occurs.



FIGURE 20-5 Photomicrograph of a carbon microneedle emitter. (Courtesy of R. P. Lattimer, BF Goodrich Research and Development Center.)

Figure 20-6 shows spectra for glutamic acid obtained by (a) electron ionization and (b) field ionization. In the EI spectrum, the molecular ion peak at $m/z = 147$ is not detectable. The highest observable peak ($m/z = 129$) is due to the loss of water by the molecular ion. The base peak at $m/z = 84$ arises from a loss

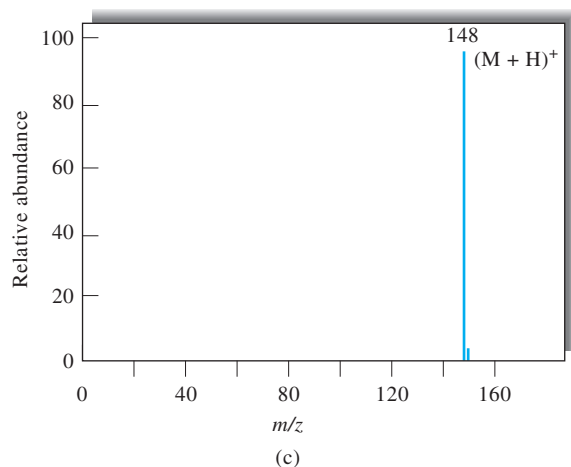
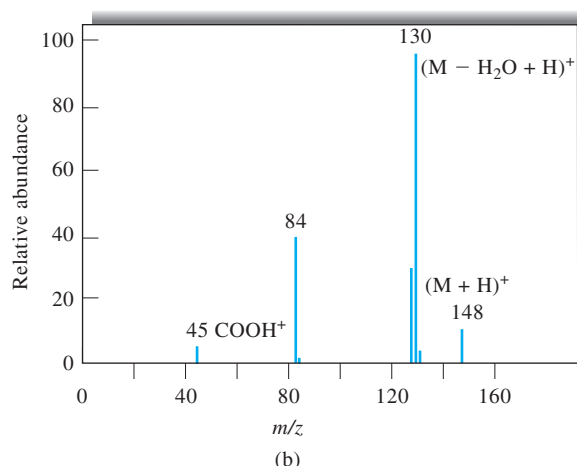
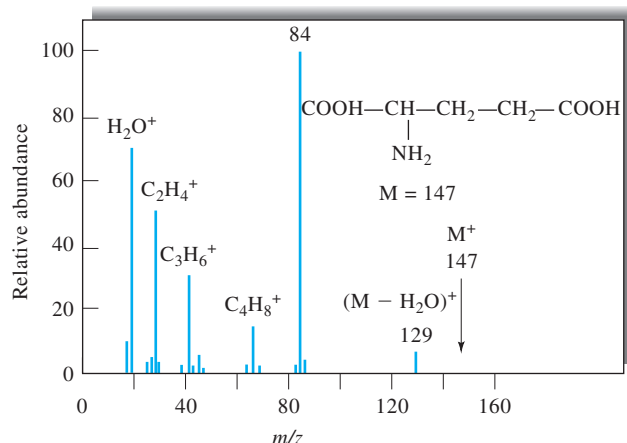


FIGURE 20-6 Mass spectra of glutamic acid with (a) EI, (b) field ionization, and (c) field desorption. (From H. D. Beckey, A. Heindrich, and H. U. Winkler, *Int. J. Mass Spec. Ion Phys.*, **1970**, **3**, A9, DOI: 10.1016/0020-7381(70)80009-2. With permission.)

of water and a $-\text{COOH}$ group. Numerous other fragments are also found at lower masses. In contrast, the field ionization spectrum is relatively simple, with an easily distinguished $(M + 1)^+$ peak at $m/z = 148$.

A limitation to field ionization is its sensitivity, which is at least an order of magnitude less than that of EI sources; maximum currents are on the order of 10^{-11} A.

20B-4 Desorption Sources

The ionization methods discussed so far require that the ionizing agents act on gaseous samples. Such methods are not applicable to nonvolatile or thermally unstable samples. A number of *desorption ionization* methods have been developed for dealing with this type of sample (see Table 20-1). These methods have enabled mass spectra to be obtained for thermally delicate biochemical species and species having molecular masses of greater than 100,000 Da.

Desorption methods dispense with volatilization followed by ionization of the gaseous analyte molecules. Instead, energy in various forms is introduced into the solid or liquid sample in such a way as to cause direct formation of gaseous ions. As a consequence, spectra are greatly simplified and often consist of only the molecular ion or the protonated molecular ion. In most cases the exact mechanism of how ions are formed without fragmentation is not well understood.

Field Desorption Methods

In field desorption, a multitipped emitter similar to that used in field ionization sources is used.⁶ In this case, the electrode is mounted on a probe that can be removed from the sample compartment and coated with a solution of the sample. After the probe is reinserted into the sample compartment, ionization again takes place by applying a high voltage to this electrode. With some samples it is necessary to heat the emitter by passing a current through the wire. As a result, thermal degradation may occur before ionization is complete.

Figure 20-6c is a field desorption spectrum for glutamic acid. It is even simpler than the spectrum from field ionization and consists of only the protonated molecular ion peak at $m/z = 148$ and an isotope peak at $m/z = 149$.

Matrix-Assisted Laser Desorption/Ionization

Matrix-assisted laser desorption/ionization (MALDI) spectrometry is an ionization method that can be used to obtain accurate molecular mass information about polar biopolymers ranging in molecular mass from a few thousand to several hundred thousand Da.⁷ The method was first described nearly simultaneously in 1988 by two research groups, one German

⁶See J. H. Gross, *Mass Spectrometry: A Textbook*, Chap. 8, Heidelberg, Germany: Springer, 2011.

⁷See reference 6, Chap. 11; J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, Chap. 9, Chichester, UK: Wiley, 2007.

and the other Japanese.⁸ Commercial instrumentation is available for MALDI.

In the MALDI technique, a low concentration of the analyte is uniformly dispersed in a solid or liquid matrix deposited on the end of a stainless steel probe or placed on a metal plate. The plate is then placed in a vacuum chamber and a laser beam is focused onto the sample. In addition to the usual vacuum-chamber MALDI, atmospheric-pressure MALDI has also been described.⁹ The MALDI matrix must strongly absorb the laser radiation. The matrix and analyte are then desorbed and ionized, creating an ion plume. The overall process is illustrated in Figure 20-7. The most common type of mass analyzer used with MALDI is the time-of-flight (TOF) analyzer. A mass spectrum from a MALDI-TOF instrument is shown in Figure 20-8. Here, the matrix material was nicotinic acid, and the analyte was a monoclonal antibody from a mouse having a molecular mass of approximately 150,000 Da. Note that the spectrum is characterized by very low background noise and a complete absence of fragmentation of the large analyte ion. Multiply charged ions are present as well as peaks for dimer and trimer species.

Although the mechanism of the formation of the MALDI ion plume is not completely understood, it is thought to involve absorption of the laser beam by the matrix, followed by transfer of the energy from the matrix to the analyte. Desorption of the analyte and the matrix then occurs. The analyte is thought to desorb as neutral molecules and then to be ionized by proton-transfer reactions with protonated matrix ions in a dense phase over the surface containing the matrix. A series of photochemical reactions may produce the protonated matrix ions.

Some of the matrix materials used for biomolecules are listed in Table 20-4 along with the laser wavelengths that have been used. Lasers used include nitrogen (337 nm), Nd-YAG (266 and 355 nm), excimer (308 nm), Er-YAG (2.94 μm), and CO_2 (10.6 μm).

The most common sample-preparation method is the dried-droplet technique in which a droplet of the matrix containing the analyte is deposited on the metal plate and then dried. Typically, the ratio of analyte to matrix is $1:10^3$ to $1:10^5$. Analyte concentrations are usually in the micromolar range.

Recently, the MALDI technique has been extended to imaging methods by scanning a localized laser beam over the dispersed sample.¹⁰ Mass spectral images of a variety of biopolymers have been obtained in this manner.

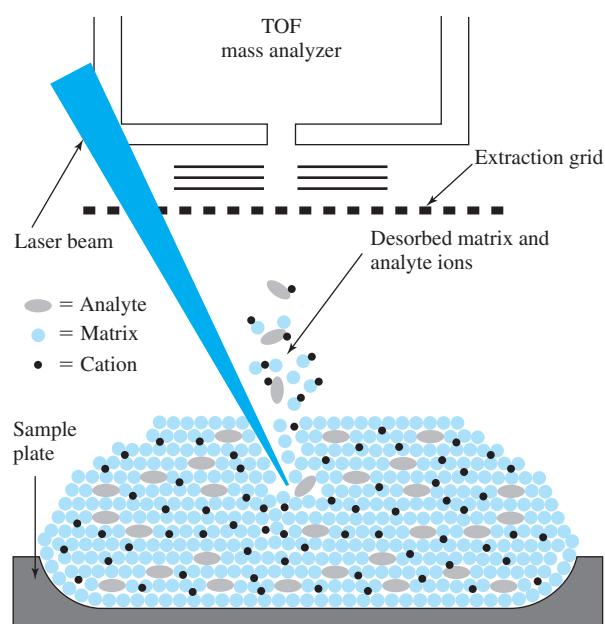


FIGURE 20-7 Diagram of the MALDI process. The analyte is uniformly dispersed in a matrix and placed on a metal sample plate. A pulsed laser beam strikes the sample causing desorption of a plume of matrix, analyte, and other ions. The analyte can be protonated, be deprotonated, or form adducts before entering the TOF analyzer.

Fast Atom Bombardment Sources

Fast atom bombardment (FAB) sources, also called *liquid secondary-ion* sources, were once very popular for producing ions for mass spectrometric studies of polar high-molecular-mass species.¹¹ With this type of source, samples in a condensed state, usually in a viscous solution matrix, are ionized by bombardment with energetic (several keV) xenon or argon *atoms*. Both positive and negative analyte ions are sputtered from the surface of the sample in a desorption process. This treatment provides very rapid sample heating, which reduces sample fragmentation. The liquid matrix helps to reduce the lattice energy, which must be overcome to desorb an ion from a condensed phase, and provides a means of “healing” the damage induced by bombardment. Successful matrices include glycerol, thioglycerol, *m*-nitrobenzyl alcohol, crown ethers (18-crown-6), sulfolane, 2-nitrophenyloctyl ether, diethanolamine, and triethanolamine.

A beam of fast atoms is obtained by passing accelerated argon or xenon ions from an ion source, or gun, through a chamber containing argon or xenon atoms at a pressure of about 10^{-3} Pa (10^{-5} torr). The high-velocity ions undergo a resonant electron-exchange reaction with the atoms without substantial loss of translational energy. Thus, a beam of energetic *atoms* is

⁸See M. Karas and F. Hillenkamp, *Anal. Chem.*, **1988**, 60, 2299, DOI: 10.1021/ac00171a028; K. Tanaka, H. Waki, Y. Ido, S. Akita, Y. Yoshida, and T. Yoshida, *Rapid Commun. Mass Spectrom.*, **1988**, 2, 151, DOI: 10.1002/rcm.1290020802; F. Hillenkamp, M. Karas, and B. T. Chait, *Anal. Chem.*, **1991**, 63, 1193A, DOI: 10.1021/ac00024a716.

⁹See Section 20B-5 for other atmospheric pressure ionization techniques.

¹⁰See for example, S. M. Weidner and J. Falkenhagen, *Rapid Commun. Mass Spectrom.*, **2009**, 23, 653, DOI: 10.1002/rcm.3919; K. A. Lunsford, G. F. Peter, and R. A. Yost, *Anal. Chem.*, **2011**, 83, 6722, DOI: 10.1021/ac2013527.

¹¹For more information, see J. H. Gross, *Mass Spectrometry: A Textbook*, Chap. 10, Heidelberg, Germany: Springer, 2011; J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, pp. 250–251, Chichester, UK: Wiley, 2007.

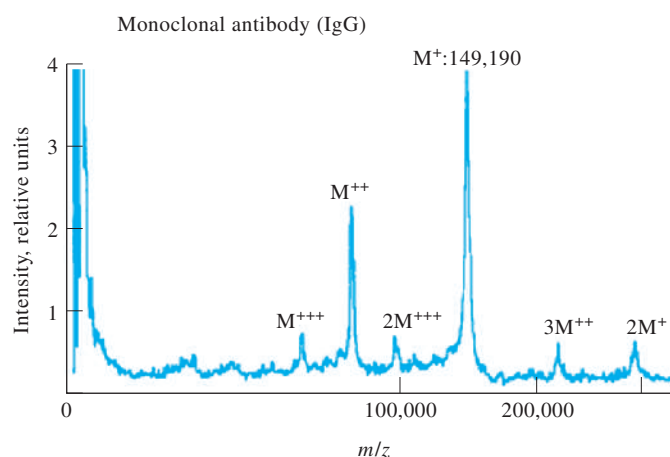


FIGURE 20-8 MALDI-TOF spectrum from nicotinic acid matrix irradiated with a 266-nm laser beam. (From M. Karas and U. Bahr, *Trends Anal. Chem. (TRAC)*, **1990**, 9, 323, DOI: 10.1016/0165-9936(90)85065-F.)

TABLE 20-4 Common Matrices for MALDI and Usable Wavelengths

Matrix	Analytes	Wavelength, nm
Nitropyridines:		
2-Amino-4-methyl-5-nitropyridine	Proteins, oligonucleotides	355
2-Amino-5-nitropyridine	Oligonucleotides	355
Nicotinic acid	Proteins, glycoproteins, oligonucleotides	266, 220–290
Benzoic acid derivatives:		
2,5-Dihydroxybenzoic acid	Proteins	266, 337, 355, 2940
Vanillic acid	Proteins	266
2-Aminobenzoic acid	Proteins	266, 337, 355
2-(4-Hydroxyphenylazo) benzoic acid	Proteins, gangliosides, polymers	266, 377
2-Pyrazinecarboxylic acid	Proteins	266
3-Aminopyrazine-2-carboxylic acid	Proteins	337
Cinnamic acid derivatives:		
Ferulic acid	Proteins, oligonucleotides	266, 337, 355, 488
Sinapinic acid	Proteins, industrial polymers	337, 355
Caffeic acid	Proteins, oligonucleotides	266, 337, 355, 10600
α -Cyano-4-hydroxy cinnamic acid	Proteins, oligosaccharides	337
3-Nitrobenzyl alcohol	Proteins	266
3-Nitrobenzyl alcohol with rhodamine 6G	Proteins	532
3-Nitrobenzyl alcohol with 1,4-diphenyl-1,3-butadiene	Proteins	337
3-Hydroxypicolinic acid	Oligonucleotides, glycoproteins	266, 308, 355
Succinic acid	Proteins	2940, 10600

formed. The lower-energy ions from the exchange are readily removed by an electrostatic deflector. Fast atom guns are available from several commercial sources, and modern spectrometers offer FAB sources as accessories.

The many drawbacks of FAB have led to its decrease in popularity. These include the limited molecular mass range of FAB, the need for larger sample quantities than many other ionization techniques and the necessity for finding an appropriate matrix in which the analyte is soluble. Also, FAB is best performed with double-focusing mass spectrometers (see Section 20C). For these reasons, electrospray (see next section) and MALDI methods have supplanted FAB techniques for many applications. Both of these are more sensitive and simpler to use than FAB methods.

20B-5 Atmospheric Pressure Ionization Methods

There are several ionization techniques that operate at atmospheric pressure, including electrospray, atmospheric pressure chemical ionization, and atmospheric pressure photoionization.

Electrospray Ionization

Electrospray ionization–mass spectrometry (ESI/MS), which was first described in 1984, has now become one of the most important techniques for analyzing biomolecules, such as polypeptides, proteins, and oligonucleotides, having molecular weights of 100,000 Da or more.¹² In addition, this method is finding more and more application to the characterization of inorganic species and synthetic polymers. For their development of soft desorption ionization methods, such as electrospray ionization, John B. Fenn and Koichi Tanaka shared the 2002 Nobel Prize in Chemistry.

Electrospray ionization takes place under atmospheric pressures and temperatures in an apparatus such as that shown in Figure 20-9. A solution of the sample is pumped through a stainless steel capillary needle at a rate of a few microliters per minute. The needle is maintained at several kilovolts with respect to a cylindrical electrode that surrounds the needle. The resulting charged spray of fine droplets then passes through a desolvating capillary, where evaporation of the solvent and attachment of charge to the analyte molecules take place. As the droplets become smaller as a consequence of evaporation of the solvent, their charge density becomes greater until, at a point called the *Rayleigh limit*, the surface tension can no longer support the charge. Here a so-called *Coulombic explosion* occurs and the droplet is torn apart into smaller droplets.

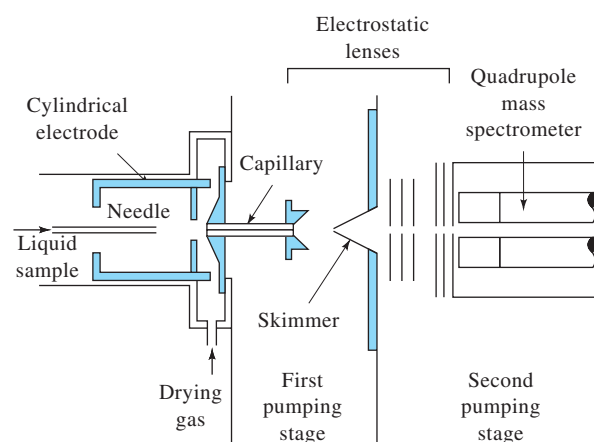


FIGURE 20-9 Apparatus for electrospray ionization. (From J. B. Fenn et al., *Science*, **1989**, 246, 64, DOI: 10.1126/science.2675315. Reprinted with permission.)

These small droplets can repeat the process until all the solvent is removed from the analyte, leaving a multiply charged analyte molecule.

An interesting and useful feature of the electrospray process is that little fragmentation of large and thermally fragile biomolecules occurs because there is little extra energy retained by the analyte upon ionization. Furthermore, the ions formed are multiply charged so that their m/z values are small enough to make them detectable with a quadrupole instrument with a range of 1500 or less. This important property is demonstrated by the mass spectra of four proteins of varying molecular mass (M) shown in Figure 20-10. In these spectra, adjacent peaks are for analyte ions that differ by one charge. A striking feature of the spectra for proteins, such as those in the figure, is that the average charge state increases in approximately linear fashion with molecular mass. The charge state corresponding to each peak can be determined from peak distribution, thus making it possible to determine the molecular mass of a protein from spectra such as those shown in Figure 20-10.

An important characteristic of electrospray ionization is that it is readily adapted to direct sample introduction from high-performance liquid chromatography and capillary electrophoresis columns. These applications are described in Chapters 28 and 30. That there is little fragmentation of the analyte makes structural elucidation a difficult task. Usually, tandem mass spectrometry (see Section 20C-5) is used for this purpose. Here, the ions from the original ionization process are separated and the ion of interest is subjected to a fragmentation step before being mass analyzed.

Atmospheric Pressure Chemical Ionization

A normal CI source, as described in Section 20B-2, operates at pressures of approximately 100 Pa. In an atmospheric pressure

¹²For additional information see *Electrospray and MALDI Mass Spectrometry*, 2nd ed., R. B. Cole, ed., Hoboken, NJ: Wiley, 2010; J. H. Gross, *Mass Spectrometry: A Textbook*, Chap. 12, Heidelberg, Germany: Springer, 2011; *Applied Electrospray Ionization Mass Spectrometry*, B. N. Pramanik, A. K. Ganguly, and M. L. Gross, eds., New York: Dekker, 2002.

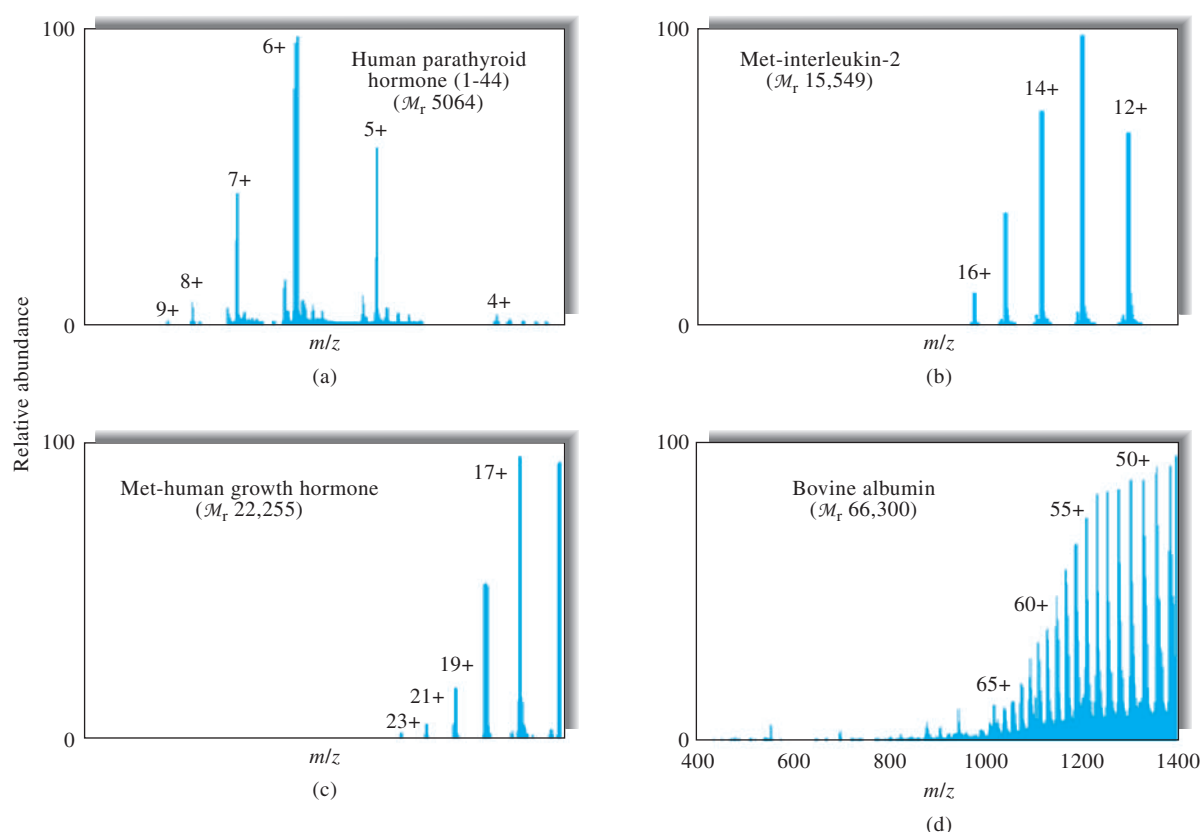


FIGURE 20-10 Typical electrospray mass spectra of proteins and peptides. The numbers above the peaks represent the molecular charge associated with each peak. (From R. D. Smith et al., *Anal. Chem.*, **1990**, 62, 882, DOI: 10.1021/ac00208a002. Copyright 1990 American Chemical Society.)

chemical ionization (APCI) source, ionization is brought about by low-energy electrons emitted by a radioactive beta source or corona discharge. Usually a reagent gas (N_2 , O_2 , H_2O) is ionized by the low-energy electrons, which leads to subsequent ionization of the analyte by means of several complex ion-molecule reactions. Because of its operation at atmospheric pressure, molecules of the reagent gas and the analyte undergo frequent collisions, which lead to high ionization efficiencies. Just as with electrospray ionization, the mass spectrometer used with APCI must be designed to input ions at atmospheric pressure. This requires “skimmer” interfaces similar to that shown in Figure 20-9. Atmospheric pressure CI is readily adaptable to samples separated by liquid chromatography (LC).

Atmospheric Pressure Photoionization

Atmospheric pressure photoionization (APPI) uses an intense ultraviolet light source instead of low-energy electrons to ionize either the analyte directly or a dopant gas. With the latter, a series of reactions then leads to the formation of analyte ions. Many of the same compounds ionized by APCI can be ionized by APPI, but the range of APPI extends to less polar species.

20B-6 Ambient Ionization Methods

Ambient desorption ionization techniques are performed in the open environment without any sample pretreatment.¹³ Almost any mass spectrometer with an atmospheric pressure sample inlet can be used. Currently, the two most popular forms of ambient mass spectrometry are desorption electrospray ionization (DESI) and direct analysis in real time (DART).¹⁴ A generalized diagram applicable to both of these methods is shown in Figure 20-11. Mechanisms of real-time sample processing in ambient ionization methods have been reviewed.¹⁵

¹³For reviews of ambient mass spectrometry methods, see R. G. Cooks, Z. Ouyang, Z. Takáts, and J. M. Wiseman, *Science*, **2007**, 311, 1566, DOI: 10.1126/science.1119426; J. T. Shelly and G. M. Hieftje, *J. Anal. At. Spectrom.*, **2011**, 26, 2153, DOI: 10.1039/c1ja10158g.

¹⁴For the original report on DESI, see Z. Takáts, J. M. Wiseman, B. Gologan, and R. G. Cooks, *Science*, **2004**, 306, 471, DOI: 10.1126/science.1104404. For the original description of DART, see R. B. Cody, J. A. Laramee, and H. D. Durst, *Anal. Chem.*, **2005**, 77, 2297, DOI: 10.1021/ac050162j.

¹⁵A. R. Venter, K. A. Douglass, J. T. Shelly, G. Hasman, Jr., and E. Honarvar, *Anal. Chem.*, **2014**, 1, 233, DOI: 10.1021/ac4038569.

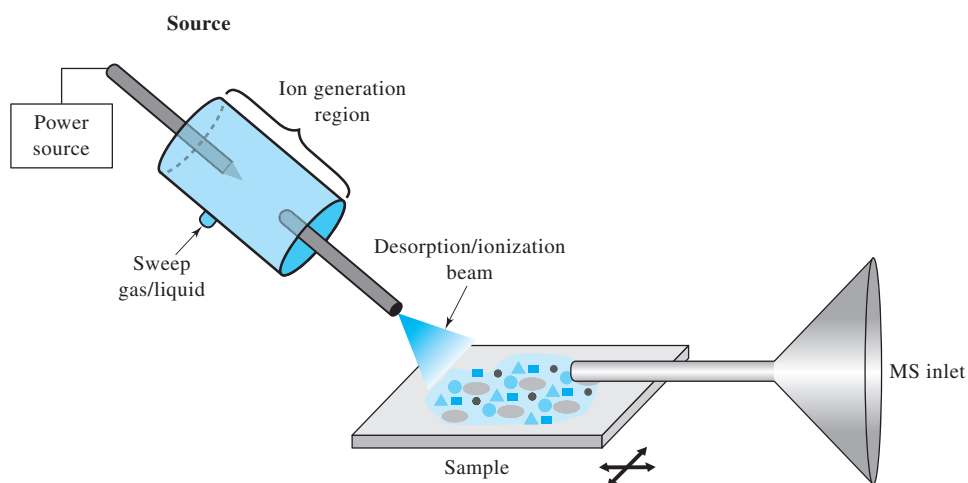


FIGURE 20-11 General diagram of an ambient mass spectrometry source. Reagent ions are generated and swept to the untreated sample by a sweep gas, liquid, or aerosol. The beam releases analytes into the gas phase and ionizes them. The ions then enter the mass spectrometer where they are separated according to their m/z ratio. (Adapted from J. T. Shelly and G. M. Hieftje, *J. Anal. At. Spectrom.*, **2011**, 26, 2153, DOI: 10.1039/c1ja10158g.)

Desorption Electrospray Ionization

The DESI source is similar to a normal electrospray source except that the droplets produced are not desolvated as they are in ESI/MS. The spray of charged droplets, the desorption/ionization beam shown in Figure 20-11, impinges on the sample from which the analyte is either released as ions or the ions are formed in the plume that results from the desorption/ionization process. In DESI, a goal is to keep the sample outside the mass spectrometer inlet and to allow only the desorbed ions to enter. Although the mechanism of the DESI phenomenon is still poorly understood, the accumulation of liquid on the surface is thought to play an important role in the process. Mass spectra of large biomolecules obtained with DESI resemble those obtained in normal ESI including the formation of multiply charged ions.

With open-air DESI, the distance between the spraying device and the sample is often important as is the angle at which the beam strikes the sample. Newer instrument designs have included a pressure-tight sample chamber that can minimize the angular dependence at the expense of eliminating the open-air advantage of DESI. Several variations of the DESI technique have been described.

Direct Analysis in Real Time

In DART, an excited stream of helium (sometimes nitrogen or neon) is directed at the surface where desorption/ionization occurs. To create the excited gas stream, He is introduced into a discharge chamber where a corona discharge produces a mixture of ions, electrons, and metastable atoms or molecules. Ions are removed by passing the stream through a set of grids of opposing polarities, leaving excited atoms and molecules in the stream.

The gas flow is usually reflected off the surface containing the sample into the mass spectrometer. The exact positioning and angle with respect to the surface and the mass spectrometer inlet are not critical. Limitations of DART include its incompatibility with LC and a somewhat limited mass range.

Other Desorption Methods

Examination of Table 20-1 reveals that several other desorption/ionization methods are available. Generally these produce spectra similar to the desorption methods we have described and have more limited applicability.

20C MASS SPECTROMETERS

Several types of instruments are currently used for molecular mass spectrometric measurements. Two of these, the quadrupole spectrometer and the TOF spectrometer, have been described in some detail in Sections 11B-2 and 11B-3. In this section, we consider other common types of mass spectrometers.

20C-1 General Description of Instrument Components

The block diagram in Figure 20-12 shows the major components of mass spectrometers. The purpose of the inlet system is to introduce a very small amount of sample (a micromole or less) into the mass spectrometer, where its components are converted to gaseous ions. Often the inlet system contains a means for volatilizing solid or liquid samples.

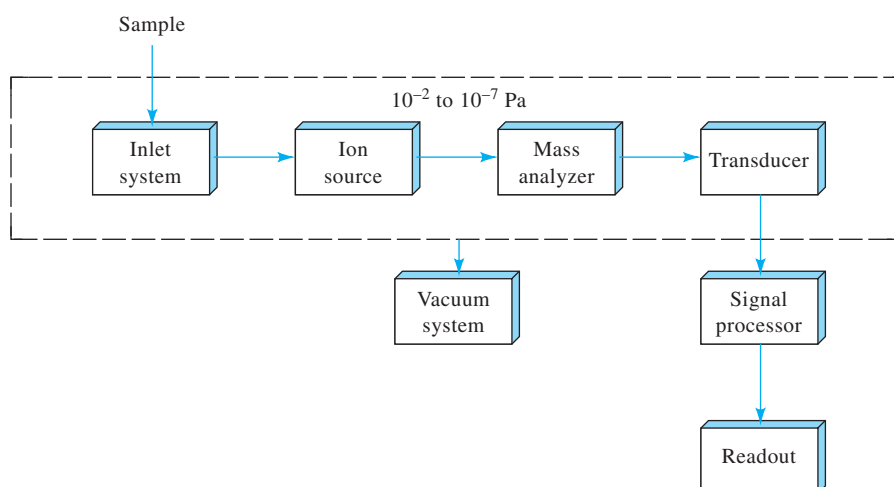


FIGURE 20-12 Components of a mass spectrometer.

Ion sources of mass spectrometers, which were discussed in some detail in the previous section and in Chapter 11, convert the components of a sample into gas-phase ions. In many cases the inlet system and the ion source are combined into a single component. In either case, the output is a stream of positive or negative ions (more commonly positive) that are then accelerated into the mass analyzer.

The function of the mass analyzer, as discussed in Chapter 11, is analogous to that of the grating in an optical spectrometer. In the mass analyzer, however, ions are dispersed according to their mass-to-charge ratios. Mass spectrometers fall into several categories, depending on the nature of the mass analyzer.

Like an optical spectrometer, a mass spectrometer contains a transducer (for ions) that converts the beam of ions into an electrical signal that can then be processed, stored in the memory of a computer, and displayed or recorded in a variety of ways. The characteristics of several types of transducers used in mass spectrometers were described in Section 11B-1.

A distinguishing feature of mass spectrometers, which is not shared by optical instruments, is the requirement of an elaborate vacuum system to create low pressures (10^{-2} to 10^{-7} Pa) in all of the instrument components except the signal processor, readout, and, in some cases, the inlet system. The need for a high vacuum arises because such conditions lead to infrequent collisions with atmospheric components and allow the production and manipulation of free electrons and ions.

In the sections that follow, we first describe inlet systems that are common to all types of mass spectrometers. We then define *resolution*, which describes the ability of a mass spectrometer to differentiate between ions with differing masses. Finally, several types of mass analyzers that lead to different categories of mass spectrometers are described.

20C-2 Sample Inlet Systems

The purpose of the inlet system is to permit introduction of a representative sample into the ion source with minimal loss of vacuum. Most modern mass spectrometers are equipped with several types of inlets to accommodate various kinds of samples; these include batch inlets, direct probe inlets, chromatographic inlets, and capillary electrophoretic inlets.

Batch Inlet Systems

The conventional (and simplest) inlet system is the batch type, in which the sample is volatilized externally and then allowed to leak into the evacuated ionization region. Figure 20-13a is a schematic of a typical system that is applicable to gaseous and liquid samples having boiling points up to about 500°C.

The first step is to obtain the mass spectrum of any residue from a previous sample. For this, the heated reservoir is evacuated, the valve to the ion source opened, and a mass spectrum scanned. The reservoir is then evacuated and isolated by closing the valves. The sample is then introduced, often through a septum, and the valve to the ion source opened. For gaseous samples a measured amount is introduced into the reservoir, sometimes with a separate introduction system. For liquid samples with boiling points greater than 150°C, the reservoir and tubing are maintained at an elevated temperature by means of an oven and/or heating tape. The sample, which is now in the gas phase, is leaked into the ionization area by means of a metal or glass diaphragm containing one or more pinholes. This simple inlet has the disadvantage of requiring rather large sample sizes. The inlet system is often lined with glass to avoid losses of polar analytes due to adsorption.

The heated reservoir inlet played an important role in the early development of mass spectrometry. This type of inlet is still used with some EI and CI sources and for the introduction of compounds for calibration of the mass scale of instruments.

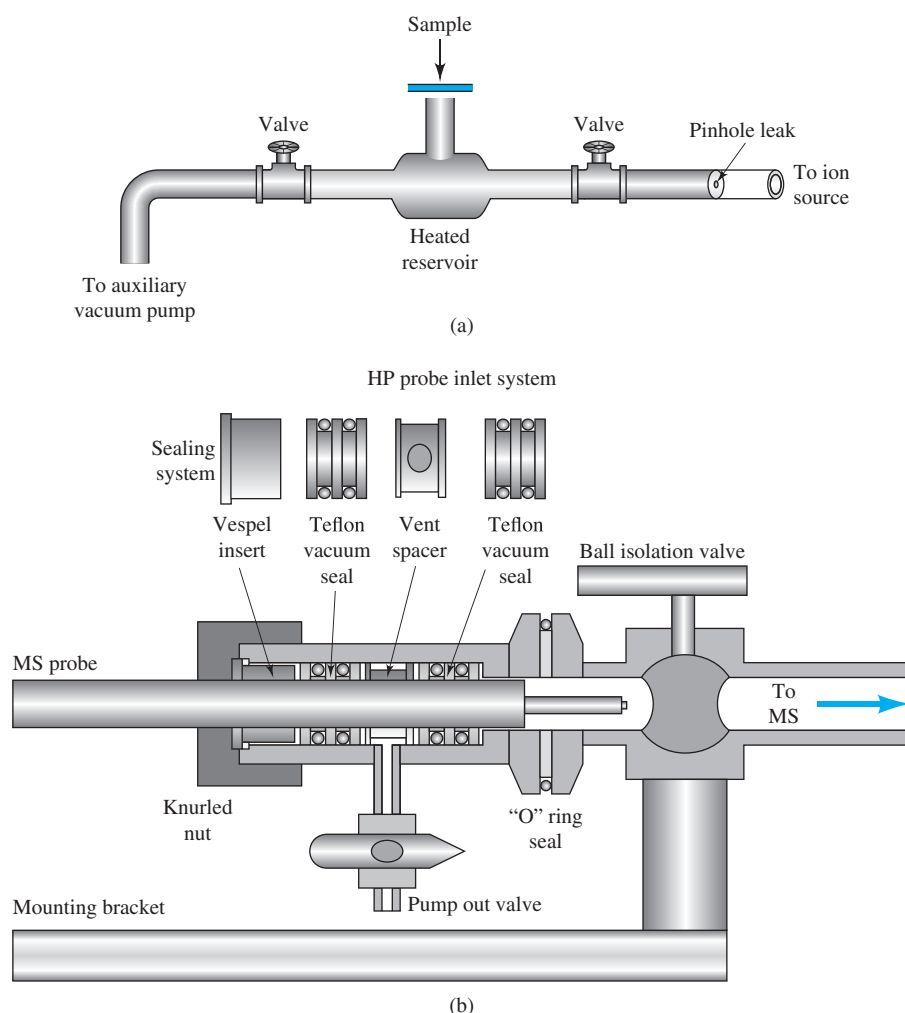


FIGURE 20-13 Schematic of a heated-reservoir sample inlet system (a) and a direct probe inlet (b). [(a) Cengage Learning; (b) Adapted from Scientific Instrument Services <http://www.sisweb.com/ms/sis/probinlt.htm>.]

The Direct Probe Inlet

Solids and nonvolatile liquids can be introduced into the ionization region by means of a sample holder, or probe, which is inserted through a vacuum lock (see Figure 20-13b). The lock system is designed to limit the volume of air that must be pumped from the system after insertion of the probe into the ionization region.

With a probe, the sample is generally held on the surface of a glass or aluminum capillary tube, a fine wire, or a small cup. The probe is positioned within a few millimeters of the ionization source and the slit leading to the spectrometer. Usually, provision is made for both cooling and heating the sample on the probe. Like batch inlets, direct probe inlets were used for EI and CI sources.

Direct probe inlets permitted the recording of mass spectra of relatively nonvolatile compounds. They were also used when sample amounts were limited since much less sample was wasted than with the batch inlet. Disadvantages included the possibility of venting to the atmosphere and the risk of contaminating the ion source if large quantities of nonvolatile compounds were

introduced. Although direct probe inlets were once standard accessories on commercial mass spectrometers, they are no longer available for most systems.

Chromatographic and Other Inlet Systems

Mass spectrometers are often coupled with gas or high-performance liquid chromatographic systems or with capillary electrophoresis units to permit the separation and determination of the components of complex mixtures. Linking a chromatographic or electrophoretic column to a mass spectrometer requires the use of specialized inlet systems, some of which are described in Sections 27B-4, 28C-6, and 30B-4.

Many additional types of inlets have been described.¹⁶ Among the most important are polymer membrane and atmospheric pressure inlets (see previous section).

¹⁶For descriptions of additional inlets, see J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, Chap. 4, Chichester, UK: Wiley, 2007.

20C-3 Mass Analyzers

Several devices are available for separating ions with different mass-to-charge ratios. Ideally, the mass analyzer should be capable of distinguishing minute mass differences. In addition, the analyzer should allow passage of a sufficient number of ions to yield readily measurable ion currents. These two properties are not entirely compatible, and design compromises must be made.

Resolution of Mass Spectrometers

The capability of a mass spectrometer to differentiate between masses is usually stated in terms of its *resolution* R , which is defined as

$$R = \frac{m}{\Delta m} \quad (20-3)$$

where Δm is the mass difference between two adjacent peaks of equal intensity that are just resolved and m is the nominal mass of the first peak (the mean mass of the two peaks is sometimes used instead). Two peaks are considered to be separated if the height of the valley between them is no more than a given fraction of their height (often 10%). Thus, a spectrometer with a resolution of 4000 would resolve peaks at m/z values of 400.0 and 400.1 (or 40.00 and 40.01).

The resolution required in a mass spectrometer depends greatly on its application. For example, discrimination among ions of the same nominal mass such as C_2H_4^+ , CH_2N^+ , N_2^+ , and CO^+ (all ions of nominal mass 28 Da but exact masses of 28.0313, 28.0187, 28.0061, and 27.9949 Da, respectively) requires an instrument with a resolution of several thousand. On the other hand, low-molecular-mass ions differing by a unit of mass

or more such as NH_3^+ ($m = 17$) and CH_4^+ ($m = 16$), for example, can be distinguished with an instrument having a resolution smaller than 50. Commercial spectrometers are available with resolutions ranging from about 500 to 500,000. Example 20-3 gives a calculation of the resolution needed to separate two species.

EXAMPLE 20-3

What resolution is needed to separate the ions C_2H_4^+ and CH_2N^+ , with masses of 28.0313 and 28.0187, respectively?

Solution

Here,

$$\Delta m = 28.0313 - 28.0187 = 0.0126$$

The average of the two masses is $(28.0313 + 28.0187)/2 = 28.0250$. Substituting into Equation 20-3 gives

$$R = m/\Delta m = 28.0250/0.0126 = 2.22 \times 10^3$$

Magnetic Sector Analyzers

Magnetic sector analyzers use a permanent magnet or an electromagnet to cause the beam from the ion source to travel in a circular path, most commonly 180° , 90° , or 60° . Figure 20-14 shows a 90° sector instrument in which ions formed by EI are accelerated through slit B into the metal analyzer tube, which is maintained at an internal pressure of about 10^{-5} Pa. Ions of different mass can be scanned across the exit slit by varying the field strength of the magnet or the accelerating potential between slits A and B. The ions passing through the exit slit

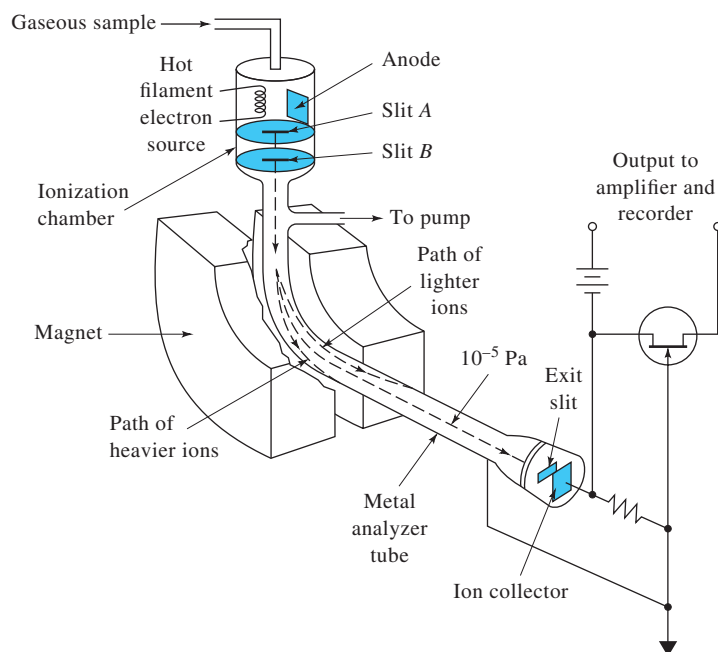


FIGURE 20-14 Schematic of a magnetic sector spectrometer.

fall on a collector electrode, resulting in an ion current that is amplified and recorded.

The translational, or kinetic, energy KE of an ion of mass m bearing a charge z on exiting slit B is given by

$$\text{KE} = zeV = \frac{1}{2}mv^2 \quad (20-4)$$

where V is the voltage difference between A and B , v is the velocity of the ion after acceleration, and e is the electronic charge ($e = 1.60 \times 10^{-19}$ C). Note that all ions having the same number of charges z are assumed to have the same kinetic energy after acceleration regardless of their mass. This assumption is only approximately true, because before acceleration, the ions possess a statistical distribution of velocities (speeds and directions), which will be reflected in a similar distribution for the accelerated ion. The limitations of this assumption are discussed in the next section when double-focusing instruments are described. Because all ions leaving the slit have approximately the same kinetic energy, the heavier ions must travel through the magnetic sector at lower velocities.

The path in the sector described by ions of a given mass and charge represents a balance between two forces acting on them. The magnetic force F_M is given by the relationship

$$F_M = Bzev \quad (20-5)$$

where B is the magnetic field strength. The balancing centripetal force F_c is given by

$$F_c = \frac{mv^2}{r} \quad (20-6)$$

where r is the radius of curvature of the magnetic sector. For an ion to traverse the circular path to the collector, F_M and F_c must be equal. Thus, equating Equations 20-5 and 20-6 leads to

$$Bzev = \frac{mv^2}{r} \quad (20-7)$$

which rearranges to

$$v = \frac{Bzer}{m} \quad (20-8)$$

Substituting Equation 20-8 into Equation 20-4 gives, after rearranging,

$$\frac{m}{z} = \frac{B^2 r^2 e}{2V} \quad (20-9)$$

Equation 20-9 reveals that mass spectra can be acquired by varying one of three variables (B , V , or r) while holding the other two constant. Most modern sector mass spectrometers contain an electromagnet in which ions are sorted by holding V and r constant while varying the current in the magnet and thus B . In older sector spectrometers that used photographic recording, B and V were constant, and r was the variable (see Figure 11-11). In Example 20-4, the use of Equation 20-9 to calculate an appropriate accelerating voltage is illustrated.

EXAMPLE 20-4

What accelerating voltage is required to direct a singly charged water molecule through the exit slit of a magnetic sector mass spectrometer if the magnet has a field strength of 0.240 T (tesla) and the radius of curvature of the ion through the magnetic field is 12.7 cm?

Solution

First, we convert all experimental variables into SI units. Thus, charge per ion $ez = 1.60 \times 10^{-19}$ C $\times$ 1

$$\text{radius } r = 0.127 \text{ m}$$

$$\begin{aligned} \text{mass } m &= \frac{18.02 \text{ g H}_2\text{O}^+/\text{mol}}{6.02 \times 10^{23} \text{ g/mol}} \times 10^{-3} \frac{\text{kg}}{\text{g}} \\ &= 2.99 \times 10^{-26} \text{ kg H}_2\text{O}^+ \end{aligned}$$

$$\text{magnetic field } B = 0.240 \text{ T} = 0.240 \text{ Vs/m}^2$$

We then substitute into Equation 20-9 and solve for the accelerating voltage V :

$$\begin{aligned} V &= \frac{B^2 r^2 ez}{2m} \\ &= \frac{[0.240 \text{ Vs/m}^2]^2 [0.127 \text{ m}]^2 [1.60 \times 10^{-19} \text{ C}]}{2 \times 2.99 \times 10^{-26} \text{ kg}} \\ &= 2.49 \times 10^3 \frac{(\text{Vs})^2 \text{C}}{\text{m}^2 \text{kg}} \\ &= 2.49 \times 10^3 \text{ V} \quad (1 \text{ volt} = 1 \text{ kg m}^2/\text{s}^2 \text{ C}) \end{aligned}$$

Magnetic sector mass spectrometers are limited in scan speed by the rate at which the magnetic field can be changed. This scan-rate limitation is a particular problem with capillary gas chromatography/mass spectrometry (see Section 27B-4), where narrow analyte peaks require rapid scanning.

Double-Focusing Spectrometers

The magnetic sector instruments discussed in the previous section are sometimes called *single-focusing* spectrometers. This terminology is used because a collection of ions exiting the source with the same mass-to-charge ratio but with small diverging directional distribution will be acted on by the magnetic field in such a way that a converging directional distribution is produced as the ions leave the field. The ability of a magnetic field to bring ions with different directional orientations to focus means that the distribution of translational energies of ions leaving the source is the factor most responsible for limiting the resolution of magnetic sector instruments ($R \leq 2000$).

The translational energy distribution of ions leaving a source arises from the Boltzmann distribution of energies of the molecules from which the ions are formed and from field inhomogeneities in the source. The spread of kinetic energies causes a broadening of the beam reaching the transducer and thus a

loss of resolution. To measure atomic and molecular masses with a precision of a few parts per million, it is necessary to design instruments that correct for both the directional distribution and energy distribution of ions leaving the source. The term *double focusing* is applied to mass spectrometers in which the directional aberrations and the energy aberrations of a population of ions are simultaneously minimized. Double focusing is usually achieved by the use of carefully selected combinations of electrostatic and magnetic fields. In the double-focusing instrument, shown schematically in Figure 20-14, the ion beam is first passed through an electrostatic analyzer (ESA) consisting of two smooth curved metallic plates across which a dc voltage is applied. This voltage has the effect of limiting the kinetic energy of the ions reaching the magnetic sector to a closely defined range. Ions with energies greater than average strike the upper side of the ESA slit and are lost to ground. Ions with energies less than average strike the lower side of the ESA slit and are thus removed.

Directional focusing in the magnetic sector occurs along the focal plane labeled d in Figure 20-14; energy focusing takes place along the plane labeled e . Thus, only ions of one m/z are double focused at the intersection of d and e for any given accelerating voltage and magnetic field strength. Therefore, the collector slit is located at this locus of the double focus.

Many different types of double-focusing mass spectrometers are available commercially. The most sophisticated of these are capable of resolution in the 10^5 range. More compact double-focusing instruments can also be purchased (for considerably less money). A typical instrument of this type will have a 6-in. electrostatic sector and a 4-in. 90° magnetic deflector.

Resolutions of about 2500 are common with such instruments. Often they are used as detection systems for gas or liquid chromatography.

The spectrometer shown in Figure 20-15 is based on the so-called *Nier-Johnson* design. Another double-focusing design, of the *Mattauch-Herzog* geometry, is shown in Figure 11-11. The geometry of this type of instrument is unique in that the energy and direction focal planes coincide; for this reason, the Mattauch-Herzog design can use an array detector or a photographic plate for recording the spectrum. The transducer is located along the focal plane, where all of the ions are in focus, regardless of mass-to-charge ratio.

Quadrupole Mass Spectrometers

Quadrupole mass spectrometers are usually less expensive and more rugged than their magnetic sector counterparts. They are generally considerably more compact than magnetic sector instruments and are commonly found in commercial bench-top mass spectrometers. They also offer the advantage of low scan times (that is, <100 ms), which is particularly useful for real-time scanning of chromatographic eluents. Quadrupole analyzers are the most common mass analyzers in use today. A detailed discussion of quadrupole mass spectrometers is found in Section 11B-2.

TOF Mass Analyzers

As shown in Section 11B-3, in TOF instruments, positive ions are produced periodically by bombardment of the sample with brief pulses of electrons, secondary ions, or laser-generated photons. The ions produced in this way are then accelerated into

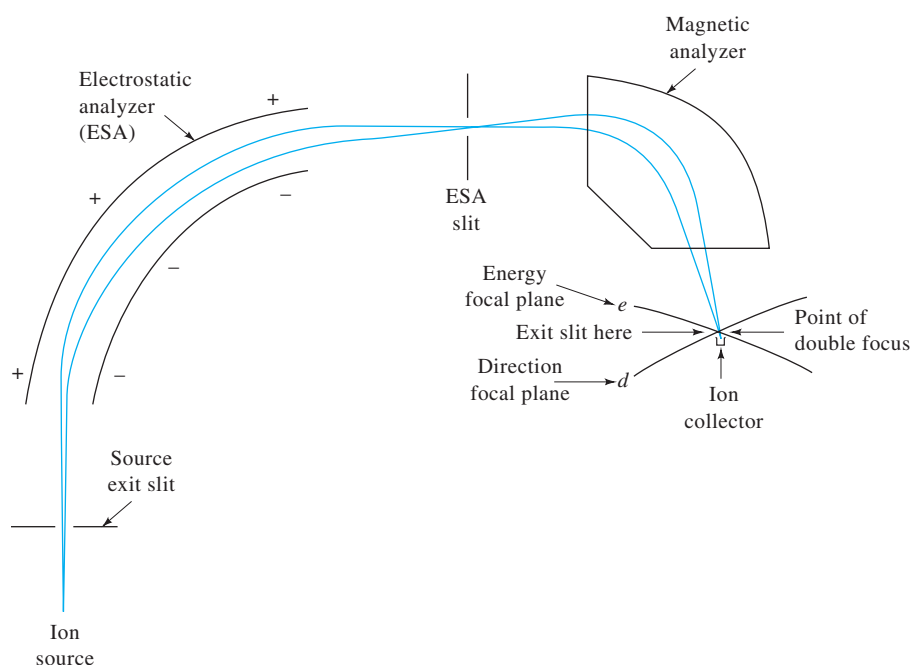


FIGURE 20-15 Nier-Johnson design of a double-focusing mass spectrometer.

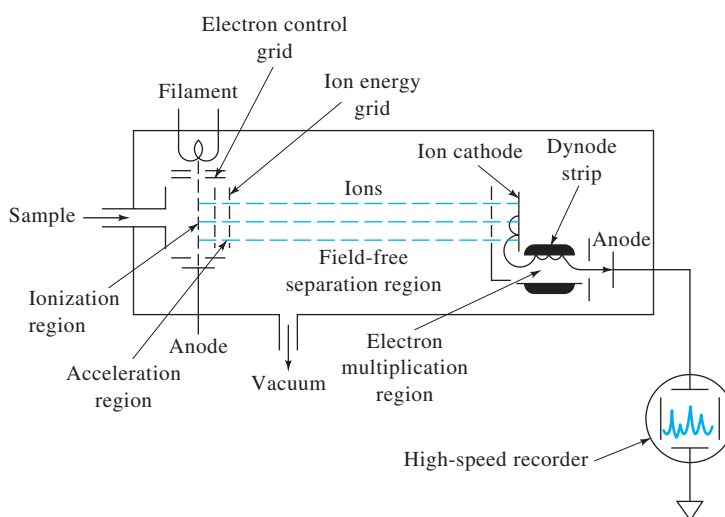


FIGURE 20-16 Schematic of a TOF mass spectrometer.

a field-free drift tube by an electric field pulse of 10^3 to 10^4 V (see Figure 20-16). Separation of ions by mass occurs during the transit of the ions to the detector located at the end of the tube. Because all ions entering the tube have the same kinetic energy, their velocities in the tube vary inversely with their masses (Equation 20-4), with the lighter particles arriving at the detector earlier than the heavier ones. The flight time t_F is given by

$$t_F = \frac{L}{v} = L\sqrt{\frac{m}{2zeV}} \quad (20-10)$$

where L is the distance from the source to the detector. Typical flight times are 1 to 50 μs .

TOF instruments offer several advantages over other types of mass spectrometers, including simplicity and ruggedness, ease of accessibility of the ion source, and virtually unlimited mass range. They suffer, however, from limited resolution and sensitivity. TOF instruments also require fast electronics because ions often arrive at the transducer only fractions of microseconds apart. Several instrument manufacturers offer TOF instruments, but they are less widely used than are magnetic sector and quadrupole mass spectrometers.

Ion-Trap Analyzers

An ion trap is a device in which gaseous anions or cations can be formed and confined for extended periods by electric and magnetic fields. The quadrupole ion trap was first introduced by Paul in 1953.¹⁷ Since that time several types of ion traps have been developed.¹⁸ Ion-trap mass spectrometers are now used

as chromatography detectors and to obtain mass spectra of a variety of analytes. For their work in developing the ion-trap technique, Wolfgang Paul and Hans Dehmelt were awarded the Nobel Prize in Physics in 1989.

Figure 20-17 is a cross-sectional view of a simple, commercially available ion trap. It consists of a central doughnut-shaped ring electrode and a pair of end-cap electrodes. A variable radio-frequency voltage is applied to the ring electrode while the two end-cap electrodes are grounded. Ions with an appropriate m/z value circulate in a stable orbit within the cavity surrounded



Tutorial: Learn more about mass analyzers at www.tinyurl.com/skoogpia7

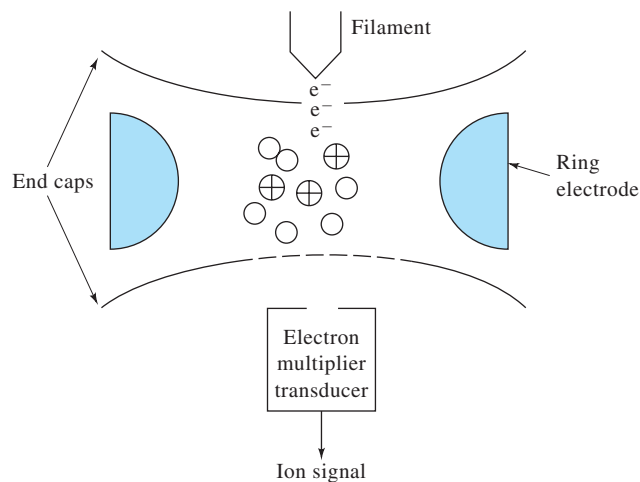


FIGURE 20-17 Ion-trap mass spectrometer. (Adapted from Watson, *Introduction to Mass Spectrometry*, p. 89, Philadelphia: Lippincott-Raven Press, 1997.)

¹⁷W. Paul and H. Steinwedel, *Z. Naturforsch.*, **1953**, 8A, 448.

¹⁸R. E. March and J. F. J. Todd, eds., *Quadrupole Ion Trap Mass Spectrometry*, 2nd ed., Hoboken, NJ: Wiley, 2005; R. E. March, *Mass Spectrom. Rev.*, **2009**, 28, 961; DOI: 10.1002/mas.20250; R. E. March, *Int. J. Mass Spectrom.*, **2000**, 200, 285, DOI: 10.1016/S1387-3806(00)00345-6; C. Hao and R. E. March, *Int. J. Mass Spectrom.*, **2001**, 212, 337, DOI: 10.1016/S1387-3806(01)00465-1.

by the ring. As the radio-frequency voltage is increased, the orbits of heavier ions become stabilized, and those for lighter ions become destabilized, causing them to collide with the wall of the ring electrode.

When the device is operated as a mass spectrometer, ions produced by an EI or a CI source are admitted through a grid in the upper end cap. The ionization source is pulsed so as to create a burst of ions. The ions over a large mass range of interest are trapped simultaneously. The ion trap can store ions for relatively long times, up to 15 minutes for some stable ions. A technique called *mass-selective ejection* is then used to sequentially eject the trapped ions in order of mass by increasing the radio-frequency voltage applied to the ring electrode in a linear ramp. As trapped ions become destabilized, they leave the ring electrode cavity via openings in the lower end cap. The emitted ions then pass into a transducer such as the electron multiplier shown in Figure 20-17.

Ion-trap mass analyzers have the advantage of being rugged, compact, and less costly than other mass analyzers. They have the potential for achieving low detection limits. In addition to EI sources, such analyzers have been interfaced to electrospray ionization sources as well as MALDI sources.

The latest development in ion-trap analyzers is the orbitrap analyzer.¹⁹ Although the principles of orbital trapping have been known since the 1920s, recent developments in ion optics and other areas led to the use of the orbitrap in mass spectrometry. The orbitrap analyzer is an electrostatic trap consisting of an inner electrode that is wide in the middle and tapered at both ends (spindle-shaped) and an outer coaxial split electrode as shown in Figure 20-18. A constant potential is applied between these two electrodes. Since the opposing surfaces of the outer electrode are not parallel to each other, the electric field varies with position reaching a minimum in the center of the trap. Prior to injection, the ions are collected in a C-trap that tightly focuses them in time and space. The ions injected into the trap follow a circular orbit around the inner electrode with an axial oscillation that is proportional to the m/z ratio of the injected ions. An image current is generated, recorded, and decoded using Fourier Transform analysis (see next section).

Orbitraps can have high resolution ($>200,000$) and a high dynamic range. Mass measurements can be made with high accuracy. The orbitrap analyzer is small and less expensive than many other high resolution analyzers. The number of ions that can be held in the analyzer is, however, limited, which can influence detection of minor components in the sample. Another disadvantage is that the orbitrap needs the lowest pressure of any mass analyzer to ensure a large mean-free path for the orbiting ions. Orbitrap analyzers have been commercially available since 2005.

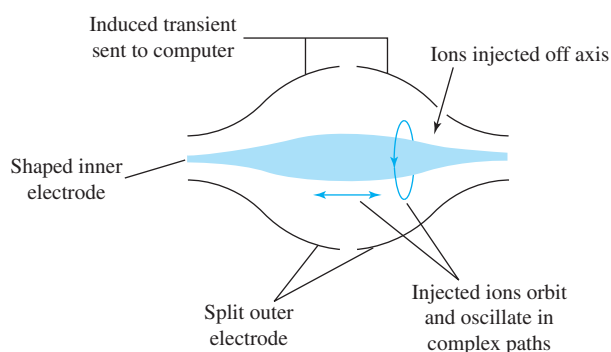


FIGURE 20-18 Orbitrap analyzer. When ions are injected into the analyzer, they follow a complex path composed of rotary motion around the inner electrode and an axial oscillation at right angles to the rotary component. Prior to injection, the ions are collected in a C-trap that tightly focuses them in time and space. (Adapted from J. Greaves and J. Roboz, *Mass Spectrometry for the Novice*, Boca Raton, FL: CRC Press, 2014, p. 84.)

20C-4 Fourier Transform Spectrometers

As was true with infrared and nuclear magnetic resonance instruments, Fourier transform mass spectrometers provide improved signal-to-noise ratios, greater speed, and higher sensitivity and resolution.²⁰ Commercial Fourier transform mass spectrometers appeared on the market in the early 1980s and are offered by a few manufacturers.

The heart of a Fourier transform instrument is an ion trap within which ions can circulate in well-defined orbits for extended periods. Such cavities are constructed to take advantage of a phenomenon known as *ion cyclotron resonance*.

Ion Cyclotron Resonance

When a gaseous ion drifts into or is formed in a strong magnetic field, its motion becomes circular in a plane perpendicular to the direction of the field. The angular frequency of this motion is called the cyclotron frequency, ω_c . Equation 20-8 can be rearranged and solved for v/r , which is the cyclotron frequency in radians per second.

$$\omega_c = \frac{v}{r} = \frac{zeB}{m} \quad (20-11)$$

Note that in a fixed field, the cyclotron frequency depends only on the inverse of the m/z value. Increases in the velocity of an ion will be accompanied by a corresponding increase in the radius of rotation of the ion. A measurement of ω_c can provide an accurate indication of z/m and thus the mass-to-charge ratio of the ion.

¹⁹For a review of orbitrap analyzers, see R. A. Zubarev and A. Makarov, *Anal. Chem.*, **2013**, 85, 5288, DOI: 10.1021/ac4001223.

²⁰For reviews of Fourier transform mass spectrometry, see R. M. A. Heeren, A. J. Kleinnijenhuis, L. A. McDonnell, and T. H. Mize, *Anal. Bioanal. Chem.*, **2004**, 378, 1048, DOI: 10.1007/s00216-003-2446-4; A. G. Marshall, *Int. J. Mass Spectrom.*, **2000**, 200, 331, DOI: 10.1016/S1387-3806(00)00324-9; A. G. Marshall, C. L. Hendrickson, and G. S. Jackson, *Mass Spectrom. Rev.*, **1998**, 17, 1, DOI: 10.1002/(sici)1098-2787(1998)17:1<1::aid-mas1>3.0.co;2-k.

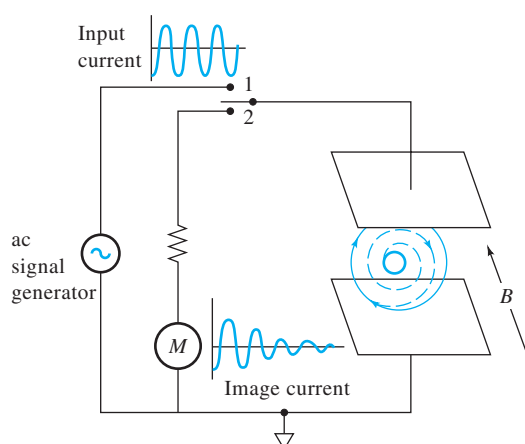


FIGURE 20-19 Path of an ion in a strong magnetic field. Inner solid line represents the original circular path of the ion. Dashed line shows spiral path when switch is moved briefly to position 1. Outer solid line is new circular path when switch is again opened.

An ion trapped in a circular path in a magnetic field is capable of absorbing energy from an ac electric field, provided the frequency of the field matches the cyclotron frequency. The absorbed energy then increases the velocity of the ion (and thus the radius of its path) without disturbing ω_c . This effect is illustrated in Figure 20-19. Here, the original path of an ion trapped in a magnetic field is depicted by the inner solid circle. Brief application of an ac voltage creates a fluctuating field between the plates that interacts with the ion, provided the frequency of the source is resonant with the cyclotron frequency of the ion. Under this circumstance, the velocity of the ion increases continuously as does the radius of its path (see dashed line). When the ac electrical signal is terminated, the radius of the path of the ion again becomes constant, as is suggested by the outer solid circle in the figure.

When the region between the plates in Figure 20-19 contains an ensemble of ions of the same mass-to-charge ratio, application of the ac signal having the cyclotron resonance frequency sets all of the particles into coherent motion in phase with the field. Ions of different cyclotron frequency, that is, those with different mass-to-charge ratios, are unaffected by the ac field.

Measurement of the ICR Signal

The coherent circular motion of resonant ions creates a so-called *image current* that can be conveniently observed after termination of the frequency sweep signal. Thus, if the switch in Figure 20-19 is moved from position 1 to position 2, a current is observed that decreases exponentially with time. This image current is a capacitor current induced by the circular movement of a packet of ions with the same mass-to-charge ratios. For example, as a packet of positive ions approaches the upper plate in Figure 20-19, electrons are attracted from circuit common to this plate, causing a momentary current. As the packet continues

around toward the other plate, the direction of external electron flow is reversed. The magnitude of the resulting alternating current depends on the number of ions in the packet. The frequency of the current (the cyclotron resonance frequency) is characteristic of the mass-to-charge value of the ions in the packet. This current is used in ion cyclotron spectrometers to measure the concentration of ions brought into resonance at various applied signal frequencies.

The induced image current just described decays over a period of a few tenths of a second to several seconds as the coherent character of the circulating packet of ions is lost. Collisions between ions provide the mechanism by which the coherently circulating ions lose energy and the ions return to a condition of thermal equilibrium. This decay of the image current provides a time-domain signal that is similar to the free-induction decay signal encountered in Fourier transform–nuclear magnetic resonance experiments (see Section 19A-3).

Fourier Transform Spectrometers

Fourier transform mass spectrometers are generally equipped with a trapped-ion analyzer cell such as that shown in Figure 20-20. Gaseous sample molecules are ionized in the center of the cell by electrons that are accelerated from the filament through the cell to a collector plate. A pulsed voltage applied at the grid serves as a gate to periodically switch the electron beam on and off. The ions are held in the cell by a 1 to 5 V potential applied to the trap plate. The ions are accelerated by a radio-frequency signal applied to the transmitter plate as shown. The receiver plate is connected to a preamplifier that amplifies the image current. This approach for confining ions is highly efficient, and storage times of up to several minutes have been observed. The dimensions of the cell are not critical but are usually a few centimeters on a side.

The basis of the Fourier transform measurement is illustrated in Figure 20-21. Ions are first generated by a brief electron beam pulse (not shown) and stored in the trapped ion cell. After a brief delay, the trapped ions are subjected to a short radio-frequency pulse that increases linearly in frequency during its lifetime. Figure 20-21a shows a pulse of 5 ms, during which time the frequency increases linearly from 0.070 to 3.6 MHz. After the frequency sweep is discontinued, the image current, induced by the various ion packets, is amplified, digitized, and stored in memory. The time-domain decay signal, shown in Figure 20-21b, is then transformed to yield a frequency-domain signal that can be converted to the mass domain via Equation 20-11. Figure 20-22 illustrates the relationship between a time-domain spectrum, its frequency-domain counterpart, and the resulting mass spectrum.

Fourier transform spectrometers can be interfaced to a variety of ionization sources, including MALDI, electrospray, FAB, and EI-CI. Resolution in Fourier transform mass spectrometry is limited by the precision of the frequency measurement rather than slits or field measurements. The resolution and mass range also depend on the magnitude and stability of the magnetic

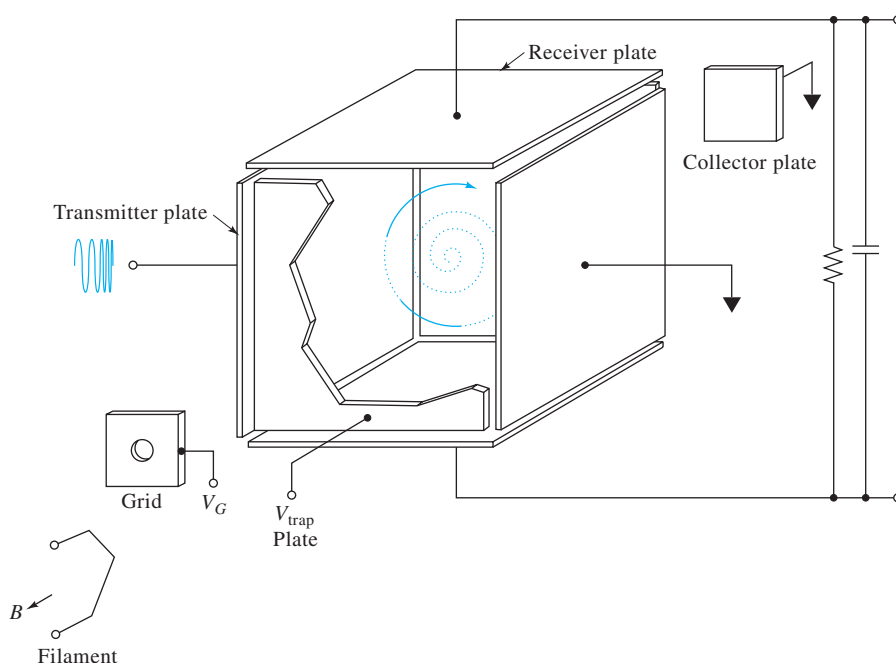


FIGURE 20-20 A trapped-ion analyzer cell. (Reprinted from E. B. Ledford Jr., R. L. White, S. Ghaderi, and C. L. Wilkins, *Anal. Chem.*, **1980**, 52, 1090, DOI: 10.1021/ac50057a022. Copyright 1980 American Chemical Society.)

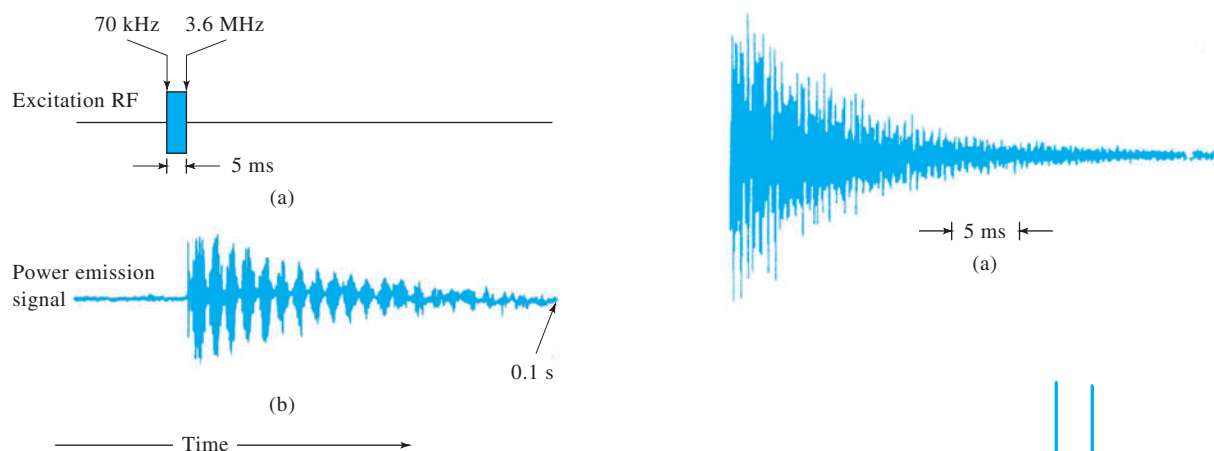


FIGURE 20-21 Schematic showing the timing of (a) the radio-frequency signal and (b) the transient image signal (lower). (Reprinted with permission from R. T. McIver Jr., *Amer. Lab.*, **1980**, 12 (11), 26. Copyright 1980 by International Scientific Communications, Inc.)

field. Because frequency measurements can be made with high precision, extremely high resolution is possible (in excess of 10^6). The accuracy with which mass measurements can be made with Fourier transform instruments is also superb. Fourier transform spectrometers can be very expensive with prices exceeding \$500,000. Commercial models are available with superconducting magnets with fields varying from 1.2 to 12 T. The high-field models are quite useful in biological applications, including proteomics.

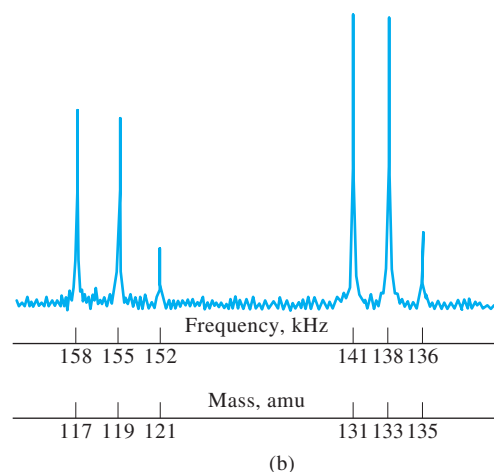


FIGURE 20-22 Time-domain (a) and (b) frequency- or mass-domain spectrum for 1,1,1,2-tetrachloroethane. (Reprinted with permission from E. B. Ledford Jr. et al., *Anal. Chem.*, **1980**, 52, 463, DOI: 10.1021/ac50053a021. Copyright 1980 American Chemical Society.)

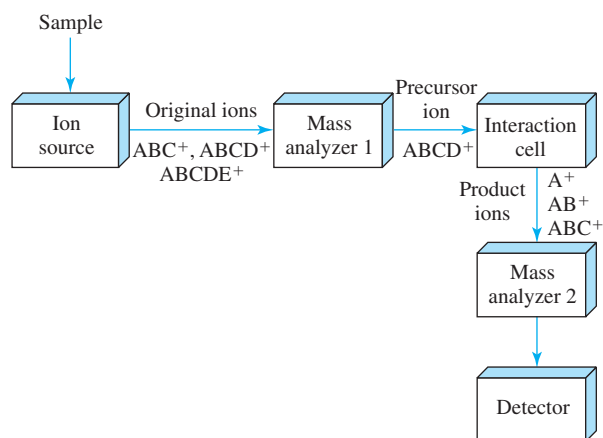


FIGURE 20-23 Block diagram of a tandem mass spectrometer.

20C-5 Tandem Mass Spectrometry

Tandem mass spectrometry, sometimes called *mass spectrometry–mass spectrometry* (MS/MS), is a method that allows the mass spectrum of preselected and fragmented ions to be obtained. The basic idea is illustrated in Figure 20-23. Here, an ionization source, often a soft ionization source, produces ions and some fragments. These are then the input to the first mass analyzer, which selects a particular ion called the *precursor ion* and sends it to the interaction cell. In the interaction cell, the precursor ion can decompose spontaneously, react with a collision gas, or interact with an intense laser beam to produce fragments, called *product ions*. These ions are then mass analyzed by the second mass analyzer and detected by the ion detector.

Types of Tandem Mass Spectra

Several different types of spectra can be obtained from the MS/MS experiment. First, the *product-ion spectrum* can be obtained by scanning mass analyzer 2, while mass analyzer 1 is held constant, acting as a mass selector to select one precursor ion. Figure 20-24 shows product-ion spectra for dibutylphthalate and sulfamethazine. Both compounds produce molecular ions with m/z values of 279. However, the product-ion spectra of the two compounds are very different.

A *precursor ion spectrum* can be obtained, in addition to product-ion spectra, by scanning the first mass analyzer while holding the second mass analyzer constant to detect a given product ion. In a mixture of compounds, those that give the same products are readily identified by precursor ion spectra. Closely related compounds often give several of the same product ions, so that this method of operation provides a measure of identity and concentration of the *members of a class* of closely related compounds. Consider, for example, a mixture of ABCD, BCDA, IJKL, and IJMN in the sample. To identify species containing the IJ group, the second analyzer is set to the mass corresponding to the IJ^+ ion and the molecular ions $ABCD^+$, $BCDA^+$, $IJKL^+$, and $IJMN^+$ are sequentially selected by the first analyzer. Ion signals in the detector would be observed

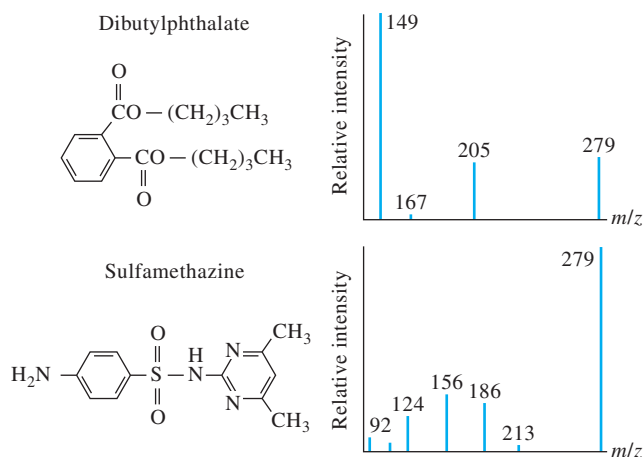


FIGURE 20-24 Product-ion spectra for dibutylphthalate and sulfamethazine obtained after the protonated precursor ion peaks at 279 Da were isolated by the first mass analyzer of an MS/MS instrument. (Reprinted from K. L. Busch and G. C. DiDonato, *Amer. Lab.*, **1986**, 18 (8), 17. Copyright 1986 by International Scientific Communications, Inc.)

only when $IJKL^+$ and $IJMN^+$ are selected by the first analyzer, indicating the presence of the IJ group.

By scanning both analyzers simultaneously with an offset in mass between them, a *neutral loss spectrum* can be obtained. This gives the identity of those precursor ions that undergo the same loss such as the loss of a H_2O or CO neutral. Finally, by scanning mass analyzer 1 and obtaining the product-ion spectrum for each selected precursor ion, a complete *three-dimensional MS/MS spectrum* can be obtained.

Dissociative Interactions in the Interaction Cell

Several types of interactions can be used to produce fragmentation in the interaction cell. In some cases, the ions selected by mass analyzer 1 in Figure 20-23 are themselves *metastable* and decompose into fragments after a certain time. In general, however, the kinetics of the decomposition process can greatly limit the applicability and sensitivity of the process. In such cases, fragmentation can be induced by adding a collision gas to the interaction cell so that interaction with precursor ions occurs, leading to decomposition into product ions. In this case, the cell is called a *collision cell*, and the interactions are termed *collisionally activated dissociation* (CAD) or alternatively *collision-induced dissociation* (CID).

Another type of interaction is *surface-induced dissociation* (SID), in which precursor ions interact with a surface to induce dissociation. Ions have been reflected off cell walls or trapping plates to increase their internal energy and promote dissociation. Chemically modified surfaces such as thin films have also been used.

Another dissociation technique that has been applied to large multiply charged ions is *electron-capture dissociation* (ECD), in which precursor ions capture a low-energy electron to

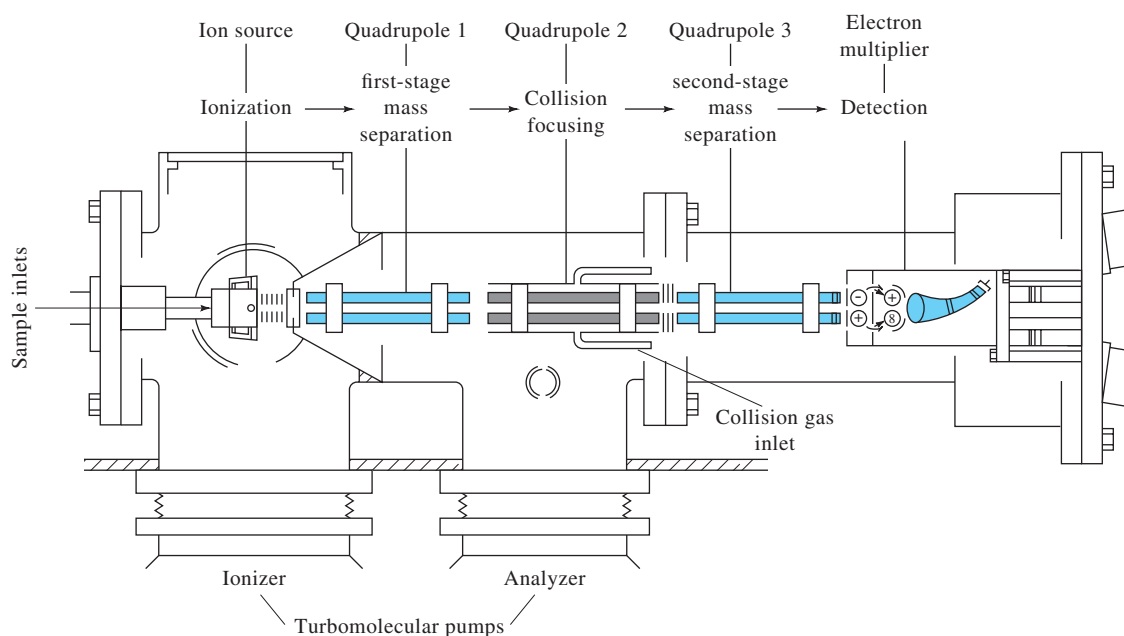


FIGURE 20-25 Schematic of a triple quadrupole mass spectrometer. (Courtesy of Thermo-Finnigan Corp.)

produce an intermediate that rapidly dissociates. In some cases, a background gas is added to aid in the dissociation process.

Photo-induced dissociation (PID) is another process to stimulate decomposition of precursor ions. In most PID methods, an intense laser beam is used in the interaction cell to promote the dissociation. A difficulty with PID is that the ion beam and photon beam must overlap in the interaction region for a time long enough for absorption and bond rupture to occur. In some cases ion-trap cells have been used to allow for long periods of overlap.

Instrumentation for Tandem Mass Spectrometry

Tandem mass spectrometry has been implemented in a number of ways.²¹ These can be classified as *tandem in space* and *tandem in time*.²²

Tandem-in-Space Spectrometers. In tandem-in-space instruments, two independent mass analyzers are used in two different regions in space. The triple quadrupole mass spectrometer is the most common of these instruments. In commercial triple quadrupole instruments, such as the instrument illustrated in Figure 20-25, the sample is introduced into a soft ionization source, such as a CI or FAB source. The ions are then accelerated into quadrupole 1 (Q), which is an ordinary quadrupole mass filter. The selected fast-moving ions pass into quadrupole 2 (q), which is a collision chamber where dissociation

of the ions selected by quadrupole 1 occurs. This quadrupole is operated in a radio-frequency-only mode in which no dc voltage is applied across the rods. This mode basically traps the precursor and product ions in a relatively high concentration of collision gas so that CAD can occur. Quadrupole 3 (Q) then allows mass analysis of the product ions formed in the collision cell. The configuration is known as the QqQ configuration. Recently, the triple-quadrupole mass spectrometer has been miniaturized to an overall size of about 9.5 inches in length.²³ Such a small analyzer could be widely used in field applications.

Sector instruments and hybrid quadrupole-sector instruments have also been used in a tandem manner. The first tandem mass spectrometers were sector instruments that combined an electric sector spectrometer with a magnetic sector spectrometer, either in forward geometry (electric sector followed by magnetic sector, or EB) or in reverse geometry (magnetic sector followed by electric sector, or BE). The reverse geometry is sometimes called a *mass-analyzed ion kinetic energy spectrometer (MIKES)*. Such instruments were not very efficient, but they allowed the principles of tandem mass spectrometry to be demonstrated.

Hybrid instruments include the BEqQ spectrometer (magnetic sector, B; electric sector, E; RF-only quadrupole, q; quadrupole mass analyzer, Q) and the BTOF (magnetic sector, B; TOF analyzer, TOF) spectrometer. The QqTOF spectrometer is similar to the triple quadrupole (QqQ) instrument except that the final quadrupole mass analyzer is replaced with a TOF analyzer. In another variant, the Qq section can be replaced by a quadrupole ion trap to yield an ion-trap-TOF instrument.

²¹For a description of these implementations, see J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., Chichester, UK: Wiley, 2007, pp. 184–196.

²²See J. W. Hager, *Anal. Bioanal. Chem.*, **2004**, 378, 845, DOI: 10.1007/s00216-003-2287-1; S. A. McLuckey and J. M. Wells, *Chem. Rev.*, **2001**, 101, 571, DOI: 10.1021/cr990087a.

²³C. Wright et al., *Anal. Chem.*, **2015**, 87, 3115, DOI: 10.1021/acs.analchem.5b00311.

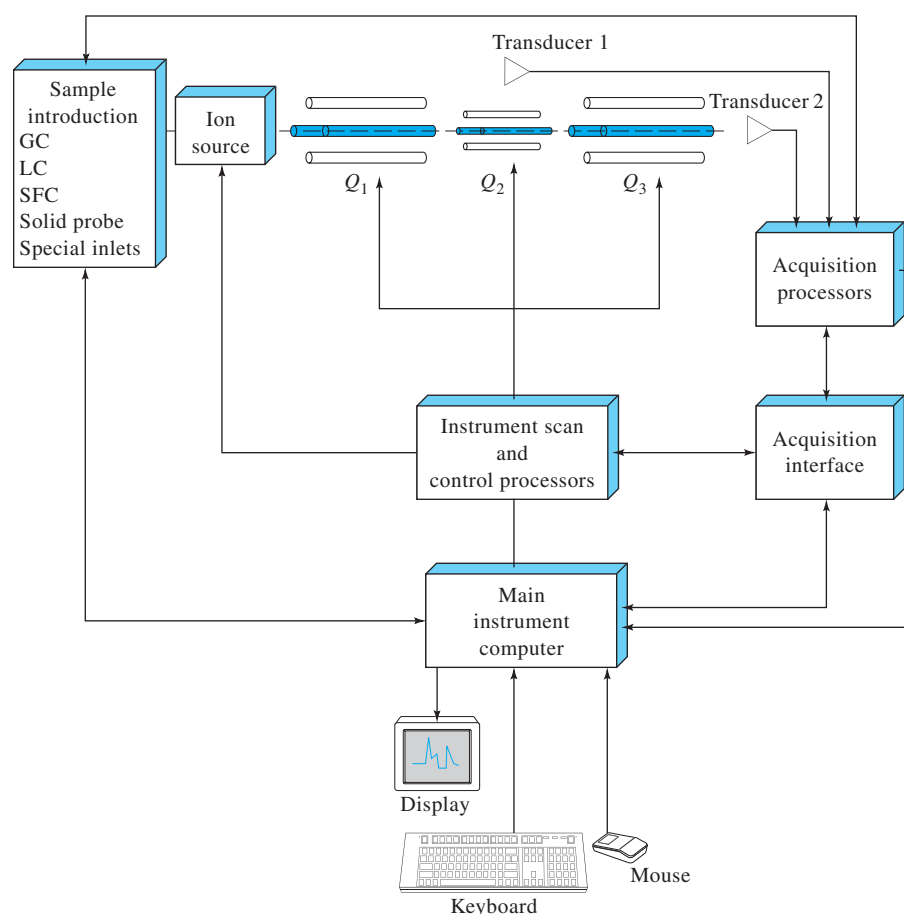


FIGURE 20-26 Instrument control and data processing for a triple quadrupole mass spectrometer. SFC = supercritical fluid chromatography (see Chap. 29).

A final type of tandem-in-space spectrometer is the TOF-TOF spectrometer, in which a TOF instrument followed by a timed ion selector separates the precursor ions. A collision cell then induces fragmentation, and the product ions are mass analyzed in the final TOF stage.²⁴ Mass resolution of several thousand was reported.

Tandem-in-Time Spectrometers. Tandem-in-time instruments form the ions in a certain spatial region and then at a later time expel the unwanted ions and leave the selected ions to be dissociated and mass analyzed in the same spatial region. This process can be repeated many times over to perform not only MS/MS experiments, but also MS/MS/MS and MSⁿ experiments. Fourier transform ICR and quadrupole ion-trap instruments are well suited for performing MSⁿ experiments. In principle, tandem-in-time spectrometers can perform MS/MS experiments much more simply than tandem-in-space instruments because of the difficulty in providing different ion focal positions in the latter. Although tandem-in-time spectrometers

can readily provide product-ion scans, other scans, such as precursor ion scans and neutral loss scans, are much more difficult to perform than they are with tandem in space instruments.

20C-6 Computerized Mass Spectrometers

Computers are an integral part of modern mass spectrometers. A characteristic of a mass spectrum is the wealth of structural data that it provides. For example, a molecule with a molecular mass of 500 may be fragmented by an EI source into 100 or more different ions, each of which leads to a discrete spectral peak. For a structural determination, the heights and mass-to-charge ratios of each peak must be determined, stored, and ultimately displayed. Because the amount of information is so large, it is essential that acquisition and processing be rapid; computers are ideally suited for these tasks. Moreover, for mass spectral data to be useful, several instrumental variables must be closely controlled or monitored during data collection. Computers are much more efficient than a human operator in exercising such controls.

Figure 20-26 is a block diagram of the computerized control and data-acquisition system of a triple quadrupole mass spectrometer. This figure shows two features encountered in any modern instrument. The first is a computer that serves as

²⁴K. F. Medzihradsky, J. M. Campbell, M. A. Baldwin, A. M. Falick, P. Juhasz, M. L. Vestal, and A. L. Burlingame, *Anal. Chem.*, **2000**, 72, 552, DOI: 10.1021/ac990809y.

the main instrument controller. The operator communicates via a keyboard with the spectrometer by selecting operating parameters and conditions via easy-to-use interactive software. The computer also controls the programs responsible for data manipulations and output. The second feature common to almost all instruments is a set of microprocessors (often as many as six) that are responsible for specific aspects of instrument control and the transmission of information between the computer and spectrometer.

The interface between a mass spectrometer and a computer usually has provisions for digitizing the amplified ion-current signal plus several other signals that are used for control of instrumental variables. Examples of the latter are source temperature, accelerating voltage, scan rate, and magnetic field strength or quadrupole voltages.

The digitized ion-current signal ordinarily requires considerable processing before it is ready for display. First, the peaks must be normalized, a process by which the height of each peak relative to some reference peak is calculated. Most often the *base peak*, which is the largest peak in a spectrum, serves as the reference and is arbitrarily assigned a peak height of 100 (sometimes 1000). The m/z value for each peak must also be determined. This assignment is frequently based on the time of the peak's appearance and the scan rate. Data are acquired as intensity versus time during a carefully controlled scan of the magnetic or electric fields. Conversion from time to m/z requires careful periodic calibration; for this purpose, perfluorotri-*n*-butylamine (PFTBA) or perfluorokerosene is often used as a standard. For high-resolution work, the standard may be admitted with the sample. The computer is programmed to then recognize and use the peaks of the standard as references for mass assignments. For low-resolution instruments, the calibration must generally be obtained separately from the sample, because of the likelihood of peak overlaps.

With most systems the computer stores all spectra and related information on a disk. In routine applications, bar graphs of the normalized spectra can be sent directly to a printer or display. However, in many cases, the user will use data-reduction software to extract specific information prior to producing a printed copy of a spectrum. Figure 20-27 is an example of the printout from a computerized mass spectrometer. The first and next-to-last columns in the table list m/z values in increasing order. The second and last columns contain the corresponding ion currents normalized to the largest peak found at mass 156. The current for this ion is assigned the number of 100, and all other peaks are expressed relative to this one. Thus, the height of the peak at mass 141 is 53% of the base peak.

As is true with infrared and nuclear magnetic resonance spectroscopy, large libraries of mass spectra (>150,000 entries) are available in computer-compatible formats.²⁵ Most commercial mass spectrometer computer systems have the ability to rapidly

Sample: Unknown compound

Date: 11/15/2004

m/z	Rel. Int.	m/z	Rel. Int.
41	9	141	53
43	14	142	4
55	5	155	6
69	4	156	100
71	5	157	21
98	5	197	5

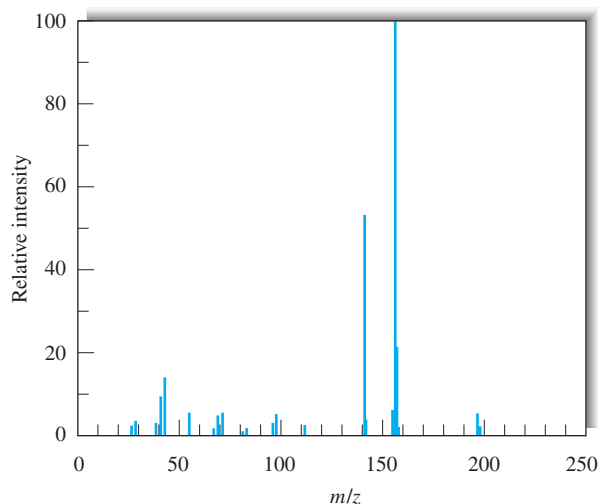


FIGURE 20-27 A computer display of mass-spectral data. The compound was isolated from a blood serum extract by chromatography. The spectrum showed it to be the barbiturate pentobarbital.

search all or part of such files for spectra that match or closely match the spectrum of an analyte.

20D APPLICATIONS OF MOLECULAR MASS SPECTROMETRY

The applications of molecular mass spectrometry are so numerous and widespread that describing them adequately in a brief space is not possible. Table 20-5 lists several of these applications to provide some idea of the capabilities of mass spectrometry. In this section we describe a few of the most widely used and important of these applications.

20D-1 Identification of Pure Compounds

The mass spectrum of a pure compound provides several kinds of data that are useful for its identification.²⁶ The first is the molecular mass of the compound, and the second is its molecular formula. In addition, study of fragmentation patterns revealed by the mass spectrum often provides information about

²⁵For example, see F. W. McLafferty, *Wiley Registry of Mass Spectral Data*, 10th ed., New York: Wiley, 2012. The registry is available online and combined with the NIST/EPA/NIH Mass Spectral Library.

²⁶R. M. Silverstein, F. X. Webster, D. Kiemle, and D. L. Bryce, *The Spectrometric Identification of Organic Compounds*, 8th ed., New York: Wiley, 2015; F. W. McLafferty and F. Turecek, *Interpretation of Mass Spectra*, 4th ed., Mill Valley, CA: University Science Books, 1993.

TABLE 20-5 Applications of Molecular Mass Spectrometry

1. Elucidation of the structure of organic and biological molecules
2. Determination of the molecular mass of peptides, proteins, and oligonucleotides
3. Identification of components in thin-layer and paper chromatograms
4. Determination of amino acid sequences in sample of polypeptides and proteins
5. Detection and identification of species separated by chromatography and capillary electrophoresis
6. Identification of drugs of abuse and metabolites of drugs of abuse in blood, urine, and saliva
7. Monitoring gases in patient's breath during surgery
8. Testing for the presence of drugs in blood in racehorses and in Olympic athletes
9. Dating archaeological specimens
10. Analyses of aerosol particles
11. Determination of pesticide residues in food
12. Monitoring volatile organic species in water supplies

the presence or absence of various functional groups. Finally, the actual identity of a compound can often be established by comparing its mass spectrum with those of known compounds until a close match is realized.

Molecular Masses from Mass Spectra

For compounds that can be ionized to give a molecular ion or a protonated or a deprotonated molecular ion by one of the methods described earlier, the mass spectrometer is an unsurpassed tool for the determination of molecular mass. This determination, of course, requires the identification of the molecular ion peak, or in some cases, the $(M + 1)^+$ or the $(M - 1)^+$ peak. The location of the peak on the abscissa then gives the molecular mass with an accuracy that cannot be realized easily by any other method.

To determine a molecular mass by mass spectrometry, the identity of the molecular ion peak must be known. Caution is therefore always advisable, particularly with EI sources, when the molecular ion peak is absent or small relative to impurity peaks. When there is doubt, additional spectra by chemical, field, and desorption ionization are particularly useful.

Molecular Formulas from Exact Molecular Masses

Molecular formulas can be determined from the mass spectrum of a compound, provided the molecular ion peak can be identified and its *exact* mass determined. This application, however, requires a high-resolution instrument capable of detecting mass differences of a few thousandths of a dalton. Consider, for example, the mass-to-charge ratios of the molecular ions of the following compounds: purine, $C_5H_4N_4$ ($\mathcal{M} = 120.044$); benzamidine, $C_7H_8N_2$ ($\mathcal{M} = 120.069$); ethyltoluene, C_9H_{12} ($\mathcal{M} = 120.096$); and acetophenone, C_8H_8O ($\mathcal{M} = 120.058$). If the measured mass of the molecular ion peak is $120.070 (\pm 0.005)$, then all but $C_7H_8N_2$ are excluded as possible formulas. Note that the precision in this example is about 40 ppm. Uncertainties on the order of a few parts per million are routinely achievable with high-resolution, double-focusing instruments. Tables that list all reasonable combinations of C, H, N, and O by molecular

mass to the third or fourth decimal place have been compiled.²⁷ A small portion of such a compilation is shown in the fifth column of Table 20-6.

Molecular Formulas from Isotopic Ratios

The data from a low-resolution instrument that can discriminate only between ions differing in mass by whole mass numbers can also yield useful information about the formula of a compound, provided that the molecular ion peak is sufficiently intense that its height and the heights of the $(M + 1)^+$ and $(M + 2)^+$ isotope peaks can be determined accurately. Example 20-5 illustrates this type of analysis.

The use of relative isotope peak heights for the determination of molecular formulas is greatly expedited by the tables referred to in note 21 and by compilations available for computer analysis. In Table 20-6, a listing of all reasonable combinations of C, H, O, and N is given for mass numbers 83 and 84 (the original tables extend to mass number 500). The heights of the $(M + 1)^+$ and $(M + 2)^+$ peaks reported as percentages of the height of the M^+ peak are tabulated. If a reasonably accurate experimental determination of these percentages can be made, a likely formula can be deduced. For example, a molecular ion peak at mass 84 with $(M + 1)^+$ and $(M + 2)^+$ values of 5.6 and 0.3% of M^+ suggests a compound having the formula C_5H_8O .

EXAMPLE 20-5

Calculate the ratios of the $(M + 1)^+$ to M^+ peak heights for the following two compounds: dinitrobenzene, $C_6H_4N_2O_4$ ($\mathcal{M} = 168$), and an olefin, $C_{12}H_{24}$ ($\mathcal{M} = 168$).

Solution

From Table 20-3, we see that for every 100 ^{12}C atoms there are 1.08 ^{13}C atoms. Because nitrobenzene contains six carbon

²⁷J. H. Beynon and A. E. Williams, *Mass and Abundance Tables for Use in Mass Spectrometry*, New York: Elsevier, 1963.

TABLE 20-6 Isotopic Abundance Percentages and Molecular Masses for Various Combinations of C, H, O, and N

Abundance, % M Peak Height				
	Formula	$\mathcal{M} + 1$	$\mathcal{M} + 2$	Molecular Mass
$\mathcal{M} = 83$	$\text{C}_2\text{HN}_3\text{O}$	3.36	0.24	83.0120
	$\text{C}_2\text{H}_3\text{N}_4$	3.74	0.06	83.0359
	C_3HNO_2	3.72	0.45	83.0007
	$\text{C}_3\text{H}_3\text{N}_2\text{O}$	4.09	0.27	83.0246
	$\text{C}_3\text{H}_5\text{N}_3$	4.47	0.08	83.0484
	$\text{C}_4\text{H}_3\text{O}_2$	4.45	0.48	83.0133
	$\text{C}_4\text{H}_5\text{NO}$	4.82	0.29	83.0371
	$\text{C}_4\text{H}_7\text{N}_2$	5.20	0.11	83.0610
	$\text{C}_5\text{H}_7\text{O}$	5.55	0.33	83.0497
	$\text{C}_5\text{H}_9\text{N}$	5.93	0.15	83.0736
	C_6H_{11}	6.66	0.19	83.0861
$\mathcal{M} = 84$	CN_4O	2.65	0.23	84.0073
	$\text{C}_2\text{N}_2\text{O}_2$	3.00	0.43	83.9960
	$\text{C}_2\text{H}_2\text{N}_3\text{O}$	3.38	0.24	84.0198
	$\text{C}_2\text{H}_4\text{N}_4$	3.75	0.06	84.0437
	C_3O_3	3.36	0.64	83.9847
	$\text{C}_3\text{H}_2\text{NO}_2$	3.73	0.45	84.0085
	$\text{C}_3\text{H}_4\text{N}_2\text{O}$	4.11	0.27	84.0324
	$\text{C}_3\text{H}_6\text{N}_3$	4.48	0.08	84.0563
	$\text{C}_4\text{H}_4\text{O}_2$	4.46	0.48	84.0211
	$\text{C}_4\text{H}_6\text{NO}$	4.84	0.29	84.0449
	$\text{C}_4\text{H}_8\text{N}_2$	5.21	0.11	84.0688
	$\text{C}_5\text{H}_8\text{O}$	5.57	0.33	84.0575
	$\text{C}_5\text{H}_{10}\text{N}$	5.94	0.15	84.0814
	C_6H_{12}	6.68	0.19	84.0939
	C_7	7.56	0.25	84.0000

Taken from R. M. Silverstein, G. C. Bassler, and T. C. Morrill, *Spectrometric Identification of Organic Compounds*, 4th ed., p. 49, New York: Wiley, 1981.

atoms, we would expect each of $6.48 = 6 \times 1.08$ molecules of nitrobenzene to have one ^{13}C atom for every 100 molecules having none. Thus, from this effect alone the $(\mathcal{M} + 1)^+$ peak will be 6.48% of the $\mathcal{M}^+$ peak. The isotopes of the other elements also contribute to this peak; we may tabulate their effects as follows:

$\text{C}_6\text{H}_4\text{N}_2\text{O}_4$	
^{13}C	$6 \times 1.08 = 6.48\%$
^2H	$4 \times 0.015 = 0.060\%$
^{15}N	$2 \times 0.37 = 0.74\%$
^{17}O	$4 \times 0.04 = 0.16\%$
$(\mathcal{M} + 1)^+/\mathcal{M}^+ = 7.44\%$	

$\text{C}_{12}\text{H}_{24}$	
^{13}C	$12 \times 1.08 = 12.96\%$
^2H	$24 \times 0.015 = 0.36\%$
$(\mathcal{M} + 1)^+/\mathcal{M}^+ = 13.32\%$	

Thus, if the heights of the $\mathcal{M}^+$ and $(\mathcal{M} + 1)^+$ peaks can be measured, it is possible to discriminate between these two compounds that have identical whole-number molecular masses.

The isotopic ratio is particularly useful for the detection and estimation of the number of sulfur, chlorine, and bromine atoms in a molecule because of the large contribution they make to the $(\mathcal{M} + 2)^+$ (see Table 20-3). For example, an $(\mathcal{M} + 2)^+$ that is about 65% of the $\mathcal{M}^+$ peak is strong evidence for a molecule

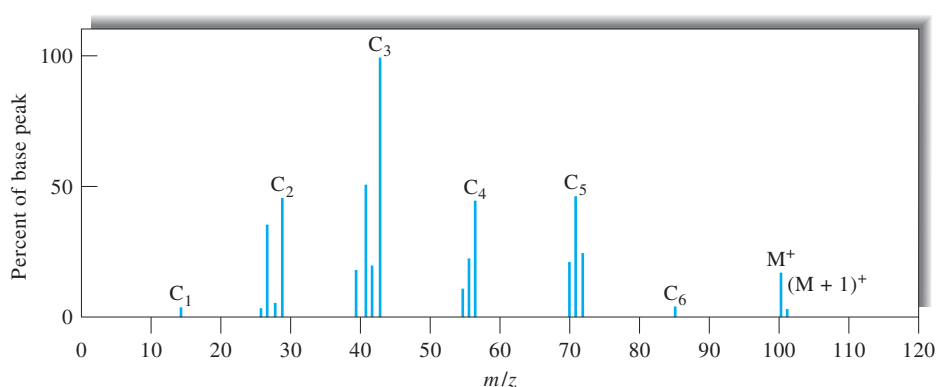


FIGURE 20-28 EI spectrum of *n*-heptanal. The peaks labeled C₆, C₅, . . . , C₁ correspond to the successive losses of a CH₂ group.

containing two chlorine atoms; an (M + 2)⁺ peak of 4%, on the other hand, suggests the presence of one atom of sulfur.

Structural Information from Fragmentation Patterns

Systematic studies of fragmentation patterns for pure substances have led to rational guidelines to predict fragmentation mechanisms and a series of general rules helpful in interpreting spectra.²⁸ It is seldom possible (or desirable) to account for all of the peaks in the spectrum. Instead, characteristic patterns of fragmentation are sought. For example, the spectrum in Figure 20-28 is characterized by clusters of peaks differing in mass by 14. Such a pattern is typical of straight-chain paraffins, in which cleavage of adjacent carbon-carbon bonds results in the loss of successive CH₂ groups having this mass. Quite generally, the most stable hydrocarbon fragments contain three or four carbon atoms, and the corresponding peaks are thus the largest.

Alcohols usually have a very weak or nonexistent molecular ion peak but often lose water to give a strong peak at (M – 18)⁺. Cleavage of the C—C bond next to an oxygen is also common, and primary alcohols always have a strong peak at mass 31 due to the CH₂OH⁺. Extensive compilations of generalizations concerning the use of mass spectral data for the identification of organic compounds are available, and the interested reader should consult the references in note 22.

Compound Identification from Comparison Spectra

Generally, after determining the molecular mass of the analyte and studying its isotopic distribution and fragmentation patterns, the experienced mass spectroscopist is able to narrow the

possible structures down to a handful. When reference compounds are available, final identification is then based on a comparison of the mass spectrum of the unknown with spectra for authentic samples of the suspected compounds. The procedure is based on the assumptions that (1) mass fragmentation patterns are unique and (2) experimental conditions can be sufficiently controlled to produce reproducible spectra. The first assumption often is not valid for spectra of stereo- and geometric isomers and occasionally is not valid for certain types of closely related compounds. The probability that different compounds will yield the same spectrum becomes markedly smaller as the number of spectral peaks increases. Assumption 2 can also be problematic. EI spectra are fairly reproducible from laboratory to laboratory. However, spectra from other sources can vary significantly. For this reason, electron-ionization is the method of choice for spectral comparison and for building spectral libraries.

Unfortunately, heights of mass spectral peaks strongly depend on such variables as the energy of the electron beam, the location of the sample with respect to the beam, the sample pressure and temperature, and the general geometry of the mass spectrometer. As a result, significant variations in relative abundances are observed for spectra acquired in different laboratories and from different instruments. Nevertheless, it has proven possible in a remarkably large number of cases to identify unknowns from library spectra obtained with a variety of instruments and operating conditions. Generally, however, it is desirable to confirm the identity of a compound by comparing its spectrum to the spectrum of an authentic compound obtained with the same instrument under identical conditions.

Computerized Library Search Systems. Although libraries of mass spectral data are available in text form,²⁹ most modern mass spectrometers are equipped with highly efficient computerized library search systems. There are two basic types of



Simulation: Learn more about MS spectral interpretation at www.tinyurl.com/skoogpia7

²⁸For example, see R. M. Silverstein, F. X. Webster, D. Kiemle, and D. L. Bryce, *The Spectrometric Identification of Organic Compounds*, 8th ed., New York: Wiley, 2015.

²⁹F. W. McLafferty and D. A. Stauffer, *The Wiley/NBS Registry of Mass Spectral Data*, 7 vols., New York: Wiley, 1989.

mass spectral libraries: large comprehensive ones and small specific ones. The largest commercially available mass spectral library (>300,000 spectra) is marketed by John Wiley and Sons.³⁰ A unique feature of this compilation is that it is available on CD-ROM and can be searched on a personal computer. Small libraries usually contain a few hundred to a few thousand spectra for application to a limited area, such as pesticide residues, drugs, or forensics. Small libraries are often part of the equipment packages offered by instrument manufacturers, and it is almost always possible for the instrument user to generate a library or to add to an existing library. Mass spectra for some 33,000 compounds are available from the National Institute of Standards and Technology (NIST) on the Internet.³¹

For large numbers of spectra, such as are obtained when a mass spectrometer is coupled with a chromatograph for identifying components of a mixture, the instrument's computer system can be used to perform a library search on all, or any subset, of the mass spectra associated with a particular sample. The results are reported to the user, and if desired, the reference spectra can be displayed on a monitor or printed for visual comparison.

20D-2 Analysis of Mixtures by Hyphenated Mass Spectral Methods

Although ordinary mass spectrometry is a powerful tool for the identification of pure compounds, its usefulness for analysis of all but the simplest mixtures is limited because of the immense number of fragments of differing m/z values produced. It is often impossible to interpret the resulting complex spectrum. For this reason, chemists have developed methods in which mass spectrometers are coupled with various efficient separation devices in so-called *hyphenated methods*.

Chromatography–Mass Spectrometry

Gas chromatography–mass spectrometry (GC/MS) has become one of the most powerful tools available for the analysis of complex organic and biochemical mixtures. In this application, spectra are collected for compounds as they exit from a chromatographic column. These spectra are then stored in a computer for subsequent processing. Mass spectrometry has also been coupled with liquid chromatography (LC/MS) for the analysis of samples that contain nonvolatile constituents. A major problem that had to be overcome in the development of both of these hyphenated methods is that the sample in the chromatographic column is highly diluted by the gas or liquid carrying it through the column. Thus, methods had to be developed for removing the diluent before introducing the sample into the mass spectrometer. Instruments and applications of GC/MS and LC/MS are described in Sections 27B-4 and 28C-6, respectively.

Capillary Electrophoresis–Mass Spectrometry

The first report on coupling capillary electrophoresis with mass spectrometry was published in 1987.³² Since then, this hyphenated method has become a powerful and important tool in the analysis of large biopolymers, such as proteins, polypeptides, and DNA species. In most of the applications reported to date, the capillary effluent is passed directly into an electrospray ionization device, and the products then enter a quadrupole mass filter for analysis. Continuous flow FAB has also been used for ionization in some applications. Capillary electrophoresis–mass spectrometry is discussed in more detail in Section 30B-4.

Applications of Tandem Mass Spectrometry

Dramatic progress in the analysis of complex organic and biological mixtures began when the mass spectrometer was first combined with gas chromatography and subsequently with liquid chromatography. Tandem mass spectrometry offers some of the same advantages as GC/MS and LC/MS but is significantly faster. Separations on a chromatographic column are achieved in a time scale of a few minutes to hours, but equally satisfactory separations in tandem mass spectrometers are complete in milliseconds. In addition, the chromatographic techniques require dilution of the sample with large excesses of a mobile phase and subsequent removal of the mobile phase, which greatly enhances the probability of introduction of interferences. Consequently, tandem mass spectrometry is potentially more sensitive than either of the hyphenated chromatographic techniques because the chemical noise associated with its use is generally smaller. A current disadvantage of tandem mass spectrometry with respect to the two chromatographic procedures is the greater cost of the required equipment; this gap appears to be narrowing as tandem mass spectrometers gain wider use.

For some complex mixtures the combination of GC or LC and MS does not provide enough resolution. In recent years, it has become feasible to couple chromatographic methods with tandem mass spectrometers to form GC/MS/MS and LC/MS/MS systems. There have also been reports of LC/MSⁿ instruments.³³

To date, tandem mass spectrometry has been applied to the qualitative and quantitative determination of the components of a wide variety of complex materials encountered in nature and industry. Some examples include the identification and determination of drug metabolites, insect pheromones, alkaloids in plants, trace contaminants in air, polymer sequences, petrochemicals, polychlorinated biphenyls, prostaglandins, diesel exhausts, and odors in air. One of the most promising areas of

³⁰See reference 21.

³¹<http://webbook.nist.gov/>.

³²J. A. Olivares, N. T. Nguyen, N. T. Yonker, and R. D. Smith, *Anal. Chem.*, **1987**, 59, 1230, DOI: 10.1021/ac00135a034. See also, D. C. Simpson and R. D. Smith, *Electrophoresis*, **2005**, 26, 1291, DOI: 10.1002/elps.200410132.

³³See, for example, J. C. A. Wuilloud, S. R. Gratz, B. M. Gamble, and K. A. Wolnik, *Analyst*, **2004**, 129, 150, DOI: 10.1039/b311030c; E. W. Taylor, W. Jia, M. Bush, and G. D. Dollinger, *Anal. Chem.*, **2002**, 74, 3232, DOI: 10.1021/ac020035c; L. Howells and M. J. Sauer, *Analyst*, **2001**, 126, 155, DOI: 10.1039/b008305o.

applications is that of *proteomics*, the study of proteins produced by a cell or by a species.³⁴

20E QUANTITATIVE APPLICATIONS OF MASS SPECTROMETRY

Applications of mass spectrometry for quantitative analyses fall into two categories. The first involves the quantitative determination of molecular species or types of molecular species in organic, biological, and occasionally inorganic samples. The second involves the determination of the concentration of elements in inorganic and, less commonly, organic and biological samples. In the first type of analysis, all of the ionization sources listed in Table 20-1 are used. Mass spectroscopic elemental analyses, which are discussed in detail in Chapter 11, are currently based largely on inductively coupled plasma sources, although glow discharge, radio-frequency spark, laser, thermal, and secondary ion sources have also found use.

20E-1 Quantitative Determination of Molecular Species

Mass spectrometry has been widely applied to the quantitative determination of one or more components of complex organic (and sometimes inorganic) systems such as those encountered in the petroleum and pharmaceutical industries and in studies of environmental problems. Currently, such analyses are usually performed by passing the sample through a chromatographic or capillary electrophoretic column and into the spectrometer. With the spectrometer set at a suitable m/z value, the ion current is then recorded as a function of time. This technique is termed *selected ion monitoring*. In some instances, currents at three or four m/z values are monitored in a cyclic manner by rapid switching from one peak to another. The plot of the data consists of a series of peaks, with each appearing at a time that is characteristic of one of the several components of the sample that yields ions of the chosen value or values for m/z . Generally, the areas under the peaks are directly proportional to the component concentrations and are used for determinations. In this type of procedure, the mass spectrometer simply serves as a sophisticated selective detector for quantitative *chromatographic* or *electrophoretic analyses*. Further details on quantitative gas and liquid chromatography are given in Sections 27B-4 and 28C-6. The use of a mass spectrometer as a detector in capillary electrophoresis is described in Section 30B-4.

In the second type of quantitative mass spectrometry for molecular species, analyte concentrations are obtained directly from the heights of the mass spectral peaks. For simple

mixtures, it is sometimes possible to find peaks at unique m/z values for each component. Under these circumstances, calibration curves of peak heights versus concentration can be prepared and used for analysis of unknowns. More accurate results can ordinarily be realized, however, by incorporating a fixed amount of an internal standard substance in both samples and calibration standards. The ratio of the peak intensity of the analyte species to that of the internal standard is then plotted as a function of analyte concentration. The internal standard tends to reduce uncertainties arising in sample preparation and introduction. These uncertainties are often a major source of indeterminate error with the small samples needed for mass spectrometry. Internal standards are also used in GC/MS and LC/MS. For these techniques, the ratio of peak areas serves as the analytical variable.

A convenient type of internal standard is a stable, isotopically labeled analog of the analyte. Usually, labeling involves preparation of samples of the analyte in which one or more atoms of deuterium, carbon-13, or nitrogen-15 have been incorporated. It is then assumed that during the analysis the labeled molecules behave in the same way as do the unlabeled ones. The mass spectrometer easily distinguishes between the two. Another type of internal standard is a homolog of the analyte that yields a reasonably intense ion peak for a fragment that is chemically similar to the analyte fragment being measured.

With low-resolution instruments, it is seldom possible to locate peaks that are unique to each component of a mixture. In this situation, it is still possible to complete an analysis by collecting intensity data at a number of m/z values that equal or exceed the number of sample components. Simultaneous equations are then developed that relate the intensity of each m/z value to the contribution made by each component to this intensity. Solving these equations then provides the desired quantitative information. Alternatively, chemometric methods such as partial least squares or principal component analysis are used.

Precision and Accuracy

The precision of quantitative mass spectral measurements by the procedure just described usually ranges between 2% and 10% relative. The analytical accuracy varies considerably depending on the complexity of the mixture being analyzed and the nature of its components. For gaseous hydrocarbon mixtures containing five to ten components, absolute errors of 0.2 to 0.8 mole percent are typical.

Applications

The early quantitative applications of mass spectrometry tended to focus on petroleum products and on industrial materials characterization. In recent years, quantitative mass spectrometry has been applied to many diverse areas, including industrial polymers, environmental and forensic samples, and increasingly to biological materials.

³⁴I. Ntai et al., *Anal. Chem.*, **2014**, 86, 4961, DOI: 10.1021/ac500395k; P. Mao and D. Wang, *J. Proteom. Res.*, **2014**, 13, 1560, DOI: 10.1021/pr401074t; N. L. Kelleher, *Biochem.*, **2013**, 52, 3794, DOI: 10.1021/bi400466p; J. P. Whitelegge, *Anal. Chem.*, **2013**, 85, 2258, DOI: 10.1021/ac303064a; See also N. L. Kelleher, *Anal. Chem.*, **2004**, 76, 196A, DOI: 10.1021/ac0415657.



Tutorial: Learn more about **quantitative applications of MS** at www.tinyurl.com/skoogpia7

Mass spectrometry is widely used for the characterization of high-molecular-mass polymeric materials. The method of choice for many of these applications is MALDI mass spectrometry.³⁵ In recent applications, MALDI-MS has been used in imaging applications to identify and determine the location and distribution of polymers on membrane substrates.³⁶ MALDI-TOF imaging mass spectrometry has been recently used as a detector for polymer LC. The individual retention behavior of single structural units of polyethylene and polypropylene oxide copolymers could be monitored.³⁷

Mass spectrometry in conjunction with liquid chromatography has become very useful in environmental analysis.³⁸ The interface of choice between LC and MS is electrospray ionization. HPLC with tandem mass spectrometry has also become quite useful in the environmental area, being particularly important in the determination of pharmaceuticals in surface and wastewaters. Ultrahigh resolution FT-ICR mass spectrometry has been shown to be very useful in identifying compositional changes that occur in the weathering of crude oil components released during oil spills.³⁹ This technique can overcome some of the volatility limitations of gas chromatography that has been traditionally used in oil spill characterizations.

In forensic science, mass spectrometry and GC/MS are widely used in identifying and characterizing evidence including explosive materials of interest in homeland security. Mass

spectrometry plays a key role in drug testing athletes and horses and in analyzing body fluids and hair in forensic investigations. In addition, paints and fibers are often subjected to mass spectral analysis.⁴⁰ Mass spectrometers have become indispensable tools in the forensic laboratory.

In the clinical laboratory, GC/MS, LC/MS, and tandem mass spectrometry are finding increasing applications. In one study, six urinary pteridines and creatinine were determined by HPLC coupled with tandem mass spectrometry in order to detect breast cancer in its early stages.⁴¹ Tandem MS methods are now the standard for screening newborn babies for metabolic disease.⁴²

Many other biological applications of mass spectrometry have appeared. Mass spectrometry has been important for many years in protein identification, particularly in the analysis of peptides derived from digestions with proteolytic enzymes such as trypsin. Recently, MS methods have been used to study intrinsically disordered proteins, that is polypeptide chains without a well-defined three-dimensional structure.⁴³ Electrospray ionization and MALDI are particularly useful. MALDI methods have found important applications in the imaging of lipids in biological materials.⁴⁴ Mass spectrometry is now playing a major role in the expanding field of proteomics as discussed in section 20D-2. It has also become of major importance in plant science to investigate the many functions and mechanism of plant biochemistry.⁴⁵

³⁵See *MALDI-MS: A Practical Guide to Instrumentation, Methods and Applications*, 2nd ed., F. Hillenkamp and J. Peter-Katalinic, eds., Weinheim: Wiley-Blackwell, 2013.

³⁶K. Krueger et al. *Anal. Chem.*, **2013**, 85, 4998, DOI: 10.1021/ac4002063.

³⁷S. M. Weidner and J. Falkenhagen, *Anal. Chem.*, **2011**, 83, 9153, DOI: 10.1021/ac202380n.

³⁸D. Barcelo and M. Petrovic, *TRAC: Trends Anal. Chem.*, **2007**, 26, 2, DOI: 10.1016/j.trac.2006.11.006.

³⁹A. M. McKenna et al., *Environ. Sci. Technol.*, **2013**, 47, 7530, DOI: 10.1021/es305284t.

⁴⁰B. Stuart, *Forensic Analytical Techniques*, Chichester, UK: Wiley, 2013.

⁴¹C. Burton, H. Shi, and Y. Ma, *Anal. Chem.*, **2013**, 85, 11137, DOI: 10.1021/ac403124a.

⁴²K. C. Dooley, *Clin. Biochem.*, **2003**, 36, 471, DOI: 10.1016/S0009-9120(03)00105-X.

⁴³R. Beveridge, Q. Chappuis, C. Macphee, and P. Barran, *Analyst*, **2013**, 138, 32, DOI: 10.1039/c2an35665a.

⁴⁴D. Gode and D. A. Volmer, *Analyst*, **2013**, 138, 1289, DOI: 10.1039/c2an36337b.

⁴⁵E. Gemperline, C. Keller, and L. Li, *Anal. Chem.*, **2016**, 88, 3422, DOI: 10.1021/acs.analchem.5b02938.

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

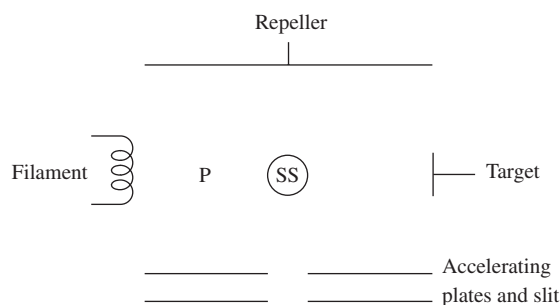
20-1 How do gaseous and desorption sources differ? What are the advantages of each?

20-2 How do the spectra for EI, field ionization, and CI sources differ from one another?

20-3 Describe the difference between gaseous field ionization sources and field desorption sources.

>> QUESTIONS AND PROBLEMS (continued)

20-4 The following figure is a simplified diagram of a commercially available EI source.



- (a) What voltage must be applied between the filament and target so that electrons interacting with molecules at the point marked SS (sample source) will have 70 eV of kinetic energy?
 - (b) What will happen to a molecule that diffuses toward the filament and is ionized at point P?
- * 20-5** When a magnetic sector instrument was operated with an accelerating voltage of 3.00×10^3 V, a field of 0.126 T was required to focus the CH_4^+ on the detector.
- (a) What range of field strengths would be required to scan the mass range between 16 and 300, for singly charged ions, if the accelerating voltage is held constant?
 - (b) What range of accelerating voltages would be required to scan the mass range between 16 and 300, for singly charged ions, if the field strength is held constant?
- * 20-6** Calculate the accelerating voltage that would be required to direct singly charged ions of mass 5000 through an instrument that is identical to the one described in Example 20-4.
- * 20-7** The ion-accelerating voltage in a particular quadrupole mass spectrometer is 10.00 V. How long will it take a singly charged cyclohexane ion to travel the 15.0 cm length of the rod assembly? Assume that the initial velocity of the ion in the z direction is zero.
- 20-8** On page 259 a qualitative discussion described how a positive ion would behave in the xz plane (positive dc potential plane) of a quadrupole mass filter. Construct a similar argument for the behavior of positive ions in the yz plane (negative dc potential plane).
- 20-9** Why do double-focusing mass spectrometers give narrower peaks and higher resolutions than single-focusing instruments?
- 20-10** Discuss the differences between quadrupole ion-trap mass spectrometers and Fourier transform ICR mass spectrometers.
- * 20-11** Calculate the resolution required to resolve peaks for
- (a) CH_2N ($\mathcal{M} = 28.0187$) and N_2^+ ($\mathcal{M} = 28.0061$).
 - (b) C_2H_4^+ ($\mathcal{M} = 28.0313$) and CO^+ ($\mathcal{M} = 27.9949$).
 - (c) $\text{C}_3\text{H}_7\text{N}_3^+$ ($\mathcal{M} = 85.0641$) and $\text{C}_5\text{H}_9\text{O}^+$ ($\mathcal{M} = 85.0653$).
 - (d) androst-4-en-3,17,-dione (M^+) at $m/z = 286.1930$ and an impurity at 286.1240.
- 20-12** What mass differences can just be resolved at m values of 100, 1000, 2000, 3000, and 5000 if the mass spectrometer has a resolution of
- (a) 500, (b) 1000, (c) 3000, (d) 5000?



*** 20-13** Calculate the ratio of the $(M + 2)^+$ to M^+ and the $(M + 4)^+$ to M^+ peak heights for

- (a) $C_{10}H_6Br_2$, (b) C_3H_7ClBr , and (c) $C_6H_4Cl_2$.

20-14 In a magnetic sector (single-focusing) mass spectrometer, it might be reasonable under some circumstances to monitor one m/z value, to then monitor a second m/z , and to repeat this pattern in a cyclic manner. Rapidly switching between two accelerating voltages while keeping all other conditions constant is called *peak matching*.

- (a) Derive a general expression that relates the ratio of the accelerating voltages to the ratio of the corresponding m/z values.
- (b) Use this equation to calculate m/z of an unknown peak if m/z of the ion used as a standard, CF_3^+ , is 69.00 and the ratio of $V_{\text{unknown}}/V_{\text{standard}}$ is 0.965035.
- (c) Based on your answer in part (b), and the assumption that the unknown is an organic compound that has a mass of 143, draw some conclusions about your answer in part (b), and about the compound.

*** 20-15** Measuring the approximate mass of an ion without using a standard can be accomplished via the following variant of the peak-matching technique described in Problem 20-14. The peak-matching technique is used to alternately cause the P^+ ion and the $(P + 1)^+$ ions to reach the detector. It is assumed that the difference in mass between P^+ and $(P + 1)^+$ is due to a single ^{13}C replacing a ^{12}C atom.

- (a) If the accelerating voltage for $(P + 1)^+$ is labeled V_2 and that for P^+ is V_1 , derive a relationship that relates the ratio V_2/V_1 to the mass of P^+ .
- (b) If $V_2/V_1 = 0.987753$, calculate the mass of the P^+ ion.

20-16 Discuss the major differences between a tandem-in-space mass spectrometer and a tandem-in-time mass spectrometer. Include the advantages and disadvantages of each type.

*** 20-17** Identify the ions responsible for the peaks in the mass spectrum shown in Figure 20-22b.

20-18 Identify the ions responsible for the four peaks having greater mass-to-charge ratios than the M^+ peak in Figure 20-4a.

Challenge Problem

20-19 Figure 20-29 (page 536) shows the mass spectrum of the same compound from an EI source and a CI source.

- (a) Which mass spectrum would be best for determining the molecular mass of the compound? Why?
- (b) Which mass spectrum would be best for determining the chemical structure? Why?
- (c) The EI source was a pulsed source used with a TOF mass analyzer. If the flight tube were 1.0 m long and the accelerating voltage were 3000 V, what would the flight time be for the ion at $m/z = 58$?
- (d) For two ions of m/z values m_1/z and m_2/z , derive an equation for the difference in flight times Δt_F as a function of the two masses, the charges, and the accelerating voltage.
- (e) For the same TOF analyzer as in part (c), calculate the difference in flight times between ions of $m/z = 59$ and $m/z = 58$.
- (f) To get more structural information, the compound of Figure 20-27 was subjected to tandem mass spectrometry. Which ionization source, EI or CI, would be most suitable for this purpose? Why?

QUESTIONS AND PROBLEMS (continued)

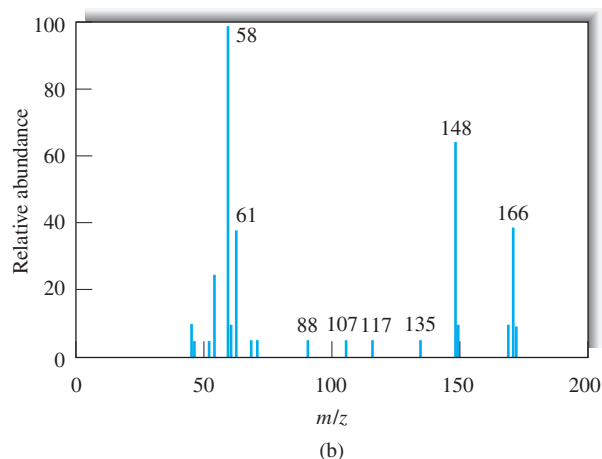
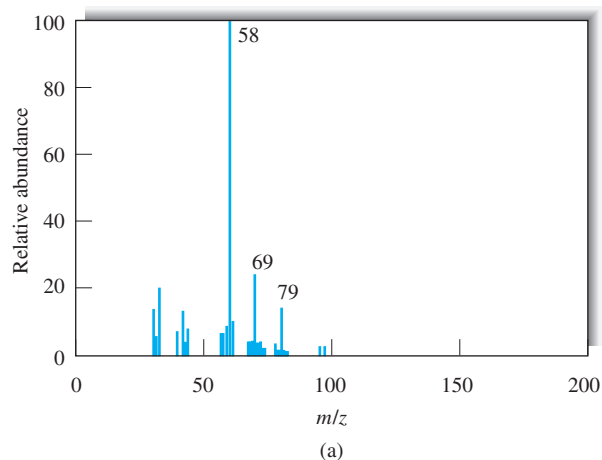


FIGURE 20-29 EI spectrum (a) and CI spectrum (b) of the same biologically important compound. (From H. M. Fales, H. A. Lloyd, and G. A. W. Milne, *J. Amer. Chem. Soc.*, **1970**, 92, 1590–1597, DOI: 10.1021/ja00709a028. American Chemical Society.)

- (g) Using the ionization source chosen in part (f), describe the types of mass spectra that could be obtained from an MS/MS experiment by:
- (1) holding the first mass analyzer constant and scanning the second analyzer.
 - (2) scanning both analyzers with a small m/z offset between them.
 - (3) scanning the first analyzer while holding the second analyzer constant.
 - (4) scanning the second mass analyzer for every mass selected by the first analyzer.

In your answer, use features of the mass spectrum of Figure 20-29 to illustrate your description.

Atomic Emission Spectrometry

This chapter covers optical **atomic emission spectrometry** (AES). Generally, the atomizers listed in Table 8-1 not only convert the components of samples to atoms or elementary ions but, in the process, excite a fraction of these species to higher electronic states. As the excited species rapidly relax back to lower states, ultraviolet and visible line spectra arise that are useful for qualitative and quantitative elemental analysis. Plasma sources have become the most important and most widely used sources for AES. These devices, including the popular inductively coupled plasma source, are discussed first. Then, emission spectroscopy based on electric arc and electric spark atomization and excitation is described. Historically, arc and spark sources were quite important in emission spectrometry, and they are still useful for the determination of some metallic elements. Finally several miscellaneous atomic emission sources, including flames, glow discharges, and lasers are presented.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

Atomic emission spectrometry (AES) offers several advantages when compared with the flame and electrothermal absorption methods considered in Chapter 9.¹ Emission methods often show lower susceptibility to chemical interferences, which is a direct result of their higher temperatures. Second, good emission spectra result for most elements under a single set of excitation conditions. Consequently, spectra for many elements can be recorded simultaneously. This property is of particular importance for the multielement analysis of very small samples. A major advantage of the more energetic plasma sources is that they permit the determination of low concentrations of elements that tend to form refractory compounds (that is, compounds that are highly resistant to thermal decomposition, such as the oxides of boron, phosphorus, tungsten, uranium, zirconium, and niobium). In addition, plasma sources permit the determination of nonmetals such as chlorine, bromine, iodine, and sulfur. Finally, plasma emission methods usually have concentration ranges of several orders of magnitude, in contrast to a two- or three-decade range for the absorption methods described in the previous chapter. Flames are less satisfactory as atomic emission sources because optimum excitation conditions vary widely from element to element; high temperatures are needed for excitation of some elements and low temperatures for others; and finally, the region of the flame that gives rise to optimum line intensities varies from element to element.

Emission spectra from high-temperature sources, such as plasmas, arcs, and sparks, are often complex and frequently comprise hundreds, or even thousands, of lines. This large number of lines, although advantageous when seeking qualitative information, increases the probability of spectral interferences in quantitative analysis. As a result, emission spectroscopy based on plasmas, arcs, and sparks requires higher resolution

¹For more extensive treatment of atomic emission spectroscopy, see J. A. C. Broekhaert, *Analytical Atomic Spectrometry with Flames and Plasmas*, 2nd ed., Chap. 5, Hoboken, NJ: Wiley-VCH, 2005; *Atomic Spectroscopy in Elemental Analysis*, M. Cullen, ed., Chaps. 3–5, Boca Raton, FL: CRC Press, 2004; L. H. J. Lajunen and P. Peramaki, *Spectrochemical Analysis by Atomic Absorption and Emission*, 2nd ed., Chaps. 4–6, Royal Society of Chemistry: Cambridge, 2004; J. D. Ingle Jr. and S. R. Crouch, *Spectrochemical Analysis*, Chaps. 7–9, 11, Upper Saddle River, NJ: Prentice-Hall, 1988.

and more expensive optical equipment than is needed for atomic absorption methods with flame or electrothermal sources.

Despite their advantages, it is unlikely that emission methods based on high-energy sources will ever completely displace flame and electrothermal atomic absorption procedures. In fact, atomic emission and absorption methods are complementary. Among the advantages of atomic absorption procedures are simpler and less expensive equipment requirements, lower operating costs, somewhat greater precision (presently, at least), and procedures that require less operator skill to yield satisfactory results.²

10A EMISSION SPECTROSCOPY BASED ON PLASMA SOURCES

Plasma sources have become extremely important in analytical spectroscopy. A plasma is an electrically conducting gaseous mixture containing a significant concentration of cations and electrons. (The concentrations of the two are such that the net charge is zero.) In the argon plasma frequently used for emission analyses, argon ions and electrons are the principal conducting species, although cations from the sample are also present in small amounts. Argon ions, once formed in a plasma, can absorb sufficient power from an external source to maintain the temperature at a level where further ionization sustains the plasma indefinitely. Such plasmas achieve temperatures as high as 10,000 K. Several different plasmas have been used in AES.³ The most important is the *inductively coupled plasma* (ICP), which is discussed in detail in section 10A-1. The *microwave-induced plasma* (MIP) is also used in some forms of emission spectroscopy. The direct current plasma (DCP) was once widely used in geochemistry, but has largely been supplanted by the ICP and other plasma sources. Microplasmas based on dc, radio-frequency, and microwave radiation have also attracted attention in recent years. Such sources have potential for use as chromatographic detectors, for forming ions in mass spectrometry, and for emission spectroscopy. Finally, the laser-induced plasma, produced by laser breakdown or by laser ablation, is also used for AES (see Section 10A-3).

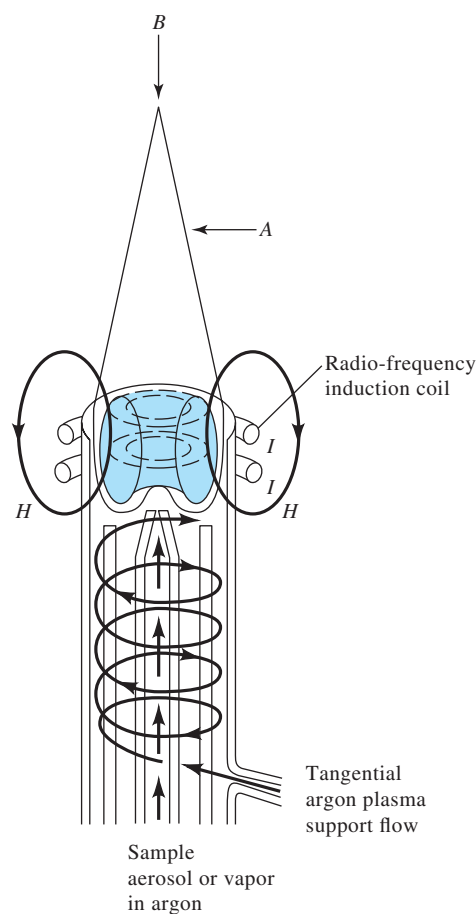


FIGURE 10-1 A typical ICP source. Position A shows radial viewing of the torch, and position B shows axial viewing. (From V. A. Fassel, *Science*, **1978**, 202, 183. With permission. Copyright 1978 by the American Association for the Advancement of Science.)

10A-1 The Inductively Coupled Plasma Source

Figure 10-1 is a schematic of a typical ICP source called a *torch*.⁴ It consists of three concentric quartz tubes through which streams of argon gas flow. Depending on the torch



Exercise: Learn more about ICP torches at www.tinyurl.com/skoogpia7

²For an excellent comparison of the advantages and disadvantages of flames, furnaces, and plasmas as sources for emission spectroscopy, see W. Slavin, *Anal. Chem.*, **1986**, 58, 589A, DOI: 10.1021/ac00295a803.

³For reviews of plasma and other atomic spectroscopy sources, see E. H. Evans et al., *J. Anal. At. Spectrom.*, **2015**, 30, 1017, DOI: 10.1039/c5ja90017d; J. Anal. At. Spectrom., **2014**, 29, 773, DOI: 10.1039/c4ja90019g; N. H. Bings, A. Bogaerts, and J. A. C. Broekaert, *Anal. Chem.*, **2013**, 85, 670, DOI: 10.1021/ac3031459; *Anal. Chem.*, **2010**, 82, 4653, DOI: 10.1021/ac1010469.

⁴For a more complete discussion on ICP sources, see V. A. Fassel, *Science*, **1978**, 202, 183; V. A. Fassel, *Anal. Chem.*, **1979**, 51, 1290A, DOI: 10.1021/ac50049a717; G. A. Meyer, *Anal. Chem.*, **1987**, 59, 1345A, DOI: 10.1021/ac00150a717; *Inductively Coupled Plasma Spectrometry and Its Applications*, 2nd ed., S. J. Hill, ed., Hoboken, NJ: Wiley-Blackwell, 2007; A. Varma, *CRC Handbook of Inductively Coupled Plasma Atomic Emission Spectroscopy*, Boca Raton, FL: CRC Press, 1990; *Inductively Coupled Plasma Mass Spectrometry*, A. E. Montaser, ed., Hoboken, NJ: Wiley-VCH, 1998; *Inductively Coupled Plasma in Analytical Atomic Spectroscopy*, 2nd ed., A. Montaser and D. W. Golightly, eds., New York: Wiley-VCH, 1992; *Inductively Coupled Plasma Atomic Emission Spectrometry: A Model Multi-Elemental Technique for Modern Analytical Laboratory*, G. A. Zachariadis, ed., Hauppauge, NY: Nova Science Publishers, Inc., 2012.

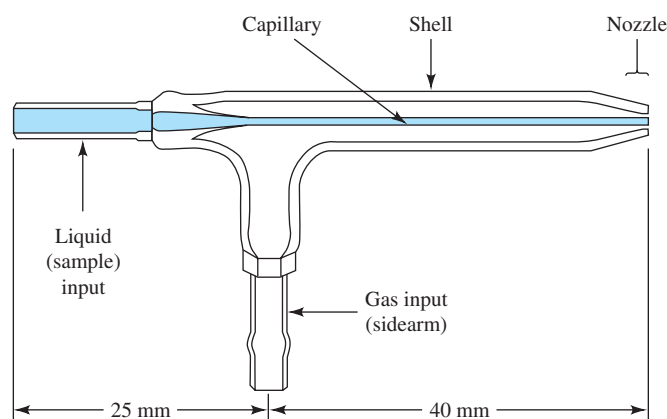


FIGURE 10-2 The Meinhard nebulizer. The nebulizing gas flows through an opening that surrounds the capillary concentrically. This causes a reduced pressure at the tip and aspiration of the sample. The high-velocity gas at the tip breaks up the solution into a mist. (Courtesy of J. Meinhard Associates, Inc.)

design, the total rate of argon consumption is 5 to 20 L/min. The diameter of the largest tube is often about 2.5 cm. Surrounding the top of this tube is a water-cooled induction coil that is powered by a radio-frequency (RF) generator, which radiates 0.5 to 2 kW of power at 27.12 MHz or 40.68 MHz.⁵ Ionization of the flowing argon is initiated by a spark from a Tesla coil. The resulting ions, and their associated electrons, then interact with the fluctuating magnetic field (labeled H in Figure 10-1) produced by the induction coil. This interaction causes the ions and electrons within the coil to flow in the closed annular paths shown in Figure 10-1. The resistance of the ions and electrons to this flow of charge causes ohmic heating of the plasma.

The temperature of the plasma formed in this way is high enough to require thermal isolation of the outer quartz cylinder. This isolation is achieved by flowing argon tangentially around the walls of the tube as indicated by the arrows in Figure 10-1. The tangential flow cools the inside walls of the center tube and centers the plasma radially.

A design offered by most manufacturers rotates the torch by 90° so that it is aligned axially with the spectrometer system (B in Figure 10-1). The radiation emitted from the center of the plasma is then used for analyses. This axial arrangement is particularly advantageous for ICP mass spectrometry (ICPMS), which we describe in Section 11C-1 (see Figure 10-6 for a spectrometer with axial viewing geometry and Figure 10-8 for radial geometry).

Note that the argon flow rate through the typical torch is great enough to produce a significant operating cost for

an ICP spectrometer (several thousand dollars annually). During the 1980s, low-flow, low-power torches appeared on the market. Typically, these torches require a total argon flow of less than 10 L/min and require less than 800 W of RF power.

Sample Introduction

Samples can be introduced into the ICP by argon flowing at about 1 L/min through the central quartz tube. The sample can be an aerosol, a thermally generated vapor, or a fine powder. The most common means of sample introduction is the concentric glass nebulizer shown in Figure 10-2. The sample is transported to the tip by the Bernoulli effect (aspiration). The high-velocity gas breaks up the liquid into fine droplets of various sizes, which are then carried into the plasma.

Another popular type of nebulizer has a cross-flow design (Figure 8-11b). Here, a high-velocity gas flows across a capillary tip at right angles, causing the same Bernoulli effect. Often, in this type of nebulizer, the liquid is pumped through the capillary with a peristaltic pump. Many other types of nebulizers are available for higher-efficiency nebulization, for nebulization of samples with high solids content, and for production of ultrafine mists.⁶

Another method of introducing liquid and solid samples into a plasma is by electrothermal vaporization. In this technique, the sample is vaporized in a furnace similar to that described in Section 8C-1 for electrothermal atomization. In plasma applications, however, the furnace is used for *sample introduction only*. Sample atomization occurs in the plasma.

⁵Most commercial instruments now operate at 40.68 MHz because better coupling efficiency between the coil and the plasma is achieved at the higher frequency and lower background emission is produced.

⁶See J. Sneedon, *Sample Introduction in Atomic Spectroscopy*, New York: Elsevier, 1990; J. L. Todili and J. M. Mermet, *Liquid Sample Introduction in ICP Spectrometry: A Practical Guide*, Amsterdam: Elsevier, 2008.

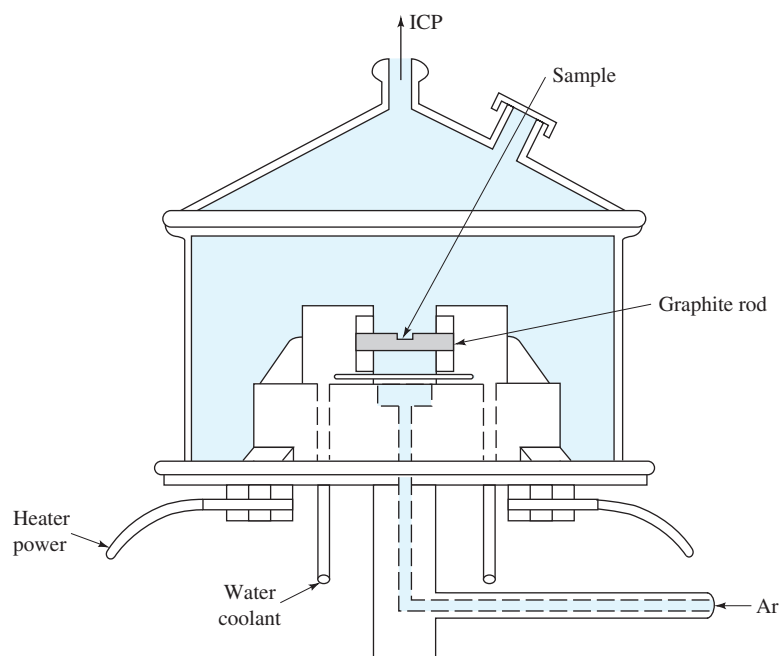


FIGURE 10-3 Device for electrothermal vaporization.

Figure 10-3 shows an electrothermal vaporizer in which vapor formation takes place on an open graphite rod. The vapor passes into a plasma torch in a stream of argon. The observed signal is a transient peak similar to the peaks obtained in electrothermal atomic absorption. Electrothermal vaporization coupled with a plasma torch offers the microsampling capabilities ($\sim 5 \mu\text{L}$) and low absolute detection limits ($\sim 1 \text{ ng}$) of electrothermal furnaces while maintaining the wide linear working range, acceptable sample-to-sample precision (5%–10%), freedom from interference, and the multielement capabilities of the ICP.⁷

The ablation devices for solids described in Section 8C-2 are also available from several makers of ICP instruments. With these types of sample-introduction systems, the plume of vapor and particulate matter produced by interaction of the sample with an electric arc or spark or with a laser beam are transported by a flow of argon into the torch where further atomization and excitation occur.

Plasma Appearance and Spectra

The typical ICP plasma has a very intense, brilliant white, non-transparent core topped by a flamelike tail. The core, which extends a few millimeters above the tube, produces the atomic spectrum of argon superimposed on a continuum spectrum. The continuum is typical of ion-electron recombination reactions and bremsstrahlung, pronounced 'brem(p)-shträ-lən, which is continuum radiation produced when charged particles are slowed or stopped. In the region 10 to 30 mm above the core,

the continuum fades and the plasma is optically transparent. Spectral observations are generally made at a height of 15 to 20 mm above the induction coil, where the temperature is 6000–6500 K. In this region, the background radiation is remarkably free of argon lines and is well suited for analysis. Many of the most sensitive analyte lines in this region of the plasma are from ions, such as Ca^+ , Cd^+ , Cr^+ , and Mn^+ .

In ICP spectrometers, the torch may be viewed radially, perpendicular to its axis (A in Figure 10-1), or axially (B in the figure) or there may be an optical system that allows viewing of either configuration. Advantages of the axial arrangement over the radial configuration include increased radiation intensity resulting from a longer path length and higher precision, which produce lower detection limits (a factor of 2 to 30 with ultrasonic nebulization). The disadvantages are that the cool plasma tail must be removed from the light path to prevent interference from oxides and that it is more difficult to prevent thermal and contaminant degradation of the spectrometer optics in the axial configuration than in the radial arrangement.⁸ The cool tail can be removed by blowing a shear gas across the plasma tip or by a conical interface device. The decision as to which viewing arrangement to use depends on the chemical behavior of the analyte in the plasma, the spectral line chosen for the analysis, the quality of the data required, and the detailed nature of the experiment. For example, the axial arrangement is especially useful in ICPMS, which is described in Section 11C.

⁷S. Kim et al., *Microchim. J.*, **2004**, 78, 127, DOI: 10.1016/j.microc.2004.03.003.

⁸L. H. J. Lajunen and P. Peramaki, *Spectrochemical Analysis by Atomic Absorption and Emission*, 2nd ed., pp. 253–254, Cambridge: Royal Society of Chemistry, 2004.

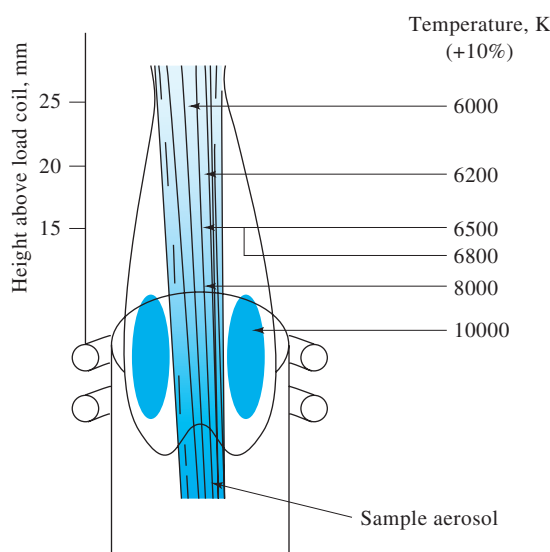


FIGURE 10-4 Temperatures in a typical ICP source. (From V. A. Fassel, *Science*, **1978**, 202, 186. With permission. Copyright 1978 by the American Association for the Advancement of Science.)

Analyte Atomization and Ionization

In Figure 10-4, isothermal contours illustrate temperatures in the plasma. Sample atoms reside in the plasma for a few milliseconds before they reach the observation point. During the residence time they experience temperatures ranging from 5500 to 8000 K. The time and temperatures are roughly two to three times greater than those found in the hottest combustion flames (acetylene–nitrous oxide) used in flame spectroscopic methods. As a result, atomization is more complete in plasmas than in flames, and fewer chemical interferences occur. Surprisingly, ionization interference effects are small, probably because the large electron concentration from ionization of the argon maintains a fairly constant electron concentration in the plasma.

Several other advantages are associated with the plasma source. First, atomization occurs in a chemically inert environment, which tends to enhance the lifetime of the analyte by preventing oxide formation. In addition, and in contrast to arcs, sparks, and flames, the temperature cross section of the plasma is relatively uniform, and effects such as self-absorption and self-reversal are not observed as often. Because of this, calibration curves are usually linear over several orders of magnitude of concentration. Finally, the plasma produces significant ionization, which makes it an excellent source for ICPMS.

10A-2 Microwave Plasmas

Microwave plasmas have been studied for many years.⁹ The MIP sustained in a discharge tube by a resonant cavity, such

as that developed by Beenakker,¹⁰ has become useful as a gas-chromatographic element-selective detector (see Section 27B-4). For AES, however, MIPs have not achieved the low detection limits of ICPs, and the plasmas are not as robust or as stable as ICPs. Introducing liquid samples with conventional nebulizers is particularly problematic.

In the past few years, however, there has been renewed interest in microwave plasmas, especially since the introduction of a commercial system that uses air for plasma operation.¹¹ The new microwave plasma system sustains the plasma by coupling energy from the magnetic field of the microwave source instead of from the electric field.¹² The magnetically excited plasma is more robust than previous electrically excited MIPs. This allows the plasma to be sustained with nitrogen or air and permits sample introduction with conventional pneumatic nebulizers. The commercial instrument generates nitrogen from an air cylinder, reducing operating costs over the argon-operated ICP. When compared to flames for AA or AES, the lack of dangerous and potentially explosive gases makes the air-operated MIP a safe and efficient source. Detection limits for several elements are comparable to those achieved with a standard ICP source.

10A-3 Other Plasma Sources

Additional plasma sources for AES included microplasmas and laser-induced plasmas. The glow-discharge devices described in Section 9A-3 have also been used successfully for AES.

Microplasmas¹³

The development of small, miniaturized plasmas is an area of active research. The plasmas described have been based on microwave coupling, radio-frequency coupling, dielectric barrier discharges, and dc discharges such as glow discharges.¹⁴ The microplasmas have the advantages of low power consumption and low gas-flow requirements as well as small sizes. Such devices have been most successfully used with gas-phase sample introduction, such as from gas chromatographic effluents and hydride generators. Discharge devices that utilize solution electrodes have also been described.¹⁵



Exercise: Learn more about the ICP at www.tinyurl.com/skoogpia7

¹⁰C. I. M. Beenakker, *Spectrochim. Acta B*, **1976**, 31, 483, DOI: 10.1016/0584-8547(76)80047-X.

¹¹Agilent Technologies, Santa Clara, CA 95051.

¹²M. R. Hammer, *Spectrochim. Acta B*, **2008**, 63, 456, DOI: 10.1016/j.sab.2007.12.007.

¹³N. H. Bings, A. Bogaerts, and J. A. C. Broekaert, *Anal. Chem.*, **2013**, 85, 670, DOI: 10.1021/ac3031459.

¹⁴For a review of miniature plasmas, see J. Franzke and M. Miclea, *Appl. Spectrosc.* **2006**, 60, 80A, DOI: 10.1366/000370206776342689.

¹⁵M. R. Webb and G. M. Hieftje, *Anal. Chem.*, **2009**, 81, 862, DOI: 10.1021/ac801561t.

⁹K. J. Jankowski and E. Rezke, *Microwave Induced Plasma Analytical Spectrometry*, Cambridge, UK: Royal Society of Chemistry Publishing, 2010.

Laser-Induced Plasmas

Over the past decade, lasers have become very useful for producing plasmas in AES. We consider here laser ablation and *laser-induced breakdown spectroscopy* (LIBS).

Laser Ablation. Ablation is the removal of material from a surface by vaporization or some other process. In laser ablation, the beam irradiates the surface, which is usually a solid but occasionally a liquid, and causes localized heating and vaporization. Usually a high powered, pulsed laser is used, but continuous wave laser beams have also been used. Double pulse ablation is sometimes used.¹⁶ With high enough laser intensity, the material ablated can be converted into a plasma.

The laser plasma can emit radiation that is useful for AES, or it can produce ions that can be analyzed by mass spectrometry. In some techniques, a laser produces only a plume of atoms and ions, while a second device provides excitation. For example, with a *laser microprobe*, the contents of the plume formed by laser irradiation are excited by a spark discharge between a pair of electrodes located immediately above the surface (see Section 10B-2). The emitted radiation is then focused on a suitable spectrometer system. With a laser microprobe source, the trace element compositions of single blood cells have been determined. With some solids, the laser can be scanned across the surface to obtain a spatial representation of surface composition.

Laser-Induced Breakdown Spectroscopy. The LIBS technique is closely related to laser ablation. Lasers used in LIBS are so powerful that dielectric breakdown of the atmosphere surrounding the ablated sample occurs, producing a highly luminous plasma. Typically, a short pulse of perhaps a few nanoseconds duration, such as that from a Nd:YAG laser, produces the plasma. Near the end of the laser pulse, the plasma cools, and radiation from excited atoms and ions can be detected. A spectrometer is used to collect the emission at the appropriate time so that the intense continuum radiation produced by the breakdown dies away before measurement of appropriate atom or ion lines.¹⁷

In addition to single-pulse LIBS, two-laser LIBS has also been successful. In two-laser LIBS, one laser ablates the sample while the second laser produces the plasma. The LIBS technique has been applied in several different areas. Metals, semiconductors, ceramics, coals, polymers, and pharmaceuticals are among these application areas. In addition to solid samples, gaseous and liquid samples have also been investigated. In fact, the first applications of LIBS were for the remote analyses of hazardous gases in industrial environments. Various process liquids, biological

solutions, aqueous environmental solutions, and pharmaceutical preparations have also been analyzed. The success of LIBS has led to commercial instrumentation devoted to the technique. A desktop LIBS analyzer is said to be able to determine trace elements, including light elements such as Al, Mg, and C, in almost any matrix.¹⁸

Recent developments have brought about miniaturization of the LIBS instrumentation, initially stimulated by the inclusion of a LIBS instrument package in the Mars Science Laboratory now active on Mars. The LIBS instrument is part of the ChemCam package on the rover Curiosity. The ChemCam system consists of a camera with a telephoto lens and the LIBS system. The LIBS laser enables the Martian soil or a rock of interest to be sampled remotely, without contamination from a robotic arm, and analyzed with the on-board spectrometer. Figure 10-5 shows an example of a spectrum obtained from the LIBS instrument. This package has enabled depth profiling by using the laser to tunnel through rock surfaces, a few laser shots at a time. ChemCam has enabled the construction of three-dimensional models of the target surface.

10A-4 Plasma Source Spectrometers

Table 10-1 lists the most important properties of the ideal instrument for plasma emission spectroscopy. The ideal spectrometer is not available today, partly because some of these properties are mutually exclusive. For example, high resolution requires the use of narrow slits, which usually reduces the signal-to-noise ratio and thus the precision of intensity readings. Nevertheless, instruments developed recently approach many of the ideals listed in the table.

A dozen or more instrument manufacturers currently offer plasma emission spectrometers. The designs, performance characteristics, and wavelength ranges of these instruments vary substantially. Most encompass the entire ultraviolet-visible spectrum, from 170 to 800 nm. A few instruments are equipped for vacuum operation, which extends the ultraviolet to 150 to 160 nm. This short-wavelength region is important because elements such as phosphorus, sulfur, and carbon have emission lines in this range.

Instruments for emission spectroscopy are of three basic types: *sequential*, *simultaneous multichannel*, and *Fourier transform*. Fourier transform instruments have not been widely used in AES. Sequential instruments are usually programmed to move from the line for one element to that of a second, pausing long enough (a few seconds) at each to measure line intensities with a satisfactory signal-to-noise ratio. In contrast, multichannel instruments are designed to measure simultaneously, or nearly so, the intensities of emission lines for a number of elements (sometimes as many as fifty or sixty). When several elements are determined, sequential instruments require significantly greater time for samples to be introduced than is required with the other

¹⁶For more information, see *Laser Ablation: Effects and Applications*, S. E. Black, ed., Hauppauge, NY: Nova Science Publishers, Inc., 2011; C. Phillips, *Laser Ablation and Its Applications*, New York: Springer, 2007.

¹⁷See D. A. Cremers and L. J. Radziemski, *Handbook of Laser-Induced Breakdown Spectroscopy*, Chichester, UK: Wiley, 2013; R. Noll, *Laser-Induced Breakdown Spectroscopy: Fundamentals and Applications*, Heidelberg: Springer-Verlag, 2012.

¹⁸TSI Incorporated, Shoreview, MN, 55126.

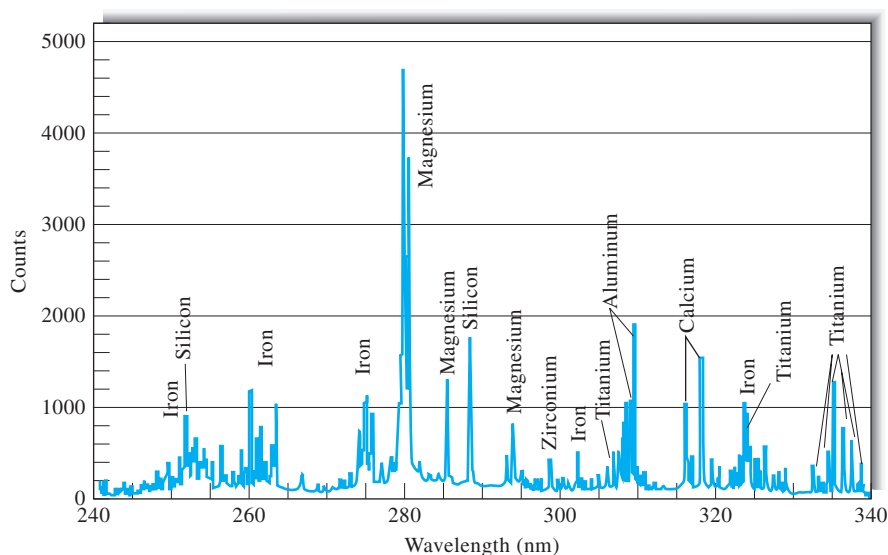


FIGURE 10-5 Spectrum of Martian rock obtained from the ChemCam LIBS system aboard the rover Curiosity. (Courtesy of NASA).

two types. Thus, these instruments, although simpler, are costly in terms of sample consumption and time.

Both sequential and multichannel emission spectrometers are of two general types, one using a classical grating spectrometer and the other an echelle spectrometer, such as that shown in Figure 7-23.

Sequential Instruments

Sequential instruments often incorporate a grating monochromator such as that shown in Figure 10-6. Usually, the grating is a holographic type having 2400 or 3600 grooves per millimeter. With some instruments of this type, scanning is accomplished by rotating the grating with a digitally controlled stepper motor so that different wavelengths are sequentially and precisely focused on the exit slit. In some designs, however, the grating is fixed and the slit and photomultiplier tube are moved along

the focal plane or curve. Instruments such as the one shown in Figure 10-6 have two sets of slits and photomultiplier tubes, one for the UV region and one for the visible. In such instruments, at an appropriate wavelength, the exit beam is switched from one photomultiplier to the other by movement of the plane mirror located between the two transducers.

Slew-Scan Spectrometers. With complex spectra made up of hundreds of lines, scanning a significant wavelength region takes too long and is thus impractical. To partially overcome this problem, *slew-scan* spectrometers were developed in which the grating (see Figure 10-6), or the transducer and slit, is driven by a two-speed (or multispeed) motor. In such instruments, the monochromator scans very rapidly, or *slews*, to a wavelength near a line of interest. The scan rate is then quickly reduced so that the instrument scans across the line in a series of small (0.01 to 0.001 nm) steps. With slew scanning, the time spent in wavelength regions containing no useful data is minimized, but sufficient time is spent at analyte lines to obtain satisfactory signal-to-noise ratios. In spectrometers, such as the one illustrated in Figure 10-6, in which the grating movement is under computer control, slewing can be accomplished very efficiently. For example, the spectrometer shown can slew to lines corresponding to fifteen elements and record their intensities in less than 5 minutes. Generally, however, these instruments are slower and consume more sample than multichannel instruments.

TABLE 10-1 Desirable Properties of an Emission Spectrometer

1. High resolution (0.01 nm or $\lambda/\Delta\lambda > 100,000$)
2. Rapid signal acquisition and recovery
3. Low stray light
4. Wide dynamic range ($>10^6$)
5. Accurate and precise wavelength identification and selection
6. Precise intensity readings ($<1\%$ relative standard deviation at $500 \times$ the detection limit)
7. High stability with respect to environmental changes
8. Easy background corrections
9. Computerized operation: readout, storage data manipulation, etc.

Scanning Echelle Spectrometers. Figure 10-7 is a schematic of an echelle spectrometer that can be operated either as a scanning instrument or as a simultaneous multichannel spectrometer. Scanning is accomplished by moving a photomultiplier tube in both x and y directions to scan an aperture plate located on the focal plane of the monochromator. The plate contains as many as 300 photo-etched slits. The time required to move from one

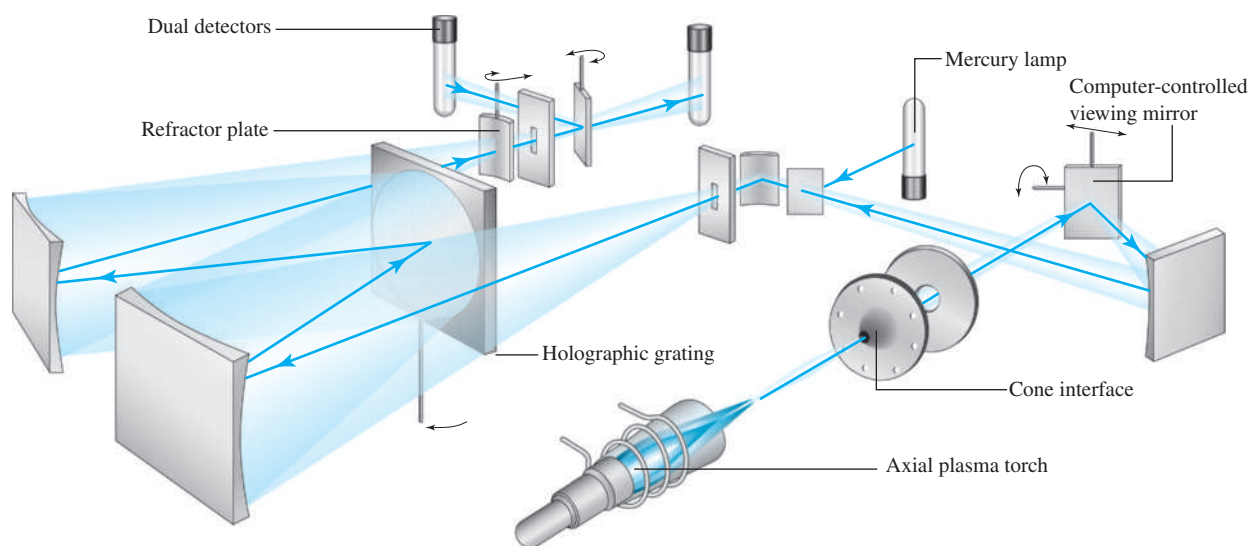


FIGURE 10-6 Optical diagram of a sequential ICP optical emission spectrometer. All moving parts are under computer control, and their modes of motion are indicated by the three-dimensional arrows. Moving parts include the grating, a mirror for transducer selection, a refractor plate for optimizing signal throughput, and a viewing mirror to optimize the plasma viewing position. The spectrometer contains a mercury lamp for automatic wavelength calibration. Notice the axial viewing geometry. (Courtesy of Agilent Technologies, Inc., Santa Clara, CA.)

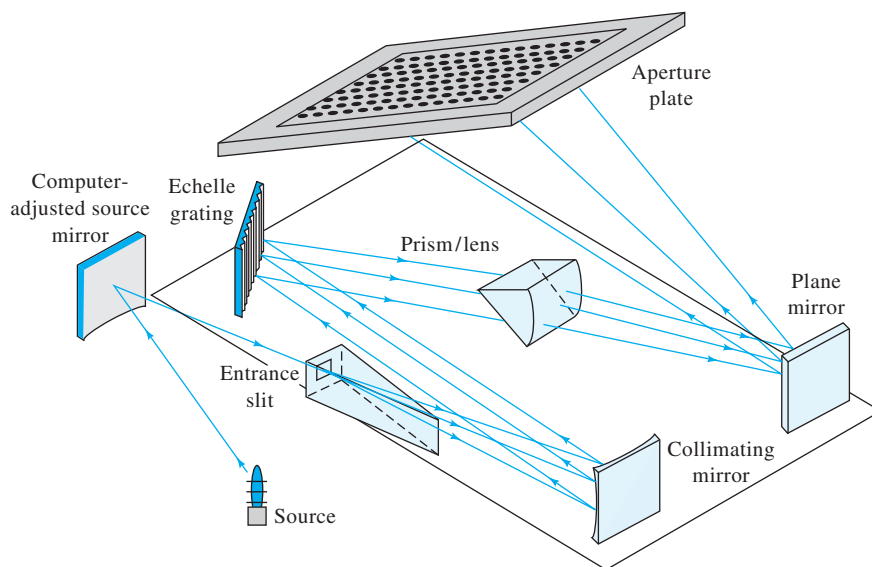


FIGURE 10-7 Schematic of an echelle spectrograph system.

slit to another is typically 1 s. The instrument can be operated in a slew-scan mode. This instrument can also be converted to a multichannel polychromator by mounting several small photomultiplier tubes behind appropriate slits in the aperture plate.

Multichannel Spectrometers

A simultaneous multichannel instrument incorporates either a *polychromator* or a *spectrograph*. Polychromators contain a series of photomultiplier tubes for detection, but spectrographs use two-dimensional charge-injection devices (CIDs) or

charge-coupled devices (CCDs) as transducers. Older spectrographs used photographic emulsions as transducers.

Polychromators. In some multichannel emission spectrometers, photomultipliers are located behind fixed slits along the focal curve of a grating polychromator such as the Paschen-Runge design shown in Figure 10-8. In these instruments, the entrance slit, the exit slits, and the grating surface are located along the circumference of a *Rowland circle*, the curvature of which corresponds to the focal curve of the concave grating.

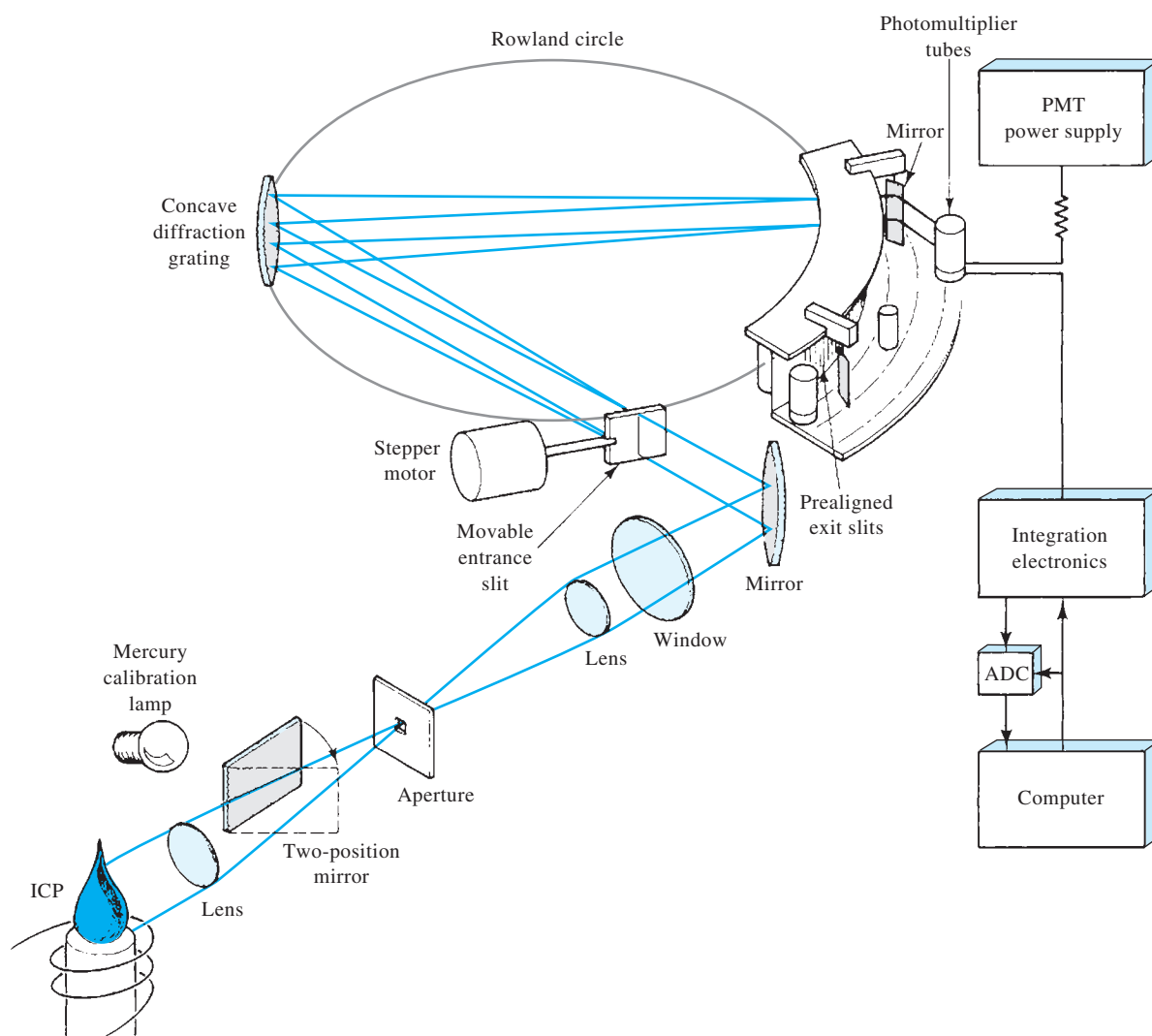


FIGURE 10-8 Direct-reading ICP emission spectrometer. The polychromator is of the Paschen-Runge design. It features a concave grating and produces a spectrum around a Rowland circle. Separate exit slits isolate each spectral line, and a separate photomultiplier tube converts the optical information from each channel into an electrical signal. Notice the radial viewing geometry. PMT = photomultiplier tube; ADC = analog-to-digital converter. (From J. D. Ingle Jr. and S. R. Crouch, *Spectrochemical Analysis*, p. 241, Upper Saddle River, NJ: Prentice-Hall, 1988, with permission.)

Radiation from each of the fixed slits impinges on the photomultiplier tubes. The slits are factory configured to transmit lines for selected elements. In these polychromators, the pattern of lines can be changed relatively inexpensively to accommodate new elements or to delete others. The signals from the photomultiplier tubes are integrated, and the output voltages are digitized. After converting to concentrations, the results are stored and displayed. The entrance slit can be moved tangentially to the Rowland circle by means of a stepper motor. This device permits scanning through peaks and provides information for background corrections.

Polychromator-based spectrometers with photomultipliers as transducers have been used both with plasma and with

older sources such as arcs and sparks. For rapid routine analyses, such instruments can be quite useful. For example, in the production of alloys, quantitative determinations of twenty or more elements can be completed within 5 minutes of receipt of a sample; close control over the composition of a final product is then possible.

In addition to speed, photoelectric multichannel spectrometers often exhibit good analytical precision. Under ideal conditions, reproducibilities of the order of 1% relative to the amount present have been demonstrated. Because other instrument components (such as the source) exhibit lower precision than the spectrometer, such high precision is not often achieved in the overall measurement. Multichannel instruments of this type

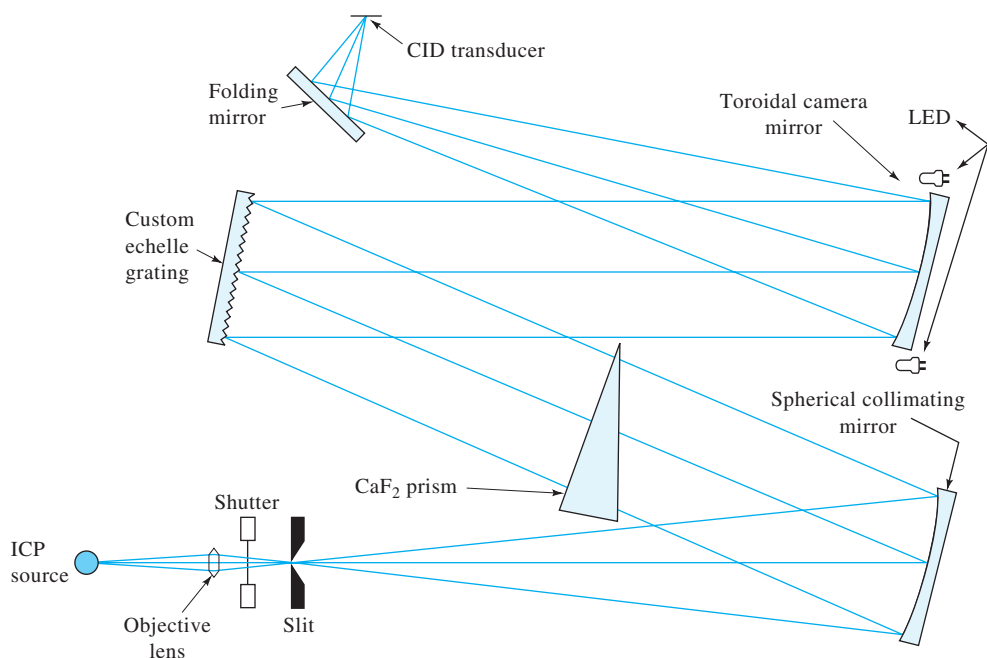


FIGURE 10-9 Optical diagram of an echelle spectrometer with a CID. (From M. J. Pilon, M. B. Denton, R. G. Schleicher, P. M. Moran, and S. B. Smith, *Appl. Spectrosc.*, **1990**, 44, 1613, DOI: 10.1366/0003702904417490. With permission.)

are generally more expensive than the sequential instruments described in the previous section and not as versatile.

A Charge-Injection Device Instrument. A number of companies offer multichannel simultaneous spectrometers based on echelle spectrometers and two-dimensional array devices. This type of instrument has replaced other types of multichannel emission spectrometers in many applications.

Figure 10-9 is an optical diagram of an echelle spectrometer that has a CID for simultaneous operation.¹⁹ It uses a calcium fluoride prism to sort the spectral orders that are subsequently formed by the echelle grating (see also Figure 7-21). The transducer is a CID (Section 7E-3) 8.7 mm by 6.6 mm and containing 94,672 transducer elements. A toroidal camera mirror focuses the slit images onto the transducer surface. To eliminate dark currents in the transducer elements, the unit is housed in a liquid nitrogen cryostat that maintains a temperature of 135 K.

A set of 39 transducer elements, called a *read window*, is used to monitor each spectral line as shown in Figure 10-10a. Normally, as shown by the projected image of one of the windows labeled “examination window,” the spectral line is focused on the 9 center elements of the window, and the 15 elements on either side of the central set provide background intensity data. Figure 10-10b shows the intensities recorded for the read

window for the iron 297.32 nm line.²⁰ Note that most of the iron radiation falls on the central elements of the window.

One of the useful features of the CID, in contrast to the CCD discussed next, is that the amount of charge accumulated in an element at any instant can be monitored nondestructively; that is, no charge is lost in the measurement process. To make the measurement of line intensity as rapid and efficient as possible, only the charge accumulated in the 9 central elements of the window are read initially to determine when sufficient charge has accumulated to provide a satisfactory signal-to-noise ratio. Only then are the remaining elements in the two 15-element sets read to correct the observed line intensity for the background radiation. This process goes on simultaneously at the read windows for each element. With an intense line, the time required to accumulate the desired charge is brief. With weak lines, the charge accumulated in a brief period is often used to estimate the integration time required to yield a satisfactory signal-to-noise ratio. In some cases integration times of 100 s or more are required.

Periodic wavelength calibration of the spectrometer just described is maintained through reference to the 253.65 nm mercury line from a small mercury lamp. Data files for line positions for more than 40 elements have been developed. The file for each element contains the wavelengths of up to ten lines and the x

¹⁹See M. J. Pilon, M. B. Denton, R. G. Schleicher, P. M. Moran, and S. B. Smith, *Appl. Spectrosc.*, **1990**, 44, 1613, DOI: 10.1366/0003702904417490.

²⁰R. B. Bilhorn and M. B. Denton, *Appl. Spectrosc.*, **1990**, 44, 1538, DOI: 10.1366/0003702904417850.

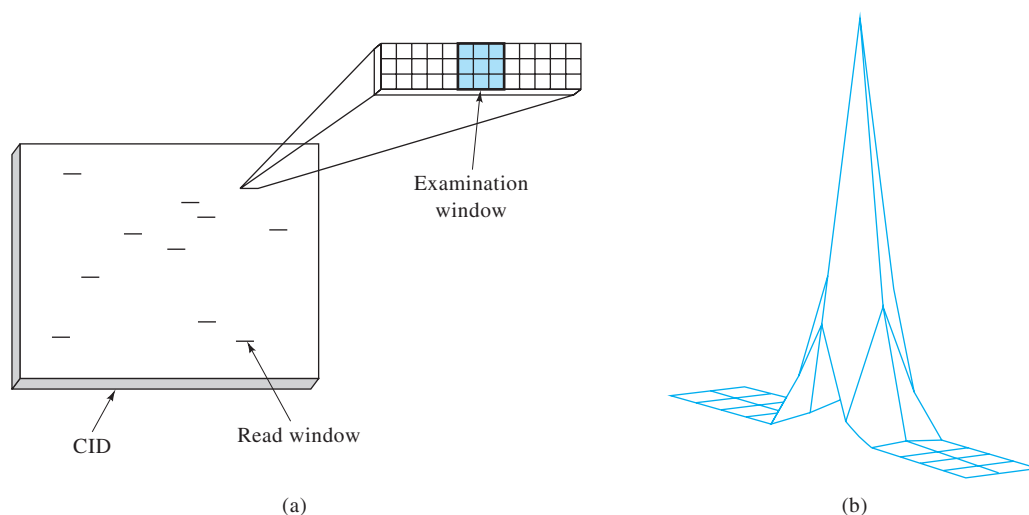


FIGURE 10-10 (a) Schematic representing the surface of a CID. The short horizontal lines represent the read windows. A magnified image of one of the read windows is also shown. The nine central elements form the examination window, where a line is positioned. (b) Intensity profile for an iron line. All of the radiation from the line falls on the 3×3 examination window. (From R. B. Bilhorn and M. B. Denton, *Appl. Spectrosc.*, **1990**, 44, 1538. With permission.)

and y coordinates of each of these spectral lines with respect to the coordinates for the mercury line. Database recalibration is seldom necessary unless something perturbs the optics of the system significantly. Identification of elements in samples is done by visual inspection with the use of a video monitor and interactive markers. With an ICP excitation source, detection limits that range from a few tenths of a nanogram to $10 \mu\text{g/mL}$ have been reported for most elements. For nonmetals such as phosphorus and arsenic, detection limits are larger by a factor of 100.

A Charge-Coupled Device Instrument. Figure 10-11 is an optical diagram of a commercial spectrometer with two echelle systems and two CCDs, one system for the 160–375-nm region and the other for the 375–782-nm range.²¹ Radiation from the plasma enters the spectrometer via a slit and is then dispersed by an echelle grating. The radiation falls on a Schmidt cross-disperser element that separates the orders of the ultraviolet radiation and also separates the ultraviolet and visible optical beams. The Schmidt element consists of a grating ruled on a spherical surface with a hole in the center to pass the visible radiation to a prism where order separation takes place as shown in Figure 7-23. The two dispersed beams are then focused onto the surface of transducer elements as shown. Note that a schematic representation of the surface of the ultraviolet transducer is shown as an insert in Figure 10-11.

These unique detector systems consist of numerous subarrays, or *array segments*, fabricated on silicon chips, with each

subarray being custom positioned so that three to four major emission lines for each of 72 elements fall on its surface. Each array segment consists of a *linear* (rather than two-dimensional) CCD that is made up of twenty to eighty pixels. Figure 10-12 is a schematic of one of these array segments, which is made up of the individual photosensitive registers, storage and output registers, and output electronics. Because each array segment can be addressed separately, charge-integration times can be varied over a wide enough range to provide dynamic ranges of 10^5 or greater. Although there are only 224 (235 in the current production version) of these array segments in the system, multiple lines fall on many of the subarrays so that about 6000 lines can be monitored simultaneously.

In the production version of this spectrometer, one of the mirrors that guides the radiation into the optical system is under computer control so that plasma viewing may be axial, radial, or mixed. This arrangement also permits optimization of the spectrometer signal. The entire optical system is contained within a purged temperature-controlled enclosure and is protected from the intense UV radiation of the plasma between samples by a pneumatically operated shutter to extend the life of the input mirror. In addition, a mercury lamp is built into the shutter mechanism to calibrate the spectrometer periodically. Different models of the spectrometer are available that cover the spectral range of 163–782 nm or segments thereof, depending on whether one or both arrays are installed.

A Combination Instrument. An interesting and useful application of both the Paschen-Runge polychromator and modular array detectors is the spectrometer shown in the photo of Figure 10-13a. Fifteen or sixteen linear CCD array modules (eight

²¹For a detailed description of this instrument, see T. W. Barnard et al., *Anal. Chem.*, **1993**, 65, 1225, DOI: 10.1021/ac00057a020; T. W. Barnard, et al., *Anal. Chem.*, **1993**, 65, 1231, DOI: 10.1021/ac00057a021.

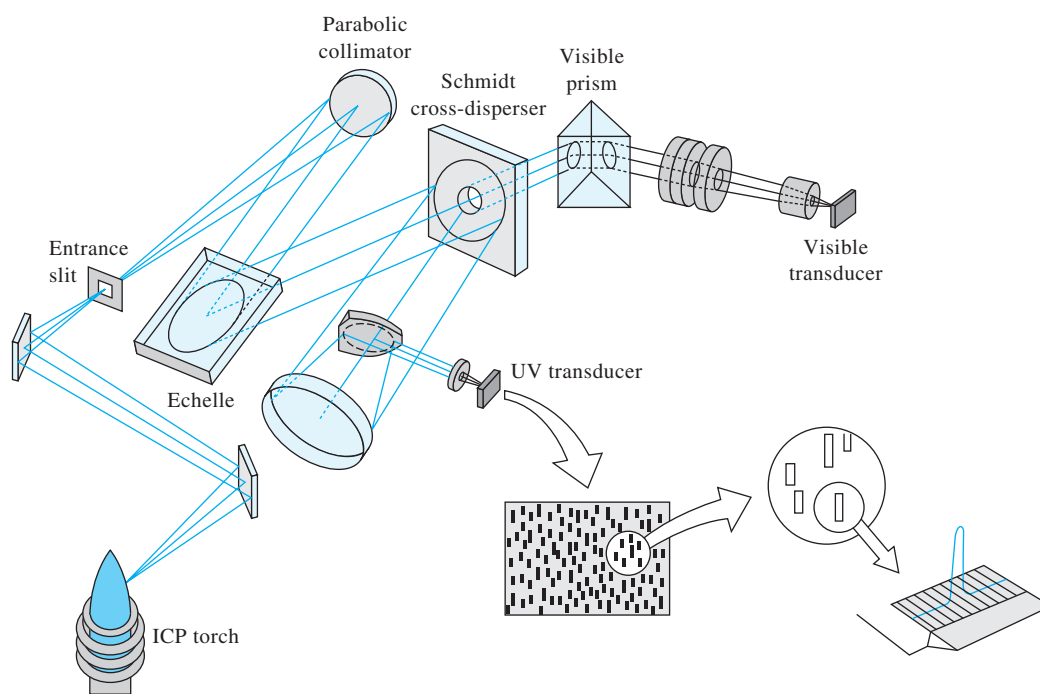


FIGURE 10-11 An echelle spectrometer with segmented array of CCDs. (From T. W. Barnard et al., *Anal. Chem.*, **1993**, 65, 1231. Figure 1, p. 1232. Copyright 1993 American Chemical Society.)

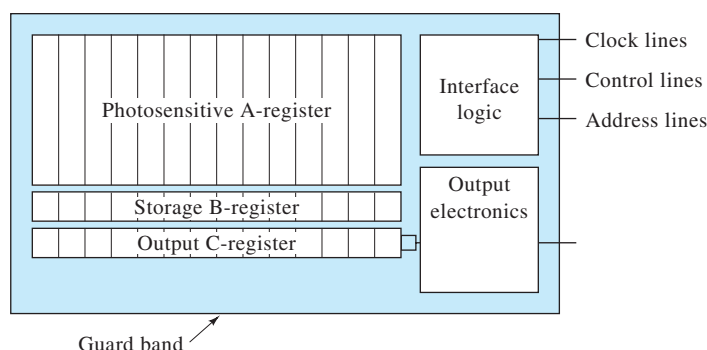


FIGURE 10-12 Schematic of an array segment showing phototransducers, storage and output registers, and readout circuitry. (From T. W. Barnard et al., *Anal. Chem.*, **1993**, 65, 1231. Copyright 1993 American Chemical Society.)

are visible) are arranged along the circumference of the Rowland circle to provide nearly complete coverage of the range from 140 to 670 nm. Each detector module (see Figure 10-13b) contains a mirror to reflect the radiation to the CCD array, which is arranged parallel to the plane of the Rowland circle. The modules are easily exchanged and positioned in the optical path. Because of its relatively compact design (115 cm wide $\times$ 70 cm deep), this spectrometer is particularly well suited to benchtop operation and routine use in industrial and environmental laboratories.

Fourier Transform Spectrometers

Since the early 1980s, several workers have described applications of Fourier transform instruments to the ultraviolet-visible region of the spectrum by using instruments that are similar in design to the infrared instruments discussed in detail in Section 16B-1.²² Much of this work has been devoted to the use of such instruments for multielement analyses with ICP sources. The advantages of Fourier transform instruments



Tutorial: Learn more about **ICP spectrometers** at www.tinyurl.com/skoogpia7

²²A. P. Thorne, *Anal. Chem.*, **1991**, 63, 57A, DOI: 10.1021/ac00002a712; L. M. Faires, *Anal. Chem.*, **1986**, 58, 1023A, DOI: 10.1021/ac00122a779.

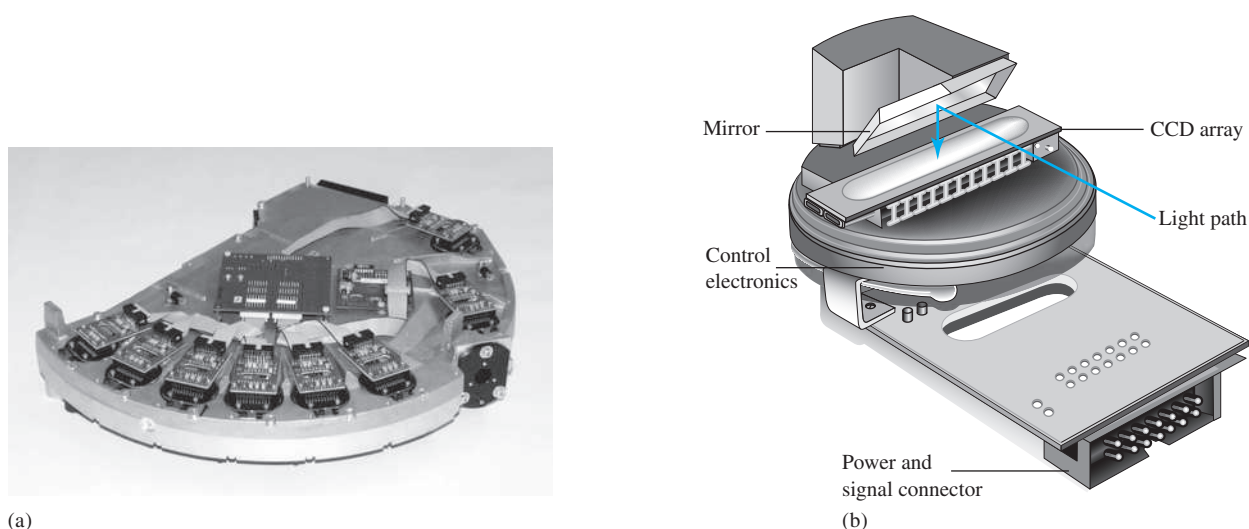


FIGURE 10-13 Components of a simultaneous CCD-ICP spectrometer: (a) Photo of the optical system. Note the Rowland circle with the grating in the rear, the slit in the front, and the detector array modules along the circle. (b) Array detector module containing a 1024-pixel linear CCD. (Courtesy of Spectro Analytical Instruments, Inc., Mahwah, NJ.)

include their wide wavelength coverage (170 nm to > 1000 nm), speed, high resolution, highly accurate wavelength measurements, large dynamic range, compact size, and large optical throughput. In contrast to Fourier transform infrared instruments, however, ultraviolet-visible instruments of this type often exhibit no multiplex advantage and indeed under some circumstances show a multiplex disadvantage (see Section 7I-1). The reason for this difference is that the performance of infrared instruments is usually limited by transducer noise, while ultraviolet-visible spectrometers are limited by shot and flicker noise associated with the source.

10A-4 Applications of Plasma Sources

Plasma sources produce spectra rich in characteristic emission lines, which makes them useful for both qualitative and quantitative elemental analysis.²³ The ICP yields significantly better quantitative analytical data than other emission sources. The quality of these results stems from its high stability, low noise, low background, and freedom from interferences when operated under appropriate experimental conditions. The newer air-operated microwave sources have substantial promise and should be less expensive to operate than the conventional ICP.

Sample Preparation

ICP emission spectroscopy is used primarily for the qualitative and quantitative analysis of samples that are dissolved or

suspended in aqueous or organic liquids. The techniques for preparation of such solutions are similar to those described in Section 9D-1 for flame absorption methods. With plasma emission, however, it is possible to analyze solid samples directly. These procedures include incorporating electrothermal vaporization, laser and spark ablation, and glow-discharge vaporization, all of which were described in Section 8C-2. Suspensions of solids in solutions can also be handled with a Babington nebulizer, such as that shown in Figure 8-11d.

Elements Determined

In principle, all metallic elements can be determined by plasma emission spectrometry. A vacuum spectrometer is necessary for the determination of boron, phosphorus, nitrogen, sulfur, and carbon because the emission lines for these elements lie at wavelengths less than 180 nm, where components of the atmosphere absorb radiation. The usefulness for the alkali metals is limited by two difficulties: (1) the compromise operating conditions that can be used to accommodate most other elements are unsuited for the alkalis, and (2) the most prominent lines of Li, K, Rb, and Cs are located at near-infrared wavelengths, which lead to detection problems with many plasma spectrometers that are designed primarily for ultraviolet radiation. Because of problems of this sort, plasma emission spectroscopy is generally limited to the determination of about 60 elements.

The periodic table of Figure 10-14 shows the applicability of ICP emission spectrometry to various elements. The detection limits for the best lines for each of the elements are indicated by the color and degree of shading. The areas of shading indicate the number of lines for each element that yields a detection limit within a factor of 3 of the best line. The more such lines that are available, the greater the chance that a usable line can be found that is free from interference when the matrix yields a line-rich spectrum.

²³For useful discussions of the applications of plasma emission sources, see *Inductively Coupled Plasma Spectrometry and Its Applications*, 2nd ed., S. J. Hill, ed., Hoboken, NJ: Wiley-Blackwell, 2007; *Inductively Coupled Plasma Atomic Emission Spectrometry: A Model Multi-Elemental Technique for Modern Analytical Laboratory*, G. A. Zachariadis, ed., Hauppauge, NY: Nova Science Publishers, Inc., 2012.

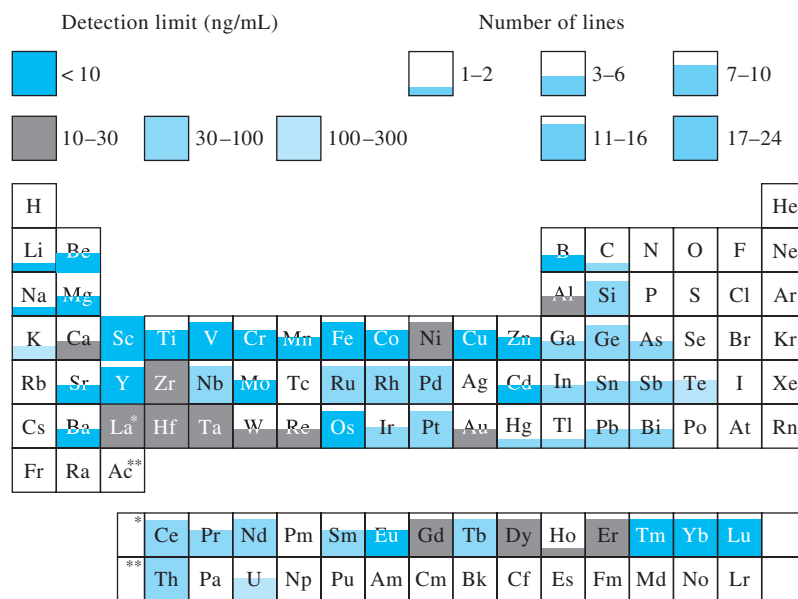


FIGURE 10-14 Periodic table characterizing the detection power and number of useful emission lines of ICP by using a pneumatic nebulizer. The color and degree of shading indicate the range of detection limits for the useful lines. The area of shading indicates the number of useful lines. (Adapted from *Inductively Coupled Plasma Emission Spectroscopy*, Part 1, p. 143, P. W. J. M. Boumans, ed., New York: Wiley, 1987. With permission.)

Line Selection

Figure 10-14 shows that most elements have several prominent lines that can be used for identification and determination purposes. Wavelength data recorded to three decimal places with appropriate intensity information for prominent lines for more than 70 elements can be found in several publications.²⁴ Thus, a suitable line for the determination of any element can usually be found. Selection depends on a consideration of what elements other than the analyte may be present in the sample and whether there is any likelihood that lines of these elements will overlap analyte lines.

Calibration Curves

Calibration curves for plasma emission spectrometry most often consist of a plot of an electrical signal proportional to line intensity versus analyte concentration. When the range of concentrations is large, log-log plots may be used instead. Figure 10-15 shows typical calibration curves for four trace elements present in steel samples. Often, calibration plots are linear, as

are the central two curves in the figure. Departures from linearity often occur, however, when large concentration ranges are covered (see the outer two lines in Figure 10-15). A major cause of nonlinearity is *self-absorption*, in which the output signal is reduced because of absorption by ground-state atoms in the medium. Self-absorption becomes evident only at high analyte concentrations and causes the calibration curve to bend toward the horizontal axis. None of the plots in Figure 10-15 show evidence of self-absorption. Nonlinearity also arises from erroneous background corrections, from ionization, and from nonlinear responses of the detection systems. The nonlinearities in the curves for niobium and thallium at low concentrations are probably a result of incorrect background corrections. Note that the departures from linearity are away from the concentration axis.

An internal standard is often used in emission spectrometry. In this case, the vertical axis of the calibration curve is the ratio or the log ratio of the detector signal for the analyte to the detector signal for the internal standard. Figure 10-16 shows calibration curves for several elements. In these experiments a fixed amount of yttrium was incorporated in all of the standards, and the relative intensity of the analyte line to that of an yttrium line at 242.2 nm served as the analytical variable. Note that all the curves are linear and cover a concentration range of nearly three orders of magnitude. Note also that some of the

²⁴See R. K. Winge, V. A. Fassel, V. J. Peterson, and M. A. Floyd, *Inductively Coupled Plasma Emission Spectroscopy: An Atlas of Spectral Information*, New York: Elsevier, 1985; P. W. J. M. Boumans, *Line Coincidence Tables for Inductively Coupled Plasma Spectrometry*, 2nd ed., Oxford: Pergamon, 1984; C. C. Wohlers, *ICP Information Newslett.*, **1985**, 10, 601.

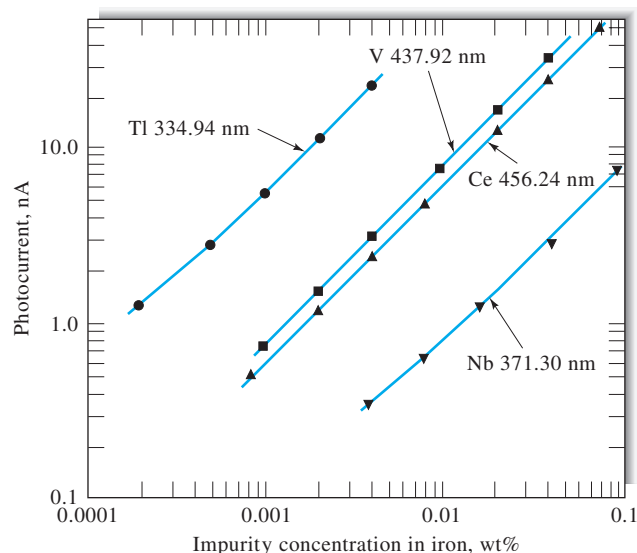


FIGURE 10-15 Typical calibration curves in ICP emission spectrometry. (From V. A. Fassel and R. N. Kniseley, *Anal. Chem.*, **1974**, *46*, 1110A, DOI: 10.1021/ac60349a023. Copyright 1974 American Chemical Society.)

data were obtained by introducing various amounts of the analyte and internal standard into pure water. Other data are for solutions that contained relatively high concentrations of different salts, thus demonstrating a freedom from interelement interference.

As in atomic absorption spectroscopy, one or more standards should be introduced periodically to correct for the effects of instrument drift. The improvement in precision that results from this procedure is illustrated by the data in Table 10-2. Note also the improved precision when higher concentrations of analyte are measured.

Interferences

As was noted previously, chemical interferences and matrix effects are significantly lower with plasma sources than with other

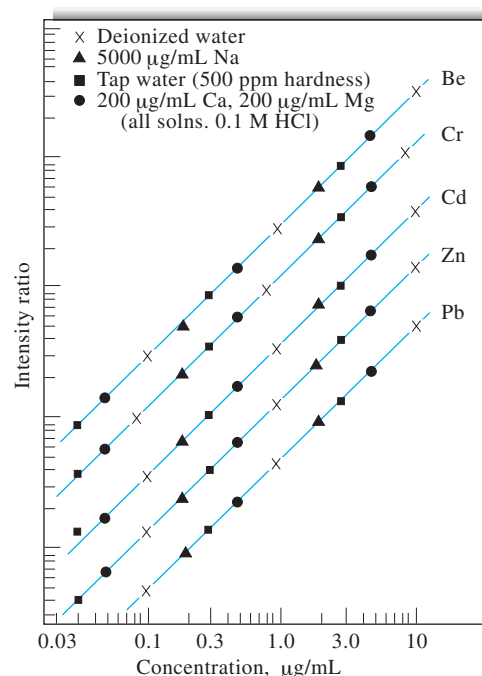


FIGURE 10-16 Internal standard calibration curves with an ICP source. Here, an yttrium line at 242.2 nm served as an internal standard. Notice the lack of interelement interference. (From V. A. Fassel, *Science*, **1978**, *202*, 183. With permission. Copyright 1978 by the American Association for the Advancement of Science.)

atomizers. At low analyte concentrations, however, the background emission due to recombination of argon ions with electrons is large enough to require careful corrections. For both single-channel and multichannel instruments, this correction is made by taking background readings on either side of the line of interest. State-of-the-art instruments include software designed to make background corrections automatically or under operator control.

Because ICP spectra for many elements are so rich in lines, spectral interferences are always possible. To avoid this type of error requires knowledge of all of the components likely to be

TABLE 10-2 Effect of Standardization Frequency on Precision of ICP Data

Frequency of Recalibration, hours	Relative Standard Deviation, %			
	10 ¹ to 10 ²	Concentration Multiple above Detection Limit 10 ² to 10 ³	10 ³ to 10 ⁴	10 ⁴ to 10 ⁵
0.5	3–7	1–3	1–2	1.5–2
2	5–10	2–6	1.5–2.5	2–3
8	8–15	3–10	3–7	4–8

Data from R. M. Barnes, in *Applications of Inductively Coupled Plasmas to Emission Spectroscopy*, R. M. Barnes, ed., p. 16, Philadelphia: The Franklin Institute Press, 1978. With permission.

TABLE 10-3 Comparison of Detection Limits for Several Atomic Spectral Methods

Method	Number of Elements Detected at Concentrations of				
	<1 ppb	1–10 ppb	11–100 ppb	101–500 ppb	>500 ppb
ICP emission	9	32	14	6	0
Flame atomic emission	4	12	19	6	19
Flame atomic fluorescence	4	14	16	4	6
Flame atomic absorption	1	14	25	3	14

Data abstracted with permission from V. A. Fassel and R. N. Kniseley, *Anal. Chem.*, **1974**, 46 (13), 1110A. Copyright 1974 American Chemical Society.

Detection limits correspond to a signal that is twice as great as the standard deviation for the background noise.

present in the sample and a careful study of the information in the reference works listed in note 23. The software for modern computerized instruments has powerful routines for wavelength and concentration calibration, spectral analysis, and deconvolution of overlapping lines. These features coupled with integrated databases of spectral lines make spotting and correcting for interferences an integral part of the analytical process.

Detection Limits

In general, detection limits with the ICP source are comparable to or better than other atomic spectral procedures. Table 10-3 compares detection limits for several of these methods. Note that more elements can be detected at levels of 10 ppb or less with plasma excitation than with other emission or absorption methods. As we shall see in Chapter 11, the ICP coupled with mass spectrometric detection improves detection limits by two to five orders of magnitude for many elements and is thus strong competition for ICP optical emission spectroscopy.

10B EMISSION SPECTROSCOPY BASED ON ARC AND SPARK SOURCES

Arc and spark source spectroscopies were the first instrumental methods to become widely used for analysis. These techniques, which began to replace the classical gravimetric and volumetric methods for elemental analysis in the 1920s, were based on excitation of emission spectra of elements with electric arcs or high-voltage sparks. These spectra permitted the qualitative and quantitative determination of metallic elements in a variety of sample types, including metals and alloys, soils, minerals, and rocks.²⁵ Arc and spark sources are still used in some

situations for qualitative and semiquantitative analysis, particularly in the metals industries. Arcs and sparks are gradually being displaced by plasma sources, and it appears likely that this trend will continue.

In arc and spark sources, sample excitation occurs in the gap between a pair of electrodes. Passage of electricity from the electrodes through the gap provides the necessary energy to atomize the sample and produce atoms or ions in electronic excited states.

10B-1 Characteristics of DC Arcs and High-Voltage Sparks

Arc and spark source methods are currently limited to the elemental analysis of solids because liquids and gaseous samples are handled much more conveniently by plasma emission methods. If the sample is a metal, electrodes can be formed from the sample by milling, by turning, or by casting the molten metal in a mold. For nonmetallic solids, the sample is often supported on an electrode. Carbon is an ideal electrode for many applications. It can be obtained in a highly pure form, is a good conductor, has good heat resistance, and is easily shaped.

Because of their instabilities, it is necessary to integrate the emission signals from arc and spark sources for at least 20 s and often for a minute or more. This requirement makes the sequential spectrometers, such as those described in section 10A-4, impractical for most applications. Simultaneous multichannel instruments are most often used. Spectrographs, polychromators, and array detector-based multichannel instruments are used.

Direct-current arc sources produce spectra that are rich in intense lines for atoms and contain a lesser number for ionic species. Direct-current arcs produced from a carbon or graphite electrode in air emit intense bands due to the presence of cyanogen radicals (CN). As a result, most of the region between 350 to 420 nm is useless for elemental analysis. High-voltage sparks are much more energetic than typical direct-current arcs. Because

²⁵For additional details, see J. D. Ingle Jr. and S. R. Crouch, *Spectrochemical Analysis*, Chap. 9, Upper Saddle River, NJ: Prentice-Hall, 1988; R. D. Sacks, in *Treatise on Analytical Chemistry*, 2nd ed., P. J. Elving, E. J. Meehan, and I. M. Kolthoff, eds., Part I, Vol. 7, Chap. 6, New York: Wiley, 1981; P. W. J. M. Boumans in *Analytical Emission Spectroscopy*, Vol. 1, Part I, E. L. Grove, ed., New York: Marcel Dekker, 1972.

of this, ionic spectra are much more pronounced in a high-voltage spark than in a direct-current arc. In fact, lines emitted by ions are often termed “spark lines” by spectroscopists, while atom lines are sometimes called “arc lines.”

10B-2 Applications of Arc and Spark Emission

Arc sources are particularly useful for qualitative and semiquantitative analysis of nonmetallic samples, such as soils, plant materials, rocks, and minerals. The precision obtainable with an arc source is usually poorer than that with a spark source and generally much poorer than that with a plasma source. Currently, the primary use for spark emission spectroscopy is for the identification and analysis of metals and other conducting materials. High-voltage sparks have also become important devices for ablating solid samples prior to introduction into plasma excitation sources.

An important use for spark and arc spectrometers is in foundry and shop floors, scrapyards, and metal-casting facilities. The instruments for these are often mobile and may be equipped with a handheld spark or arc gun that the operator can touch to the metal surface to excite atoms of the sample. These devices can be used for the rapid identification of alloy types and for sorting metals by type in scrapyards and recycling centers.

10C OTHER SOURCES FOR OPTICAL EMISSION SPECTROSCOPY

In addition to plasma, arc, and spark sources, two additional sources are also used: flame emission sources and the glow-discharge source.

10C-1 Flame Emission Sources

For many years, flames have been used to excite emission spectra for various elements, and most modern atomic absorption spectrometers may be adapted for flame emission measurements. Flames are not widely used for this purpose, however, because for most single-element determinations, absorption methods provide as good or better accuracy, convenience, and detection limits. For multielement analyses, plasma sources are far superior to flames in most regards. For these reasons flame emission spectrometry is little used except for the determination of the alkali metals and occasionally calcium. These elements are excited at the relatively low temperatures of flames to give spectra that are remarkably simple and free of interference from other metallic species. Alkali-metal spectra generally consist of a relatively few intense lines, many of which are in the visible region and are well suited to quantitative emission measurements.

Because these spectra are simple, basic filter photometers can be quite adequate for routine determinations of the alkali and alkaline-earth metals. A low-temperature flame is used to avoid excitation of most other metals. As a result, interference filters can isolate the appropriate emission line.

Several instrument manufacturers supply flame photometers designed specifically for the determination of sodium, potassium, lithium, and sometimes calcium in blood serum, urine, and other biological fluids. Single-channel and multichannel (two to four channels) instruments are available for these determinations. In the multichannel instruments, each channel can be used to determine a separate element without an internal standard, or one of the channels can be reserved for an internal standard such as lithium. The ratios of the signals from the other channels to the signal of the lithium channel are then taken to compensate for flame noise and noise from fluctuations in reagent flow rate. Flame photometers such as these have been coupled with flow injection systems to automate the sample-introduction process (see Section 33B-3). Typically, relative standard deviations for flow-injection-analysis-based flame photometric determinations of lithium, sodium, and potassium in serum are on the order of a few percent or less. Automated flow injection procedures require 1/100 the amount of sample and 1/10 the time of batch procedures.²⁶

Although the alkali metals are determined daily in a huge number of samples throughout the world, most are clinical samples that are analyzed potentiometrically (see Chapter 23). Flame photometry is currently used for only a fraction of these samples.

10C-2 Glow-Discharge Sources

The glow discharge, which is described in Section 8C-2 and Section 9A-3, has proven to be a useful source for exciting emission spectra of metals, alloys, and other solid materials.²⁷ Glow-discharge optical emission spectroscopy (GDOES) has become a mature technique in recent years, and several instrument manufacturers offer glow-discharge sources as well as complete spectrometers configured for this type of analysis. Glow-discharge spectroscopy is a versatile technique in that it is capable of bulk analysis and depth profiling of solids.

Figure 10-17 shows a typical cell for GDOES, and it is similar in many respects to the absorption cells described in Sections 8C-2 and 9A-3. A dc voltage of up to 1 kV applied between the electrodes produces sputtering of the solid sample

²⁶G. N. Doku and V. P. Y. Gadzekpo, *Talanta*, **1996**, *43*, 735, DOI: 10.1016/0039-9140(95)01808-5.

²⁷*Glow Discharge Plasmas in Analytical Spectroscopy*, R. K. E. Marcus and J. A. C. E. Broekaert, eds., Hoboken, NJ: Wiley, 2003.

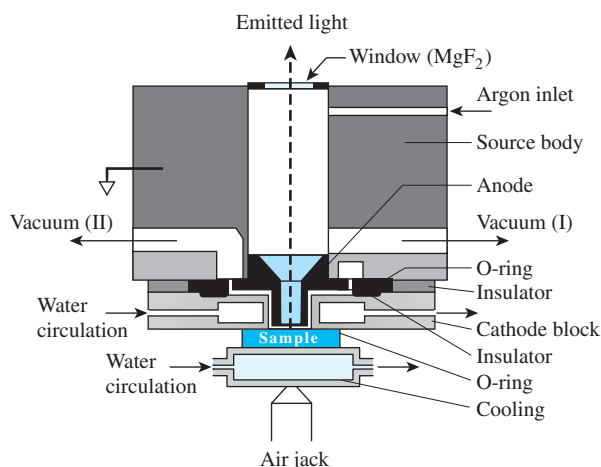


FIGURE 10-17 Diagram of a Grimm-type glow-discharge source. (From M. Boucharcourt and F. Schwoeher, in *Glow Discharge Optical Emission Spectrometry*, R. Payling, D. G. Jones, and A. Bengtson, eds., p. 54, New York: Wiley, 1997, with permission.)

at a current of 40–200 mA. In the glow discharge, at the cathode surface, ground-state analyte atoms become excited by collision with high-energy electrons, relax, and emit their characteristic radiation. Radio-frequency (RF) excitation, which permits nonconducting materials to be analyzed by GDOES, is available

in some commercial instruments, and pulsed dc and pulsed RF modes have been explored to increase line intensities.²⁸

GDOES depth profiling is illustrated in the curves of Figure 10-18 for a sample of brass. The plots show profiles for seven monitored elements as a function of time from the initiation of the glow discharge. During the pre-integration period, surface contaminants volatilize, and over a period of 60 s, the signals reach a relatively constant level corresponding to the composition of the bulk material. The length of the pre-integration period is best determined by the precision of the signal at various times during the sputtering process. For the sample illustrated, the precision was optimal at about 3% relative over the period indicated by I_1 . This period was thus chosen for the determination of the composition of similar samples. Depending on the nature of the analysis, the power, the pressure, and the pre-integration and integration periods can be selected to optimize results.²⁹ The relatively constant signal levels for all of the elements indicate uniform composition throughout the

²⁸N. Jakubowski, A. Bogaerts, and V. Hoffmann, *Atomic Spectroscopy in Elemental Analysis*, M. Cullen, ed., p. 120, Boca Raton, FL: CRC Press, 2004.

²⁹T. A. Nelis and R. A. Payling, *Glow Discharge Optical Emission Spectroscopy*, pp. 23–24, New York: Springer, 2004.

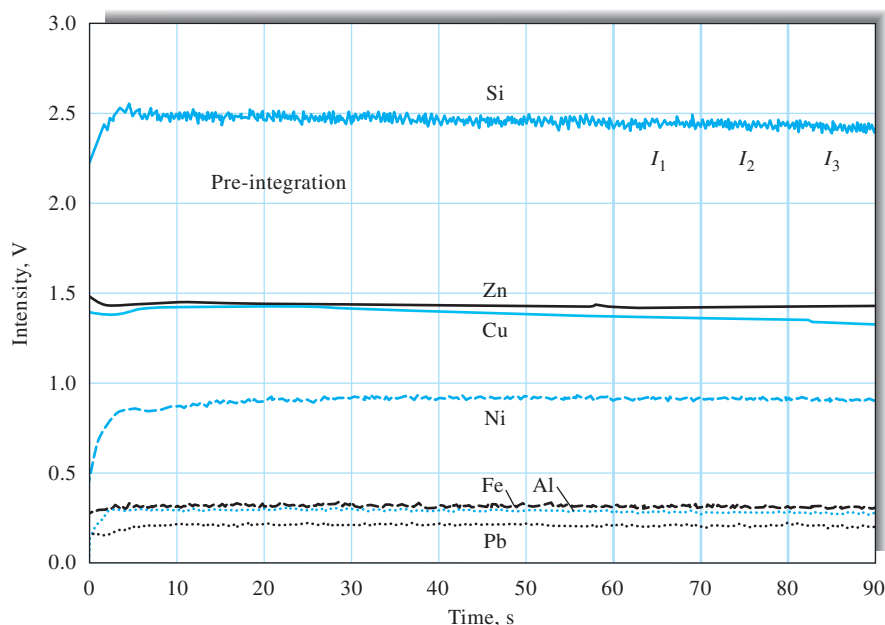


FIGURE 10-18 Qualitative depth profile of a brass sample, showing major and minor elements, indicating the time regions to be used for pre-integration (60 s) and three 10-s integration periods I_1 , I_2 , and I_3 . (From T. A. Nelis and R. A. Payling, *Glow Discharge Optical Emission Spectroscopy*, p. 23, New York: Springer, 2004. With permission.)

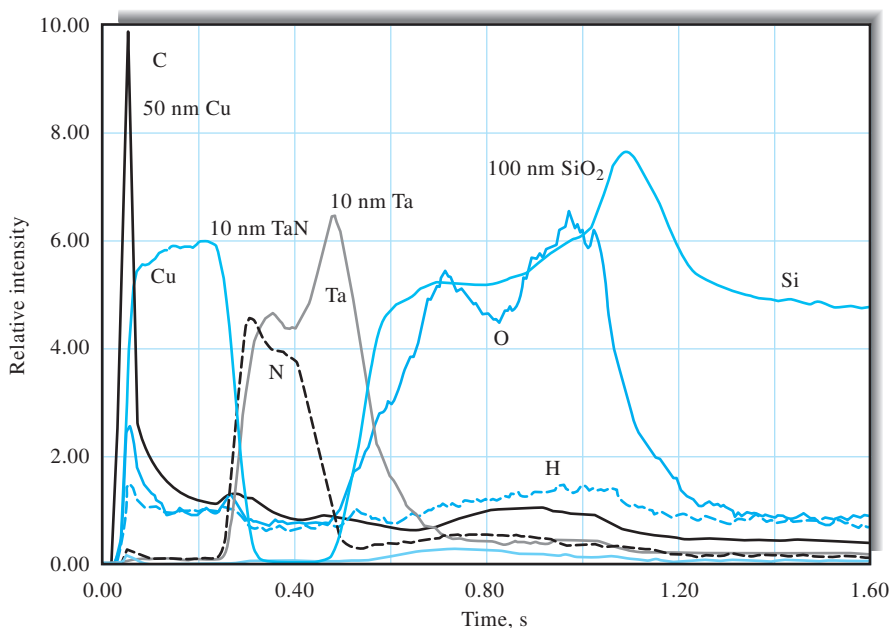


FIGURE 10-19 RF-GDOES depth profile of a microelectronic multilayer system. Note the thickness and composition of each layer shown above each peak or band and the elemental composition indicated by the curves. (From N. Jakubowski, A. Bogaerts, and V. Hoffmann, *Atomic Spectroscopy in Elemental Analysis*, M. Cullen, ed., p. 129, Boca Raton, FL: CRC Press, 2004. With permission.)

sample. In Figure 10-19, we see a depth profile of an electronic integrated circuit obtained using GDOES with RF excitation. The emergence with time of a peak or a band indicates the appearance of the elements contained in each successive layer of the circuit materials. Labels above each curve indicate the identity and thickness of each layer.³⁰

Because of the low background levels of GDOES, detection limits on the order of parts per million are typical using the

Grimm source of Figure 10-17. The dynamic range is relatively large compared to arc and spark sources, and relative standard deviations of 1% or lower are common with these devices.³¹

Recent work with glow discharges has developed the solution-cathode technique that has several advantages.³² Unlike plasma sources, the solution-cathode glow discharge requires no gas or fuel and can sample analytes directly without a nebulizer.

³⁰N. Jakubowski, A. Bogaerts, and V. Hoffmann, *Atomic Spectroscopy in Elemental Analysis*, M. Cullen, ed., p. 120, Boca Raton, FL: CRC Press, 2004.

³¹J. A. C. Broekaert, *Analytical Atomic Spectrometry with Flames and Plasmas*, pp. 244–246. Hoboken, NJ: Wiley-VCH, 2002.

³²Wang, Zheng, A. J. Schwartz, S. J. Ray, and G. M. Hieftje, *J. Anal. Atom. Spectrosc.*, **2013**, 28, 234, DOI: 10.1039/c2ja30253e.

QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.

 Problems with this icon are best solved using spreadsheets.

10-1 What is an internal standard and why is it used?

10-2 Why are atomic emission methods with an ICP source better suited for multielement analysis than are flame atomic absorption methods?

10-3 Why do ion lines predominate in spark spectra and atom lines in arc spectra?

* **10-4** Calculate the theoretical reciprocal linear dispersion of an echelle grating with a focal length of 0.85 m, a groove density of 120 grooves/mm, and a diffraction angle of $61^{\circ}16'$ when the diffraction order is (a) 30 and (b) 90.

>> QUESTIONS AND PROBLEMS (continued)

- 10-5** Why are arc sources often blanketed with a stream of an inert gas?
- 10-6** Describe three ways of introducing a sample into an ICP torch.
- 10-7** What are the relative advantages and disadvantages of ICP plasmas compared to microwave plasmas?
- 10-8** Why are ionization interferences less severe in ICP than in flame emission spectroscopy?
- 10-9** What are some of the advantages of plasma sources compared with flame sources for emission spectrometry?
- 10-10** Discuss the advantages and disadvantages of sequential versus simultaneous multichannel ICP spectrometers.
- 10-11** Gold can be determined in solutions containing high concentrations of diverse ions by ICP-AES.³³ Aliquots of 5.00 mL of the sample solution were transferred to each of four 50.0 mL volumetric flasks. A standard solution was prepared containing 10.0 mg/L Au in 20% H₂SO₄, and the following quantities of this solution were added to the sample solutions: 0.00, 2.50, 5.00, and 10.00 mL added Au in each of the flasks. The solutions were made up to a total volume of 50.0 mL, mixed, and analyzed by ICP-AES. The resulting data are presented in the following table.

Volume of Standard, mL	Emission Intensity, Counts
0.00	12,568
2.50	19,324
5.00	26,622
10.00	40,021

- (a) Use a spreadsheet to perform a least-squares analysis to determine the slope, intercept, and regression statistics, including the standard deviation about regression.
- (b) Use your results to find the concentration of gold in the sample solution in mg/L.
- (c) The known concentration of gold in the sample is 8.51 mg/L. Test the hypothesis that your result is equal to this value at the 95% confidence level.

 Challenge Problem

- 10-12** Watters et al. have discussed uncertainties associated with calibration curves for ICP-OES and procedures for optimizing the results of least-squares analysis of such data.³⁴
- (a) The central issue addressed by these workers is whether a weighted or an unweighted least-squares procedure is appropriate for ICP calibration. What is the primary criterion for deciding which procedure to use?
- (b) What do the terms *homoscedasticity* and *heteroscedasticity* mean?
- (c) The model used by these workers for the ICP curves is represented by the following equation:

$$Y_{ij} = a + bx_i + \text{error}_{ij}$$

Define each variable in the equation, and describe the significance of the relationship embodied in it.

- (d) Watters et al. chose to model the error in ICP working curves in terms of concentration rather than intensity. What is their rationale for this choice?

³³J. A. Whitehead, G. A. Lawrance, and A. McCluskey, *Aust. J. Chem.*, **2004**, 57, 151, DOI: 10.1071/CH03198.

³⁴R. L. Watters Jr., R. J. Carroll, and C. H. Spiegelman, *Anal. Chem.*, **1987**, 59, 1639, DOI: 10.1021/ac00140a013.

- (e) One of the suggested models for the error in the calibration curves is

$$\sigma(x) = c + dx + ex^2$$

Describe the significance of each variable, and characterize the nature of the model.

- (f) How is $\sigma(x)$ determined in practice?
- (g) Shot noise and source flicker noise correspond to which variables in the expression shown in (e)?
- (h) What experimental noise sources are constant? Which variable in the expression shown in (e) corresponds to these sources (or source)?
- (i) What is the relationship of the following alternative model to the model in (e)?

$$\sigma^2(x) = g + hx + kx^2$$

- (j) What is the purpose of the models in (e) and (i)?
- (k) What is the significance of each of the variables $\hat{a}$ and $\hat{b}$?
- (l) The authors conclude that “if heteroscedasticity is ignored, confidence intervals will be too narrow at the high end and too wide at the low end of the ICP calibration curve. The magnitude of these effects will depend on the particular dilution scheme used to make the calibration standard solutions.” How does the dilution scheme for standards affect the results of least-squares analysis on ICP calibration curves?
- (m) Create an Excel spreadsheet similar to the one shown next containing data from Table I of the paper by Watters et al., and perform both unweighted and weighted least-squares analysis of the data. First use LINEST to perform the unweighted analysis, and then perform an unweighted analysis using Solver to minimize cell C25 by varying B14 and C14. Compare the results obtained by both methods. Compare and contrast advantages of each method.³⁵

	A	B	C	D	E	F	G	H	I
1	Concn,	Intensity	s_i	s	s				
2	$\mu\text{g}/\mu\text{L}$	(counts)	(obsd)	(10 reps)	(4 reps)			LINEST	
3	0.00	11.33	8.54	7.88	6.97			b	a
4	0.0101	16.60	7.88	7.98	7.03		Value		
5	0.0251	37.92	9.06	8.12	7.12		Uncertainty		
6	0.0503	57.00	8.46	8.36	7.28		R / Sy		
7	0.101	149.88	6.13	8.84	7.58		F / dF		
8	0.251	369.24	11.57	10.25	8.50		SSreg/SSres		
9	0.503	763.36	11.94	12.48	10.02				
10	2.51	3688.46	26.24	25.42	22.25				
11	5.03	7431.08	29.12	29.30	37.55				
12									
13	Unweighted	b	a	Weighted(10)	b	a	Weighted(4)	b	a
14									
15	$bx + a$	Resid	Resid ²	$bx + a$	Resid	Resid ² / s^2	$bx + a$	Resid	Resid ² / s^2
16									
17	Cell A16=\$B\$14*A3+\$C\$14				Cell B14=Initial estimate of slope				
18	Cell B16=B3-A16				Cell C14=Initial estimate of intercept				
19	Cell C16=B16^2								
20	Cell D16=\$E\$14*A3+\$F\$14								
21	Cell F16=E16^2*(1/D3^2)								
22	Cell C25=SUM(C16:C24)								
23	Cell C26=SQRT(C25/(COUNT(C16:C24)-2))								
24									
25		SS	0.0000		SS	0.0000		SS	0.000
26		s_r	0.0000		s_r	0.0000		s_r	0.0000

³⁵S. R. Crouch and F. J. Holler, *Applications of Microsoft® Excel in Analytical Chemistry*, 3rd ed., pp. 95, 131–138, Belmont, CA: Cengage Learning, 2017.

>> QUESTIONS AND PROBLEMS (continued)

Repeat the analysis for the ten-replicate data using the formulas provided in the spreadsheet documentation. Formulas entered in row 16 must be copied into rows 17–24. Minimize cell F25 by varying cells E14 and F14 using Solver to obtain estimates of the slope and intercept. Repeat the analysis for the four-replicate data using Solver to minimize cell I25 by varying cells H14 and I14. Compare your procedures and results to those of Watters et al. and comment on any differences. What advantage does weighted least-squares analysis have over unweighted analysis?

- (n) Add a section to your Excel spreadsheet to compute the mean and standard deviation of the concentration of an analyte given a number of measurements of sample ICP emission intensity.³⁶
- (o) There are a number of commercial and noncommercial sources on the Internet for Excel add-ins, which are programs or function packs that supplement the built-in functions of Excel. For example, Solver is actually an add-in that is produced by an independent contractor and that is available from the vendor in an enhanced version. Use a search engine such as Google to locate the websites of programmers and vendors that offer add-ins to perform weighted least-squares analysis. One such vendor is XLSTAT (www.xlstat.com). Download the demonstration version of XLSTAT, install it on your computer, and use it to perform weighted and unweighted least-squares analysis on the data of (n). Note that you must calculate the weighting factors from the standard deviations given in the paper and that you may need to scale them. You should calculate columns in your spreadsheet containing $1/s^2$, and then divide each cell by the largest value in the column. Weighting factors need be only proportional to the variance, so you can scale them in any convenient way. Compare the results from XLSTAT with the results from your spreadsheet, and comment on the ease of use and functionality of XLSTAT. XLSTAT contains many other useful statistics and numerical analysis functions and is available at modest cost to students who desire permanent use of the add-in.

³⁶D. A. Skoog, D. M. West, F. J. Holler, and S. R. Crouch, *Fundamentals of Analytical Chemistry*, 9th ed., pp. 173–174, Belmont, CA: Brooks/Cole, 2014.

Atomic Mass Spectrometry

Atomic mass spectrometry is a versatile and widely used tool for identifying the elements present in samples of matter and for determining their concentrations. Nearly all the elements in the periodic table can be determined by mass spectrometry. Atomic mass spectrometry offers a number of advantages over the atomic optical spectrometric methods considered in the preceding chapters, including (1) detection limits that are, for many elements, as great as three orders of magnitude better than optical methods; (2) remarkably simple spectra that are usually unique and often easily interpretable; and (3) the ability to measure atomic isotopic ratios. Disadvantages include (1) instrument costs that are two to three times that of optical atomic instruments, (2) instrument drift that can be as high as 5% to 10% per hour, and (3) certain types of interference effects that are discussed later.



Throughout this chapter, this logo indicates an opportunity for online self-study at www.tinyurl.com/skoogpia7, linking you to interactive tutorials, simulations, and exercises.

11A SOME GENERAL FEATURES OF ATOMIC MASS SPECTROMETRY

An atomic mass spectrometric analysis¹ contains the following steps: (1) atomization, (2) conversion of a substantial fraction of the atoms formed in step 1 to a stream of ions (usually singly charged positive ions), (3) separating the ions formed in step 2 on the basis of their mass-to-charge ratio (m/z), where m is the mass number of the ion and z is the number of fundamental charges that it bears,² and (4) counting the number of ions of each type or measuring the ion current produced when the ions formed from the sample strike a suitable transducer. Because most of the ions formed in step 2 are singly charged ($z = +1$), m/z is usually simply the mass number of the ion. Steps 1 and 2 require the same techniques that were discussed in Section 8C for atomic optical spectroscopy. Steps 3 and 4 are carried out with a mass spectrometer. The data from mass spectrometry are usually presented as a plot of relative intensity or ion abundance versus m/z .

11A-1 Atomic Masses in Mass Spectrometry

At the outset, it is worthwhile to point out that the atomic masses in the literature of mass spectrometry, and in this chapter, differ from those used in most other subdisciplines of analytical chemistry, because mass spectrometers discriminate among the masses of isotopes but other analytical instruments generally do not. We review briefly some terms related to atomic (and molecular) masses.

¹For general treatments of atomic mass spectrometry, see S. Becker, *Inorganic Mass Spectrometry*, Chichester, UK: Wiley, 2007; J. R. A. De Laeter, *Applications of Inorganic Mass Spectrometry*, Hoboken, NJ: Wiley-Interscience, 2001; C. M. A. Barshick, D. C. A. Duckworth, and D. H. A. Smith, *Inorganic Mass Spectrometry: Fundamentals and Applications*, New York: Dekker, 2000. For general discussion of mass spectrometry, see J. H. Gross, *Mass Spectrometry: A Textbook*, 2nd ed., Berlin: Springer-Verlag, 2011; C. Dass, *Fundamentals of Contemporary Mass Spectrometry*, Hoboken, NJ: Wiley, 2007; E. A. de Hoffmann and V. A. Stroobant, *Mass Spectrometry: Principles and Applications*, 3rd ed., Hoboken, NJ: Wiley, 2007; J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., Chichester, UK: Wiley, 2007.

²Strictly, the mass number m is unitless as is z , the number of fundamental charges on an ion. The quantity z , which is an integer, is equal to q/e , where q is the charge on the ion and e is the charge on the electron, both measured in the same units (e.g., coulombs). For a review of mass spectrometry definitions and terms, see K. K. Murray, et al., *Pure Appl. Chem.*, **2013**, 85, 1515, DOI: 10.1351/PAC-REC-06-04-06.

Atomic and molecular masses are generally expressed on the *atomic mass scale*, which is based on a specific isotope of carbon. One *unified atomic mass unit* on this scale is equal to 1/12 the mass of a neutral ^{12}C atom. The unified atomic mass unit is given the symbol u. One unified atomic mass unit is commonly termed one *dalton* (Da), which has become the accepted term even though it is not an official SI unit. The older term, atomic mass unit (amu) is discouraged since it was based on the most abundant stable isotope of oxygen ^{16}O . Thus, *one unified atomic mass unit, or Da, is*

$$\begin{aligned}\text{mass of 1 atom } ^{12}\text{C} &= \frac{12 \text{ g } ^{12}\text{C/mol } ^{12}\text{C}}{6.022142 \times 10^{23} \text{ atoms } ^{12}\text{C/mol } ^{12}\text{C}} \\ &= 1.992646 \times 10^{-23} \text{ g/atom } ^{12}\text{C} \\ &= 1.992646 \times 10^{-26} \text{ kg/atom } ^{12}\text{C}\end{aligned}$$

The unified atomic mass unit is then

$$\begin{aligned}1 \text{ u} &= 1 \text{ Da} \\ &= \frac{1}{12} (1.992646 \times 10^{-23} \text{ g}) \\ &= 1.6605387 \times 10^{-24} \text{ g} \\ &= 1.6605387 \times 10^{-27} \text{ kg}\end{aligned}$$

The relative atomic mass of an isotope such as ^{35}Cl is then measured with respect to the mass of the reference ^{12}C atom. Chlorine-35 has a mass that is 2.914071 times greater than the mass of the carbon isotope. Therefore, the atomic mass of the chlorine isotope is

$$\begin{aligned}\text{atomic mass } ^{35}\text{Cl} &= 2.914071 \times 12.000000 \text{ Da} \\ &= 34.968853 \text{ Da}\end{aligned}$$

Because 1 mol of ^{12}C weighs 12.000000 g, the molar mass of ^{35}Cl is 34.968853 g/mol.³

In mass spectrometry, in contrast to most types of chemistry, we are often interested in the exact mass *m* of particular isotopes of an element or the exact mass of compounds containing a particular set of isotopes. Thus, we may need to distinguish between the masses of compounds such as

$$\begin{aligned}^{12}\text{C}^1\text{H}_4 \quad m &= (12.000000 \times 1) + (1.008 \times 4) \\ &= 16.03200 \text{ Da} \\ ^{13}\text{C}^1\text{H}_4 \quad m &= (13.0000 \times 1) + (1.008 \times 4) \\ &= 17.0320 \text{ Da} \\ ^{12}\text{C}^1\text{H}_3^2\text{H}_1 \quad m &= (12.0000 \times 1) + (1.008 \times 3) \\ &\quad + (2.0160 \times 1) \\ &= 17.0400 \text{ Da}\end{aligned}$$

We have presented the isotopic masses in the aforementioned calculations with four digits to the right of the decimal point. Normally, exact masses are quoted to three or four figures to the right of the decimal point because typical high-resolution mass spectrometers make measurements at this level of precision.

In other contexts, we use the term *nominal mass*, which implies a whole-number precision in a mass measurement. Thus, the nominal masses of the three isomers just cited are 16, 17, and 17 Da, respectively.

The *chemical atomic mass*, or the *average atomic mass* (*A*), of an element in nature is given by the equation

$$A = A_1p_1 + A_2p_2 + \cdots + A_np_n = \sum_{i=1}^n A_ip_i$$

where $A_1, A_2, \dots, A_n$ are the atomic masses in daltons of the *n* isotopes of the element and $p_1, p_2, \dots, p_n$ are the fractional abundances of these isotopes in nature. The chemical atomic mass is, of course, the type of mass of interest to chemists for most purposes. The average, or chemical, molecular mass of a compound is then the sum of the chemical atomic masses for the atoms appearing in the formula of the compound. Thus, the chemical molecular mass of CH_4 is $12.011 + (4 \times 1.008) = 16.043 \text{ Da}$. The atomic or molecular mass expressed without units is the *mass number*.

11A-2 Mass-to-Charge Ratio

One other term that is used throughout this chapter is the *mass-to-charge ratio* of an atomic or molecular ion. The mass-to-charge ratio of an ion is the unitless ratio of its mass number to the number of fundamental charges *z* on the ion. Thus, for $^{12}\text{C}^1\text{H}_4^+$, $m/z = 16.032/1 = 16.032$. For $^{13}\text{C}^1\text{H}_4^{2+}$, $m/z = 17.032/2 = 8.516$. Because most ions in mass spectrometry are singly charged, the term mass-to-charge ratio is sometimes shortened to the more convenient term *mass*. Strictly speaking, this abbreviation is not correct, but it appears widely in the mass spectrometry literature.

11A-3 Types of Atomic Mass Spectrometry

Table 11-1 lists the most important types of atomic mass spectrometry. Historically, thermal ionization mass spectrometry and spark source mass spectrometry (SSMS) were the first mass spectrometric methods developed for qualitative and quantitative elemental analysis, and these types of procedures still find applications, although they are now overshadowed by some of the other methods listed in Table 11-1, particularly inductively coupled plasma mass spectrometry (ICPMS). Note that the first three entries in the table are *hyphenated methods*, combinations of two instrumental techniques that produce analytical results superior in some way to the results from either of the original individual methods. We shall encounter a number of hyphenated methods elsewhere in this book.

³A listing of the isotopes of all of the elements and their atomic masses can be found in several handbooks, such as *Handbook of Chemistry and Physics*, 96th ed., Boca Raton, FL: CRC Press, 2015, also available online through many libraries.

TABLE 11-1 Types of Atomic Mass Spectrometry

Name	Initialism or Acronym	Atomic Ion Sources	Typical Mass Analyzer
Inductively coupled plasma	ICPMS	High-temperature argon plasma	Quadrupole
Direct current plasma	DCPMS	High-temperature argon plasma	Quadrupole
Microwave-induced plasma	MIPMS	High-temperature argon plasma	Quadrupole
Spark source	SSMS	Radio-frequency electric spark	Double-focusing
Thermal ionization	TIMS	Electrically heated plasma	Double-focusing
Glow discharge	GDMS	Glow-discharge plasma	Double-focusing
Laser microprobe	LMMS	Focused laser beam	Time-of-flight
Secondary ion	SIMS	Accelerated ion bombardment	Double-focusing

Before examining the various methods for atomization and ionization found in the table, we describe briefly how mass spectrometers are used to separate and measure ionic species.

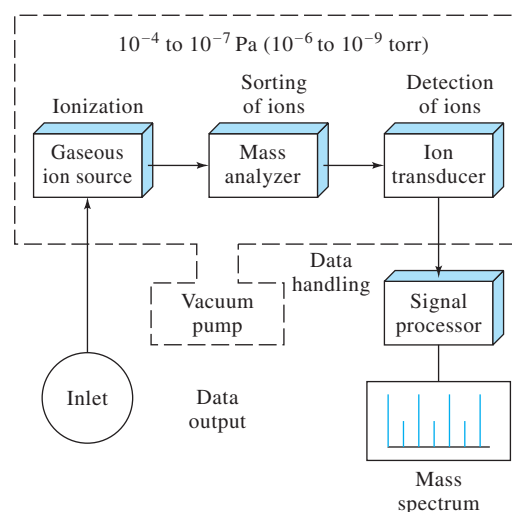
11B MASS SPECTROMETERS

A mass spectrometer is an instrument that produces ions and separates them according to their mass-to-charge ratios, m/z . Most of the ions we will discuss are singly charged so that the ratio is simply equal to the mass number of the ion. Several types of mass spectrometers are currently available from instrument manufacturers. In this chapter, we describe the three types that are used in atomic mass spectrometry: the *quadrupole mass spectrometer*, the *time-of-flight mass spectrometer*, and the *double-focusing mass spectrometer*. Other types of mass spectrometers are considered in Chapter 20, which is devoted to molecular mass spectrometry. The first column in Table 11-1 indicates the types of atomic mass spectrometry in which each of the three kinds of mass spectrometer is usually applied.

The block diagram in Figure 11-1 shows the principal components of all types of mass spectrometers. The purpose of the inlet system is to introduce a micro amount of sample into the ion source where the components of the sample are converted into gaseous ions by bombardment with electrons, photons, ions, or molecules. Alternatively, ionization is accomplished by applying thermal or electrical energy. The output of the ion source is a stream of positive (most common) or negative gaseous ions that are then accelerated into the mass analyzer.

The function of the mass analyzer is analogous to the monochromator in an optical spectrometer. In a mass analyzer, however, dispersion depends on the mass-to-charge ratio of analyte ions rather than on the wavelength of photons.

Like an optical spectrometer, a mass spectrometer contains a transducer that converts the beam of ions into an electrical signal that can then be processed, stored in the memory of


FIGURE 11-1 Components of a mass spectrometer.

a computer, and displayed or stored. Unlike most optical spectrometers, mass spectrometers require an elaborate vacuum system to maintain a low pressure in all of the components except the signal processor and readout. Low pressure ensures infrequent collisions in the mass spectrometer to produce and maintain free ions and electrons.⁴

In the sections that follow, we first describe the various transducer systems that are used in mass spectrometers. Then we consider the three types of mass analyzers that are commonly used in atomic mass spectrometers. Later sections contain material on the nature and operation of common ion sources for atomic mass spectrometers.

⁴The SI unit of pressure is the pascal (Pa), which is one newton of force per square meter of area. It is common in mass spectrometry to find pressures quoted in torr (mm of mercury). These manometric units are readily converted to Pa since 1 torr = 133.3 Pa (very roughly 100 Pa/torr).

11B-1 Transducers for Mass Spectrometry

Several types of transducers are commercially available for mass spectrometers. The *electron multiplier* is the transducer of choice for most routine experiments.

Electron Multipliers

Figure 11-2a is a schematic of a discrete-dynode electron multiplier designed for collecting and converting positive ions into an electrical signal. This device is similar to the photomultiplier transducer for ultraviolet-visible radiation, with each dynode held at a successively higher voltage. When energetic ions or electrons strike the Cu-Be surfaces of the cathode and the dynodes, bursts of electrons are emitted. The electrons are then attracted to the next dynode down the chain until, at the

last dynode, a huge number of electrons appear for every ion that strikes the cathode. Electron multipliers with up to twenty dynodes are available that typically provide a current gain of 10^7 .

Figure 11-2b illustrates a continuous-dynode electron multiplier. The device is shaped like a cornucopia and is made of glass heavily doped with lead to give the material a small conductivity. A voltage of 1.8 to 2 kV applied across the length of the transducer produces a voltage gradient from one end to the other. Ions that strike the surface near the entrance eject electrons that are then attracted to higher-voltage points farther along the tube. These *secondary electrons* skip along the surface, ejecting more electrons with each impact. Transducers of this type typically have gains of 10^5 , but in certain applications gains as high as 10^8 can be achieved.

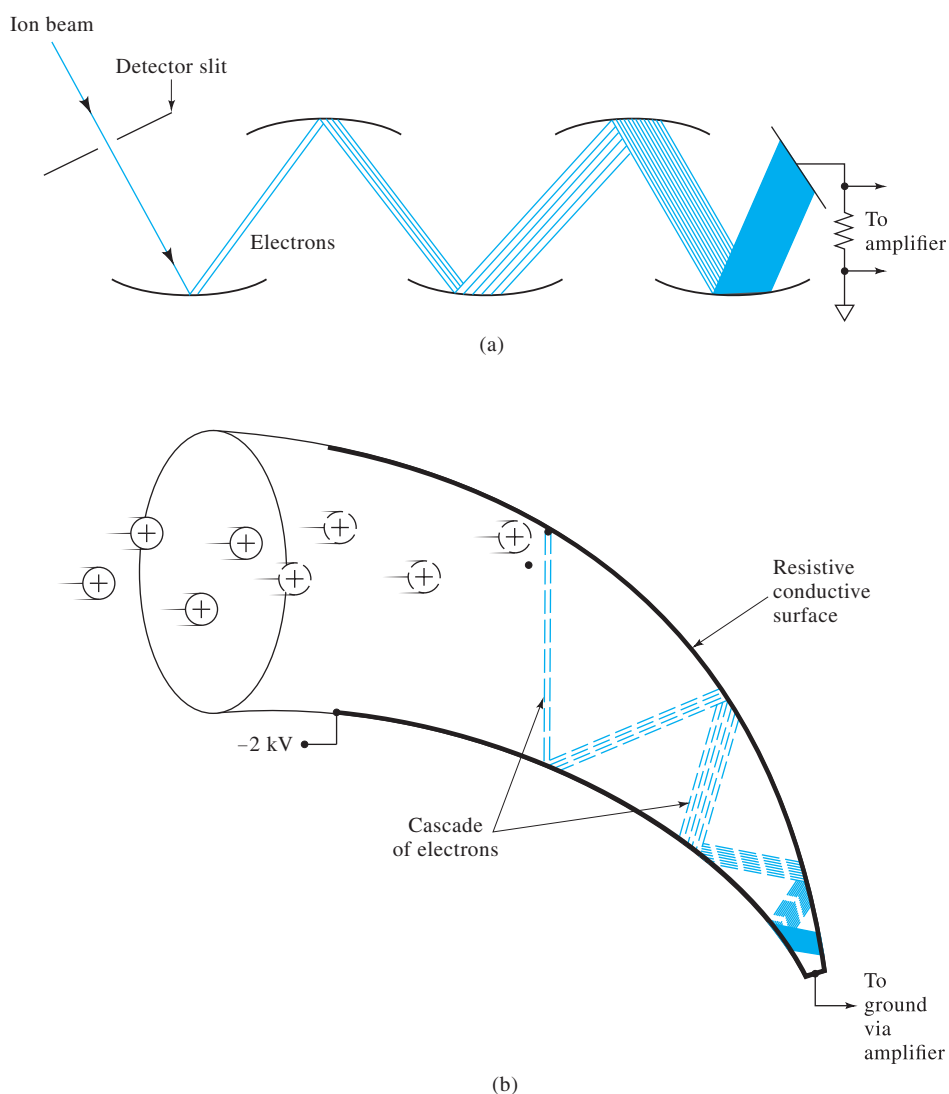


FIGURE 11-2 (a) Discrete-dynode electron multiplier. Dynodes are kept at successively higher voltages via a multistage voltage divider. (b) Continuous-dynode electron multiplier. (Adapted from J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., p. 141, Chichester, UK: Wiley 2007. With permission.)

In general, electron multipliers are rugged and reliable and are capable of providing high-current gains and nanosecond response times. These transducers can be placed directly behind the exit slit of a magnetic sector mass spectrometer, because the ions reaching the transducer usually have enough kinetic energy to eject electrons from the first stage of the device. Electron multipliers can also be used with mass analyzers that use low-kinetic-energy ion beams (that is, quadrupoles), but in these applications the ion beam exiting the analyzer is accelerated to several thousand electron volts prior to striking the first stage.

The Faraday Cup

Figure 11-3 is a schematic of a Faraday cup collector. The transducer is aligned so that ions exiting the analyzer strike the collector electrode. This electrode is surrounded by a cage that prevents the escape of reflected ions and ejected secondary electrons. The collector electrode is inclined with respect to the path of the entering ions so that particles striking or leaving the electrode are reflected from the entrance to the cup. The collector electrode and cage are connected to ground through a large resistor. The charge of the positive ions striking the plate is neutralized by a flow of electrons from ground through the resistor. The resulting voltage drop across the resistor is amplified by a high-impedance amplifier. The response of this transducer is independent of the energy, the mass, and the chemical nature of the ion. The Faraday cup is inexpensive and simple mechanically and electrically; its main disadvantage is the need for a high-impedance amplifier, which limits the speed at which a spectrum can be scanned (see Section 3B-4). The Faraday cup transducer is also less sensitive than an electron multiplier, because it provides no internal amplification.

Array Transducers

The analogy between optical spectrometers and mass spectrometers extends to detection systems. As we have seen, array transducers have revolutionized optical spectrometry for systems in which dispersed electromagnetic radiation is focused at a plane or curved surface. An array transducer positioned in the focal plane of a mass spectrometer (see Figure 11-11) provides several significant advantages over single-channel detection; the most important is the simultaneous detection of multiple resolution

elements. Others include greater duty cycle, improved precision using ratio measurements and internal standards, and improved detection of fast transient signals.

Microchannel Plates. One type of array transducer for mass spectrometry is the electrooptical ion detector (EOID) shown in Figure 11-4a. The key element in the EOID is the microchannel electron multiplier, or microchannel plate, which is referred to in some optical applications as an image intensifier. The design and basic principles of operation of the microchannel plate are illustrated in Figure 11-4b. The plate consists of an array of tiny tubes (diameters as small as 6 μm) made of lead glass. Metallic electrodes are deposited on both sides of the array, and a voltage of about 1500 V is applied between the ends of the channels. Each tube acts as an electron multiplier so that for every particle that strikes the inside wall of a channel, 1000 electrons emerge from the other end of the channel. If greater gain is required, two or more microchannel plates may be cascaded to provide signals appropriate for the experiment.

In the EOID detector of Figure 11-4a, a phosphorescent screen is placed behind the microchannel plate so that the cascade of electrons produces a flash of light that is directed to an optical array detector via fiber optics. The data collected from the array detector then provide two-dimensional resolution of ions appearing at the focal plane of a mass spectrometer.

Micro-Faraday Array Transducer. Micro-Faraday array transducers show promise for the simultaneous detection of ions at the focal plane of Mattauch-Herzog-type mass spectrometers and other similar applications.⁵ An image of the prototype Faraday cup array is shown in Figure 11-5. The device contains gold Faraday electrodes bonded to capacitive transimpedance amplifiers. Each amplifier integrates the charge on the electrode, and the output voltage is switched via a multiplexer to a data system for analog-to-digital conversion. The elements in the image are 145 μm wide, but in principle, they can be as narrow as 10 μm . Faraday-strip array detectors have also been developed containing 512 detector elements in the third generation instrument and 1696 detectors in the fourth generation.⁶ The latter has a linear dynamic range of at least nine orders of magnitude.

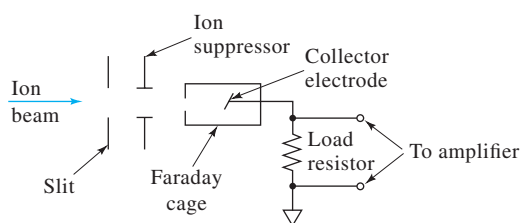
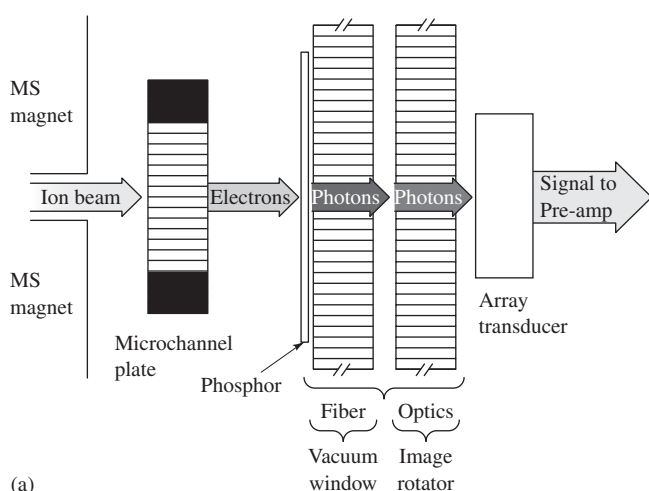


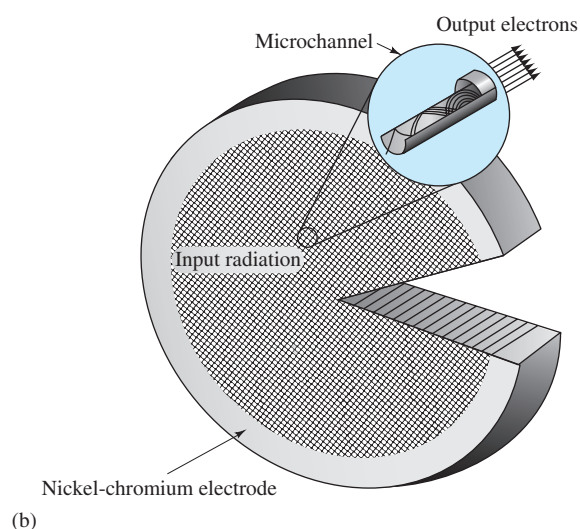
FIGURE 11-3 Faraday cup detector. The voltage on the ion suppressor plates is adjusted to minimize differential response as a function of mass.

⁵A. K. Knight IV, R. P. Sperline, G. M. Hieftje, E. Young, C. J. Barinaga, D. W. Koppenaal, and M. B. Denton, *Int. J. Mass Spectrom.*, **2002**, 215, 131, DOI: 10.1016/S1387-3806(02)00528-6; J. H. Barnes IV, G. D. Schilling, G. M. Hieftje, R. P. Sperline, M. B. Denton, C. J. Barinaga, D. W. Koppenaal, *J. Am. Soc. Mass Spectrom.*, **2004**, 15, 769–776, DOI: 10.1016/j.jasms.2004.01.010; J. H. Barnes IV, G. D. Schilling, R. P. Sperline, M. B. Denton, E. T. Young, C. J. Barinaga, D. W. Koppenaal, G. M. Hieftje, *Anal. Chem.*, **2004**, 76, 2531–2536, DOI: 10.1021/ac030337u; J. H. Barnes IV et al., *J. Anal. At. Spectrom.*, **2004**, 19, 751–756, DOI: 10.1039/b311353a; G. D. Schilling et al., *Anal. Chem.*, **2006**, 78, 4319, DOI: 10.1021/ac052026k.

⁶A. A. Rubinshtein, et al., *J. Anal. At. Spec.*, **2010**, 25, 735, DOI: 10.1039/B913118C; J. A. Felton, et al., *J. Anal. At. Spec.*, **2011**, 26, 300, DOI: 10.1039/C0JA00087F.



(a)



(b)

FIGURE 11-4 (a) Electrooptical ion detector. Ions strike the microchannel plate, which produces a cascade of electrons that strike the phosphorescent screen. Radiation from the screen passes to the array transducer via fiber optics, and the transducer then converts the optical signal to an electrical signal for processing. (Reprinted with permission from J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., p. 145, Chichester, UK: Wiley 2007. With permission.) (b) Microchannel plate. The tiny channels, which are oriented at a small angle from the normal of the array surface, are individual electron multipliers. Particles striking the inner surface of a channel cascade through the channel and produce about 1000 electrons for every incoming particle. (Diagram courtesy of Photonis, Inc.)

Other Types of Detection Systems

Scintillation-type transducers consist of a crystalline phosphor dispersed on a thin aluminum sheet that is mounted on the window of a photomultiplier tube. When ions (or electrons produced when the ions strike a cathode) impinge on the phosphor, they produce flashes of light that are detected by the photomultiplier. A specialized version of this type of device is the Daly

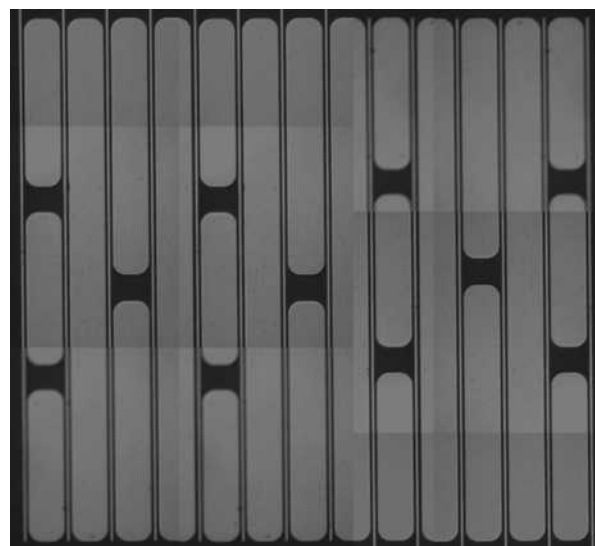


FIGURE 11-5 A micro-Faraday array ion transducer. The transducer elements are 5, 2.45, and 1.60 mm long and 145 μm wide with 10 μm spacing. There is also a 10- μm guard electrode between each element. (Reprinted from A. K. Knight IV, R. P. Sperline, G. M. Hieftje, E. Young, C. J. Barinaga, D. W. Koppenaal, and M. B. Denton, *Int. J. Mass Spectrom.*, **2002**, 215, 131. Copyright 2002 Wiley.)

detector, which consists of an aluminized cathode in the shape of a knob (the Daly knob) held at a very large negative voltage opposite a scintillation transducer. Analyte ions collide with the cathode producing secondary electrons that are then attracted to the surface of a scintillation transducer. The Daly detector has the advantage that all important components of the system except the knob are located outside the vacuum chamber and so can be serviced without disrupting the vacuum. The detector is especially useful for species with high masses.⁷

Just as in optical spectroscopy, where for many years photographic plates and film were used as photon transducers, photographic emulsions are sensitive to energetic ions, and thus they were the original and for a time the only transducers for mass spectrometry. This type of detection is well suited to the simultaneous observation of a wide range of m/z values in instruments that focus ions along a plane (Figure 11-11). As electronic detectors have evolved, photographic detection has become rare. Undoubtedly, some obscure applications still require photographic recording of mass spectra, but with the advent of array transducers for mass spectrometry and the potential elimination of the tedium of developing film or plates, this method of detection is likely to become extinct.



Tutorial: Learn more about **transducers for mass spectrometry** at www.tinyurl.com/skoogpia7

⁷J. T. Watson and O. D. Sparkman, *Introduction to Mass Spectrometry*, 4th ed., pp. 145–146, Chichester, UK: Wiley, 2007.

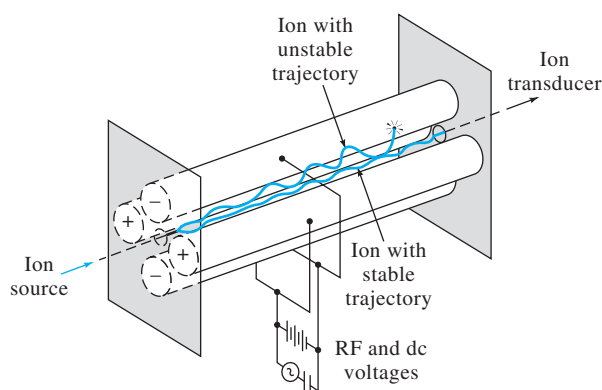


FIGURE 11-6 A quadrupole mass spectrometer.

11B-2 Quadrupole Mass Analyzers

The most common type of mass spectrometer used in atomic mass spectroscopy is the quadrupole mass analyzer shown in Figure 11-6. This instrument is more compact, less expensive, and more rugged than most other types of mass spectrometers. It also has the advantage of high scan rates so that an entire mass spectrum can be obtained in less than 100 ms.⁸

The heart of a quadrupole instrument is the four parallel cylindrical (originally hyperbolic) rods that serve as electrodes. Opposite rods are connected electrically, one pair being attached to the positive side of a variable dc source and the other pair to the negative terminal. In addition, variable radio-frequency ac voltages, which are 180° out of phase, are applied to each pair of rods. To obtain a mass spectrum with this device, ions are accelerated into the space between the rods by a potential difference of 5 to 10 V. Meanwhile, the ac and dc voltages on the rods are increased simultaneously while maintaining their ratio constant. At any given moment, all of the ions except those having a certain m/z value strike the rods and are converted to neutral molecules. Only ions having a limited range of m/z values reach the transducer.

Note that a quadrupole mass spectrometer is more analogous to an optical variable-band filter photometer than to an optical spectrometer in which a grating simultaneously disperses a spectrum of electromagnetic radiation. For this reason, the device is sometimes referred to as a mass filter rather than a mass analyzer.

Ion Trajectories in a Quadrupole

To understand the filtering capability of a quadrupole, we need to consider the effect of the dc and ac voltages on the trajectory of ions as they pass through the channel between the rods.⁹ Let

us first focus on the pair of positive rods, which are shown in Figure 11-7 as lying in the xz plane. In the absence of a dc voltage, ions in the channel will tend to converge in the center of the channel during the positive half of the ac cycle and will tend to diverge during the negative half. This behavior is illustrated at points A and B in the figure. If during the negative half cycle an ion strikes the rod, the positive charge will be neutralized, and the resulting molecule will be carried away. Whether a positive ion strikes the rod depends on the rate of movement of the ion along the z -axis, its mass-to-charge ratio, and the frequency and magnitude of the ac signal.

Now let us consider the effect of a positive dc voltage that is superimposed on the ac signal. From Newtonian physics, the momentum of ions of equal kinetic energy is directly proportional to the square root of mass. It is therefore more difficult to deflect a heavier ion than to deflect a lighter one. If an ion in the channel is heavy or the frequency of the ac voltage is large, the ion will not respond significantly to the alternating voltage and will be influenced largely by the dc voltage. Under these circumstances, the ion will tend to remain in the space between the rods. In contrast, if the ion is light or the frequency is low, the ion may collide with the rod and be eliminated during the negative excursion of the ac voltage. Thus, as shown in Figure 11-8a, the pair of positive rods forms a high-pass mass filter for positive ions traveling in the xz plane.

Now let us turn to the pair of rods that are maintained at a negative dc voltage. In the absence of the ac field, all positive ions will tend to be drawn to the rods, where they are annihilated. For the lighter ions, however, this movement may be offset by the excursion of the ac voltage. Thus, as shown in Figure 11-8b, the rods in the yz plane operate as a low-pass filter.

For an ion to travel through the quadrupole to the detector, it must have a stable trajectory in both the xz and yz planes. Thus, the ion must be sufficiently heavy so that it will not be eliminated by the high-mass filter in the xz plane and must be

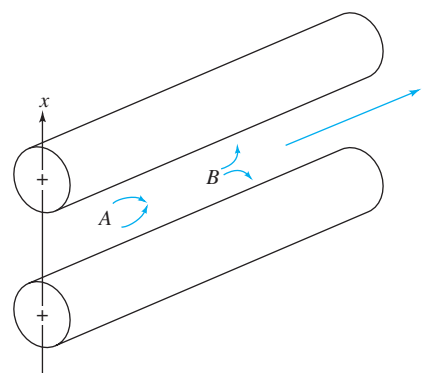


FIGURE 11-7 Operation of a quadrupole in the xz plane. A: ions are focused toward the z -axis; B: ions are attracted toward the x rods.

⁸P. E. Miller and M. B. Denton, *J. Chem. Educ.*, **1986**, 63, 617, DOI: 10.1021/ed063p617.

⁹The discussion of quadrupole mass filters that follows is largely based on P. E. Miller and M. B. Denton, *J. Chem. Educ.*, **1986**, 63, 617, DOI: 10.1021/ed063p617. See also R. E. Marchand and R. J. Hughes, *Quadrupole Storage Mass Spectrometry*, New York: Wiley, 1989; J. J. Leary and R. L. Schmidt, *J. Chem. Educ.*, **1996**, 73, 1142, DOI: 10.1021/ed073p1142.

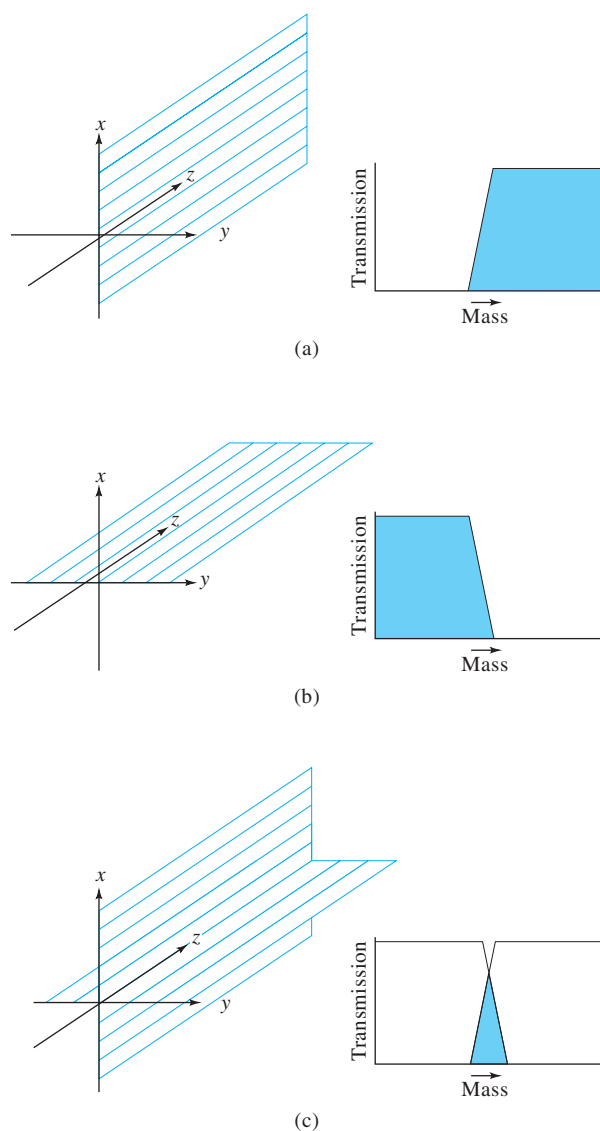


FIGURE 11-8 Quadrupole acts as (a) a high-pass mass filter in the xz plane, (b) a low-pass mass filter in the yz plane, and (c) a narrow-band filter when high-pass and low-pass filters are both in operation. (Reprinted with permission from P. E. Miller and M. B. Denton, *J. Chem. Educ.*, **1986**, 63, 619, Figures 6, 7, and 8. Copyright 1986 Journal of Chemical Education.)

sufficiently light so that it will not be removed by the low-mass filter in the yz plane. Therefore, as is shown in Figure 11-8c, the entire quadrupole transmits a band of ions that have a limited range of m/z values. The center of this band can be varied by adjusting the ac and dc voltages.

Scanning with a Quadrupole Filter

The differential equations necessary to describe the behavior of ions of different masses in a quadrupole are complex and beyond the scope of this book. These equations reveal, however, that the oscillations of charged particles in a quadrupole

fall into two categories: (1) those in which the amplitudes of the oscillations are finite and (2) those in which the oscillations grow exponentially and ultimately approach infinity. The variables in these equations are mass-to-charge ratio, the dc voltage, the frequency and magnitude of the ac voltage, and the distance between the rods. The resolution of a quadrupole is determined by the ratio of the ac to dc voltage and becomes a maximum when this ratio is just slightly less than 6. Thus, quadrupole spectrometers are operated at a constant voltage ratio of this value.

To scan a mass spectrum with a quadrupole instrument, the ac voltage V and the dc voltage U are increased simultaneously from zero to some maximum value while their ratio is maintained at slightly less than 6. The changes in voltage during a typical scan are shown in Figure 11-9. The two diverging straight lines show the variation in the two dc voltages as a function of time. The time for a single sweep is a few milliseconds. While the dc voltages are varied from 0 to about ± 250 V, the ac voltages increase linearly from zero to approximately 1500 V. Note that the two ac voltages are 180° out of phase.

Quadrupole mass spectrometers with ranges that extend up to 3000 to 4000 m/z are available from several instrument manufacturers. These instruments easily resolve ions that differ in mass by one unit. Generally, quadrupole instruments are equipped with a circular aperture, rather than a slit (see Figure 11-6), to introduce the sample into the dispersing region. The aperture provides a much greater sample throughput than can be tolerated in magnetic sector instruments, in which resolution is inversely related to slit width.



Simulation: Learn more about **quadrupole mass analyzers** at www.tinyurl.com/skoogpia7

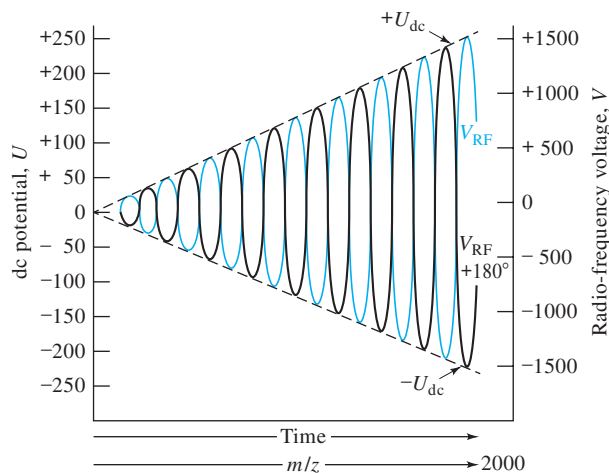


FIGURE 11-9 Voltage relationships during a mass scan with a quadrupole analyzer.

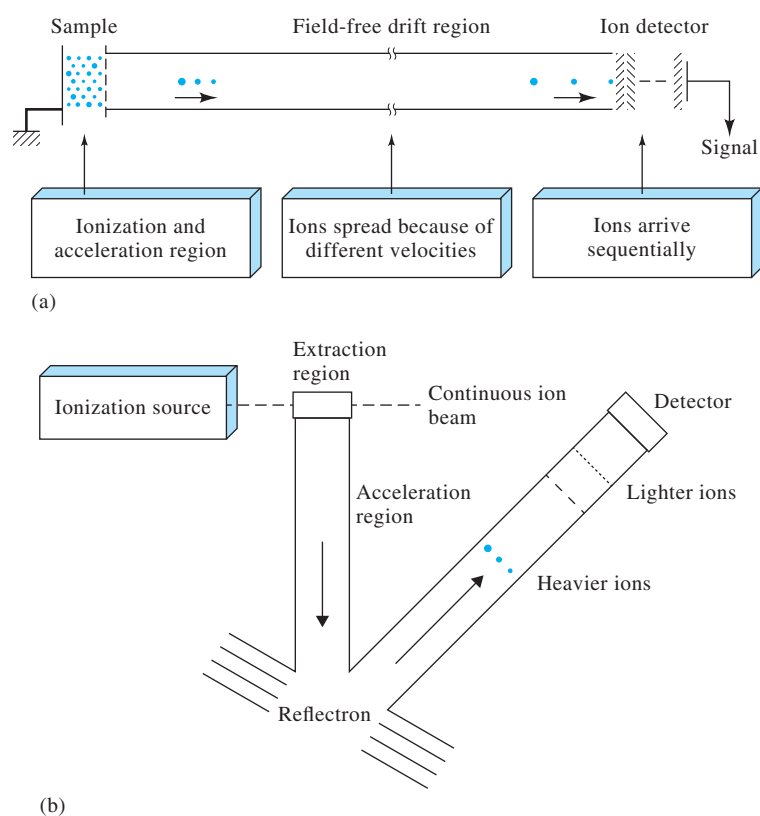


FIGURE 11-10 (a) Principle of a linear TOF mass spectrometer. A spatially tightly bunched group of ions is accelerated into the drift tube where separation occurs because of different velocities. The ions then arrive at the detector sequentially with lighter ions arriving before heavier ions. (Adapted from A. H. Verbueken, F. J. Bruynseels, R. Van Grieken, and F. Adams, in *Inorganic Mass Spectrometry*, p. 186, F. Adams, R. Gijbels, and R. Van Grieken, eds., New York: Wiley, 1988. With permission.) (b) Orthogonal acceleration TOF mass spectrometer. The ion beam continuously traverses an extraction region in which an electrostatic pulse injects the ions into the acceleration region. An ion mirror (reflectron) reflects the ions into the flight tube toward the detector.

11B-3 Time-of-Flight Mass Analyzers

In time-of-flight (TOF) instruments, the time required for positive ions to travel from an ionization source to a detector is measured. The ions receive approximately the same kinetic energy during ionization and acceleration. The accelerated particles pass into a field-free *drift tube* 1–2 m in length (Figure 11-10a). Because all ions entering the drift region ideally have the same kinetic energy, their velocities in the field-free region must vary inversely with their masses, with the lighter ions arriving at the detector earlier than the heavier ones. Typical flight times are in the microsecond range for a 1-m flight tube.¹⁰ The equations governing TOF mass spectrometry are given in Section 20C-3. Ions can be produced by bombardment of the sample with

pulses of electrons, secondary ions, or laser-generated photons. An electric field pulse is used to accelerate the ions.

The linear TOF mass spectrometer has relatively low resolving power that is limited by flight-tube length, acceleration field strength, and the spatial and velocity distributions of the ion packet produced by the ionization source. Use of an ion mirror, or reflectron, to reflect ions can compensate for differences in kinetic energy because more energetic ions penetrate deeper into the reflectron and take a slightly longer path than less energetic ions. This results in ions of the same m/z , but with differing kinetic energies, arriving at the detector at essentially the same time, thus improving the resolving power. Also, with a reflectron, a longer flight path can be achieved in a shorter distance. Several different geometries of reflectron mass spectrometers have been proposed. A very popular design is the right angle or orthogonal acceleration geometry shown in Figure 11-10b. In this system, a continuous beam of ions is produced and extracted by a pulsed

¹⁰W. N. Delgass and R. G. Cooks, *Science*, **1987**, 235, 545.

electric field into one arm of the flight tube. The design minimizes energy dispersion and provides high resolving power. The reflectron mass spectrometer is also capable of acquiring spectra very rapidly (μs) with good sensitivity.

The transducer in a TOF mass spectrometer is usually a microchannel plate electron multiplier whose output is displayed on the vertical deflection plates of an oscilloscope or on the y -axis of a fast xy recording system. The y -axis is synchronized with the accelerator pulses. Because typical flight times are in the ns to μs range, digital data acquisition requires extremely fast electronics.

The inferior resolution and reproducibility of TOF mass analyzers often make them less satisfactory relative to quadrupole and magnetic analyzers for many applications. ICP-TOF mass spectrometers, for example, typically are an order of magnitude poorer in sensitivity and in detection limits than comparable quadrupole systems.¹¹ Several advantages partially offset these limitations, however, including simplicity and ruggedness, ease of accessibility to the ion source, virtually unlimited mass range, and rapid data-acquisition rate. Several instrument manufacturers now offer TOF instruments, but they are less widely used than are quadrupole mass spectrometers.

Recently, a new type of mass spectrometer, called a distance-of-flight instrument, has been introduced and characterized.¹² In this type of spectrometer, the m/z values of the ions produced

are determined by the distance the ions travel in a fixed time interval. The mass spectrum is calculated from the instrument response function and the positions of the ions obtained with a position-sensitive detector. The distance-of-flight technique has shown the ability to detect ions *simultaneously* over a wide range of m/z values.¹³

11B-4 Double-Focusing Analyzers

As shown in Figure 11-11, a double-focusing mass spectrometer contains two devices for focusing a beam of ions: an *electrostatic analyzer* and a *magnetic sector analyzer*. In this instrument, the ions from a source are accelerated through a slit into a curved electrostatic field that focuses a beam of ions having a narrow band of kinetic energies onto a slit that leads to a curved magnetic field. In this field, the lightest ions are deflected the most and the heaviest the least. The dispersed ions then fall on a photographic plate or an array transducer and are thus recorded. A detailed discussion of how the electrostatic and magnetic analyzers function is in Section 20C-3. As shown in Table 11-1, analyzers of this type are used in several types of atomic mass spectrometry.



Simulation: Learn more about **time-of-flight mass analyzers** at www.tinyurl.com/skoogpia7



Simulation: Learn more about **double-focusing mass analyzers** at www.tinyurl.com/skoogpia7

¹¹For a review of the principles and applications of mass analyzers for ICP-TOF mass spectrometry, see S. J. Ray and G. M. Hieftje, *J. Anal. At. Spectrom.*, **2001**, 16, 1206, DOI: 10.1039/b102857j.

¹²C. G. Enke and G. S. Dobson, *Anal. Chem.*, **2007**, 79, 8650, DOI: 10.1021/ac070638u; A. W. G. Graham, S. J. Ray, C. G. Enke, C. J. Barinaga, D. W. Koppenaal, and G. M. Hieftje, *J. Am. Soc. Mass Spectrom.*, **2011**, 22, 110, DOI: 10.1007/s13361-010-0005-8.

¹³A. Gundlach-Graham, et al., *J. Am. Soc. Mass Spectrom.*, **2013**, 24, 1736, DOI: 10.1007/s13361-013-0718-6.

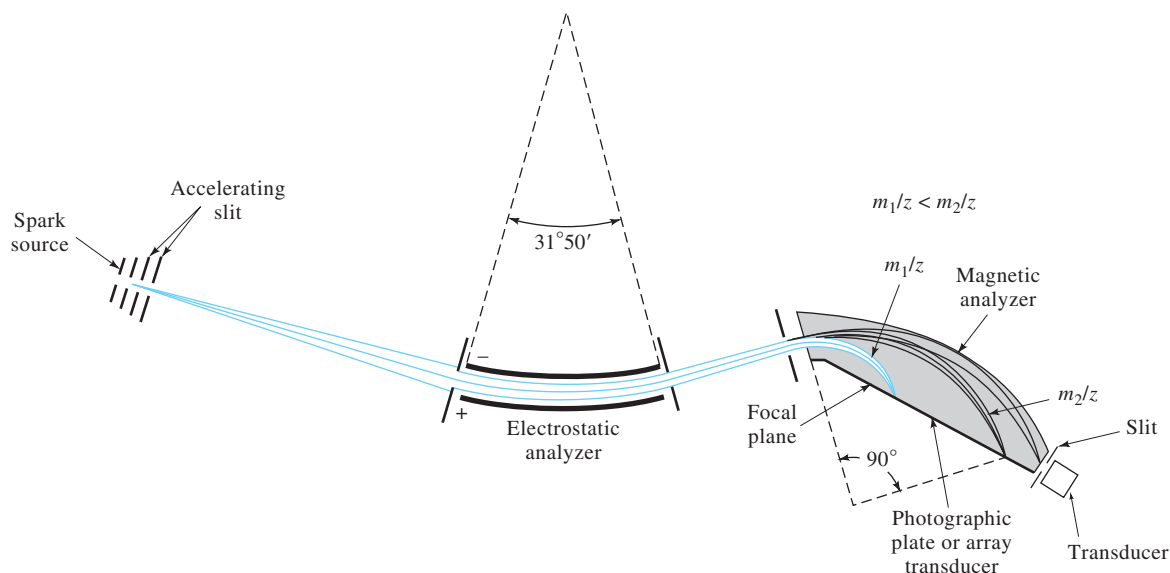


FIGURE 11-11 Mattauch-Herzog-type double-focusing mass spectrometer. Resolution $> 10^5$ has been achieved with instruments based on this design.

11C INDUCTIVELY COUPLED PLASMA MASS SPECTROMETRY

Since its introduction in the early 1980s, ICPMS has grown to be one of the most important techniques for elemental analysis because of its low detection limits for most elements, its high degree of selectivity, and its reasonably good precision and accuracy.¹⁴ In these applications an ICP torch serves as an atomizer and ionizer. For solutions, sample introduction is accomplished by a conventional or an ultrasonic nebulizer. For solids, one of the other sample-introduction techniques discussed in Section 8C-2, such as spark, laser ablation, or glow discharge, are used. Commercial versions of instruments for these various techniques have been on the market since the 1980s. In these instruments, positive metal ions, produced in a conventional ICP torch, are sampled through a differentially pumped interface linked to a mass analyzer, usually a quadrupole. Spectra produced in this way, which are remarkably simple compared with conventional ICP optical spectra, consist of a simple series of isotope peaks for each element present. These spectra are used to identify the elements present in the sample and for their quantitative determination. Usually, quantitative analyses are based on calibration

curves in which the ratio of the ion count for the analyte to the count for an internal standard is plotted as a function of concentration. Analyses can also be performed by the isotope dilution technique, which is described in detail in Section 32D.

11C-1 Instruments for ICPMS

Figure 11-12 shows schematically the components of a commercial ICPMS system.¹⁵ A critical part of the instrument is the interface that couples the ICP torch, which operates at atmospheric pressure, with the mass spectrometer that requires a pressure of less than 10^{-2} Pa (10^{-4} torr). This coupling is accomplished by a differentially pumped interface coupler that consists of a sampling cone, which is a water-cooled nickel cone with a small orifice (<1.0 mm) in its center. The hot plasma gas is transmitted through this orifice into a region that is maintained at a pressure of about 1 torr by means of a mechanical pump. In this region, rapid expansion of the gas occurs, which results in its being cooled. A fraction of the gas in this region then passes through a small hole in a second cone called a skimmer and into a chamber maintained at the pressure of the mass spectrometer. Here, the positive ions are separated from electrons and molecular species by a negative voltage. They are then accelerated and focused by a magnetic ion lens onto the entrance orifice of a quadrupole mass analyzer.

¹⁴For detailed descriptions of this technique, see R. A. Thomas, *Practical Guide to ICP-MS*, 3rd ed, Boca Raton, FL: CRC Press, 2013; *Inductively Coupled Plasma Spectrometry and Its Applications*, 2nd ed., S. J. Hill, ed., Hoboken, NJ: Wiley-Blackwell, 2007; H. E. Taylor, *Inductively Coupled Plasma-Mass Spectrometry: Practices and Techniques*, San Diego, CA: Elsevier, 2001; *Inductively Coupled Plasma Mass Spectrometry*, A. E. Montaser, ed., Hoboken, NJ: Wiley-VCH, 1998.

¹⁵For a review of instrumentation and developments in ICPMS, see N. H. Bings, A. Bogaerts, and J. A. C. Broekaert, *Anal. Chem.*, **2013**, 85, 670, DOI: 10.1021/ac3031459.

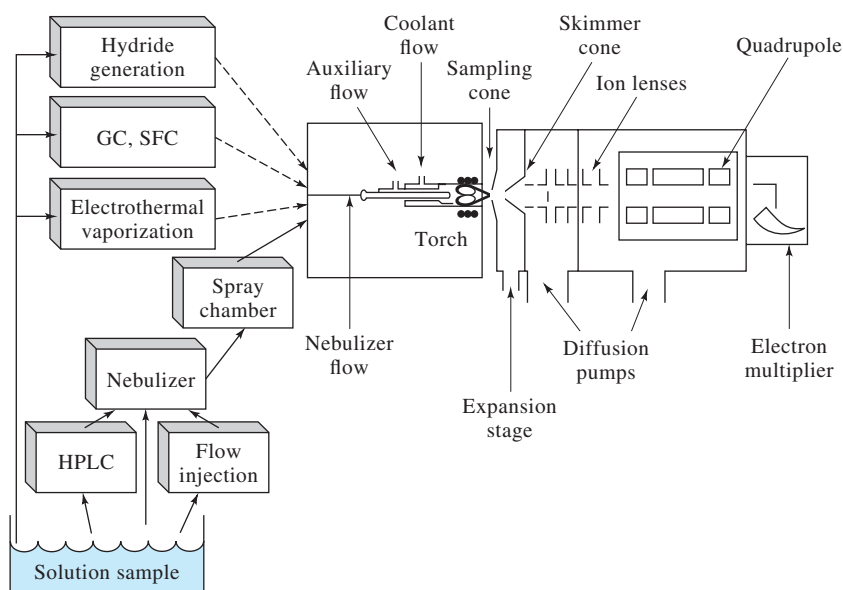


FIGURE 11-12 Schematic of an ICPMS system. Dotted lines show introduction of gaseous samples; solid lines show introduction of liquid samples. HPLC = high-performance liquid chromatography, SFC = supercritical fluid chromatography. (From N. P. Vela, L. K. Olson, and J. A. Caruso, *Anal. Chem.*, **1993**, 65, 585A, DOI: 10.1021/ac00061a719. Copyright 1993 American Chemical Society.)

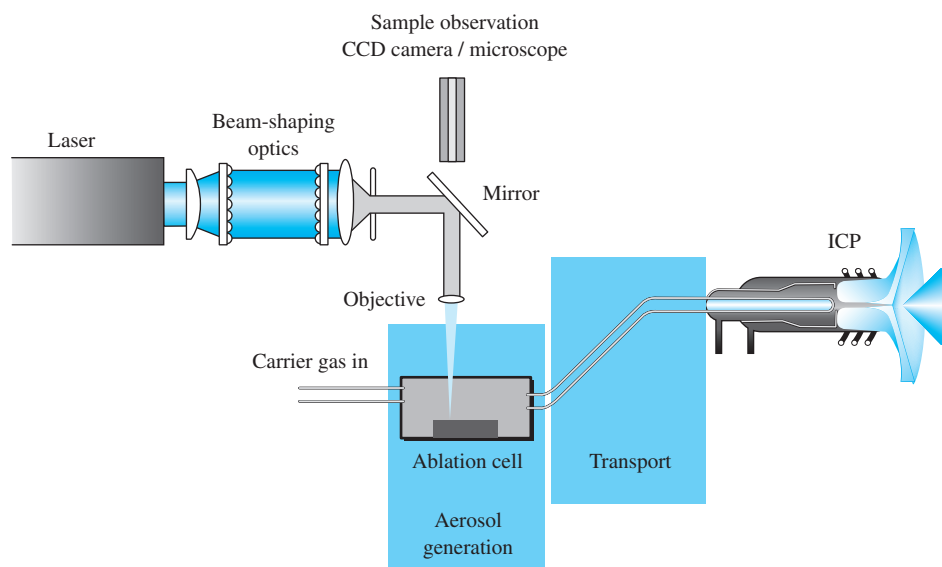


FIGURE 11-13 Laser ablation sample-introduction system for mass spectrometry. The device is similar to most commercially available instruments for spatially resolved elemental analysis. CCD = charge-coupled device. (Reprinted from B. Hattendorf, C. Latkoczy, and D. Gunther, *Anal. Chem.*, **2003**, 75, 341A. Copyright 2003 American Chemical Society.)

Performance specifications for typical commercial atomic mass spectrometers equipped with an ICP torch include a mass range of 3 to 300, the ability to resolve ions differing in m/z by 1, and a dynamic range of six orders of magnitude. More than 90% of the elements in the periodic table have been determined by ICPMS. Measurement times of 10 s per element, with detection limits in the range of 0.1 to 10 ppb for most elements, are achieved. Relative standard deviations of 2% to 4% for concentrations in the middle regions of calibration curves are reported.

In recent years, laser ablation has come into its own as a method of solid-sample introduction with minimal preparation for ICPMS.¹⁶ In this technique, shown in Figure 11-13, a pulsed laser beam is focused onto a few square micrometers of a solid, giving power densities of as great as 10^{12} W/cm². This high-intensity radiation rapidly vaporizes most materials, even refractory ones. A flow of argon then carries the vaporized sample into an ICP torch where atomization and ionization take place. The resulting plasma then passes into the mass spectrometer. Instruments of this type are particularly well suited for semiquantitative analysis (or quantitative analysis if an internal standard can be used) of samples that are difficult to decompose or dissolve, such as geological materials, alloys, glasses, gemstones, agricultural products, urban particulates, and soils. Figure 11-14 is a mass spectrum of a standard rock sample obtained by laser ablation–ICPMS. To obtain this spectrum, the rock was ground into a fine powder and formed

into a small disk under pressure. After ablation with the laser beam, the resulting vapor was introduced into an ICP torch by injector gas flow, where ionization took place. The gaseous ions were then analyzed in a quadrupole mass spectrometer.

Although detection limits for laser ablation in ICPMS depend on a broad range of experimental variables, including instrument variables as well as sample type and quality, typical limits are below the microgram-per-gram level; under certain experimental conditions, for certain elements such as thallium and uranium, detection limits are in the nanogram-per-gram range. Standard deviations on the order of a few percent relative are typical. A major advantage of laser ablation is that the beam can be directed at various locations on the sample to provide mass spectra with spatial resolution.

Accessories that permit use of most of the other ion sources listed in Table 11-1 in conjunction with an ICP torch are available commercially.

11C-2 Atomic Mass Spectra and Interferences

One of the advantages of using mass spectrometric detection with ICPs as opposed to optical detection is that mass spectra are usually much simpler and easier to interpret than corresponding optical spectra. This property is especially true for those elements such as the rare earths that may exhibit thousands of emission

¹⁶See B. Hattendorf, C. Latkoczy, D. Gunther, *Anal. Chem.*, **2003**, 75, 341A, DOI: 10.1021/ac031283r; A. Montaser et al., in *Inductively Coupled Plasma Mass Spectrometry*, A. Montaser, ed., Hoboken, NJ: Wiley-VCH, 1998, pp. 194–218.



Tutorial: Learn more about ICPMS at www.tinyurl.com/skoogpia7

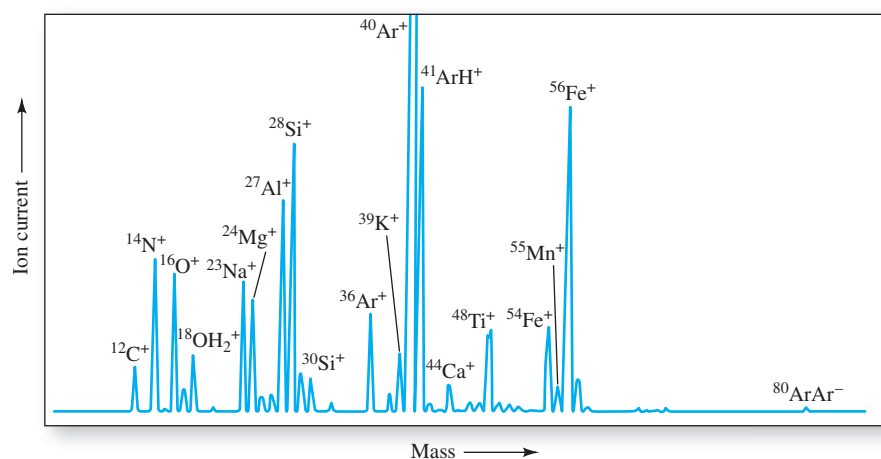


FIGURE 11-14 The mass spectrum of a geological sample obtained by laser ablation/ICP-MS. Ion current on the y-axis is proportional to the number of ions (ion abundance). Mass on the x-axis is proportional to the mass-to-charge ratio for singly charged ions. Major components (%): Na, 1.80; Mg, 3.62; Al, 4.82.1; Si, 26.61; K, 0.37; Ti, 0.65; Fe, 9.53; Mn 0.15. (Reproduced (adapted) from A. L. Gray, *Analyst*, **1985**, 110, 551, DOI: 10.1039/AN9851000551, with permission of the Royal Society of Chemistry.)

lines. Figure 11-15 illustrates this advantage for a solution containing cerium. Figure 11-15a is the optical emission spectrum of a solution containing 100 ppm cerium; the lower plot is the mass spectrum of a solution having a concentration of 10 ppm of the element. The optical spectrum contains a dozen or more strong cerium lines and hundreds of weak lines for the element, all of which are superimposed on a complicated background spectrum. The optical background is made up of molecular bands from atmospheric contaminants, such as NH, OH, N₂, and H₂, and a continuum from the recombination of argon and other ions with electrons. In contrast, the mass spectrum for cerium, shown in Figure 11-15b, is simpler, consisting of isotope peaks for Ce⁺ and peaks for background ions such as Ar⁺, ArO⁺, ArH⁺, O⁺, O₂⁺, and Ar₂⁺. Note that most of the background peaks occur at low *m/z* values, well separated from the Ce⁺ peaks.

Spectra such as that in Figure 11-15b led early workers in the field of ICPMS to have hopes of an interference-free method. Unfortunately, this hope was not realized in further studies, and serious interference problems are sometimes encountered in atomic mass spectrometry just as in optical atomic spectroscopy. Interference effects in atomic mass spectrometry fall into two broad categories: spectroscopic interferences and nonspectroscopic interferences. The latter effects are similar to the matrix effects encountered in optical emission and absorption methods and generally can be managed by means of the various methods described in the two previous chapters.

Spectroscopic Interferences

Spectroscopic interferences occur when an ionic species in the plasma has the same *m/z* values as an analyte ion. Such interferences

fall into four categories: (1) isobaric ions, (2) polyatomic or adduct ions, (3) doubly charged ions, and (4) refractory oxide ions.¹⁷

Isobaric Interferences. Isobaric species are two elements that have isotopes of essentially the same mass. For atomic mass spectrometry with a quadrupole mass spectrometer, isobaric species are isotopes that differ in mass by less than one unit. With higher-resolution instruments, smaller differences can be tolerated.

Most elements in the periodic table have one, two, or even three isotopes that are free from isobaric overlap. An exception is indium, which has two stable isotopes, ¹¹³In⁺ and ¹¹⁵In⁺. The former overlaps with ¹¹³Cd⁺ and the latter with ¹¹⁵Sn⁺. More often, an isobaric interference occurs with the most abundant and thus the most sensitive isotope. For example, the very large peak for ⁴⁰Ar⁺ (see Figure 11-15b) overlaps the peak for the most abundant calcium isotope ⁴⁰Ca⁺ (97%), making it necessary to use the second-most abundant isotope ⁴⁴Ca⁺ (2.1%). As another example, the most abundant nickel isotope, ⁵⁸Ni⁺, suffers from an isobaric overlap by ⁵⁸Fe⁺. This interference can be corrected by measuring the peak for ⁵⁶Fe⁺. From the natural-abundance ratio of the ⁵⁶Fe⁺ isotope to that of ⁵⁸Fe⁺ isotope, the contribution of Fe to the peak at *m/z* 58 can be determined and the

¹⁷For comprehensive discussions of interferences in ICPMS, see H. E. Taylor, *Inductively Coupled Plasma-Mass Spectrometry: Practices and Techniques*, San Diego, CA: Elsevier, 2001; G. Horlick and A. Montaser in *Inductively Coupled Plasma Mass Spectrometry*, A. Montaser, ed., Chap. 7, Hoboken, NJ: Wiley-VCH, 1998; K. E. Jarvis, A. L. Gray, and R. S. Houk, *Handbook of Inductively Coupled Plasma Mass Spectrometry*, Chap. 5, New York: Blackie, 1992.

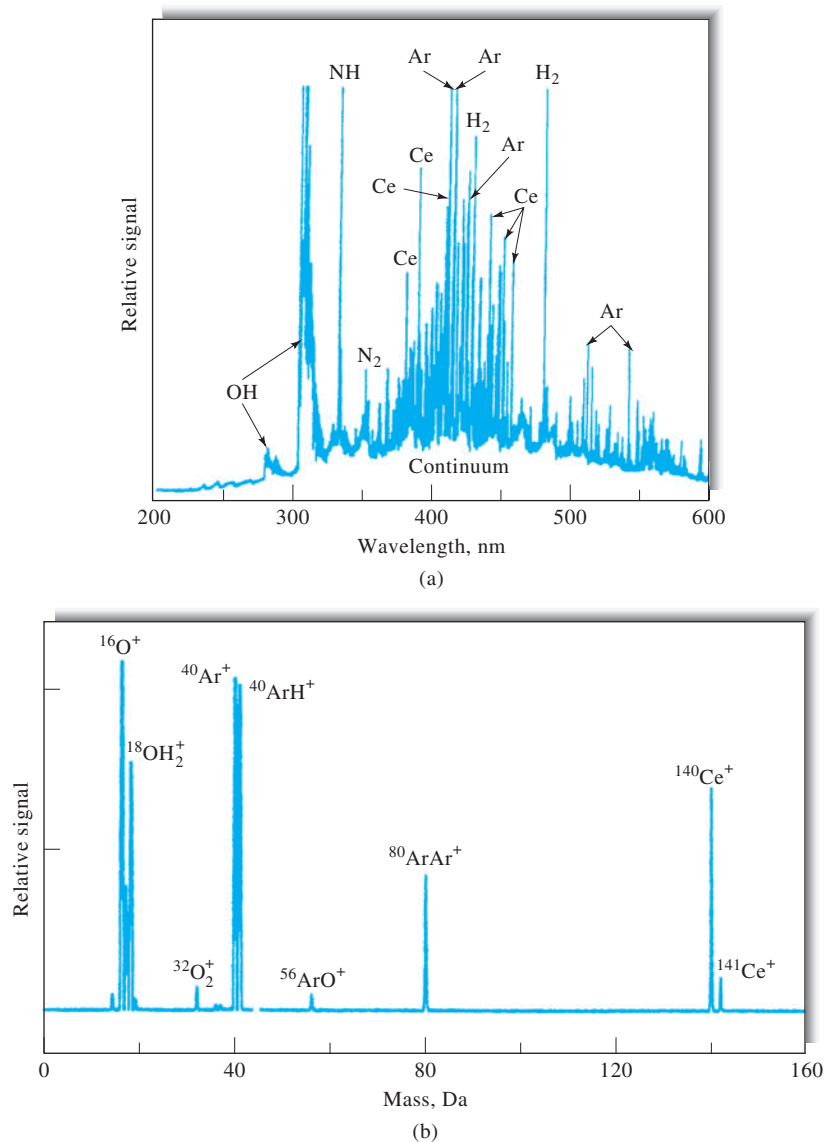


FIGURE 11-15 Comparison of (a) optical ICP spectrum for 100 ppm cerium and (b) mass ICP spectrum for 10 ppm cerium. (Adapted from M. Selby and G. M. Hieftje, *Amer. Lab.*, **1987**, 19, 16.)

correction made. Because isobaric overlaps are exactly predictable from abundance tables, corrections for the problem can be carried out with appropriate software. Some current instruments are capable of automatically making such corrections.

Polyatomic Ion Interferences. An interference problem that is usually more serious than that caused by isobaric elements occurs when polyatomic species form from interactions between species in the plasma and species in the matrix or the atmosphere. These species may then produce several molecular ions that interfere. This type of interference is found usually at m/z values of below about 82. The presence of several of these species is shown in Figure 11-15b. Among the potential interferents are

$^{40}\text{Ar}^+$, $^{40}\text{ArH}^+$, $^{16}\text{O}_2^+$, $^{16}\text{OH}_2^+$, $^{16}\text{OH}^+$, $^{14}\text{N}^+$, and several others. Some of these can cause serious problems. Examples of serious polyatomic interferences include $^{14}\text{N}_2^+$ with $^{28}\text{Si}^+$, NOH^+ with $^{31}\text{P}^+$, $^{16}\text{O}_2^+$ with $^{32}\text{S}^+$, ArO^+ with $^{56}\text{Fe}^+$, and $^{40}\text{Ar}_2^+$ with $^{80}\text{Se}^+$. A blank can be used to correct for some of these. With others, a different analyte isotope may have to be used.

Oxide and Hydroxide Species' Interference. The most serious general class of interferences in ICPMS includes oxides and hydroxides formed from the analyte itself, the matrix components, the solvent, and the plasma gases. Some of these oxides are apparent in Figure 11-15b. The most serious interferences of this type arise from the oxides and hydroxides of the analytes

TABLE 11-2 Calcium Oxide and Hydroxide Species and Other Potential Interferences in the Mass Region for Ni Determination

<i>m/z</i>	Element ^a	Interferences
56	Fe (91.66)	⁴⁰ ArO, ⁴⁰ CaO
57	Fe (2.19)	⁴⁰ ArOH, ⁴⁰ CaOH
58	Ni (67.77), Fe (0.33)	⁴² CaO, NaCl
59	Co (100)	⁴³ CaO, ⁴² CaOH
60	Ni (26.16)	⁴³ CaOH, ⁴⁴ CaO
61	Ni (1.25)	⁴⁴ CaOH
62	Ni (3.66)	⁴⁶ CaO, Na ₂ O, NaK
63	Cu (69.1)	⁴⁶ CaOH, ⁴⁰ ArNa
64	Ni (1.16), Zn (48.89)	³² SO ₂ , ³² S ₂ , ⁴⁸ CaO
65	Cu (30.9)	³³ S, ³² S, ³³ SO ₂ , ⁴⁸ CaOH

From M. A. Vaughan and D. M. Templeton, *Appl. Spectrosc.*, **1990**, *44*, 1685, DOI: 10.1366/0003702904417634. With permission.

^aPercentage natural abundance in parentheses.

and of the matrix components. Almost all of these species tend to form MO⁺ and MOH⁺ ions, where M represents the analyte or matrix element. The peaks from these species have the potential to overlap a peak of one of the analyte ions. For example, the singly charged oxide ions of the five naturally occurring titanium isotopes have masses of 62, 63, 64, 65, and 66. The peaks for these ions have the potential to interfere with the analyte peaks for ⁶²Ni⁺, ⁶³Cu⁺, ⁶⁴Zn⁺, ⁶⁵Cu⁺, and ⁶⁶Zn⁺. As a further example, the potential problems created by the oxide and hydroxide isotopes of calcium in the determination of several metallic species are shown in Table 11-2.

Much research has been focused on reducing oxide and hydroxide formation in plasmas. Oxide formation depends on such experimental variables as injector flow rate, radio-frequency power, sampler skimmer spacing, sample orifice size, plasma gas composition, oxygen elimination, and solvent removal efficiencies. All of these variables can be adjusted to address specific oxide and hydroxide overlap problems.

Some spectroscopic interferences can be reduced or eliminated by using high-resolution mass spectrometers, such as double focusing instruments. Such systems are, however, much more expensive than quadrupole-based instruments.

Matrix Effects

In ICPMS, matrix effects become noticeable at concomitant concentrations of greater than about 500 to 1000 mg/mL. Usually these effects cause a reduction in the analyte signal, although under certain experimental conditions signal enhancement is observed. The matrix effect is quite general in that a high concentration of nearly any concomitant element results in a matrix effect. Generally, matrix effects can be minimized by using more

dilute solutions, by altering the sample introduction procedure, or by separating out the offending species. The effects can also be largely eliminated by the use of an appropriate internal standard, that is, by introduction of an internal standard element that has about the same mass and ionization potential as those of the analyte.

11C-3 Applications of ICPMS

ICPMS can be used for qualitative, semiquantitative, and quantitative determination of one or more elements in a broad range of sample types.

Qualitative and Semiquantitative Applications

Because ICPMS is easily adapted to multielement analyses, it is well suited to the rapid characterization and semiquantitative analysis of various types of naturally occurring and manufactured complex materials. Generally, detection limits are better than those for optical emission ICP and compete with detection limits for electrothermal atomic absorption spectroscopy (ETAAS).

Usually, atomic mass spectra are considerably simpler and easier to interpret than optical emission spectra. This property is particularly important for samples that contain rare earth elements and other heavy metals, such as iron, that yield complex emission spectra. Figure 11-15b illustrates this advantage. This spectral simplicity is further illustrated in Figure 11-16, which is the atomic mass spectrum for a mixture of fourteen rare earth elements that range in atomic mass number from 139 to 175. The optical emission spectrum for such a mixture would be so complex that interpretation would be tedious, time-consuming, and perhaps impossible.

Semiquantitative analysis for one or more of the components in a mixture such as the one that yielded Figure 11-16 is possible by measuring the peak ion current or intensity for a solution having a known concentration of the element in question. The analyte concentration in the sample is then computed by assuming that the ion current is proportional to concentration. Concentrations calculated by this crude but simple procedure are generally accurate to $\pm 100\%$ relative.

Detection Limits

One of the main attractions of ICPMS lies with the lower detection limits attainable with mass spectrometric detection than with optical detection. These limits in many cases equal and sometimes exceed those that can be realized by electrothermal atomic absorption methods. The ICPMS procedure, of course, offers the great advantages of speed and multielement capability. Figure 11-17 compares detection limits for ICPMS with those for ICP optical emission spectroscopy (ICP-OES) and those for ETAAS for selected elements. These data are typical for most other elements in the periodic table. Generally, detection limits with mass spectrometric detection range from 0.02 to 0.7 ppb with the majority of elements in the range of 0.02 to 0.1 ppb.

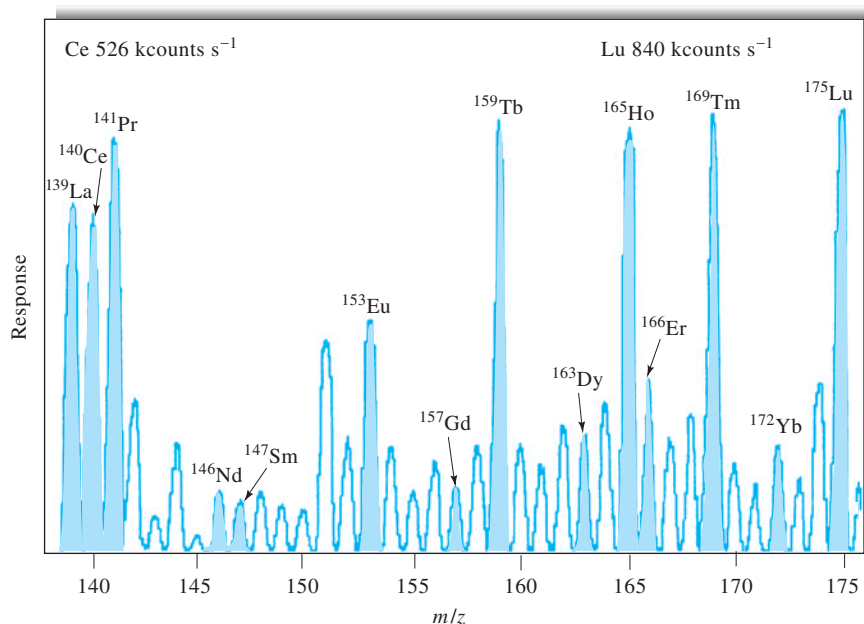


FIGURE 11-16 ICPMS spectrum for the rare earth elements. Solutions contain 1 $\mu\text{g/mL}$ of each element. (From K. E. Jarvis, *J. Anal. Atom. Spectrom.*, **1989**, 4, 563, DOI: 10.1039/ja9890400563. With permission.)

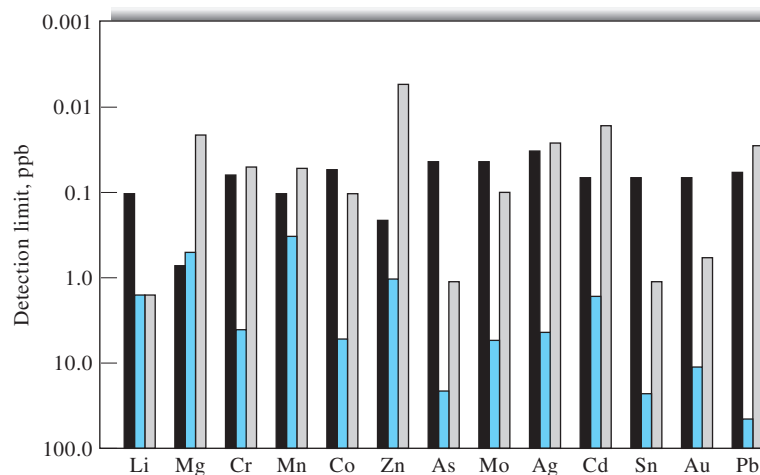


FIGURE 11-17 Detection limits for selected elements by ICPMS (black bars) compared with those for ICP-OES (blue bars) and ETAAS (gray bars), plotted on a logarithmic scale in concentrations of ppb (or $\mu\text{g/L}$). Because ETAAS detection limits are inherently in mass units (pg), they have been converted to concentration by assuming a 20- μL sample. (From M. Selby and G. M. Hieftje, *Amer. Lab.*, **1987**, 19, 16. With permission.)

Quantitative Analyses

The most widely used quantitative method of ICPMS uses the external-standard calibration method (see Section 1D-2). Simple aqueous standards are usually adequate if the unknown solutions are sufficiently dilute—less than 2000 $\mu\text{g/mL}$ of total dissolved solids. With higher concentrations of matrix elements,

attempts are often made to match the matrix elements in the samples with those in the standards. To compensate for instrument drift, instabilities, and matrix effects, an internal standard can be introduced into the standards and the unknowns. The internal standard is an element that is absent from the samples and that has an atomic mass and ionization potential near those

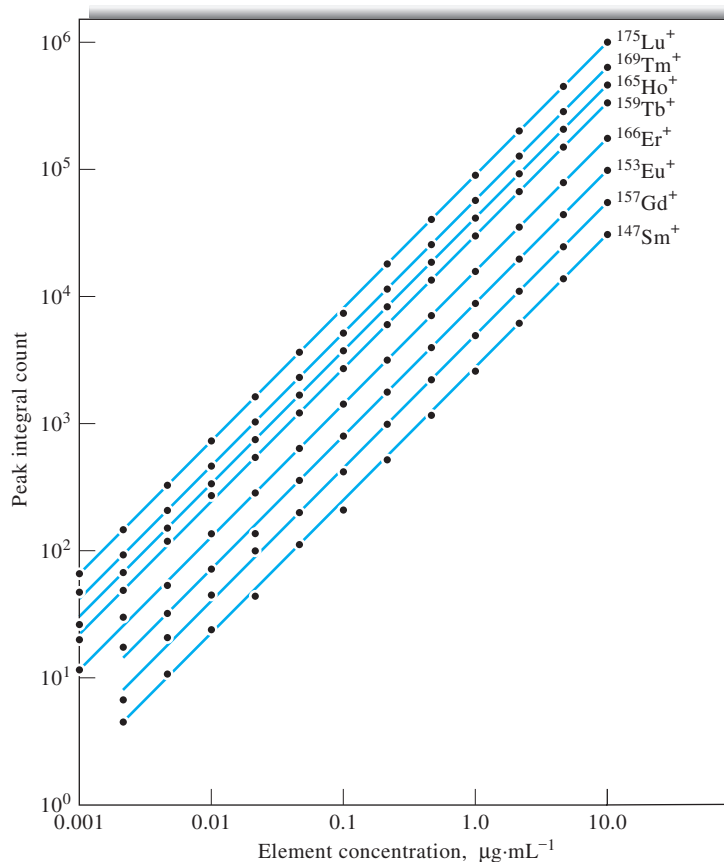


FIGURE 11-18 ICPMS calibration curves for several rare earth elements. Data from S. Abbey, *Geological Survey of Canada*, papers 80-14, 1980 and 83-15, 1983.

of the analytes. Two elements frequently used for internal standards are indium and rhodium. Both produce ions in the central part of the mass range (^{115}In , ^{113}In , and ^{103}Rh) and are not often found naturally occurring in samples. Generally, log-log plots of ion current, ion count, or intensity ratios for sample and internal standards are linear over several orders of magnitude of concentration. Figure 11-18 shows calibration curves for the determination of several rare earth elements. Note that a linear relationship between peak count and concentration is observed over four orders of magnitude in concentration.

Table 11-3 illustrates typical results obtained for the quantitative analysis of trace elements in a natural water sample from the National Bureau of Standards (NBS), now the National Institute of Standards and Technology (NIST). Note that the concentrations are in parts per billion. The thirteen analyses were completed in an elapsed time of only 15 min. For most of the elements the results show remarkably good agreement with the standard sample data.

For the most accurate quantitative analyses, an isotope dilution method may be used. The basis of this technique is the measurement of the change in ratio of signal intensities for two isotopes of an element that result from the addition of a known

quantity, or a spike, of a standard solution that is enriched in one of the isotopes. The principles and mathematical relationships that relate analyte concentration to the quantity of spike added and the change in signal ratio are described in detail in Section 32D, which deals with the use of this technique with radioactive isotopes. The advantages of the isotope dilution method are improved accuracy and immunity to a wide variety of chemical and physical interferences. Its main limitation is the time required to complete this type of analysis.

Isotope Ratio Measurements

The measurement of isotope ratios is of considerable importance in several fields of science and medicine. For example, archeologists and geologists use such data to establish the age of artifacts and various types of deposits. Chemists and clinicians use isotopically enriched materials as tracers in various types of studies. The outcome of these studies is based on isotope ratio measurements. Historically, isotope ratio measurements were based on thermal atomization and ionization in which the sample was decomposed, atomized, and ionized on one or more electrically heated filaments. The ions formed in this way were then directed into a double-focusing mass spectrometer, where isotopic ratios were measured. These measurements were time-consuming but

TABLE 11-3 Quantitative Determination of Trace Elements in a Standard Sample of Water

Element	Ion	NBS ^a	ICPMS ^a	
			Mean ^b	RSD (%) ^b
Beryllium	⁹ Be ⁺	19	21	20
Vanadium	⁵¹ V ⁺	54	52	6
Chromium	⁵² Cr ⁺	17	18	12
Manganese	⁵⁵ Mn ⁺	32	34	5
Cobalt	⁵⁹ Co ⁺	19	21	7
Zinc	⁶⁶ Zn ⁺	69	57	11
Arsenic	⁷⁵ As ⁺	77	76	5
Strontium	⁸⁸ Sr ⁺	243	297	7
Molybdenum	⁹⁸ Mo ⁺	97	134	9
Silver	¹⁰⁷ Ag ⁺	2.8	3.5	16
Cadmium	¹¹⁴ Cd ⁺	10	13	22
Barium	¹³⁸ Ba ⁺	47	74	17
Lead	²⁰⁸ Pb ⁺	27	31	8

^aConcentration in parts per billion.^bBased on ten determinations.From A. R. Date and A. L. Gray, *Spectrochim. Acta*, **1985**, 40B, 115, DOI: 10.1016/0584-8547(85)80015-X.

quite precise, with relative standard deviations on the order of 0.01%. Now, isotope ratios are often determined by ICPMS, with which measurements can often be completed in 5 or 10 minutes. The precision in this case, however, is generally poorer by a factor of 10 to 100. For many purposes, reproducibility of this order is satisfactory.

11D SPARK SOURCE MASS SPECTROMETRY

Spark source mass spectrometry (SSMS)¹⁸ was first introduced in the 1930s as a general tool for multielement and isotope trace analyses. It was not until 1958, however, that the first commercial spark source atomic mass spectrometer appeared on the market. After a period of rapid development in the 1960s, the use of this technique leveled off and then declined with the appearance of ICPMS and some of the other mass spectrometric methods discussed in this chapter. Currently, SSMS is still applied to samples that are not easily dissolved and analyzed by plasma methods.

In SSMS, the atomic constituents of a sample are converted by a high-voltage (~30 kV), radio-frequency spark to gaseous

ions for mass analysis. The spark is housed in a vacuum chamber located immediately adjacent to the mass analyzer. The chamber is equipped with a separate high-speed pumping system that quickly reduces the internal pressure to about 10⁻⁶ Pa after sample changes. Often, the sample serves as one or both electrodes. Alternatively, it is mixed with graphite and loaded into a cup-shaped electrode. The gaseous positive ions formed in the spark plasma are drawn into the analyzer by a dc voltage.

A spark source produces ions with a wide range of kinetic energies. Consequently, double-focusing mass spectrometers are required for mass analysis of the ions. The Mattauch-Herzog type shown in Figure 11-11 is often used. Spark source mass spectrometers have been designed for both photographic and electrical detection systems. The latter are generally based on electron multipliers, which are described in Section 11B-1. When electron multipliers are used with double-focusing instruments, the spectrum is scanned by varying the magnetic field of the magnetic analyzer. Generally, the resolution of the double-focusing mass spectrometers used with spark sources is orders of magnitude greater than that of the quadrupole instruments used with ICPMS. Thus, the number of isobaric interferences is significantly smaller. For example, in the determination of iron in a silicate matrix, as found in samples of rock or glass, there is potential for interference by silicon. Here, the determination is based on the peak for ⁵⁶Fe⁺ ($m/z = 55.93494$), but a detectable peak for ²⁸Si₂⁺ ($m/z = 55.95386$) is also observed. The resolving power of most double-focusing mass spectrometers is sufficient to separate these peaks, but with a quadrupole instrument, an isobaric interference would result from the silicon dimer.

11D-1 Spectra

Like ICP mass spectra, spark source mass spectra are much simpler than atomic emission spectra, consisting of one major peak for each isotope of an element as well as a few weaker lines corresponding to multiply charged ions and ionized oxide and hydroxide species. The presence of these additional ions creates the potential for interference just as in ICPMS.

11D-2 Qualitative Applications

SSMS is a powerful tool for qualitative and semiquantitative analysis. All elements in the periodic table from ⁷Li through ²³⁸U can be identified in a single excitation. By varying data-acquisition parameters, it is possible to determine order of magnitude concentrations for major constituents of a sample as well as for constituents in the parts-per-billion concentration range. Interpretation of spectra does require skill and experience, however, because of the presence of multiply charged species, polymeric species, and molecular ions.

11D-3 Quantitative Applications

A radio-frequency spark is not a very reproducible source over short periods. As a result, to collect good quantitative data, it is

¹⁸See G. Ramendik, J. Verlinden, and R. Gijbels, in *Inorganic Mass Spectrometry*, F. Adams, R. Gijbels, and R. Van Grieken, eds., Chap. 2, New York: Wiley, 1988.

necessary to integrate the output signals from a spark for periods that range from several seconds to hundreds of seconds. The detection system must be capable of electronic signal integration, either in hardware or software. In addition to integration, it is common practice to improve reproducibility by using the ratio of the analyte signal to that of an internal standard as the analytical variable. Often, one of the major elements of the sample matrix is chosen as the standard; alternatively, a fixed amount of a pure compound is added to each sample and each standard used for calibration. In the latter case, the internal standard substance must be absent from the samples. With these precautions, relative standard deviations that range from a few percent to as much as 20% can be realized.

The advantages of SSMS include its high sensitivity, its applicability to a wide range of sample matrices, the large linear dynamic range of the output signal (often several orders of magnitude), and the wide range of elements that can be detected and measured quantitatively.

11E GLOW-DISCHARGE MASS SPECTROMETRY

As shown in Section 8C-2, a glow-discharge source is used as an atomization device for various types of atomic spectroscopy.¹⁹ In addition to atomizing samples, it also produces a cloud of positive analyte ions from solid samples. This device consists of a simple two-electrode closed system containing argon at a pressure of 0.1 to 10 torr. A voltage of 5 to 15 kV from a pulsed dc power supply is applied between the electrodes, causing the formation of positive argon ions, which are then accelerated toward the cathode. The cathode is fabricated from the sample, or alternatively, the sample is deposited on an inert metal cathode. Just as in the hollow-cathode lamp (Section 9B-1), atoms of the sample are sputtered from the cathode into the region between the two electrodes, where they are converted to positive ions by collision with electrons or positive argon ions. Analyte ions are then drawn into the mass spectrometer by differential pumping. The ions are then filtered in a quadrupole analyzer or dispersed with a magnetic sector analyzer for detection and determination.

As mentioned earlier, glow-discharge sources are often used with ICP torches. The glow discharge serves as the atomizer and the ICP torch as the ionizer.

The glow-discharge source appears to be more stable than the spark source and is less expensive to purchase and operate. It is particularly useful for the direct analysis of solid samples. Commercial quadrupole and double-focusing mass spectrometers with glow-discharge sources are now on the market.

¹⁹See N. H. Bings, A. Bogaerts, and J. A. C. Broekaert, *Anal. Chem.*, **2013**, 85, 670, DOI: 10.1021/ac3031459; R. K. Marcus and J. A. C. Broekaert, *Glow Discharge Plasmas in Analytical Spectroscopy*, New York: Wiley, 2003; W. W. Harrison, in *Inorganic Mass Spectrometry*, Chap. 3, F. Adams, R. Gijbels, and R. Van Grieken, eds., New York: Wiley, 1988.

11F OTHER MASS SPECTROMETRIC METHODS

Over the past quarter century, the science and technology of mass spectrometry has grown explosively. In the last section of this chapter we refer briefly to a few important developments in this burgeoning field.

11F-1 Accelerator Mass Spectrometry

In recent years, a variation on the isotope-ratio method called *accelerator mass spectrometry* (AMS) has been developed. In AMS, the target element is first chemically separated from the sample before it is placed in a sample holder in the AMS instrument. The sample element is then bombarded by cesium ions to sputter the analyte element from the sample as negative ions. The analyte ions are then accelerated down a beam tube by a positive potential difference of several million volts, passed through an electron stripper to convert them to positive ions, and accelerated back down the beam tube toward common potential where ion velocities approach a few percent of the speed of light. Using a series of magnetic and electrostatic mass filters, the ion beam containing all isotopes of the analyte element is then separated into separate beams containing the (usually unstable) isotope of interest and other isotopes, and each of the isotopes is counted by a separate detector. As an example, in carbon dating, a single ¹⁴C nucleus can be separated and counted in the presence of as many as 10¹⁵ nuclei of ¹³C and ¹²C.

Although there are a relatively small number of AMS facilities in the world, the technique is invaluable for determining the concentration of an unstable isotope in the presence of a massive amount of a second stable isotope, particularly in radiocarbon dating and the determination of nuclides produced from the interaction of cosmic rays with both extraterrestrial and terrestrial sources. These nuclides include ¹⁰Be, ¹⁴C, ²⁶Al, ³⁶Cl, ⁴¹Ca, and ¹²⁹I.²⁰

11F-2 Methods for Elemental Surface Analysis

In certain areas of science and engineering, the elemental composition of a surface layer of solid that is a few angstroms to a few tens of angstroms thick is of much greater interest than the bulk elemental composition. Fields where surface composition is extremely important include heterogeneous catalysis, corrosion and adhesion studies, embrittlement properties, and studies of the behavior and functions of biological membranes.

Two atomic mass spectrometric methods are often used to determine the elemental composition of solid surfaces: secondary-ion mass spectrometry and laser microprobe mass spectrometry. These techniques are discussed in detail in Sections 21D-1 and 21D-3.

²⁰For several interesting descriptions of AMS facilities, do a Google search on "accelerator mass spectrometry."

» QUESTIONS AND PROBLEMS

*Answers are provided at the end of the book for problems marked with an asterisk.



Problems with this icon are best solved using spreadsheets.

- 11-1** What types of mass spectrometers are used in ICPMS? How do they differ from one another?
- 11-2** Describe the differences between a linear TOF mass spectrometer and an orthogonal TOF instrument? What are the advantages and limitations of these two geometries?
- * **11-3** What function does the ICP torch have in ICPMS?
- * **11-4** What are the ordinate and the abscissa of an ordinary atomic mass spectrum?
- 11-5** Why has ICPMS become an important and widely used analytical method?
- 11-6** Describe the interface between the ICP torch and the mass spectrometer in an ICPMS instrument.
- 11-7** What is laser ablation and how can it be used to sample solids for ICPMS?
- 11-8** What is a distance-of-flight mass spectrometer and how does it differ from a TOF instrument?
- 11-9** Discuss the difference between an isobaric interference and a polyatomic ion interference in ICPMS.
- * **11-10** Why is an internal standard often used in quantitative analysis by ICPMS?
- 11-11** Describe how glow-discharge mass spectrometry is carried out. What are the advantages of this technique?
- 11-12** What is accelerator mass spectrometry? What type of information does this method provide?

Challenge Problem

- 11-13** Lezius has described a B-TOF mass spectrometer, presented the theoretical background for the instrument, and provided early experimental results.²¹
- (a) What is a B-TOF mass spectrometer?
 - (b) Explain how the instrument works.
 - (c) For what types of experiments is the instrument useful?
 - (d) Define the term *start-up energy*.
 - (e) The *deflection velocity* $\dot{x}$ of an ion in the flight tube is given by

$$\dot{x} = \frac{qdB}{m}$$

where $q = ze$, and B and m have their usual significance, and d , shown in Figure 1 of the paper, is the width of the magnetic field region. The deflection velocity is the component of the velocity perpendicular to the initial flight path of the ion. Derive this expression. What is curious about the deflection velocity?

- (f) Show that the deflection x of the ion perpendicular to the initial flight path of the ion is given by

$$x = (l + d/2) \frac{qdB}{mv \cos \alpha}$$

²¹M. Lezius, *J. Mass Spectrom.*, **2002**, 37, 305, DOI: 10.1002/jms.286.

where l is the distance traveled by an ion in its initial direction along the axis of the spectrometer after it leaves the magnetic field region, and v is the velocity of the ion along the deflected path. Why are the distances x and l important in the B-TOF experiment?

- (g) The TOF t_{MD} of an ion to traverse the distance l from its exit from the magnetic field is given by

$$t_{\text{MD}} = \frac{1}{\sqrt{\frac{2E_{\text{tot}}}{m} - x^2}}$$

What is E_{tot} ? There is an error in equation 6 of the paper. Find and explain the error.

- (h) Using the conservation of energy, derive the expression in (g).
- (i) For small deflection angles, $\cos \alpha \approx 1$. What are the consequences of this condition for ions of the same charge and mass?
- (j) Look up some typical values for E_{tot} , l , and v , and use Excel to calculate typical values for t_{MD} for a realistic range of values for m . Produce a plot of t_{MD} versus m .